

# Catapult HLS User View Scheduling Rules

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## Abstract

This document defines a user-visible scheduling contract and modeling methodology for high-level synthesis (HLS). Its practical objective is to support a proof-backed flow in which much of the system-level functional verification and debug is performed on the pre-HLS SystemC model, for designs and implementations that satisfy the stated applicability and conformance conditions. The contract treats message-passing operations, signal I/O, and explicit synchronization through a small set of uniform rules while still allowing important implementation transformations such as loop pipelining, added FSM states, permitted memory-access reordering, direct-input late sampling, and bounded or more complex channel buffering.

The main formal safety guarantee compares observable system behavior: every covered post-HLS RTL behavior has a matching behavior in the prescribed pre-HLS reference model. This supports source-level functional verification without requiring the RTL to preserve every source-level cycle or internal state. The converse claim is deliberately narrower: showing that a particular source execution is reproduced by the RTL requires the additional alignment and witness-selection conditions described later in the document. Deadlock preservation is a separate result for internal message-passing deadlock. Its primary practical route checks one selected complete deterministic serial-issue pre-HLS execution together with finite response bounds for the relevant blocking operations. In the standard qualified-synchronous scope, a checked one-cycle source/RTL correspondence argument can be extended across all reachable implementation cycles. For functional safety properties that are prefix-closed in the Appendix-J finite-ideal sense, the completed-cycle coverage also covers finite observable prefixes within a cycle; the same cycle-by-cycle framework supplies the implementation-wide coverage used by the universal deadlock result. These guarantees remain conditional on the stated channel-capacity, fairness, progress, environment, reset, structural, and implementation-correspondence conditions.

The proofs do not by themselves certify a particular HLS invocation or RTL instance. The applicable source well-formedness, scheduling, channel realization and capacity, reset, environment, progress, and implementation-mapping conditions must be established by the tool or verification flow through documentation, generated certificates, static checks, targeted RTL/interface checks, formal verification, or equivalent evidence. Subject to those conditions, the document provides a Catapult-compatible, tool-neutral basis for industrial HLS verification and for possible scheduling-contract standardization.

## Introduction

HLS loses much of its practical advantage when design teams must return to machine-generated RTL as the primary level for functional verification and debug. RTL verification can be one of the most costly and time-consuming parts of a hardware development flow, and generated RTL may differ substantially in structure and timing from the source model. A useful HLS methodology should therefore let designers and verification engineers reason primarily at the source level while retaining a precise way to qualify the resulting implementation.

This document presents a user-visible scheduling contract intended to provide "no surprises" at process and interface boundaries. It reduces interface scheduling to uniform rules for message-passing operations, signal I/O, and explicit synchronization. A common verification intent can then be reused across the prescribed pre-HLS source reference model and the post-HLS implementation, without requiring the implementation to preserve every source-level cycle or internal state.

Subject to the conditions stated in the main text and formal appendices, the methodology accommodates sequential and combinational logic, pipelined designs, shared and external memories through the stated mapping conditions, permitted RAM-access reordering, direct-input late sampling, finite channel buffering, and localized latency-sensitive protocols. These capabilities reflect requirements encountered across a broad range of industrial HLS designs and are important for achieving competitive hardware quality of results (QOR).

The methodology builds on established verification practices such as SystemVerilog UVM, constrained-random stimulus, and functional and code coverage. The system under test is modeled and verified in SystemC, while implementation-level verification focuses on establishing that the synthesized or otherwise implemented RTL falls within the scope of the applicable theorem instance. When the selected abstract channel boundaries and back-annotated capacities faithfully represent the RTL storage or credit at those boundaries, the pre-HLS model can also be used to assess capacity and throughput effects rather than deferring those questions entirely to RTL simulation.

The formal architecture separates safety from liveness. Safety is largely compositional: design-specific evidence is supplied for individual processes, explicit channel boundaries, named atomic interactions, the environment, and reset behavior, and a generic theorem composes that evidence into a system-level observable trace-compatibility result. Liveness and deadlock preservation have a more global dependency structure and therefore require additional cutpoint-coverage, progress, fairness, drain, environment, and source-side premises.

The most important default deadlock-avoidance rule can be stated plainly. HLS may overlap or shift message-passing operations in time, but it may not reverse the dynamic source order of message-passing calls under the default scheduling contract. It therefore cannot introduce a new deadlock merely by reversing the order of source calls. This rule is not a complete deadlock theorem: overlapped loop iterations, eager issue, finite capacities, and environment behavior can still affect deadlock, and those effects are addressed by the separate liveness machinery, whose primary practical route checks one selected, complete serial-issue execution of the pre-HLS model with total bounded-response coverage (Appendix K.4b and K.4c).

The guarantees remain conditional. The mathematical proofs do not establish that a particular HLS tool invocation or generated RTL instance satisfies the source-side and implementation-side obligations. Those obligations must be discharged by the tool or flow through documentation, generated certificates, lint or static analysis, targeted RTL/interface checks, formal verification, or another

appropriate validation method. The same framework can also be applied to hand-written, modified, or otherwise generated RTL when equivalent evidence is supplied.

Because the rules are expressed as a user-visible, tool-neutral scheduling contract rather than as the internal behavior of a particular HLS implementation, they can also serve as a starting point for standardization, for example within the Accellera Synthesis Working Group. The next section summarizes the guarantees without exposing the internal proof API. A separate Guide to the Formal Appendices appears immediately before Appendix G, Appendix F provides user-oriented modeling guidelines, and Appendix M provides a more detailed Formal Results Overview.

## Formal Guarantees at a Glance

The statements below provide a user-level orientation. The formal definitions, assumptions, scope conditions, and evidence requirements appear in the appendices.

**Implementation safety.** Within the stated scheduling, channel, environment, reset, and qualification scope, the generated RTL does not introduce new observable functional behavior that is absent from the prescribed pre-HLS reference model. This allows most functional verification and debug to remain at the source level when those conditions are established.

**Use in the reverse direction.** A particular pre-HLS simulation is not automatically guaranteed to be reproduced cycle for cycle, or even as the directly corresponding execution, by the RTL. Direct comparison of a selected source execution with RTL requires the alignment and witness-selection conditions described in Appendices G and L.

**Internal message-passing deadlock.** The primary practical method checks one complete deterministic serial-issue execution of the pre-HLS model and verifies finite response bounds for the blocking operations relevant to the proof. For a terminating design, a clean complete execution can provide the required source-side deadlock evidence without exploring alternative scheduler interleavings, provided the tool or verification flow separately establishes that the method applies to the design and implementation. The method is deliberately conservative: it may not apply when a design avoids an internal message-passing deadlock only because communication requests happen to overlap favorably, and testbenches that provide inputs at particular clock times or react to design outputs may require additional evidence or a different deadlock argument. The result is specifically about internal message-passing deadlock; it does not by itself cover barrier stalls, reset-induced stalls, environment-only stalls, starvation outside the modeled internal wait structure, or protocols whose correctness depends on timing behavior outside the method's scope. Qualification may need to include internal message-passing channels as well as the design's external interfaces.

**RTL coverage and qualification.** A clean pre-HLS execution is only one part of the overall verification argument. A claim covering all relevant reachable RTL behavior also requires implementation-side evidence that the generated RTL follows the assumed scheduling, channel, reset, and correspondence rules, and that persistent blocking behavior cannot remain hidden indefinitely in transient internal activity. In the standard qualified-synchronous scope, the formal results allow a checked one-cycle source/RTL correspondence to be established initially and then preserved from one reachable implementation cycle to the next. The safety result needs only the functional-correspondence part of this evidence, while the universal deadlock result requires additional blocking and progress evidence. For functional safety properties that are prefix-closed in the Appendix-J finite-ideal sense, the cycle-by-cycle safety evidence also covers finite observable prefixes within each covered cycle; it is not necessary to construct a separate source/RTL package for every such intra-cycle ideal. A flow that establishes the stronger deadlock argument therefore also supplies the cycle-by-cycle correspondence evidence used by the safety result. This must cover all reachable cycle transitions, not just one observed simulation path. Some of the more advanced deadlock-preservation proof and certificate machinery is not yet fully

mechanized and qualified. Appendices G, I, and J provide the detailed safety and source/RTL correspondence results; Appendix K provides the deadlock results; and Appendix M.7 summarizes the division of verification responsibilities and current framework status.

## Document Goals

This document presents HLS tool scheduling rules and a modeling methodology. The goals are:

1. To provide easy-to-understand rules to HLS users.
2. To ensure precise and consistent rules in both SystemC and C++.
3. To offer rules that are effectively compatible with how Catapult currently operates.
4. To enable the best possible *quality of results* (QOR) in Catapult synthesis.
5. To cover all known user requirements and scenarios.
6. To serve as a suitable starting point for a standardization proposal (e.g., in Accellera SWG).

To illustrate the goals of this document more specifically, consider what an engineer writing a testbench for an HLS model needs to understand about how HLS tools operate. This engineer may be using SystemVerilog UVM or may be writing a testbench in C++/SystemC.

They likely are not an expert in any specific HLS tool (and may not want to be), but their testbench should be usable across both the pre-HLS model and the post-HLS model, subject to the latency-insensitivity, observation-boundary, and witness-selection conditions stated in this document. Thus, it is crucial for the DV engineer to have a precise understanding of how the HLS tool will transform the design while still enabling the resulting RTL behavior to be checked against the prescribed source reference model. This document describes what transformations the HLS tool is allowed to perform so that the pre-HLS and post-HLS models can be effectively verified with a common verification intent, and in the deterministic or annotated RTL-consistent cases with the same directly corresponding testbench execution. The overarching philosophy of the scheduling rules is to present “no surprises” to such a DV engineer, while still giving the HLS tool ample freedom to optimize the design.

The main body of this document is intended to be sufficiently precise and complete to support the Accellera standardization effort while still being understandable to HW architects. The formal appendices are intended to have a very high degree of precision and completeness sufficient to support AI-driven Lean mechanization of the formal theorems.

## Background

HLS tools generate RTL from C++ models. Broadly speaking, this conversion takes a sequential C++ model and turns it into concurrent hardware that maintains the same behavior. HLS tools identify concurrent processes within the C++/SystemC model and then independently synthesize each process. Briefly, some of the techniques that HLS tools use to achieve good HW QOR when synthesizing each process include:

- Optimized scheduling based on the selected silicon target technology.
- Automatic HW pipeline construction according to the user’s specifications.
- Automatic HW resource sharing.
- Automatic scheduling of memory accesses.

The internal behavior of each process is specified by the control and dataflow behavior of the C++ code within the process. However, the external communication that each process has with other processes and HW blocks is specified via IO operations that are coded within the model. To enable a reliable, scalable, and verifiable HLS flow that generates high quality hardware, the scheduling behavior of these IO operations needs to be precisely handled at all steps of the flow. This document specifies the rules that govern the scheduling behavior for these IO operations within HLS models.

These rules are specified with respect to an individual process, but the intent of the rules is to enable reliable and verifiable behavior of large sets of interacting processes operating as a system in real-world designs. (Appendix G provides the formal trace-compatibility guarantees between the prescribed source reference model and the post-HLS system, including the unrestricted Post  $\rightarrow$  Ref direction and the restricted annotated RTL-consistent / witness-selected reverse use.)

By default, HLS tools can insert additional clock cycles (or latency) anywhere within a process -- for example, when pipelining a loop or to enable HW resource sharing. The overall approach used in this document is to make the entire design and testbench system *latency insensitive* to the maximum extent possible, while still fully enabling key HW optimizations. In addition, the overall approach enables the pre-HLS system simulation to be capacity-accurate and throughput-accurate with respect to the post-HLS RTL system.

In some cases, designs or testbenches may use protocols which are latency-sensitive. These situations can be handled by isolating the latency-sensitive portions to small, self-contained parts of the design or testbench, and then keeping the rest of the design and testbench latency insensitive. See Appendix D for more information.

In many cases the overall design will need to satisfy end-to-end latency requirements. For these designs it is still highly advantageous to use a latency-insensitive modeling approach and verify in the post-HLS model that the overall design latency requirements have been satisfied, since this is typically easy to do.

This document focuses on sequential HW processes. Combinational HW processes are supported but mostly not discussed since their synthesis is straightforward. All the examples and discussion in this document are for SystemC processes that are sensitive only to a single rising clock edge. (Appendix O discusses support for multiple clock domains.)

This document distinguishes between the following:

1. The *conceptual model* for the scheduling rules.
2. The simulation behavior of a model using the rules in C++ or SystemC.
3. The synthesis of a C++/SystemC model using the rules in an HLS tool such as Catapult.

The goal is to align each of these three cases as closely as possible, so that the user has easy-to-understand rules, while simulation and synthesis work without surprises. However, as we will see, there are practical considerations which may in certain cases cause small deviations from the conceptual model in either simulation or HLS.

A simple real-world example that illustrates the motivation for the scheduling rules is provided in Appendix A of this document.

The examples referred to in this document are available here:

<https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/tree/master>

The most up-to-date version of this document is available here:

[https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib\\_examples/doc/catapult\\_user\\_view\\_scheduling\\_rules.pdf](https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf)

A summary of the scheduling rules is available here:

[https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib\\_examples/doc/brief\\_scheduling\\_rules.txt](https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/brief_scheduling_rules.txt)

A formal architecture guide is available here:

[https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib\\_examples/doc/formal\\_architecture\\_guide.pdf](https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/formal_architecture_guide.pdf)

A white paper describing the motivation for the methodology is available here:

<https://hls.academy/topics/systemc-modeling/verifying-hardware-at-the-source-why-a-user-visible-hls-scheduling-contract-matters-in-an-age-of-ai-generated-design/>

## An Analogy from RTL Synthesis

To better understand the specific purpose of this document, let's consider how RTL synthesis works. Say you have a sequential block that you are modeling in Verilog RTL, and it has an output port coded like:

```
Out1 <= #some_delay new_val;
```

In Verilog simulation, if `some_delay` is less than the clock period of the block, then it will probably not affect the overall cycle-level behavior of the system during simulation. However, if `some_delay` is more than the clock period, it probably will.

During RTL synthesis, all RTL synthesis tools will ignore all delays in the input model, in this case even if `some_delay` is greater than the clock period. Some RTL synthesis tools might give a warning or error for code like the above such as "Simulation and synthesis results are likely to mismatch because the delay in model is greater than clock period." Some RTL synthesis tools might outright reject a model containing such delays.

One might argue that RTL synthesis tools should always match the Verilog simulation behavior of the input model. But the overall approach works well because RTL is a good and simple *conceptual model* that users and tool vendors can align around. The slight differences between the *simulation model* and the *conceptual model* used by RTL synthesis tools can be easily managed.

We'll return to this example later in this document.

Next, let's clarify some terminology related to signal IO. Say you have a Verilog model like:

```
forever begin
  @(posedge clk); // wait for 1st rising clock edge
  output1 <= input1 + 10;
  @(posedge clk); // wait for 2nd rising clock edge
end
```

In this example, `input1` is sampled when the process wakes on the first `@(posedge clk)` and evaluates the RHS. For the purposes of the scheduling rules, we anchor that `Read(input1,·)` to the most recent synchronization point. The resulting `Write(output1,·)` is anchored to the next synchronization point (the cycle in which the written value is treated as stable for others to observe).

**Cycle-level anchoring convention (used throughout this document).** For cycle-level reasoning, each signal IO is assigned an anchor cycle relative to surrounding synchronization calls (e.g., `wait()` / `SyncChannel` / `@(posedge clk)`):

- `Read(sig, val)` is anchored to the closest preceding synchronization call, i.e. `pred_S(Read)`.
- `Write(sig, val)` is anchored to the closest succeeding synchronization call, i.e. `succ_S(Write)`.

This convention is only for cycle-level reasoning and does not depend on simulator update regions or delta-cycle ordering.

## Catapult HLS Status Concerning Rules in this Document

A separate document provides a list of clarifications related to support within Catapult HLS for the rules described in this document. The items in the list are named *Cat#*, so that each item has a unique number. This document annotates certain rules with *Cat#* to refer to the items in the separate document. These *Cat#* annotations are not part of the formal proof itself.

## Terms Used in this Document

*Latency-Insensitive*: In digital hardware design, *latency-insensitive* refers to a system that operates correctly despite variable communication delays between components. This is achieved by using mechanisms to decouple computation from communication timing. Such designs improve scalability and reliability in complex systems with unpredictable or variable latencies. ARM's AXI4 and APB are examples of latency-insensitive protocols. ARM's AMBA 5 CHI credit-based NOC protocol is also an example of a latency-insensitive protocol.

*Process*: In Verilog, a process is an *always block* and its equivalent constructs. In SystemC, a process is an instance of an `SC_THREAD` or `SC_METHOD`.

*Message-passing Interface*: A *message-passing interface* reliably delivers messages (or transactions) from one process to another. This document uses this term to denote the type of communication found in Kahn Process Networks. See [https://en.wikipedia.org/wiki/Kahn\\_process\\_networks](https://en.wikipedia.org/wiki/Kahn_process_networks)

Message-passing read interfaces are always separate from message-passing write interfaces – there are no bidirectional message-passing interfaces. In this document, message-passing channels are assumed to be point-to-point: for each channel `c`, exactly one producer performs all `Push_c` operations and exactly one consumer performs all `Pop_c` operations.

*Synchronization Interface:* A *synchronization interface* synchronizes one process with another and/or with a global clock. For an example of a synchronization interface, see [https://en.wikipedia.org/wiki/Barrier\\_\(computer\\_science\)](https://en.wikipedia.org/wiki/Barrier_(computer_science))

*Signal IO:* In digital HW design, signals are the fundamental communication mechanism. Signals enable communication between two HW blocks/processes, but communication with signals in real HW always incurs at least some delay because communication cannot be faster than the speed of light. In HDLs and in SystemC, signal delays are modeled with the *delayed update* semantics.

*blocking / non-blocking:* In this document, a blocking message-passing operation is modeled as issuing a persistent request that remains asserted (and with stable payload, if applicable) until the operation commits (the message is sent or received). A non-blocking message-passing operation returns immediately with a status indicating whether the operation committed in that cycle.

*One-way / two-way handshake protocols:* A one-way handshake protocol only has one signal between a sender and receiver to synchronize communication. A two-way handshake protocol has two signals (one in each direction) between a sender and receiver to synchronize communication.

*shall:* This term indicates that a compliant tool or flow is required to follow the indicated rule.

*may:* This term indicates that a compliant tool or flow is allowed to follow the indicated rule but is not required to do so.

## Classes of Operations Involved in Scheduling Rules

There are three classes of operations involved in the scheduling rules:

1. Calls to message-passing interfaces (which are all `ac_channel` methods, all SystemC Push/PushNB/Pop/PopNB calls except calls to `SyncChannel`)
2. Calls to synchronization interfaces (which are calls to `ac_sync` and Matchlib `SyncChannel`, also `ac_wait` and SystemC `wait`)
3. Signal IO (which are SystemC signal reads and writes, also C++ model *direct inputs*)

All the operations above are referred to as *IO operations*.

## Basic Conceptual Model

The basic conceptual model encompasses processes that have no loop pipelining but may have preserved loops. If a process has a preserved loop, then the user may place a wait statement in the loop. Wait statements explicitly placed in the model are called explicit wait statements in this document and are classified as calls to a synchronization interface. However, for a loop to be treated as an HLS-pipelined loop under the formal guarantees in this document, the loop body shall not contain any `wait/ac_wait` statement or other explicit synchronization call used as an in-loop wait boundary. Such synchronization boundaries must be placed outside the pipelined loop, or the loop must be treated as non-pipelined/preserved or outside the theorem instance.

The basic conceptual model rules are:

1. Synchronization interface calls within a process always remain in the source code order.
2. Signal read operations occur at the closest preceding call to a synchronization interface. (Cat1)
3. Signal write operations occur at the closest succeeding call to a synchronization interface.
4. Message-passing operations may be shifted in time, issued in the same cycle, or overlapped by simulation or synthesis, subject to the following constraints:
  - All message-passing operations before a call to a synchronization interface shall be issued before or in the same cycle in which the synchronization interface commits. (Here “the synchronization interface commits” refers to the commit of that synchronization call in the calling process’s trace. For blocking message-passing operations (Push/Pop), the process cannot complete the synchronization call until any earlier issued blocking message requests in that interval have committed.)
  - All message-passing operations after a call to a synchronization interface shall be issued after the cycle in which the synchronization interface call commits.
  - Message-passing operations on the same message-passing interface shall be issued in source order; they shall not be reversed. This same-interface order requirement applies both within a single source interval and across overlapped iterations of a pipelined loop.
  - Two message-passing operations on separate interfaces which appear in sequence in the model are source-ordered. They may be issued in the same source sequence, and the later operation may in some permitted modes issue before the earlier operation commits, including in the same clock cycle where allowed by the issue policy. They shall not be issued in the reverse sequence. (Cat2)

Here “parallel” issue means that requests may overlap or be asserted in the same cycle while the calls remain ordered by their dynamic source order. Preserving that order prevents HLS from introducing a deadlock solely by reversing source-ordered message calls. It does not by itself guarantee deadlock preservation, because loop overlap, request timing, and finite channel capacity can still affect deadlock behavior. The formal appendices address those additional cases.

## Pipelined Loops

When a loop is pipelined in HLS, the body of the loop is split into pipeline stages. HLS may start the next iteration of the loop before the current iteration has completed, thus increasing throughput as compared to a non-pipelined implementation.

A loop selected for HLS pipelining shall not contain `wait/ac_wait` statements, `SyncChannel/ac_sync` calls, or any other explicit synchronization interface call inside the loop body. The pipelined-loop rules in this document therefore apply to wait-free pipelined loop bodies, with ordinary synchronization boundaries only outside the pipelined loop or around the pipelined region. If a design requires a synchronization call inside the loop body, the flow shall either reject that loop for pipelining under this contract, require the source to be rewritten so that the synchronization boundary is outside the pipelined loop, or exclude that pipelined-loop instance from the Appendices G-K formal guarantees unless a separate specialized theorem/certificate is supplied.

To guarantee equivalent behavior between the pre-HLS and post-HLS systems, pipelined loops must be implemented with automatic flush to prevent deadlocks. See Appendix B for a formal presentation of pipelined loops.

When HLS pipelines a loop, multiple iterations of the loop are overlapped and execute at the same time. During loop pipelining, for all IO operations, HLS shall ensure that an access to a specific message-passing interface, signal, or synchronization interface shall not be reordered relative to same-interface accesses from other iterations.

During pipelining, signal reads and writes are physically embedded in specific pipeline stages, but the loop body itself contains no explicit wait/ac\_wait or other in-body synchronization call under the formal contract above. The logical anchor of an ordinary signal Read/Write is therefore still determined by the surrounding real synchronization calls selected into S under R2/E2, not by an in-body wait. Considering the entire set of signals read or written by the process, if HLS pipelining would cause the order of signal IO to differ between the pre-HLS and post-HLS models, or if any two signal IO operations that occur on the same clock edge in the pre-HLS model would not occur on the same clock edge in the post-HLS model, then the HLS tool shall detect such a situation and require the user to explicitly indicate in the model source (e.g., via a pragma) that HLS pipelining can still be used. (Cat7) In other words, if a process communicates with other processes using only latency-insensitive protocols, then HLS pipelining by default shall not break any of the protocols.

### Synchronization boundary for pipelined loops

When a pipelined region ends at an explicit synchronization call, work already accepted before the boundary may continue to drain. However, the implementation and the back-annotated source model must not expose capacity for work belonging to the next synchronization interval until the synchronization call completes. This prevents upstream transfers from entering the next interval prematurely. See Appendix B for the formal rule and modeling treatment.

### Direct Inputs

Ordinary SystemC signal reads occur at the closest preceding synchronization interface call (e.g., wait statement) in both the pre-HLS and post-HLS models. (Cat1). If the HLS tool adds states to the process, or if it pipelines the process, then this implies that the HLS tool must add registers for each such read operation such that the read occurs where specified in the pre-HLS model, and the value is stored until the point where it is consumed in the post-HLS model. The area cost of such registers may be high if there are a lot of signals, and in some cases, it may be unneeded area since the value of the signals may not actually need to be stored internally to the process.

The simplest case is if such external signals are held stable after the process comes out of reset. In this case, HLS may assume that it is free to physically sample the signal values as late as possible, with no need for register storage, while the logical  $\Sigma$ -visible Read event remains anchored as specified by E2. This case is handled with the following pragma on SystemC signals and ports (Cat6):

```
#pragma hls_direct_input
```

To ensure that there are no pre-HLS versus post-HLS simulation mismatches, the environment that drives the signal shall hold it stable within each relevant reset epoch: after all receiving processes that use this signal with this pragma have come out of a reset boundary, the signal value shall remain stable until the next reset boundary that affects those receiving processes, unless the signal is instead governed by an explicit `hls_direct_input_sync` update contract as described below. A *dynamic reset* starts a new stability epoch for the affected receiving logic and its associated direct-input signals. Thus it remains allowable to have *dynamic resets*, i.e. activation of block resets and associated resetting of

direct input signals as part of the normal operation of the HW. Use of dynamic reset with the Appendix K Policy-S deadlock reduction is additionally subject to K.ResetEpoch; see Appendix M.6.

A related but somewhat more complex case is where input signals to the HW block may only be changed at “agreed upon” times, typically while the portion of the HW block that relies on them is temporarily idle. For example, a block may process 2D images. At the start of each new image, it may be desirable for the TB or external environment to update the control signals for how the block will process the next image. Typically HLS designs are pipelined, and the HW pipeline for the current iteration must be fully *ramped down* before the input signals can be updated to affect the next iteration. In the pipelined-loop case, this relies on the general synchronization-boundary rule for pipelined loops: while the process is pending at the designated sync, any back-annotated message-passing capacity whose accepted data would belong to the next post-sync interval is temporarily unavailable to upstream producers until the sync commits. In this case we can use the SystemC *SyncChannel* or C++ *ac\_sync* primitives to precisely synchronize the DUT with the TB/environment to enable the input signals to be updated at the correct time. The `#pragma hls_direct_input_sync` directive shown below associates the sync operation with the direct inputs that it controls. The precise synchronization scheme shown here ensures that there are no pre-HLS versus post-HLS simulation mismatches even though we are using direct inputs and also changing their values while the design is executing.

When a signal or port covered by `#pragma hls_direct_input` is also included in the scope of a later `#pragma hls_direct_input_sync` directive, the sync directive selects the effective update contract for that controlled signal set. Such signals remain direct inputs for purposes of late physical sampling, but they are not required to remain stable for the entire reset epoch; instead, they must satisfy the sync-controlled discipline described here: updates may occur only at the designated synchronization handshake, and the values must remain stable between permitted sync-controlled updates and reset boundaries.

```
// This is example 61 in Catapult Matchlib examples
sc_in<bool> SC_NAMED(clk);
sc_in<bool> SC_NAMED(rst_bar);

Connections::Out<uint32_t> SC_NAMED(out1);
Connections::In<uint32_t> sample_in[num_samples];
Connections::SyncIn SC_NAMED(sync_in);
#pragma hls_direct_input
sc_in<uint32_t> direct_inputs[num_direct_inputs];

void main() {
    out1.Reset();
    sync_in.Reset();

#pragma hls_unroll yes
    for (int i=0; i < num_samples; i++) {
        sample_in[i].Reset();
    }

    wait(); // reset state

    while (1) {
#pragma hls_direct_input_sync all
        sync_in.sync_in();

#pragma hls_pipeline_init_interval 1
```

```

#pragma hls_stall_mode flush
    for (uint32_t x=0; x < direct_inputs[0]; x++) {
        for (uint32_t y=0; y < direct_inputs[1]; y++) {
            uint32_t sum = 0;
#pragma hls_unroll yes
            for (uint32_t s=0; s < num_samples; s++) {
                sum += sample_in[s].Pop() * direct_inputs[2 + s];
            }
            ac_int<32, false> ac_sum = sum;
            ac_int<32, false> sqrt = 0;
            ac_math::ac_sqrt(ac_sum, sqrt); // internal loop unrolled in cat .tcl
file
            if (sqrt > direct_inputs[7])
                out1.Push(sqrt);
        }
    }
}
};

```

From the perspective of the testbench or the environment, the updating of the signals controlled by the `hls_direct_input_sync` directive shall only occur at a precise point. The TB shall first wait for the `rdy` signal for the sync to be asserted by the DUT, and then the TB shall update all the input signals it wishes to change while simultaneously driving the `sync_vld` signal high for one cycle.

It is important to note that the only safe operation to use to synchronize the updating of direct inputs is the sync operation as shown above. Other operations such as Push/Pop or `ac_channel` operations should not be used for this. In terms of the Appendix G trace model, the  $\Sigma$ -visible Read(sig,val) events remain anchored to their synchronization points; any additional internal sampling made by the implementation between anchors is treated as an  $\epsilon$ -step and must not change the recorded value label.

In the example above the DUT block that is being synthesized determines when to call sync, and thus it determines when the direct inputs will be updated. In some cases, it may be necessary for the environment around the DUT to determine when the direct inputs should be updated. In this case the same approach as shown above should be used; however a separate input from the environment to the DUT (either using a signal or a message-passing interface) should request that the DUT call sync as soon as feasible. This will ensure that the DUT has properly ramped down its pipeline and is ready to receive the newly updated direct inputs as per the overall synchronization scheme described above.

## Additional Options for Scheduling Message-passing Interfaces

The following option may be added during HLS (Cat2):

```
STRICT_IO_SCHEDULING=relaxed
```

When this is specified, the HLS tool is allowed to reorder message-passing interface calls freely on distinct interfaces (still not moving calls across synchronization interface calls). This relaxed mode may violate the default no-reverse discipline and therefore may introduce deadlocks that are not present under the default rules. It is recommended that this relaxed option only be used when the user wants to see what order the HLS tool prefers to schedule message-passing interface calls (e.g., to achieve best QOR).

Once the user knows the preferred order, it is recommended that the user modify the pre-HLS source code to reflect the preferred order, and then return to the use of the default scheduling modes. This methodology ensures that HLS cannot introduce new deadlocks *via reversal of source order on distinct message interfaces* as described earlier. Deadlock behavior may still depend on bounded-capacity effects and on overlapped execution (e.g., loop pipelining); those cases are addressed separately in this document via the pipelined-loop discussion (automatic flush / WFG-inertness) and the later deadlock reasoning. The formal equivalence and deadlock-preservation results in Appendices G–K assume STRICT\_IO\_SCHEDULING is not set to relaxed.

## Scheduling of Array Accesses

Arrays may appear in HLS models, and they may be preserved through synthesis and mapped to RAMs. Pointers may also appear in HLS models, and pointer dereferences are resolved to array accesses during HLS.

There are two cases to consider for arrays for the purposes of the scheduling rules:

1. Array instantiation in the HW is internal to the process
  - The array accesses are not visible outside the process, and thus their scheduling is also not visible externally.
  - All the scheduling rules described elsewhere in this document remain unaffected in this case.
2. Array instantiation in the HW is external to the process
  - In this case the user model shall indicate how array accesses are mapped onto IO operations that are external to the process.
  - We call this the *array access mapping layer*. The *array access mapping layer* maps array accesses onto IO operations described above (signal IO, message-passing interface calls, and synchronization calls).
  - The user model may indicate that it is allowable for HLS to transform array accesses, for example, to cache, merge, split, or reorder array accesses (e.g., to improve QOR). These transformed operations, if allowed, are an outcome of using the *array access mapping layer*.
  - In all cases the scheduling rules described elsewhere in this document for the core IO operations (signal IO, message-passing calls, synchronization calls) remain unaffected.
  - In all cases array accesses shall remain constrained by any explicit synchronization interface calls present in the process in the source model. Precisely speaking, all array reads or writes before a synchronization call shall commit before or in the same cycle at which the synchronization call commits, and all array reads or writes after a synchronization call shall commit in a cycle after the synchronization call commits. This rule is stated in terms of commit cycles (the cycle the mapped memory storage is accessed); the scheduler enforces it by choosing issue cycles consistent with the fixed access latency defined by the array access mapping layer. The corresponding formal audit obligation is Core.MemSyncOrder in the Normative Formal Core.
  - Note that if transformed operations occur and array accesses are visible externally in both the pre-HLS and post-HLS model, then comparison of pre-HLS and post-HLS behaviors may need to account for the transformed operations.

When a loop is pipelined, multiple iterations of the loop are overlapped and execute at the same time. During loop pipelining, an access to a memory interface (or array) may be moved over or in parallel with an access to the same memory if the HLS tool can prove the reordering is conflict-free. If the

array/memory is external to the process, the *array access mapping layer* shall indicate that such reordering is allowable if such reordering is to occur during loop pipelining.

## Definitions: Issue vs. Commit

Informally, we can define **commit** and **issue** as follows:

### Commit (message channel) — informal operational reading

For the RTL\_B implementation reading at a mapped ready/valid boundary, a message commit is observed in the cycle in which an external clocked observer sees the transfer complete:

- The mapped boundary commit is observed when the observer sees both valid and ready high in that cycle.
- The committed payload is the data value seen in that same cycle. This is an implementation-facing observation rule; for the G–K theorem stack, the formal abstract commit and  $\text{Commit}(q)/\text{CommitCycle}(q)$  definitions are in Definition Core.8.

### Definition: Issue (message channel) — message request assertion

Issue is the cycle in which the calling process begins requesting the transfer at the selected message-channel boundary. For a Push, its boundary manifestation is assertion of vld; for a Pop, it is assertion of rdy. Appendix C defines the precise attribution of an observed request to a dynamic source operation.

For RAM reads/writes as presented above, **commit** denotes the precise cycle the memory storage is accessed; since the access latency is fixed and known to the HLS scheduler, it can schedule the operation's issue cycle so that the commit cycle satisfies the required ordering relative to Sync. The "Scheduling of Array Accesses" constraints above are therefore written in commit-time.

**Sequential processes use registered cycle semantics:** a result returned by a message-passing call at a cycle boundary cannot be used to generate a dependent request in that same cycle. Such a dependent request may first issue in a later cycle. Appendix C and Definition Core.RegSeq give the precise rule, including its treatment of failed non-blocking calls and independently active pipeline iterations.

## Additional States Added by HLS Synthesis

By default, HLS synthesis tools may add additional states to processes (e.g. add latency to enable resource sharing), which may introduce latency differences in the interface behavior between the pre-HLS and post-HLS models. These additional states are never included in the set of *synchronization interface calls* as described above.

When the directive `IMPLICIT_FSM=true` is set on a process, the HLS synthesis tool shall ensure that the cycle-level behavior of the interfaces of the pre-HLS and post-HLS models shall be identical. With this option, the internal state machines of the pre-HLS and post-HLS models will be the same.

When the directive `IO_MODE=FIXED` is set on a process, the HLS synthesis tool shall ensure that the cycle-level behavior of the interfaces of the pre-HLS and post-HLS models shall be identical. With this option, it is still possible that the state machine internal to the process is different between the pre-HLS and post-HLS models (e.g. the post-HLS model may choose to use a pipelined multiplier where the pre-HLS model did not.)

## Avoiding Pre-HLS and Post-HLS Simulation Mismatches

The scheduling rules described in this document are designed to be easy to understand, while providing good QOR via HLS and generally avoiding any mismatches between the pre-HLS and post-HLS simulation behaviors.

Non-blocking message-passing operations (PushNB/PopNB in SystemC, C++ `ac_channel nb_read/nb_write`) are a potential source of mismatches between pre-HLS and post-HLS simulations since their behavior is inherently dependent on the latency within the model, which often changes during HLS. Because of this, non-blocking message-passing interfaces should only be used when no alternative approach is possible. For example, non-blocking message-passing interfaces are required to model time-based arbitration of multiple message streams which access a shared resource. A full discussion of recommended guidelines on the use and verification of non-blocking message-passing interfaces is provided in Appendix L. Note that the scheduling rules described previously in this document fully specify how HLS tools are required to schedule such operations.

Unidirectional message-passing between two processes should not be relied on to achieve synchronization between the two processes, since in general the message latency and storage capacity between the processes may be variable. Also, depending on buffering/pipelining in the realization, the time between a producer's committed Push and the consumer's committed Pop (and the transient in-flight occupancy) may vary. Two unidirectional message-passing channels in opposite directions can be relied upon for synchronization between two processes. (An example of this is the AXI4 `ar` and `r`, and `aw` and `b`, channels). Note that such synchronization is weaker than explicit synchronization like `SyncChannel` or signal IO that uses a two-way handshake.

SystemC signal IO operations are a potential source of pre-HLS versus post-HLS simulation mismatches since timing behaviors may change between the two models. The following section provides guidance and rules to help avoid potential mismatches due to signal IO.

When signal IO operations are synchronized with a wait statement, there generally should be a proper two-way handshake associated with the wait statement so that the signal IO is latency insensitive. (Note that this statement does not apply to the signal synchronization approaches described in the Direct Inputs section.)

For example, here's a simple two-way handshake protocol when writing the signal `out_dat`:

```
out_dat = value;
out_vld = 1;
do {
    wait();
} while (out_rdy != 1);
out_vld = 0;
```

And here's a two-way handshake example when reading signal `in_dat`:

```
in_rdy = 1;
do {
    wait();
} while (in_vld != 1);
value = in_dat;
```

```
in_rdy = 0;
```

Some signal-level protocols have different two-way handshaking approaches (e.g. ARM APB), but they are still latency insensitive.

If signal IO operations are associated with a wait statement and that wait statement does not have a proper two-way handshake, then the signal IO is likely to be latency-sensitive and may result in pre-HLS versus post-HLS simulation mismatches. In some systems a one-way signal handshake is sufficient for reliable system operation. See Appendix E for further discussion.

The scheduling rules state that signal IO operations occur at either SystemC wait statements or SyncChannel calls (sync\_in and sync\_out). In the remainder of this section, we will use *wait* statements to refer to both.

**RULE 1:** It is always best coding style to group signal write operations just before their corresponding wait statement, and signal read operations just after their corresponding wait statement. (Cat3). An example is below:

```
sc_in<int> i1;
sc_in<bool> go;
sc_out<int> o1;
void my_thread() {
    int new_val=0;
    while (1) {
        o1.write(new_val);
        do {
            wait();
        } while (!go.read());
        new_val = i1.read();
        new_val = some_function(new_val); // function has no internal IO
    }
}
```

By placing the signal IO operations as close as possible to their corresponding wait statement, the HW intent is very clear. And there is no benefit either in terms of simulation performance or HLS QOR if they are placed further away from their corresponding wait statement.

Let's look at another similar example, which now also uses a Matchlib Connections blocking Pop operation:

```
sc_in<int> i1;
sc_in<bool> go;
sc_out<int> o1;
Connections::In<int> pop1;
void my_thread() {
    int new_val=0;
    while (1) {
        o1.write(new_val);
        do {
            wait();
        } while (!go.read());
        int pop_val = pop1.Pop();
        new_val = i1.read();
    }
}
```

```

        new_val = some_function(new_val + pop_val); // function has no internal
IO    }
    }

```

Under the anchoring rules, `i1.read()` remains associated with the same synchronization point regardless of the intervening Pop. However, in raw pre-HLS simulation, a blocking Pop may stall for one or more cycles, so `i1` can change before the `i1.read()` executes—creating behavior that differs from the anchored interpretation (and potentially from post-HLS scheduling). The fix is simply to place `i1.read()` immediately after its intended `wait()` anchor; this removes the mismatch risk without affecting QOR or simulation performance.

For a proof-backed flow, RULE 1 conformance is a source well-formedness obligation, not merely a recommended diagnostic. If a blocking message-passing operation separates a signal read or write operation from its corresponding synchronization interface call, then the flow shall either reject the model for purposes of applying the formal guarantees, require the source to be rewritten so the signal IO is adjacent to its intended anchor, or discharge an explicit source-level well-formedness argument showing that the signal IO has a unique unconditional real synchronization anchor on every dynamic path and cannot observe a value inconsistent with that anchor. HLS tools may implement this requirement using errors, warnings, lint checks, or other diagnostics, but an unresolved warning is not sufficient to claim the RULE 1 / RULE 2-based guarantees.

Another scenario in which RULE 1 applies is shown below:

```

sc_in<bool> go;
sc_out<int> o1;
void my_thread() {

    while (1) {
        wait();                // WAIT 1
        o1.write(some_value);
        if (go.read()) {
            some_value = some_function();
        }
        else {
            wait();            // WAIT 2
        }
        some_other_function();
        wait();                // WAIT 3
    }
}

```

The signal read of `go` is clearly and uniquely associated with WAIT 1. However, the signal write of `o1` associates with WAIT 2 if `go` is false and WAIT 3 if it is not. This is a violation of RULE 1 and should be flagged as an error during HLS. The fix, as before, is to move the signal IO operation as close as possible to its intended wait statement so that the association is unconditional.

Next, let's consider *rolled* (or *preserved*) loops that perform signal IO within the loop body. Consider the following example:

```

sc_in<int> i1;
sc_out<int> o1;
void my_thread() {

```

```

    wait(); // reset state
    while (1) {
        wait(); // start of while loop
        #pragma hls_unroll no
        for (int i=0; i < 10; i++) {
            o1.write(i1.read() * i);
        }
    }
}

```

Note that the `i1.read()` operation is located inside the `for` loop, so presumably the user's intent is that it should be read as the loop iterates. *If that is not the user's intent, they simply should move the `i1.read()` operation before the loop start.*

In the post-HLS simulation, each iteration of the loop will consume at least one clock cycle, and a new value for `i1` will be read (and a new value for `o1` written) on each iteration. Again, this is the user's intent as per the code. In the pre-HLS simulation, the `for` loop body will execute in zero time, and only the last write to `o1` will have any effect. The solution to avoid this mismatch is to manually place a `wait()` statement within the `for` loop body so that the signal IO synchronization is explicit in the pre-HLS simulation.

**RULE 2:** If you have signal IO operations within *rolled* (or *preserved*) loops, manually place a `wait` statement within the body of the loop to avoid pre-HLS versus post-HLS simulation mismatches, and while doing so also follow **RULE 1**.

**Scope of RULE 2 for pipelining.** RULE 2 applies only to rolled or preserved loops that are not selected for pipelining under this contract. A loop body selected for pipelining shall not contain `wait/ac_wait` statements, `SyncChannel/ac_sync` calls, or any other explicit synchronization interface call. Therefore, if ordinary per-iteration signal IO inside a loop requires a per-iteration wait anchor, that loop is outside the pipelined-loop contract unless the source is rewritten: move the signal IO outside the pipelined loop, model the interaction through message-passing, use `hls_direct_input` / `hls_direct_input_sync` under their stated contracts, or forgo pipelining for that loop. Without an in-body synchronization anchor, repeated per-iteration signal reads or writes inside one surrounding source interval do not become distinct per-iteration  $\Sigma$ -visible Read/Write events merely because HLS later pipelines the loop; their logical anchors remain the surrounding synchronization calls selected into `S` under `R2/E2`.

For a proof-backed flow, a rolled loop with signal IO operations but no `wait` statement in the loop body shall be treated as a source well-formedness failure unless an explicit source-level argument shows that each observable signal Read/Write still has a unique real synchronization anchor on every dynamic path. HLS tools may implement this requirement using errors, warnings, lint checks, or other diagnostics. However, if a diagnostic is emitted as a warning rather than an error, the warning must be resolved, waived with an explicit well-formedness justification, or treated as excluding that model from the formal guarantees. (Cat4)

If a pre-HLS model has established RULE 1 and RULE 2 conformance, then all signal IO in the post-HLS model shall occur only at clock cycles that correspond to explicit `wait` statements or explicit synchronization statements. For direct-input signals/ports (e.g., `#pragma hls_direct_input` and `#pragma hls_direct_input_sync`), this requirement refers to the logical ( $\Sigma$ -visible) anchor cycle of the Read/Write: the event is still considered to occur at its synchronization point even if HLS physically samples a stable signal later; such late-sampling micro-steps are  $\epsilon$ -steps and are not  $\Sigma$ -visible. User designs that do not adhere to both RULE 1 and RULE 2 are ill-formed and are outside the formal guarantees unless the

nonconformance has been eliminated or explicitly discharged by a valid source-level well-formedness argument.

## Returning to the Analogy from RTL Synthesis

At the beginning of this document, we presented the example of a Verilog sequential block with an output coded like:

```
Out1 <= #some_delay new_val;
```

Recall that in Verilog simulation, if `some_delay` is less than the clock period of the block, then it will probably not affect the overall cycle-level behavior of the system during simulation. However, if `some_delay` is more than the clock period, it probably will.

During RTL synthesis, all RTL synthesis tools will ignore all delays in the input model, in this case even if `some_delay` is greater than the clock period. Some RTL synthesis tools might give a warning for the code above like “Simulation and synthesis results are likely to mismatch because delay in model is greater than clock period.”

HLS tools or associated lint/qualification flows that adhere closely to the conceptual model presented in this document should provide a concrete mechanism for detecting or discharging violations of RULE 1 and RULE 2 as described in the section above. This is analogous to the error or warning that the RTL synthesis tool would provide in the example directly above.

However, the formal guarantees in this document do not depend on any particular diagnostic mechanism. If an HLS synthesis tool does not report a RULE 1 / RULE 2 issue, the design is not thereby made well-formed. The proof-backed flow must still establish that the source model satisfies the RULE 1 / RULE 2 well-formedness obligation before applying the trace-compatibility guarantees.

## Summary

At the beginning of this document, we said that the intent was to present “no surprises” to a DV engineer who wants a common verification methodology for the pre-HLS and post-HLS models. The key aspects of the document which support this are:

- Three groups of IO operations are defined (message-passing, signal IO, and synchronization calls) and each is treated uniformly. These IO operations are easy for verification engineers to understand because they are already using them in their testbenches.
- The document specifically avoids complex constructs such as *protocol regions* used in some HLS tools.
- The document preserves the ability of the pre-HLS SystemC model to be *throughput accurate* by using a library such as Matchlib.
- Synchronization calls can affect the scheduling (anchoring) of signal IO operations, and synchronization calls can affect the scheduling of message-passing calls. In the conceptual anchoring semantics of this document, message-passing calls do not affect the anchor cycle of

signal IO operations (and signal IO does not affect the legality/order of committed message transfers) except through explicit synchronization. Practically, in raw pre-HLS SystemC simulation a blocking message operation may introduce additional cycles of waiting, so signal IO statements that are lexically separated from their intended anchor can observe different values; RULE 1 addresses this by recommending placement of signal IO adjacent to its anchor.

- HLS cannot by default reverse the order of message-passing calls, so it cannot introduce new deadlocks *via call-order reversal*. For pipelined loops (overlapped iterations) and other transformations that change the timing of request/commit under bounded capacities, deadlock preservation depends on the additional pipelined-loop discipline described in this document (automatic flush / WFG-inertness), not on the no-reverse rule alone.
- HLS pipelining is largely a *don't care* from the perspective of the verification engineer. If the design and the testbench are insensitive to changes in latency, and if external array accesses are not reordered or rearranged during loop pipelining, then the possible use of HLS pipelining will not affect verification, provided the pipelines flush automatically. Even if the design or testbench are sensitive to changes in latency, or if they are sensitive to reordering or rearranging of external memory accesses due to the use of HLS loop pipelining, then the behavior of the DUT will only change in expected (rather than unexpected) ways, provided the applicable observation-boundary and witness-selection conditions are satisfied. (Appendix G provides formal trace-compatibility guarantees rather than an unrestricted bidirectional equivalence theorem.)
- For sequential processes, signal IO operations in the post-HLS model always occur exactly at synchronization points (e.g., wait statements) that are explicit in the pre-HLS source. For direct-input signals/ports, “occur at synchronization points” is meant in the logical/anchoring sense: the  $\Sigma$ -visible Read/Write is timestamped at the synchronization point even if the implementation samples the stable signal later; those internal late samples are  $\epsilon$ -steps. For combinational processes, observable signal Read/Write events are instead interpreted at the cycle's  $\epsilon$ -quiescent combinational fixed point, as specified by R2C.

The principal formal safety and deadlock-preservation results, their scope limitations, and the evidence required to apply them are summarized in Appendix M. Appendix F provides user-oriented modeling guidelines.

## Appendix A – Factory Analogy

The scheduling rules described in this document apply to pre-HLS and post-HLS HW models. Although the rules may seem abstract and perhaps even arbitrary, they are shaped by the need to model systems in the real world that are required to have predictable behavior.

To understand the motivation behind the rules, it might be helpful to draw a simple real-world analogy and its correspondence to the rules described earlier.

Consider a factory that produces various types of wooden furniture:

The factory consists of people (processes) stationed at workbenches with various tools.

Each person is given written instructions about the specific tasks they are to perform (C++ code within a process).

People are instructed to send or receive objects (messages) to or from other people in the factory.

Sending or receiving objects may be blocking or non-blocking from the perspective of a person.

There is a clock with a second hand on the factory wall that everyone can see. (HW clock).

People have colored flags they can raise or lower to communicate with other people (signals).

If someone raises or lowers a flag, this is only seen by others the next time the clock second hand is at the top of the clock. (propagation delay of signals between sequential processes)

A person can choose to pipeline the tasks that they were assigned by hiring subordinates (pipeline stages) and having each one do a subtask. In general, this will improve the throughput for the tasks that the person was assigned.

It is possible for two people to explicitly synchronize their work by communicating via a synchronization protocol such as a barrier (synchronization).

It is possible to restart the work of some or all the people by raising a reset flag (HW resets).

Assume:

1. That the time that people take to complete their various tasks is in general variable, and similarly that the time that objects (messages) take to pass between people is in general variable.
2. That each person in general wants to complete their tasks as quickly and efficiently as possible.
3. That the factory needs to be able to make multiple types of furniture at the same time. For example, it might make chairs that need to be different colors or have different styles.

To reliably produce output, each person in the factory will need to adhere to rules like those described in this document.

Here's a sketch of a specific way the above example relates to the scheduling rules:

Person 1 sends chair seats and chair backs to Person 2. This is done with blocking operations, and there is no storage capacity between the people when the objects are sent. (This means that a Push operation cannot complete until the corresponding Pop operation is performed.) Assume that the fastest an object can be sent between people is 1 minute.

The written instructions that person 1 is given are:

```
while (1) {  
    // internal processing code for seats and backs ...  
    seats.Push(seat_object);  
    backs.Push(back_object);  
}
```

The written instructions that person 2 is given are:

```
while (1) {
    seat_object = seats.Pop();
    back_object = backs.Pop();
    // internal processing code for seats and backs ...
}
```

If both person 1 and person 2 choose to perform their tasks sequentially as written, then objects will be passed over time as:

|           | Person 1           | Person 2             |
|-----------|--------------------|----------------------|
| Minute 1: | seats.Push(seat1); |                      |
| Minute 2: | backs.Push(back1); | seat1 = seats.Pop(); |
| Minute 3: | seats.Push(seat2); | back1 = backs.Pop(); |
| Minute 4: | backs.Push(back2); | seat2 = seats.Pop(); |
| Minute 5: |                    | back2 = backs.Pop(); |

If both person 1 and person 2 choose to perform their IO operations in parallel, then objects will be passed over time as:

|           | Person 1                              | Person 2                                  |
|-----------|---------------------------------------|-------------------------------------------|
| Minute 1: | seats.Push(seat1); backs.Push(back1); |                                           |
| Minute 2: | seats.Push(seat2); backs.Push(back2); | seat1 = seats.Pop(); back1 = backs.Pop(); |
| Minute 3: |                                       | seat2 = seats.Pop(); back2 = backs.Pop(); |

If person 1 chooses to perform their IO operations in parallel, and person 2 chooses to perform their IO operations sequentially as written, then objects will be passed over time as:

|           | Person 1                              | Person 2             |
|-----------|---------------------------------------|----------------------|
| Minute 1: | seats.Push(seat1); backs.Push(back1); |                      |
| Minute 2: |                                       | seat1 = seats.Pop(); |
| Minute 3: | seats.Push(seat2); backs.Push(back2); | back1 = backs.Pop(); |
| Minute 4: |                                       | seat2 = seats.Pop(); |
| Minute 5: |                                       | back2 = backs.Pop(); |

If person 1 chooses to perform their IO operations sequentially as written, but person 2 chooses to perform their IO operations sequentially in the reverse order as it was written, then objects will be passed over time as:

|           | Person 1           | Person 2             |
|-----------|--------------------|----------------------|
| Minute 1: | seats.Push(seat1); |                      |
| Minute 2: | backs.Push(back1); | back1 = backs.Pop(); |
| Minute 3: |                    | seat1 = seats.Pop(); |

In this case the person 1 seats.Push(seat1) operation at minute 1 will not complete at the start of minute 2. This means that the person 1 backs.Push(back1) operation will never start, and thus the Person 2 back1 backs.Pop() operation will never complete. So, the system will be in deadlock.

This example directly corresponds to the ordering rules related to message-passing operations in the *basic conceptual model* within this document, and in particular the specific rule that disallows reversing the order of message-passing operations.

## Appendix B – Formal Presentation of Pipelined Loops

### Modeling convention for pipelined-loop overlap.

Pipelined-loop body no-wait condition. The formal pipelined-loop model in this appendix applies only to HLS-pipelined loop bodies that contain no `wait/ac_wait` statement, `SyncChannel/ac_sync` call, or other explicit synchronization interface call inside the loop body. All ordinary synchronization calls selected into  $S$  must be outside the pipelined loop body or at explicit boundaries around the pipelined region. This is a well-formedness condition for applying the Appendices G-K theorem stack to a pipelined-loop instance; an in-loop synchronization call requires source rewriting, non-pipelined/preserved-loop treatment, or a separate proof/certificate outside this appendix.

The post-HLS pipelined-loop implementation may overlap iterations and may internally accept/stage up to the finite bound  $B(c_{in})$  of input values for younger overlapped iterations before older iterations' outputs are produced. This internal accept/stage behavior is part of the back-annotated channel realization counted in  $B(c_{in})$  between the  $\Sigma$ -visible endpoints of  $c_{in}$  (producer  $Push_{\{c_{in}\}}$  and consumer  $Pop_{\{c_{in}\}}$ ). Here  $B(c_{in})$  is an effective abstract-boundary capacity/credit bound, not a claim that the RTL contains a single physical FIFO named  $c_{in}$ ; the proof obligation is that the chosen  $Sys\_ref$  boundary exposes E4-consistent `ChannelEnabled/ChannelDisabled` behavior, with hidden physical storage represented as  $\epsilon$ -state unless it is explicitly elaborated into  $Sys\_B^{elab}$  channels. These internal accept/stage movements are modeled as  $\epsilon$ -steps.  **$\Sigma$  records only the committed boundary transfers at the  $\Sigma$ -visible endpoints.** In particular, the consumer-side boundary  $Pop_{\{c_{in}\}}$  is the loop-body  $Pop_{\{c_{in}\}}$  operation from the pre-HLS source when that  $Pop$  is included in the chosen  $\Sigma / \Sigma\_DUT$  projection. The externally relevant effect of pipelining is therefore captured entirely by  $B(c_{in})$  (via boundary backpressure/acceptance behavior and overlap timing at the  $\Sigma$ -visible boundary), without introducing any new auxiliary or duplicate  $\Sigma$ -visible boundary Pops: the  $\Sigma$ -visible consumer-side  $Pop_{\{c_{in}\}}$  commits are still the same loop-body  $Pop_{\{c_{in}\}}$  operations from the pre-HLS source, though they may commit at different times (and thus at different prefixes) than in an unpipelined execution. Once the pipeline reaches a quiescent blocked state under the automatic flush discipline below, it does not initiate new blocking request assertions for younger overlapped iterations.

### B-faithfulness obligation

The use of  $B(c_{in})$  in this modeling convention assumes that the selected abstract boundary and the reported capacity faithfully summarize the corresponding RTL realization: all storage/credit that can accept, hold, or backpressure messages between the chosen producer and consumer endpoints is either counted in  $B(c_{in})$  or represented as an explicit  $Sys\_B^{elab}$  boundary, and movements inside that realization are  $\epsilon$ -steps that do not create, delete, duplicate, reorder, or relabel  $\Sigma$ -visible boundary transfers. This is a tool/RTL-flow compliance obligation, not a fact derived by Appendix B.

### Multi-input Pop points and coupler policies.

When a pipelined loop consumes from multiple input channels (e.g.,  $c1, c2$ ), back-annotation still assigns a separate finite bound  $B(c_i)$  and cycle-boundary occupancy  $occ_{ci}(t)$  to each outer  $\Sigma$ -visible boundary channel  $c_i$ . However, in the multi-input / coupler cases of Appendix Q the formal source-side proof target is no longer merely the outer-boundary  $Sys\_B$ ; it is the elaborated reference model  $Sys\_B^{elab}$  of Definition Core.1 and Appendix J, Remark 2. In  $Sys\_B^{elab}$ ,  $B(\cdot)$ ,  $occ_{\cdot}(t)$ , E4, `BoundaryReq`, `ChannelEnabled/ChannelDisabled`, and the Appendix K blocking/WFG machinery are interpreted per

explicit abstract channel, including any introduced staged/bundle/join channels. Any conjunctive/join dependency (e.g., “join not met”) shall therefore be represented only via such explicit staged/bundle/join boundary channels in  $\text{Sys\_B}^{\text{elab}}$  (not as cross-channel disablement at the outer  $\Sigma$ -visible endpoints), so that for Appendix K the blocking point is that explicit boundary and the relevant  $\text{Front\_P}(\sigma)$  members there are jointly channel-disabled (“ALL disabled  $\Rightarrow$  stall”). For the detailed elaboration patterns and examples, see Appendix Q.

### Practical note (outer $\Sigma_{\text{DUT}}$ projection and proof-internal $\Sigma$ )

The loop-body  $\text{Pop}_{\{c_{\text{in}}\}}$  in the pre-HLS source (i.e., the consumer-side  $\text{Pop}_{\{c_{\text{in}}\}}$  operation, conceptually after the in-flight capacity  $B(c_{\text{in}})$ ) may be exposed in the outer DUT/testbench-visible alphabet  $\Sigma_{\text{DUT}}$  if desired, but is typically omitted since it is latency-insensitive, and it is often practically difficult to expose/collect in the post-HLS RTL. The liveness/deadlock reasoning still accounts for this blocking point without requiring it to be  $\Sigma_{\text{DUT}}$ -visible: Appendix K defines WFG edges using the  $\text{ChannelEnabled}/\text{ChannelDisabled}$  status of the corresponding pending boundary operation instance  $q$  with  $\text{kind}(q)=\text{Pop}_{\{c_{\text{in}}\}}$  (via  $E4/\text{occ}_c/B(c)$ ). In ordinary non-elaborated  $\text{Sys\_B}$  cases, this does not require introducing any additional proof-internal abstract channel. In elaborated cases  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$ , however, any introduced staged/bundle/join channel needed to represent a blocking point is an explicit proof-internal  $\Sigma$  boundary for Appendix K, even if its committed Push/Pop events are projected away from the outer  $\Sigma_{\text{DUT}}$  trace by  $\Pi_{\text{elab}}$ .

### Drain-only blocked-frontier discipline and automatic flush.

Potential deadlocks are avoided by having the pipeline automatically flush. In this document, “automatic flush” does not mean a global pipeline clear or a mandatory drain-to-empty phase. It is the pipelined or otherwise overlapped implementation of the more general drain-only blocked-frontier discipline used by Appendix K. Once the minimal pending blocking frontier is fully channel-disabled after  $\varepsilon$ -settling, the process is blocked at that frontier; no new later source work may issue past that frontier; already asserted blocking requests remain asserted until commit; and already-issued downstream work may continue to drain only insofar as it does not create new WFG-relevant blocking causes. For ordinary non-overlapped control, the same Appendix K discipline may instead be discharged structurally, as described below.

### Terminology (pending blocking frontier / $\text{Front\_P}(\sigma)$ )

At a settled boundary-evaluation cutpoint, consider the set of pending source-level blocking message-operation frontier items  $x \in \Omega_{\text{msg},P}$  in the current interval between adjacent synchronization calls, where each such item has an owning dynamic message-call instance  $q = \text{MsgOf\_P}(x)$  whose  $\text{Issue}(q)$  has occurred and whose  $\text{Commit}(q)$  has not yet occurred. By the  $\text{Issue}/\text{Commit}$  linkage (Appendix C) together with  $\text{BoundaryReq}$ ’s definition at the  $\Sigma$ -visible boundary (as given in “Normative Formal Core for Appendices G–K”), each such pending blocking frontier item  $x$  has its endpoint-owned boundary request asserted through  $q$  (Push: vld, Pop: rdy) until  $q$  commits; purely combinational channel-side structure (including couplers) may affect only the complementary mirror signal (Push: rdy, Pop: vld) and thus whether  $x$  commits in that cycle after  $\varepsilon$ -settling. Define  $\text{Front\_P}(\sigma)$  to be the set of such pending blocking message-operation frontier items that are minimal with respect to the source-induced order  $\leq_P$  within that interval. By Lemma Core.OrdFrontSingleton, an ordinary non-elaborated source interval has at most one minimal pending blocking message-operation frontier item. If  $\text{Front\_P}(\sigma)$  has multiple message-operation members, that multiplicity must come from explicit  $\text{Sys\_B}^{\text{elab}}$  proof-internal structure such as staged/bundle/join/epoch-gate abstract channels; it is not inferred merely from the

fact that ordinary source calls use distinct interfaces. We refer to  $\text{Front\_P}(\sigma)$  as the pending blocking frontier (or “minimal pending blocking frontier”). This  $\text{Front\_P}(\sigma)$  notion is distinct from  $\text{NextFront\_P}$  (Definition Core.6; reconstructed from KObs by Lemma J.KObs-Proc and Corollary J.1), which denotes the next source-level frontier-item set over  $\Omega\_P$  and may include synchronization-call items, signal-anchored action items, and message-operation items. A message-operation item in  $\text{NextFront\_P}$  may be pending before it contributes any committed  $\Sigma$ -event.

A pipelined loop supplies the drain-only blocked-frontier discipline by automatically flushing if, whenever the pending blocking frontier  $\text{Front\_P}(\sigma)$  is non-empty and every frontier item  $x \in \text{Front\_P}(\sigma)$  cannot make progress because it is not channel-enabled (evaluated after combinational ready/valid/backpressure has  $\varepsilon$ -settled for the cycle), then:

1. **Block point.** The process is blocked on message passing at that pending blocking frontier  $\text{Front\_P}(\sigma)$ , as in the unpipelined source.
2. **No new later work.** While blocked at that frontier (i.e., while  $\text{Front\_P}(\sigma)$  remains non-empty and all of its members are channel-disabled), the implementation does not issue any new blocking Pop/Push message-operation frontier item  $y \in \Omega\_msg, P$  that is later in source order than some  $x \in \text{Front\_P}(\sigma)$ , including any younger overlapped iteration, and therefore does not begin asserting  $\text{BoundaryReq}(y, \cdot)$  for such later work. Here issue is read through the owning message instance  $q\_y = \text{MsgOf\_P}(y)$ .
3. **No withdrawal.** While blocked, the implementation does not withdraw any already-asserted blocking Pop/Push request (no “try-and-withdraw”); outstanding blocking requests remain asserted until their owning message instances commit.
4. **Clarification (Frontier-minimality vs. other asserted requests).** The Wait-For Graph in Appendix K is intentionally defined from the minimal pending blocking frontier  $\text{Front\_P}(\sigma)$ , i.e., the current blocking causes. Accordingly, an already-asserted blocking request whose frontier item is not in  $\text{Front\_P}(\sigma)$  (for example, a request from a younger overlapped iteration that was issued before the blocked condition, or a downstream request draining work already past the frontier) may persist as protocol state under the “no withdrawal” rule, but it is not treated as a wait-for source while an earlier pending blocking frontier item remains frontier-minimal. Such a request contributes WFG blocking only if/when its frontier item later becomes a member of  $\text{Front\_P}(\sigma)$ .
5. **Drain-only downstream progress.** Work that has already passed the blocked frontier point(s) may continue to drain, and any already-asserted resulting output-Push requests may commit when channel-enabled, provided no work advances past the blocked frontier point(s). (Formally: “no work advances past” is meant in the  $\leq\_P$  / issue sense—while  $\text{Front\_P}(\sigma)$  is fully disabled,  $P$  does not issue any new blocking message-operation frontier item  $y$ , with owning instance  $q\_y = \text{MsgOf\_P}(y)$ , such that, for some  $x \in \text{Front\_P}(\sigma)$ ,  $x \leq\_P y$  and  $x \neq y$ ; downstream drain may only complete/commit operations whose  $\text{BoundaryReq}$  was already asserted prior to reaching this blocked condition.) (Equivalently: this is the “ALL disabled  $\Rightarrow$  stall” policy— $P$  stalls only when none of the frontier items in  $\text{Front\_P}(\sigma)$  are channel-enabled; downstream work may drain while  $\text{Front\_P}(\sigma)$  is fully disabled.)
6. **Finite explicit boundary storage/credit.** Every selected abstract boundary that can hold or credit already accepted work has a finite declared capacity  $B(c)$ , or is represented by an explicit finite  $\text{Sys\_B}^{\text{elab}}$  boundary chain whose component capacities are finite and  $B$ -faithful. Together with “no new later work,” this makes the set of drainable work finite and non-replenishing.
7. **Synchronization-boundary withdrawal of future-epoch acceptance.**  
Let  $s$  be an explicit synchronization interface call that separates the current source interval from later work. While the process is pending at  $s$ , the implementation shall not present producer-facing availability for any back-annotated capacity intended only for  $\text{suff\_S}(s)$ ; equivalently, any

asserted upstream producer request targeting such capacity shall remain channel-disabled until  $s$  commits. This does not prevent drain of work already accepted before reaching  $s$ . **Formal elaboration requirement.** For purposes of Appendices J–K, this withdrawal shall be represented in  $\text{Sys\_ref}$  by the elaborated source model  $\text{Sys\_B}^{\text{elab}}$ , not by adding a hidden extra enablement predicate to an otherwise E4-enabled outer channel.  $\text{Sys\_B}^{\text{elab}}$  shall introduce explicit sync-boundary / epoch-gate abstract channels, or equivalent explicit staged realization structure, so that capacity usable only by  $\text{suff\_S}(s)$  is separated from capacity available to the current interval. While  $s$  is pending, the gate/boundary for future-epoch acceptance is channel-disabled in the ordinary E4 sense at that explicit abstract boundary. After  $s$  commits, the corresponding post-sync capacity may again become producer-facing available according to the ordinary ChannelEnabled rules for the elaborated channels.

## General drain-only blocked-frontier obligation used by Appendix K

Appendix K applies the drain-only blocked-frontier discipline to every source interval in which Policy L can leave source-later same-interval blocking work issued while an earlier minimal blocking frontier remains uncommitted. This includes pipelined-loop overlap and other explicitly modeled eager/overlapped issue admitted by the selected  $\text{Sys\_ref}$  transition system; it is not a license to treat ordinary source-sequenced operations as incomparable, and it is not limited to physically pipelined control.

For pipelined or otherwise overlapped control, Appendix B automatic flush is the implementation-side discharge. For ordinary non-overlapped control, the obligation may be discharged by proving that, after the process reaches a channel-disabled blocked point, its control cannot launch any new source-later dynamic operation until the blocker commits. In both cases, Appendix C supplies the non-withdrawal discipline, and E4/B-faithfulness plus explicit  $\text{Sys\_B}^{\text{elab}}$  accounting supply finite boundary storage or credit. The resulting drain set may shrink through already-issued commits but cannot be replenished across the blocked frontier.

**Example (II=1, latency=4, automatic flush):** a loop that performs one blocking  $\text{Pop}_{\{c\_in\}}$  and one blocking  $\text{Push}_{\{c\_out\}}$  per iteration in the pre-HLS source may, after pipelining in the post-HLS RTL, internally accept/stage up to four input values into pipeline staging (pipeline overlap) before the first output  $\text{Push}_{\{c\_out\}}$  commits, subject to the back-annotated bound  $B(c\_in)$  (E4). Values may then reside for several cycles in internal pipeline storage; such internal staging/hand-off is  $\epsilon$ -state within the back-annotated realization and does not create any new auxiliary or duplicate  $\Sigma$ -visible  $\text{Push}_c/\text{Pop}_c$  events beyond the boundary commits at the  $\Sigma$ -visible endpoints (it may change only the timing of those boundary commits under overlapped execution). The drain-only blocking behavior is as defined above under “automatic flush”; in particular, if the blocked operation is in a late pipeline stage (e.g., the final Push), there may be little or no downstream work to drain.

## Appendix C – Issue/Commit Terminology and Linkage

### Scope / shared definitions

The definitions below are with respect to the  $\Sigma$ -visible endpoints of the chosen source reference model  $\text{Sys\_ref}$  (Definition Core.1 and Appendix J, Remark 2)—i.e., the ready/valid interfaces at which  $\Sigma$  records message-transfer events in the raw rendezvous case  $\text{Sys}$ , the ordinary bounded-FIFO case  $\text{Sys\_B}$ , or the elaborated back-annotated case  $\text{Sys\_B}^{\text{elab}}$ . All shared notions about (i)  $\Sigma$  vs.  $\epsilon$  steps, (ii) BoundaryReq / ChannelEnabled / ChannelDisabled, (iii) non-blocking polls as  $\epsilon$  on failure, and (iv) the  $B(c)$  /  $\text{occ}_c(t)$  (cycle-boundary, non-fall-through) interpretation of E4 are as defined in “Normative Formal Core for Appendices G–K” and are to be interpreted over the explicit abstract channels of  $\text{Sys\_ref}$ . Appendix C

uses Definition Core.8 for the formal  $\text{Commit}(q)/\text{CommitCycle}(q)$  meaning and defines  $\text{Issue}$ ,  $\text{ReturnedAtCycle}$ , and the  $\text{Issue}/\text{Commit}$  linkage conditions that make  $\text{Issue}(q)$  well-defined.

### **Commit (message channel) — informal operational reading**

Informally, on  $\text{RTL\_B}$  at a mapped ready/valid boundary, a message commit is observed when an external clocked observer sees both valid and ready high in the same cycle, with the committed payload equal to the mapped data value seen in that cycle. This is the RTL-facing realization of the abstract boundary commit; for the G–K theorem stack, Definition Core.8 is authoritative and defines the abstract E4 commit transition,  $\text{Commit}(q)$  as the source operation’s process-facing commit occurrence, and  $\text{CommitCycle}(q)$  as its cycle index.

**Definition: Issue (message channel)** — as an auxiliary (scheduler/witness) timestamp

**Issue(q)** is an auxiliary timestamp used to state scheduling constraints (R4/E3/E5). Intuitively, it is the first cycle in which the calling process begins requesting the transfer corresponding to the source-level message operation instance  $q$  at the chosen explicit abstract boundary endpoint.

**Definition: ReturnedAtCycle (message result availability).**  $\text{ReturnedAtCycle}(q, t)$  means that the process-visible result of message-call instance  $q$  becomes available at the cycle- $t$  boundary. For a successful blocking or non-blocking call, this is the transfer-commit cycle. For a failed PushNB/PopNB call, no message commits, but the success/failure status is nevertheless returned at the cycle- $t$  boundary and is therefore a cycle- $t$  result.  $\text{ReturnedAtCycle}$  is an operational timestamp used for sequential-result availability; it is not a KObs field.

**Same-cycle rendezvous clarification.** For a rendezvous channel  $B(c)=0$ , if a matched Push\_c operation instance and Pop\_c operation instance first assert their endpoint-owned requests in the same cycle  $t$ , and the ready/valid rendezvous is selected/observed in that cycle, then both operation instances have  $\text{Issue} = t$  and both committed transfer events occur in cycle  $t$ . There is no implicit extra cycle between same-cycle matching endpoint requests and the corresponding rendezvous commit. If one endpoint request was already asserted from an earlier blocking operation, then its  $\text{Issue}$  may be earlier than  $t$ , but the matched Push\_c/Pop\_c commit is still a same-cycle commit in cycle  $t$ .

### **Issue/Commit linkage (well-formedness conditions)**

Fix an endpoint direction at an explicit abstract boundary, and let  $\text{req}(t)$  denote the endpoint-owned boundary request signal in cycle  $t$  (valid for Push, ready for Pop), i.e., the process-driven request level captured by BoundaryReq at that boundary (per “Normative Formal Core for Appendices G–K”). For each source-level message operation instance  $q$  on that endpoint direction:

- Boundary-request soundness (blocking requests are non-withdrawable).  $\text{req}(\text{Issue}(q)) = 1$ . If  $q$  is a blocking Push/Pop, then within its reset epoch, once issued it remains the unique outstanding request on that endpoint direction until it commits:  $\text{req}(t)=1$  in every cycle  $t$  from  $\text{Issue}(q)$  through  $\text{CommitCycle}(q)$  (inclusive), with stable payload while asserted for Push. If an affecting RESET occurs first, RW2 ends that dynamic instance rather than withdrawing it within the epoch. (This is the “no deassert-before-commit” discipline used throughout the document.)
- Stable attribution for a single outstanding source-level request. If  $\text{req}(t)=1$  and no transfer commits on that endpoint direction in cycle  $t$ , then any continued assertion  $\text{req}(t+1)=1$  is attributed to the same outstanding source-level message-operation instance  $q$  only so long as the requesting source-level operation remains that same outstanding instance. For blocking Push/Pop, this is the mandatory non-withdrawable case above, so attribution cannot change before commit. For non-blocking PushNB/PopNB, continuous assertion of the endpoint request signal, by itself, does not identify later executed non-blocking call instances as the same  $q$ ; a

new executed source-level non-blocking call creates a new dynamic instance  $q' \in Q_{\text{exec}}, P$  for the owning process  $P$ , even if  $\text{req}$  does not deassert between cycles.

- **Back-to-back request attribution under continuously asserted req.** Suppose  $q_0$  commits in cycle  $t$  and  $\text{req}(t+1)=1$  on the same endpoint direction. The continued level alone does not imply that the next source-level operation  $q_1$  has issued. If the operational witness/certificate identifies  $q_1$  as the next same-direction source-level operation in the same source interval between adjacent ordinary synchronization calls selected into  $S$  and attributes  $\text{req}(t+1)$  to  $q_1$  beginning in cycle  $t+1$ , then  $\text{Issue}(q_1)=t+1$  even though  $\text{req}$  did not deassert. If  $q_1$  does not become the attributed owner until a later cycle  $u>t+1$ , then  $\text{Issue}(q_1)=u$ . This is a request-attribution rule, not a requirement that  $q_1$  issue immediately after  $q_0$  commits.
- **Same-endpoint blocking issue serialization.** Let  $q_0$  and  $q_1$  be distinct blocking source-level message operations on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint in one reset epoch, with  $q_0 <_{\text{src}} q_1$ . If both issue, then  $\text{Issue}(q_0) < \text{Issue}(q_1)$ . A scalar endpoint has one operation-attributed process-owned request direction at a time; two distinct blocking dynamic instances may not acquire that same request direction in the same issue cycle. This is the strict same-endpoint fact used by mechanization; R4/E3 retain their general non-strict form because they also cover distinct interfaces and non-blocking cases.
- **Sequential-result availability (no same-cycle dependent continuation).** Let  $\text{ReturnedAtCycle}(q_0, t)$  hold in a sequential process  $P$ . If a dynamically succeeding operation  $q_1$  is reachable only after  $q_0$  returns, or if  $q_1$ 's control choice, endpoint, address, payload, or other required operand depends on the value or success/failure status returned by  $q_0$ , then  $\text{Issue}(q_1) > t$ . This rule applies to successful commits and failed non-blocking returns, across distinct channels and endpoint directions as well as on the same endpoint. Same-cycle issue or commit remains permitted only when  $q_1$  was independently reached and all required inputs, payload values, and control predecessors were available before the cycle- $t$  boundary under  $\text{Core.RegSeq}$ ; overlap between independently active pipeline iterations is not prohibited, but an unresolved NB result in an earlier iteration places any dependent later-iteration action in a strictly later Horizon.
- **Sync-boundary discipline.** If  $q_0 \in \text{pref}_S(s)$  and  $q_1 \in \text{suff}_S(s)$  for an ordinary synchronization call  $s$  selected into  $S$ , then  $\text{Issue}(q_1)$  is strictly after  $s$ 's commit in the same trace (i.e.,  $\text{Issue}(q_1) > \text{clk\_pre}(s)$  in  $\tau_{\text{pre}}$  and  $\text{Issue}(q_1) > \text{clk\_post}(s)$  in  $\tau_{\text{post}}$ ). If  $q_0$  is a blocking Push/Pop,  $q_0$  must process-facing commit no later than  $s$ ; no issued-but-uncommitted blocking operation in  $\text{pref}_S(s)$  may survive the commit of  $s$  into the succeeding source interval. Same-cycle completion is permitted:  $q_0$  and  $s$  may both commit in cycle  $t$  when  $q_0$ 's process-facing commit is part of the selected cycle- $t$  boundary actions and is accounted for before the interval advance caused by  $s$ . Any boundary request assertion that persists while the process is still pending at  $s$  is attributed to the outstanding operation(s) in  $\text{pref}_S(s)$  and does not constitute issuance of any operation in  $\text{suff}_S(s)$ . Under the pipelined-loop body no-wait condition, there are no in-body wait events that create additional sync-boundary cuts inside an HLS-pipelined loop.

For non-blocking message passing (PushNB/PopNB), a failed poll produces no  $\Sigma$ -event and is modeled as an  $\epsilon$ -step, while a successful poll contributes the corresponding committed  $\text{Push}_c(v)$  /  $\text{Pop}_c(v)$  event; see “Non-blocking message-passing” in “Normative Formal Core for Appendices G–K” and Appendix I.2. Each executed non-blocking call is a distinct dynamic source-level message-call instance  $q \in Q_{\text{exec}}, P$  for the owning process  $P$ . Thus, if repeated PopNB/PushNB polls occur in later loop iterations or at later source-level call instances, they are distinct  $q$ 's even if the endpoint request remains continuously asserted and no intervening commit occurs. Stable attribution therefore applies to a single outstanding source-level request instance; it does not collapse distinct non-blocking dynamic call instances across

iterations into one  $q$ . Successful non-blocking calls that commit induce message-operation frontier items  $x \in \Omega_{\text{msg},P}$  when their committed transfer is relevant to  $\Omega_P$ , with owning message instances  $q = \text{MsgOf}_P(x)$  in  $Q_{\Omega,P}$ . Failed non-blocking polls remain only in  $Q_{\text{exec},P}$  and contribute no  $\Omega_P$  or  $\Omega_{\text{msg},P}$  item.

## Appendix D – Modeling Latency-Sensitive Protocols

In some cases, designs or testbenches may use protocols or local control policies which are latency-sensitive. These situations are handled by isolating the latency-sensitive portions into small, self-contained parts, and then keeping the rest of the design and testbench latency insensitive. A latency-sensitive portion is any local block of logic whose functional or externally visible behavior depends on cycle-level timing that is not represented by the ordinary latency-insensitive  $\Sigma$  interface. This includes ordinary signal-level protocols with fixed cycle timing, global interrupt/request ordering logic, non-blocking arbitration whose winner depends on request arrival timing, and non-blocking retry/timeout logic whose later behavior depends on the number or cadence of failed PushNB/PopNB polls before success.

For such logic, the intended methodology is not to treat the hidden timing dependence as part of the ordinary latency-insensitive source-core proof. Instead, the latency-sensitive portion should be isolated behind an explicit boundary. On one side of the boundary, the transactor or local protocol block may be modeled cycle-accurately, snooped, or otherwise verified with its detailed timing behavior. On the other side of the boundary, the rest of the system should see a latency-insensitive interface, such as message-passing transactions, synchronization events, or a properly handshaken signal protocol.

Consider a case where a signal-level protocol is latency-sensitive. The protocol will have specific timing behavior that it must meet. To handle this, we create a transactor that has two sides: the first side interacts with the signals and handles the detailed timing requirements, and the second side sends and receives messages with the rest of the system. The message-passing side is latency insensitive. The same pattern applies to cadence-sensitive non-blocking polling logic: if retry counters, timeout counters, or arbitration decisions depend on the number or cadence of failed non-blocking polls, that polling loop is part of the latency-sensitive local protocol block. Its detailed failed-poll cadence should be contained inside the isolated block or aligned by the Appendix L snooping mechanism; it should not leak as an unmodeled hidden timing dependence into the ordinary Appendix I/J latency-insensitive proof. To properly model the detailed timing behavior, the latency-sensitive transactor or local protocol block is modeled at the cycle-accurate level in SystemC, or is otherwise covered by an explicit snooping/alignment argument. There is an example of such a transactor model in the following document and example:

[https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib\\_examples/examples/53\\_transactor\\_modeling/transactor\\_modeling.pdf](https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/examples/53_transactor_modeling/transactor_modeling.pdf)

For purposes of the formal appendices, once the latency-sensitive portion has been isolated, the ordinary Appendix I/J proof is applied at the chosen observable boundary of that isolated block. The failed-poll cadence internal to the isolated block remains local timing behavior; the rest of the system observes only the latency-insensitive or explicitly modeled boundary behavior.

## Appendix E – One-Way Handshake Protocols

In some systems a one-way signal handshake is sufficient for reliable system operation. When using a one-way handshake, the implicit assumption is that the “missing” handshake isn’t required since it is always true.

For example, in time-domain signal processing hardware designs, signal processing HW blocks may input new samples at a fixed rate. The HW blocks are always ready to receive new samples on each clock, but they need to know if the samples are valid. These types of designs can be modeled with the one-way handshake dat/vld protocol demonstrated in this example:

[https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/tree/master/matchlib\\_examples/examples/32\\_dat\\_vld](https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/tree/master/matchlib_examples/examples/32_dat_vld)

## Appendix F - Modeling Guidelines and Design Constraints

The following are simple modeling guidelines that align with the modeling methodology described in this document.

- 1) Prefer to use `Connections::In/Out + SyncChannel` over signal IO and `wait()`.
- 2) Prefer to use `Pop()/Push()` over `PopNB()/PushNB()`.
- 3) When pipelining a loop, prefer to use `hls_stall_mode flush`.

### When using signal IO:

- 4) If modeling a cycle-accurate process, use `disable_spawn` and follow the example `53_transactor_modeling` style and do not use `Push/Pop` in the process.
- 5) If modeling a *direct input*, use `hls_direct_input` and possibly `hls_direct_input_sync`, and only change the signal at allowed times.
- 6) If combining signal IO with `Connections::In/Out` in the same process, use proper signal IO handshake that does not rely on `In/Out` ports for process synchronization. Place signal IO operations very close to their `wait()` statements.
- 7) Unless you are modeling a cycle-accurate process, you should expect that latency will change between the pre-HLS and post-HLS models.

## Design Constraints for Applying the Formal Guarantees

This section consolidates the design choices that most directly determine whether the formal guarantees in this document can be applied. It is written for hardware architects and HLS users rather than for the tool or proof implementer. A design can synthesize and function correctly while still lying outside the stated theorem scope if one of these behaviors is left implicit.

This section is informative. Each constraint restates a normative requirement located in the referenced sections; the referenced text is authoritative.

The architect is responsible for choosing interfaces, reset behavior, and communication structures that admit the required correspondence. The HLS tool and verification flow are responsible for constructing and checking the detailed process, channel, attribution, coverage, and other certificates. The architect does not need to produce those certificates, but the design must expose enough explicit structure for the flow to do so.

The first list applies to the implementation-safety and observable-correspondence result. The second list contains additional constraints for the internal message-passing deadlock-preservation result. A design outside the simple route is not necessarily prohibited; it generally requires explicit modeling, witness alignment, or a separate proof route.

### Design constraints for the safety and correspondence guarantee

These constraints support the Post -> Ref guarantee that the covered RTL behavior introduces no new observable functional behavior absent from the prescribed source reference model.

1. **Give every signal read and write an unambiguous synchronization anchor.** For a sequential process, place signal reads immediately after the synchronization call that defines their observation point and place signal writes immediately before the synchronization call at which they become visible. Branches must not allow one write to be associated with different later waits on different paths. Signal I/O inside a rolled or preserved loop needs an explicit per-iteration wait when each iteration is intended to create a distinct observation. A loop selected for HLS pipelining cannot use an in-body wait under this contract; move the signal interaction outside the loop, use a supported message-passing or direct-input mechanism, or do not pipeline that loop.
2. **Keep direct inputs stable, or update them through the supported synchronization scheme.** A direct input may be physically sampled later than its logical source-level read only when its value remains stable for the applicable reset epoch. If software or the testbench must change control inputs while the design is running, use the documented `hls_direct_input_sync` handshake and change the values only at the designated synchronization point. An ordinary Push or Pop is not a substitute for that update contract.
3. **Make timing-sensitive environment behavior explicit.** The ordinary safety result assumes that the same external behavior is selected from the same prior observable history, not from hidden differences in absolute cycle count. Testbench branches, watchdogs, stimulus changes, or pass/fail decisions that depend on raw latency must be isolated as latency-sensitive logic, made observable through an explicit protocol, or checked at RTL rather than assumed to transfer automatically from an arbitrary pre-HLS run.
4. **Treat non-blocking polling as an architectural proof-route decision.** The simple route applies when a failed PopNB or PushNB does not change the next observable action, request, payload, endpoint choice, or control path. Typical arbitration code, retry counters, timeouts, backoff, and "work on B if A is unavailable" behavior do change what happens next. Such designs require an approved witness-alignment mechanism, commonly the Appendix L snooping route, or an isolated and explicitly modeled latency-sensitive protocol block. Prefer blocking interfaces when the architecture does not genuinely require poll-dependent control.

5. **Do not encode an unbounded wait as an invisible busy-poll loop in the ordinary proof fragment.** A loop that repeatedly executes failed non-blocking polls continues to take local internal steps and is not a stable waiting state. Use a blocking operation, an explicit stable-wait or handshake construct, or isolate and model the polling loop as timing-sensitive logic. A finite, statically bounded polling sequence can be handled when its outcomes and later effects satisfy the applicable witness and dependency conditions.
6. **Expose cross-process shared-memory value flow through a supported interface.** Synchronization alone can order two processes' memory accesses, but the present safety proof does not carry an independent global shared-memory state that proves both models read the same values. Use the supported memory library or mapping layer so commands, responses, or value-carrying accesses appear as covered message-passing or anchored-signal events. A design using an ordinary shared array between processes instead needs a separate memory-state correspondence proof that is not developed in this document.
7. **Model the channel behavior that the implementation actually provides.** The default buffered-channel abstraction is a bounded FIFO whose availability is evaluated at the beginning of the cycle and which does not permit first-word fall-through. If the RTL permits a value pushed into an empty FIFO to be consumed in the same cycle, or includes a bypass, debug drain, hidden credit path, or side entrance, that behavior must be represented explicitly in the selected source reference model. Back-annotated capacity alone is not sufficient when hidden structure changes enablement or transfer ordering.
8. **Use an elaborated model for coupled multi-input pipelines and other joined structures.** When acceptance on one input depends on another input, a bundle, join, shared stage, coupler, or epoch gate, do not interpret the outer capacity of each input as independent slack. The verification flow must expose the relevant staged, rendezvous, bundle, or join boundaries in the elaborated source reference model. The outer interface remains the user-visible boundary, but proof-relevant occupancy and enablement may be tracked at the introduced internal abstract boundaries.
9. **Give every dynamic reset a channel-consistent disposition.** Reset both endpoints of a live channel together with the channel state, or define an explicit flush, abort, cancellation, or end-of-epoch protocol that determines what happens to outstanding requests and in-flight data. A reset of only one endpoint cannot silently discard, retain, or reattribute a transaction across the reset boundary. The protocol must be represented in both the source model and RTL using ordinary observable actions or another explicitly verified reset-state correspondence.

### Additional constraints for the internal message-passing deadlock guarantee

The Appendix K result excludes the defined reachable closed internal wait-cycle cutpoints. It therefore needs the safety constraints above plus the following structural and operational conditions. These are additional requirements for the deadlock claim; they should not be read as extra premises for every use of the Appendix J safety theorem.

1. **Give each analyzed logical channel one producer and one consumer.** A shared bus, crossbar, multi-client queue, or resource with multiple requesters should be represented by an explicit arbiter process and point-to-point channels. This makes the complementary endpoint and wait-for ownership unambiguous. A custom shared-channel abstraction requires its own ownership and arbitration proof rather than being treated as an ordinary point-to-point channel.

2. **Represent multi-lane, burst, and multi-transfer resources explicitly.** The standard scalar-channel model permits at most one Push and one Pop commit per channel direction in one cycle. A resource that transfers several elements, lanes, or beats at once must expose that behavior in its abstract interface and capacity model. It should not be encoded as several hidden transfers on a channel that the proof treats as scalar.
3. **Keep blocking requests asserted and Push payloads stable until completion.** A blocking Push or Pop is a persistent request, not a speculative try that may be withdrawn when back-pressure lasts longer than expected. Request ownership must remain with the same dynamic operation until commit or an explicit reset. A separate nonterminal cancellation/withdrawal action belongs to a different modeled protocol and is not part of the ordinary blocking wait-for interpretation used by the G–K theorem stack.
4. **Ensure that a blocked pipelined process can only drain previously accepted work.** Once a process has reached a disabled blocking frontier, later blocking work from that process must not continue to enter the system behind the blocked operation. Previously accepted pipeline work may finish and release finite downstream state, but the pipeline must not replenish that work indefinitely or expose next-interval capacity before the synchronization boundary completes. This is the practical meaning of the automatic-flush and drain-only discipline used by the deadlock proof.
5. **Do not make deadlock freedom depend on favorable overlap or same-cycle issue.** The primary practical check deliberately uses a conservative serial-issue source execution. If the design avoids deadlock only because two source-ordered blocking operations overlap, issue in the same cycle, or are arbitrated in a particularly favorable order, a clean serial execution cannot justify scheduler-robust deadlock freedom. Rewrite or buffer the protocol, add explicit synchronization, or use a different scheduler-specific proof.
6. **Check deadlock against the actual capacities and explicit boundary structure.** The relevant source model is the back-annotated bounded or elaborated model, not an idealized unbounded network and not necessarily the raw rendezvous model. A capacity-induced deadlock is real for the selected capacity configuration even if it disappears at a larger depth. Conversely, a rendezvous-only run can be a useful early screen but may report a deadlock that the intended buffered or elaborated design does not have.
7. **Account explicitly for reset during the serial comparison.** For a candidate blocked component, the primary serial-reduction route requires either a comparison continuation with no affecting reset or an already-triggered, non-waived response monitor whose exact pending operation is cleared by the matched reset. A reset that occurs before such a monitor exists, or whose scope cannot be matched to the pending operation, is outside that route and needs another deadlock argument.
8. **Keep other kinds of stalls separate from the internal message-passing claim.** Barrier participation errors, a process waiting for a peer that has terminated without an explicit end-of-stream protocol, environment-only stalls, and timed or reactive multi-frontier behavior are not automatically proved safe by the internal wait-cycle theorem. Model EOF, done, abort, and barrier protocols explicitly, and use the supported environment/frontier fragment or a separate proof when external timing can release a candidate wait cycle.

Formal references. The detailed rules are in RULE 1 and RULE 2, Direct Inputs, Scheduling of Array Accesses, Appendices B, D, I, J, K, L, and Q, and the shared channel, memory, reset, and registered-interface definitions in the Normative Formal Core. Appendix M.7a remains the definitive per-instance evidence checklist for the tool and verification flow.



## Guide to the Formal Appendices

This guide is informative. It explains how the formal appendices fit together and suggests reading paths; it does not introduce new assumptions, definitions, or proof obligations. The appendix letters and theorem identifiers remain authoritative.

**How to interpret the qualification detail.** The formal appendices deliberately make explicit many assumptions and implementation facts that are often left implicit in hardware verification, whether the RTL is synthesized from HLS source or written manually. The underlying engineering conditions were largely already there: a claimed model-to-implementation result already depends on facts about scheduling, interfaces, buffering, reset, environment behavior, progress, and implementation correspondence, whether or not those dependencies have been individually named. The formalization makes those dependencies explicit and auditable. It should not be read as requiring every design user to understand or manually discharge every formal condition; many conditions are expected to be established automatically, satisfied by simple design restrictions, or discharged once through reusable tool, library, or implementation-flow qualification. The methodology may add explicit evidence and accounting where a theorem-backed guarantee requires it, but that should not be mistaken for creating all of the underlying engineering conditions.

### Recommended reading paths

**HLS users and hardware architects.** Read the main body through Appendix F for the scheduling contract and modeling guidance, then read Appendix M for the formal conclusions and their practical limits. Consult Appendix L when the design uses non-blocking operations or cadence-sensitive behavior. If you need the Policy-S deadlock-preservation route, read Appendix M.6 followed by Appendix K.4b, K.4c, and K.4f. These sections explain the selected serial-mode run, the bounded-response check, and the standard applicability fragment. The design-user activity is one complete instrumented Policy-S execution; the flow separately establishes simulator determinism, applicability, response coverage, and the implementation-side QSync/correspondence evidence from which RTL cutpoint coverage is derived or otherwise discharged. Tool, checker, and formal-proof implementers should additionally read Appendix K.4e and Appendix K-P.

**Safety reviewers.** Read the Normative Formal Core for Appendices G–K, then Appendix G, Appendix I, and Appendix J. Appendix J.2b is optional for a one-prefix Theorem J.1 safety review or an independently discharged J.TraceCertCov domain, but it is used by the standard QSync implementation-wide safety route: J.QSyncTraceInd/J.TraceCertCov-QSyncCycle plus J.CycleIdealCover/J.1-Safe-QSync extend safety to all reachable finite observable prefixes. Its richer J.QSyncCertInd route additionally supplies the K-facing package-totality result used by the universal Appendix K conclusion. Appendix R provides a compact derived companion, and Appendix T highlights important examples of the scheduling-specific obligations that remain beyond generic transition-system and simulation theory.

**Liveness and deadlock reviewers.** Begin with the Normative Formal Core and Appendix G, then read Appendix J for the source/RTL cutpoint connection and Appendix J.2b for the standard QSync canonical-cycle package-totality route, Appendix K for the preservation theorem, Appendix K-P for the serial-reduction proof companion, Appendix L for witness selection and environment-sensitive cases, and Appendix N for fairness and progress assumptions. Appendix Q is required when the selected source reference model uses elaborated staged, bundle, join, coupler, or epoch-gate structure.

**Mechanization work.** Use the Normative Formal Core as the semantic skeleton and Appendix R as its derived compact map, Appendix S for the Lean proof strategy and mechanization decomposition/import anchors, and Appendix T as a selective explanation of why some obligations require scheduling-specific machinery beyond standard libraries. The Normative Formal Core and Appendices G–K are the

normative sources for the definitions and theorem interfaces; Appendix M.7a supplies the definitive consolidated side-condition discharge record; Appendix R is derived.

## Appendix map

**Normative Formal Core for Appendices G–K.** Defines the shared reference model, operational issue/admissibility semantics, observation and frontier objects, channel/request predicates, drain discipline, KObs record, capacity conventions, reset treatment, and terminology used by the G–K theorem stack. Appendix M.7a supplies the consolidated side-condition discharge record. Appendix M.7b gives the standard QSync logical certificate-profile packaging of that record.

**Appendix G - Trace Compatibility Rules.** Defines the observable trace-compatibility framework, the E1-E5 rules, the selected source-reference/RTL observation boundary, and the asymmetric Post  $\rightarrow$  Ref and restricted Ref  $\rightarrow$  Post uses.

**Appendix H - Possible Criticisms.** Collects anticipated objections, limitations, and responses. It is useful for understanding why the formal contract has its present scope and why certain stronger-looking claims are not made.

**Appendix I - Per-Process Replay and Preparation Proof.** Establishes the process-local replay and finite preparation machinery used by the compositional safety construction, including the treatment of deferred non-blocking decisions and registered phase behavior.

**Appendix J - System-Level Trace Compatibility Proof.** Composes local process, channel, atomic-interaction, environment, reset, footprint, request-snapshot, dependency, agreement, and coverage evidence into a global source witness and the system-level Post  $\rightarrow$  Ref trace-compatibility result. Section J.2b additionally gives the QSync-specialized base-and-step routes used to derive cycle-aligned safety package coverage and total K-facing package coverage for Appendix K. The cycle-aligned route discharges J.TraceCertCov on J.CycleReach\_post; Lemma J.CycleIdealCover and Corollary J.1-Safe-QSync then extend ideal-prefix-closed safety to all reachable finite outer observable prefixes without asserting a separate J.TraceCertCov package for each non-cycle-aligned prefix. General safety-side J.TraceCertCov remains the interface when an application needs direct per-prefix correspondence packages or uses a broader non-QSync route.

**Appendix K - Preservation of Internal Wait-Cycle Freedom.** States the certified-cutpoint deadlock-preservation result, the exact admissibility and drain bridge, the A5-L premise, and K.CertCov for a universal reachable-cutpoint conclusion. In the standard QSync normal-compositional scope, J.QSyncCertInd/J.CertExist-QSync plus K.CertCov-QSync provide the preferred theorem-derived K.CertCov route; alternate independently checked coverage routes remain permitted.

**Appendix K-P - Proofs Companion for the Serial-Reduction Route.** Develops the principal Policy-S/A5-S route for discharging A5-L, including deterministic serial-source dominance and bounded-response evidence, while recording the route's environment/frontier applicability limits.

**Appendix L - Snooping Introduction, Implementation and Concerns.** Explains witness selection, non-blocking and latency-sensitive cases, and the conditions under which a source execution can be aligned with the RTL for restricted reverse use.

**Appendix M - Formal Results Overview.** Provides a readable but technically precise summary of the models, principal safety theorem, restricted reverse direction, liveness theorem, coverage requirements, and implementation-qualification obligations.

**Appendix N - Scheduler and Environment Fairness Assumptions.** Restates the scheduler, channel-progress, and quiescence assumptions in operational ready/valid form and explains that these are execution-environment obligations rather than automatic HLS-tool guarantees.

**Appendices O and P.** Appendix O gives an informative sketch for multiple clock domains. Appendix P records alternative formal frameworks and their relationship to the chosen architecture.

**Appendix Q - Multiple Input Pipelines Back-Annotation.** Defines the local elaboration structures and projection/accounting rules used when ordinary bounded-FIFO abstraction is insufficient for staged, bundled, joined, coupled, or sync-gated implementations.

**Appendix R - Compact Derived Formal Core and Mechanization View.** Provides a downstream review and mechanization map of the Core  $\rightarrow I \rightarrow J \rightarrow K$  dependency chain; it introduces no normative definition or theorem interface used by earlier sections.

**Appendices S and T.** Appendix S records the Lean proof strategy. Appendix T identifies the scheduling-specific proof obligations that cannot be obtained merely by importing generic labeled-transition-system, simulation, or temporal-logic libraries.

**Appendix U - Revision Record.** Records material changes to the formal interfaces and explanatory text so that reviewers can trace the evolution of the document.

## Normative Formal Core for Appendices G–K

Scope of comparison. Appendix G first presents the raw rendezvous source model Sys and the R/E scheduling rules. For the Appendices I–K proof stack, and for the Sys\_ref-specific uses in Appendix G, the formal comparison is between RTL\_B and Sys\_ref (Definition Core.1 and Appendix J, Remark 2), i.e., the same source processes interpreted under the channel semantics of E4 with capacities B(c) matching the RTL. The E4 channel semantics are case-split: B(c)=0 is interpreted as a capacity-zero rendezvous channel, and B(c)>0 is interpreted as a cycle-boundary, non-fall-through bounded FIFO. The rendezvous case is therefore recovered as the B(c)=0 case of the channel model, not by applying the B(c)>0 FIFO-availability inequalities with B(c)=0.

Authority of the Core. The shared definitions in this Core are normative for the G–K theorem stack. Later appendices may refine these objects and prove properties about them, but they do not redefine them. Material that merely explains the completed J/K architecture or records certificate-discharge choices is placed downstream in Appendix R or Appendix M rather than inside this definitional layer.

| Symbol / Rule                                                                                               | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BoundaryReq(x, σ)</b><br>(shorthand<br>BoundaryReq(q, σ)<br>only in ordinary<br>unique-boundary<br>case) | For any dynamic source-level message-operation frontier item $x \in \Omega_{\text{msg},P}$ on an explicit abstract channel $c = \text{BoundaryOf\_P}(x)$ of Sys_ref, with owning source-level message instance $q = \text{MsgOf\_P}(x)$ and $\text{kind}(q) \in \{\text{Push\_c}, \text{Pop\_c}\}$ , BoundaryReq(x,σ) means: at that chosen abstract boundary and a BoundaryEvalPoint σ, q's endpoint-owned boundary request for item x is asserted. Operationally (ready/valid): <ul style="list-style-type: none"> <li>· for <math>\text{kind}(q)=\text{Push\_c}</math>: the producer endpoint at BoundaryOf_P(x) asserts vld_c, with stable payload as required; and</li> <li>· for <math>\text{kind}(q)=\text{Pop\_c}</math>: the consumer endpoint at BoundaryOf_P(x) asserts rdy_c.</li> </ul> BoundaryReq is therefore a predicate about the endpoint's boundary-visible request assertion for a specific dynamic operation/frontier item, not about an already-committed Σ-event label. In the ordinary unelaborated case, where q has a unique evaluated boundary and frontier item, BoundaryReq(q,σ) is accepted as shorthand for BoundaryReq(x,σ). In Sys_B^elab, the frontier-item form is authoritative because one source-level call q may own a chain of proof-internal frontier items at multiple explicit boundaries. Purely combinational |

| Symbol / Rule                                                                                                                           | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                                                                                                                                         | realization structure on the channel side may affect only the complementary mirror signal at that boundary (Push: rdy; Pop: vld) and thus the commit condition after $\epsilon$ -settling, but it does not change the endpoint request assertion captured by $\text{BoundaryReq}(x, \sigma)$ . When a cycle-index notation is needed, $\text{BoundaryReq}(x, t)$ is shorthand for $\text{BoundaryReq}(x, \sigma_t)$ , where $\sigma_t$ is the selected $\text{BoundaryEvalPoint}$ for cycle $t$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>ChannelEnabled(x, <math>\sigma</math>)</b><br>(shorthand $\text{ChannelEnabled}(q, \sigma)$ only in ordinary unique-boundary case)   | <p><math>\text{ChannelEnabled}(x, \sigma)</math> means: <math>\text{BoundaryReq}(x, \sigma)</math> holds, and at the settled <math>\text{BoundaryEvalPoint}</math> <math>\sigma</math>, the channel-side commit-enabling condition for the owning operation <math>q = \text{MsgOf\_P}(x)</math> holds at the chosen boundary. Concretely, for bounded-FIFO channels <math>B(c) &gt; 0</math>, the channel-side condition is exactly the E4 availability predicate at that boundary:</p> <ul style="list-style-type: none"> <li>· for <math>\text{kind}(q) = \text{Push\_c}</math>: <math>\text{occ\_c}(\sigma) &lt; B(c)</math>, i.e. space is available; and</li> <li>· for <math>\text{kind}(q) = \text{Pop\_c}</math>: <math>\text{occ\_c}(\sigma) &gt; 0</math>, i.e. data is available.</li> </ul> <p>For rendezvous channels <math>B(c) = 0</math>, the channel-side condition is the matching complementary endpoint participation in the same settled boundary-evaluation cycle. If the owning operation represented by <math>x</math> commits, it contributes the corresponding committed <math>\Sigma</math>-event <math>m(q) \in M</math>; before it commits, <math>m(q)</math> is absent, but <math>\text{BoundaryReq}(x, \sigma)</math> and <math>\text{ChannelEnabled}(x, \sigma)</math> are still well-defined for the pending frontier item. In the ordinary unique-boundary case, <math>\text{ChannelEnabled}(q, \sigma)</math> is shorthand for <math>\text{ChannelEnabled}(x, \sigma)</math>; in <math>\text{Sys\_B}^{\text{elab}}</math>, the frontier-item/boundary-specific form <math>\text{ChannelEnabled}(x, \sigma)</math> is authoritative.</p> |
| <b>ChannelDisabled(x, <math>\sigma</math>)</b><br>(shorthand $\text{ChannelDisabled}(q, \sigma)$ only in ordinary unique-boundary case) | <p><math>\text{ChannelDisabled}(x, \sigma)</math> means: <math>\text{BoundaryReq}(x, \sigma)</math> holds but <math>\text{ChannelEnabled}(x, \sigma)</math> does not. Intuitively: the endpoint is requesting the transfer for dynamic operation instance <math>q</math>, but the transfer is blocked by channel-side conditions, such as empty/full status or the rendezvous partner not simultaneously requesting, as evaluated at the settled <math>\text{BoundaryEvalPoint}</math> used by the Core enablement definitions.</p> <p>In the ordinary unique-boundary case, <math>\text{ChannelDisabled}(q, \sigma)</math> is shorthand; in <math>\text{Sys\_B}^{\text{elab}}</math>, the frontier-item/boundary-specific form <math>\text{ChannelDisabled}(x, \sigma)</math> is authoritative.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b><math>\Sigma</math></b>                                                                                                              | <p>Set of observable actions for the equivalence check (committed IO events) at the chosen observable boundary (often the DUT IO boundary for functional equivalence, but optionally expanded—e.g., for Appendix K deadlock reasoning):</p> <ul style="list-style-type: none"> <li>• channel commit events <math>\text{Push\_c}(v)</math> and <math>\text{Pop\_c}(v)</math> on channels designated as observable, labeled with the transferred message value (or abstract message identity) <math>v</math></li> <li>• synchronization events <math>\text{wait}()</math>, <math>\text{Sync}</math>, <math>\text{START\_P}</math>, <math>\text{FINISH\_P}</math></li> <li>• single-cycle signal actions <math>\text{Write}(\text{sig}, \text{val})</math> and <math>\text{Read}(\text{sig}, \text{val})</math> on signals designated as observable, labeled with the value written/read.</li> </ul> <p>Internal-only channels/signals may be excluded from <math>\Sigma</math> (treated as <math>\epsilon</math>-microstate) when they are not part of the chosen observable boundary. If such channels/signals are later brought into the observable boundary (e.g., by snooping for debug or analysis), then they become <math>\Sigma</math>-events and must match</p>                                                                                                                                                                                                                                                                                                                                                                                                     |

| Symbol / Rule | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|               | <p>under <math>\approx</math>.</p> <p>Deadlock-analysis requirement (Appendix K): for Appendix K, <math>\Sigma</math> is instantiated to include every message-passing channel whose endpoint transfers at the chosen Sys_ref boundary can contribute a dependency edge in the Appendix K Wait-For Graph, either because the channel is on the declared observable boundary or because it is snooped. The WFG itself is defined over pending blocking message-operation frontier items <math>x \in \Omega\_msg, P</math> for the relevant owning process <math>P</math>, with owning dynamic message-call instance <math>q = \text{MsgOf\_P}(x)</math>, not over already-committed <math>\Sigma</math>-event labels and not over failed non-blocking polls in <math>Q\_exec, P</math>. Thus, when Appendix K refers to a blocked Push or Pop at a boundary, the relevant frontier object is <math>x</math>, and its owning pending source-level operation instance is <math>q</math> with <math>\text{kind}(q) \in \{\text{Push\_c}, \text{Pop\_c}\}</math>; the edge is determined by <math>\text{BoundaryReq}(x, \sigma)</math> together with <math>\text{ChannelEnabled}(x, \sigma) / \text{ChannelDisabled}(x, \sigma)</math> at the selected Core.KCut <math>\sigma</math> (and hence at a settled BoundaryEvalPoint). Internal micro-interfaces that lie within the back-annotated realization counted in <math>B(c)</math> (e.g., post-HLS RTL pipeline stage implementation) are not treated as separate channels for this requirement: they may be exposed/snooped for debug, but Appendix K's WFG edges are defined using the ChannelEnabled/ChannelDisabled status of the corresponding boundary operation instance <math>q</math> via <math>E4/occ\_c/B(c)</math>, so such internal points can participate logically in the deadlock argument without being explicitly <math>\Sigma</math>-observable.</p> <p><b>Non-blocking message-passing (PushNB/PopNB):</b> A non-blocking call that returns “not completed” produces no <math>\Sigma</math>-event and is modeled as an internal <math>\epsilon</math>-step.</p> <p>When a non-blocking call succeeds (returns “completed”), that success is represented in <math>\Sigma</math> as the corresponding committed Push_c(v) or Pop_c(v) event on that channel, labeled with the same transferred value/message <math>v</math> (i.e., <math>\Sigma</math> records transfers and their payloads, not failed polls). A non-blocking call is considered issued in the cycle in which it executes (its Issue cycle). A call that returns “completed” commits in that same cycle and therefore yields a committed transfer <math>m(q) \in M</math>; a call that returns “not completed” produces no committed transfer event (<math>m(q)</math> undefined / absent) and may be treated as an <math>\epsilon</math>-step with respect to <math>\Sigma</math>-visible message-transfer events.</p> <p>Each executed non-blocking call is still a distinct dynamic source-level message-call instance <math>q \in Q\_exec, P</math> for the owning process <math>P</math>. Since failed non-blocking <math>q</math>'s have no <math>\Sigma</math>-label, their cross-trace matching is witness-relative: it is supplied by the chosen <math>\mu\_Q / WC</math> witness, by NB-CF when failed polls cannot affect the next <math>\Sigma</math>-visible control flow or frontier, or by Appendix L's snooping / witness-selection mechanism. Continuous assertion of an endpoint request does not by itself identify multiple executed non-blocking calls as one <math>q</math>, and the <math>\Sigma</math> trace alone is not required to determine how failed non-blocking <math>q</math>'s are matched. A successful non-blocking call may also be a member of <math>Q\_Q, P</math> when it contributes a committed Push_c(v) / Pop_c(v) event; a failed non-blocking poll remains in <math>Q\_exec, P \setminus Q\_Q, P</math>.</p> |

| Symbol / Rule                     | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>START_P</b>                    | First observable action emitted by process P after reset. Marks the moment P begins executing user code. A new START_P is emitted each time P exits reset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>FINISH_P</b>                   | Last observable action of process P in the current reset epoch. If P executes an unbounded loop within that reset epoch, FINISH_P never occurs for that epoch, and the epoch-local trace is treated as infinite until a later reset boundary, if any. If a later reset affecting P occurs and P exits reset again, a new dynamic START_P begins a fresh reset epoch; any later termination is represented by a distinct dynamic FINISH_P occurrence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b><math>\epsilon</math>-step</b> | An <i>internal</i> , unobservable RTL state transition (pipeline advance, FSM micro-state, etc.).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>B(c)</b>                       | <p>Back-annotated effective capacity of channel c at the chosen explicit abstract Sys_ref boundary used by E4 and Appendix K. In the ordinary non-elaborated case this is the chosen <math>\Sigma</math>-observable Sys_B boundary; in elaborated cases it is the corresponding explicit abstract channel boundary of Sys_B^elab, which may be proof-internal and later projected away from the outer <math>\Sigma</math>_DUT trace. Finite, <math>B(c) \geq 0</math>. It reflects the total bounded “composite buffer” storage/credit in the RTL realization represented by that explicit abstract channel boundary, including FIFO storage, skid/elastic buffering, HLS-inserted pipeline staging that can hold in-flight messages, and any explicit staged/bundle/join channel capacity introduced by Sys_B^elab.</p> <ul style="list-style-type: none"> <li>• <math>B(c)=0 \Rightarrow</math> rendezvous (capacity-zero) channel.</li> <li>• <math>B(c)&gt;0 \Rightarrow</math> bounded FIFO (of that effective capacity).</li> </ul> <p><b>B-faithfulness.</b> For the formal results to apply to a concrete RTL_B instance, each back-annotated B(c) must be faithful to the chosen explicit abstract boundary. This means:</p> <ul style="list-style-type: none"> <li>• <b>Boundary completeness:</b> every RTL storage element, credit, skid/elastic buffer, pipeline stage, or elaborated staged/bundle/join component that can accept, hold, release, or backpressure messages across the abstract boundary c is either included in the effective capacity B(c) for that boundary or represented as a separate explicit Sys_B^elab boundary with its own capacity and E4 semantics.</li> <li>• <b>No hidden capacity or hidden enablement:</b> the RTL realization does not permit a producer-facing accept or consumer-facing delivery at boundary c except in cycles allowed by E4 using the corresponding <math>\text{occ}_c(t)</math> and B(c), and any additional enablement restriction that affects boundary behavior is represented by an explicit Sys_B^elab boundary rather than by an unmodeled predicate.</li> <li>• <b><math>\epsilon</math>-opacity of internal movement:</b> movement of an already accepted message within the RTL realization counted by B(c) is internal <math>\epsilon</math>-state. Such movement does not create additional <math>\Sigma</math>-visible Push_c/Pop_c events and does not create, delete, duplicate, reorder, or relabel the <math>\Sigma</math>-visible boundary transfers.</li> <li>• <b>Evidence obligation:</b> B-faithfulness is a tool/RTL-flow compliance obligation. It may be discharged by a trusted back-annotation algorithm, generated</li> </ul> |

| Symbol / Rule   | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                 | <p>certificate, structural RTL check, interface monitor, formal equivalence/checking flow, or other validation method appropriate to the implementation flow. It is not derived by E4, Proposition J.0, or Theorem J.1.</p> <p><b>Point-to-point endpoint assumption (single-producer/single-consumer):</b> For every message-passing channel <math>c</math> (including both buffered channels with <math>B(c)&gt;0</math> and rendezvous channels with <math>B(c)=0</math>), there is exactly one producer process that may execute <math>\text{Push}_c</math> on <math>c</math>, and exactly one consumer process that may execute <math>\text{Pop}_c</math> on <math>c</math>. Hence the complementary endpoint relevant to any blocked <math>\text{Push}_c/\text{Pop}_c</math> on <math>c</math> is unique.</p> <p><b>Modeling note (shared resources / arbitration):</b> If an implementation conceptually has multiple producers and/or multiple consumers for a logical resource, that sharing must be modeled explicitly (e.g., an arbiter process plus per-client point-to-point channels), rather than as a single multi-endpoint “channel.” This keeps WFG wait dependencies well-defined and unique per edge.</p> <p><b>Single-transfer-per-cycle endpoint constraint (scalar channel interface):</b> For every message-passing channel <math>c</math>, in any clock cycle <math>t</math>, at most one <math>\text{Push}_c</math> may commit and at most one <math>\text{Pop}_c</math> may commit. Equivalently, the set of commits on a fixed channel in a single cycle contains <math>\leq 1</math> Push and <math>\leq 1</math> Pop (it may contain one of each in the same cycle). For rendezvous channels (<math>B(c)=0</math>), a committed transfer occurs only as a matching same-cycle <math>\text{Push}_c(v)</math> and <math>\text{Pop}_c(v)</math> pair. For buffered channels (<math>B(c)&gt;0</math>), a same-cycle <math>\text{Push}_c</math> and <math>\text{Pop}_c</math> do not imply fall-through; the channel semantics are cycle-boundary, non-fall-through as stated in E4.</p> <p><b>Reset-empty FIFO convention:</b> after each reset boundary at which the compared execution begins or restarts, every buffered message-passing channel with <math>B(c)&gt;0</math> has logical FIFO occupancy 0 at the chosen <math>\Sigma</math> boundary. Equivalently, the reset state contains no preloaded in-flight messages in the abstract channel realization used by <math>\text{Sys\_ref}</math> / <math>\text{RTL\_B}</math>. If a design requires preloaded channel contents, those contents must be modeled explicitly as post-reset producer actions or by a separate initialized-state extension outside the present empty-FIFO formal model.</p> <p><b>Modeling note (multi-lane / burst transfers):</b> If an implementation can transfer <math>k&gt;1</math> messages per cycle on a logical resource, model it as <math>k</math> parallel point-to-point channels (or as a single widened message transferred by one <math>\text{Push}_c/\text{Pop}_c</math> per cycle) so that the <math>\Sigma</math>-level event model continues to satisfy the scalar constraint above.</p> |
| <b>occ_c(t)</b> | <p>Abstract FIFO occupancy at the chosen explicit abstract <math>\text{Sys\_ref}</math> channel boundary: number of items committed by <math>\text{Push}_c</math> but not yet committed by <math>\text{Pop}_c</math> at that boundary, measured from the most recent reset boundary under the reset-empty FIFO convention. In the ordinary non-elaborated case this is the chosen <math>\text{Sys\_B}</math> channel boundary; in elaborated cases <math>\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}</math>, this is the occupancy of the relevant explicit abstract staged/bundle/join channel boundary.</p> <p>Equivalently (E4), <math>\text{occ}_c(t) = \text{push}(\pi_t) - \text{pop}(\pi_t)</math>, where <math>\pi_t</math> is the <math>\Sigma</math>-visible</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Symbol / Rule              | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|                            | <p>boundary-commit prefix for that explicit abstract channel strictly before cycle <math>t</math> (i.e., excluding any Push<sub>c</sub>/Pop<sub>c</sub> commits that occur in cycle <math>t</math>), so <math>\text{occ}_c(t)</math> is the abstract occupancy at the beginning of cycle <math>t</math>. For pure-FIFO channels with <math>B(c) &gt; 0</math>, the FIFO availability predicates (not-full/not-empty) are evaluated against this begin-of-cycle occupancy; hence a Pop<sub>c</sub> cannot commit in a cycle that begins empty solely because a Push<sub>c</sub> also commits in that same cycle (no fall-through), and likewise a Push<sub>c</sub> cannot commit in a cycle that begins full solely because a Pop<sub>c</sub> also commits in that same cycle.</p> <p>Clarification (no “second Push/Pop”): movements of an item within the RTL channel realization (e.g., FIFO→staging/skid buffering, staging/skid buffering→pipeline-stage registers, pipeline-stage→pipeline-stage handoff) are internal <math>\epsilon</math>-steps; they do not constitute additional <math>\Sigma</math>-visible committed Push<sub>c</sub>/Pop<sub>c</sub> events beyond the boundary events counted by <math>\text{push}(\pi_t)/\text{pop}(\pi_t)</math>.</p> <p>Derived cutpoint shorthand (used in the Core and Appendix K): if <math>\sigma</math> is a BoundaryEvalPoint occurring in cycle <math>t_\sigma</math>, write <math>\text{occ}_c(\sigma) \triangleq \text{occ}_c(t_\sigma)</math>. Likewise write <math>\text{ChannelEnabled}(q, \sigma) \triangleq \text{ChannelEnabled}(q, t_\sigma)</math> and <math>\text{ChannelDisabled}(q, \sigma) \triangleq \text{ChannelDisabled}(q, t_\sigma)</math> when enablement is evaluated at that settled boundary point. Every source SettlePoint and every Core.KCut used by Appendix K is a BoundaryEvalPoint. This is only notational shorthand for cutpoint reasoning; the primary semantic definition of occupancy remains the cycle-boundary quantity <math>\text{occ}_c(t)</math>.</p> <p>Note: <math>\text{occ}_c(t)</math> is an abstract analysis quantity derived from <math>\Sigma</math>-visible boundary commits; in both Sys<sub>B</sub> and RTL<sub>B</sub>, processes do not directly read <math>\text{occ}_c(t)</math> and instead proceed solely based on the per-channel ready/valid outcome of their own Push/Pop attempts.</p> |
| <b>P<sub>push(c)</sub></b> | <p><i>Finite-progress invariant (producer)</i>: this is a persistent-request premise, not a ChannelEnabled premise. In this paragraph, “the producer’s request remains asserted continuously” means that the producer presents a persistent, non-withdrawable Push<sub>c</sub> endpoint request from the starting cycle through the matching discharge event, with no deassert-before-commit and with stable payload while asserted.</p> <ul style="list-style-type: none"> <li>• Buffered case (<math>B(c) &gt; 0</math>): If the producer’s Push<sub>c</sub> request remains asserted continuously from some cycle <math>t_0</math> onward while the channel is full (<math>\text{occ}_c = B(c)</math>) —with stable payload while asserted—and the complementary endpoint continuously requests the matching Pop<sub>c</sub> (persistent request asserted; Pop<sub>c</sub> boundary request remains true), then some cycle <math>t' \geq t_0</math> commits a complementary Pop<sub>c</sub> (i.e., the full condition is eventually discharged).</li> <li>• Rendezvous case (<math>B(c) = 0</math>): If the producer’s Push<sub>c</sub> request remains asserted continuously from some cycle <math>t_0</math> onward (with stable payload while asserted) and the complementary endpoint continuously requests the matching Pop<sub>c</sub> (persistent request asserted; Pop<sub>c</sub> boundary request</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Symbol / Rule                            | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                                          | remains true), then some cycle $t' \geq t_0$ commits the rendezvous transfer—i.e., a matching Push_c(v) and Pop_c(v) commit in the same cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>P_pop(c)</b>                          | <p><i>Finite-progress invariant (consumer)</i>: this is a persistent-request premise, not a ChannelEnabled premise. In this paragraph, “the consumer’s request remains asserted continuously” means that the consumer presents a persistent, non-withdrawable Pop_c endpoint request from the starting cycle through the matching discharge event, with no deassert-before-commit.</p> <ul style="list-style-type: none"> <li>• Buffered case (<math>B(c) &gt; 0</math>): If the consumer’s Pop_c request remains asserted continuously from some cycle <math>t_0</math> onward while the channel is empty (<math>occ_c = 0</math>)—and the complementary endpoint continuously requests the matching Push_c (persistent request asserted with stable payload; Push_c boundary request remains true)—then some cycle <math>t' \geq t_0</math> commits a complementary Push_c (i.e., the empty condition is eventually discharged).</li> <li>• Rendezvous case (<math>B(c) = 0</math>): If the consumer’s Pop_c request remains asserted continuously from some cycle <math>t_0</math> onward and the complementary endpoint continuously requests the matching Push_c (persistent request asserted with stable payload; Push_c boundary request remains true), then some cycle <math>t' \geq t_0</math> commits the rendezvous transfer—i.e., a matching Push_c(v) and Pop_c(v) commit in the same cycle.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Trace-compatibility rules (E1–E5)</b> | <p>E1 Synchronization-order preservation: For each process P, the sequence of synchronization events executed by P (wait(), SyncChannel, START_P, FINISH_P) appears in the same source order in the pre-HLS and post-HLS traces.</p> <p>E2 Signal visibility:<br/> For each sequential process P, each signal Read in P occurs at the closest preceding synchronization event in P, and each signal Write in P occurs at the closest succeeding synchronization event in P, in both the pre-HLS and post-HLS traces (per R2/E2).<br/> For each combinational process or explicitly modeled combinational network component C_comb, observable combinational signal Read/Write labels appear only inside emitted R2C fixed-point signal-observation units for the selected R2C observation component, as defined by R2C. Each emitted R2C unit is evaluated at one <math>\epsilon</math>-quiescent combinational fixed point, and all Read/Write labels inside that unit are atomic with respect to that fixed point. Emitted R2C units are matched by component identity and local emitted-occurrence ordinal, not by global observable-cycle bucket number. R2C stutter cycles whose selected observable slice is unchanged do not create separate <math>\Sigma</math>-visible units unless an explicitly modeled observer/sampler requires a fresh emitted unit.</p> <p>Interpretation (logical timestamping). For sequential processes, the equations in E2 define the logical timestamps of Read/Write <math>\Sigma</math>-events: the anchor points at which compatibility is checked. An implementation may physically sample a Read later than <math>pred\_S(r)</math>, or physically apply a Write earlier than <math>succ\_S(w)</math>, provided the value is stable across the anchor interval and the trace emitted</p> |

| Symbol / Rule | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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|               | <p>for compatibility records the <math>\Sigma</math>-event at the anchor point. For combinational processes, the logical observation point is the emitted R2C fixed-point signal-observation unit, not a global observable-cycle bucket number; internal delta-cycle propagation before the fixed point and R2C stutter cycles with unchanged selected observable slices are <math>\epsilon</math>/stutter and are not separately <math>\Sigma</math>-visible.</p> <p>Direct-input pragmas (e.g., <code>#pragma hls_direct_input</code> and <code>#pragma hls_direct_input_sync</code>) do not change E2 and are not exceptions to E2. For sequential processes, they impose stability/timing contracts that allow later physical sampling, or controlled updates at a synchronization boundary, while the logical Read/Write <math>\Sigma</math>-events used for <math>\approx</math> checking remain timestamped at the E2 synchronization anchor points and retain the anchored value labels. For combinational processes, ordinary direct-input pragmas do not introduce a separate anchoring rule; any observable combinational signal Read/Write label remains part of an emitted R2C fixed-point signal-observation unit under the R2C unit-generation and matching conventions. Thus the pragma affects only implementation sampling freedom under its stated contract, not the logical <math>\Sigma</math> observation used for compatibility.</p> <p>E3 (Safe message issue ordering). (Post-HLS preservation of R4).<br/> For any two executed message-call instances <math>q_1, q_2 \in Q_{\text{exec}, P}</math> of the same process <math>P</math>:</p> <ul style="list-style-type: none"> <li>(i) Same-interface order is always preserved: if <math>q_1</math> and <math>q_2</math> access the same message-passing interface/channel and <math>q_1 &lt;_{\text{src}} q_2</math>, then <math>\text{issue\_post}(q_1) \leq \text{issue\_post}(q_2)</math>.</li> <li>(ii) Distinct-interface no-reverse: if <math>q_1</math> and <math>q_2</math> access distinct message-passing interfaces and appear in sequence in the source (<math>q_1 &lt;_{\text{src}} q_2</math>), then <math>\text{issue\_post}(q_1) \leq \text{issue\_post}(q_2)</math> (i.e., they shall not be issued in the reverse sequence).</li> <li>(iii) Prefix-closed issue obligation: at every cycle-aligned cutpoint <math>\sigma</math> of the post-HLS execution / witness, if <math>q_1 &lt;_{\text{src}} q_2</math> and <math>q_2</math> has issued by <math>\sigma</math>, then <math>q_1</math> has also issued by <math>\sigma</math> and <math>\text{Issue}(q_1) \leq \text{Issue}(q_2)</math>. This is an implementation-side obligation on issued sets, not merely an inequality over pairs whose timestamps are already defined.</li> </ul> <p>(Note: any pipelined-loop internal staging/dequeue discussed in “Pipelined Loops” is <math>\epsilon</math>-microstate within the back-annotated channel realization and is not a <math>\Sigma</math>-visible boundary issue event associated with any <math>q \in Q_{\text{exec}, P}</math> nor a committed-transfer event in <math>M</math>; therefore it does not constitute a counterexample to (ii) or (iii).)</p> <p>E4 FIFO legality: for each channel <math>c</math>, the post-HLS committed (<code>Push_c(v)</code>, <code>Pop_c(v)</code>) history is a legal bounded-capacity execution consistent with <code>Sys_B</code> for that channel (capacity <math>B(c)</math>), and reduces to rendezvous consistency when <math>B(c)=0</math>. In particular, on each channel, the committed <code>Pop_c(v)</code> events return exactly the FIFO-ordered sequence of values <math>v</math> previously committed by <code>Push_c(v)</code>, with no drops, duplications, or reordering.</p> <p>E5 (Messages cannot cross their immediate synchronization boundary).</p> |

| Symbol / Rule | Definition — concise but complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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|               | <p>For every synchronization call <math>s</math> (in the source order used by R3 / <math>\text{pref\_S}</math> / <math>\text{suff\_S}</math>):</p> <p><math>\forall q \in \text{pref\_S}(s) : \text{issue\_post}(q) \leq \text{clk\_post}(s)</math></p> <p><math>\forall q \in \text{suff\_S}(s) : \text{issue\_post}(q) &gt; \text{clk\_post}(s)</math></p> <p>For every blocking <math>q \in \text{pref\_S}(s)</math>: <math>m(q)</math> is defined and <math>\text{clk\_post}(m(q)) \leq \text{clk\_post}(s)</math>.</p> |

Frontier-item predicate convention.  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled}$ , and  $\text{ChannelDisabled}$  are authoritative in their frontier-item form. For a message-operation frontier item  $x \in \Omega_{\text{msg},P}$ , let  $q = \text{MsgOf\_P}(x)$  and  $c = \text{BoundaryOf\_P}(x)$ .  $\text{BoundaryReq}(x,\sigma)$ ,  $\text{ChannelEnabled}(x,\sigma)$ , and  $\text{ChannelDisabled}(x,\sigma)$  are evaluated at that specific explicit abstract boundary and are undefined for non-message frontier items unless the statement explicitly restricts  $x \in \Omega_{\text{msg},P}$ . In the ordinary unelaborated case, where  $q$  has a unique frontier item and a unique evaluated boundary, the  $q$ -forms  $\text{BoundaryReq}(q,\sigma)$ ,  $\text{ChannelEnabled}(q,\sigma)$ , and  $\text{ChannelDisabled}(q,\sigma)$  may be used as shorthands for the corresponding  $x$ -forms. In  $\text{Sys\_B}^{\text{elab}}$ , one source-level call  $q$  may own several frontier items at different explicit boundaries, so the  $x$ -form is authoritative.

**No hidden cross-channel disablement.** Any cross-channel dependency introduced by combinational couplers must be reflected only at explicit internal staged/bundle/join boundaries introduced by the elaboration, i.e., via  $\text{BoundaryReq}/\text{ChannelEnabled}/\text{ChannelDisabled}$  on message-operation frontier items  $x \in \Omega_{\text{msg},P}$ , with  $\text{MsgOf\_P}(x)=q$ , at those explicit staged/bundle channels. At the chosen  $\Sigma$ -visible channel endpoints,  $\text{BoundaryReq}(x,\sigma)$  and  $\text{ChannelEnabled}(x,\sigma)/\text{ChannelDisabled}(x,\sigma)$  for a message-operation frontier item  $x \in \Omega_{\text{msg},P}$  on channel  $c$  may not depend on other channels' state beyond the per-channel E4 predicate for  $c$ , and no extra hidden predicate may block an otherwise E4-enabled boundary transfer.

## Prescribed reference model

### Definition Core.1 (Prescribed source reference model $\text{Sys\_ref}$ ).

For a fixed design, fixed external stimulus, and selected elaboration package, define the prescribed source reference model  $\text{Sys\_ref}$  as follows: •  $\text{Sys\_ref} = \text{Sys\_B}$  in the ordinary abstract back-annotated presentation using the case-split channel semantics of E4:  $B(c)=0$  is rendezvous, and  $B(c)>0$  is cycle-boundary non-fall-through bounded FIFO. •  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$  in the elaborated/back-annotated case when explicit staged, bundle, join, guard, coupler, or epoch-gate boundaries are required. All source-side semantic notions in Appendices G–K are interpreted with respect to this prescribed source reference model. Appendix J, Remark 2 explains the model ladder and practical back-annotation interpretation but does not redefine  $\text{Sys\_ref}$ .

### Definition Core.ElabProj (Elaboration projection/accounting map $\Pi_{\text{elab}}$ ).

When  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$ , fix the projection/accounting map  $\Pi_{\text{elab}}$  supplied by the selected elaboration package E.  $\Pi_{\text{elab}}$  maps proof-internal executions, committed events, and explicit-channel occupancy state of  $\text{Sys\_B}^{\text{elab}}$  to the chosen outer DUT/testbench-visible boundary view. It has four roles:

- (1) Trace projection.  $\Pi_{\text{elab}}$  preserves the labels and order of the selected outer  $\Sigma_{\text{DUT}}$ -visible boundary events and hides or projects away introduced staged/bundle/join events that are not part of that outer observation boundary.
- (2) Occupancy projection. For every selected outer abstract channel  $c$ ,  $\Pi_{\text{elab}}$  maps the occupancies of

the explicit internal channels in the elaboration to the outer occupancy  $\text{occ}_c(\sigma)$  used by the outer E4 interpretation.

(3) Conservation/accounting. Internal  $\epsilon$ -steps and committed transfers across introduced proof-internal channels preserve the projected outer history and occupancy except when E designates a transfer as the corresponding outer-boundary commit. Each projected outer Push\_c/Pop\_c commit updates projected occupancy exactly as required by E4; bundled or joined items use the accounting declared by E.

(4) WFG visibility. Projection does not make a proof-internal blocking boundary invisible. Every staged/bundle/join/guard/epoch-gate boundary used for BoundaryReq, ChannelEnabled/ChannelDisabled, Pending\_P, Front\_P, or WFG reasoning remains an explicit abstract channel of  $\text{Sys\_B}^{\text{elab}}$  even when its committed events are projected away from the outer  $\Sigma_{\text{DUT}}$  trace. Appendix Q supplies the permitted elaboration constructions and the detailed well-formedness, projection-fiber, join-boundary, and locality obligations that instantiate this Core interface.

### Remark Core.1 (Interpretation at explicit abstract channels).

Whenever  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$ , the quantities  $B(\cdot)$ ,  $\text{occ}_\cdot(t)$ , E4, BoundaryReq, ChannelEnabled, ChannelDisabled, and the Appendix K blocking / wait-for-graph machinery are interpreted per explicit abstract channel, including any staged, bundle, or join channels introduced by elaboration. Any conjunctive/join dependency shall therefore be represented only via such explicit abstract channels, not as cross-channel disablement at the outer  $\Sigma$ -visible endpoints.

### Remark Core.1a (Operational Sys\_ref transition system and issue policies).

$\text{Sys\_ref}$  is the selected source-reference transition system defined operationally in this Core. Appendix G constrains that system through R1–R5 and E1–E5. Its states contain process-local control/variable state, reached/issued/committed dynamic message-call instances, explicit abstract channel state for every  $\text{Sys\_B}$  or  $\text{Sys\_B}^{\text{elab}}$  boundary, operation-attributive pending endpoint requests, synchronization/signal-anchor state, and  $\epsilon$ -microstate. Its steps are pure source-evaluation  $\epsilon$ -steps, message issue steps, E4 channel-commit steps, synchronization/signal-anchor steps, and admissible  $\epsilon$ -settling/drain steps. Core.Phase is part of the legality of this transition system: after the selected L1 window closes, L1-only process evaluation and issue steps are phase-disabled until the next phase permits them. This phase restriction may be represented by an explicit phase tag or by a phase-indexed transition relation; it is not inferred merely from which enabled step a scheduler happened to select. The phase-advance markers that close L1 and select L2, advance L2 to L3, advance L3 to L4, and begin the next cycle are meta-level control transitions of the cycle semantics: they are neither  $\epsilon$ -steps nor  $\Sigma$ -events and do not themselves change the proof-visible state or history; they only select which phase-indexed operational subrelation is active. State changes associated with L3 boundary actions and L4 returned-result/registered updates occur through the operational actions already named by Core.Phase, not through the phase-advance marker itself. Policy L and Policy S are two issue policies over this same transition system. Policy L permits source-later blocking requests to be issued before source-earlier same-interval blockers process-facing commit, subject to source-order prefix closure and all R/E/E4/E5 obligations. Policy S adds the conservative serial-blocking restriction: a later source-order blocking Push/Pop may issue only after every earlier source-order blocking Push/Pop in that interval has process-facing committed in a strictly earlier cycle. In  $\text{Sys\_B}^{\text{elab}}$ , process-facing commit is the commit that releases the owning source-level operation at its selected boundary, not an arbitrary internal staged/bundle/join transfer. The authoritative definitions are Definition Core.LLegal for Policy-L step legality and fixed-instance binding, Definition Core.Phase for the L0–L4 cycle phases and pre-settle/settled-frontier boundary, and Definition Core.LAdm for finite-prefix admissibility and complete admissible executions; this remark introduces no alternate legality, phase, or admissibility relation.

## Observable alphabet and executions

### Definition Core.2 (Observable alphabet $\Sigma$ ).

Let  $\Sigma$  be the observable event alphabet consisting of:

- (1) committed message-transfer events  $\text{Push}_c(v)$  and  $\text{Pop}_c(v)$ ,
- (2) synchronization events selected into  $S$ , and
- (3) signal events  $\text{Read}(\text{sig}, v)$  and  $\text{Write}(\text{sig}, v)$ , interpreted as anchored signal events for sequential processes under R2 and as labels carried by emitted  $\epsilon$ -fixed-point observation units for combinational processes under R2C; those R2C units are identified by component identity and local emitted-occurrence ordinal, not by global cycle bucket.

Under the pipelined-loop body no-wait condition, an HLS-pipelined loop body contributes no additional in-body wait/synchronization events to  $S$  or  $\Sigma$ .

**Observable-unit carrier convention.** Core.17  $\Sigma\text{Match}$  and downstream references to the selected observable-unit projection range over the selected  $\Sigma$ -events together with any additional proof-internal observable units explicitly included by the theorem instance, notably  $\text{RESET}(S_p, S_c)$ . Such proof-internal units need not be members of  $\Sigma$  and may be projected away from the outer  $\Sigma_{\text{DUT}}$  view. Their matching and ordering rules are stated explicitly where introduced; including them in the observable-unit carrier does not convert them into ordinary  $\Sigma$  commits.

### Definition Core.3 ( $\epsilon$ -steps and compact trace notation).

Let  $\epsilon$  denote internal micro-steps that are not observable at the proof-internal alphabet  $\Sigma$ , including internal staging/hand-off within the concrete realization of a single explicit abstract channel, late direct-input sampling between anchors, and failed non-blocking polls. In elaborated cases  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$ , transfers across introduced staged/bundle/join channels are not  $\epsilon$  merely because  $\Pi_{\text{elab}}$  hides them from the outer DUT/testbench-visible alphabet  $\Sigma_{\text{DUT}}$ ; if such a channel is an explicit abstract boundary used by Appendix K, its committed Push/Pop transfers are  $\Sigma$ -events for the proof-internal comparison.

A finite execution prefix  $\tau$  is understood in the same sense as in Appendix J: it carries the inherited cycle partition / per-cycle observable buckets of the execution, together with any  $\epsilon$ -steps between successive  $\Sigma$ -visible cutpoints. For compact notation one may write a linearization over  $(\Sigma \cup \{\epsilon\})$ , but only modulo permutation of  $\Sigma$ -events that occur in the same cycle and are not additionally ordered by the document's explicit  $\Sigma$ -visible constraints. Accordingly,  $\tau|_{\Sigma}$  denotes the observable projection together with that inherited cycle partition; same-cycle observable events are not assigned any extra order merely by this compact notation.

### Definition Core.4 ( $\epsilon$ -quiescent cutpoint).

For any proof model  $M$  in this formal stack with a declared  $\Sigma/\epsilon$  transition relation (including  $\text{Sys\_ref}$ ,  $\text{Sys\_K}$ ,  $\text{RTL\_B}$ , and explicit lowered/elaborated variants), write  $\text{Quiescent\_M}(s)$  when no  $\epsilon$ -step of  $M$ 's full proof transition relation is enabled from state  $s$  at its current cycle/phase. For any finite prefix  $\tau$  of  $M$ ,  $\text{Norm\_M}(\tau, v, s)$  means that  $v$  is a finite  $\epsilon$ -only extension of  $\tau$  ending at  $s$ ,  $s$  satisfies  $\text{Quiescent\_M}$ , and  $v$  preserves the inherited  $\Sigma$ -visible history and cycle-bucket structure of  $\tau$ . A selected maximal finite normalization may still be written  $\bar{\tau}$ . Core.4 quiescence excludes enabled  $\epsilon$ -work only; it does not exclude enabled  $\Sigma$ -actions and is therefore weaker than a B5/global fixed point. For  $\text{Sys\_ref}/\text{Sys\_K}$ , Core.Phase is part of the full transition legality used by  $\text{Quiescent\_M}$ , and the meta-level phase-advance markers of Remark Core.1a are not  $\epsilon$ -steps.  $\text{RTL\_B}$  has its own cycle/ $\epsilon$  relation and is not assigned the source-side L0–L4 phases. Appendix J/K comparison cutpoints use the model-specific Quiescent predicate through Core.KCut below.

### Definition Core.Settle (L2 settled boundary-evaluation point).

For a Policy-L or Policy-S cycle  $t$ , after the selected L1 process-owned pre-settle window closes, let  $\rightarrow_{\epsilon, L2}$  denote the complete phase-L2  $\epsilon$  subrelation: process-owned L1 source-evaluation and message-issue steps are disabled; process-owned request assertions and Push payloads that were fixed at the L1 endpoint remain fixed; and the cycle- $t$  exogenous/environment-driven values treated as inputs to the selected G–K system remain fixed throughout L2. Every  $\epsilon$ -step that remains phase-legal at L2 is a combinational, explicit-boundary-evaluation, or other cycle-local L2 settling step in  $\rightarrow_{\epsilon, L2}$ .

L2 frame (normative). Every  $\rightarrow_{\epsilon, L2}$  step preserves the cycle index; registered process/FSM/pipeline state; registered explicit-channel storage, occupancy, and credit state; and the selected cycle- $t$  exogenous/environment-driven input values. It emits no  $\Sigma$ -visible commit and performs no L4 or other next-cycle registered update. It may modify only cycle-local  $\epsilon$ /settling microstate and derived combinational boundary values such as complementary ready/valid, guard, join/coupler, and ChannelEnabled/ChannelDisabled evaluation results. A same-cycle reactive component whose value is intentionally combinationaly dependent on DUT outputs is not a changing exogenous input under this clause: it must be included inside the same qualified settling network or isolated behind an explicitly verified boundary.

SettlePoint\_I( $t, \sigma$ ) holds when  $\sigma$  is the first fixed point of  $\rightarrow_{\epsilon, L2}$  reached from that cycle's selected L1 endpoint. This definition is conditional on such a fixed point being reached; it does not by itself prove termination or existence. Define BoundaryEvalPoint\_I( $\sigma$ ) to mean a state at which the explicit boundary request and ChannelEnabled/ChannelDisabled network has reached the applicable combinational fixed point; every SettlePoint is a BoundaryEvalPoint. SettlePoint is the L2 activation/evaluation notion used by Core.16 before L3 boundary actions; Lemma Core.SettleIsKCut below records its relationship to Core.4 quiescence for the phase-bearing source models. The standard existence/uniqueness result is supplied by Definition Core.QSync and Theorem Core.QSync-Ref below.

### Definition Core.KCut (K-facing normalized cutpoint domain).

Fix a theorem instance  $I$  and the explicit abstract boundaries/elaboration applicable to the model being evaluated. For any proof model  $M$  used by Appendix K, KCut\_M,  $I(s)$  holds when  $s$  is reachable, Quiescent\_M( $s$ ) holds, and every explicit boundary used for BoundaryReq/ChannelEnabled/ChannelDisabled has reached its selected combinational fixed point. Write KCutReach\_M( $I$ ) for the reachable domain. For the prescribed source reference model use the aliases KCut\_ref,  $I$  and KCutReach\_ref( $I$ ); for RTL\_B use KCut\_post,  $I$  and KCutReach\_post( $I$ ). In Appendices J–K, “K-relevant cutpoint” means a member of the appropriate KCutReach\_M domain. For a reachable RTL\_B prefix  $\tau_{\text{post}}$ , NormPost\_I( $\tau_{\text{post}}, v_{\text{post}}, p_{\text{post}}$ ) abbreviates Norm\_RTL\_B( $\tau_{\text{post}}, v_{\text{post}}, p_{\text{post}}$ )  $\wedge$  KCut\_post,  $I(p_{\text{post}}$ ); the analogous prescribed-source normalization is NormRef\_I. By Lemma Core.SettleIsKCut, every source-side Sys\_ref/Sys\_K L2 SettlePoint is itself a K-facing cutpoint. Existence of the source SettlePoint is supplied by the qualified synchronous-settling theorem Core.QSync-Ref in the standard theorem scope. RTL\_B has no L0–L4 phases; its KCut existence remains expressed through Core.KCutClosure\_post, which Theorem Core.QSync-Post derives from the same qualified synchronous-model discipline in the standard scope.

### Lemma Core.SettleIsKCut (source L2 settle implies K-facing cutpoint).

For each phase-bearing source proof model  $M \in \{\text{Sys\_ref}, \text{Sys\_K}\}$  and its explicit lowered/elaborated variants, SettlePoint\_I( $t, \sigma$ ) implies KCut\_M,  $I(\sigma)$ . Proof. The SettlePoint is reachable from the selected L1 endpoint. At L2, all L1-only  $\epsilon$ /evaluation/issue steps are phase-disabled; by Remark Core.1a the phase-advance marker is not an  $\epsilon$ -step; and by Definition Core.Settle,  $\rightarrow_{\epsilon, L2}$  contains every  $\epsilon$ -step that remains

phase-legal at L2 and has reached its fixed point. Hence  $\text{Quiescent\_M}(\sigma)$ . Definition `Core.Settle` also gives  $\text{BoundaryEvalPoint\_I}(\sigma)$ , so the remaining `Core.KCut` condition holds.  $\square$

**Hardware interpretation of the qualified synchronous model (informative).** At the architectural level, the intended pre-HLS and post-HLS comparison models are ordinary single-clock, cycle-level synchronous hardware systems. At the beginning of cycle  $t$ , registered process, pipeline, and explicit-channel state is fixed. Sequential processes determine their cycle- $t$  request, payload, and other process-owned outputs from that registered state and from operands fixed for the cycle; they do not provide a same-cycle combinational through-path from a value newly returned or sampled at an interface in cycle  $t$  to a dependent process-owned output in that same cycle. Ordinary combinational derivation of outputs from registered state is permitted. The complete system-level same-cycle network—including R2C combinational processes, ready/valid and enablement logic, and stateless elaboration logic—then settles before the boundary action to one deterministic valuation. Pure combinational feedback is outside the standard theorem scope; feedback that affects later behavior passes through explicit registered or finite-capacity state. Boundary actions are selected from the settled valuation, after which the next registered state is formed.

**Relation to the proof vocabulary.** Definition `Core.QSync` formalizes this hardware discipline rather than introducing an additional hardware mechanism. On the phase-bearing source side, `Core.Phase` presents the cycle as L1 reach/issue, L2 settle, L3 boundary commit/decision, and L4 registered update; `RTL_B` need not expose those proof phases. `SettlePoint` and `BoundaryEvalPoint` name the ordinary settled source-cycle snapshot, and `Core.KCut` names that same kind of settled snapshot when it is used by the J–K proof stack. These are proof views/cutpoints of the cycle-level machine, not additional designer-visible hardware states or extra physical timing phases. HLS-added silent cycles remain real clock cycles and are hidden from the selected observable behavior except where the contract declares timing significant. WC/latency-sensitive choices, `Policy-L` issue latitude, arbitration, and other explicitly modeled choices remain available, so this interpretation does not assert that `Sys_ref` has one unique execution.

#### Definition `Core.QSync` (qualified synchronous same-cycle settling model).

Fix a theorem instance  $I$  and a proof model  $M$  used by the G–K stack. For a phase-bearing source model, the model-specific same-cycle settling relation is  $\rightarrow_{\epsilon, L2}$  from Definition `Core.Settle`. For `RTL_B`, it is the complete K-facing same-cycle  $\epsilon$ /combinational settling relation beginning at a reachable K-origin after the current cycle’s registered/FSM/pipeline state and process-owned registered request outputs have been established and ending before any cycle-advancing registered update.  $\text{QualifiedSynchronousModel\_M}(I)$  holds when all of the following are true.

(QS1) Single-clock scope. There is one selected global clock/cycle relation shared by every process and explicit abstract channel inside the G–K theorem instance. The cycle indices used by K-origins, boundary evaluation, commits, and registered updates are all defined relative to that clock, and no unmodeled cross-clock-domain boundary lies inside the selected instance. The Appendix M.7a `ClockScope` entry records the concrete evidence for this condition.

(QS2) Registered-phase/frame separation. The same-cycle settling relation satisfies the L2/K-origin frame: the cycle index, registered process/FSM/pipeline state, registered explicit-channel storage/occupancy/credit state, and selected cycle- $t$  exogenous inputs are invariant during settling; no  $\Sigma$ -visible commit and no next-cycle registered update occurs in the settling relation. Sequential process request/payload drives covered by `Core.RegSeq` remain fixed after their selected pre-settle value is established. R2C combinational-process outputs, and stateless elaboration logic such as couplers, are instead part of the settling network. (QS2a) RTL KCut origin recoverability. For `RTL_B` in the standard K-

facing scope, every reachable state  $p\_post$  satisfying  $Core.KCut\_post, I$  belongs to one cycle- $t$  QS2 registered/input frame and has a designated reachable K-origin  $o\_post$  for that same frame after the cycle's registered/FSM/pipeline state and process-owned registered request outputs are established and before K-facing same-cycle settling. The reachable suffix from  $o\_post$  to  $p\_post$  before any later cycle-advancing registered update consists only of the qualified same-cycle settling relation. The designated origin/frame is unique for that KCut under the selected single-clock operational semantics. This clause is vacuous for the phase-bearing source models, whose L1/L2 phase boundary already names the source K-origin explicitly.

(QS3) Settling-network realization and deterministic local evaluation. The complete same-cycle settling relation is realized by the qualified settling network: every state component that may change during settling is represented by a cycle-local settling node or by a derived value of such nodes; each settling step changes only such represented values by evaluating one or more node functions from their predecessor values; every phase-legal same-cycle settling step is accounted for by this representation; and no additional settling transition exists outside it. A state is a fixed point of the settling relation iff all represented node equations are satisfied. Every combinational or other cycle-local settling node has a deterministic value for fixed predecessor values. Any state that can affect a later cycle is represented at an explicit registered/process/channel state boundary rather than being hidden inside the settling relation.

(QS4) Finite acyclic settling dependency. After cutting the network at those explicit registered/storage boundaries, the dependency graph for the complete same-cycle settling network is finite and acyclic. Pure combinational feedback is outside the standard theorem scope; required feedback shall pass through explicit state. An equivalent independently checked structural theorem establishing the same finite acyclic dependency property may be used.

(QS5) Explicit-boundary completeness. Every value used by `BoundaryReq`, `ChannelEnabled/ChannelDisabled`, rendezvous pairing, join/coupler/guard/epoch-gate evaluation, or Appendix K WFG construction at a selected boundary is produced by that qualified settling network from the fixed cycle- $t$  state/inputs. No hidden enablement or storage predicate may bypass E4/Core.ElabProj/B-faithfulness. For  $Sys\_B^{elab}(E)$ , Appendix Q supplies the elaboration-specific obligations; for  $RTL\_B$ , the selected boundary mapping and B-faithfulness evidence supply the implementation side.

`QualifiedSynchronousModel` is a same-cycle structural property. It does not assert one unique  $Sys\_ref$  execution: latency-sensitive/WC choices, Policy-L issue latitude, arbitration/environment choices across cycles, and other explicitly modeled choices may still admit multiple source executions. Nor does it imply B1; B1 remains the separate observable RTL trace-determinism assumption.

**Theorem Core.QSync-Ref (qualified source/elaboration settling gives a unique source KCut).**

For  $M \in \{Sys\_ref, Sys\_K\}$  and its explicit lowered/elaborated variants, assume `QualifiedSynchronousModel_M(I)`. Then every reachable source K-origin has exactly one L2 fixed-point endpoint  $\sigma\_set$  in that cycle. That endpoint satisfies `SettlePoint_I(t,  $\sigma\_set$ )`; by Lemma `Core.SettleIsKCut` it lies in `KCutReach_M(I)`. Hence `KCutClosure_M(I)` holds, and the K-facing portion of `NormRef` may select this unique settled endpoint canonically.

Proof sketch. QS2 freezes registered state and cycle- $t$  inputs and excludes  $\Sigma/L4$  updates during settling. QS3 realizes the complete settling relation as evaluation of the represented cycle-local nodes, makes

each node functional, and identifies relation fixed points with satisfaction of all node equations. QS4 gives a finite acyclic dependency order, so evaluation terminates and produces one order-independent valuation. QS5 makes that valuation complete for every K-facing boundary predicate. Therefore the complete  $\rightarrow_{\epsilon, L2}$  relation reaches one fixed point; Lemma Core.SettleIsKCut supplies the KCut conclusion. In the elaborated case, Appendix Q supplies or cites the settling facts for the elaboration-owned template subnetwork. Those template-side facts are not by themselves the whole-source QS2–QS5 discharge: the concrete theorem instance must also establish the applicable design-owned source-settling facts and a checked template/design settling-composition condition, including QS4 for the complete combined same-cycle settling network. Definition Core.Settle alone is not used as an existence axiom.

### Theorem Core.QSync-Post (qualified RTL settling gives a unique post KCut and KCut closure).

Assume QualifiedSynchronousModel\_RTL\_B(I). Then every reachable RTL K-origin has exactly one same-cycle settled endpoint  $p\_post$  before the next cycle-advancing registered update. There is a finite  $\epsilon$ /combinational normalization from that origin to  $p\_post$ ,  $p\_post \in KCutReach\_post(I)$ , and therefore Core.KCutClosure\_post(I) holds. The theorem also supplies a canonical K-facing NormPost normalization/endpoint for that origin. With  $K\epsilon$ , Lemma Core.KCutObserve then gives the existing persistent-blocker observability consequence.

This canonicalization is limited to K-facing same-cycle RTL settling. It does not eliminate Appendix I/J ReplayNormalizationConvention or the selected source replay-normalization/congruence machinery, which may include witness-fixed replay-changing  $\epsilon$  behavior and preparation steps unrelated to combinational settling. In particular, Core.QSync-Post does not prove that a later clock cycle exists, that a temporarily idle RTL state advances to the next registered state, or that a cross-cycle RTL execution satisfies B2/B3. Lemma Core.RTLCycleCarrier below is separate: it supplies only the carrier/completion fact for a legal RTL real-cycle transition that a reachable prefix has already entered; it is not a progress theorem from an arbitrary K-origin. The finite-prefix Appendix J correspondence and Appendix K RTL cutpoint-closure results do not require a generic RTL fair-extension premise; any later theorem that needs complete RTL continuation must state its own continuation/progress assumptions.

### Lemma Core.QSync-KCutCanonical (every reachable post KCut is the canonical endpoint of its K-origin).

Assume QualifiedSynchronousModel\_RTL\_B(I). Let  $p\_post \in KCutReach\_post(I)$  in the standard K-facing theorem scope. By QS2a,  $p\_post$  determines a unique cycle-t QS2 registered/input frame and a designated reachable K-origin  $o\_post$  for that frame, and the reachable suffix from  $o\_post$  to  $p\_post$  before any later registered update consists only of the qualified same-cycle settling relation. Then  $p\_post$  is exactly the unique settled endpoint selected for  $o\_post$  by Theorem Core.QSync-Post. Consequently, in the standard QSync scope,  $KCutReach\_post(I)$  is the image of the reachable RTL K-origin domain under the canonical QSync-Post endpoint map; every member of  $KCutReach\_post(I)$  therefore admits the canonical finite NormPost witness supplied by Core.QSync-Post. Proof. QS2a supplies the origin/frame recovery that is not implied by Core.KCut quiescence alone. Because  $p\_post$  is a KCut reached by settling from that origin, it is  $\epsilon$ /combinationally quiescent at the selected explicit boundaries in the fixed QS2 frame. QS3 identifies such a fixed point with satisfaction of the complete settling-network equations, and QS4 makes that fixed point unique. Core.QSync-Post produces the unique fixed point for  $o\_post$ , so  $p\_post$  must be that endpoint. The argument is same-cycle only and adds no cross-cycle progress or fairness premise.  $\square$

### Definition Core.CycleResult, CompleteCycle, SilentCycle, and terminal state.

For a proof model  $M$  and theorem instance  $I$ ,  $\text{Core.CycleState\_M,I}$  is the full cycle-entry theorem state: registered/internal machine state, time-aware environment state, global cycle index, and every cycle-valued auxiliary or monitor component declared by  $I$ .  $\text{CycleResult\_M,I}(s_t, r)$  is a typed legal result  $r = \langle \text{nextState}, \text{committedUnits}, \text{resetUnits} \rangle$ . The  $\text{nextState}$  field is a  $\text{Core.CycleState\_M,I}$  at cycle  $t+1$ , so the global clock advances exactly once; its environment component is related to the cycle- $t$  environment by  $\text{EnvStep\_I}$  using the applicable canonical  $\Sigma$ -causal history, and its auxiliary/monitor components advance according to their definitions. In particular, Appendix K's  $K.\text{Resp}$  age is part of the cycle-valued auxiliary state and advances in  $\text{nextState}$  on each selected  $\text{Policy-S}$  cycle while its monitor remains pending.  $\text{committedUnits}$  is the finite collection of selected ordinary complete observable units /  $\Sigma$  commits produced in cycle  $t$ .  $\text{resetUnits}$  is the finite collection of dynamic proof-internal  $\text{RESET}(S_p, S_c)$  occurrences in that cycle; each occurrence carries its reset scope and corresponds exactly to the state-changing reset transition governed by RW1-RW5. For a phase-bearing source model, a legal result is witnessed by a finite L1 preparation/issue segment, finite L2 settling, the selected legal L3 boundary actions/decisions and reset boundary units, and the L4 registered/result update. For  $\text{RTL\_B}$  it is witnessed by the model-specific real clock transition through its  $K$ -facing same-cycle settling/boundary behavior and next registered update;  $\text{RTL\_B}$  is not assigned source L0-L4 phases.

$\text{CompleteCycle\_M,I}(s_t, s_{t+1})$  holds exactly when there exists  $r$  with  $\text{CycleResult\_M,I}(s_t, r)$  and  $r.\text{nextState} = s_{t+1}$ . Thus  $\text{CompleteCycle}$  remains the next-state projection used by existing theorem statements;  $\text{CycleResult}$  is the primary typed object carrying the full cycle evidence.

$\text{SilentCycle\_M,I}(s_t, s_{t+1})$  holds when such a result exists with no selected  $\Sigma$  event in  $r.\text{committedUnits}$ . A  $\text{SilentCycle}$  is a real cycle, not an  $\epsilon$ -step and not an identity transition: registered/internal machine state, environment state, and cycle-valued monitor state may change. A reset-only cycle may be  $\text{SilentCycle}$  with respect to the selected outer  $\Sigma$  alphabet when  $\text{RESET}$  is proof-internal/projected away; that does not make the cycle event-free, because the corresponding occurrence remains in  $r.\text{resetUnits}$  and in the proof-internal history.  $\text{RESET}$  is therefore not an alternative cycle outcome: a partial reset changes only its named scope while the unaffected state still proceeds to the same result's  $\text{nextState}$  under RW5. When the fixed semantics permits more than one reset occurrence in one cycle,  $\text{resetUnits}$  retains each dynamic occurrence and only the order required by RW1-RW5 and  $\prec_J$  is imposed; independent disjoint resets are not artificially serialized. Lemma J.RS continues to consume the emitted  $\text{RESET}$  unit itself, not the enclosing cycle. Persistent blocking requests follow  $\text{Core.IC}$ : once issued, an outstanding blocking dynamic instance remains asserted and attributed until its process-facing commit or an affecting  $\text{RESET}$  governed by RW2; a  $\text{DeclaredTerminal}$  state may instead end the execution while the request remains pending. No other nonterminal transition in the  $G-K$  theorem model may deassert, retire, reattribute, cancel, or replace that outstanding blocking instance. Protocols with independent nonterminal cancellation, withdrawal, or retry require a separately defined operational extension and are outside this theorem stack. An activated  $\text{Core.16}$  no-new-later-work restriction remains in force while its fully disabled frontier persists.

### Lemma Core.RTLCycleCarrier (RTL finite prefixes inherit complete real-cycle carriers).

In the standard  $\text{RTL\_B}$  semantics,  $\text{Reach\_post}$  is carried by legal real-cycle transitions represented by  $\text{Core.CycleResult}$  together with their permitted intra-cycle settling/boundary prefixes. Let  $\tau_{\text{post}} \in \text{Reach\_post}(I)$  be a reachable finite  $\text{RTL\_B}$  prefix that, after the cycle- $t$   $K$ -origin, contains at least one complete selected observable unit attributed to real cycle  $t$ . Then the enclosing legal cycle- $t$   $\text{CycleResult}$  exists in that same execution carrier:  $\tau_{\text{post}}$  is a prefix of that already-entered real-cycle transition, and the same carrier contains its completion through  $r.\text{nextState}$  and the successor cycle- $(t+1)$  registered/input/request  $K$ -origin frame. If that successor state is  $\text{DeclaredTerminal}$ , it may still be the final successor  $K$ -origin; terminality removes the obligation for a further  $\text{CycleResult}$ , not the

nextState/K-origin carrier produced by the cycle already in progress. This is a typing/completion fact for an already-entered legal RTL cycle, not a generic cross-cycle progress, fairness, or totality premise: it does not assert that an arbitrary reachable K-origin starts another real cycle. Proof. Definition Core.3 and Definition J.Reach make  $\tau_{\text{post}}$  a prefix of a legal RTL\_B execution with its inherited real-cycle bucket structure. Once  $\tau_{\text{post}}$  contains a complete selected cycle-t unit, that unit belongs to the enclosing model-specific real clock transition represented by Definition Core.CycleResult; the transition carrier includes its nextState and successor registered/input/request frame.  $\square$

DeclaredTerminal\_M,I(s) remains a predicate on the full Core.CycleState, not a field or alternative constructor of CycleResult. It means that the selected machine/test contract explicitly declares the execution terminated, aborted, or at end-of-test so no ordinary CycleResult / CompleteCycle successor is required. Only a disposition modeled as part of the selected theorem instance's machine/test semantics counts as DeclaredTerminal. An external verification-harness watchdog that merely stops host simulation or reports a timeout, without being a modeled transition or termination disposition of M, is not DeclaredTerminal and does not truncate Core.CycleResult or Core.CompleteCycle. A modeled cycle-count, watchdog, abort, or end-of-test rule that is part of the selected machine/test semantics does count. Mere absence of a currently enabled  $\Sigma$ -action or  $\epsilon$ -step is not DeclaredTerminal when the time-aware environment or registered machine can evolve on a later cycle. TerminalIdle\_M,I(s,H,t,e) is the stronger B6 eligibility condition: s is a genuine B5/global fixed point and every future machine/environment evolution permitted by the fixed theorem instance preserves the no-future-enabling condition, so no registered/internal update and no environment step can change non-clock state or re-enable work. Core.EnvTerminal below supplies the environment component of this obligation; the machine-side permanent-idle fact must also be established. A stable DeclaredTerminal state is one sufficient machine-side route, but B6 may also apply to a nonterminating service model proved permanently idle after stimulus exhaustion.

#### Definition Core.EnvStep and Core.EnvTerminal (time-aware environment evolution).

Let EnvStep\_I(H,t,e,e') be the fixed theorem-instance environment transition from environment state e at cycle t and canonical  $\Sigma$ -causal history H to the next-cycle environment state e'. Every legal CycleResult uses this relation for the environment component of nextState. The requirement that a compatible next-cycle environment evolution exist at every reachable nonterminal source cycle-entry state is folded into Core.CycleClosure below rather than exposed as a separate environment-totally premise. The cycle/environment state is part of the relation: identical  $\Sigma$  history at two different cycles need not imply identical future availability. Core.EnvTerminal\_I(H,t,e) is the environment component of Core.TerminalIdle: every future environment evolution allowed by the fixed instance preserves the no-future-enabling condition required for an indefinite B6 idle suffix. StandardEnv stimulus exhaustion is one sufficient environment-side discharge route. A timed environment may satisfy Core.CycleClosure's environment-definedness clause while delivering a new input later and therefore fail EnvTerminal at the earlier idle cycle.

#### Definition Core.CycleClosure (source full-cycle qualification record).

For a selected source theorem instance I, Core.CycleClosure\_I is the normative full-cycle qualification record used whenever a theorem constructs a complete/fair source continuation, including Lemma K.2b.E. It packages the following obligations. (C1) Full-state cycle typing: every legal CycleResult has a nextState with cycle index advanced exactly once, environment state related by EnvStep\_I, every declared cycle-valued auxiliary/monitor component advanced according to its semantics, and committedUnits/resetUnits consistent with the selected source transition semantics. Each resetUnits occurrence corresponds exactly to one scoped state-changing RESET transition and satisfies RW1-RW5; no reset is represented as an alternative to the enclosing cycle result. (C2) Nonterminal source-cycle

existence, finite phase execution, and partial-cycle extension: every reachable nonterminal source cycle-entry state admits at least one legal `CycleResult` / `CompleteCycle` successor; its L1 part reaches a finite preparation/issue endpoint under `Core.ReachPrep/Core.Phase` and its L2 part reaches the required settled endpoint before L3/L4. In addition, every finite phase-legal partial source-cycle segment selected from the operational transition relation—respecting `Core.Phase/Core.16`, the fixed reset contract, and an `EnvStep`-compatible environment choice—admits a legal completion to a `CycleResult` whose phase projection contains that selected segment in the chosen order. This extension clause is what permits a later proof to complete a finite phase-legal issue/boundary selection to a typed complete cycle; bare successor existence is insufficient. It does not itself construct a fair selection or prove that a proposed issue/L3 combination is jointly phase-legal. When Lemma K.2b.E constructs a complete fair continuation, the existing `Core.IssueFair` and B2/B3 discharges are therefore supplied in the constructive `Core.FairPick` form below and combined with C2 by Lemma `Core.FairCycleDovetail`. The standard discharge combines finite source-control/FPD evidence for L1 with `Core.QSync-Ref` for L2 and an operational cycle-completion theorem, or an allowed equivalent source-cycle theorem. (C3) Environment definedness: at every reachable nonterminal source cycle-entry state, the fixed time-aware environment supplies at least one `EnvStep` compatible with the fixed stimulus/history and with some legal `CycleResult`. The former standalone source-cycle-closure and environment-totality obligations are exactly the source-cycle-existence/finiteness and environment-definedness clauses of this record and are no longer independent premises. `Core.CycleClosure` is an existence/typing qualification, not a fairness premise: `Core.IssueFair` and B2/B3/B5 remain separate where a complete/fair continuation is required. No generic RTL cross-cycle totality/fair-extension premise is part of `Core.CycleClosure`: finite-prefix Appendix J uses `Reach_post`, while the universal Appendix K RTL cutpoint result uses `Core.QSync-Post/Core.KCutClosure_post`. Any infinite B6 suffix is governed separately by `Core.TerminalIdle` and B6.

#### Lemma `Core.CycleChoiceExt` (phase-legal source-cycle selection extends to a `CycleResult`).

Assume `Core.CycleClosure_I`. Let  $s$  be a reachable nonterminal source cycle-entry state and let  $\alpha$  be a finite phase-legal partial source-cycle segment from  $s$  whose steps are operationally legal in sequence, satisfy `Core.Phase/Core.16`, respect the selected reset contract, and are compatible with a selected `EnvStep_I` environment evolution. Then clause C2 supplies a legal `CycleResult`  $r$  from  $s$  whose phase projection extends  $\alpha$  and completes the remaining L2/L3/L4 work required by the source semantics. Consequently, a later proof may extend any already-constructed finite phase-legal selection, but it may invoke dependent choice only after obtaining a typed `CycleResult` for each selected cycle. This lemma neither constructs the fair selection nor proves that independently proposed L1 and L3 choices compose legally; `Core.IssueFair` and B2/B3 remain separate obligations. Definition `Core.FairPick` and Lemma `Core.FairCycleDovetail` below provide the constructive bridge used by Appendix K.  $\square$

#### Definition `Core.FairPick` (constructive discharge of source issue/boundary fairness).

Fix a phase-bearing source theorem instance  $I$  and issue policy  $\Pi$ . `Core.FairPick_I,Π` is the constructive witness form used when a theorem must build, rather than merely assume, a complete execution satisfying `Core.IssueFair` and B2/B3. It does not add a new behavioral fairness property beyond those obligations, but it is a stronger constructive evidence obligation than merely citing fairness of an already-supplied execution: it packages the issue-fairness and boundary-progress discharges as prefix-local selection functions plus a starvation-freedom proof for the particular execution being constructed. The witness contains the following. (FP1) Issue selection. At each reachable nonterminal cycle-entry prefix, `IssuePick` returns a finite phase-legal L1 segment  $\alpha_{\text{issue}}$  (possibly empty). Any fairness-selected issue in  $\alpha_{\text{issue}}$  is a currently `IssueEnabled_Π` operation;  $\alpha_{\text{issue}}$  may also contain permitted preparation or other operationally required L1 work. The whole segment is operationally legal in sequence, respects `Core.Phase/Core.16` and source-order prefix closure, uses the fixed

witness/reset/scheduling semantics, and is compatible with at least one  $\text{EnvStep}_I$  environment evolution for that cycle, so that  $\text{Core.CycleChoiceExt}$  is applicable. (FP2) Boundary selection. After an actual finite L2 settling prefix of that same cycle has reached its  $\text{SettlePoint}$ ,  $\text{BoundaryPick}$  returns a finite phase-legal L3 segment  $\gamma_{\text{bdy}}$  (possibly empty). Any fairness-selected boundary action in  $\gamma_{\text{bdy}}$  is a currently B2-enabled Push/Pop/rendezvous/Sync action. The combined prefix through L2 followed by  $\gamma_{\text{bdy}}$  is operationally legal in sequence, respects E4,  $\text{Core.SyncFence}$ , the fixed arbitration/reset semantics and endpoint cardinality, and is compatible with at least one  $\text{EnvStep}_I$  evolution under the fixed environment semantics. Any same-cycle predecessor commits required by  $\text{Core.SyncFence}$  are included in  $\gamma_{\text{bdy}}$  before the selected Sync. (FP3) Local finiteness and stable obligation identity. At every finite operational prefix, the currently live issue and boundary obligations presented to the selectors are finite and have stable dynamic identities until issue/commit, RESET, or  $\text{DeclaredTerminal}$ . In the standard hardware scope this follows from the finite process and explicit-boundary sets together with finite generation in every finite source prefix and the endpoint-cardinality rules; an equivalent checked locally-finite enumeration is allowed. (FP4) Starvation freedom and fixed-choice discipline. Along any nonterminal suffix within one applicable reset epoch, repeated per-cycle  $\text{IssuePick}$  selections eventually select every operation that remains  $\text{IssueEnabled}_I$  at every applicable L1 opportunity, and repeated  $\text{BoundaryPick}$  selections eventually select every B2 action whose enabling predicate remains continuously true at the applicable settled boundary opportunities. Where the selected Policy-L operational semantics leave issue-selection latitude proof-selectable, an oldest-enabled/round-robin queue over the finite live obligations, with consecutive-enable age reset when an obligation ceases to be enabled or is discharged, is a sufficient construction. Instance-fixed choices—including any fixed scheduler component, arbitration/grant choice, environment evolution, witness outcome, reset behavior, or other choice bound by  $\text{Core.LLegal}_I$ —are not proof-selectable:  $\text{IssuePick}/\text{BoundaryPick}$  must respect those actual choices and the discharge must prove their required starvation-freedom, for example by a well-founded round-robin, deficit, or scheduler-rank argument. In particular,  $\text{BoundaryPick}$  may not replace a fixed unfair arbiter with an invented fair one. (FP5) B3 exposure. For every B3  $P_{\text{push}}/P_{\text{pop}}$  or rendezvous antecedent used by the theorem, the corresponding discharge action is exposed to  $\text{BoundaryPick}$  as a continuously B2-enabled action under the selected explicit-boundary semantics, or the witness supplies the equivalent elaborated-boundary progress segment and proves it is selected under FP4. Thus the same constructive witness discharges the B3 progress cases used by the continuation. For theorem applications that only assume an already supplied complete fair execution,  $\text{Core.FairPick}$  need not be materialized; it is required only at proof sites, such as Lemma K.2b.E, that construct the fair execution from a finite prefix.

#### Lemma $\text{Core.FairCycleDovetail}$ (typed fair per-cycle construction).

Assume  $\text{Core.CycleClosure}_I$  and  $\text{Core.FairPick}_I, \Pi$ . Let  $\pi$  end at a reachable nonterminal source cycle-entry state  $s$ . One fair typed successor cycle is constructed in two C2 applications. First take the finite L1 segment  $\alpha_{\text{issue}}$  returned by  $\text{IssuePick}$ . By FP1 it is operationally and phase legal and is compatible with some  $\text{EnvStep}_I$  evolution, so Lemma  $\text{Core.CycleChoiceExt}$  yields a provisional  $\text{CycleResult}$   $r_0$  extending  $\alpha_{\text{issue}}$ . Let  $\beta$  be the finite phase prefix of  $r_0$  from  $s$  through the end of its L2 settling segment and its actual  $\text{SettlePoint}$ ;  $\beta$  is finite and phase legal because it is a prefix of a legal  $\text{CycleResult}$ . Evaluate  $\text{BoundaryPick}$  only at that actual settled state. Let  $\gamma_{\text{bdy}}$  be its returned finite L3 segment. By FP2,  $\beta \cdot \gamma_{\text{bdy}}$  is a finite operationally legal, phase-legal partial cycle and FP2 supplies compatibility with an  $\text{EnvStep}_I$  evolution under the same fixed environment semantics. That environment evolution need not be the provisional  $r_0$  next-state choice: it is the one compatible with the final selected  $\beta \cdot \gamma_{\text{bdy}}$  history. Apply Lemma  $\text{Core.CycleChoiceExt}$  a second time to  $\beta \cdot \gamma_{\text{bdy}}$  to obtain the final  $\text{CycleResult}$   $r$  for this cycle. The second application is the step that proves the selected L1 fairness action, the actual L2 settle path, and the selected L3 fairness action are jointly realizable in one typed cycle; no boundary choice is

guessed before settling. Additional legal actions inserted by the C2 completion do not remove or reorder the selected segment. Now suppose a prefix invariant  $Inv$  is preserved by these selected/completed cycles and  $Inv$  implies that every constructed cycle-entry state remains nonterminal and that  $Core.FairPick$  remains applicable. Repeating the construction and applying dependent choice only to the resulting typed  $CycleResult$  successors yields an infinite  $CompleteCycle$  continuation. FP4 proves  $Core.IssueFair\_P$  and B2 on that continuation: a continuously enabled obligation cannot be skipped forever by the certified selectors. FP5 gives the B3 progress obligations. Any separately assumed B5 quiescence-closure property continues to apply to the constructed operational continuation; it is not manufactured by  $CycleChoiceExt$  or by  $FairPick$ .  $\square$

#### Remark $Core.Settle\_a$ ( $\delta$ -settling, silent $\tau$ -cycles, and B6 idle ticks).

The following names separate three phenomena that are easy to conflate but are different objects in the formal model.

- $\delta$ -settling is the cycle-local same-cycle settling relation. On the phase-bearing source side, a  $\delta\_ref$  step is exactly a step of  $\rightarrow_{\epsilon, L2}$ ; on  $RTL\_B$  in the standard qualified synchronous scope,  $\delta\_post$  denotes the same-cycle settling relation of Definition  $Core.QSync$ . By the  $L2/QSync$  frame,  $\delta$ -steps preserve the cycle index and registered state. The converse is not intended: other  $\epsilon$ -steps, including  $L1$  source-evaluation activity, may also preserve the cycle index but are not  $\delta$ -settling steps.  $\delta$  is therefore a settling classification, not a renaming of all  $\epsilon$ .
- A silent  $\tau$ -cycle is the explanatory name for a  $Core.SilentCycle$ : a real  $CompleteCycle$  in which no selected  $\Sigma$  event commits. Hidden registered/internal progress may occur, but need not; a completely idle-looking yet nonterminal cycle is still a  $SilentCycle$  if the clock/environment advances. Silent cycles carry the  $CompleteCycle$  state updates described above, including time-aware environment evolution and any active cycle-valued monitor aging. They are neither  $\epsilon$ -steps nor identity stutter.
- A B6 idle tick is the special real clock tick used only when  $Core.TerminalIdle$  holds: a genuine B5/global fixed point together with machine-side permanent-idle evidence and  $Core.EnvTerminal$ . It performs no  $\epsilon$ -prefix or  $\epsilon$ -suffix work and leaves all non-clock state unchanged. It is treated as  $\epsilon$ -stutter only for E1–E5 prefix/trace matching.  $Core.CycleClosure$ 's environment-definedness clause or  $Core.EnvTerminal$  alone does not justify this branch.

The  $\tau$  name remains explanatory cycle-level terminology and should not be confused with execution-prefix variables such as  $\tau\_ref/\tau\_post$ ; a mechanization may use the constructor name  $SilentCycle$ . No theorem may replace the model-specific  $\epsilon$  relation by  $\delta$  or by  $SilentCycle$  without the stated typing.

#### Remark $Core.Det$ (determinism taxonomy and discharge status).

The document uses several distinct determinism notions; they shall not be conflated. (1) Same-cycle settling determinism is the model-class restriction captured by  $R2C/Core.QSync$ : fixed registered state, fixed cycle- $t$  inputs, and fixed explicit issue/request drives yield one settled boundary valuation. (2) Conditional synchronous-step determinism is the corresponding functional-cycle view after the currently legal witness/issue/environment choices for that cycle have been fixed; it does not remove Policy-L scheduling latitude or WC-sensitive source nondeterminism. (3) B1 is observable RTL trace determinism under fixed stimulus and initial state; internal  $\epsilon$  behavior may still differ. (4)  $I.DR/I.DR'$  are derived replay theorems, not assumptions to be asserted in a discharge record. (5)  $K.CV$  is the Appendix K causal/value-determinism premise. (6)  $SDet$  fixes the selected Policy-S simulator/scheduler/arbitration semantics and excludes a second incompatible committed history for that selected instance. (7) Normalization

selection/congruence records which proof-relevant normalization is used and why the replay quotient is preserved; Core.QSync canonicalizes only the K-facing same-cycle settling part, not the Appendix I/J replay normalization.

For any item that is a theorem premise, Appendix M.7a's normative evidence axis applies directly: the record states whether the premise is inapplicable to the selected theorem scope, satisfied by a simple syntactic/design restriction, discharged by tool/RTL evidence, or discharged by a source/RTL witness construction. Derived theorems such as I.DR are not separate premises and therefore require no independent M.7a discharge entry.

#### Definition Core.KCutClosure (K-facing cutpoint existence / non-vacuity).

For a proof model M used by Appendix K, a K-origin is a reachable per-cycle state from which the K-facing boundary network is to be settled without first advancing to the next cycle: for Sys\_ref/Sys\_K it is the selected L1 endpoint immediately before  $\rightarrow_{\epsilon, L2}$  settling; for RTL\_B it is the corresponding reachable cycle-evaluation origin after the current cycle's registered/FSM/pipeline state and process-owned request outputs are established and before K-facing same-cycle settling. KCutClosure\_M(I) holds when every reachable K-origin admits a finite  $\epsilon$ -only/same-cycle settling normalization, within that same cycle and before any later cycle-advancing registered update, to some  $\sigma_K \in \text{KCutReach\_M}(I)$ . For the RTL use  $\text{KCutClosure\_post}(I) := \text{KCutClosure\_RTL\_B}(I)$ .

In the standard K-facing/cutpoint theorem scope, Core.KCutClosure is a derived interface rather than an independent modeling freedom. Theorem Core.QSync-Ref derives source-side KCutClosure\_M from QualifiedSynchronousModel\_M(I), including existence of the source SettlePoint; Theorem Core.QSync-Post derives Core.KCutClosure\_post from QualifiedSynchronousModel\_RTL\_B(I). Downstream Appendix K statements continue to consume Core.KCutClosure\_post so that the liveness proof is factored from the particular structural settling proof. A future non-QSync semantic extension could reuse that downstream interface only by separately proving the same closure property and explicitly extending the theorem scope; it is not part of the standard user-facing G-K model defined here.

#### Lemma Core.KCutObserve (persistent blocker observability).

Assume KCutClosure\_M(I) and  $K\epsilon$  for the selected model. Let  $s_0$  be a K-origin and let  $\sigma_K$  be a KCut supplied by the closure witness. If a K-relevant boundary request/frontier/enableness configuration is present at  $s_0$  after the owning requests are fixed and persists through the selected  $\epsilon$ -normalization, then the same attributed K-facing configuration is present at  $\sigma_K$ ; WFG-relevant facts cannot be changed solely by the intervening  $\epsilon$  segment. Therefore a persistent K-relevant blocker cannot be excluded from the Appendix K observation domain merely because unrelated  $\epsilon$ -work would otherwise continue indefinitely. A configuration that commits, withdraws under an allowed reset disposition, or otherwise ceases to persist before the KCut is transient and is not asserted to satisfy Dead\_K. Proof. KCutClosure supplies the finite endpoint;  $K\epsilon$  supplies WFG-inertness of the intervening  $\epsilon$  segment; BoundaryEvalPoint/Core.KCut supply settled enablement at the endpoint.  $\square$

#### Definition Core.SettleKBridge (L2 trigger to K-facing cutpoint).

Fix a Core.LAdm-admissible source prefix or continuation. Core.SettleKBridge\_I holds when every Core.16 activation at an L2 SettlePoint  $\sigma_{\text{set}}$  is related to every later K-facing cutpoint  $\sigma_K$  used to evaluate the persisting blocked condition as follows: while the same fully disabled frontier persists, the only intervening changes are phase-legal  $\epsilon$ -normalization and Core.16-permitted downstream drain commits; no new source-later work is launched; blocking requests are nonwithdrawable;  $K\epsilon$  makes each  $\epsilon$ -only segment WFG-inert; and every intervening  $\Sigma$ -visible drain commit is represented explicitly in H/E4

before *Front*, *ChannelEnabled/Disabled*, and *WFG* are re-evaluated at  $\sigma_K$ . By Lemma *Core.SettleIsKCut*,  $\sigma_{\text{set}}$  is itself a source *KCut*; the bridge is nevertheless needed because *Dead\_K* additionally uses drain closure and may therefore be evaluated at a later *KCut* after already-issued work has drained. If the blocked frontier commits or otherwise disappears, no *Dead\_K* conclusion is drawn from the earlier activation. If it persists to a later drain-closed  $\sigma_K \in \text{KCutReach\_ref}(I)$ , Definition *K.Dead* is evaluated there. *K.DrainReach* shall discharge this bridge for the exact source witness used by Appendix *K*.

## Source order and source-level frontier items

**Definition Core.5 (Per-process execution-call universe, frontier-item universe, observable projection, and source order).**

For each process *P*, let  $Q_{\text{exec},P}$  denote its execution-level set of dynamic source-level message-call instances for the particular execution / witness under the fixed external stimulus being analyzed.  $Q_{\text{exec},P}$  includes blocking *Push/Pop* calls, successful non-blocking *PushNB/PopNB* calls, and failed non-blocking *PushNB/PopNB* polls.

Let  $\Omega_P$  denote *P*'s local dynamic source-level frontier-item universe for that same execution / witness. Thus  $\Omega_P$  is stimulus-dependent: it contains exactly the local synchronization-call items, signal-action items, and message-operation frontier items that belong to the realized execution being compared, and excludes items on untaken control-flow branches, unentered cases, and unexecuted loop iterations. For sequential processes, signal-action items are positioned by the *R2* anchoring rule; for combinational processes, signal-action items are positioned in the observable cycle bucket determined by the *R2C*  $\epsilon$ -fixed-point rule. Message-operation frontier items include reached blocking *Push/Pop* instances, whether pending or committed, and successful non-blocking *PushNB/PopNB* instances that contribute committed *Push\_c(v)* / *Pop\_c(v)* events. Failed non-blocking polls are  $\epsilon$ -steps: they are members of  $Q_{\text{exec},P}$  when executed, but they are not members of  $\Omega_P$  merely because they executed.

Because  $Q_{\text{exec},P}$  is a universe of dynamic message-call instances while  $\Omega_P$  is a universe of source-level frontier items, their message-operation overlap is not a literal set intersection. Let  $\text{MsgOf}_P(x)$  be the partial projection from a frontier item  $x \in \Omega_P$  to its owning dynamic message-call instance, defined exactly for message-operation frontier items and undefined for pure synchronization or signal-anchor items. Define the message-frontier slice

$$\Omega_{\text{msg},P} := \{ x \in \Omega_P \mid \exists q \in Q_{\text{exec},P} : \text{MsgOf}_P(x) = q \}.$$

For any message-operation frontier item  $x \in \Omega_{\text{msg},P}$ , let  $\text{BoundaryOf}_P(x)$  denote the explicit abstract channel boundary at which *x* is evaluated for *BoundaryReq*, *ChannelEnabled*, *ChannelDisabled*, *Front\_P*, and Appendix *K* *WFG* reasoning. In the ordinary unelaborated case, each source-level message-call instance *q* has a unique corresponding frontier item *x* and a unique evaluated boundary. In *Sys\_B<sup>elab</sup>*,  $\text{MsgOf}_P$  need not be injective: one owning source-level message-call instance *q* may project to a chain or bundle of proof-internal frontier items *x* at several explicit abstract boundaries. The elaboration package must then provide  $\text{BoundaryOf}_P(x)$ ,  $\text{MsgOf}_P(x)$ , and the partial order  $\leq_P$  for each introduced item.

Define the corresponding message-instance slice induced by  $\Omega_P$  as

$$Q_{\Omega,P} := \{ q \in Q_{\text{exec},P} \mid \exists x \in \Omega_{\text{msg},P} : \text{MsgOf}_P(x) = q \}.$$

Let  $\Sigma_P$  denote the corresponding observable event/label projection: synchronization and signal items contribute their  $\Sigma$  labels when observed, and a message-operation frontier item  $x \in \Omega_{\text{msg},P}$  with  $\text{MsgOf}_P(x)=q$  contributes the committed  $\Sigma$ -event  $m(q)$  only if  $q$  commits. Pending blocking message-operation frontier items may therefore be in  $\Omega_P$  and  $\Omega_{\text{msg},P}$ , and their owning message-call instances may be in  $Q_{\Omega,P}$ , without yet contributing any committed  $\Sigma$ -event. Let  $\leq_P$  denote the source order on  $\Omega_P$ , i.e. the order generated by the document's scheduling rules for synchronization calls, signal actions as positioned by R2 or R2C, message-operation frontier items, and any explicit partial-order structure declared by the selected  $\text{Sys\_ref}$  elaboration. The execution-level order constraints E3/E5 use the corresponding source-induced order on  $Q_{\text{exec},P}$ .

#### Definition Core.6 (Done\_P, Remain\_P, and NextFront\_P relative to the fixed witness).

Fix the particular execution / witness under analysis, hence the associated  $Q_{\text{exec},P}$ ,  $\Omega_P$ ,  $\Omega_{\text{msg},P}$ ,  $Q_{\Omega,P}$ ,  $\Sigma_P$ , and  $\leq_P$  from Definition Core.5. For a state  $\sigma$  occurring as a cutpoint of that compared execution, let

$\text{Done}_P(\sigma) \subseteq \Omega_P$

be the set of local frontier items already realized by  $\sigma$ . For synchronization and signal items, “realized” means that the corresponding  $\Sigma$ -event has occurred. For a message-operation frontier item  $x \in \Omega_{\text{msg},P}$  with  $\text{MsgOf}_P(x)=q$ , “realized” means that  $q$  has committed and its committed  $\Sigma$ -event  $m(q)$  has occurred. A pending issued-but-uncommitted blocking message-operation frontier item  $x$  is therefore not in  $\text{Done}_P(\sigma)$ , even though  $\text{BoundaryReq}(x,\sigma)$ ,  $\text{ChannelEnabled}(x,\sigma)$ , and  $\text{ChannelDisabled}(x,\sigma)$  may be defined at  $\sigma$ , read operation-attributively through  $q = \text{MsgOf}_P(x)$ . A failed non-blocking poll  $q \in Q_{\text{exec},P}$  that induces no frontier item in  $\Omega_{\text{msg},P}$  is not considered by  $\text{Done}_P$ ,  $\text{Remain}_P$ , or  $\text{NextFront}_P$ , because it is an  $\varepsilon$ -step with no frontier item and no committed  $\Sigma$ -event.

Let

$\text{Remain}_P(\sigma) := \Omega_P \setminus \text{Done}_P(\sigma)$ .

Then define the next source-level frontier-item set by

$\text{NextFront}_P(\sigma) := \min_{\leq_P}(\text{Remain}_P(\sigma))$ .

This is the shared next-frontier notion used by Theorem J.1 and reconstructed from KObs by Lemma J.KObs-Proc, interpreted relative to the fixed witness universe  $\Omega_P$ . It may contain synchronization-call items, signal-anchored action items, and message-operation items. It is a frontier of source-level items, not a set of already-committed  $\Sigma$ -event labels.

#### Definition Core.7 (Message-operation slice of the next frontier).

For the same fixed compared execution / witness, define

$\text{MsgNextFront}_P(\sigma) := \{ x \in \text{NextFront}_P(\sigma) \mid x \in \Omega_{\text{msg},P} \text{ and, for } q = \text{MsgOf}_P(x), q \text{ is defined and } \text{kind}(q) \in \{\text{Push}_c, \text{Pop}_c\} \text{ for some channel } c \}$ .

This is only the message-operation frontier-item slice of the candidate next frontier; it is not yet the pending blocking frontier of Appendix B / K.

#### Definition Core.7a (Blocking message-operation slice of the next frontier).

For the same fixed compared execution / witness, define

$\text{BlockingMsgNextFront}_P(\sigma) := \{ x \in \text{NextFront}_P(\sigma) \mid x \in \Omega_{\text{msg},P} \text{ and, for } q = \text{MsgOf}_P(x), q \text{ is defined and } q \text{ is a blocking Push or Pop instance} \}$ .

This is the blocking-message candidate-frontier slice used by the Appendix K Front/NextFront bridge and WFG classification. It may contain a reached blocking operation before that operation has issued;  $\text{BoundaryReq}$  and  $\text{Pending}$  determine whether such an item is currently an outstanding request and therefore a member of  $\text{Front}_P$ . Successful non-blocking transfer items may belong to  $\text{MsgNextFront}_P$  but never to  $\text{BlockingMsgNextFront}_P$ .

### Clarification Core.5a (ordinary source order versus elaborated partial order).

For ordinary user-written sequential source code, dynamic message-call instances that appear in sequence in one process are ordered by dynamic program order, including distinct-interface calls, repeated loop iterations, and unrolled instances whose selected source expansion is ordered. Same-cycle issue, same-cycle commit, or issue before an earlier operation commits is a timing/issue-policy property, not incomparability. A message-operation multi-frontier exists only when an explicit  $\text{Sys\_B}^{\text{elab}}$  elaboration package introduces incomparable proof-internal frontier items and supplies the corresponding projection, boundary, order, and attribution facts. Ordinary distinct-interface source calls are not made incomparable merely because they access different channels.

### Lemma Core.OrdFrontSingleton (ordinary-source frontier uniqueness).

Fix a process  $P$ , a particular execution/witness, and an ordinary non-elaborated source interval. Assume every message-operation frontier item in that interval is induced by a unique ordinary dynamic source-call instance and no  $\text{Sys\_B}^{\text{elab}}$  partial-order structure is introduced. Then the restriction of  $\leq_P$  to the message-operation frontier items in that interval is total. Consequently, for every cutpoint  $\sigma$  in the interval,  $|\text{Front\_P}(\sigma)| \leq 1$ .

Proof. By Clarification Core.5a, all ordinary dynamic source-call instances encountered by  $P$  in the fixed execution are ordered by dynamic program order, including distinct-interface calls, repeated loop iterations, and ordered unrolled instances. In the ordinary unelaborated case, each such call induces one frontier item and  $\leq_P$  is the corresponding reflexive source-induced order. By Definitions Core.14 and Core.15, every member of  $\text{Front\_P}(\sigma)$  belongs to  $\text{Pending\_P}(\sigma)$  and hence is a blocking message-operation frontier item in the same interval. Hence any two members of  $\text{Front\_P}(\sigma)$  are comparable. Two distinct elements therefore cannot both be  $\leq_P$ -minimal. Thus  $\text{Front\_P}(\sigma)$  contains at most one member.  $\square$

### Operational interpretation of $\text{Sys\_ref}$ and source issue policies

The R-rules stated in Appendix G are constraints on a selected source-reference transition system, not merely constraints on a finished list of labels. For a fixed design, fixed stimulus, and selected reference-model choice,  $\text{Sys\_ref}$  has the following operational reading.

A  $\text{Sys\_ref}$  state consists of: (1) one local source state per process, including control location, local variables, call-stack/loop state, the current synchronization interval, and the reached/issued/committed status of dynamic message-call instances; (2) one state for each explicit abstract channel of the selected reference model, including  $B(c)$ ,  $\text{occ\_c}(t)$ , FIFO order when  $B(c) > 0$ , rendezvous endpoint requests when  $B(c) = 0$ , and any explicit  $\text{Sys\_B}^{\text{elab}}$  staged/bundle/join/epoch-gate channel state; (3) signal and synchronization state at the selected observable anchors; and (4)  $\epsilon$ -microstate for internal staging, direct-input sampling between anchors, and other proof-hidden activity. In  $\text{Sys\_B}^{\text{elab}}$ , introduced abstract channels are explicit proof-internal boundaries: their commits may be  $\Sigma$ -events for Appendices J-K even when projected away from the outer DUT/testbench-visible alphabet  $\Sigma_{\text{DUT}}$  by  $\Pi_{\text{elab}}$ .

The transition relation contains the following kinds of steps. Pure source-evaluation  $\epsilon$ -steps advance ordinary C++/SystemC computation and allocate dynamic operation instances in dynamic source order. For a sequential process, ordinary call-return state and the L1 preparation/reach state are distinguished by Definition Core.ReachPrep below: a later operation may become reached before an earlier blocking operation has process-facing committed only through a legal dependency-independent preparation segment, and this does not make the operations unordered. A message-issue step for  $q$  may occur only after  $q$  has been reached, the endpoint request can be attributed to  $q$ , the R3 synchronization-boundary constraints are respected, and the issued set is prefix-closed under  $\leq_{\text{src}}$ . Issuing a blocking Push or Pop asserts the endpoint-owned persistent request for  $q$  and records  $\text{Issue}(q)$ ; issuing PushNB/PopNB asserts its one-cycle attributed attempt. A channel-commit step for a bounded FIFO boundary  $B(c) > 0$  may occur

when  $\text{BoundaryReq}(q, \sigma)$  and  $\text{ChannelEnabled}(q, \sigma)$  hold at the selected abstract boundary; it emits the corresponding  $\text{Push}_c(v)$  or  $\text{Pop}_c(v)$  event, updates the E4 channel state, and releases the owning call at the process-facing boundary. For a rendezvous boundary  $B(c)=0$ , commit is one atomic two-endpoint transfer step with matching producer and consumer endpoint requests in the same settled cycle and one shared payload value. A failed non-blocking boundary decision emits no  $\Sigma$  event but produces a  $\text{ReturnedAtCycle}$  status and retires its one-cycle attempt. A synchronization boundary action for call  $s$  may be selected only when  $\text{Core.SyncFence}$  is satisfied by the already-completed and same-cycle selected process-facing message commits; the interval advance caused by  $s$  occurs only after those predecessor commits have been accounted for. Synchronization and signal-anchor steps then emit  $\text{wait/Sync}$  and anchored  $\text{Read/Write}$  events according to  $R1/R2/R2C$ .  $\varepsilon$ -settling and drain steps update only proof-hidden or explicitly elaborated internal state and must respect  $K\varepsilon$  /  $B$ -faithfulness when used by Appendices J–K. For a sequential process, Definition  $\text{Core.RegSeq}$  below governs which process-owned evaluation, request, and payload actions may share one Horizon.

#### Definition $\text{Core.ReachPrep}$ (overlap-aware source reach/preparation semantics).

Fix a sequential process  $P$ , its current source interval, and a cycle  $t$ .  $\text{Sys\_ref}$  retains the logical source/replay control position together with reached/issued/committed status for dynamic calls. During the L1 pre-settle phase, an operational preparation cursor may advance through a finite source-control segment and mark a later dynamic operation  $q$  as reached without claiming that an earlier blocking operation  $q_0$  has committed or returned. Such preparation is legal only when: (1)  $q$  remains in the current synchronization interval and all dynamic source-order relations are preserved; (2) the reachability, control choice, endpoint, address, payload, and required operands of every prepared action are independent of any unresolved earlier returned data/status and satisfy  $\text{Core.RegSeq}$  and the applicable field-sensitive  $\text{NoDependentUse/footprint}$  restrictions; (3) no synchronization boundary is crossed; (4) no already-activated  $\text{Core.16}$  blocked-frontier discipline forbids launching that source-later work; and (5) the selected source/tool witness or certificate validates the preparation segment. Reaching  $q$  does not itself assert a request. Issue remains a separate transition and remains prefix-closed under  $<_{\text{src}}$ .

This permission applies both to overlapped/pipelined iterations and to ordinary straight-line source sequence  $q_0; q_1$  when  $q_1$  is dependency-independent of the unresolved  $q_0$  result and the selected  $\text{Sys\_ref}$  theorem instance admits eager preparation. It does not make  $q_0$  and  $q_1$  incomparable, and it does not assert that a raw serial  $\text{SystemC}$  execution has returned from  $q_0$  early; it is the source-reference scheduling permission used to represent allowed RTL-matched eager issue. Examples  $K.CE$  and  $K.CE-2$ , together with the Appendix K A5-L blocking-rendezvous example, use this straight-line case. Coverage direction for  $\text{Post} \rightarrow \text{Ref}$ : for every  $\text{RTL\_B}$  prefix included in a  $\text{Post} \rightarrow \text{Ref}$  coverage claim, the selected  $\text{Sys\_ref}$   $\text{ReachPrep}$  semantics shall admit every dependency-independent preparation/reach segment required by the accepted  $\text{J.LocalGWR}$  or  $\text{J.GWR}$  construction for that prefix. The instance may be more permissive than the observed RTL scheduling, but it shall not exclude preparation needed to realize a covered RTL witness. Once the L2  $\text{SettlePoint}$  exposes a nonempty fully disabled pending frontier,  $\text{Core.16}$  prohibits every new source-later launch while that frontier persists. Every L1 preparation segment used by a theorem instance must be finite. Lemma I.3/FPD supplies that fact for its certified finite construction segments;  $\text{Core.CycleClosure}$ 's source-cycle-existence/finiteness clause supplies the generic per-cycle finiteness obligation when a complete/fair source continuation is required.

#### Definition $\text{Core.RegSeq}$ (registered sequential-interface semantics).

The  $\text{Core}$  classifies each process for the Appendices I–K proof stack as either combinational under  $R2C$  or sequential under  $\text{Core.RegSeq}$ . For a sequential process  $P$ , every process-driven endpoint request and

Push-data output assigned to cycle/Horizon  $t$  is a function of registered process or pipeline state and operands fixed before the L2 settle point of cycle  $t$ . Combinational derivation from that registered state is permitted, but the request and payload must remain stable for the selected boundary action. Direct inputs count as fixed operands only when their applicable E2, `hls_direct_input`, or `hls_direct_input_sync` contract makes their value stable at the relevant anchor. If `ReturnedAtCycle(q,t)` holds, any P-owned evaluation, control selection, endpoint/address selection, payload computation, or message issue that depends on  $q$ 's returned data or success/failure status has Horizon strictly greater than  $t$ . A same-Horizon source-later action is permitted only when its reachability, control, operands, endpoint, address, and payload are independent of every unresolved cycle- $t$  NB outcome and when its writes do not change the `ReqSnapshot` of that unresolved decision except through an explicitly ordered dependency. This is a normative theorem-scope restriction, not a consequence of the word “clocked” alone. Tool-side Cat# or equivalent conformance evidence and source-side simulation/lint/witness evidence shall discharge it. Intentional zero-time outcome-dependent NB cascades or retry loops shall be identified by checked source/tool analysis and then rejected or explicitly excluded from the theorem instance, isolated as latency-sensitive logic, remodeled as one explicit atomic multi-way decision, or aligned by Appendix L; silent scope exclusion is not permitted.

#### Definition Core.PolicyL (liberal, witness-relative issue).

Policy L is the most permissive source-side issue policy used by the safety witness and by the K.2b dominance/extraction bridge. It may issue a reached source-level message operation whenever the transition-system preconditions above, R3/R4/R5, E4, E5, `Core.RegSeq`, the selected non-blocking outcome witness, and the source-order prefix-closure requirement are satisfied. A source-later operation may have become reached through `Core.ReachPrep` even while a source-earlier same-interval blocker remains uncommitted. Policy L may therefore leave the source-later blocking request issued in that state. This is how the reference model represents loop overlap, ordinary dependency-independent eager preparation, RTL-matched eager issue, and other allowed same-cycle/overlapped request timing. Policy L is not a claim that every raw pre-HLS simulator mode can generate every such witness state. Policy L does not override returned-result availability: it cannot issue an operation in cycle  $t$  when its `Core.ReachPrep` reachability, control selection, endpoint or address selection, payload, or any required operand depends on data/status first returned at the cycle- $t$  boundary. `Core.LLegal` fixes Policy-L step legality, `Core.Phase` fixes the cycle-settle boundary and the trigger for Core.16, and `Core.LAdm` below combines those pieces with the activated `Core.Drain` qualification used by liveness admissibility.

#### Definition Core.LLegal (Policy-L step legality and fixed-instance binding).

Fix a selected liveness instance  $I$  consisting of `Sys_ref`, the initial/reset state, capacities and elaboration/lowering, the fixed stimulus and reactive environment, witness-selected non-blocking and other  $\epsilon$  outcomes, arbitration choices, and the selected synchronization, signal-anchor, and  $\epsilon$ -normalization conventions. A finite or infinite path  $p$  is Policy-L-legal for  $I$  iff it begins in the selected initial/reset state, every transition is a transition of the operational `Sys_ref` relation above permitted by Policy L, and every fixed choice in  $I$  is respected. R1–R5, E4/E5, dynamic-operation identity, request attribution and persistence, endpoint cardinality, reset-epoch typing, and sequential-result availability are part of this step legality.

#### Definition Core.Phase (Policy-L cycle phases and settled-frontier trigger).

Each cycle  $t$  of a Policy-L-legal path is interpreted using the following normative phase boundary. (L0/L1) Cycle entry and pre-settle issue window. Starting from the cycle-entry state after all prior-cycle boundary updates, sequential processes may take a finite selected sequence of pure source-evaluation  $\epsilon$ -steps and message-issue steps for operations that were already reached at cycle entry or become

reached during this phase through Core.ReachPrep and evaluation independent of every cycle- $t$  boundary result. Their control choice, endpoint, address, payload, status inputs, direct-input values, and other required operands must be available under Core.RegSeq independently of any cycle- $t$  commit or failed/non-failed NB decision outcome. These issue steps remain subject to R3–R5, source-order prefix closure, stable payload, and the deferred-hole restrictions of Lemma I.3. Thus a source-later outcome-independent operation, including an ordinary straight-line successor admitted by Core.ReachPrep, may issue during this window while a source-earlier blocker or unresolved NB attempt remains uncommitted/undecided. A value or status first returned at the cycle- $t$  boundary cannot be used in this window. Any theorem that constructs a complete/fair source continuation additionally requires the selected L1 preparation/issue segment to terminate finitely as part of Core.CycleClosure.

(L2) settle point. After the selected process-owned pre-settle steps for cycle  $t$ , the L1 window closes. Process-owned request assertions and Push payloads are then held fixed while the explicit channel network and every other phase-legal L2  $\epsilon$ -step settle under  $\rightarrow_{\epsilon, L2}$  to the first  $\text{SettlePoint}_I(t, \sigma_t^{\text{set}})$  of Definition Core.Settle. The phrase “the frontier settles” and every Appendix K test of whether a frontier is fully channel-disabled at the activation boundary refer to this L2 SettlePoint. By Lemma Core.SettlesKCut, the settled source state is also Core.4-quiescent and lies in the applicable source KCut domain; its special role here is the Core.16 activation boundary, not necessarily the later drain-closed Dead\_K evaluation point.

(L3) Freeze and boundary action.  $\text{Front}_P$  and ChannelEnabled/ChannelDisabled are evaluated at  $\sigma_t^{\text{set}}$ . If  $\text{Front}_P$  is non-empty and every frontier item is disabled there, Definition Core.16 activates immediately at  $\sigma_t^{\text{set}}$ : no new source-later operation may issue after that snapshot in cycle  $t$  or in a later cycle while the same fully disabled frontier persists, although already-issued downstream work may drain as Core.16 permits. If the frontier is not fully disabled, the selected E4/rendezvous, synchronization, signal-anchor, arbitration, and endpoint-cardinality rules determine the boundary actions that commit in cycle  $t$ . Any synchronization action  $s$  in that selected L3 boundary-action set is additionally gated by Core.SyncFence: every blocking  $q \in \text{pref}_S(s)$  must have process-facing committed earlier or be selected to process-facing commit in the same cycle before  $s$  advances  $P$  into the next source interval.

(L4) Next-cycle result availability. Boundary commits and failed non-blocking decisions produce their process-visible ReturnedAtCycle results and update registered process/channel state for the next cycle. Process-owned evaluation or issue whose reachability, control, endpoint, address, payload, or required operand depends on a value or success/failure status returned in cycle  $t$  may begin only in a strictly later cycle. Every deferred NB hole created for Horizon  $t$  is resolved or retired before the exported post-boundary cutpoint for that Horizon.

The L2/L3 freeze is not retroactive. An issue that was legal in the L1 pre-settle window is not invalidated merely because the subsequent settled snapshot exposes an earlier fully disabled frontier. Conversely, once that settled snapshot exists, an operation not already issued before it cannot use the same-cycle pre-settle window to bypass the activated Core.16 block.

Dependency convention. Core.Phase defines the settled-frontier trigger and phase ordering without assuming the continuation discipline. Core.16 separately defines the drain-only continuation obligations activated by that trigger. Core.LAdm, stated after Core.16, combines Policy-L legality, Core.Phase, nonwithdrawal,  $K\epsilon$ , and the activated Core.Drain qualification; there is therefore no circular definitional dependency between admissibility and the drain discipline.

### Definition Core.PolicyS (conservative serial-blocking issue).

Policy S uses the same Sys\_ref state space, channels, values, witness choices, and transition rules as Policy L, but adds one process-side restriction: within a source interval between adjacent ordinary synchronization calls selected into  $S$ , a later source-order blocking Push/Pop operation may not issue

until every earlier source-order blocking Push/Pop operation in that same interval has committed in a strictly earlier clock cycle. In  $\text{Sys\_B}^{\text{elab}}$ , the relevant commit for this restriction is the process-facing commit that completes, accepts, or releases the owning source-level blocking operation at its selected boundary; internal staged/bundle/join/epoch-gate transfers do not by themselves satisfy the serial condition unless the elaboration package identifies that transfer as the process-facing commit for the source operation. Policy S serializes only process-side issue of source-level blocking calls. It does not serialize independent peer processes, E4 channel-side drainage, FIFO movement, or  $\epsilon$ -steps that are otherwise admissible.

#### Remark Core.PolicyS-Sim (Relation to practical pre-HLS simulation modes).

A raw serial source simulation mode implements the Policy-S discipline for ordinary unelaborated blocking calls. A Matchlib-style eager or skidbuffered source simulation mode may execute source calls in dynamic program order while issuing requests without consuming a clock cycle until data or space is unavailable; that mode is Policy-L-compatible only to the extent that its request attribution, channel states, and non-blocking outcomes satisfy the transition-system conditions above. Neither mode treats ordinary distinct-interface source calls as unordered. The difference between serial and eager simulation modes is a difference in issue timing and buffering, not a difference in  $\_src$ .

### Commit, issue, and boundary requests

#### Definition Core.8 (Commit).

A boundary-commit occurrence at an explicit abstract boundary endpoint is the corresponding legal E4 channel-commit transition of the selected proof model: it produces the applicable committed  $\text{Push\_c}(v)$  or  $\text{Pop\_c}(v)$  observable unit and the channel-state update prescribed by the selected rendezvous or bounded-FIFO semantics. For  $\text{RTL\_B}$ , the selected explicit-boundary mapping and B-faithfulness/conformance evidence relate that abstract commit to the implementation clock cycle in which the mapped valid and ready conditions are simultaneously satisfied; the committed payload is the mapped data value observed at that boundary in that cycle. For a source-level message operation  $q$ ,  $\text{Commit}(q)$  denotes  $q$ 's process-facing commit occurrence. In the ordinary unelaborated case, this is the boundary-commit occurrence at  $q$ 's unique selected explicit abstract boundary. In  $\text{Sys\_B}^{\text{elab}}$ , it is the commit/accept/release occurrence designated by the elaboration package as completing or releasing  $q$ ; proof-internal boundary commits that do not release  $q$  are not  $\text{Commit}(q)$ . Let  $\text{CommitCycle}(q)$  denote the cycle index of  $\text{Commit}(q)$ . When a rule compares  $q$ 's process-facing commit with a clock cycle, it compares  $\text{CommitCycle}(q)$ . Per-boundary E4 commit occurrences at explicit abstract channels retain the cycle index of their own boundary-commit occurrence and are not mapped to  $\text{CommitCycle}(q)$  unless that boundary commit is the designated process-facing commit for  $q$ . Process-facing commit identifies completion/release of the source operation; it does not imply that returned data or status is available for dependent use in that same cycle, which remains governed by Core.Phase L4 and Core.RegSeq.

#### Definition Core.9 (Issue).

For a source-level message operation instance  $q$  at a chosen explicit abstract boundary endpoint,  $\text{Issue}(q)$  is the first cycle in which the calling process begins requesting the transfer corresponding to  $q$  at that endpoint.  $\text{Issue}(q)$  is an auxiliary scheduler/witness timestamp used for rules such as R4, E3, and E5.

#### Definition Core.IssueFair (weak fairness of source issue).

Fix an issue policy  $\Pi \in \{\text{Policy L}, \text{Policy S}\}$  and a complete phase-bearing source execution  $p$ .  $\text{IssueEnabled\_}\Pi(q, t)$  holds at an applicable L1 opportunity when  $q$  has been reached, has not yet issued or been retired/reset, and its message-issue transition is legal under the operational  $\text{Sys\_ref}$  relation,

Core.Phase,  $\Pi$ , R3–R5/E4/E5, Core.RegSeq, source-order prefix closure, Core.16, and the fixed witness/environment/reset/arbitration choices.  $\text{IssueFair}_{\Pi}(\rho)$  holds iff, whenever from some cycle  $t_0$  onward within one reset epoch  $q$  remains  $\text{IssueEnabled}_{\Pi}$  at every applicable L1 opportunity until it issues or the execution reaches an affecting RESET for  $q$  or an explicit DeclaredTerminal disposition,  $q$  is issued after finitely many such opportunities. This is weak fairness: an operation whose issue legality merely recurs intermittently does not satisfy the antecedent. An operation forbidden by Core.16 is not  $\text{IssueEnabled}_{\Pi}$  and imposes no  $\text{IssueFair}$  obligation.  $\text{IssueFair}$  governs assertion/issue of a reached source operation; B2/B3 separately govern selection/commit of already-issued boundary actions. A constructive complete-source continuation theorem may use a stronger recurrent-opportunity dovetailing scheduler, but the required property remains Core.IssueFair itself. Appendix K consumes Core.IssueFair as a premise of complete admissible continuations.

#### Definition Core.10 (Endpoint-owned boundary request).

Fix an endpoint direction at an explicit abstract boundary. For any message-operation frontier item  $x \in \Omega_{\text{msg},P}$  with owning source-level message instance  $q = \text{MsgOf}_P(x)$  and evaluated boundary  $c = \text{BoundaryOf}_P(x)$ , let  $\text{req}_x(t)$  denote the endpoint-owned boundary request signal for that endpoint in cycle  $t$ , namely valid for a Push endpoint and ready for a Pop endpoint. The predicate  $\text{BoundaryReq}(x, \sigma)$  records that this endpoint-owned request is asserted for frontier item  $x$  at state  $\sigma$ .  $\text{BoundaryReq}(x, \sigma)$  is operation- and boundary-attributive: it is true only when the asserted endpoint-owned request at  $\sigma$  is attributed to the specific frontier item  $x$ , its owning source-level operation  $q$ , and its evaluated boundary  $\text{BoundaryOf}_P(x)$ . In particular, if  $\text{Issue}(q)$  has not yet occurred at  $\sigma$ , then  $\text{BoundaryReq}(x, \sigma)$  is false. In the ordinary unelaborated case, where  $q$  has a unique frontier item and a unique evaluated boundary,  $\text{BoundaryReq}(q, \sigma)$  may be used as shorthand for  $\text{BoundaryReq}(x, \sigma)$ . In  $\text{Sys}_B^{\text{elab}}$ , the  $x$ -form is authoritative.

#### Definition Core.1C (Issue/Commit linkage invariant).

Fix an endpoint direction at an explicit abstract boundary. For source-level message operation instances on that endpoint direction, the following all hold:

(1) Boundary-request soundness / non-withdrawal.

If  $q$  is blocking, then once issued its endpoint-owned request remains asserted, with stable payload while asserted for Push, until  $q$  process-facing commits or an affecting RESET governed by RW2 clears that dynamic instance. Ordinary back-pressure, scheduler delay, a silent cycle, or any other nonterminal transition cannot deassert, retire, reattribute, cancel, or replace it. These are the only nonterminal mechanisms in the G–K theorem model by which an outstanding blocking dynamic instance may cease to be outstanding: process-facing commit or an affecting RW2 RESET. A DeclaredTerminal disposition may instead end the execution with  $q$  still pending. Protocols that require independent nonterminal cancellation, withdrawal, retry, or reattribution of a blocking request are outside the G–K theorem model unless an explicit operational extension and its correspondence/progress obligations are separately defined and proved.

(2) Stable attribution while the same source-level request remains outstanding.

If a blocking request remains asserted across a cycle in which no transfer commits on that endpoint direction, the continued assertion remains attributed to the same outstanding  $q$  until its process-facing commit or an affecting RW2 RESET. For non-blocking PushNB/PopNB, continuous assertion of the endpoint request signal does not collapse later executed non-blocking call instances into the same  $q$ ; each later executed source-level poll is a distinct  $q' \in Q_{\text{exec},P}$  unless the witness explicitly identifies the same still-outstanding source-level request instance.

(3) Back-to-back request attribution.

If  $q_0$  commits in cycle  $t$  and the same endpoint request remains asserted in cycle  $t+1$ , the continued level

alone does not issue a next operation  $q_1$ . If  $q_1$  is the next same-direction source-level message operation in the same source interval and the operational witness/certificate attributes the cycle- $(t+1)$  request to  $q_1$ , then  $\text{Issue}(q_1)=t+1$ . If  $q_1$  first becomes the attributed owner in a later cycle  $u>t+1$ , then  $\text{Issue}(q_1)=u$ . Thus this clause permits arbitrary legal scheduler/HLS delay before  $q_1$  issues.

(4) Same-endpoint blocking issue serialization.

For distinct blocking  $q_0$  and  $q_1$  on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint within one reset epoch, if  $q_0 <_{\text{src}} q_1$  and both issue, then  $\text{Issue}(q_0) < \text{Issue}(q_1)$ . Equal issue cycles are not legal for two such scalar blocking instances.

(5) Sync-boundary discipline.

If  $q_0$  and  $q_1$  are same-direction source-level message operation instances and  $q_0 \in \text{pref\_S}(s)$  and  $q_1 \in \text{suff\_S}(s)$  for a synchronization call  $s$ , then  $\text{Issue}(q_1)$  occurs strictly after the commit of  $s$ . Any request assertion that persists while the process is still pending at  $s$  is attributed to the outstanding operation(s) in  $\text{pref\_S}(s)$ , not to issuance of any operation in  $\text{suff\_S}(s)$ . Synchronization-completion legality itself is governed by Core.SyncFence below.

### Definition Core.SyncFence (synchronization-completion legality).

Let  $s$  be a dynamic synchronization-call instance of process  $P$  selected into  $S$ , and let  $\text{BPref\_S}(s)$  be the blocking Push/Pop instances in  $\text{pref\_S}(s)$ . A boundary action that commits  $s$  is legal only if every  $q \in \text{BPref\_S}(s)$  has process-facing committed no later than that same cycle. Equivalently, after  $s$  commits and  $P$  advances into the succeeding source interval, no  $q \in \text{BPref\_S}(s)$  may remain issued and uncommitted. Same-cycle completion is permitted:  $q$  and  $s$  may both commit in cycle  $t$  provided  $q$ 's process-facing commit is included in the selected L3 boundary-action set for  $t$  and is accounted for before the interval advance caused by  $s$ . This "accounted for before" convention is an ordering of the source operational update and of the proof/certificate attribution of the same-cycle boundary batch; on RTL\_B it does not require an observable physical sub-cycle ordering between two events that commit on the same active clock edge. In  $\text{Sys\_B}^{\text{elab}}$ , "process-facing commit" means the commit/accept/release event designated by the elaboration package as completing the owning source-level operation  $q$ ; an internal staged, bundle, join, guard, or epoch-gate transfer that does not release  $q$  does not satisfy this fence. Appendix G rules R3 and E5 state the corresponding source-side and implementation-side trace constraints explicitly.

Status and conformance split. Core.IC and Core.SyncFence are constitutive legality/invariant clauses of the selected  $\text{Sys\_ref}$  operational transition system: source executions outside these rules are not executions of the theorem's prescribed source model. They are not global proof axioms to be admitted independently of that semantics. On RTL\_B, the corresponding request-lifecycle, attribution, process-facing completion, and synchronization-fence behavior is not obtained by definition; it is an implementation-side conformance obligation established through E3/E5, boundary mapping/B-faithfulness, request-attribution evidence, and the named Appendix M.7a Core.IC/Core.SyncFence audit entries as applicable.

### Remark Core.10a (Non-blocking polls).

For non-blocking message passing, a failed poll produces no  $\Sigma$  event and is modeled as an  $\epsilon$ -step, while a successful poll contributes the corresponding committed  $\text{Push\_c}(v)$  or  $\text{Pop\_c}(v)$  event. Each executed  $\text{PushNB/PopNB}$  call is nevertheless a distinct dynamic source-level instance  $q \in Q_{\text{exec},P}$  for its owning process  $P$ ; thus repeated failed polls in later loop iterations are distinct  $q$ 's even if the endpoint request remains continuously asserted. The  $\epsilon$ -modeling concerns observability in  $\Sigma$ , not the dynamic-instance identity of the underlying source-level calls.

## Channel enablement

### Definition Core.11 (ChannelEnabled / ChannelDisabled).

Let  $x$  be a message-operation frontier item on channel  $c = \text{BoundaryOf\_P}(x)$ , and write  $q = \text{MsgOf\_P}(x)$  with  $\text{kind}(q) \in \{\text{Push\_c}, \text{Pop\_c}\}$  for its endpoint direction. At a  $\text{BoundaryEvalPoint\_I}(\sigma)$  of Definition Core.Settle, let  $t_\sigma$  denote the cycle of that cutpoint. Define:

$\text{ChannelEnabled}(x, \sigma)$  to mean that  $\text{BoundaryReq}(x, \sigma)$  holds and that the channel-side condition required for  $x$ 's owning operation  $q$  to commit at  $\text{BoundaryOf\_P}(x)$  is satisfied after combinational settling at that cutpoint, and

$\text{ChannelDisabled}(x, \sigma)$  to mean that  $\text{BoundaryReq}(x, \sigma)$  holds but  $\text{ChannelEnabled}(x, \sigma)$  does not.

In the ordinary unique-boundary case,  $\text{ChannelEnabled}(q, \sigma)$  and  $\text{ChannelDisabled}(q, \sigma)$  may be used as shorthand for the corresponding  $x$ -form. Every L2 SettlePoint and every Core.KCut cutpoint used by Appendix K is a  $\text{BoundaryEvalPoint}$ ; the definition therefore has one boundary-evaluation meaning without identifying those two kinds of cutpoint.

### Definition Core.12 (Buffered channels).

If  $B(c) > 0$  and  $x$  is a source-level or elaborated message-operation frontier item on channel  $c = \text{BoundaryOf\_P}(x)$ , with  $q = \text{MsgOf\_P}(x)$ , then the buffered-channel enabling condition is:

- $\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge \text{occ\_c}(\sigma) < B(c)$ , when  $\text{kind}(q) = \text{Push\_c}$
- $\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge \text{occ\_c}(\sigma) > 0$ , when  $\text{kind}(q) = \text{Pop\_c}$

Thus, if  $x$  is a buffered  $\text{Push\_c}$  frontier item that is channel-disabled, then  $\text{BoundaryReq}(x, \sigma)$  holds and  $\text{occ\_c}(\sigma) = B(c)$ ; and if  $x$  is a buffered  $\text{Pop\_c}$  frontier item that is channel-disabled, then  $\text{BoundaryReq}(x, \sigma)$  holds and  $\text{occ\_c}(\sigma) = 0$ .

### Definition Core.13 (Rendezvous channels).

If  $B(c) = 0$  and  $x$  is a source-level or elaborated message-operation frontier item on channel  $c = \text{BoundaryOf\_P}(x)$ , with  $q = \text{MsgOf\_P}(x)$ , then the rendezvous enabling condition is:

$\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge$  (the complementary endpoint request on  $c$  is asserted at  $\sigma$ ).

Equivalently:

- if  $\text{kind}(q) = \text{Push\_c}$ , then

$\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge$  (the complementary  $\text{Pop\_c}$  endpoint request is asserted at  $\sigma$ )

- if  $\text{kind}(q) = \text{Pop\_c}$ , then

$\text{ChannelEnabled}(x, \sigma) \Leftrightarrow \text{BoundaryReq}(x, \sigma) \wedge$  (the complementary  $\text{Push\_c}$  endpoint request is asserted at  $\sigma$ )

Hence, if a rendezvous frontier item  $x$  is channel-disabled, its own  $\text{BoundaryReq}$  holds but the complementary endpoint is not simultaneously requesting the matching partner action at that cutpoint. In the ordinary unique-boundary case, the equivalent  $q$ -form may be used as shorthand.

## Pending blocking frontier

### Definition Core.14 (Pending blocking operations).

Fix a  $\text{BoundaryEvalPoint\_I}(\sigma)$  for process  $P$  within the current interval  $I\_P(\sigma)$  between the adjacent synchronization calls that bracket  $\sigma$  in  $P$ 's current execution. Define

$\text{Pending\_P}(\sigma) := \{ x \mid x \in \Omega\_msg, P \text{ is a blocking source-level message-operation frontier item in that} \}$

same interval  $I_P(\sigma)$ , and for  $q = \text{MsgOf\_P}(x)$ ,  $q$  is defined,  $\text{Issue}(q)$  has occurred, and  $\text{Commit}(q)$  has not yet occurred in  $\sigma$  }.

By Appendix C's Issue/Commit linkage together with the definition of  $\text{BoundaryReq}$ , each member  $x$  of  $\text{Pending\_P}(\sigma)$  has its endpoint-owned request asserted at the cutpoint, read operation-attributively through its owning dynamic message-call instance  $q = \text{MsgOf\_P}(x)$ . By  $\text{Core.SyncFence}$ , an issued blocking operation from the preceding interval cannot survive the synchronization commit that advances  $P$  into  $I_P(\sigma)$ ; therefore the interval restriction cannot discard an outstanding blocker. This definition is valid both at an L2  $\text{SettlePoint}$  used by  $\text{Core.16}$  and at a  $\text{Core.KCut}$  comparison cutpoint;  $\text{WFG}$  and  $\text{Dead\_K}$  use only the latter.

### Definition Core.15 (Pending blocking frontier $\text{Front\_P}(\sigma)$ ).

Define

$\text{Front\_P}(\sigma) := \min_{\leq P}(\text{Pending\_P}(\sigma))$

If several  $\leq P$ -minimal pending blocking message-operation frontier items are simultaneously outstanding, they form a frontier. Lemma  $\text{Core.OrdFrontSingleton}$  proves that this set has at most one member for an ordinary non-elaborated source interval. This is the Appendix B / K notion of the pending blocking frontier (or minimal pending blocking frontier). It is distinct from  $\text{NextFront\_P}(\sigma)$ , which is the broader source-level frontier-item set over  $\Omega\_P$ .

## Wait-for graph

### Definition Core.WFG (Wait-for graph).

At a  $K$ -facing cutpoint  $\sigma \in \text{KCutReach\_ref}(I)$ , or at the corresponding certified RTL  $K$ -facing view, define the wait-for graph  $\text{WFG}(\sigma)$  as follows. Its vertices are the modeled processes, and its directed edges are induced from frontier members  $\text{Front\_P}(\sigma)$  by the corresponding channel-disabled dependencies on internal message-passing channels only. Equivalently, a directed edge  $P \rightarrow Q$  is present when some frontier item  $x \in \text{Front\_P}(\sigma)$ , with owning message instance  $q = \text{MsgOf\_P}(x)$ , is channel-disabled at the chosen boundary and the unique complementary endpoint needed to enable  $q$  belongs to process  $Q$ . Here “internal” means that both endpoints of the channel are modeled processes of  $\text{Sys\_ref}$  /  $\text{RTL\_B}$ . Channels whose complementary endpoint is the external environment/testbench at the DUT boundary are excluded from  $\text{WFG}$  and from the internal message-passing deadlock notion used in Appendix K. Buffered and rendezvous channels are both included, with  $B(c)$ ,  $\text{occ\_c}(\sigma)$ ,  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled}$ , and  $\text{ChannelDisabled}$  interpreted at the chosen  $\text{Core.KCut}$  boundary-evaluation point on the explicit abstract channels of  $\text{Sys\_ref}$  or reconstructed certified RTL view. A single  $\text{WFG}$  may therefore contain a mix of buffered-channel and rendezvous-channel edges. Appendix K consumes this  $\text{Core}$  definition and the clarifications below and adds the deadlock, closure, waiver, and preservation predicates; it does not redefine  $\text{WFG}$ .  $\text{Core.16}$  itself does not require a  $\text{WFG}$  at its L2  $\text{SettlePoint}$ ; it uses  $\text{Front\_P}$  and the ALL-disabled boundary test there and reaches the  $K$ -facing  $\text{WFG}$  domain only through  $\text{Core.SettleKBridge/K.DrainReach}$ .

### Clarification Core.WFGa (WFG edge versus blocked process).

A directed  $\text{WFG}$  edge records a channel-disabled dependency of one frontier item; it does not by itself imply that the owning process  $P$  is blocked on message passing. If  $\text{Front\_P}(\sigma)$  contains another explicitly incomparable channel-enabled member,  $P$  has an outgoing  $\text{WFG}$  edge but is not blocked under Appendix K's ALL-disabled test. The internal message-passing-deadlock predicate therefore quantifies only over nonempty sets  $D$  in which every  $P \in D$  is blocked, so every edge used to close a deadlocked component originates from a process with no channel-enabled frontier item.

### Clarification Core.WFGb (Multi-frontier outgoing edges).

When  $\text{Front\_P}(\sigma)$  has multiple members,  $P$  may have multiple outgoing WFG edges, one for each channel-disabled frontier item and its unique complementary endpoint. By Lemma  $\text{Core.OrdFrontSingleton}$ , such multi-frontier behavior cannot arise from an ordinary non-elaborated source interval. It is admitted only when the explicit  $\text{Sys\_B}^{\text{elab}}$  package introduces and types the corresponding incomparable proof-internal frontier items; ordinary source-sequenced message calls do not become incomparable merely because they use distinct channels.

### Drain-only blocked-frontier discipline and automatic flush

#### Definition Core.16 (Drain-only blocked-frontier discipline).

A source interval satisfies the drain-only blocked-frontier discipline if, at the L2  $\text{SettlePoint\_I}(t, \sigma\_t^{\text{set}})$  of Definitions  $\text{Core.Phase/Core.Settle}$ ,  $\text{Front\_P}(\sigma\_t^{\text{set}})$  is non-empty and every frontier item  $x \in \text{Front\_P}(\sigma\_t^{\text{set}})$  is channel-disabled. The discipline activates at that first settled snapshot. It does not retroactively invalidate source-later operations legally issued during the preceding L1 pre-settle window, but it forbids any new such issue after  $\sigma\_t^{\text{set}}$  in that cycle and while the same fully disabled frontier persists. Set  $\sigma := \sigma\_t^{\text{set}}$  at activation. This activation state is a  $\text{BoundaryEvalPoint}$  and, by Lemma  $\text{Core.SettleIsKCut}$ , is itself a source  $\text{Core.KCut}$  in the applicable phase-bearing source proof model; it is not necessarily the later drain-closed  $\text{KCut}$  at which  $\text{Dead\_K}$  is evaluated, and that connection is supplied by  $\text{Core.SettleKBridge/K.DrainReach}$ . Once activated, the following all hold:

(1) Block point.

$P$  is blocked on message passing at  $\text{Front\_P}(\sigma)$ .

(2) No new later work.

While  $\text{Front\_P}(\sigma)$  remains non-empty and fully disabled, no new later blocking message-operation frontier item  $y \in \Omega_{\text{msg}, P}$  is issued, where “later” means that, for some  $x \in \text{Front\_P}(\sigma)$ ,  $x \leq_P y$  and  $x \neq y$ , with issue understood through the owning message instance  $q_y = \text{MsgOf\_P}(y)$ . More generally, no new source-later dynamic operation is launched past the blocked point. Here “launched” is intentionally broader than  $\text{Issue}(q)$ : it includes a source-evaluation or message-issue transition that advances new source-later dynamic work past the blocked point. The specific no-new-Issue clause and this broader no-new-launch clause are retained as distinct obligations.

(3) No withdrawal.

Already-asserted blocking requests are not withdrawn before the owning message instances commit.

(4) Frontier-minimality.

An already-asserted blocking request whose frontier item is not in  $\text{Front\_P}(\sigma)$  is not treated as a wait-for source unless and until that frontier item later becomes frontier-minimal.

(5) Drain-only downstream progress.

Work already past the blocked frontier may drain and commit when channel-enabled, but no new work advances past the blocked frontier while that frontier remains fully disabled.

(6) Finite non-replenishing drain.

Every selected explicit abstract boundary that can hold or credit drainable work has finite  $B$ -faithful storage or credit, directly or through a finite  $\text{Sys\_B}^{\text{elab}}$  boundary chain. Together with clause (2), this makes the drain set finite and non-replenishing.

(7) Sync-boundary capacity withdrawal — elaborated-boundary interpretation.

While a process is pending at an explicit synchronization call  $s$ , producer-facing availability for any back-annotated capacity belonging only to  $\text{suff\_S}(s)$  is withheld until  $s$  commits; work already accepted before  $s$  may still drain. For the  $\text{Sys\_ref}$  proof model, this withdrawal is not an additional hidden enablement predicate on an otherwise ordinary E4 channel. It is represented only by choosing  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$

with explicit sync-boundary / epoch-gate abstract channels, or equivalent explicit staged realization structure, so that the refusal appears as ordinary ChannelDisabled behavior at an explicit abstract boundary.

For a pipelined or otherwise overlapped implementation, Appendix B’s automatic-flush rules are the implementation-side construction that supplies Definition Core.16. For ordinary non-overlapped control, the same definition may be discharged by a structural proof that no new source-later dynamic operation can be launched after the blocked point is reached. Thus the K.2b dominance/extraction premise ranges over every liberal interval that can contain eager issue, not only loop pipelines. In particular, eager issue permitted by Policy L must occur in Core.Phase’s L1 pre-settle window; the no-new-launch obligation begins at the L2 settled snapshot, not at the earlier issue of the frontier request.

### Definition Core.Drain (activated drain-discipline qualification).

For a fixed liveness instance  $I$  and a Policy-L path or prefix  $p$ ,  $\text{Core.Drain}_I(p)$  holds when every Core.Phase L2 SettlePoint at which Definition Core.16 activates obeys all seven Core.16 clauses for as long as the same fully disabled frontier persists. The qualification is bound to the fixed instance, witness, environment, arbitration, reset, elaboration, and boundary attribution; its finite-drain clause is interpreted over the selected B-faithful explicit capacities or credits. For Appendix K applications, the same prefix/continuation must additionally satisfy  $\text{Core.SettleKBridge}_I$  so that an L2 activation is never silently treated as the later global K-facing comparison state. Core.Drain is the Core-level continuation qualification consumed by Core.LAdm. Later theorem layers may require design- or cutpoint-specific evidence that discharges Core.Drain and Core.SettleKBridge, but that evidence does not redefine either qualification.

### Remark Core.16a (ALL disabled $\Rightarrow$ stall).

The document’s intended policy is exactly the “ALL disabled  $\Rightarrow$  stall” interpretation: the process stalls only when none of the frontier items in  $\text{Front}_P(\sigma)$  is channel-enabled. Downstream drain is compatible with that stall provided no new later work is issued past the blocked frontier.

### Definition Core.LAdm (admissible Policy-L prefixes and complete executions).

For Appendix K, a finite Policy-L prefix is admissible under Core.LAdm iff it is Policy-L-legal, follows L0–L4, respects nonwithdrawal and  $K\epsilon$ , satisfies Core.Drain from every activation point reached in the prefix, and satisfies Core.SettleKBridge for every K-facing cutpoint selected to evaluate a persisting blocked frontier. Write  $\text{Pref}_L^{\text{adm}}(I)$  for these finite prefixes. A complete Policy-L-admissible execution follows Core.CompleteCycle for every real cycle and is either infinite or ends at an explicitly DeclaredTerminal state; it additionally satisfies the selected B2/B3/B5 progress/quiescence obligations and Core.IssueFair for Policy L. A state is not finite-maximal merely because no  $\Sigma$ -action or  $\epsilon$ -step is enabled at the present cycle: if it is nonterminal, Core.CycleClosure/CompleteCycle continues it, possibly through a SilentCycle. If a complete execution is extended indefinitely by the B6 convention, the terminal state must also satisfy Core.TerminalIdle (including its Core.EnvTerminal component). Write  $\text{Exec}_L^{\text{adm}}(I)$  for this set. A5-L’s reachability clause quantifies over the source K-facing cutpoints  $\text{KCutReach\_ref}(I)$  reached by members of  $\text{Pref}_L^{\text{adm}}(I)$ , not over L2 SettlePoints merely because they are combinationally settled. K.2b.Q’s continuation, stabilization, and limit arguments additionally require the existence of a member of  $\text{Exec}_L^{\text{adm}}(I)$  extending the finite counterexample prefix. Lemma K.2b.E constructs such an extension from its stated premises, using Core.FairPick and Lemma Core.FairCycleDovetail to turn the existing Core.IssueFair/B2/B3 discharges into typed per-cycle choices; the extension is not a theorem-application certificate field or an additional witness-package assumption.

Fairness is therefore not decided by a finite prefix, but a finite prefix that has already violated the phase, freeze, or settle-to-K-cutpoint bridge rules is not admissible.

Normative-use convention. Elsewhere in this document, “Policy-L execution” means a Policy-L-legal path unless the context says otherwise. “Admissible Policy-L execution/prefix” in A5-L, K.1A, K.2b, K.Ser, the K.CE examples, and Appendix K-P means Core.LAdm-admissible. Core.Drain is therefore not silently part of the basic Policy-L issue rule; it is an explicit qualification of liveness admissibility. Policy S uses the same L0–L4 cycle phases, with its stricter serial-blocking issue condition substituted for the Policy-L issue permission in L1.

## Observable compatibility and cutpoint correspondence

### Definition Core.17 (System-level observable compatibility).

For a witness prefix  $\tau_{\text{ref}}$  of Sys\_ref and a post-HLS prefix  $\tau_{\text{post}}$  of RTL\_B, define  $\Sigma\text{Match}(\tau_{\text{ref}}, \tau_{\text{post}})$  to mean that the selected  $\Sigma$ -visible events/observable units match under the canonical occurrence convention of this theorem stack. For each message-passing channel  $c$ , the projected sequences of committed Push\_c events and of committed Pop\_c events have the same length; the  $k$ -th Push\_c on one side matches the  $k$ -th Push\_c on the other, and likewise for Pop\_c, with the same endpoint and payload/value label. For each observable signal sig of a sequential process P, the canonicalized sequences of Read(sig, ·) and Write(sig, ·) occurrences have the same length and the  $k$ -th occurrences match with the same process/signal/direction and value label; within each source anchor interval, canonicalization records at most one  $\Sigma$ -visible Read and at most one  $\Sigma$ -visible Write for each (P, sig) at the E2 anchor, while redundant physical sampling or re-driving is  $\epsilon$ -internal. For each R2C combinational process or explicitly modeled combinational network component C\_comb, the emitted fixed-point observation-unit sequences have the same length and the  $k$ -th units match by component identity, included process/signal/direction labels, and fixed-point values, including any cycle-locked participation requirement. For proof-internal RESET(S\_p, S\_c) units included in the selected theorem comparison, project each trace to the RESET occurrences of each reset scope (S\_p, S\_c); the  $k$ -th RESET occurrence for a scope on one side matches the  $k$ -th RESET occurrence for the same scope on the other, so matched RESET units have identical S\_p and S\_c. Independent disjoint RESET units may commute except where the reset well-formedness rules or another selected observable-order constraint imposes an order. Synchronization events match by their per-process occurrence index in source order. Independent matched  $\Sigma$  events may commute or occupy different global cycle buckets except where E1–E5 or the selected observable-unit semantics impose an ordering or same-cycle constraint. An unmatched selected  $\Sigma$  event/unit on either side makes  $\Sigma\text{Match}$  false. Define

$$\tau_{\text{ref}} \approx \tau_{\text{post}} :\Leftrightarrow \Sigma\text{Match}(\tau_{\text{ref}}, \tau_{\text{post}}) \wedge E1 \wedge E2 \wedge E3 \wedge E4 \wedge E5$$

This is an ordered compatibility predicate, not a mathematical equivalence relation. The left argument supplies the Sys\_ref witness prefix, and the right argument supplies the RTL\_B post-HLS prefix.  $\Sigma\text{Match}$  and E1–E5 are normative conjuncts of the Core definition. Appendix G expands the canonical  $\Sigma$ -event matching convention and the E1–E5 rules but does not redefine this relation or add a separate matching conjunct. In particular, the R2C matching statement in E2 below is the Appendix-G expansion of the R2C clause already contained in  $\Sigma\text{Match}$ , not a second R2C matching requirement.

### Definition Core.18 (Cross-model KObs correspondence).

For an exact selected source/RTL package with common typed record  $\langle E, w, H, O \rangle$ , cross-model KObs correspondence is equality of the source K-observation record with a post-side K-observation record of that same Core.18c type. On the source side,  $\text{KObs}_{E, w}(\sigma_{\text{ref}}) = \langle E, w, H, O \rangle$ . The Core fixes only the type and equality notion of this correspondence; it does not independently define how an RTL\_B cutpoint yields such a record. Appendix J supplies that post-side representation as  $\text{KObsPost}_J(p_{\text{post}}; E, w, H, O)$

and, for the selected package of Corollary J.1, proves by  $J.KObsConf$  that  $KObsPost\_J(\rho\_post; E, w, H, O) = KObs\_E, w(\sigma\_ref)$ . This definition introduces no additional state clauses. Process, channel, frontier, request, enablement, WFG, and snapshot-drain facts are obtained by applying Definitions Core.18d–Core.18e to the common record. Appendix K separately defines and factors its deadlock, waiver, environment-terminal, and diagnostic predicates from those reconstructed inputs.

## K-observation quotient and residual-offer layer

Status of this subsection. Definitions Core.18a–Core.18e provide the source-side  $KObs$  record and its pure Core reconstruction functions. Appendix J proves the source-side reconstruction lemmas  $J.KObsProc$ ,  $J.KObsChan$ ,  $J.KObsOffers$ , and  $J.KObsWFG$ , constructs the corresponding J-facing RTL record, and uses  $J.OfferReach/J.OfferConf/J.KObsConf$  to prove equal  $KObs$  at matched cutpoints. Appendix K then defines and proves the deadlock-, waiver-, and diagnostic-predicate factoring results over that record. This subsection does not characterize the image of arbitrary well-typed records beyond the explicit  $J.OfferReach$  predicate, and it does not alter the operational Issue, Policy-L/Policy-S, or DrainDiscipline semantics.

### Definition Core.18a (Enriched replay history H).

Fix a theorem instance  $I$  consisting of the design, fixed stimulus, declared environment model, reset policy, chosen observation boundaries, and  $KObs$ -facing waiver set  $W$ . For the Policy-S A5-L discharge,  $I$  may additionally reference a response-monitor schema, finite-bound function, reset/end-of-test disposition, and response-waiver policy; those are operational certificate data and are not fields of  $KObs$ .  $I$  also records an environment-reconstruction mode for optional downstream diagnostics: either  $StandardEnv$ , meaning the normative B1-avail pull-based, consumption-ordered input model with default always-ready outputs, or  $EnvReplayAdequate$ , meaning a certificate that supplies a typed reconstructed environment state  $EnvStateRec\_I(H, w)$  and proves that permanent next-interaction availability is determined by  $H$ ,  $w$ , and that reconstructed state. The exact environment-terminal and diagnostic predicates are defined in Appendix K. Fix the full elaboration package  $E$  and witness  $w$ . For a reachable source  $K$ -facing cutpoint  $\sigma \in KCutReach\_ref(I)$ , let  $H\_E, w(\sigma)$  be the finite canonical annotated replay history through  $\sigma$ . It contains the observable units and annotations needed by Theorem J.1 and  $I.DR/I.DR'$ : committed  $Push\_c(v)/Pop\_c(v)$  events with dynamic identities and per-channel ordinals; synchronization and wait units; anchored Read/Write units;  $START\_P$ ,  $FINISH\_P$ , and  $RESET(S\_p, S\_c)$  units where applicable; and the  $WC$ /witness-selected non-blocking, arbitration, signal, and  $\epsilon$ -choice outcomes required to replay the selected execution.  $H$  retains explicitly atomic unit grouping, per-process replay order, per-channel FIFO order, reset-epoch structure, and the required-prefix constraints used by  $\leq\_J$ . It quotients away incidental model-specific clock-bucket placement and ordering among independent units not required by those constraints. Raw source or RTL cycle buckets may be retained as certificate metadata, but they are not fields compared by  $KObs$  equality.  $H\_E, w(\sigma)$  contains no historical  $Issue(q)$  timestamps merely because requests were asserted before commit.

### Definition Core.18b (Attributed residual offer and residual-offer set).

At a reachable source  $K$ -facing cutpoint  $\sigma \in KCutReach\_ref(I)$ , an attributed residual offer is a tuple  $o = (P, x, q, c, dir, e)$  satisfying all of the following:  $P$  owns  $x$ ;  $x \in \Omega\_msg, P$ ;  $q = MsgOf\_P(x)$ ;  $c = BoundaryOf\_P(x)$ ;  $q$  is a blocking  $Push$  or  $Pop$  instance that has issued but not committed in reset epoch  $e$ ;  $dir$  identifies the endpoint-owned producer-valid or consumer-ready direction; and  $BoundaryReq(x, \sigma)$  is asserted and attributed to that same  $x$ ,  $q$ ,  $c$ , direction, and epoch. Define the derived operation-kind projection  $kind(o) := kind(q)$ , with the typing invariant  $dir = \text{producer-valid}$  iff  $kind(q) = Push$  and  $dir = \text{consumer-ready}$  iff  $kind(q) = Pop$ . Let  $ResOffers\_E, w(\sigma)$  be exactly the set of such tuples.

$\text{ResOffers\_E,w}(\sigma)$  is well typed and issue-prefix consistent under the existing E3/Core.IC semantics. On reachable source cutpoints,  $\text{Core.SyncFence}$  additionally ensures that no residual blocking offer owned by  $P$  belongs to a source interval earlier than  $P$ 's current interval: such an offer would have had to process-facing commit before the intervening synchronization could commit. In particular: (a) its item and boundary fields are supplied by  $E$ ; (b) if a source-later residual offer exists while an earlier blocking item in the applicable source prefix is also issued, uncommitted, and still requesting at an evaluated boundary, that earlier offer is also represented; (c) reset clears all offers in the ending epoch under RW2; and (d) for each explicit abstract boundary and endpoint direction there is at most one attributed outstanding source-level offer. A concrete token or transfer already accounted for as explicit channel occupancy is not additionally represented as an independent source-level residual offer unless  $E$  declares a distinct source-attributed frontier item at that boundary.

#### Definition Core.18c (K-observation KObs).

For the fixed instance  $I$ , elaboration  $E$ , witness  $w$ , and reachable source  $K$ -facing cutpoint  $\sigma \in \text{KCutReach\_ref}(I)$ , define the  $K$ -observation record by  $\text{KObs\_E,w}(\sigma) := \langle E, w, H\_E,w(\sigma), \text{ResOffers\_E,w}(\sigma) \rangle$ .

The fixed stimulus, environment model and reconstruction mode, reset policy, observation-boundary selection, and  $\text{KObs}$ -facing waiver set  $W$  are ambient parameters of the theorem instance  $I$  rather than duplicated in every record.  $K.\text{Resp}$  configuration and monitor ages are likewise ambient or CF-S operational data, not  $\text{KObs}$  fields. Equality of  $\text{KObs}$  records means typed equality of  $E$  and  $w$  and equality of the canonical replay-history quotient and residual-offer components. It does not require equality of the source and RTL clock-bucket decompositions permitted to differ by Theorem J.1. It is not equality of arbitrary process, scheduler, channel-realization, environment, response-monitor, or RTL microstate.

#### Definition Core.18c-1 (Observational waiver closure).

For fixed  $I, E, w$ , define  $\text{KObsClosed\_I,E,w}(W)$  to mean: for all  $\sigma_1, \sigma_2 \in \text{KCutReach\_ref}(I)$  of the same fixed instance,  $\text{KObs\_E,w}(\sigma_1) = \text{KObs\_E,w}(\sigma_2)$  implies  $(\sigma_1 \in W \text{ iff } \sigma_2 \in W)$ . This is an observational-closure premise for any waiver-parameterized cutpoint predicate later claimed to be determined by  $\text{KObs}$ ; it is not a claim that every well-typed  $\text{KObs}$  record is reachable. Environment-terminal adequacy and the optional diagnostic reconstructions are defined in Appendix K, where the relevant terminal predicates are introduced.

#### Definition Core.18d (Process and channel reconstruction from KObs).

For  $K = \langle E, w, H, O \rangle$ , define  $\text{ProcRec\_P}(K)$  to be the replay-derived process slice reconstructed by deterministic replay of  $P$ 's projection of  $H$  under  $w$ : the source control location and reset epoch, source-visible local values needed by future predicates and payloads, completed/realized-item status, candidate  $\text{NextFront}$ , branch/guard outcomes, operation identities and per-channel ordinals, returned values, anchored signal facts, and source-level completion status.  $\text{ProcRec}$  does not contain the current asserted blocking-request inventory;  $\text{PendingRec}$ ,  $\text{BoundaryReqRec}$ , and  $\text{FrontRec}$  are reconstructed separately from  $O$ . Define  $\text{ChanRec\_c}(K)$  for every explicit abstract channel  $c$  declared by  $E$  to be the E4 abstract state reconstructed from its reset/initial state and the committed Push/Pop history in  $H$ : for  $B(c) > 0$ , occupancy, empty/full status, and logical ordered contents/head information; for  $B(c) = 0$ , capacity-zero occupancy together with endpoint-request information supplied separately by  $O$ . For an outer channel represented by  $\text{Sys\_B}^{\text{elab}}$ , projected outer occupancy is obtained through  $\Pi_{\text{elab}}$ , while every proof-internal boundary used by Appendix K retains its own explicit  $\text{ChanRec}$  state.

#### Definition Core.18e (K-relevant derived functions).

For  $K = \langle E, w, H, O \rangle$  define the following typed functions.  $\text{DoneRec\_P}(K)$ ,  $\text{RemainRec\_P}(K)$ , and  $\text{NextFrontRec\_P}(K)$  are computed from  $\Omega\_P / \leq\_P$  supplied by  $E$  and the realized-item history

reconstructed from  $H$ .  $\text{PendingRec\_P}(K)$  is the set of frontier items  $x$  appearing in  $O$  for  $P$ . For records extracted from reachable source cutpoints,  $\text{Core.SyncFence}$  guarantees that every such outstanding blocking offer belongs to  $P$ 's current synchronization interval, so this unqualified offer projection is extensionally the interval-scoped  $\text{Pending\_P}$  set.  $\text{BoundaryReqRec}(x, K)$  holds exactly when  $O$  contains the attributed offer for  $x$ .  $\text{FrontRec\_P}(K) := \min_{\leq P}(\text{PendingRec\_P}(K))$ .  $\text{ChannelEnabledRec}(x, K)$  and  $\text{ChannelDisabledRec}(x, K)$  are computed at  $\text{BoundaryOf\_P}(x)$  from  $\text{ChanRec}$  and the complementary endpoint offers in  $O$  using Definitions  $\text{Core.11}$ – $\text{Core.13}$ .  $\text{WFGRec}(K)$  is obtained by applying Definition  $\text{Core.WFG}$  to  $\text{FrontRec}$  and  $\text{ChannelDisabledRec}$ .  $\text{DrainClosedNowRec}(D, K)$  holds exactly when no already-offered non-frontier blocking item owned by a process  $P \in D$  is  $\text{ChannelEnabledRec}$  at the cutpoint. Appendix K defines  $\text{DeadKRec}$ , waiver-parameterized deadlock reconstruction, environment-terminal reconstruction,  $\text{WFG+}$ ,  $\text{Dead\_KS}$ , and response-monitor predicates from these Core functions and its own  $K$ -specific definitions; none of those later predicates is defined by the Core record alone. At this stage the Core functions are interpreted on records extracted from reachable source cutpoints; no image characterization is asserted.

Historical issue-time quotient. The record deliberately retains whether an operation survives as a current attributed offer, but not the earlier cycle in which that request first became asserted. Therefore two reachable source cutpoints with the same  $E$ ,  $w$ ,  $H$ , and  $O$  have the same  $\text{KOb}$ s even if their operational executions used different legal historical issue times. This is only an observation equivalence at the selected cutpoint; it does not assert that the two executions have the same admissible continuations or satisfy the same transition-level  $\text{DrainDiscipline}$  for reasons independent of the common cutpoint.

## Shared-memory and reset side conditions

**Array/synchronization commit-order conformance ( $\text{Core.MemSyncOrder}$ ).**  $\text{Core.MemSyncOrder}$  holds when the selected source/HLS/mapping-layer schedules satisfy the commit-time fence stated in “Scheduling of Array Accesses”: for every dynamic array or memory access  $a$  and every synchronization call  $s$  selected into  $S$  that is ordered with  $a$  in the process's dynamic source execution, every mapped storage commit representing  $a$  occurs no later than the corresponding commit of  $s$  when  $a$  is source-before  $s$ , and occurs strictly after  $s$  when  $a$  is source-after  $s$ . Any allowed cache/merge/split/reorder transformation shall carry mapping-layer evidence that preserves those synchronization fences. This is a local scheduling-conformance obligation whether the storage is process-internal or externally mapped. It does not add memory read/write events to  $\Sigma$  and does not extend the Theorem J.1 replay state. Accordingly,  $\text{Core.MemSyncOrder}$  is a separate scheduling-contract/tool-conformance obligation recorded in Appendix M.7a; it is not a premise consumed by Appendix I replay or Theorem J.1. For an application of J.1 with cross-process shared-memory value flow,  $\text{J.Mem}$  is separately required; the non-lowered direct memory-state extension remains outside J.1 as stated below.

**Shared-memory lowering condition ( $\text{J.Mem}$ ).** The proof-internal observable alphabet  $\Sigma$  of Appendices I–K contains committed channel transfers, synchronization events, and signal Read/Write actions; it does not contain mapped memory read/write events, and the Theorem J.1 invariant carries per-process replay state and per-channel commit histories but no global shared-memory state. Accordingly, the formal results of Appendices I–K apply to cross-process shared-memory communication only under the following lowering condition, named  $\text{J.Mem}$ : every shared-memory dependency through which one process's memory write can affect another process's control flow, payload/value expressions, or the identity or order of its later operations shall be lowered by the array-access mapping layer onto operations already covered by the proof alphabet — explicit point-to-point message-passing channels governed by E4 (memory command/response channels), and/or  $\Sigma$ -visible anchored signal IO governed by E2 with the producing Writes observable — so that every memory-carried value is carried by a matched, value-labeled  $\Sigma$  event. Synchronization alone (E1) fixes access order but does not, within the

present invariant, establish that the two models contain the same memory value; J.Mem closes that gap by making the value flow  $\Sigma$ -visible. Designs that instead rely on the non-lowered “declared memory semantics” branch of the shared-memory transformation entry in Appendix G are outside Theorem J.1, Corollary J.1, and Appendix K as proved here; for them, agreement under the declared memory semantics is a separate verification obligation (see the informative note below). J.Mem is a named entry in the side-condition discharge record. K.CV’s “declared memory serialization” clause remains a liveness-side premise and does not substitute for J.Mem in the safety proof.

**Informative note (direct memory-state extension, not developed here).** The alternative repair — adding mapped memory Read(addr, val)/Write(addr, val) events to the proof-internal  $\Sigma$ , extending the required-order relation  $\prec_J$  with the mapping layer’s declared serialization constraints, and adding a new global memory-state agreement clause to the Theorem J.1 invariant together with a memory-state component in the enriched replay/KObs reconstruction and the corresponding J.OfferReach/J.OfferConf interface — is compatible with the present proof architecture but is not developed in this document. A flow that requires direct coverage of non-lowered shared memory shall supply that extension and re-verify Lemmas I.DR/I.DR’, J.HCanon, and J.KObs-Proc–J.KObs-WFG and the corresponding Appendix K deadlock-factorization result, together with Corollary J.1, with the added clause.

**Reset-boundary observable unit (RESET) and reset well-formedness (RW1–RW5).** When dynamic reset is used, each dynamic reset occurrence is represented in the proof-internal event model as an explicitly atomic observable unit RESET( $S_p, S_c$ ): a proof-internal boundary marker whose scope names the set  $S_p$  of processes and the set  $S_c$  of explicit abstract channels affected by that reset occurrence. RESET units may be projected away from the outer  $\Sigma_{DUT}$  alphabet, like other proof-internal units. The unit is ordered strictly after every observable unit of the ending epoch owned by the affected scope and strictly before every observable unit of the new epoch of that scope, and the first observable action of each  $P \in S_p$  in the new epoch is its fresh dynamic START\_P. The state-changing reset transition itself — which occurs before the new epoch’s START\_P — is exactly what the RESET unit represents. In Core.CycleResult that dynamic occurrence is carried in resetUnits of the enclosing real cycle rather than replacing the cycle with a reset outcome; it therefore retains its own correspondence obligation (Lemma J.RS in Appendix J) rather than being folded into START\_P matching. The following reset well-formedness conditions are implementation-side obligations for designs using dynamic reset:

**RW1 (channel-consistent scope).** For every explicit abstract channel  $c$ , either both endpoint processes of  $c$  are in  $S_p$  and  $c \in S_c$  — in which case  $c$  re-initializes to reset-empty occupancy under the Core reset-empty FIFO convention at the boundary and any in-flight (accepted but not yet delivered) contents are discarded consistently on both sides in both models — or neither endpoint is in  $S_p$  and  $c \notin S_c$ , in which case  $c$ ’s state and history are unaffected. A reset that would affect exactly one endpoint of a live channel is permitted only if the design models the cross-epoch disposition explicitly (for example, an explicit flush/abort protocol on that channel); that protocol’s events are ordinary  $\Sigma$  events, and the channel is treated per this clause after the protocol completes.

**RW2 (request disposition).** Every outstanding endpoint-owned boundary request of a process in  $S_p$  is cleared at the RESET unit; its owning dynamic instance does not commit, is not re-attributed across the boundary, and any post-reset re-execution is a fresh dynamic instance in the new epoch’s universe. The Appendix C non-withdrawal discipline is scoped per reset epoch.

**RW3 (corresponding reset transitions).** At the RESET unit, RTL\_B’s post-reset state at the chosen abstract boundary corresponds to Sys\_ref’s prescribed epoch-initial state: control at epoch-initial locations for every  $P \in S_p$ , source-visible values at their reset values, reset-empty occupancy on every  $c \in S_c$ , no stale synchronization state, no unaccounted pending endpoint request, no hidden accepted message absent from the source reference model, and direct-input signals obeying their per-epoch stability or sync-controlled contracts (main text, Direct Inputs).

**RW4 (atomic boundary).** The reset state transition is not interleaved with same-cycle observable actions of the affected scope. If the implementation asserts reset in a cycle in which affected-scope observable actions also commit, those actions belong to the ending epoch and precede the matching RESET occurrence in `Core.CycleResult.resetUnits` in the required order; the two reset epochs of the affected scope share no observable cycle bucket ambiguity at the boundary. The enclosing `CycleResult` may still carry unaffected-scope progress because RW5 frames state outside the reset scope.

**RW5 (locality).** The RESET unit changes no state of processes outside `S_p` and no history or occupancy of channels outside `S_c`. Cross-scope effects, if any, must be expressed by ordinary modeled events.

## Additional Semantic Assumptions (B-rules)

The following B rules are process assumptions used within the formal proofs. These B rules are in addition to the R rules presented in Appendix G.

| ID                                             | Basic-process property (informal statement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Why it is needed / how it is used                                                                                                                                                 |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B1<br/>Deterministic<br/>Trace Property</b> | <p>For any fixed test stimulus and initial state, the sequence of observable actions emitted by the post-HLS design is unique, including the channel identity and payload/value of each committed <code>Push_c(v)/Pop_c(v)</code> and the value of each <code>Write(sig,val)/Read(sig,val)</code>. Internal <math>\epsilon</math>-steps may differ between runs, but the externally visible <math>\Sigma</math>-trace (with labels) cannot.</p> <p><b>Clarification:</b> B1 is assumed of <b>RTL_B</b> (post-HLS) only. The source-level models <b>Sys/Sys_B</b> may admit multiple <math>\epsilon</math>-/latency-different executions with the same <math>\Sigma</math>-projection. When a single concrete witness execution of <code>Sys_B</code> is required for simulation/debug, Appendix L's snooping wrapper may be used to <b>select</b> one admissible witness whose latency-sensitive events align with the unique RTL <math>\Sigma</math>-trace.</p> <p>Derived per-process consequence. For every process P, the projection of the unique RTL_B <math>\Sigma</math>-trace onto the observable units owned by or attributed to P is also unique. This follows from the system-level B1 property; it does not assert that P is deterministic when considered in isolation under arbitrary</p> | Ensures the per-process and system-level trace-compatibility proofs (App. I & J) can match one post-HLS trace to one pre-HLS trace without branching on scheduler nondeterminism. |

| ID                                            | Basic-process property (informal statement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Why it is needed / how it is used                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | peer, channel, arbitration, or environment behavior.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                             |
| <b>B2 Weak Fairness of the Scheduler (WF)</b> | <p>If an action's enabling predicate remains continuously true from cycle <math>t</math> onward, the scheduler/arbitrator must eventually select that action, and the selected action then commits in the selected cycle, within a finite but unspecified number of cycles. Applies to Push, Pop, and Sync operations.</p> <p>For message-passing operations, "enabled" means ChannelEnabled under E4 at the chosen abstract boundary. Enablement is eligibility for selection; it is not itself a same-cycle commit. Thus B2 is not vacuous: an enabled Push/Pop may be delayed by fair arbitration, but it may not be delayed forever while it remains continuously enabled.</p>                                                                                                                                                                                                                    | Required for all progress arguments, especially the liveness lemmas in Appendix K and the channel-progress corollaries.                                                                                                                                                                                                                                                     |
| <b>B3 Channel Progress Invariants</b>         | <p>B3 is exactly the following family of per-channel progress obligations. For every explicit abstract channel <math>c</math> with capacity <math>B(c)</math>, B3 consists of both <math>P\_push(c)</math> and <math>P\_pop(c)</math>, as defined below. Thus references such as "B3," "Appendix N's B3," and "the <math>P\_push/P\_pop</math> obligations" name the same channel-progress assumption; <math>P\_push(c)</math> and <math>P\_pop(c)</math> are not additional assumptions separate from B3.</p> <p>B3 is stated separately from B2 because the proofs often reason from the perspective of a currently stalled request. A blocked <math>Push\_c</math> at <math>occ\_c = B(c)</math>, a blocked <math>Pop\_c</math> at <math>occ\_c = 0</math>, or a rendezvous endpoint waiting for its peer is not itself able to complete until the relevant complementary discharge occurs. B3</p> | Required by complete/fair progress arguments, principally Appendix K, to rule out permanent back-pressure loops in which both endpoints continuously request a transfer until the matching discharge event but the matching discharge never occurs; such progress is logically independent of the cycle-by-cycle R-rules. It is not a premise of finite-prefix Theorem J.1. |

| ID | Basic-process property (informal statement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Why it is needed / how it is used |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|    | <p>packages that discharge fact as a named per-channel</p> <p>progress obligation: under the persistent-request premises below, the full/empty/rendezvous-blocked condition is eventually discharged, after which ordinary E4 availability and B2 can be used</p> <p>for any still-pending enabled transfer.</p> <p>For a single explicit abstract boundary with ordinary E4 buffered or rendezvous semantics, the corresponding B3 discharge is implied by B2 plus E4 once the complementary endpoint request in the B3 premise remains continuously asserted. In the buffered full case, <math>occ\_c = B(c) &gt; 0</math> makes the complementary Pop_c continuously ChannelEnabled; B2 then forces eventual selection and commit of that Pop_c, freeing space. In the buffered empty case, <math>occ\_c = 0 &lt; B(c)</math> makes the complementary Push_c continuously ChannelEnabled; B2 then forces eventual selection and commit of that Push_c, providing data. In the rendezvous case, the two persistent endpoint requests make the rendezvous pair continuously ChannelEnabled, so B2 similarly supplies eventual rendezvous selection and commit.</p> <p>B3 is nevertheless retained as an explicit named obligation because Lemma I.2, Appendix K, and the channel-validation methodology consume that modular form directly. It is also the form checked per explicit abstract channel in elaborated <math>Sys\_B^{elab}</math> boundary chains. Thus</p> <p>B3 should be read as the modular</p> |                                   |

| ID | Basic-process property (informal statement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Why it is needed / how it is used |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|    | <p>channel-progress assumption used by the proof and checked against channel realizations, not as a claim that the ordinary single-boundary buffered clauses are logically independent of B2+E4.</p> <p>In the definitions below, “persistent,” “continuously requests,” and “remains true” are interval-scoped. The relevant endpoint request must remain asserted from the starting cycle <math>t_0</math> through the first matching discharge event, if such a discharge occurs. If no matching discharge occurs, the request must remain asserted for all later cycles. A Push request also keeps its payload stable throughout the outstanding interval. B3 does not require an endpoint request or Push payload to remain asserted forever after the matching transfer has already committed.</p> <ul style="list-style-type: none"> <li>• <math>P\_push(c)</math>: If a process <math>P</math> is stalled at a blocking <math>Push(c)</math> with its boundary request true and a persistent request asserted with stable payload while the channel is full (<math>occ(c) = B(c)</math>), and the complementary endpoint process continuously requests the matching <math>Pop(c)</math> until the matching discharge event (<math>Pop(c)</math> endpoint request asserted; <math>Pop(c)</math> boundary request remains true), then the matching full/rendezvous discharge must eventually occur. In particular, for <math>B(c) &gt; 0</math>, some <math>Pop(c)</math> eventually commits, freeing space; for <math>B(c) = 0</math>, the rendezvous completion eventually occurs. In the buffered case this conclusion is the complementary <math>Pop(c)</math> commit, not a direct assertion that the original stalled <math>Push(c)</math> commits; E4 and B2 govern that still-pending</li> </ul> |                                   |

| ID                                  | Basic-process property (informal statement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Why it is needed / how it is used                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Push<sub>(c)</sub> after space is freed.</p> <ul style="list-style-type: none"> <li>• P<sub>pop(c)</sub>: If a process P is stalled at a blocking Pop<sub>(c)</sub> with its boundary request true and a persistent request asserted while the channel is empty (occ<sub>(c)</sub> = 0), and the complementary endpoint process continuously requests the matching Push<sub>(c)</sub> until the matching discharge event (Push<sub>(c)</sub> endpoint request asserted with stable payload; Push<sub>(c)</sub> boundary request remains true), then the matching empty/rendezvous discharge must eventually occur. In particular, for B(c)&gt;0, some Push<sub>(c)</sub> eventually commits, providing data; for B(c)=0, the rendezvous completion eventually occurs. In the buffered case this conclusion is the complementary Push<sub>(c)</sub> commit, not a direct assertion that the original stalled Pop<sub>(c)</sub> commits; E4 and B2 govern that still-pending Pop<sub>(c)</sub> after data is provided.</li> </ul> |                                                                                                                                                                                                                                                                                                                                               |
| <b>B4 Finite Stutter Bound</b>      | <p>Every process P has a finite constant depth D<sub>P</sub> such that whenever P is not externally stalled (i.e., P is advancing internal micro-state toward its next observable <math>\Sigma</math>-action, or its next <math>\Sigma</math>-action is locally enabled), P can execute at most D<sub>P</sub> consecutive <math>\epsilon</math>-steps before either (i) executing a <math>\Sigma</math>-action, or (ii) reaching a stable waiting state in which P has no further <math>\epsilon</math>-step available until some external/peer/environment condition changes the enabling predicates.</p>                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>This guarantees the <math>\epsilon</math>-stutter closure needed to construct witness mappings in Appendix I and to bound internal micro-latency (pipeline/FSM bookkeeping) by <math>\leq D_P</math>. Unbounded waiting due to back-pressure or missing peer stimulus is governed by WF/B2 and the progress obligations B3, not by B4.</p> |
| <b>B5 System Quiescence Closure</b> | <p>B5 carries a finite system-level cycle bound C<sub>sys</sub> for the selected bucket semantics. A standard discharge defines per-process cycle bounds C<sub>P</sub> and takes C<sub>sys</sub> = max<sub>P</sub> C<sub>P</sub>. Each C<sub>P</sub> may be supplied independently or derived from the B4 epsilon-step bound D<sub>P</sub> by a proved charging lemma showing that every relevant silent cycle in which P</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Needed to finish the starvation-vs-deadlock analysis in Appendix K: ensures an all-disabled state cannot hide behind an infinite tail of unobservable activity.</p>                                                                                                                                                                        |

| ID                                                  | Basic-process property (informal statement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Why it is needed / how it is used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <p>remains internally advancing consumes at least one P-owned epsilon-step; under that lemma <math>C_P \leq D_P</math>. If, at some cycle <math>t_0</math>, no observable actions are enabled in any process, and no external or environmental change from <math>t_0</math> through the closure interval changes the relevant enabling predicates, then within at most <math>C_{sys}</math> further cycles the system reaches an epsilon-fixed point. Thereafter, while no external or environmental change re-enables observable or internal work, the system executes no additional epsilon-steps.</p>                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>B6 Time-Divergence / Idle-Stutter Convention</b> | <p>After <code>Core.TerminalIdle</code> is established — a genuine B5/global fixed point together with proof that neither future registered/internal machine evolution nor any permitted future environment evolution can re-enable work — the global clock continues to advance. <code>Core.EnvTerminal</code> supplies the environment component of this obligation. We model each subsequent B6 idle clock tick as a stutter step that produces no <math>\Sigma</math>-event, performs no <math>\epsilon</math>-prefix or <math>\epsilon</math>-suffix work, and leaves all non-clock state unchanged. A present-instant observation that no action is enabled is not sufficient to establish this antecedent.</p> | <p>When a theorem explicitly represents a genuinely terminal-idle execution by an infinite B6 suffix, those idle ticks are treated as <math>\epsilon</math>-stutter for the applicable E1–E5 prefix/trace matching. Thus B6 gives a time-divergent infinite extension only for a reachable state already proved to satisfy <code>Core.TerminalIdle</code>. <code>Core.CycleClosure</code>'s environment-definedness clause or <code>Core.EnvTerminal</code> alone is insufficient: a total timed/reactive environment may legally deliver new stimulus at a later cycle.</p> <p>B6 is not an arbitrary deadlock-cutpoint or temporary-idle extension principle. Ordinary nonterminal cycles in which no <math>\Sigma</math> event commits are represented by <code>Core.SilentCycle/CompleteCycle</code> and may advance registered state, environment state, and cycle-valued monitors. A partial internal message-passing deadlock cutpoint in Appendix K need not be a global fixed point because outside-D activity, environment activity, or already-issued non-frontier drain work may still be possible. Such cutpoints are handled by the separate Appendix K complete-cycle/fair drain continuation; B6 is used only for a genuine terminal-idle suffix.</p> |

## Appendix G – Trace Compatibility Rules

This document informally states that a primary goal is to present “no surprises” to a DV engineer using the same testbench to verify the pre-HLS and post-HLS models.

Somewhat more formally, we can show how the scheduling rules within this document establish a set of trace-compatibility rules between the pre-HLS and post-HLS models that provide strong guarantees about their behaviors.

The *basic conceptual model* establishes the base behavior for a single process:

- 1) Synchronization interface calls always remain in source code order.
- 2) Signal reads occur at closest preceding synchronization interface call.
- 3) Signal writes occur at closest succeeding synchronization interface call.
- 4) All message-passing operations before a call to a synchronization interface shall be issued before or in the same cycle in which the synchronization interface commits.  
(For blocking message-passing operations (Push/Pop), this implies the corresponding commits occur no later than the synchronization call’s commit cycle.)
- 5) All message-passing operations after a call to a synchronization interface shall be issued after the cycle in which the synchronization interface call commits.
- 6) Two message-passing operations on separate interfaces which appear in sequence in the model are source-ordered. They may issue with timing overlap or in the same clock cycle where the selected issue policy allows, but they shall not be issued in the reverse sequence. Operations on the same message-passing interface are issued in source order (they are never reversed).
- 7) Synchronization calls can affect the scheduling of signal IO operations, and synchronization calls can affect the scheduling of message-passing calls, but message-passing calls cannot affect the scheduling of signal IO operations and vice-versa.
- 8) The STRICT\_IO\_SCHEDULING=relaxed option is not used.
- 9) We distinguish the rendezvous source model Sys (B(c)=0 for all channels) from the buffered source interpretation Sys\_B, which uses bounded-FIFO channel capacities B(c) matching the post-HLS RTL (Appendices I–K).

Conceptually, systems are composed of many such processes which follow the basic conceptual model, and which communicate using only synchronization interface calls, latency-insensitive signal-level protocols, and committed message-passing transfers. If a design uses non-blocking message-passing calls (PushNB/PopNB), then unsuccessful polls are treated as internal  $\epsilon$ -steps (no  $\Sigma$ -event), and each successful completion is represented in  $\Sigma$  as the corresponding committed Push\_c/Pop\_c transfer ( $\Sigma$  records transfers, not failed polls). If the resulting latency-sensitive behavior is externally visible at the DUT boundary, the snooping wrapper of Appendix L may be used to align the selected admissible pre-HLS execution with the RTL. Here we call this system of processes the *conceptual system*.

Practically, the modeling approach described above is too restrictive for real-world systems with optimized HW, so the scheduling rules in this document allow for key HW optimizations. But this is done in a carefully controlled manner, such that the HW optimizations do not break equivalent behavior with the *conceptual system*.

Here is a summary of the HW optimizations that the scheduling rules allow, and a brief description of how we maintain equivalent behavior with the *conceptual system*:

1. Pipelined loops: In the post-HLS RTL, the pipeline may overlap iterations to improve throughput. See Appendix B for a formal presentation of pipelined loops.
2. Pipelined loops: HLS by default will not break any latency-insensitive signal IO protocols when pipelining loops.
3. Direct inputs: Signals that do not change after a process comes out of reset can be read as late as possible using *hls\_direct\_input*, saving register area.
4. Direct inputs: When *hls\_direct\_input\_sync* is used, signals read by a process are updated at the precise point at which the HW pipeline is guaranteed to be ramped down, so the signal values do not need to be saved within pipeline registers, saving register area.
5. Memory accesses: HLS can reorder memory accesses within a process only if it can prove that the reordering is conflict-free.
6. Shared memories: Memories shared between processes are modeled as simple C arrays but always include explicit synchronization between processes in both the pre-HLS and post-HLS models. For the formal results of Appendices I–K, the value-carrying accesses must additionally satisfy the J.Mem lowering condition (Normative Formal Core).
7. Non-blocking message-passing interfaces: They are modeled at the level of committed transfers (successful completion), with unsuccessful polls treated as  $\epsilon$ -steps and thus not part of  $\Sigma$ . If latency differences in these committed-transfer events are externally visible at the DUT boundary and must be aligned for verification, equivalent behavior between pre-HLS and post-HLS systems can be maintained by snooping the post-HLS system and using this to delay pre-HLS message arrival (Appendix L).
8. Latency-sensitive global signal IO: If latency differences are externally visible to the DUT, equivalent behavior between pre-HLS and post-HLS systems can be maintained by snooping the post-HLS system and using this to delay pre-HLS signals.
9. Latency-sensitive local signal IO: Isolate local latency-sensitive protocols to small, dedicated transactors, and use latency-insensitive signal IO or message-passing to communicate with the rest of the system.
10. One-way signal handshake protocols: Only use in cases where backpressure is not possible and embed assertions into the model to check that this is always true.
11. Combinational processes: If the pre-HLS model contains combinational processes, their observable signal Read/Write behavior is interpreted at each cycle's  $\epsilon$ -quiescent combinational fixed point. Under the well-formedness requirement that the combinational network is deterministic and reaches a unique fixed point, the RTL/tool-flow conformance obligation is to preserve the fixed-point signal relation, so combinational signal IO can be included in  $\Sigma$  without using synchronization anchors.

Taken as a whole, these rules support a verification methodology in which full-system verification is performed primarily on the pre-HLS system model, and post-HLS RTL verification can be reduced to targeted checks that the formal side conditions are met. The precise guarantee is the execution-level guarantee of Appendix J: for every reachable RTL\_B prefix whose local certificate package satisfies Definition J.LocalGWR and whose selected source model satisfies the source-side Core.QSync premise required by J.GWR-Comp, Theorem J.GWR-Comp constructs a globally witness-realizable package and a matching Sys\_ref behavior (Post  $\rightarrow$  Ref). A directly validated J.GWR package carrying equivalent per-Horizon settling evidence is an alternate route. The reverse direction applies only to annotated RTL-consistent source-side prefixes or to source-side executions selected by the Appendix L snooping / witness-selection mechanism. Thus a source-side pass justifies the corresponding RTL\_B behavior only within that annotated RTL-consistent / witness-selected subset. For liveness/deadlock guarantees, the results additionally assume the Appendix K/N progress, fairness, drain, and A5 premises; the cycle-by-cycle R/E rules and the safety construction alone do not imply those liveness properties.

## Formalization of the Trace Compatibility Rules in Appendix G

The following notation is used in this section.

| Symbol     | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P          | A process (thread or method) in the design                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\Sigma$   | The alphabet of observable IO actions                                                                                                                                                                                                                                                                                                                                                                                                                         |
| $\tau_P$   | <b>A (finite or infinite) per-cycle trace of observable actions executed by process P:</b> for each global clock cycle $t$ , $\tau_P[t]$ is the (possibly empty) set of $\Sigma$ -actions committed by P at $t$ . Actions committed in the same cycle are treated as simultaneous (unordered); only cross-cycle order (and the explicit E1–E5 constraints) is semantically significant.                                                                       |
| S          | The subset of $\Sigma$ that are <b>synchronization calls</b> (explicit wait, SyncChannel, START_P/FINISH_P indicating process start/finish)                                                                                                                                                                                                                                                                                                                   |
| R          | The subset of $\Sigma$ that are <b>signal reads</b>                                                                                                                                                                                                                                                                                                                                                                                                           |
| W          | The subset of $\Sigma$ that are <b>signal writes</b>                                                                                                                                                                                                                                                                                                                                                                                                          |
| M          | The subset of $\Sigma$ that are <b>committed message-passing transfers</b> (i.e., committed Push_c / Pop_c events on message-passing channels, including successful non-blocking transfers). Unsuccessful non-blocking polls contribute no $\Sigma$ -event and are modeled as $\epsilon$ -steps, and therefore are not members of M.                                                                                                                          |
| Q_exec     | The execution-level set of dynamic source-level message-call instances executed by a process at the $\Sigma$ -visible endpoints (blocking Push/Pop, and non-blocking PushNB/PopNB calls). Elements of Q_exec may or may not commit. In particular, failed non-blocking polls are elements of Q_exec, have no committed $\Sigma$ -event, and are modeled as $\epsilon$ -steps. The frontier-item universe $\Omega_P$ , defined below, is separate from Q_exec. |
| $\Omega_P$ | The local dynamic frontier-item universe for process P in the selected execution / witness. It contains synchronization-call items, signal-action items, and                                                                                                                                                                                                                                                                                                  |

| Symbol                   | Meaning                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | message-operation frontier items; it is distinct from $Q_{\text{exec},P}$ , the universe of dynamic source-level message-call instances.                                                                                                                                                                                                                                                 |
| $\Omega_{\text{msg},P}$  | The message-operation slice of $\Omega_P$ : frontier items $x$ for which $\text{MsgOf}_P(x)$ is defined.                                                                                                                                                                                                                                                                                 |
| $\text{MsgOf}_P(x)$      | Partial projection from a message-operation frontier item $x \in \Omega_{\text{msg},P}$ to its owning dynamic source-level message-call instance $q \in Q_{\text{exec},P}$ . In $\text{Sys}_B^{\text{elab}}$ , $\text{MsgOf}_P$ need not be injective: one source-level $q$ may own several elaborated frontier items.                                                                   |
| $\text{BoundaryOf}_P(x)$ | The explicit abstract channel boundary at which frontier item $x$ is evaluated for $\text{BoundaryReq}$ , $\text{ChannelEnabled}$ , $\text{ChannelDisabled}$ , $\text{Front}_P$ , and Appendix K WFG reasoning. In ordinary unelaborated cases this boundary is unique for the owning source-level operation; in $\text{Sys}_B^{\text{elab}}$ it is declared by the elaboration package. |

A *system trace* is the tuple  
 $\tau = (\tau_P)_P$ , one component per process.

For any action  $a \in \Sigma$ , let  $\text{clk\_pre}(a) / \text{clk\_post}(a)$  be the clock cycle at which  $a$  is committed.

For any issued message-call instance  $q \in Q_{\text{exec}}$ ,  $\text{issue\_pre}(q) / \text{issue\_post}(q)$  denotes the auxiliary Issue timestamp for  $q$  in the pre-HLS / post-HLS trace, respectively, as defined in Appendix C and required to satisfy the associated Issue/Commit linkage (well-formedness) conditions with respect to the  $\Sigma$ -visible boundary request signal on  $q$ 's endpoint direction.

If  $q$  commits, it contributes a unique committed transfer  $m(q) \in M$  with commit cycle  $\text{clk\_pre}(m(q)) / \text{clk\_post}(m(q))$ . If  $q$  does not commit (e.g., an unsuccessful non-blocking call), then  $m(q)$  is undefined / absent and no  $\Sigma$ -event is added to  $M$ .

**( $Q_{\text{exec}}$ -matching / witness convention.)** Because unsuccessful non-blocking polls are  $\epsilon$ -steps and produce no  $\Sigma$ -event,  $Q_{\text{exec}}$  is not in general reconstructible from the  $\Sigma$ -projection alone. Whenever E3, E5, or any later proof compares executed message-call instances across  $\text{Sys\_ref}$  and  $\text{RTL}_B$ , the comparison is made relative to the chosen execution witness. That witness supplies a partial matching  $\mu_Q$  between dynamic source-level message-call instances in the compared traces, preserving process identity, endpoint/interface identity, dynamic source-order position, and source-level call-instance identity where applicable. If  $q$  commits, then  $\mu_Q(q)$  commits the corresponding matched transfer  $m(\mu_Q(q))$  with the same  $\Sigma$ -label/value under the event-matching convention. If  $q$  is an unsuccessful non-blocking poll, then  $q$  has no  $m(q)$  and no  $\Sigma$ -label; its presence and its match are supplied only by the witness-compatibility condition WC, by the sufficient NB-CF condition when applicable, or by the Appendix L snooping / witness-selection construction. Thus E3/E5 quantify over  $Q_{\text{exec}}$  only relative to this chosen witness; they do not require failed non-blocking polls to be inferable from  $\Sigma$  alone.

**(No deassert-before-commit assumption.)** For blocking message requests  $q \in Q_{\text{exec}}$ , once  $q$  is issued, its  $\Sigma$ -visible boundary request is not withdrawable within that reset epoch: it remains asserted in every cycle from  $\text{issue\_pre}(q)$  (resp.  $\text{issue\_post}(q)$ ) through  $q$ 's commit cycle if it commits, and for Push the payload remains stable while the request is outstanding. If an affecting RESET occurs first, RW2 ends the dynamic instance at the epoch boundary; no other nonterminal transition may withdraw or reattribute it.

**Notation convention (Issue/Commit functions).** The function-like forms  $\text{clk\_pre}/\text{clk\_post}$ ,  $\text{issue\_pre}/\text{issue\_post}$ , and  $m(q)$  abbreviate the unique values supplied when the corresponding Appendix C existence/uniqueness and Issue/Commit-linkage conditions hold. In particular,  $m(q)$  remains partial: a rule that requires a committed transfer must also require or entail that  $m(q)$  is defined. This notation does not replace the underlying relational Issue/Commit semantics.

### Dynamic source-order convention

The order relation used below is over dynamic source-level operation/frontier items in the compared execution / witness, not over all syntactic call sites in the source text.

For a process  $P$ , let  $Q_{\text{exec},P}$  denote the execution-level set of dynamic source-level message-call instances of  $P$  at the chosen abstract boundary.  $Q_{\text{exec},P}$  includes blocking Push/Pop calls, successful non-blocking PushNB/PopNB calls, and failed non-blocking PushNB/PopNB polls.

Separately, let  $\Omega_P$  denote the witness-specific dynamic universe of source-level frontier items that are actually reached under the fixed stimulus / selected witness and that are relevant to observable/frontier/deadlock reasoning. Items in  $\Omega_P$  include synchronization-call instances, anchored signal Read/Write action instances, and message-operation frontier items at the chosen abstract boundaries. A message-operation frontier item is either (i) a blocking Push/Pop item that has been reached, whether pending or committed, or (ii) a successful non-blocking PushNB/PopNB item that contributes a committed  $\text{Push}_c(v)$  /  $\text{Pop}_c(v)$  event. A failed non-blocking poll  $q$  is an element of  $Q_{\text{exec},P}$ , has no committed  $\Sigma$ -event  $m(q)$ , and is not an element of  $\Omega_P$  merely because it executed.

Because  $Q_{\text{exec},P}$  is a universe of dynamic message-call instances while  $\Omega_P$  is a universe of source-level frontier items, their message-operation overlap is not a literal set intersection. Let  $\text{MsgOf}_P(x)$  be the partial projection from a frontier item  $x \in \Omega_P$  to its owning dynamic message-call instance, defined exactly for message-operation frontier items and undefined for pure synchronization or signal-anchor items. Define the message-frontier slice

$$\Omega_{\text{msg},P} := \{ x \in \Omega_P \mid \exists q \in Q_{\text{exec},P} : \text{MsgOf}_P(x) = q \}.$$

Define the corresponding message-instance slice induced by  $\Omega_P$  as

$$Q_{\Omega,P} := \{ q \in Q_{\text{exec},P} \mid \exists x \in \Omega_{\text{msg},P} : \text{MsgOf}_P(x) = q \}.$$

Let  $\Sigma_P$  denote the corresponding observable labels/events contributed by items of  $\Omega_P$  when they are  $\Sigma$ -visible. For a message-call instance  $q \in Q_{\text{exec},P}$ , the committed  $\Sigma$ -event  $m(q) \in M$  exists only if  $q$  commits; if  $q$  is a failed non-blocking poll,  $m(q)$  is absent and  $q$  contributes no  $\Sigma$ -event. For a message-operation frontier item  $x \in \Omega_{\text{msg},P}$  with  $\text{MsgOf}_P(x)=q$ , item  $x$  contributes the committed  $\Sigma$ -event  $m(q)$  only if  $q$  commits. A pending blocking message-operation frontier item  $x$  may therefore be in  $\Omega_P$  and  $\Omega_{\text{msg},P}$ , and its owning message instance  $q$  may be in  $Q_{\Omega,P}$ , even though  $m(q)$  is not yet present.

Operations on untaken branches and unexecuted loop iterations are not members of  $\Omega_P$ ,  $\Omega_{\text{msg},P}$ ,  $Q_{\Omega,P}$ , or  $Q_{\text{exec},P}$ . Distinct loop iterations, unrolled instances, and repeated calls are represented by distinct dynamic instances.

Default source order for ordinary message calls. For dynamic items  $x$  and  $y$  of process  $P$ , write  $x <_{\text{src}} y$ ,

or equivalently  $x <_P y$  when the process must be explicit, to mean that  $P$ 's source control flow and the R-rules force  $x$  to precede  $y$  in that dynamic execution / witness. For ordinary user-written source-level message-call instances in a sequential process, this is the dynamic program order along the executed control path. Thus, if two executed message-call instances  $q1, q2 \in Q_{\text{exec}, P}$  appear in sequence in the same process, exactly one of  $q1 <_{\text{src}} q2$  or  $q2 <_{\text{src}} q1$  holds according to that dynamic program order, including when  $q1$  and  $q2$  use distinct message-passing interfaces, occur in the same synchronization interval, arise from repeated loop iterations, or arise from unrolled instances whose source-level expansion is ordered by the selected source elaboration. Same-cycle issue, same-cycle commit, or issue of a later operation before an earlier operation commits is a timing/issue-policy freedom; it does not make ordinary source message-call instances incomparable. The no-reverse rule R4/E3 is therefore stated over this ordered dynamic source relation.

Limited sources of incomparability. The relation  $\leq_P$  used for frontier reasoning is the reflexive closure of the restriction of this source-induced order to  $\Omega_P$ , extended only by explicitly modeled partial-order structure. Ordinary distinct-interface source calls are not incomparable merely because they are on different interfaces. A message-operation multi-frontier may arise only from explicit  $\text{Sys\_B}^{\text{elab}}$  proof-internal structure such as staged/bundle/join/epoch-gate abstract channels whose elaboration package declares the relevant frontier items, evaluated boundaries, partial order, and projection/attribution facts. Such elaborated frontier items are part of the selected  $\text{Sys\_ref}$  proof model; they are not an implicit weakening of source order among the ordinary user-written message calls.

For sequential-process signal IO,  $<_{\text{src}} / <_P$  is interpreted through the R2 anchor semantics rather than through the raw lexical statement location of the signal read or write. Thus a signal Read is ordered, for formal frontier and trace-compatibility purposes, at its  $\text{pred\_S}$  anchor, and a signal Write is ordered, for formal frontier and trace-compatibility purposes, at its  $\text{succ\_S}$  anchor. A lexically earlier signal Write whose  $\text{succ\_S}$  anchor is a later synchronization call does not, merely by lexical placement, precede or block intervening message-operation instances in the same source interval; conversely, a lexically later signal Read whose  $\text{pred\_S}$  anchor is an earlier synchronization call does not, merely by lexical placement, wait behind intervening message-operation instances. Signal IO may constrain message IO only through the explicit synchronization anchors and the stated R/E rules, not by raw lexical adjacency inside an interval.

## Conceptual Model for a Single Process

For every process  $P$  the pre-HLS trace  $\tau_P$  must satisfy the following *partial-order constraints* over these dynamic operation instances:

### R1 (Program order of S).

$$\forall s1, s2 \in S : s1 <_{\text{src}} s2 \Rightarrow \text{clk\_pre}(s1) < \text{clk\_pre}(s2)$$

### R2 (Location of R/W for sequential processes).

For a sequential process  $P$ , ordinary signal Read/Write events are anchored to synchronization events as follows:

$$\forall r \in R_P : \text{clk\_pre}(r) = \text{clk\_pre}(\text{pred\_S}(r))$$

$$\forall w \in W_P : \text{clk\_pre}(w) = \text{clk\_pre}(\text{succ\_S}(w))$$

where  $\text{pred\_S}(r)$  (resp.  $\text{succ\_S}(w)$ ) is the closest dynamic synchronization-call instance that precedes (resp. follows) the signal Read/Write on the dynamic source path of the current execution / witness, after applying the anchoring convention above. This anchoring determines the signal event's formal position for  $\leq_P$  /  $<_P$ , Done\_P, Remain\_P, and NextFront\_P reasoning. In particular, for a sequential-process signal Write  $w$ , the fact that the write statement appears lexically before a message operation  $q$  in the same synchronization interval does not imply  $w <_P q$  when  $w$  is anchored to the succeeding synchronization call; the write's formal observable event is placed at  $\text{succ\_S}(w)$ . Likewise, a signal Read anchored to  $\text{pred\_S}(r)$  is not delayed by intervening message operations merely because the read statement appears lexically after them.

Every observable signal Read/Write of a sequential process must have a real bounding synchronization anchor on the dynamic path on which it occurs. A process-start synchronization event  $\text{START\_P}$  may serve as the initial predecessor anchor when the modeling convention explicitly includes  $\text{START\_P}$  in  $S$ . A terminating process event  $\text{FINISH\_P}$  may serve as the terminal successor anchor when the process actually terminates and  $\text{FINISH\_P}$  occurs. If no real predecessor anchor exists for a Read, or no real successor anchor exists for a Write on the dynamic path being modeled, the sequential-process signal IO is ill-formed for this formal model or lies outside the finite observable prefix being compared. No observable Read/Write event is assigned clock time  $\infty$ .

### R2C (Combinational signal IO at the $\epsilon$ -fixed point).

A combinational process  $P$  has no process-owned synchronization events and no message-passing events in this formal clause; it may, however, contribute observable signal Read(sig,val) and Write(sig,val) events to  $\Sigma$  through explicitly selected R2C observation components. These events are not anchored by  $\text{pred\_S}/\text{succ\_S}$ . In every clock cycle  $t$ , the global state first reaches the cycle's  $\epsilon$ -quiescent combinational fixed point after all sequential clock-boundary updates and all internal combinational propagation for that cycle have settled. Let  $\mu_t$  denote this unique fixed-point valuation of the signal network.

**R2C cycle-entry valuation.** The fixed point  $\mu_t$  is reached from the cycle- $t$  entry valuation established after the applicable sequential clock-boundary updates, with the registered/process-owned request and signal drives and the external/direct-input values selected for cycle  $t$  fixed. Internal combinational propagation then settles from that valuation without advancing the architectural cycle.

### R2C unit-generation convention

Fix an explicitly modeled combinational process or combinational network component  $C_{\text{comb}}$  and its selected observable slice  $\text{Obs}(C_{\text{comb}})$ , consisting of the process/signal/direction labels whose fixed-point values are included in that component's R2C observation. Define  $\text{ComponentParticipatesInCycle}(C_{\text{comb}}, t)$  to mean that  $C_{\text{comb}}$  is active in the selected reset epoch and  $\text{Obs}(C_{\text{comb}})$  is defined at the cycle- $t$   $\epsilon$ -quiescent fixed point  $\mu_t$ ; when cycle-locked observation is used, the selected observation/elaboration package shall define this participation predicate explicitly. The fixed-point valuation  $\mu_t$  exists every cycle, but for an ordinary non-cycle-locked component a  $\Sigma$ -visible R2C fixed-point signal-observation unit is emitted only at canonical observation occurrences: (i) the first cycle  $t$  for which  $\text{ComponentParticipatesInCycle}(C_{\text{comb}}, t)$ ; (ii) any later participating cycle in which the vector of fixed-point values on  $\text{Obs}(C_{\text{comb}})$  differs from the vector in the most recent emitted R2C unit; or (iii) any participating cycle in which an explicitly modeled observer/sampler requires a fresh observation unit. Participating cycles in which the observable fixed-point slice is identical to the most recent emitted unit and no explicit observer/sampler requires a fresh unit are R2C stutter cycles for  $C_{\text{comb}}$  and emit no new  $\Sigma$ -visible unit. Cycles in which the component does not participate likewise emit no  $C_{\text{comb}}$  unit. Delta-cycle re-evaluation, intermediate combinational values,

redundant re-drive before the fixed point, and ordinary R2C stutter cycles are  $\epsilon$ /stutter for this component and are not separately  $\Sigma$ -visible.

**R2C cycle-locked completeness.**

If  $C\_comb$  is declared cycle-locked, then for every cycle  $t$  satisfying

$ComponentParticipatesInCycle(C\_comb, t)$ , exactly one R2C fixed-point observation unit for  $C\_comb$  shall be emitted from  $\mu\_t$ , even when  $Obs(C\_comb)$  is unchanged and no separate observer/sampler trigger occurs. Conversely, no cycle-locked unit is emitted for a cycle in which  $ComponentParticipatesInCycle(C\_comb, t)$  is false. This is a positive completeness obligation as well as an emission guard: proving only that every emitted unit occurs in a permitted cycle is insufficient if a required participating-cycle unit is omitted.

If a combinational process  $P$  reads  $sig$  in an emitted R2C unit of  $C\_comb$ , the corresponding label is  $Read(sig, \mu\_t(sig))$  for the cycle  $t$  at which that unit is emitted. If  $P$  drives  $sig$  in an emitted R2C unit of  $C\_comb$ , the corresponding label is  $Write(sig, \mu\_t(sig))$ , where  $\mu\_t(sig)$  is the final resolved fixed-point value of  $sig$  at the chosen observation boundary. All Read/Write labels inside one emitted R2C unit are interpreted at the same  $\epsilon$ -quiescent fixed point and may not be split.

Combinational-process signal IO is well formed only when the relevant combinational network is deterministic and reaches a unique  $\epsilon$ -fixed point in each cycle being compared, and when each observable R2C component has a well-defined selected observable slice  $Obs(C\_comb)$ . For a multi-input combinational component whose inputs may be updated by independent latency-insensitive sources, the component is  $\Sigma$ -observable under the stutter-canonicalized R2C rule only if the chosen observation boundary provides a single explicit anchoring domain for the slice, or if an additional component-specific well-formedness proof shows that all admissible interleavings induce the same emitted R2C unit sequence. Otherwise the component's internal combinational propagation must remain  $\epsilon$ -hidden, or the design must expose the relevant behavior through explicit synchronization, message passing, a cycle-accurate transactor, or a cycle-locked R2C observation boundary. Pure combinational oscillations, ambiguous multiple-driver resolution, hidden stateful behavior, or intentional combinational-loop behavior outside a unique fixed-point interpretation are outside this R2C model unless represented by explicit state or synchronization elsewhere in the formal model.

**R2C determinism clarification.** Here “deterministic” means that fixed cycle-entry registered state/drives and fixed cycle- $t$  inputs determine one  $\epsilon$ -quiescent fixed-point valuation for the selected observation network. It does not require the composed internal combinational microstep relation or evaluation order itself to be functional. Value-preserving re-evaluation, intermediate values, and redundant re-drive may be  $\epsilon$ /stutter provided they do not change the unique settled result.

### R3 (Isolation around S).

Let  $s$  be a dynamic synchronization-call instance of process  $P$ . Let  $pref\_S(s)$  be the set of executed dynamic message-call instances  $q \in Q\_exec, P$  in the immediately preceding source interval of  $P$  whose source-induced order places  $q$  before or at the boundary controlled by  $s$ . Let  $suff\_S(s)$  be the set of executed dynamic message-call instances  $q \in Q\_exec, P$  in the immediately following source interval of  $P$  whose source-induced order places  $q$  after  $s$ . Then

$\forall q \in pref\_S(s) : issue\_pre(q) \leq clk\_pre(s) \quad \text{and} \quad \forall q \in suff\_S(s) : issue\_pre(q) > clk\_pre(s).$

$\forall q \in pref\_S(s)$  with  $q$  a blocking Push/Pop :  $m(q)$  is defined and  $clk\_pre(m(q)) \leq clk\_pre(s).$

The third clause is the synchronization-completion fence of  $Core.SyncFence$ ; it permits a blocking transfer and  $s$  to commit in the same cycle. In  $Sys\_B^{elab}$ ,  $m(q)$  here is the designated process-facing commit that completes/releases  $q$ , not an arbitrary internal elaboration transfer. Failed non-blocking polls in  $Q\_exec, P$  satisfy the issue-side synchronization-boundary rule, but they contribute no  $\Sigma$ -event and do not become  $\Omega\_P$  frontier items merely because they executed.

#### R4 (Safe message issue ordering).

For every pair  $q1, q2 \in Q\_exec, P$  within the same process and within the compared dynamic execution / witness,

$q1 <_{src} q2 \Rightarrow issue\_pre(q1) \leq issue\_pre(q2)$ .

(For same-channel operations, this states that dynamic calls on a single interface are issued in source-induced program order. For distinct channels, it forbids issuing dynamic message-call instances in the reverse of their source-induced order. This rule does not impose ordering between syntactic operations on untaken branches or between dynamic instances that are not ordered by the source-induced partial order of the selected witness. Failed non-blocking polls are included in this issue-order universe when they are executed, but remain  $\epsilon$ -steps with no  $\Sigma$ -event.) Strict scalar same-endpoint case: when  $q1$  and  $q2$  are distinct blocking instances on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint in one reset epoch, Core.IC(4) strengthens this to  $issue\_pre(q1) < issue\_pre(q2)$ .

Prefix-closed reading of issue order. R4 and E3 are to be read not only as inequalities between already-defined issue timestamps, but also as closure conditions on issue itself. Within a compared dynamic execution / witness, if  $q1 <_{src} q2$  and  $q2$  has issued by cutpoint  $\sigma$ , then  $q1$  has also issued by  $\sigma$ , and  $Issue(q1) \leq Issue(q2)$  on the same side of the comparison ( $Issue\_pre$  for R4/source-side prefixes,  $Issue\_post$  for E3/implementation-side prefixes). Equivalently, in each source interval, the set of issued message-operation instances is downward closed under the source-induced order used by R4/E3. This rule is what excludes the case where a source-later message operation has issued while a required source-earlier message operation in the same interval remains merely reached but unissued.

#### R5 (Channel capacity semantics; Sys vs. Sys\_B).

Appendix G defines the raw rendezvous source model Sys, in which every channel has effective capacity 0. Concretely: for every channel  $c$  and any clock cycle  $t$ , the number of unmatched committed Push\_c actions is zero and the number of unmatched committed Pop\_c actions is zero; equivalently, no Push\_c shall commit unless a matching Pop\_c also commits at  $t$ , and no Pop\_c shall commit unless a matching Push\_c also commits at  $t$ .

In Appendices I–K we additionally use the back-annotated source interpretation Sys\_B: the same source processes as Sys, but interpreted under the case-split channel semantics of E4 with capacities  $B(c)$  matching RTL\_B. Thus:

- $B(c)=0$  is interpreted as the rendezvous case of E4: matching Push\_c( $v$ ) and Pop\_c( $v$ ) endpoint events commit in the same cycle, and no unmatched committed endpoint event may remain after any cycle boundary.
- $B(c)>0$  is interpreted as the cycle-boundary, non-fall-through bounded-FIFO case of E4, using the occupancy predicate  $occ\_c(t)$  and the FIFO availability predicates  $occ\_c(t) < B(c)$  for Push\_c and  $occ\_c(t) > 0$  for Pop\_c.

The source conceptual semantics for a process is thus a labeled partial order  $(\Sigma, \leq_P)$  generated by R1–R4 plus the applicable E4 channel semantics: raw rendezvous for Sys, or the case-split back-annotated channel semantics of Sys\_B / Sys\_ref.

#### Operational semantics location

The selected Sys\_ref transition system, registered sequential-interface semantics, Policy L and Policy S, and the admissible Policy-L execution domain are defined normatively in the preceding “Normative

Formal Core for Appendices G–K.” Appendix G uses that core while stating the R1–R5 and E1–E5 trace-compatibility rules; it introduces no second operational semantics.

## Trace Compatibility Relation

Let  $\tau_{\text{pre}}$  and  $\tau_{\text{post}}$  be the  $\Sigma$ -traces (finite or infinite) of the same process before and after HLS, where each  $\Sigma$ -event is labeled exactly as in “Normative Formal Core for Appendices G–K” (e.g.,  $\text{Push}_c(v)$ ,  $\text{Pop}_c(v)$ ,  $\text{Read}(\text{sig}, \text{val})$ ,  $\text{Write}(\text{sig}, \text{val})$ , and synchronization events).

Notation note. We write  $\tau_{\text{pre}} \approx \tau_{\text{post}}$  as a named, ordered trace-compatibility predicate. The symbol  $\approx$  is local to this document and is not used here to assert a mathematical equivalence relation; in particular, reflexivity, symmetry, and transitivity are not being claimed. Its arguments are ordered: the left trace supplies the pre-HLS/source reference witness, and the right trace supplies the post-HLS implementation trace constrained by E1–E5. In later appendices the same convention applies to  $\tau_{\text{ref}} \approx \tau_{\text{post}}$ , with  $\tau_{\text{ref}}$  as the prescribed  $\text{Sys}_{\text{ref}}$  witness.

Convention (S): all references to  $S$ ,  $\text{pred}_S/\text{succ}_S$ ,  $\text{pref}_S/\text{suff}_S$ , and  $<_{\text{src}}$  in E1–E5 use the dynamic source-induced order defined in R1–R4 for the compared execution / witness. Thus  $\tau_{\text{pre}}$  denotes the  $\Sigma$ -trace of the pre-HLS source over the dynamic operation instances reached under the fixed stimulus / selected witness.

$\Sigma$ -event matching convention (expansion of Definition Core.17  $\Sigma\text{Match}$ ). We use a canonical per-endpoint occurrence matching to make “the same  $\Sigma$ -events” precise. For each channel  $c$ , match committed  $\text{Push}_c$  and  $\text{Pop}_c$  events by their ordinal occurrence on that channel (e.g., the  $k$ -th committed  $\text{Push}$  on  $c$  in  $\tau_{\text{pre}}$  matches the  $k$ -th committed  $\text{Push}$  on  $c$  in  $\tau_{\text{post}}$ ; likewise for  $\text{Pop}_c$ ). For each such matched pair, the endpoint ( $c$ ) and the payload/value label must agree. For the ordinal index  $k$  to be well-defined under the per-cycle “set/bucket” trace representation used in these appendices, we assume that for any fixed channel endpoint  $c$ , at most one committed  $\text{Push}_c$  and at most one committed  $\text{Pop}_c$  may occur in any single clock cycle.

For each observable signal  $\text{sig}$  of a sequential process, match  $\text{Read}(\text{sig}, \cdot)$  and  $\text{Write}(\text{sig}, \cdot)$  by their ordinal occurrence on that signal within the process trace, again requiring the value labels to agree.

For sequential processes, we apply the following canonicalization for observable signals: within each anchor interval between consecutive synchronization events in the order  $S$ , we record at most one  $\Sigma$ -visible  $\text{Read}(\text{sig}, \text{val})$  and at most one  $\Sigma$ -visible  $\text{Write}(\text{sig}, \text{val})$  for each (process,  $\text{sig}$ ), with the anchor cycle given by E2. Any additional physical sampling of  $\text{sig}$  by the implementation, or redundant re-driving of  $\text{sig}$  within the same anchor interval, is treated as an internal  $\epsilon$ -step and must not change the recorded value label. If a design intentionally relies on intra-interval changes of  $\text{sig}$ , or on distinguishing multiple reads/writes of  $\text{sig}$  within the same anchor interval, then  $\text{sig}$  must be modeled using an explicit synchronized interface rather than as an ordinary sequential-process observable signal.

For combinational processes, the canonical observable unit is the emitted R2C fixed-point signal-observation unit for the relevant combinational process/network component  $C_{\text{comb}}$ : the set of  $\text{Read}(\text{sig}, \text{val})$  and  $\text{Write}(\text{sig}, \text{val})$  labels for the selected observable slice  $\text{Obs}(C_{\text{comb}})$ , recorded at one  $\epsilon$ -quiescent combinational fixed point under the R2C unit-generation convention. Intermediate delta-cycle reads/writes, repeated re-evaluations before the fixed point, and stutter cycles in which

$\text{Obs}(C\_comb)$  is unchanged and no explicit observer/sampler requires a fresh unit are  $\epsilon$ /stutter and are not separately  $\Sigma$ -visible.

### R2C unit matching/index convention

For each combinational process or explicitly modeled combinational network component  $C\_comb$ , project each trace to the subsequence of emitted R2C fixed-point signal-observation units for  $C\_comb$ . The  $k$ -th emitted such unit in  $\text{Sys\_ref}$  is matched with the  $k$ -th emitted such unit in  $\text{RTL\_B}$ . This local emitted-occurrence ordinal, together with the component identity  $C\_comb$ , is the correspondence rule for R2C fixed-point units; it is not the global observable-cycle bucket number of the whole-system trace. Thus repeated identical fixed-point observations across stutter cycles do not create additional units, while repeated equal-valued units created by explicit observer/sampler occurrences are distinguished by their local emitted-occurrence ordinal for  $C\_comb$ .

Matched pre-HLS and post-HLS fixed-point units must agree on the included process/signal/direction labels and fixed-point values. The same-cycle agreement required here is local to the matched emitted R2C fixed-point unit: all Read/Write labels inside that unit are interpreted at one  $\epsilon$ -quiescent fixed point and may not be split. This local atomicity requirement is one of the explicit same-cycle semantic constraints and does not require unrelated  $\text{Sys\_ref}$  and  $\text{RTL\_B}$  events to share the same global bucket decomposition. If a component is cycle-locked, its local emitted-occurrence sequence is the ordered sequence of cycles  $t$  satisfying  $\text{ComponentParticipatesInCycle}(C\_comb, t)$ , with one required unit per such cycle. The pre-HLS and post-HLS traces must therefore agree both on the participation-indexed unit values and on the absence of omitted required units. Cycle-locked components fall outside the bucket-stutter freedom claimed for ordinary latency-insensitive R2C components.

**R2C cross-model/cycle-lock clarification.**  $\text{Sys\_ref}$  and  $\text{RTL\_B}$  are not assumed to share internal combinational state, a common combinational microstep model, or the same global cycle-bucket placement; each side settles its own cycle-entry valuation. For a cycle-locked component, both sides must satisfy the per-side participation/completeness rule above, and the  $k$ -th matched units are compared within those respective component-local participation-indexed emitted sequences; equality of absolute  $\text{Sys\_ref}$  and  $\text{RTL\_B}$  cycle numbers is not required. If  $\text{RTL\_B}$  does not expose an explicit combinational-network model from which participation can be derived, the implementation/certificate evidence for the selected annotated prefix shall justify that its emitted sequence obeys the declared cycle-lock/participation rule. The R2C matching statement in E2 is the Appendix-G expansion of this same R2C clause of Core.17  $\Sigma\text{Match}$ , not an additional independent matching conjunct.

For proof-internal  $\text{RESET}(S\_p, S\_c)$  units included in the selected theorem comparison, match by reset scope and local occurrence within that scope's reset-epoch history: the  $k$ -th  $\text{RESET}$  for a given  $(S\_p, S\_c)$  scope in  $\tau\_pre$  matches the  $k$ -th  $\text{RESET}$  for the same scope in  $\tau\_post$ . Matched  $\text{RESET}$  units therefore have the same scope. Independent disjoint  $\text{RESET}$  occurrences may commute unless an explicit reset-boundary or other observable-order constraint orders them.

For synchronization events, match by their per-process occurrence index in the source order ( $R1$ ; preserved on the post-HLS side by E1 below). Independent  $\Sigma$ -events on different endpoints (including distinct message-passing interfaces) may commute and/or be grouped differently within a clock cycle as permitted by the R-rules and E3. Here “independent” is meant in the  $\Sigma$ -equivalence sense: the pair is not additionally ordered by any explicit  $\Sigma$ -visible ordering/timestamp constraint in E1–E5 (notably E3/E5); it is not a statement about (in)comparability under the Appendix G causal-dependency relation ( $\rightarrow$ ). Accordingly,  $\tau\_pre$  and  $\tau\_post$  are not required to be identical as a single total order, nor to agree

on same-cycle “grouping” of distinct-interface Push/Pop operations, beyond the explicit ordering/timestamp constraints in E1–E5 below.

Consistently with Definition Core.17, we say  $\tau_{\text{pre}} \approx \tau_{\text{post}}$  iff  $\Sigma\text{Match}(\tau_{\text{pre}}, \tau_{\text{post}})$  holds under the convention above and all the following hold:

### E1 (Per-process synchronization-order preservation).

For every process  $P$ ,

$\forall s_1, s_2 \in S_P : s_1 <_P s_2 \Rightarrow \text{clk\_post}(s_1) < \text{clk\_post}(s_2)$ .

Equivalently, synchronization events are matched and ordered by their per-process source occurrence index. E1 does not preserve incidental global clock ordering between synchronization events of different processes unless that ordering is induced by an explicit inter-process synchronization relation already represented in  $\Sigma$ .

### E2 (Signal-visibility preservation).

Partition signal Read/Write events by process kind:

- $R_{\text{seq}}$  and  $W_{\text{seq}}$  are the ordinary signal Read/Write events of sequential processes.
- $R_{\text{comb}}$  and  $W_{\text{comb}}$  are the signal Read/Write events of combinational processes, interpreted by R2C.

For sequential-process signal IO:

$\forall r \in R_{\text{seq}} : \text{clk\_post}(r) = \text{clk\_post}(\text{pred}_S(r))$

$\forall w \in W_{\text{seq}} : \text{clk\_post}(w) = \text{clk\_post}(\text{succ}_S(w))$

Equivalently, ordinary signal Reads/Writes of sequential processes are logically timestamped at their synchronization anchors, as in R2. The same anchoring convention is used on the pre-HLS side when forming the matched trace.

For combinational-process signal IO,  $\text{pred}_S/\text{succ}_S$  anchoring does not apply. Instead, each R2C fixed-point signal-observation unit is matched by the component-local occurrence ordinal defined above: for each combinational process or explicitly modeled combinational network component  $C_{\text{comb}}$ , the  $k$ -th R2C fixed-point unit of  $C_{\text{comb}}$  in  $\text{Sys\_ref}$  matches the  $k$ -th R2C fixed-point unit of  $C_{\text{comb}}$  in  $\text{RTL\_B}$ . The matched units must contain the same corresponding combinational Read/Write events and fixed-point values at their respective  $\epsilon$ -quiescent combinational fixed points, as defined by R2C. Any delta-cycle re-evaluation, intermediate combinational value, or redundant re-drive before the fixed point is an  $\epsilon$ -step and is not separately  $\Sigma$ -visible. This condition does not require the matched R2C units to occur in the same global observable-cycle bucket number, nor to be grouped with the same unrelated events, in the two executions.

### E3 (Safe message issue ordering). (Post-HLS preservation of R4).

For every pair of executed message-call instances  $q_1, q_2 \in Q_{\text{exec}}, P$  of the same process  $P$ :

- Same-interface source order: if  $q_1$  and  $q_2$  access the same message-passing interface and  $q_1 <_{\text{src}} q_2$ , then  $\text{issue\_post}(q_1) \leq \text{issue\_post}(q_2)$ . This preserves source order at cycle granularity; any stricter serialization required by the concrete interface is supplied by the channel/interface semantics, not by E3 itself.
- Distinct-interface no-reverse: if  $q_1$  and  $q_2$  access distinct message-passing interfaces and appear in sequence in the source ( $q_1 <_{\text{src}} q_2$ ), then  $\text{issue\_post}(q_1) \leq \text{issue\_post}(q_2)$  (i.e., they shall not be issued

in the reverse sequence).

(iii) Prefix-closed issue obligation: for every cycle-aligned cutpoint  $\sigma$  of the post-HLS execution / witness, if  $q1 <_{\text{src}} q2$  and  $q2$  has issued by  $\sigma$ , then  $q1$  has also issued by  $\sigma$  and  $\text{Issue}(q1) \leq \text{Issue}(q2)$ . This applies across same-interface and distinct-interface source order. It is an implementation-side obligation on issued sets, not merely an inequality over pairs whose timestamps are already defined.

(Note: HLS internal pipeline staging, dequeue, and hand-off inside the back-annotated process/channel realization are not, by themselves,  $\Sigma$ -visible boundary issue events in  $Q_{\text{exec}}, P$  nor committed-transfer events in  $M$ ; therefore such internal movement does not constitute a counterexample to (ii) or (iii).)

Strict scalar same-endpoint case: when  $q1$  and  $q2$  are distinct blocking instances on the same scalar explicit abstract boundary and endpoint direction governed by the Core single-transfer-per-cycle endpoint constraint in one reset epoch,  $\text{Core.IC}(4)$  strengthens clause (i) to  $\text{issue\_post}(q1) < \text{issue\_post}(q2)$ .

#### E4 (Per-channel channel semantics: rendezvous and bounded FIFO).

For every observable channel  $c$  with capacity  $B(c)$ , the projection of  $\tau_{\text{post}}$  to events on  $c$  is legal for that channel's capacity model. The channel model splits into two cases.

- Rendezvous case,  $B(c)=0$ . Channel  $c$  is a capacity-zero rendezvous channel. A committed transfer on  $c$  consists of a matching same-cycle pair  $\text{Push}_c(v)$  and  $\text{Pop}_c(v)$  with the same payload value  $v$ . No unmatched committed  $\text{Push}_c$  or  $\text{Pop}_c$  may exist after any cycle boundary. Equivalently, the  $k$ -th committed  $\text{Push}_c(v)$  and the  $k$ -th committed  $\text{Pop}_c(v)$  occur in the same clock cycle and carry the same value. A  $\text{Push}_c$  endpoint can commit in cycle  $t$  only when the matching  $\text{Pop}_c$  endpoint boundary request is also asserted in cycle  $t$ , and symmetrically a  $\text{Pop}_c$  endpoint can commit in cycle  $t$  only when the matching  $\text{Push}_c$  endpoint boundary request is also asserted in cycle  $t$ .

- Buffered case,  $B(c)>0$ . Channel  $c$  is a bounded FIFO with cycle-boundary, non-fall-through semantics. The committed  $\text{Pop}_c(v)$  events return exactly the FIFO-ordered sequence of values  $v$  previously committed by  $\text{Push}_c(v)$ , with no drops, duplications, or reordering. For every cycle-aligned prefix  $\pi_t$  of the  $c$ -projection—i.e.,  $\pi_t$  contains exactly the events on  $c$  committed in cycles  $< t$ —letting  $\text{push}(\pi_t)$  and  $\text{pop}(\pi_t)$  be the number of  $\text{Push}_c$  and  $\text{Pop}_c$  events in  $\pi_t$ , we have  

$$0 \leq \text{push}(\pi_t) - \text{pop}(\pi_t) \leq B(c).$$

Equivalently, the abstract occupancy after cycle  $t-1$  is  $\text{occ}_c(t) = \text{push}(\pi_t) - \text{pop}(\pi_t)$ . Availability for commits in cycle  $t$  is evaluated against  $\text{occ}_c(t)$  at the start of cycle  $t$ :  $\text{Pop}_c$  may commit in cycle  $t$  only if  $\text{occ}_c(t) > 0$ ,  $\text{Push}_c$  may commit in cycle  $t$  only if  $\text{occ}_c(t) < B(c)$ , and simultaneous  $\text{Push}_c$  and  $\text{Pop}_c$  commits in the same cycle are permitted iff  $0 < \text{occ}_c(t) < B(c)$ .

Thus the inequality-based FIFO availability predicates are used only for  $B(c)>0$ . The  $B(c)=0$  case is not interpreted by substituting  $B(c)=0$  into the bounded-FIFO inequalities; it is interpreted by the rendezvous rule above.

Note:  $\text{occ}_c(t)$  is the logical occupancy of channel  $c$  at the chosen  $\Sigma$  boundary, induced solely by the  $\Sigma$ -visible boundary-commit history ( $\text{Push}_c/\text{Pop}_c$ ) and the annotated capacity  $B(c)$  in the buffered case  $B(c)>0$ . For  $B(c)=0$ , the corresponding channel fact is the same-cycle rendezvous request/commit predicate rather than a nonzero FIFO occupancy predicate.

#### E5 (Messages cannot cross their immediate synchronization boundary).

For every synchronization call  $s$  (in the source order used by  $R3$  /  $\text{pref}_S$  /  $\text{suff}_S$  over  $Q_{\text{exec}}, P$ ):  
 $\forall q \in \text{pref}_S(s) : \text{issue\_post}(q) \leq \text{clk\_post}(s)$

$\forall q \in \text{suff\_S}(s) : \text{issue\_post}(q) > \text{clk\_post}(s)$

$\forall q \in \text{pref\_S}(s)$  with  $q$  a blocking Push/Pop :  $m(q)$  is defined and  $\text{clk\_post}(m(q)) \leq \text{clk\_post}(s)$ .

The third clause is the implementation-side synchronization-completion fence: no earlier blocking request may survive  $s$  into the succeeding source interval, although  $q$  and  $s$  may commit in the same cycle. In  $\text{Sys\_B}^{\text{elab}}$ ,  $m(q)$  denotes the designated process-facing commit that completes/releases  $q$ . For a same-cycle  $q/s$  completion, the  $\text{Core.SyncFence}$  statement that  $q$  is “accounted for before” the interval advance is an abstract trace/certificate accounting convention used to preserve operation and interval attribution; E5 does not require a separately observable physical sub-cycle ordering between  $q$  and  $s$  on  $\text{RTL\_B}$ . Under the Issue/Commit linkage above, any boundary request assertion that persists while the process is still pending at  $s$  reflects only already-issued outstanding operation(s) in  $\text{pref\_S}(s)$ ; no operation  $q \in \text{suff\_S}(s)$  may have  $\text{Issue}(q) \leq \text{clk\_post}(s)$ , and any back-to-back request handoff on that endpoint direction, if present, is attributed in the first cycle strictly after  $\text{clk\_post}(s)$ . For non-blocking calls, a later executed source-level PushNB/PopNB in  $\text{suff\_S}(s)$  is a distinct  $q$  and cannot be identified with an earlier  $\text{pref\_S}(s)$  call solely because the endpoint request signal remained asserted.

See Appendix I – Per-Process Replay and Preparation Proof for the local result used by the system theorem:

For any process  $P$  participating in a selected finite  $\text{RTL\_B}$  prefix, the fixed stimulus, WC witness, dynamic-operation matching, and  $P$ -local observable-unit history determine the corresponding immediate source replay state. Appendix I proves that, from that state, a finite  $P$ -owned  $\epsilon$  preparation path reaches the exact dynamic source operation for the next selected unit—either as next-frontier or as an already issued persistent request—while preserving E3/E5 and committing no unmatched observable unit. This is a local readiness theorem, not an independent channel-commit or global-trace existence theorem. Appendix J combines these local paths with channel and atomic-interaction certificates to construct the complete  $\text{Sys\_ref}$  witness. The reverse direction remains restricted to  $\text{RTLConsistentRefPrefix}$  / Appendix L witness-selected executions.

## Allowed Transformations and Why They Preserve $\approx$

| Transformation permitted by the rules                                                      | Formal justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Loop pipelining</b> (overlaps iterations)                                               | Introduces internal pipeline stages (modeled as $\epsilon$ -steps) and overlaps iterations. R1–R5 (and hence E1–E5) are applied to the explicit $S$ set. Loop pipelining may commute independent $\Sigma$ -actions across iterations (e.g., across distinct channels)—where “independent” here means “not additionally ordered by any explicit $\Sigma$ -visible constraint in E1–E5 between the pair (notably E3/E5),” rather than “incomparable under Appendix G’s causal-dependency relation ( $\rightarrow$ )”—but it must still preserve: (i) per-channel FIFO legality (E4), (ii) the required issue discipline (E3) for source-ordered pairs, and (iii) the sync-boundary constraints (E1/E5) with respect to the (explicit + implicit) $S$ -actions. |
| <b>Overlap of internal dequeue/staging vs. later pipeline work (<math>\epsilon</math>)</b> | Permitted because the internal staging/acceptance activity is $\epsilon$ -internal within the back-annotated channel realization (as defined in “Appendix B - Pipelined Loops”) and does not                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Transformation permitted by the rules                             | Formal justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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|                                                                   | constitute a $\Sigma$ -visible boundary message-issue reorder. $\Sigma$ -visible boundary message issues must still satisfy E3, and per-channel committed-transfer legality is ensured by E4; automatic flush / Appendix K addresses deadlock preservation for overlap timing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>#pragma hls_direct_input</b> (late sampling of static signals) | This pragma imposes a designer/environment stability contract: the signal's value must remain stable after reset (or after its last permitted update) while the receiving process may consume it. The logical signal Read actions $r$ remain anchored exactly as in E2 ( $\text{clk\_post}(r) = \text{clk\_post}(\text{pred\_S}(r))$ ). HLS may implement this by sampling the signal later (instead of inserting internal storage), but because the signal is stable the sampled value equals the value at $\text{pred\_S}(r)$ ; therefore E2 and $\approx$ are preserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>#pragma hls_direct_input_sync</b> (controlled updates)         | <p>This pragma imposes an explicit timing contract on the environment: the controlled direct-input signals may change only on the cycle of the designated synchronization call <math>s^*</math> (i.e., during the sync handshake). With that contract, the logical anchoring required by E2 is preserved: for the receiving process, the relevant Reads are anchored at the closest preceding synchronization call <math>\text{pred\_S}(r) = s^*</math>, and any environment update Write <math>w\_update</math> is aligned with synchronization (<math>\text{succ\_S}(w\_update) = s^*</math>). HLS may still implement late sampling internally, but the value observed is the same as the value at the E2 anchor point, so <math>\approx</math> is preserved without modifying E2.</p> <p><b>Instrumentation note (logical signal timestamping).</b> In the <math>\Sigma</math>-traces used for <math>\approx</math> checking, sequential-process signal Read/Write events are timestamped at their E2 anchor points (e.g., <math>\text{clk\_post}(\text{pred\_S}(r))</math> for Reads), not at any later physical RTL sample cycle that HLS may introduce. Any internal late-sampling micro-steps are treated as <math>\epsilon</math>-steps. Combinational-process signal Read/Write labels are handled by R2C rather than by <math>\text{pred\_S}/\text{succ\_S}</math> anchoring or by global observable-cycle bucket number. A monitor records a combinational fixed-point value only after the simulator/RTL has reached an <math>\epsilon</math>-quiescent combinational fixed point, and that value contributes to <math>\Sigma</math> only when the relevant R2C observation component emits an R2C fixed-point signal-observation unit under the R2C unit-generation convention. Intermediate delta-cycle reads/writes, redundant re-evaluations, transient combinational values before the fixed point, and R2C stutter cycles whose selected observable slice is unchanged are <math>\epsilon</math>/stutter and are not separately <math>\Sigma</math>-visible.</p> |

| Transformation permitted by the rules                                            | Formal justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                  | Therefore, equivalence checking must instrument/record the appropriate logical Read/Write event: the anchored event for sequential-process signal IO, or the emitted R2C fixed-point signal-observation unit for combinational-process signal IO. A wrapper/transactor may be used when needed to expose only those logical events rather than naive physical sample, delta-cycle, or per-cycle stutter events.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Memory-access reordering proven conflict-free</b>                             | Let $\text{addr}(a)$ be the symbolic address of access $a$ . If the tool proves $\text{addr}(a_1) \neq \text{addr}(a_2)$ for the overlapped window, then the commutation of $a_1$ and $a_2$ is observationally silent; no external signal/message depends on the internal order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Implicit FSM states / added latency</b>                                       | Extra states introduce <i>silent</i> $\epsilon$ -steps between observable actions; the partial order on $\Sigma$ is unchanged, so E1–E5 are unaffected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Shared-memory arrays with explicit synchronization</b>                        | When a memory is accessed by multiple processes, the designer explicitly coordinates access with synchronization operations modeled as S-actions, and the externally relevant memory behavior is defined by the array access mapping layer. E1 preserves the order of the synchronization actions that bracket or authorize the shared-memory accesses. The array access mapping layer then defines the mapped memory operations, their commit cycles, and the required serialization/coherence rule for the shared storage, and the pre-HLS and post-HLS traces must agree under that declared memory semantics. E2 is not used here except for ordinary signal IO. E4 is relevant only if the mapping layer represents memory commands/responses as message-passing channels; otherwise memory legality is discharged by the mapping layer's memory semantics, not by FIFO channel semantics. Under these conditions, the shared-memory transformation preserves the observable behavior required by $\approx$ . Scope (J.Mem): the formal theorems of Appendices I–K discharge this entry only when the value-carrying memory traffic is lowered per the J.Mem shared-memory lowering condition (Normative Formal Core); the non-lowered “declared memory semantics” branch states an agreement obligation that Theorem J.1 as proved does not discharge. |
| <b>Non-blocking message-passing (PushNB/PopNB) verified by post-HLS snooping</b> | If the latency differences are externally visible to the DUT, a verification wrapper delays the pre-HLS side so that the committed transfer events (successful completions) occur in the same order/timing as observed on the post-HLS channel. This wrapper is itself purely synchronising (S), so it cannot violate R1–R5. Once the wrapper is assumed, $\tau_{\text{pre}}$ and $\tau_{\text{post}}$ coincide on the committed message-transfer history recorded in $\Sigma$ (i.e., the Push_c/Pop_c commit events),                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Transformation permitted by the rules                                                  | Formal justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                        | so E3–E5 are preserved. Unsuccessful non-blocking polls remain $\epsilon$ -steps and are not required to match. This is not a transformation that inherently preserves $\approx$ , but rather a verification technique to enforce equivalence for inherently latency-sensitive operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Latency-sensitive <i>global</i> signal IO matched by snooping</b>                   | If the latency differences are externally visible to the DUT, the same “mirror-and-delay” wrapper used for NB channels is applied to any globally-visible signal whose timing matters. Because the wrapper inserts only S-actions, the partial-order constraints are unchanged. This is not a transformation that inherently preserves $\approx$ , but rather a verification technique to enforce equivalence for inherently latency-sensitive operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Latency-sensitive <i>local</i> signal IO isolated in cycle-accurate transactors</b> | Local timing-exact protocols are confined to dedicated transactor processes that expose only latency-insensitive M or S operations to the rest of the design. Since the transactor boundary is now the observable interface, R1–R5 apply <i>outside</i> the timing-sensitive region, so system-level $\approx$ still holds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>One-way signal-handshake protocols with run-time assertions</b>                     | For single-direction vld or rdy signals, an assertion proves that back-pressure can <i>never</i> occur. That assertion establishes a refinement in which the missing handshake is equivalent to a permanent logical ‘1’, so the protocol is observationally identical to the corresponding two-way handshake with the missing side tied true. E2 then applies only to the logical timestamping of the signal Read/Write events, not to the protocol-refinement step itself.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Combinational processes</b>                                                         | Combinational processes have no process-owned S actions and no message-passing M actions in this formal clause, but they may have observable signal Read/Write events in $\Sigma$ through explicitly selected R2C observation components. Their signal IO is interpreted at $\epsilon$ -quiescent combinational fixed points as specified by R2C: intermediate delta-cycle propagation is $\epsilon$ , and only emitted fixed-point Read/Write labels in the selected observable slice are $\Sigma$ -visible. Repeated cycles in which the selected fixed-point slice is unchanged are R2C stutter for that component unless an explicitly modeled observer/sampler requires a fresh emitted unit. Under the well-formedness requirement that the combinational network is deterministic, reaches a unique fixed point in each compared cycle, and has a well-defined R2C unit-generation rule for the selected component, the RTL/tool-flow conformance obligation is to preserve the emitted fixed-point signal-observation unit sequence at the chosen observation |

| Transformation permitted by the rules | Formal justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | boundary. Thus the required correspondence is not equality of simulator delta-cycle behavior, nor equality of global cycle-bucket numbers, nor an automatic consequence of ordinary combinational synthesis; it is equality of matched emitted R2C fixed-point units by component identity and local emitted-occurrence ordinal. Designs with oscillation, ambiguous multiple-driver resolution, hidden stateful behavior, intentional combinational-loop behavior outside a unique fixed-point interpretation, or multi-input observation slices whose interleavings can change the emitted unit sequence are outside R2C unless represented by explicit state, synchronization, a cycle-accurate transactor, a cycle-locked R2C observation boundary, or another formally modeled boundary. |

For the purposes of the formal analysis, non-blocking message-passing is modeled at the level of committed transfers: a successful PushNB/PopNB contributes the same committed Push<sub>c</sub>/Pop<sub>c</sub> event to  $\Sigma$  as a blocking Push/Pop, while unsuccessful polls are  $\epsilon$ -steps.

If a design contains latency-sensitive global signals or non-blocking transfers whose timing or ordering is externally visible at the DUT boundary and must be reproduced in the selected comparison, an Appendix L snooping/witness-selection wrapper may be used. When a witness satisfying WC and L.Dir exists, the wrapper selects an admissible pre-HLS execution whose relevant  $\Sigma$ -events align with the observed RTL execution. The wrapper does not enlarge the set of admissible pre-HLS behaviors; it selects one admissible execution. If the wrapper cannot realize such a witness, the compared run is outside the theorem scope and must be handled according to the Appendix L triage rules.

Appendix L defines the witness-existence conditions, monitoring, and triage required to establish this alignment for a concrete compared run.

## Proof Structure

The proofs proceed in three stages:

First, Appendix I establishes process-local deterministic replay and finite preparation. For each process participation in a selected RTL<sub>B</sub> observable unit, it identifies the exact dynamic source operation, proves the relevant issue/order and witness facts, and constructs a finite process-owned  $\epsilon$  path that reaches the operation or its persistent issued request without committing an unmatched unit. These facts are packaged by J.PCert. Appendix I deliberately does not assert that the complementary endpoint is present, that a channel transfer is globally enabled, or that the local paths already form one Sys<sub>ref</sub> execution.

Second, Appendix J combines J.PCert with J.CCcert, J.ACcert, environment/reset certificates, the normative field-sensitive footprint vocabulary, deferred-hole records, Dep<sub>J</sub>, and LocalAgree. J.CanonicalRank is derived from the graph. J.Frame/J.LocalLift/J.PrepareCommutate and J.RegPhaseNF prove that outcome-independent actions can be moved into L1, requests freeze before the L2 snapshot,

and A3/A4 decisions and units execute in L3 modulo J.HCanon. Horizon-batched J.UnitExt and J.GWR-Comp construct J.GWR and J.RegPhaseReach; Proposition J.0 proves observable compatibility. This safety construction does not by itself prove K.DrainReach, A5-L, or RTL deadlock freedom.

Finally, Appendix K addresses a separate liveness-preservation obligation for internal message-passing deadlock. Appendix K does not discharge the Appendix I/J weak-fairness, finite-depth, channel-progress, realization-correctness, K.Drain,  $K\epsilon$ , J.KObsConf, K.RLAdmReach, QualifiedSynchronousModel/Core.KCutClosure\_post, K.CertCov, or A6–A11 side conditions. Nor does it derive SDet, A5-L, A5-S, K.RespCov, K.RespClean, K.EnvMF, K.InterruptionGate/K.ResetEpoch/K.TerminalDisposition applicability, or any alternate A5-L discharge from the local R-rules. The core Theorem K.1A consumes A5-L and one K.CertifiedCutpoint package directly; Corollary K.1A-Total additionally consumes Core.KCutClosure\_post and K.CertCov; in the standard scope Core.QSync-Post derives the closure premise from the qualified synchronous structural evidence for the universal conclusion. The primary user-facing route fixes the actual Policy-S pre-HLS simulator and its selected execution through SDet, satisfies K.EnvMF and the shared K.InterruptionGate applicability, instantiated first as K.ResetEpoch and then, only on K.ResetEpoch.Continue, as K.TerminalDisposition, supplies maximality/fairness/completeness and complete bounded-response evidence through A5-S/CF-S, uses Lemma K.2b and Corollary K.2c to discharge A5-L, and uses Core.KCutClosure\_post plus K.CertCov to justify the universal reachable K-facing RTL conclusion. Structural, invariant, exploration, direct Policy-L, and tool-certificate routes remain alternatives only for instances in which A5-L is actually true; they cannot rescue a design containing an admissible serial deadlock prefix.

Thus the architecture is non-circular. Appendix I supplies only local replay/preparation facts; J.GWR-Comp proves, rather than assumes, their global coexistence and complete-unit extension; Proposition J.0 consumes an already constructed trace pair; and Appendix K separately preserves internal message-passing deadlock freedom under A5-L and its recorded discharge route. The liveness theorem remains conditional on the named fairness, progress, drain, and serial-dominance premises and does not retroactively justify the safety construction.

## Causal Dependency Between Processes

To prove system-level trace compatibility, we must distinguish between incidental timing and functionally significant ordering.

### Formal Definition of Causal Dependency

To formalize the notion of functionally significant dependency and resolve ambiguity, we define a causal-dependency relation, denoted by  $\rightarrow$ , on the set of observable IO actions ( $\Sigma$ ) across all processes in the system. The relation is causal/information-flow rather than purely temporal: events related by  $\rightarrow$  may commit in the same clock cycle, and the direction of  $\rightarrow$  indicates which event may be depended on by the other.

**Important:**  $\rightarrow$  is a causal-dependency graph, not by itself the source order used for the Appendix J induction. In particular, because the methodology permits distinct-interface message operations to be issued with timing overlap or in the same cycle while still preserving dynamic source order, and because rendezvous transfers are same-cycle atomic Push/Pop transactions, the unquotiented dependency graph  $\rightarrow$  need not be acyclic in every legal execution. A cycle in  $\rightarrow$  can represent a legal same-cycle mutually enabling dependency among atomic transactions; it is not, by itself, an observable mismatch.

When Appendix J needs a well-founded induction order, it uses the separate required-order relation  $\prec_J$  over observable units. That relation is generated only by the explicit ordering, timestamp, FIFO, atomicity, and synchronization constraints that must be preserved by the trace-compatibility predicate, not by the full causal-dependency graph  $\rightarrow$  and not by raw per-process source order on distinct-interface message commits.

**Well-formedness use of causal-dependency.** The well-formedness rule below uses  $\rightarrow$  as a design-intent and source-admissibility relation: if correctness depends on the relative timing, value, availability, or cross-process observation of two events, that dependence must be represented by an explicit modeled causal mechanism such as message passing, synchronization, signal handshakes/anchors, declared memory serialization, or an Appendix L snooped/witness-selected boundary. This use of  $\rightarrow$  does not add proof-preserved observable ordering beyond  $\prec_J$ , FIFO legality, rendezvous/synchronization atomicity, signal-anchor constraints, synchronization-boundary constraints, and declared memory-serialization constraints.

The causal-dependency relation  $\rightarrow$  is the smallest directed relation satisfying the conditions below:

1. **Intra-Process Order:** If actions  $a$  and  $b$  occur within the same process  $P$  and  $a$  precedes  $b$  in the program's execution trace, then  $a \rightarrow b$ . This directly reflects the sequential nature of the code within a single thread. **Clarification (non-temporal).** Here “precedes” refers to the order of IO action *instances* in the process's trace/program order (i.e., the order induced by the process's execution), **not** the relative order of their **commit cycles**; distinct-interface message actions may be **issued** in order while **committing** in either order depending on backpressure.
2. **Inter-Process Communication:** The relation is established by the direct exchange of information or synchronization between processes.
  - **Message-passing (committed transfer):** If  $a$  is the commit of a send on channel  $c$  in process  $P_1$  (a committed  $\text{Push}_c$ , including a successful  $\text{PushNB}$ ), and  $b$  is the commit of the corresponding receive on  $c$  in process  $P_2$  (a committed  $\text{Pop}_c$ , including a successful  $\text{PopNB}$ ), for the same logical message instance, then  $a \rightarrow b$ . Here “corresponding receive” is defined by the channel's FIFO semantics:  $b$  is the  $\text{Pop}_c$  commit that consumes the element produced by  $a$  (matching is by FIFO position / transfer instance, not by payload-value equality; duplicate payload values do not introduce ambiguity). Equivalently, if we enumerate committed Pushes on  $c$  as  $\text{Push}_c^1, \text{Push}_c^2, \dots$  in commit order and committed Pops as  $\text{Pop}_c^1, \text{Pop}_c^2, \dots$  in commit order, then  $\text{Push}_c^k \rightarrow \text{Pop}_c^k$ . For buffered channels with  $B(c) > 0$ , E4 also imposes the required FIFO legality used by Appendix J's required-order relation  $\prec_J$ . For rendezvous channels with  $B(c) = 0$ , the matching  $\text{Push}_c$  and  $\text{Pop}_c$  commits occur in the same clock cycle and are treated in Appendix J as one explicitly atomic observable unit. The directed dependency  $a \rightarrow b$  records unidirectional information flow — the Pop observes the value supplied by the Push — but it does not make the two rendezvous endpoints separately orderable for the Appendix J induction, and it does not imply an earlier commit cycle. No reverse edge  $b \rightarrow a$  is implied unless there is an explicit reverse-direction communication, such as another channel or a signal handshake. Unsuccessful non-blocking polls are  $\epsilon$ -steps and do not participate in  $\rightarrow$ .

- **Signal Handshake:** If  $a$  is a signal write action  $w(\text{sig})$  in  $P1$  and  $b$  is a signal read action  $r(\text{sig})$  in  $P2$  that reads the value written by  $a$  as part of an explicit handshake protocol, then  $a \rightarrow b$ . A complete two-way handshake (e.g.,  $\text{vld}/\text{rdy}$ ) creates a causal chain, such as  $w(\text{vld})@P1 \rightarrow r(\text{vld})@P2 \rightarrow w(\text{rdy})@P2 \rightarrow r(\text{rdy})@P1$ .
- **Explicit Synchronization (two-party SyncChannel / ac\_sync):** Let  $\text{sout}$  be the commit of a SyncChannel  $\text{sync\_out}$  (or equivalent  $\text{ac\_sync}$  call) in process  $P1$ , and let  $\text{sin}$  be the commit of the matching  $\text{sync\_in}$  in process  $P2$  for the same synchronization instance. Then this synchronization acts as a two-process barrier:
  - Every observable action in  $P1$  that occurs before  $\text{sout}$  (in  $P1$ 's trace/program order) satisfies  $a \rightarrow b$  for every observable action  $b$  in  $P2$  that occurs after  $\text{sin}$ .
  - Symmetrically, every observable action in  $P2$  that occurs before  $\text{sin}$  satisfies  $a \rightarrow b$  for every observable action  $b$  in  $P1$  that occurs after  $\text{sout}$ .
  - (The two sync commits  $\text{sout}$  and  $\text{sin}$  are treated as simultaneous; we do not impose  $\text{sout} \rightarrow \text{sin}$  or  $\text{sin} \rightarrow \text{sout}$ .)

3. **Transitivity:** The relation is transitive. If  $a \rightarrow b$  and  $b \rightarrow c$ , then  $a \rightarrow c$ .

Using this relation, we now provide a formal definition for causal dependency between synchronization events, which are the fundamental ordering anchors in the scheduling model.

#### Definition: Causal Dependency

A synchronization event  $s2$  in process  $P2$  is **causally dependent** on a synchronization event  $s1$  in process  $P1$  if and only if  $s1 \rightarrow s2$ .

If neither  $s1 \rightarrow s2$  nor  $s2 \rightarrow s1$  holds true, the events  $s1$  and  $s2$  are considered **causally independent** (or concurrent). Any change in the observed temporal ordering of causally independent events between the pre-HLS and post-HLS simulations is considered an incidental, functionally insignificant variation in latency. A **well-formed design**, under this methodology, **shall not rely** on a specific ordering of causally independent events for functional correctness.

To be clear, the shall-not-rely requirement is a design rule. To adhere to this design rule, designs must express functionally significant cross-process dependencies through explicit modeled mechanisms such as message passing, SyncChannel/ac\_sync operations, signal handshakes/anchors, declared memory-serialization constraints in the array-access mapping layer, or Appendix L snooping/witness-selection boundaries. Designs that violate this design rule are outside the formal guarantees. This rule concerns design well-formedness; it does not convert every causal-dependency edge into a strict proof-order edge.

#### System-Level Theorem (B1-Specialized Appendix G Statement)

Appendix G gives the high-level system theorem in the same standing-assumption regime used by Appendices J–K: B1 is assumed for RTL\_B under the fixed stimulus, so the practical “unique RTL\_B trace” reading is available throughout the main theorem stack. The authoritative finite-prefix trace-set theorem, including the exact reachable-prefix quantification and the conditional use of B4/B5 when  $\epsilon$ -normalization or uniqueness up to finite  $\epsilon$ -stutter is invoked, is Theorem J.1. Therefore, if this Appendix G statement and Theorem J.1 are read together, Theorem J.1 controls; Appendix G is the high-level summary of the same B1-standing finite-prefix theorem regime.

Fix an initial state and a fixed external stimulus/environment behavior for both the prescribed source reference model  $\text{Sys\_ref}$  and  $\text{RTL\_B}$ , with  $\text{Sys\_ref}$  as defined in Definition Core.1 and Appendix J, Remark 2. (The raw rendezvous model  $\text{Sys}$  is the special case obtained when  $\text{Sys\_ref} = \text{Sys\_B}$  and all effective capacities are zero.)

Interpret “fixed stimulus” per Appendix I.2, “Fixed-stimulus interpretation (reactive environments).” Let  $\text{Reach\_post}$  /  $\text{Reach\_ref}$  denote the corresponding finite reachability sets with no fair-extension qualifier (Definition J.Reach, Appendix J).

Assume the Appendix I / Appendix L witness machinery and WC are available, together with the local certificate package of Definition J.LocalGWR: J.PCert for every process, J.CCert for every explicit abstract channel, J.ACert for every explicitly atomic unit, the applicable environment/reset certificates, and J.LocalAgree. Also assume the capacities/occupancies/enableness semantics of  $\text{Sys\_ref}$ , finite capacities and B-faithfulness at every explicit abstract boundary, the standing  $\text{RTL\_B}$  determinism assumption B1 under the fixed stimulus, and—on the normal compositional route—QualifiedSynchronousModel\_ $\text{Sys\_ref}$ (I), so Core.QSync-Ref supplies the finite L2 settled snapshot used by J.RegPhaseNF/J.UnitExt. A directly validated J.GWR package may instead carry equivalent per-Horizon settling evidence. No B2/B3 or generic fair-extension premise is used for either finite-prefix direction; use B4/B5 only for replay  $\epsilon$ -normalization or uniqueness up to finite  $\epsilon$ -stutter not discharged by the QSync same-cycle settle. Complete/fair execution obligations belong to later theorems that explicitly require them.

Then the B1-specialized system-level result is:

(Post  $\rightarrow$  Ref existence) For every  $\tau_{\text{post}} \in \text{Reach\_post}$  equipped with a local certificate package satisfying Definition J.LocalGWR and the standard source-side QualifiedSynchronousModel\_ $\text{Sys\_ref}$ (I) premise of J.GWR-Comp, Theorem J.GWR-Comp constructs  $W$  with GlobalWitnessRealizable( $\tau_{\text{post}}, W$ ), and there exists  $\tau_{\text{ref}} \in \text{Reach\_ref}$  such that  $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}, \text{system}}$  under E1–E5. A directly validated J.GWR package may instead supply the equivalent checked per-Horizon settling evidence.

(Ref  $\rightarrow$  Post existence, annotated RTL-consistent prefixes only) For every  $\tau_{\text{ref}} \in \text{Reach\_ref}$  equipped with a selected  $\text{RTL\_B}$  prefix  $\tau_{\text{post}}^* \in \text{Reach\_post}$  and a compatible witness/annotation package satisfying the annotated RTL-consistency requirement of Theorem J.1 (RTLConsistentRefPrefix), there exists  $\tau_{\text{post}} \in \text{Reach\_post}$  such that  $\tau_{\text{ref}, \text{system}} \approx \tau_{\text{post}, \text{system}}$ .

Moreover, because B1 holds in this Appendix G specialization, the matching  $\Sigma$ -label sequence is unique (up to insertion/removal of finite  $\epsilon$ -steps permitted by B4/B5). That uniqueness is only projection uniqueness; it does not by itself determine  $\mu_Q$ , failed-poll instances, issue annotations, request attribution, E4 boundary histories, or the corresponding source-side witness prefix/state in  $\text{Sys\_ref}$ .

Proof sketch. Appendix I proves the process-local replay and finite-preparation facts packaged by J.PCert. Theorem J.GWR-Comp combines those facts with J.CCert, J.ACert, environment/reset evidence, and the operational frame/commutation lemmas to construct one reachable  $\text{Sys\_ref}$  witness and J.GWR package. Proposition J.0 then lifts the chosen compatible trace pair to system-level  $\approx_{\text{sys}}$ , and Theorem J.1 supplies the finite reachable-prefix trace-set statement. Thus the main result is explicitly about sets of finite reachable execution prefixes under fixed stimulus, not about fair-extension-qualified prefixes. The Ref  $\rightarrow$  Post direction is not a symmetric arbitrary-source construction; it applies only to annotated

RTL-consistent source prefixes, and under this Appendix G specialization B1 supplies only the unique RTL\_B  $\Sigma$ -trace against which the projection component of that consistency is judged. The compatible annotation package remains a separate witness obligation and is not inferred from bare  $\Sigma$ -label matching.

## Practical Implication (“No Surprises” Guarantee)

If a verification environment exercises only the alphabet  $\Sigma$  (signals, message ports, synchronization calls) and does not test internal latency, then—under the assumptions of Theorem J.1 in Appendix J—the prescribed source reference model Sys\_ref and the post-HLS RTL agree at the chosen observable boundary in the following qualified finite-prefix sense:

- for every  $\tau_{\text{post}} \in \text{Reach\_post}$  whose local certificate package satisfies Definition J.LocalGWR and whose selected Sys\_ref satisfies the source-side Core.QSync premise of J.GWR-Comp, Theorem J.GWR-Comp constructs a J.GWR package and a matching  $\tau_{\text{ref}} \in \text{Reach\_ref}$ ; a directly validated J.GWR package carrying equivalent per-Horizon settling evidence may be used instead; and
- in the reverse direction, only those source-side  $\tau_{\text{ref}} \in \text{Reach\_ref}$  that are annotated RTL-consistent in the sense of Theorem J.1—i.e., equipped with a selected RTL\_B prefix  $\tau_{\text{post}}^* \in \text{Reach\_post}$  and a compatible witness/annotation package satisfying RTLConsistentRefPrefix, not merely a bare  $\Sigma$ -projection match—are guaranteed to have a matching reachable  $\tau_{\text{post}}$ .

**Direction-of-proof clarification for source-side safety signoff.** Although users often describe this use case as transferring a pre-HLS result “from Ref to Post,” the deductive direction ordinarily used for safety is the Post  $\rightarrow$  Ref clause. If every reachable RTL\_B prefix has a matching Sys\_ref prefix with the same canonical  $\Sigma_{\text{DUT}}$  observation, then RTL\_B introduces no  $\Sigma_{\text{DUT}}$ -observable safety behavior absent from Sys\_ref. A safety property proved for every covered Sys\_ref prefix therefore transfers without the restricted Ref  $\rightarrow$  Post clause. When Definition J.RefProjCover holds, one complete pre-HLS execution covers those source-reference observations and can support the same transfer. The restricted Ref  $\rightarrow$  Post clause instead concerns realization of a selected source behavior by RTL\_B and remains annotation-dependent.

## Latency-sensitive testbench caveat

A testbench that branches on raw cycle counts, uses timeout watchdogs, checks performance/latency bounds, or otherwise changes its stimulus or pass/fail behavior based on timing not represented in  $\Sigma$  is not covered by the same-stimulus trace-compatibility guarantee unless that timing dependence is itself modeled through explicit  $\Sigma$ -visible causal mechanisms such as synchronization events, handshakes, or message-passing transactions. Such checks may still be useful, but they should be treated as latency-sensitive testbench logic, isolated as described in Appendix D, or performed as post-HLS / RTL-level performance checks rather than as properties transferred automatically from an arbitrary pre-HLS simulation run.

Accordingly, the precise “no surprises” reading distinguishes two source-side uses. For safety, no  $\Sigma$ -observable behavior can appear in RTL\_B under the fixed stimulus without a matching behavior in Sys\_ref; therefore universal Sys\_ref safety, or one complete source run satisfying Definition J.RefProjCover, transfers through Post  $\rightarrow$  Ref as stated by Corollary J.1-Safe. Separately, a claim that a particular source-side behavior is realized by RTL\_B uses the restricted Ref  $\rightarrow$  Post clause and requires RTLConsistentRefPrefix or the applicable Appendix L snooping / witness-selection construction.

Thus, for a  $\Sigma$ -only testbench under the same fixed stimulus, any mismatch found on RTL\_B corresponds

to a matching mismatch on `Sys_ref`. A source-side safety pass justifies `RTL_B` through Corollary J.1-Safe under universal source verification or `J.RefProjCover`, provided `J.TraceCertCov` supplies the applicable `Post → Ref` construction evidence, including source-side `Core.QSync` on the normal compositional route or equivalent checked per-Horizon settling evidence on the direct `J.GWR` route; realization of a selected source trace still uses the restricted reverse direction. In the standard qualified-synchronous normal-compositional route, `J.QSyncTraceInd` supplies the cycle-aligned `J.TraceCertCov` evidence and Lemma `J.CycleIdealCover` / Corollary J.1-Safe-`QSync` extend safety predicates satisfying Definition `J.IdealPrefixClosed` to finite observable prefixes within those cycles, without requiring an independent correspondence package for each such intra-cycle ideal. Here `Sys_ref` means `Sys_B` in the ordinary bounded-FIFO case and `Sys_B^elab` in the elaborated back-annotated cases of Appendix B / Appendix Q. See Appendix J / Theorem J.1, Lemma `J.CycleIdealCover`, and Corollaries J.1-Safe/J.1-Safe-`QSync` for the full execution-level statements.

Appendix G states the trace-compatibility rules and a B1-specialized high-level implication. Appendix I supplies process-local deterministic replay and finite preparation certificates; Appendix J proves that these local process facts, together with local channel and atomic-interaction certificates, construct a single globally realizable source witness, and then proves the system theorem for the prescribed source reference model. Taken together, these appendices establish the stated execution-level correspondence at the chosen observable interfaces under fixed stimulus and the named local/tool obligations. The unrestricted guarantee is `Post → Ref`; the reverse direction ranges only over annotated RTL-consistent source-side behaviors covered by Theorem J.1.

## Appendix H – Possible Criticisms

This section highlights several potential challenges in the verification approach of Appendix G.

### 1. Designer-Managed Shared-Memory Synchronization

Appendix G requires that any memory shared between processes use explicit synchronization primitives inserted by the designer. The HLS tool does not perform static verification of those primitives. In practice, we mitigate this risk by encapsulating shared memories within parameterized library classes (for example, see examples `12_ping_pong_mem` and `15_native_ram_fifo`) that are pre-verified for correct memory access coordination in both the pre-HLS and post-HLS models. Typically such classes will be parameterized for aspects such as element type and memory size. Formally, the Appendix I–K theorems cover shared memories only under the `J.Mem` lowering condition; such library classes should present their value-carrying memory traffic at boundaries satisfying `J.Mem`, or carry their own separate correctness argument outside the Appendix I–K theorems.

### 2. Latency-Sensitive Signal Alignment (“Snooping”)

See Appendix L for detailed discussion on snooping.

### 3. By Default, Matchlib Connections Model a Skidbuffer inside In<> Ports

(Note that this overall document and specifically Appendix G are not intended to be strictly tied to Matchlib. Instead, this document is intended to be applicable to any pre-HLS model written in SystemC using signals, message-passing channels, and synchronization calls.)

Matchlib Connections supports throughput accurate modeling in the pre-HLS simulation. Currently the throughput accurate modeling mechanism relies on the `Pre()` and the `Post()` methods using a 1-place buffer to store transactions that are in flight on `Connections::In<>` ports. This means that by default `In<>` ports function like a skidbuffer. `Connections::Out<>` ports do not introduce any additional storage capacity into the pre-HLS simulation.

By default, Catapult adds skidbuffers on input ports into the post-HLS design, so the formal trace-compatibility relationship described in this document still holds. In other words, the default pre-HLS and post-HLS designs both still model rendezvous semantics, since both explicitly include a structural instantiation of a skidbuffer on input ports. In effect, we define the rendezvous boundary *before* the skidbuffer.

It is possible to remove some or all the skidbuffers during Catapult HLS. Because the abstract rendezvous boundary is defined before the skidbuffer, this choice is orthogonal to the trace-compatibility rules (R1–R5/E1–E5) as long as skidbuffer-internal behavior is not included in  $\Sigma$ . However, to keep the pre-HLS simulation’s port micro-architecture aligned with the RTL configuration, the following compile flag must then be used in this case:

```
-DFORCE_AUTO_PORT=Connections::SYN_PORT
```

This will remove all the skidbuffers from the pre-HLS model. In this case, if some of the skidbuffers are still inserted during HLS, the capacity effects of those specific skidbuffers should be added into `Sys_B` so that the formal trace-compatibility relationship still holds.

In this case the recommended methodology is to use both approaches:

- Use the default throughput accurate mode in Matchlib for most analysis and debug, and for pre-HLS performance verification.
- To keep the pre-HLS simulation’s *port micro-architecture* strictly aligned with the RTL configuration, then use `-DFORCE_AUTO_PORT=Connections::SYN_PORT`, and rerun final verification tests.

## Appendix I – Per-Process Replay and Preparation Proof

### I.1 Overview

This appendix establishes the process-local replay and preparation results used by Appendix J. Under a fixed initial state, fixed external stimulus/environment behavior, and WC-compatible witness, a selected `RTL_B` observable-unit prefix determines the corresponding source-local replay state for each process and a finite process-owned  $\epsilon$  preparation path to the exact dynamic operation participating in the next selected unit. Appendix I does not by itself commit a channel transfer, pair rendezvous or synchronization endpoints, or construct a complete global `Sys_ref` execution; those composition steps are proved in Appendix J by Lemmas `J.Frame–J.UnitExt` and Theorem `J.GWR-Comp`.

Downstream citation discipline. Later appendices may cite Appendix I for Implementation assumption I.A0; Definitions I.ReplayObs and I.CutpointOfferObs; Lemmas I.1, I.ReplayObs-Congruence, I.DR, I.DR', and I.3; and the process-local certificate obligations later packaged as J.PCert. Appendix I does not justify an unrestricted per-process or system-level existence theorem in isolation. A complete matching Sys\_ref prefix is obtained only after Appendix J combines the process-local paths with the common explicit-channel state, endpoint co-presence, atomic grouping, environment/reset evidence, current request-issue evidence, and interface agreement. The converse direction likewise remains restricted to RTLConsistentRefPrefix / Appendix L witness-selected executions.

## I.2 Formal Preconditions

- **Observable alphabet  $\Sigma$**  is as defined in *Normative Formal Core for Appendices G–K*: it consists of the event schemas {Push\_c, Pop\_c, Sync, wait, START\_P, FINISH\_P, Write, Read} instantiated only for channels/signals designated observable at the chosen proof boundary. For Appendices I–K,  $\Sigma$  is the proof-internal observable alphabet of the prescribed Sys\_ref / RTL\_B comparison. Thus all message-passing channels that can participate in Appendix K deadlock reasoning are designated observable at that proof boundary and hence included in  $\Sigma$ . In elaborated cases Sys\_ref = Sys\_B^elab, this proof-internal  $\Sigma$  may include introduced staged/bundle/join channels even when the projection  $\Pi_{elab}$  hides those internal events from the outer DUT/testbench-visible alphabet  $\Sigma_{DUT}$ . Such events are therefore  $\Sigma$ -visible for the proof-internal comparison, but may be projected away from the outer observation boundary.
- **Non-blocking message-passing interpretation.** In both RTL\_B and Sys\_ref, a non-blocking call (PushNB / PopNB) is treated as a poll. A successful poll contributes the same observable event as the corresponding committed Push\_c / Pop\_c on that channel. An unsuccessful non-blocking call (returns “false” / “not completed”) contributes no  $\Sigma$ -event and is modeled as an  $\epsilon$ -step. A non-blocking poll may still assert a  $\Sigma$ -visible endpoint request in the cycle of that call. However, each executed non-blocking call is a distinct dynamic source-level instance  $q \in Q_{exec,P}$  for its owning process P. Accordingly, continuous assertion of an endpoint request with no intervening commit does not by itself collapse later executed non-blocking call instances into an earlier q. Only a poll that succeeds contributes the corresponding committed transfer  $m(q) \in M$  for that call instance. If a request assertion persists without a new executed source-level call instance being entered, its attribution remains with that same currently outstanding q; but repeated non-blocking polls in later loop iterations or later source-level call instances create additional  $q \in Q_{exec,P}$  even if req does not deassert.
- **Silent step** Any internal RTL microstate transition is written  $\epsilon$ . In particular, unsuccessful non-blocking polls, arbitration bookkeeping, and pipeline advances are  $\epsilon$ -steps.
- **Finite-pipeline-depth premise (FPD)** There exists a finite constant  $pipe\_depth(P) = D_P < \infty$  such that once P begins internal micro-progress toward its next observable  $\Sigma$ -action (or that action is locally enabled), P executes at most  $D_P$  consecutive  $\epsilon$ -steps before committing a  $\Sigma$ -action or reaching a stable wait with no further  $\epsilon$ -step available until external/peer conditions change. In particular, a process cannot perform an unbounded number of  $\epsilon$ -only failed non-blocking polls while making no progress toward any  $\Sigma$ -action; designs that can spin indefinitely without reaching either a  $\Sigma$ -action or a stable wait state violate FPD/B4 and are outside the model.
- **Weak-fairness premise (WF)** If a micro-operation remains continuously enabled, the scheduler eventually selects it for progress. For an already-issued blocking message request whose BoundaryReq is asserted and whose ChannelEnabled predicate remains true, this means the enabled transfer is eventually allowed to commit; it does not mean that the same source-level operation is “issued” again. This is an assumption on the execution/scheduling environment (see

Appendix N), not an automatic guarantee of “clock-synchronous RTL.” In practice it requires that any arbiters/back-pressure logic that can affect whether an enabled operation is selected or allowed to commit must be fair in the sense of B2/B3.

- **Finite-progress invariants (B3)** For every channel  $c$ , assume producer and consumer obligations  $P\_push(c)$  and  $P\_pop(c)$  hold. In a buffered full or empty condition, these obligations guarantee the matching complementary discharge under their persistent two-endpoint-request premises:  $P\_push(c)$  yields a  $Pop\_c$  commit that frees space, and  $P\_pop(c)$  yields a  $Push\_c$  commit that provides data. They do not directly assert that the original stalled  $Push\_c$  or  $Pop\_c$  commits. After the complementary discharge, E4 determines whether the still-pending original operation is  $ChannelEnabled$ , and B2 supplies eventual selection/commit while that enablement remains continuous. For  $B(c)=0$ , the discharge is the rendezvous completion itself. B3 is an assumption on the execution/scheduling environment (e.g., fairness of arbiters/back-pressure; see Appendix N), not a consequence of the cycle-by-cycle R-rules.
- **Channel enablement semantics.** Let  $q \in Q\_exec, P$  be a dynamic source-level message-operation instance of its owning process  $P$  on message-passing channel  $c$ , and let  $kind(q) \in \{Push\_c, Pop\_c\}$  denote its endpoint direction. If  $q$  has issued a persistent blocking request to transfer on  $c$ , then  $BoundaryReq(q, \sigma)$  holds at a settled boundary-evaluation point  $\sigma$  ( $BoundaryEvalPoint$  in the Core sense): the endpoint-owned boundary request remains asserted, and the payload is stable if  $q$  is a  $Push$ . The channel-side commit-enabling predicate for  $q$  is determined as follows.

Rendezvous case,  $B(c)=0$ . A  $Push\_c$  operation instance  $q$  is channel-enabled exactly when  $BoundaryReq(q, \sigma)$  holds and the matching  $Pop\_c$  endpoint boundary request is asserted in the same settled boundary-evaluation cycle. A  $Pop\_c$  operation instance  $q$  is channel-enabled exactly when  $BoundaryReq(q, \sigma)$  holds and the matching  $Push\_c$  endpoint boundary request is asserted in the same settled boundary-evaluation cycle. Thus, when both endpoint requests are asserted in that cycle, the matched  $Push\_c(v)$  and  $Pop\_c(v)$  operation instances are eligible for selection as a rendezvous pair. If the scheduler selects that enabled rendezvous pair in that cycle, the matched  $Push\_c(v)$  and  $Pop\_c(v)$  commit as a same-cycle rendezvous pair and contribute the corresponding  $\Sigma$ -events. If the pair remains continuously enabled, B2 supplies eventual selection.

Buffered case,  $B(c)>0$ . The channel-side eligibility predicate for  $q$  is exactly the FIFO availability predicate conjoined with  $BoundaryReq(q, \sigma)$ :

for  $kind(q)=Push\_c$ :  $ChannelEnabled(q, \sigma) \Leftrightarrow BoundaryReq(q, \sigma) \wedge occ\_c(\sigma) < B(c)$  (“not full”),  
and

for  $kind(q)=Pop\_c$ :  $ChannelEnabled(q, \sigma) \Leftrightarrow BoundaryReq(q, \sigma) \wedge occ\_c(\sigma) > 0$  (“not empty”).

Here  $ChannelEnabled$  means that the abstract channel boundary permits the transfer to be selected; it does not by itself assert that the transfer has already committed in the current cycle.

Modeling note (fall-through / simultaneous empty/full cases). The buffered abstraction for  $B(c)>0$  is cycle-boundary, non-fall-through. A  $Push$  and a  $Pop$  may both be selected and commit in the same cycle when the FIFO begins the cycle neither empty nor full ( $0 < occ\_c(t) < B(c)$ ); in that case both commits are legal and the occupancy is unchanged. The abstraction does not permit a  $Pop$  to be selected in a cycle that begins empty solely because a  $Push$  is also selected in that same cycle, nor does it permit a  $Push$  to be selected in a cycle that begins full solely because a  $Pop$  is also selected in that same cycle. If an implementation uses a fall-through FIFO

(allowing same-cycle pass-through when empty), model that behavior explicitly (e.g., as an explicit bypass/rendezvous path composed with the bounded FIFO), so that the  $\Sigma$ -level committed-transfer trace matches the intended cycle semantics.

Operational interpretation (availability, selection, and ready/valid):  $\text{occ}_c(t)$  is an abstract quantity derived from  $\Sigma$ -visible boundary commits for  $B(c)>0$  buffered channels; processes do not read  $\text{occ}_c(t)$  directly. For  $B(c)=0$  rendezvous channels, the corresponding channel-side availability fact is the matching endpoint request predicate in the same settled boundary-evaluation cycle. In both  $\text{Sys\_ref}$  and  $\text{RTL\_B}$ ,  $\text{ChannelEnabled/ChannelDisabled}$  records the settled abstract boundary availability/eligibility predicate. A commit occurs only when an enabled operation, or enabled rendezvous pair, is selected by the scheduler/arbiter for that cycle.

Here  $\text{Sys\_ref}$  denotes the prescribed source reference model at the chosen abstract boundary:  $\text{Sys\_B}$  in the ordinary case, or  $\text{Sys\_B}^{\text{elab}}$  in the elaborated back-annotated cases. A concrete realization is correct only if its committed ready/valid transfers are legal with respect to the corresponding rendezvous or buffered availability predicate at that chosen boundary, and if every transfer that remains continuously eligible is eventually selected as required by B2. Thus E4 is a safety/legality condition on which transfers may commit, while B2 is the progress/fairness condition that prevents continuously eligible transfers from starving.

No additional hidden availability predicate is part of the channel semantics. Any cross-channel coordination, arbitration, grant, or back-pressure policy in the realization must be represented either

(1) as scheduler/arbiter selection among already  $\text{ChannelEnabled}$  actions, governed by B2, or  
 (2) as explicit abstract boundary structure in  $\text{Sys\_B}^{\text{elab}}$  with its own  $B(\cdot)$ ,  $\text{occ}_\cdot(t)$ ,  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled/ChannelDisabled}$ , and B3 obligations. Such coordination must not be encoded as an unmentioned extra conjunct that silently changes the E4 availability predicate at an otherwise ordinary channel boundary.

Therefore, when the boundary request is true and the relevant rendezvous or FIFO availability predicate holds continuously, the transfer is continuously enabled; by weak fairness (B2), the corresponding commit occurs after a finite, though not necessarily bounded, delay.

- **No ill-formed signal IO.** Every observable signal Read/Write has a unique real bounding synchronization operation on each dynamic path on which it occurs; otherwise, the design is ill-formed.  $\text{START\_P}$  may be used as an initial anchor only when it is included as a real synchronization event in  $S$ , and  $\text{FINISH\_P}$  may be used as a terminal anchor only when the process actually terminates. The formal model does not use a virtual synchronization event at  $\infty$ . (From RULE 1 and RULE 2.)
- **Fixed-stimulus interpretation (reactive environments).** If the “same stimulus” includes a reactive testbench and/or peer processes, it is interpreted as the same global  $\Sigma$ -causal behavior (same externally supplied inputs, and the same peer/environment responses to the same prior  $\Sigma$ -visible history), not as a function of P-local history alone. In Appendix I this interpretation is used only as a witness-replay condition: when a next  $\text{RTL\_B}$  event  $e$  is already fixed in the chosen post trace prefix, any peer/environment enabling needed for  $e$  is part of that same fixed global  $\Sigma$ -causal behavior and is therefore replayed consistently for the matching  $\text{Sys\_ref}$  construction. This is not a standalone assumption that arbitrary  $\text{Sys\_ref}$  continuations are globally live.

- **WC (Witness-compatible non-blocking outcomes).** Because failed polls are modeled as  $\varepsilon$  (unobservable), all correspondence results in Appendices I–K are stated for witness-compatible comparisons: any  $\varepsilon$ -modeled non-blocking poll outcome (and any other  $\varepsilon$ -modeled internal choice that can affect the next  $\Sigma$ -visible control flow / next  $\Sigma$ -visible blocking frontier) is fixed/selected to be consistent with the compared RTL\_B execution prefix  $\tau_{\text{post}}$  (e.g., via the Appendix L witness-selection/snooping wrapper). WC also supplies the  $\mu_Q$  matching for  $\varepsilon$ -modeled failed non-blocking call instances when E3/E5 or the proof construction refers to  $q \in Q_{\text{exec},P}$ ; those failed  $q$ 's are not inferred from  $\Sigma$  alone.
- **Remark (NB-CF as a sufficient condition).** If a design satisfies the following non-blocking control-flow property—failed PopNB/PushNB polls do not influence the next  $\Sigma$ -visible control flow/frontier—then the witness is trivial and WC holds automatically. Equivalently: between two consecutive  $\Sigma$ -events of a process, inserting/removing any finite sequence of failed PopNB/PushNB polls may change only internal  $\varepsilon$ -behavior/timing, not which  $\Sigma$ -visible action (Push/Pop/synchronization anchor) is next in program order.

### I.3 Auxiliary Lemmas

Implementation assumption I.A0 (RTL message-issue discipline).

The post-HLS RTL\_B execution satisfies the full E3 obligation as an implementation-side scheduling obligation: (i) per-interface issue order is preserved for all message operations; (ii) for distinct interfaces, the no-reverse rule holds; and (iii) issue is prefix-closed under source order, so at every compared cutpoint a source-later issued message operation implies that every required source-earlier message operation has also issued, with Issue timestamps ordered accordingly. This condition is not derived inside Appendix I; it is the RTL/tool-compliance obligation corresponding to the scheduling rules stated in the main text and Appendix G. For hand-written RTL or other non-HLS implementations, I.A0 is the corresponding proof/qualification obligation on that implementation.

#### **Lemma I.1 (Bounded $\varepsilon$ -chain to $\Sigma$ -action or stable wait).**

Assume the finite-pipeline-depth / finite-internal-stutter premise FPD/B4. From any reachable well-formed cycle-boundary state of P in the Appendix I construction, if P is not already stably waiting on an external/peer enabling condition, then within at most  $D_P$  consecutive P-owned epsilon-steps of internal progress, P either:

- (i) commits its next observable Sigma-action, or
- (ii) reaches a stable waiting state in which no further P-local epsilon-step is available until some external, peer, channel, or stimulus condition changes the relevant enabling predicates.

Equivalently, P cannot produce an infinite P-local epsilon-only execution segment from a reachable well-formed state while remaining internally enabled. Any unbounded delay that remains after the stable-wait point is therefore attributable to external/channel/stimulus enabling and is governed by WF/B2 and B3, not by local epsilon-stutter. When a clock-cycle bound is required, the proof uses the separate B5 bound  $C_{\text{sys}}$ , or a proved bucket-semantic charging lemma deriving a per-process cycle bound  $C_P$  from  $D_P$ ; Lemma I.1 does not identify one epsilon-step with one clock cycle.

Proof. Direct from FPD/B4 for the epsilon-step conclusion, together with the separately stated B5 or bucket-semantic bridge only when a cycle-count conclusion is used. ■

Lemma I.2 (Eventual space / data). Assume P is blocked on either:

- Push\_c with  $\text{occ}_c = B(c)$ , P's Push\_c request remains asserted with stable payload for the outstanding transfer, and the complementary endpoint continuously requests the matching Pop\_c until the matching discharge event (Pop\_c endpoint request asserted; Pop\_c boundary request remains true); or
- Pop\_c with  $\text{occ}_c = 0$ , P's Pop\_c request remains asserted for the outstanding transfer, and the

complementary endpoint continuously requests the matching Push\_c until the matching discharge event (Push\_c endpoint request asserted with stable payload; Push\_c boundary request remains true). Then a complementary Pop\_c (respectively Push\_c) commits within finite time.

Proof. (By B3 / Appendix N.) In either case, P is stalled at a blocking channel call while its persistent request for that outstanding transfer remains asserted until commit, with stable payload if it is a Push. By the additional premise, the complementary endpoint also continuously requests the matching operation until the corresponding discharge event, with its boundary request remaining true. Therefore the interval-scoped premises of P\_push/P\_pop (Appendix N) hold, and the matching complementary discharge must eventually occur: for Push\_c at full, some complementary Pop\_c eventually commits, freeing space; for Pop\_c at empty, some complementary Push\_c eventually commits, providing data. The premise does not require either endpoint request or Push payload to remain asserted after that matching transfer has committed.

■

**Definition I.ReplayObs (replay-derived  $\Sigma$ -relevant source-state slice).**

Fix a process P, a fixed external stimulus, and a fixed witness w satisfying WC for the chosen observable boundary. Define  $\text{ReplayObs}_{\{\Sigma, P, w\}}(\sigma)$  to be the logical P-local replay checkpoint represented at operational state  $\sigma$ : the tuple of source-level facts determined by the complete P-local observable-unit replay history under that stimulus and witness and needed to determine later P-local  $\Sigma$ -visible behavior or replay-derived Appendix I–K predicates. ReplayObs is a projection of  $\sigma$ , not the complete raw operational source state at  $\sigma$ . Its control-position field is the logical replay control position immediately after the last matched observable unit (or at the selected initial/reset checkpoint), and its value fields contain only replay-relevant logical source variables and latched values reconstructed from committed history, the fixed stimulus, and w. Optional outcome-independent L1 preparation may advance the raw operational cursor, calculate prepared temporaries, or assert source-later requests without changing this logical checkpoint, but only when the field-sensitive footprint, alias-closure, and NoDependentUse obligations certify that the segment is replay-inert.

The tuple includes: the logical replay control position and reset epoch; replay-relevant source variables and latched values used to evaluate future control predicates  $\text{Pred}_x$ , payload/value expressions, anchored Write values, and anchored Read labels; witness-selected epsilon outcomes that can affect later  $\Sigma$ -visible control flow;  $Q_{\text{exec}, P} / Q_{\Omega, P}$  dynamic-operation attribution facts determined by the replay prefix; completed/realized-item status; candidate  $\text{NextFront}_P$ ; anchored signal values and returned Pop/NB results reconstructed from the committed history and witness; and P's normal-completion status. Raw preparation progress—including the operational cursor, prepared local temporaries, current issue attribution, and current request assertions—is not part of ReplayObs. In particular,  $\text{Pending}_P$ ,  $\text{Front}_P$ , and  $\text{BoundaryReq}$  are not ReplayObs fields. Define  $\sigma \sim^{\text{replay}}_{\{\Sigma, P, w\}} \sigma'$  iff  $\text{ReplayObs}_{\{\Sigma, P, w\}}(\sigma) = \text{ReplayObs}_{\{\Sigma, P, w\}}(\sigma')$ . The replay-derived source-state slice of  $\sigma$  is its equivalence class  $[\sigma]^{\text{replay}}_{\{\Sigma, P, w\}}$ .

**Replay-abstraction and selected-normalization convention.** Not every epsilon-step is replay-inert. A witness-fixed epsilon transition, such as a failed non-blocking poll that writes a returned status or another replay-relevant field, may change ReplayObs. WC fixes the selected outcome; it does not make that transition neutral. Deterministic normalized replay therefore uses paired congruence: from equal logical replay checkpoints, corresponding transitions under the same witness produce equal successor checkpoints. A selected finite epsilon-normalization may contain both (a) certified replay-inert preparation segments, which leave each logical checkpoint unchanged, and (b) witness-fixed replay-changing transitions, which must map equal checkpoints to equal successors. The selected normalization convention is proof-relevant and must satisfy this congruence requirement; the document does not

assert that every legal normalization has the same raw operational endpoint or that each normalization leaves one execution's `ReplayObs` unchanged.

**Definition I.CutpointOfferObs (current blocking-offer slice).**

For a reachable source state  $\sigma$  at the selected abstract boundaries, define  $\text{CutpointOfferObs}_{\{P,E\}}(\sigma)$  to be the  $P$ -owned current blocking-offer facts: the uncommitted blocking message-operation frontier items whose endpoint requests are asserted, their `Pending_P` and `Front_P` membership, `BoundaryReq` values, owning dynamic message-call instances, evaluated explicit boundaries, endpoint directions, reset epochs, and the stable Push payload obligation where applicable. `CutpointOfferObs` is indexed by the actual operational issue schedule and raw L1 preparation progress and is not determined by the committed observable-unit history alone: two legal source executions can share the same logical `ReplayObs` checkpoint while differing in how far outcome-independent preparation has advanced and whether a source-later blocking operation has already issued. Appendix J constructs these request assertions through its phase-normal preparation/issue actions, represents the normalized cutpoint slice by the residual-offer set  $O$ , proves source reachability through `J.OfferReach`, and audits the corresponding RTL request inventory through `J.OfferConf`. The complete  $K$ -facing cutpoint observation therefore combines replay history  $H$  with current residual offers  $O$ ; deterministic replay never manufactures  $O$  from  $H$ .

This quotient intentionally ignores purely internal diagnostic state and epsilon-only micro-timing state that cannot affect any future  $\Sigma$ -visible label, replay-derived frontier predicate, payload/value expression, or Appendix I–K predicate. `ChannelEnabled` / `ChannelDisabled` facts are not independently added to `ReplayObs`. They are obtained by combining the logical `ReplayObs` checkpoint with the operational I.CutpointOfferObs slice—including `BoundaryReq`, `Pending_P`, and `Front_P`—and with the corresponding channel-state facts in the chosen `Sys_ref` boundary model, such as `B(c)`, `occ_c(\sigma)`, rendezvous complementary endpoint requests, and any explicit `Sys_B^elab` staged/bundle/join boundary state. For cadence-sensitive non-blocking polling sites, such as `retry`/`timeout` counters whose externally visible behavior depends on the number or cadence of failed polls, this definition is applied only after the site has been handled as an isolated or snoop-aligned latency-sensitive local protocol block under Appendix D / Appendix L. The ordinary Appendix I source-core proof does not derive `retry`/`timeout` behavior from an unconstrained hidden failed-poll cadence.

**Lemma I.ReplayObs-Congruence (one-step closure of the replay-derived slice).**

Fix  $P$ , the external stimulus, the chosen observable boundary, and a witness  $w$  satisfying `WC`. Let  $\sigma_1$  and  $\sigma_2$  be source states of  $P$  such that  $\text{ReplayObs}_{\{\Sigma,P,w\}}(\sigma_1) = \text{ReplayObs}_{\{\Sigma,P,w\}}(\sigma_2)$ . Suppose that, from  $\sigma_1$  and  $\sigma_2$ , the same next  $P$ -local observable unit  $u$  is produced under the same witness  $w$  and the same fixed-stimulus interpretation, where  $u$  may be a committed Push/Pop value, successful non-blocking transfer, synchronization/wait unit, anchored Read/Write unit, `START_P`/`FINISH_P` unit, `R2C` fixed-point observation unit, or another explicitly atomic  $\Sigma$ -visible boundary unit. Let  $\hat{\sigma}_1$  and  $\hat{\sigma}_2$  be the corresponding canonical immediate post-unit states. Then their `ReplayObs` checkpoints are equal. If  $\sigma_1'$  and  $\sigma_2'$  are instead the epsilon-quiescent endpoints selected from  $\hat{\sigma}_1$  and  $\hat{\sigma}_2$  by the fixed proof-relevant normalization convention, require that convention to preserve replay equivalence as specified above: replay-inert preparation segments leave the logical checkpoint unchanged, while corresponding witness-fixed replay-changing epsilon transitions map equal checkpoints to equal successors. Under that selected convention:

$$\text{ReplayObs}_{\{\Sigma,P,w\}}(\sigma_1') = \text{ReplayObs}_{\{\Sigma,P,w\}}(\sigma_2').$$

*Proof.* By case analysis on the next source construct and on the kind of observable unit  $u$ .

For deterministic local computation, equality of the logical replay control position and of all replay-relevant variables/latched values feeding future predicates, payloads, anchored writes/reads, guards, and replay-derived proof predicates implies equality of the updated `ReplayObs` fields. Raw L1 cursor

movement, prepared temporaries, and request assertions are outside ReplayObs and may differ only under the certified replay/preparation noninterference obligations.

For anchored sequential signal reads/writes and emitted R2C fixed-point units, the relevant value and component-local observation identity are fixed by the common stimulus, witness, anchor/fixed-point semantics, and common unit  $u$ . Therefore all future-relevant latched values, anchored-label facts, and R2C replay facts agree after the unit.

For committed blocking Push/Pop and synchronization units, the common unit  $u$ , together with equality of the logical replay control position, relevant payload variables, replay-prefix dynamic-operation attribution, and completed-item history, determines the same replay-derived control/data update and candidate NextFront\_P. Whether a not-yet-committed blocking operation was already issued before the unit is not a ReplayObs fact and is not concluded equal here; that current assertion state belongs to CutpointOfferObs and is established by explicit operational issue evidence in Appendix J.

For successful non-blocking PushNB/PopNB, the successful transfer is included in  $u$  as the corresponding committed Push\_c(v) / Pop\_c(v) unit. The common unit and equal current ReplayObs determine the same control, returned-data, completed-item, dynamic-attribution, and candidate-next-front updates.

For failed non-blocking PushNB/PopNB, the failed poll is an epsilon-step. If NB-CF holds, failed polls cannot affect the next  $\Sigma$ -visible control flow, dynamic-operation selection, guard values, payload values, or candidate NextFront, so any difference is outside ReplayObs. If NB-CF does not hold but WC is supplied by Appendix L, the selected failed-poll outcome may update returned-status, witness, or other replay fields. WC fixes that outcome; it does not make the step replay-neutral. The normalized proof therefore pairs the corresponding failed-poll transitions and uses equal prior checkpoints plus the same witness-selected outcome to obtain equal successor ReplayObs. Cadence-sensitive sites remain subject to the Appendix D / Appendix L isolation or snoop-alignment requirement.

For any other epsilon-only implementation choice, WC requires that every outcome capable of affecting later  $\Sigma$ -visible behavior or replay-derived Appendix I–K predicates is selected by  $w$  and is causally consistent with the matched prefix. A certified replay-inert preparation step preserves each logical checkpoint by framing. A witness-fixed replay-changing step need not be neutral, but equal prior ReplayObs checkpoints and the same selected outcome must produce equal successor checkpoints. Current request assertions remain outside this conclusion.

Therefore equality of ReplayObs is preserved by the canonical immediate same-unit replay step and, for normalized cutpoints, by the two-sided congruence of the exact selected normalization paths. This is not a claim that each selected normalization preserves one side's ReplayObs individually, nor a claim of global normalization confluence. No equality of CutpointOfferObs is asserted. ■

**Lemma I.DR (Deterministic replay / replay-derived source-state determinacy under fixed stimulus + witness).** Fix a process  $P$ , a fixed external stimulus, a fixed witness  $w$  satisfying WC, and the proof-relevant epsilon-normalization convention selected for the theorem instance. Consider two executions of the same  $P$  source code that start from the same initial source-level state and are run with that same witness. If their chosen  $P$ -local observable-unit replay projections are the same, then for every  $k$ ,  $\text{ReplayObs}_{\{\Sigma, P, w\}}$  at the selected epsilon-quiescent cutpoint after the first  $k$   $P$ -local observable units is identical in the two executions. The selected normalization convention must satisfy the paired replay-congruence requirement of Definition I.ReplayObs and Lemma I.ReplayObs-Congruence; arbitrary maximal epsilon-extensions are not quantified here. “ $P$ -local observable-unit replay projection” means the sequence obtained from the document's event-matching / Appendix J observable-unit convention: explicitly atomic same-cycle transactions are grouped as single units; independent same-cycle bucket members are not given an intrinsic order by the bucket itself; and any ordering used for deterministic replay is the selected witness/replay order, together with  $w$  and  $\mu_Q$  for epsilon-modeled failed non-blocking  $q \in Q_{\text{exec}, P}$  when those instances are needed by E3/E5. Thus Lemma I.DR is neither a

determinacy theorem from a bare unordered  $\Sigma$ -bucket projection nor a theorem that reconstructs current `CutpointOfferObs/Pending_P/Front_P/BoundaryReq` from committed history.

Proof. At  $k = 0$  the executions start from the same initial source-level state, so their logical `ReplayObs_ $\{\Sigma, P, w\}$`  checkpoints agree. For the induction step, assume the checkpoints agree after  $k$  P-local observable units. The two executions have the same  $(k+1)$ -st P-local observable unit and are run under the same fixed stimulus and witness. Lemma I.ReplayObs-Congruence first gives equal canonical immediate post-unit checkpoints and then applies the paired congruence of the two exact selected normalization paths. Therefore the selected normalized `ReplayObs` checkpoint agrees after every finite P-local observable-unit prefix. ■

Lemma I.DR' (Immediate post-unit deterministic replay under fixed stimulus + witness). Fix a process  $P$ , a fixed external stimulus, and a fixed witness  $w$  satisfying WC for the chosen observable boundary.

Consider two executions of the same  $P$  source code that start from the same initial source-level state and are run with that same witness. Suppose that their chosen P-local observable-unit replay projections through the first  $k$  units are the same. Stop each execution at the canonical immediate post-unit semantic state: after the transition, or explicitly atomic transition group, that realizes the  $k$ th P-local observable unit, and before any optional following L1 preparation or epsilon-normalization. Then the two executions agree on the P-owned logical `ReplayObs` checkpoint at that point, including:

- the logical replay control position, not the raw post-preparation operational cursor;
- all replay-relevant source variables and latched values that can influence future `Predx` predicates, payload/value expressions, or anchored Read/Write values, excluding prepared temporaries that are certified outside the replay projection;
- the dynamic-operation attribution and completed/realized-item facts determined by the replay prefix;
- the candidate `NextFrontP`, returned-result, anchored-signal, reset-epoch, and normal-completion facts determined by that prefix.

This lemma does not assert that the immediate post-unit state is epsilon-quiescent or drain-closed, and it does not identify that state with a later epsilon-normalized cutpoint. Lemma I.DR remains the corresponding theorem for endpoints selected by the fixed paired-congruent normalization convention.

Neither lemma determines raw L1 preparation progress or the current asserted blocking-request inventory: `PendingP`, `FrontP`, and `BoundaryReq` belong to `I.CutpointOfferObs` and, at the  $K$ -facing normalized cutpoint, are represented by `O` and justified by `J.OfferReach/J.OfferConf`.

Proof. By induction on  $k$ . The base case follows from equality of the initial source-level state. For the induction step, equality of the prior immediate `ReplayObs` state, together with the same fixed stimulus, witness-selected  $\epsilon$  outcomes, dynamic-instance attribution, and next observable unit, determines the same finite source transition segment through the transition or explicitly atomic transition group that realizes the next unit. Therefore the two canonical immediate post-unit states agree on every listed replay-derived fact. The proof makes no claim about issue-schedule-dependent `CutpointOfferObs`, and no maximal  $\epsilon$ -closure is invoked. ■

#### I.4 Local Preparation Construction for Finite Observable-Unit Prefixes (Post $\rightarrow$ Ref support)

Fix a process  $P$ , a finite selected `RTLB` prefix  $\tau_{\text{post}}$ , and a finite rank-independent required-prefix candidate  $H$  as defined by `J.RequiredPrefixCandidate` below. Let `ReplayP(H,w)` be the deterministic immediate post-unit source replay state obtained from the P-local projection of  $H$  under the fixed stimulus, WC witness  $w$ , and  $\mu_Q$  dynamic-instance matching. Let  $\gamma$  be a not-yet-matched observable unit in which  $P$  participates, and let  $q_P(\gamma)$  denote  $P$ 's matched dynamic source operation, if  $\gamma$  has a process-owned operation. This local construction is indexed without reference to `DepJ`, `J.DepAcyclic`, or `J.CanonicalRank`; Appendix J later instantiates it at the prefixes selected by the validated construction order.

The process-local input to the construction consists only of P's source code and state, P-owned endpoint request state, the fixed witness and dynamic identities, and read-only summaries of the incident explicit boundaries named by  $\gamma$ . It does not include a premise that  $\gamma$  is globally enabled or that a complete  $\text{Sys\_ref}$  extension exists.

**Definition I.DeferredNBHole (carried non-blocking decision obligation).**

A deferred hole for dynamic call  $q$  at boundary  $c$  is the proof-construction record  $\text{NBHole}(q, c, \kappa, t, \text{dst}, e, \chi)$ , where  $\kappa$  is the snapshot key,  $t$  is the selected Horizon,  $\text{dst}$  identifies the returned status/data destinations,  $e$  is the reset epoch, and  $\chi$  records the WC-selected expected outcome together with  $\text{NoDependentUse}(q)$ .  $\text{NoDependentUse}(q)$  is a locally checkable assertion that no action executed while the hole remains unresolved reads  $\text{NBResult}(q)$ , uses  $q$ 's outcome to select reachability/control/endpoint/address/payload, or otherwise depends on the returned value/status. The dependence test is semantic and transitively closed: it includes aliases and pointer-derived accesses, loads from result-containing locations, derived dataflow expressions, and control, address, endpoint-selection, and payload dependence. The local certificate or checker shall justify this closure using the normative field-sensitive location/footprint model rather than a purely syntactic scan for  $\text{NBResult}(q)$ . The record also retains the attributed one-cycle attempt and every location needed to reconstruct  $\text{ReqSnapshot\_c}(q)$ . It is ghost construction state, not  $\text{KObs}$  state. All deferred holes for Horizon  $t$  must be resolved or retired before the exported post-boundary cutpoint of  $t$ .

Lemma I.3 (Finite process-local preparation script with carried shared-decision holes). Assume I.A0, WC,  $\text{Core.RegSeq}$  for sequential  $P$ , the source well-formedness conditions, and equality between  $P$ 's current source state and  $\text{Replay\_P}(H, w)$ . Assume the  $P$ -local certificate identifies  $q\_P(\gamma)$ , the finite dynamic source-control segment leading toward it, and every locally generated  $\text{Dep\_J}$  predecessor required by that segment. Then there exists a finite preparation script  $\text{Prep\_P}(H, \gamma)$  consisting of  $P$ -owned  $\epsilon$ /request actions plus zero or more carried I.DeferredNBHole records. Each action has the field-sensitive footprint and Horizon metadata required by  $\text{J.ConstructionAction}$ , respects E3/E5, synchronization boundaries, same-interface order, issue prefix closure, persistence, stable payload, and commits no unmatched  $\Sigma$  unit. When a witness-significant  $\text{PushNB/PopNB}$  decision is reached, the script asserts/records the one-cycle attempt, creates  $\text{NBHole}(q, c, \kappa, t, \text{dst}, e, \chi)$ , and does not locally choose success or failure. The script may nevertheless continue within Horizon  $t$  through source-later actions certified by  $\text{NoDependentUse}(q)$  and by disjoint snapshot/result footprints; this includes a later pipeline iteration when it is outcome-independent. It stops at the first action that depends on  $q$ 's result, would change  $\text{ReqSnapshot\_c}(q)$  without an ordered dependency, crosses a synchronization boundary, or leaves the L1 pre-settle phase. Every outcome-dependent continuation is assigned a Horizon strictly greater than  $t$ . A successful deferred decision remains a complete observable unit; a failed deferred decision is an  $\epsilon$  boundary result. Filling all holes and completing the selected Horizon yields the applicable next-frontier/issued-pending state for  $\gamma$ , with no unresolved hole exported at the post-boundary cutpoint.

Proof. Induct over the finite source-control segment certified for  $q\_P(\gamma)$ , using  $\text{Core.RegSeq}$  and the typed location footprints. Deterministic local statements preserve the replay slice by Lemma I.ReplayObs-Congruence; anchored and direct-input values are fixed by their E2/stability contracts. Blocking requests and NB attempts may be asserted in the certified L1/source-order prefix and remain attributed with stable payload. At a witness-significant NB call, emit the enriched deferred-hole record instead of forcing a global stop. Continue only through actions whose reachability and operands are independent of the unresolved result and whose writes are disjoint from the hole's snapshot/result locations; record  $\text{NoDependentUse}(q)$  for each such step. The first dependent use receives a strictly later Horizon by  $\text{ReturnedAtCycle/Core.RegSeq}$ . Appendix J later reconstructs the common settled snapshot and applies  $\text{J.NBDecision}$ . Success is a complete unit; failure writes the returned status and retires the attempt without channel commit. FPD/B4 makes each L1 segment and the complete per-

Horizon script finite. Field-sensitive framing ensures that carried holes and independent continuation coexist without changing one another’s semantics. ■

Corollary I.4 (J.PCert construction). Applying Lemma I.3 to every process participation and every applicable finite J.RequiredPrefixCandidate yields the process-local records required by Definition J.PCert: deterministic replay state, exact dynamic-operation identity, finite per-Horizon preparation scripts, issued/pending attribution, stable-payload facts, enriched deferred NB holes with result destinations and NoDependentUse evidence, field-sensitive action footprints, ReturnedAtCycle/Horizon facts, and locally generated dependency facts. This corollary remains local; J.CCert and J.NBDecision must evaluate each shared-state poll from the common settled boundary snapshot, J.RegPhaseNF must normalize the complete Horizon, and J.ACert/J.UnitEnable must establish the complete observable units.

#### I.5 Countable Local Replay Chains and Scope

For a process whose selected RTL\_B projection contains countably many observable units, repeated application of Lemma I.3 and Lemma I.DR’ constructs a countable family of process-local replay/preparation scripts. Each finite record is indexed by a finite rank-independent J.RequiredPrefixCandidate. J.GWR-Comp later instantiates that family only at the candidates that occur as prefixes of the theorem-derived validated construction order. The construction does not independently evaluate shared-boundary non-blocking decisions, select channel commits, or choose an infinite global scheduler.

If P eventually emits no further observable unit, its local replay record simply stabilizes after its last participation; this does not justify appending a global B6 idle suffix. Appendix I constructs only the countable family of finite local replay/preparation records and makes no existence claim for an infinite global Sys\_ref execution. If a later theorem requires a complete/fair source execution, that theorem must supply or construct it under its own continuation/progress premises; Appendix K, for example, uses Definition Core.LAdm, Exec\_L^adm, and Lemma K.2b.E. B6 may supply global idle stutter only when such a theorem explicitly uses an infinite terminal-idle suffix and the complete system satisfies Core.TerminalIdle, including both the genuine B5/global-fixed-point condition and its machine/environment no-future-enabling proof.

Thus Appendix I proves local replay determinacy and finite local preparation. Appendix J is solely responsible for gluing the local paths into one reachable global execution, validating shared channel and atomic interaction state, and establishing the Post  $\rightarrow$  Ref system witness. ■

## Appendix J – System-Level Trace Compatibility Proof

### Synchronization-pair well-formedness for $\approx$ .

For SyncChannel/ac\_sync operations with paired endpoints, the compared Sys\_ref and RTL\_B executions use the same dynamic synchronization-transaction pairing. That is, the two endpoint events of a matched sync\_out/sync\_in transaction are matched by dynamic sync-instance identity, commit as one two-party barrier transaction, and are treated as simultaneous events in the same observable cycle bucket. This synchronization transaction induces the same cross-process before/after ordering as described in the causal-dependency definition.

Equivalently, E1 preserves the per-process source order of synchronization endpoint events, but E1 does not weaken or replace the paired-barrier semantics of SyncChannel/ac\_sync. The system-level trace-compatibility predicate  $\tau_{\text{ref}, \text{system}} \approx \tau_{\text{post}, \text{system}}$  is defined only over synchronization-well-formed traces in this sense: if one endpoint of a dynamic SyncChannel/ac\_sync transaction appears in the trace, then the matching endpoint appears in the same observable synchronization transaction, with the same dynamic pairing, in both Sys\_ref and RTL\_B.

## Theorem J.1 (Execution-Level / Trace-Set Compositional Compatibility)

Fix an initial state and a fixed external stimulus/environment behavior (e.g., the same testbench-driven inputs) for both the prescribed source reference model  $\text{Sys\_ref}$  and  $\text{RTL\_B}$ , with  $\text{Sys\_ref}$  as defined in Remark 2. The raw rendezvous model  $\text{Sys}$  is the special case obtained when  $\text{Sys\_ref} = \text{Sys\_B}$  and all effective capacities are zero.

Clarification (reactive environments). Interpret “fixed stimulus” per Appendix I.2, “Fixed-stimulus interpretation (reactive environments).”

**Definition J.Reach (finite reachability sets  $\text{Reach\_post}$  /  $\text{Reach\_ref}$ ).** Let  $\text{Reach\_post}$  denote the set of all finite execution prefixes of  $\text{RTL\_B}$  reachable under the fixed stimulus, with no fair-extension qualifier; define  $\text{Reach\_ref}$  analogously for  $\text{Sys\_ref}$ . Both finite-prefix directions of Theorem J.1 and the cutpoint correspondence of Corollary J.1 are stated over  $\text{Reach\_post}$  /  $\text{Reach\_ref}$ . Their finite-ideal/selected-pair arguments never use the existence of a fair infinite extension, so a genuinely bad reachable prefix (for example, one ending at a deadlock cutpoint) remains in scope regardless of whether it can be embedded in a fair infinite execution. Appendix J therefore introduces no generic fair-extension-qualified execution domain and no generic reachable-prefix-to-fair-execution theorem. Any later theorem that requires a complete/fair execution states its own execution domain and continuation premises; Appendix K uses  $\text{Exec\_L}^{\text{adm}}$  and Lemma K.2b.E for the liberal source continuation and  $\text{K.CompleteS}$  for the selected Policy-S run.

Cycle-bucket and required-prefix convention for this theorem. A system execution may be represented as a sequence of observable cycle buckets  $\beta_1 \beta_2 \dots$ , where  $\beta_i$  is the finite set of  $\Sigma$ -events committed in one observable clock cycle of that particular execution. Same-cycle events inside a bucket are unordered except for the explicit constraints imposed by E1–E5, the channel semantics, signal-anchor semantics, and explicit synchronization constructs.

The bucket decomposition records the clock timing of one execution; it is not, by itself, part of the cross-model compatibility obligation. The relation  $\tau_{\text{ref},\text{system}} \approx \tau_{\text{post},\text{system}}$  does not require  $\text{Sys\_ref}$  and  $\text{RTL\_B}$  to use the same observable cycle-bucket decomposition. Events that are not ordered by the explicit required constraints may be placed in different cycles or grouped into different buckets in the two executions, provided the matched executions preserve all ordering, timestamp, FIFO-legality, and atomicity constraints required by E1–E5 and by the explicitly modeled channel, signal, and synchronization semantics.

The prefix order used by Theorem J.1 is required-prefix order over matched observable units, not  $\text{RTL\_B}$  bucket-prefix order and not the full Appendix G causal-dependency graph  $\rightarrow$ . An observable unit is either:

- a singleton  $\Sigma$ -event; or
- an explicitly atomic same-cycle transaction that the semantics requires not to be split, including a rendezvous  $\text{Push\_c}(v)/\text{Pop\_c}(v)$  transfer, the two endpoints of a paired  $\text{SyncChannel}/\text{ac\_sync}$  transaction, an R2C combinational fixed-point signal-observation unit, a proof-internal RESET boundary unit when dynamic reset is used (Normative Formal Core, reset well-formedness RW1–RW5), and any other same-cycle transaction explicitly identified by the signal/channel semantics.

E2 signal-anchor grouping convention: a synchronization commit together with the sequential-process signal Read/Write events that E2/R2 anchor to it (and therefore require to share its commit cycle) may — and, for mechanization, should — be grouped as one explicitly atomic same-cycle unit. This grouping

replaces the same-cycle equality constraint between separately enumerated units by unit-internal atomicity and is consistent with the J.RegPhaseNF/J.UnitExt treatment of a synchronization commit and its anchored signal actions as one same-cycle observable unit. When anchored signal events are instead enumerated as singleton units, their E2 timestamp equality with the anchor is enforced by the Sys\_ref realization semantics (R2) and the within-cycle listing convention, not by the strict order  $\prec_J$ , which never orders same-cycle-aligned events.

Let  $\prec_J$  be the strict required-order relation over observable units generated only by the explicit constraints that must be preserved by  $\tau_{\text{ref,system}} \approx \tau_{\text{post,system}}$ :

- E1 synchronization source-order constraints within each process;
- E2 signal-anchor timestamp constraints, interpreted as required alignment with the relevant synchronization or R2C fixed-point unit rather than as raw ordering among all source actions;
- E4 channel/FIFO legality for explicit abstract channels, including same-channel FIFO order for  $B(c)>0$  and atomic rendezvous grouping for  $B(c)=0$ ;
- E5 immediate synchronization-boundary constraints;
- paired SyncChannel/ac\_sync barrier semantics; and
- any other explicitly stated signal/channel/synchronization atomicity constraint.

Per-process source order contributes to  $\prec_J$  only where it is reflected in one of these explicit required constraints, such as synchronization order, same-endpoint/same-channel order, signal anchoring, or immediate synchronization-boundary legality. E3 contributes issue-order and request-attribution obligations, but it does not by itself create a strict predecessor edge between distinct-interface committed observable units merely because their source calls are lexically ordered. The Appendix G causal-dependency relation  $\rightarrow$  is not a generator of  $\prec_J$ .

A theorem prefix is a finite ideal of observable units under  $\prec_J$ , i.e. a finite down-closed set  $H$  such that  $\gamma \in H$  and  $\gamma' \prec_J \gamma$  imply  $\gamma' \in H$ . The proof below proceeds by induction along the finite ideal chain induced by the selected J.ValidatedConstructionOrder. On the normal compositional route, that order is derived from Definition J.DepJ, Lemma J.DepRank, Lemma J.DepAcyclic, and J.CanonicalRank. On the optional direct route, it is supplied with an independently checked well-founded construction order satisfying the same required-predecessor and ideal-prefix obligations. Acyclicity of  $\prec_J$  on the fixed execution prefix. For the chosen legal RTL\_B execution prefix, assign each observable unit  $\gamma \in U(\tau_{\text{post}})$  its RTL\_B commit cycle  $\text{clk}(\gamma)$ . If  $\gamma$  is an explicitly atomic same-cycle unit, such as a rendezvous Push\_c(v)/Pop\_c(v) transfer or a paired SyncChannel/ac\_sync transaction,  $\text{clk}(\gamma)$  is the common commit cycle of the endpoints in that atomic unit.

Every strict generator edge  $\gamma_1 \prec_J \gamma_2$  implies a strict cycle inequality  $\text{clk}(\gamma_1) < \text{clk}(\gamma_2)$  in that same legal execution prefix:

- E1 synchronization source-order constraints impose strict ordering between the relevant synchronization commits.
- E4 buffered-channel FIFO/order/capacity constraints impose strict commit-cycle ordering through reset-empty FIFO order, bounded occupancy, and non-fall-through cycle-boundary availability. In particular, for  $B(c)>0$ , a Pop\_c can consume only a value already present at the relevant cycle boundary, and a Push\_c that depends on space becoming available after a full condition is enabled only in a later cycle after the required Pop\_c effect has occurred.
- E5 immediate synchronization-boundary constraints impose strict separation between a synchronization commit and later suff\_S work whose issue/commit is after that synchronization boundary.
- Paired synchronization/barrier before/after constraints impose the corresponding strict before/after

cycle relation between distinct observable units when such a relation is explicitly part of the synchronization semantics.

- E2 signal-anchor constraints contribute timestamp equality/alignment to the relevant synchronization or R2C fixed-point unit, or same-cycle atomicity when explicitly specified; they do not by themselves introduce an independent strict edge between distinct observable units.
- Rendezvous Push/Pop endpoints, paired same-cycle synchronization endpoints, and other explicitly atomic same-cycle transactions are represented as single observable units, not as strict internal edges.
- Non-strict issue simultaneity permitted by E3 does not create a strict  $\prec_J$  edge.

Therefore any cycle  $\gamma_0 \prec_J \gamma_1 \prec_J \dots \prec_J \gamma_0$  over  $U(\tau_{\text{post}})$  would imply  $\text{clk}(\gamma_0) < \text{clk}(\gamma_1) < \dots < \text{clk}(\gamma_0)$ , an impossibility. Hence  $\prec_J$  is acyclic on  $U(\tau_{\text{post}})$ , which is the only acyclicity needed for the finite-ideal induction. For the proof of Theorem J.1, Definition J.DepJ and Lemmas J.DepSound–J.DepAcyclic derive J.CanonicalRank from the chosen RTL\_B witness’s concrete issue/decision/commit annotations and the local operational footprints. The induced observable-unit enumeration refines commit-cycle order and  $\prec_J$ . Its prefixes are ideals under  $\prec_J$ , but stable tie ordering between commuting actions has no semantic significance and must not be read as adding an observable order among independent events. There is no semantic cutpoint after only one endpoint of an explicitly atomic multi-endpoint unit has been taken; such units are split neither in Sys\_ref nor in RTL\_B.

Scope note (intentional). The finite-prefix Post  $\rightarrow$  Ref clause of Theorem J.1 and the cutpoint correspondence of Corollary J.1 are stated over Reach\_post / Reach\_ref and do not require an infinite fair extension. Their finite-ideal construction uses the process-local preparation records and the local channel/atomic certificates of Definition J.LocalGWR, combined by Lemma J.UnitExt and Theorem J.GWR-Comp with the Sys\_ref operational transition semantics. On the normal compositional route, J.RegPhaseNF/J.UnitExt also require a finite source L2 settle for every constructed Horizon; in the standard scope that same-cycle existence obligation is supplied by QualifiedSynchronousModel\_Sys\_ref(I) and Theorem Core.QSync-Ref. A directly supplied J.GWR package may instead carry independently checked finite L2-settled segments for its Horizons and need not separately assert global source-side Core.QSync solely for Theorem J.1. Weak fairness B2, channel progress B3, and the time-divergence / idle-stutter convention B6 are not premises of either finite-prefix direction of Theorem J.1. They remain available only to later theorems that explicitly construct or validate complete/fair runs, such as Core.LAdm/K.2b.E and K.CompleteS as applicable. B4/B5 remain separate normalization premises for replay/quiescence uses not discharged by the same-cycle Core.QSync theorem.

Assume the Appendix I / Appendix L local replay machinery is available—namely I.A0, WC for  $\varepsilon$ -modeled choices, deterministic replay, and the finite process-local preparation theorem—with NB-CF as a sufficient WC route where applicable. Also assume the capacities/occupancies/enablement semantics of Sys\_ref, finite capacities and B-faithful E4 behavior on every explicit abstract channel, the complete local certificate package of Definition J.LocalGWR, and the standing RTL\_B determinism assumption B1 under the fixed stimulus. For the normal compositional Post  $\rightarrow$  Ref route, additionally assume QualifiedSynchronousModel\_Sys\_ref(I); Theorem Core.QSync-Ref then supplies the finite L2 SettlePoint required by J.RegPhaseNF and J.UnitExt for every reachable constructed Horizon. The direct semantic route may instead supply a J.GWR package whose checked reachable phase-normal chain already contains the required finite L2-settled segment for each Horizon; such a package need not separately discharge global source-side Core.QSync merely to apply finite-prefix Theorem J.1. Core.QSync-Post is not a premise of plain finite-prefix J.1 unless a post-side KCut/NormPost interface is additionally invoked. No B2/B3 premise is used by the finite-prefix clause. Use B4/B5 additionally when replay  $\varepsilon$ -

normalization or uniqueness up to finite  $\varepsilon$ -stutter outside the QSync same-cycle settle is invoked. When a later theorem explicitly asserts a complete/fair execution, separately require that theorem's applicable B2/B3 and continuation/closure premises. Additionally assume the shared-memory lowering condition J.Mem for every cross-process shared-memory dependency in the compared design, and, when dynamic reset is used, the reset well-formedness conditions RW1–RW5 (all in Normative Formal Core). Then:

Compatibility-relation typing convention. The symbol  $\approx$  is an ordered cross-model compatibility predicate, not a mathematical equivalence relation. It is intentionally directed from a Sys\_ref execution/projection to an RTL\_B execution/projection, and no symmetry or transitivity property is implied. Its carrier is not a bare  $\Sigma$ -label trace alone.

For each process P, the per-process carrier is an annotated observable projection record. Such a record contains the P-local bucketed  $\Sigma$  projection, the relevant observable-unit projection, the dynamic source/witness universe  $\Omega_P$  and  $Q_{\text{exec},P}$  used for that comparison, the Issue/Commit annotation for dynamic message-call instances, the selected witness w, and the  $\mu_Q$  matching supplied by WC / Appendix L when  $\varepsilon$ -modeled failed non-blocking calls or other witness-selected choices are relevant. Forgetting these annotations yields the ordinary P-local observable  $\Sigma$  projection, but E3, E5, BoundaryReq attribution, and the witness construction may depend on the annotations and are not reconstructed from the  $\Sigma$  projection alone.

Accordingly, write

$$\approx_P \subseteq \text{AnnTraces}_{\{\Sigma, P, w\}}(\text{Sys\_ref}) \times \text{AnnTraces}_{\{\Sigma, P, w\}}(\text{RTL\_B})$$

for the per-process annotated observable-compatibility predicate induced by the P-local R/E obligations: synchronization ordering, signal anchoring or R2C fixed-point agreement, message issue-order constraints, synchronization-boundary issue constraints, and the P-local witness/request-attribution facts needed for those obligations. Here  $\text{AnnTraces}_{\{\Sigma, P, w\}}(\cdot)$  denotes the annotated projection records just described, not merely  $\text{Traces}_{\{\Sigma, P\}}(\cdot)$ .

Similarly, write

$$\approx_{\text{sys}} \subseteq \text{AnnTraces}_{\{\Sigma, w\}}(\text{Sys\_ref}) \times \text{AnnTraces}_{\{\Sigma, w\}}(\text{RTL\_B})$$

for system-level annotated observable compatibility. A system-level annotated record includes the per-process annotated projections plus the explicit abstract-channel histories and boundary-state facts needed to check E4 at the chosen Sys\_ref / RTL\_B boundaries.

For system executions  $\tau_{\text{ref}, \text{system}}$  and  $\tau_{\text{post}, \text{system}}$ , the assertion  $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}, \text{system}}$  is therefore an abbreviation: there exist compatible annotation/witness records for the two executions such that every process projection satisfies the corresponding  $\tau_{\text{ref}, P} \approx_P \tau_{\text{post}, P}$  obligation and every explicit abstract channel in the chosen Sys\_ref / RTL\_B comparison satisfies the applicable E4 channel semantics at the agreed abstract boundary. When the annotations are clear from context, the same  $\tau_{\text{ref}} \approx \tau_{\text{post}}$  notation may be used for readability; it should not be read as typing  $\approx$  over bare  $\Sigma$ -traces alone.

**Definition J.RTLConsistentRefPrefix** (annotated RTL-consistent source prefix). A source-side prefix  $\tau_{\text{ref}} \in \text{Reach\_ref}$  is RTL-consistent for the Ref  $\rightarrow$  Post clause only when it is equipped with a selected RTL\_B prefix  $\tau_{\text{post}}^* \in \text{Reach\_post}$  and a witness/annotation package W such that: (1)  $\tau_{\text{post}}^*$  is admitted by RTL\_B under the same fixed stimulus; (2) the complete selected observable-unit projection of  $\tau_{\text{ref}}$  matches the complete selected observable-unit projection of  $\tau_{\text{post}}^*$  under the canonical occurrence matching of Definition Core.17; in particular, neither selected projection may contain an additional unmatched observable unit beyond the other; (3) W provides compatible annotated projection records for  $\tau_{\text{ref}}$  and  $\tau_{\text{post}}^*$ , including the dynamic operation universes  $\Omega_P$  and  $Q_{\text{exec},P}$ ,  $\mu_Q$  matching, the operational Issue/Commit and E3/E5 annotations needed by the execution-level construction, request-attribution facts (including failed non-blocking poll witnesses where relevant), and the applicable per-

channel E4 histories and abstract boundary facts; and (4) whenever Corollary J.1, Appendix K, or any cutpoint transfer uses the selected pair, the same selected witness and normalized source prefix satisfy J.OfferReach, the selected RTL cutpoint satisfies J.OfferConf, and the composite J.KObsConf relation holds for the same H and O. Label-projection matching alone is therefore not sufficient to establish  $\text{Ref} \rightarrow \text{Post}$ ; the compatible annotations and, when needed, the KObs cutpoint package are part of the  $\text{RTLConsistentRefPrefix}$  witness.

Then:

(Post  $\rightarrow$  Ref existence, compositional route) For every  $\tau_{\text{post}} \in \text{Reach\_post}$  together with a local certificate package L satisfying  $\text{LocalGWRPackage}(\tau_{\text{post}}, L)$  (Definition J.LocalGWR), there exist a package W and a prefix  $\tau_{\text{ref}} \in \text{Reach\_ref}$  such that  $\text{GlobalWitnessRealizable}(\tau_{\text{post}}, W)$  holds and the annotated system projections satisfy  $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}, \text{system}}$  under E1–E5.

(Post  $\rightarrow$  Ref existence, direct semantic route) The same core conclusion holds when the application directly supplies a validated package W satisfying  $\text{GlobalWitnessRealizable}(\tau_{\text{post}}, W)$ . The compositional route is the normal theorem path; the direct route permits translation validation, symbolic replay, or another independently checked global certificate to target the stable J.GWR interface.

(Ref  $\rightarrow$  Post existence, annotated RTL-consistent prefixes only) For every  $\tau_{\text{ref}} \in \text{Reach\_ref}$  for which there exist  $\tau_{\text{post}}^* \in \text{Reach\_post}$  and a witness/annotation package W satisfying  $\text{RTLConsistentRefPrefix}(\tau_{\text{ref}}, \tau_{\text{post}}^*, W)$  under the same fixed stimulus, Definition J.RTLConsistentRefPrefix supplies the already-selected annotated source/RTL pair and its compatible per-process projection records. The theorem-wide finite-capacity and B-faithful abstract-boundary assumptions, together with the per-channel E4 histories and abstract-boundary facts carried by W, supply the corresponding per-channel well-formedness premises of Proposition J.0. Proposition J.0 therefore applies to this selected pair. Hence there exists  $\tau_{\text{post}} \in \text{Reach\_post}$  with the required compatibility, witnessed by  $\tau_{\text{post}} := \tau_{\text{post}}^*$ , and the annotated system projections satisfy  $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}, \text{system}}$ . In Appendix L snooping/witness-selection applications, the selected  $\tau_{\text{post}}^*$  and W are the  $\text{RTL\_B}$  reachable prefix and witness package selected by the snooping wrapper/witness.

Moreover, because B1 is a standing assumption in this theorem instance (deterministic  $\Sigma$ -projected  $\text{RTL\_B}$  trace under fixed stimulus), the matching observable  $\Sigma$  projection of a selected  $\text{RTL\_B}$  prefix is unique (up to insertion/removal of finite  $\epsilon$ -steps permitted by B4/B5). This uniqueness is only a uniqueness statement about the observable projection; it does not by itself determine  $\mu_Q$ , failed-poll instances in  $Q_{\text{exec}}$ , issue annotations, request-attribution facts, the selected package W, or the full stateful  $\text{Sys\_ref}$  witness. The corresponding execution prefix witness  $\tau_{\text{ref}}$  need not be unique as a stateful prefix, since  $\text{Sys\_ref}$  may admit multiple  $\epsilon$ -different realizations with the same  $\Sigma$  projection.  $\text{Sys\_ref}$  may also admit executions whose  $\Sigma$  projection does not match  $\text{RTL\_B}$ 's  $\Sigma$  trace when latency-sensitive / non-blocking effects are  $\Sigma$ -visible; such executions are outside the scope of (Ref  $\rightarrow$  Post) unless constrained by  $\text{RTLConsistentRefPrefix}$  / the snooping/witness-selection technique (Appendix L).

#### Definition J.RefProjCover (selected-reference-run observable coverage)

Fix a theorem instance I, a DUT-boundary observable projection  $\Sigma_{\text{DUT}}$ , and a selected complete source-reference execution  $\rho_{\text{ref}}^0$ . Define  $\text{RefProjCover\_I, } \Sigma_{\text{DUT}}(\rho_{\text{ref}}^0)$  to hold iff, for every  $\tau_{\text{ref}} \in \text{Reach\_ref}$  of I, the finite canonical  $\Sigma_{\text{DUT}}$  history  $\text{CanonHist\_}\Sigma_{\text{DUT}}(\tau_{\text{ref}})$  is a finite ideal (a down-closed canonical prefix) of  $\text{CanonHist\_}\Sigma_{\text{DUT}}(\rho_{\text{ref}}^0)$  under the required observable-unit order, modulo J.HCanon and the permitted finite  $\epsilon$ -stutter. The definition covers observable ideals/down-

closed prefixes; it does not require identical cycle buckets or a fixed linear order among independent observable units.

#### Definition J.IdealPrefixClosed (ideal-prefix closure for safety predicates)

Fix theorem instance  $I$  and outer observation projection  $\Sigma\_DUT$ . For a predicate  $\Phi$  on finite canonical  $\Sigma\_DUT$  histories,  $J.IdealPrefixClosed\_I, \Sigma\_DUT(\Phi)$  holds iff, for all finite canonical histories  $H$  and  $H'$  in the selected theorem instance, whenever  $H'$  is a finite ideal (down-closed canonical prefix) of  $H$  under the required observable-unit order used by Theorem J.1 and Definition J.RefProjCover,  $\Phi(H)$  implies  $\Phi(H')$ . In the safety results below, “prefix-closed” means this ideal-prefix notion, not merely closure under deleting a suffix from one chosen linearization of otherwise independent events.

**Practical scope.** Typical qualifying properties include assertion-style “no bad observable event has occurred” invariants, per-channel FIFO/order/value-legality invariants, and protocol-legality predicates explicitly formulated to be downward closed over the retained required-order history. A cross-interface transaction checker need not satisfy  $J.IdealPrefixClosed$  when it requires an explanatory event that the retained  $\prec\_J$  order permits an ideal to omit—for example, an output on one channel checked against an otherwise independent input on another. Such a property is not transferred to arbitrary intra-cycle ideals by  $J.CycleIdealCover$  alone; use Definition J.TraceCertCov on the relevant arbitrary-prefix domain, or refine the selected observation/order so that the needed dependency is retained.

#### Definition J.TraceCertCov (Post $\rightarrow$ Ref certificate coverage)

Fix a theorem instance  $I$  and a coverage domain  $D \subseteq Reach\_post$ .  $TraceCertCov\_I(D)$  holds iff every  $\tau\_post \in D$  is equipped with one of the following route-complete Post  $\rightarrow$  Ref packages: (a) a  $J.LocalGWR$  package accepted by  $J.GWR-Comp$ , together with  $QualifiedSynchronousModel\_Sys\_ref(I)$  so that  $Core.QSync-Ref$  supplies the finite L2 settle required by  $J.RegPhaseNF/J.UnitExt$ , and the remaining applicable Post  $\rightarrow$  Ref premises of Theorem J.1; or (b) a directly validated  $J.GWR$  package whose checked reachable phase-normal chain supplies the required finite L2-settled segment for every represented Horizon, together with the remaining applicable Post  $\rightarrow$  Ref premises of Theorem J.1. All evidence shall be bound to the same theorem instance, stimulus, observation boundary, and selected RTL\_B prefix. The definition names only safety-side certificate/package coverage; it does not strengthen Theorem J.1 and does not imply  $K.CertCov$ .

#### Corollary J.1-Safe (source-side safety transfer through Post $\rightarrow$ Ref)

Assume  $J.TraceCertCov\_I(D)$  for a selected coverage domain  $D \subseteq Reach\_post$  containing every reachable RTL\_B prefix relevant to the claimed property.  $J.TraceCertCov$  includes the source-settling evidence appropriate to the selected Post  $\rightarrow$  Ref construction route: source-side  $Core.QSync$  for the normal compositional route, or equivalent checked per-Horizon settling evidence for a direct  $J.GWR$  route. Let  $\Phi$  be a predicate on finite canonical  $\Sigma\_DUT$  histories satisfying  $J.IdealPrefixClosed\_I, \Sigma\_DUT(\Phi)$  and invariant under  $J.HCanon$ . Then:

- (1) If every  $\tau\_ref \in Reach\_ref$  satisfies  $\Phi$ , then every  $\tau\_post \in D$  satisfies  $\Phi$ .
- (2) If one selected complete pre-HLS execution  $\rho\_ref^0$  satisfies  $\Phi$  at every finite canonical ideal (down-closed prefix) and  $RefProjCover\_I, \Sigma\_DUT(\rho\_ref^0)$  holds, then every  $\tau\_post \in D$  satisfies  $\Phi$ .

**Proof.** For any  $\tau\_post \in D$ ,  $J.TraceCertCov$  supplies the applicable Post  $\rightarrow$  Ref package, and Theorem J.1 constructs a reachable matching  $Sys\_ref$  prefix with the same canonical  $\Sigma\_DUT$  observation. Clause (1) follows from the universal  $Sys\_ref$  premise. For clause (2),  $J.RefProjCover$  places that matching observation among the finite canonical ideals/down-closed prefixes covered by  $\rho\_ref^0$ . Prefix closure and  $J.HCanon$ -invariance of  $\Phi$  yield the RTL\_B conclusion.  $\square$

**Common-case discharge note.** For blocking deterministic dataflow, or for designs in which NB-CF makes failed non-blocking polls irrelevant to later  $\Sigma_{\text{DUT}}$  behavior, J.RefProjCover may be discharged from fixed stimulus, deterministic/value-consistent process behavior, deterministic or witness-fixed arbitration and environment choices, and completeness of the selected source execution. Lemma K.2b.P1 supplies the payload and dynamic-operation-identity determinacy ingredient in its blocking-only scope, but it is not by itself the full observable-prefix coverage theorem. Static extraction of B(c) and any elaboration package define Sys\_ref; B-faithfulness and J.TraceCertCov remain separate obligations.

**Scope.** Corollary J.1-Safe transfers safety predicates only. It does not transfer raw cycle latency or performance properties, prove eventual RTL realization of every source event, or establish unrestricted equality of the two behavior sets.

**Proof strategy.** Theorem J.1 is not obtained from pairwise composition alone. Appendix I proves the local replay/preparation scripts with carried deferred NB holes, packaged as J.PCert. Definition J.LocalGWR adds per-channel, atomic-unit, environment/reset, field-sensitive footprint, request-snapshot, generated-dependency, and interface-agreement certificates. Lemmas J.DepSound/J.HorizonOrderSafe/J.DepComplete/J.DepRank/J.DepAcyclic derive a canonical action order rather than accepting one as a premise. Lemma J.RegPhaseNF then normalizes each Horizon into the single Core.Phase L1/L2/L3 phase vocabulary, preserving the J.HCanon history quotient. Horizon-batched J.UnitExt and Theorem J.GWR-Comp construct one coherent source execution and the theorem-derived J.ValidatedConstructionOrder/J.GWR package. J.GWR-Direct instead supplies an equivalent directly validated order, phase-normal chain, and cutpoint invariants. Proposition J.0 remains only the conditional lifting step for an already selected compatible trace pair.

### Definition J.ConstructionAction and operational footprints

Fix a finite RTL\_B prefix  $\tau_{\text{post}}$  and a candidate local package L.  $\text{Act}_L(\tau_{\text{post}})$  is the finite set of primitive source-construction actions named by the process scripts, channel records, deferred-hole records, and complete observable units. An action is one of: (A1) a process-owned  $\epsilon$  evaluation/control transition, including outcome-independent continuation past a carried NB hole; (A2) assertion or retention of an endpoint-owned blocking request, or assertion of a one-cycle attributed NB attempt; (A3) a shared-boundary NB decision/return that reads the settled ReqSnapshot, writes NBResult(q), retires NBPending(q)/the attempt, and on success is contracted with the corresponding transfer commit; (A4) any other complete explicit-channel or observable atomic-unit commit; or (A5) an environment, reset, or explicit Sys\_B^elab transition. A3 does not write the ordinary control/operand locations used by certified same-Horizon outcome-independent actions; dependent continuation is an A1 action at a strictly later Horizon. Every action has a typed identity, owner set, reset epoch, Horizon, associated unit, semantic stage, local ordinal, and finite ReadSet/WriteSet over a normative location vocabulary. That vocabulary distinguishes at least Ctrl(P), Local(P,x)/Operand(P,e), NBResult(q), NBPending(q), endpoint request and Push-payload locations, each explicit channel/elaboration location, environment locations, and reset locations. Certificates may realize these locations as components, fields, or lenses, but shall map them to this common vocabulary and prove the corresponding disjointness/frame laws. The vocabulary is intentionally finer than a monolithic “process-local state” component. Custom Sys\_B^elab components require local footprint/commutation and well-founded ordinal certificates.

### Definition J.ReqSnapshot and J.NBDecision

For every witness-significant NB dynamic call q at explicit boundary c, let  $\kappa(q)$  be its unique snapshot key and let NBPending(q) be the carried I.DeferredNBHole record. ReqSnapshot\_c(q) is the common source-boundary state at the L2 settled point of q’s Horizon, after every L1 request/attempt write and every other Dep\_J predecessor of the decision has executed, but before q’s A3 success/failure return and before any channel-state update caused by that decision. The snapshot contains the reset epoch; E4

channel state; every attributed persistent request and one-cycle attempt present at  $c$ , including  $q$ 's own attempt with stable Push payload; and every explicit gate/join/arbitration/environment field affecting  $c$ .  $J.NBDecision(q, ReqSnapshot\_c(q))$  is the unique result prescribed by the source call semantics and E4. For  $B(c) > 0$ , PopNB succeeds exactly when nonempty and PushNB exactly when nonfull; for  $B(c) = 0$ , success requires the matching complementary attributed request/attempt with value agreement. Failure is an  $\epsilon$  A3 return with  $ReturnedAtCycle(q, t)$  and no channel commit; success is the contracted A3/committed-transfer unit. WC selects the implementation outcome and J.CCerts proves equality. Process-local preparation may create NBPending( $q$ ) and may continue through certified outcome-independent actions, but it may not execute A3 or read the settled snapshot locally.

#### Lemma J.BufSameCycle (same-cycle buffered Push/Pop sequentialization)

Let  $c$  have  $B(c) > 0$ , let the pre-cycle logical FIFO contents be  $Q$  with  $n = |Q|$ , and suppose one  $Push\_c(v)$  and one  $Pop\_c(w)$  both commit in the same RTL cycle under the non-fall-through E4 semantics. Then  $0 < n < B(c)$ ,  $w = head(Q)$ , and both source serializations  $Push\_c(v); Pop\_c(w)$  and  $Pop\_c(w); Push\_c(v)$  are enabled from  $Q$ . Both terminate with logical contents  $tail(Q)$  followed by  $v$ , occupancy  $n$ , the same producer and consumer ordinals, and the same unordered same-cycle observable bucket. The proof may therefore execute the two source transitions in either order while retaining their common RTL-cycle annotation. If  $n = 0$ , the Pop cannot be enabled by the same-cycle Push; if  $n = B(c)$ , the Push cannot be enabled by the same-cycle Pop. For  $B(c) = 0$  the endpoints remain one indivisible rendezvous action and this lemma does not apply. Proof: direct case analysis on the E4 enablement predicates and FIFO update equations. ■

#### Definition J.DepJ (generated construction-dependency relation)

After contracting every explicitly atomic action, let  $a \prec\_D^{sem} b$  denote a direct semantic-prerequisite edge. The semantic relation is generated only by (D1) same-process dynamic control, data, source-issue, synchronization-boundary, and  $ReturnedAtCycle$  dependencies; (D2) request/attempt-before-settle/snapshot-before-A3-decision edges for every location read by  $ReqSnapshot\_c(q)$ , plus A3-decision-before-continuation edges only for actions whose reachability or operands depend on  $NBResult(q)$ ; (D3) per-channel FIFO/history, occupancy/head, and non-commuting same-boundary update predecessors, with J.BufSameCycle as the named buffered exception; and (D4) reset, environment, and explicit elaboration-component predecessors. Let  $a \prec\_D b$  denote the full construction-dependency relation obtained by adding (D5) concrete earlier-Horizon proof-stratification edges from the selected RTL annotation. A D5 edge that is not already implied by  $(\prec\_D^{sem})^+$  is admissible only when the local footprint/frame certificate proves that the two actions commute modulo J.HCanon, so choosing the earlier-Horizon action first preserves transition legality, complete units, and the post-settle semantic state. Thus D5 selects a canonical construction representative; it is not by itself a universal Sys\_ref happens-before fact. An outcome-independent same-Horizon action reached past an unresolved  $q$  acquires no A3-to-action edge: its  $I.DeferredNBHole$  NoDependentUse fact and field-sensitive footprints must instead show that it is independent of  $NBResult(q)$  and of  $q$ 's settled snapshot, so ordinary framing/commutation applies. An edge may not be inserted merely for replay convenience. Mere source order between distinct-interface operations creates the applicable issue-order obligations, but not a decision/commit-to-later-request edge when Core.RegSeq and the local certificate establish independent co-issue.

#### Lemma J.DepSound (semantic generated edges are operational prerequisites)

If  $a \prec\_D^{sem} b$ , then no legal Sys\_ref construction segment realizing both actions can execute  $b$  before  $a$  unless the actions are contracted into one explicit atomic unit. Proof: D1 follows from source control/data/issue rules and  $ReturnedAtCycle/Core.RegSeq$ ; D2 from the settled-snapshot read and true

NB-result dependence; D3 from E4/FIFO state dependence except J.BufSameCycle; and D4 from the named reset/environment/elaboration rule. Soundness of every declared footprint against the normative location semantics, including frame and disjointness laws, is part of the local-certificate obligation.  $\square$

#### Lemma J.HorizonOrderSafe (D5 proof stratification is construction-safe).

Let  $a \prec_D b$  be a D5 edge with no path  $a (\prec_D^{\text{sem}})^+ b$ . The LocalGWR package must then supply the D5 commutation certificate required by Definition J.DepJ:  $a$  and  $b$  may be exchanged without changing transition legality, the complete observable units, the terminal post-settle semantic state, or the J.HCanon quotient. Hence the canonical construction may choose  $a$  before  $b$  even though  $a$  is not an operational prerequisite of  $b$  in every legal Sys\_ref execution. If the pair does not commute, the actual overlap must instead be represented by D1–D4 and is covered by J.DepSound.  $\square$

#### Lemma J.DepComplete (all non-commuting construction actions are ordered)

Let  $a$  and  $b$  be distinct construction actions that are not endpoints of one contracted atomic action. If neither  $a \prec_D^+ b$  nor  $b \prec_D^+ a$ , then either (i) their field-sensitive ReadSet/WriteSet interference tests are disjoint and the transitions commute by J.Frame, or (ii) they are the legal same-cycle buffered Push/Pop dual update of J.BufSameCycle. Every pair that cannot be swapped while preserving transition legality, the same complete observable units, the same post-settle semantic state, and the same J.HCanon history quotient is therefore ordered in at least one direction. In particular, narrowed A3( $q$ ) commutes with a same-Horizon owner action carried past  $q$  only when NoDependentUse( $q$ ) and the location footprints show disjointness from NBResult( $q$ ), NBPending( $q$ ),  $q$ 's attempt, and every location in ReqSnapshot\_c( $q$ ). The finite A1–A5 case analysis invokes D1–D4 for every actual overlap; D5 may additionally order already-commuting cross-Horizon pairs under Lemma J.HorizonOrderSafe, and custom components use their local certificate. ■

#### Definition J.FoundationRank (rank independent of graph acyclicity)

For each construction action  $a$ , define FR( $a$ ) lexicographically as  $\langle \text{EpochIndex}(a), \text{Horizon}(a), \text{SemanticStage}(a), \text{LocalOrdinal}(a), \text{ComponentTie}(a) \rangle$ . Horizon is the selected RTL cycle/decision horizon or enclosing unit cycle. If an action consumes a value or status with ReturnedAtCycle( $q, t$ ), Core.RegSeq requires  $\text{Horizon}(a) > t$ . Same-Horizon actions carried past  $q$  must instead satisfy NoDependentUse( $q$ ). SemanticStage is the fixed order: reset/environment prerequisites; local evaluation plus request/attempt writes (Core.Phase L1); completed settled-snapshot formation (L2); shared A3 decision or contracted/other boundary commit (L3); and a reserved continuation-bookkeeping stage usable only for within-Horizon sinks whose outgoing same-Horizon edges never return to an earlier stage. Outcome-dependent process continuation after an NB result is not assigned to that reserved stage; it is an ordinary evaluation/request action at a later Horizon. LocalOrdinal is the action-kind-specific script, request/snapshot, transfer, environment, reset, or component ordinal. ComponentTie is used only when all prior coordinates agree and a local certificate proves commutation or a component-local order. FR is computed before Dep\_J and does not use Height or J.CanonicalRank.

#### Lemma J.DepRank (every generated dependency advances the foundation rank)

For every well-formed LocalGWRPackage and generated direct edge  $a \prec_D b$ ,  $\text{FR}(a) < \text{FR}(b)$ . D5 strictly increases Horizon; reset edges increase EpochIndex. D1 either advances the within-L1 local ordinal/stage, or—when  $b$  depends on a boundary result ReturnedAtCycle( $q, t$ )—strictly increases Horizon under Core.RegSeq. Independently reached same-Horizon actions carried past an unresolved  $q$  have no A3-to-action edge; their true control/data/issue predecessors advance the L1 ordinal or stage. D2 advances request/attempt  $\rightarrow$  settled snapshot  $\rightarrow$  A3 decision/contracted commit. A decision-to-dependent-continuation edge strictly increases Horizon; any retained within-Horizon continuation-

bookkeeping target is a certified sink and therefore cannot generate a later stage regression. D3 advances transfer/history ordinal or a non-commuting substage, with J.BufSameCycle removing the legal dual-update ordering; D4 advances the certified component ordinal/substage. Thus the prior failed-NB  $\text{decision}(t) \rightarrow \text{dependent request}(t)$  case is impossible by construction, and the  $\text{decision}(t)$  /  $\text{independent request}(t)$  pair is unordered and commuting. ■

### Lemma J.DepAcyclic (well-foundedness of the generated dependency graph)

For every well-formed LocalGWRPackage, the graph  $\text{Dep\_J}$  is acyclic after atomic contraction. Proof. Suppose a directed cycle  $a_0 <_D a_1 <_D \dots <_D a_0$  existed. Repeated application of Lemma J.DepRank would give  $\text{FR}(a_0) < \text{FR}(a_1) < \dots < \text{FR}(a_0)$ , contradicting irreflexivity of the well-founded lexicographic order on the finite rank components. This argument uses one global semantic rank that every D1–D5 edge advances; it does not infer acyclicity from a union of separately acyclic process, channel, request, environment, reset, or component orders. Height and J.CanonicalRank are defined only after this lemma and play no role in proving it. ■

### Definition J.CanonicalRank (derived construction order)

Because  $\text{Act\_L}(\tau_{\text{post}})$  is finite and  $\text{Dep\_J}$  is acyclic by the independent FR argument, define  $\text{Height}(a)=0$  for actions with no incoming  $<_D$  edge and  $\text{Height}(a)=1+\max\{\text{Height}(b) \mid b <_D a\}$  otherwise. Define  $\text{CanonicalRank}(a):=\langle \text{FR}(a), \text{Height}(a), \text{KindKey}(a), \text{StableId}(a) \rangle$  lexicographically. FR preserves the concrete epoch/horizon and semantic prerequisite structure; Height is a theorem-derived topological depth used only after acyclicity has been established; and KindKey/StableId only totalize otherwise commuting actions and create no observable  $<_J$  edge. The rank of an observable unit is the rank of its contracted commit action. J.CanonicalRank is reconstructed from checked action identities, local operational coordinates, and  $\text{Dep\_J}$ ; no phase or replay rank is supplied by the implementation flow.

### Definition J.RequiredPrefixCandidate (rank-independent local-certificate index)

For a finite RTL\_B prefix  $\tau_{\text{post}}$ ,  $\text{RequiredPrefixCandidate}(\tau_{\text{post}}, H)$  means that  $H$  is a finite set of complete observable units of  $\tau_{\text{post}}$  satisfying only order-neutral prefix conditions: (R1)  $H$  is down-closed under the execution-specific required-order relation  $<_J$ ; (R2) for every process  $P$ , the  $P$ -local projection of  $H$  is a prefix of the selected  $P$ -local observable-unit projection, with no dynamic operation identity used twice and with reset epochs respected; (R3) every explicitly atomic unit is either wholly present or wholly absent; and (R4) each explicit-channel projection is a reset-epoch-consistent prefix of the selected rendezvous or FIFO history. The definition does not mention  $\text{Dep\_J}$ , J.DepAcyclic, J.CanonicalRank, a proposed global replay order, or reachability of a common  $\text{Sys\_ref}$  state. It is only the finite index domain on which Appendix I can compute process-local replay states and preparation scripts. Theorem J.GWR-Comp later selects from this domain the candidates that arise along its derived construction chain.

### Definition J.PCert (process-local replay and preparation certificate)

For a process  $P$  and finite RTL\_B prefix  $\tau_{\text{post}}$ ,  $\text{PCert\_P}(\tau_{\text{post}}, L_P)$  holds when  $L_P$  supplies, for every relevant  $\text{RequiredPrefixCandidate}$  and process participation: (P1)  $\text{WC}/\mu_Q$  identity, reset epoch, and matched dynamic operation; (P2)  $\text{Replay\_P}(H, w)$  with future- $\Sigma$ -relevant values, control, completion, and operation ordinals; (P3) the finite per-Horizon Lemma I.3 script, every enriched I.DeferredNBHole, its NBResult/NBPending/result-destination locations, and NoDependentUse evidence for all same-Horizon actions carried past it; (P4) Issue/E3/E5, prefix closure, persistence, stable payload, Core.RegSeq, direct-input stability, and synchronization facts; (P5) exact dynamic NB attempt identity, desired WC outcome, and snapshot key, with no local A3 decision; (P6) complete A1/A2/A3 inventory, normative field-sensitive footprints, Horizon/stage/local ordinal, and every D1/D2 generator; and (P7) proof that local actions modify only their declared locations and export no unmatched commit or unresolved same-

Horizon hole at a cutpoint. PCert contains no J.CanonicalRank and does not assume the global order or phase-normal chain.

#### Definition J.CCert (explicit-channel history, request, and enablement certificate)

For every explicit abstract channel  $c$  of  $E$ , CCert<sub>c</sub>( $\tau_{\text{post}}, L_c$ ) supplies: (C1) boundary,  $B(c)$ , ownership, epochs, and  $B$ -faithfulness/ $\epsilon$ -opacity; (C2) complete prefix-indexed E4 history, ordinals, payloads, occupancy/head, and reset-empty restarts; (C3) producer/consumer agreement on transfer identity, value, ordinal, direction, and epoch; (C4) for every Horizon and witness-significant  $q$ , the complete L2 settled ReqSnapshot<sub>c</sub>( $q$ ), including all absent/present requests/attempts, stable payloads, and explicit gate/join/environment fields; (C5) proof that  $WC(q) = J.NBDecision(q, \text{ReqSnapshot}_c(q))$ ; (C6) exact local channel/A3 state updates with no hidden availability; (C7) J.BufSameCycle evidence for every legal same-cycle buffered pair; (C8) every D2/D3 edge and field-sensitive channel/snapshot/result footprint; and (C9) proof that every one-cycle NB attempt is retired and every deferred-hole record for the Horizon is resolved before the exported post-boundary cutpoint.

#### Definition J.ACert, J.EnvCert, and J.ResetCert

ACert<sub>y</sub> identifies the complete participant set, dynamic identities, values, reset epoch, and atomic transition rule for every rendezvous transfer, paired SyncChannel/ac\_sync transaction, E2 anchor/signal group, R2C fixed-point unit, RESET unit, and other explicitly atomic unit. It proves that every participant record names the same contracted action and supplies the endpoint requests, values, guards, and component state from which the generic Sys\_ref rule can derive enablement; it does not take complete-unit enablement itself as an unexplained certificate field. An endpoint cannot be certified independently of its counterpart. EnvCert proves, for every selected external interaction, that the source and RTL records use the same response of the fixed reactive environment to the same preceding  $\Sigma$ -causal history and supplies the corresponding footprints, horizon/stage/ordinal fields, and dependencies. ResetCert supplies RW1–RW5 for each reset occurrence and scope, including request clearing, reset-empty channel state, source/RTL reset-state correspondence, atomic reset splicing, framing outside the reset scope, and the D4 dependency and J.DepRank coordinate facts.

#### Definition J.LocalAgree and J.LocalGWR

LocalAgree(L) is the finite conjunction of equality, uniqueness, ownership, and coverage checks on duplicated local-record fields: shared action/unit identity; endpoint pairing; boundary identity; payload/value; transfer ordinal; request/attempt attribution; deferred-hole identity and result destination; snapshot key/contents; NoDependentUse facts; reset epoch; witness/elaboration identity; Horizon and Core.LAdm phase; semantic stage/local ordinal; the normative field-sensitive ReadSet/WriteSet locations; and every generated dependency edge. Every process, explicit boundary, construction action, atomic unit, deferred hole, and witness-significant  $\epsilon$  choice in  $\tau_{\text{post}}$  must be covered, and each endpoint request/attempt has at most one dynamic owner. LocalAgree does not assert a global execution, acyclicity, phase-normality, or extension.

Definition J.LocalGWR. LocalGWRRPackage( $\tau_{\text{post}}, L$ ) consists of PCert<sub>P</sub> for every process, CCert<sub>c</sub> for every explicit boundary, ACert for each atomic unit, applicable EnvCert/ResetCert, the complete A1–A5 action/deferred-hole inventory, the normative field-sensitive footprints and footprint-soundness proofs, Core.RegSeq/direct-input conformance facts, locally checkable epoch/Horizon/stage/ordinal metadata, every local fact from which Dep<sub>J</sub> is generated, custom-component certificates, and LocalAgree, all bound to one instance/stimulus/E/w/observation boundary/RTL prefix. J.DepRank, J.DepAcyclic, J.CanonicalRank, J.RegPhaseNF, J.RegPhaseReach, and the global source chain are theorem-derived, not supplied as unexplained local premises.

### Definition J.ConstructionOrderProvenance and J.ValidatedConstructionOrder

For fixed  $\tau_{\text{post}}$  and  $W$ , a `ValidatedConstructionOrder` contains a complete unit enumeration  $\gamma_1 \dots \gamma_m$ , the finite predecessor relation  $<_W$ , and the partition of actions/units into nondecreasing Horizon buckets. It proves: (V1) well-foundedness; (V2) refinement of  $<_J$ , reset epochs, atomic grouping, and every strict Horizon constraint; (V3) exact action/unit/deferred-hole coverage; (V4) coverage of every preparation, request, settled snapshot, A3 decision, channel, environment, reset, and atomic prerequisite; (V5) every flattened enumeration prefix is a finite required-prefix ideal; and (V6) within each Horizon, the order admits the `Core.LAdm` phase normal form of `J.RegPhaseNF`, with all L1 requests fixed before L2 settle, all boundary decisions/commits in L3, and no deferred hole surviving the bucket. The theorem-derived form uses `LocalGWR`, `Dep_J`, `J.CanonicalRank`, and `J.RegPhaseNF`. The direct form independently validates V1–V6. Stable keys order only actions/units that commute modulo `J.HCanon`.

### Definition J.GWR (GlobalWitnessRealizable( $\tau_{\text{post}}, W$ ))

A selected package  $W$  is globally witness-realizable when it contains one coherent annotation package—`WC/μ_Q`, `Issue/ReturnedAtCycle/Commit` attribution, E4 histories, atomic groups, environment/reset identities, dynamic operations, deferred-hole/snapshot identities, and construction actions—and there exist: (G1) one `J.ValidatedConstructionOrder` with finite Horizon buckets and a flattened complete-unit enumeration; (G2) required-prefix ideals for the flattened prefixes and matched cycle-entry/post-boundary bucket prefixes; and (G3) reachable `Sys_ref` states such that each Horizon bucket extends by one finite legal phase-normal segment: L1 process-owned A1/A2 evaluation and request/attempt assertion, then L2 freeze/ε-settle to the common snapshots, then L3 A3/A4 boundary decisions and complete units in commit-action rank order, with `J.BufSameCycle` used for legal buffered dual updates. No new L1 request occurs after the bucket's L2 settle; every action depending on a `ReturnedAtCycle` result is placed in a later Horizon; and every deferred hole is resolved or retired before the exported post-boundary state. The segment has the same complete units and terminal post-settle state and the same `CanonHist/J.HCanon` quotient as the selected construction order, even when same-Horizon independent actions are permuted. The chain preserves replay, channel, witness, attribution, reset, phase, and snapshot invariants and is compatible with  $\tau_{\text{post}}$  under E1–E5. The direct route supplies the same bucketed interface. Historical operational data remains evidence rather than `KObs` fields. `J.GWR` is a stable semantic interface. The normal route derives its `ValidatedConstructionOrder` and the complete witness chain by Theorem `J.GWR-Comp` from Definition `J.LocalGWR`. A flow may instead establish the same interface directly through the direct-validated construction-order provenance and a checked reachable chain. No theorem below infers `J.GWR` from bare  $\Sigma$ -label equality, and the direct route may not refer to `J.CanonicalRank` unless it also supplies the `LocalGWR` package and theorem-derived provenance that define that rank.

### Lemma J.Frame (local transition framing)

Let `Sys_ref` state be viewed through the normative typed location vocabulary of `J.ConstructionAction`. A transition segment whose certified `WriteSet` is  $F$  changes only locations in  $F$ . Every location disjoint from  $F$  is equal before and after, and every guard/expression whose `ReadSet` is disjoint from  $F$  has the same value. This includes separate framing of `Ctrl/Local/Operand`, `NBResult`, `NBPending`, endpoint request/payload, channel/elaboration, environment, and reset locations. Each certificate must prove soundness of its declared locations against the actual `Sys_ref` transition semantics; the representation may use components, fields, or lenses. ■

### Lemma J.LocalLift (lifting a certified process-local preparation path)

If the P-owned locations of a reachable global state equal the start state of a `PCert_P` segment, every shared location read by that segment equals its certified summary, and the field-sensitive footprint

conditions hold, then the segment lifts to a finite global  $\varepsilon$ /request path. The lifted path has the same P-local result and endpoint writes, commits no unmatched  $\Sigma$  unit, and frames every unrelated location. At an NB call it creates/carries  $I$ .DeferredNBHole and performs no A3 decision; unlike the previous stop-at-hole formulation, it may continue through certified NoDependentUse same-Horizon actions. Proof: induction over the segment using J.Frame. ■

#### Lemma J.PrepareCommute (commutation of local preparation paths)

Consider the executable L1 preparation actions and other construction actions in one Horizon. Field-sensitive disjoint ReadSet/WriteSet actions commute by J.Frame. Separately owned request writes may accumulate before the common settle when neither reads a snapshot whose fields remain mutable. A narrowed A3(q) never moves before an action whose write appears in ReqSnapshot\_c(q), but it commutes with a carried same-Horizon owner action when NoDependentUse(q) and the footprints establish disjointness from NBResult(q), NBPending(q), q's attempt, and all snapshot locations. J.DepComplete orders every other non-commuting pair; J.BufSameCycle handles the legal buffered L3 dual update. Thus topological executions can be permuted into the normal form stated next, preserving terminal state and the J.HCanon quotient. No fairness premise is used. ■

#### Lemma J.RegPhaseNF (registered sequential-interface phase normal form)

Assume Core.RegSeq, a well-formed LocalGWRPackage, and that the frozen L2 state of the selected Horizon admits a finite  $\rightarrow_{\varepsilon, L2}$  path to a SettlePoint. In the standard compositional scope, QualifiedSynchronousModel\_Sys\_ref(I) and Theorem Core.QSync-Ref discharge this existence premise uniformly for every reachable Horizon; a directly validated J.GWR route may instead carry the equivalent checked per-Horizon settling evidence. For every finite construction fragment whose actions have one selected Horizon t, there is a permutation respecting Dep\_J and atomic contraction with the following order: (1) all required A1 local evaluation/control and A2 request/attempt assertions permitted by Core.Phase L1, including outcome-independent actions carried past deferred NB holes; (2) one L2 request freeze and  $\varepsilon$ -settle producing every ReqSnapshot for t; and (3) the A3/A4 boundary decisions and complete units of L3, ordered by commit-action rank except for explicitly atomic units and J.BufSameCycle sequentializations. Actions depending on a value/status ReturnedAtCycle in t occur only at a later Horizon. The normalized fragment contains the same complete observable units, reaches the same post-boundary semantic state, and yields the same canonical enriched history under J.HCanon; literal ordering of independent same-Horizon labels need not be identical. Every deferred hole for t is resolved or retired at the end. Proof: repeatedly commute adjacent unordered actions using J.Frame/J.PrepareCommute, use NoDependentUse and field-sensitive footprints for decision/independent-continuation swaps, use J.BufSameCycle within L3, and use J.DepRank to exclude any required edge from a later phase back to an earlier phase. Termination follows from finiteness. ■

#### Lemma J.NBDecision (globally valid non-blocking decision step)

Suppose the matched Horizon has completed its J.RegPhaseNF L1 actions and L2 settle, every Dep\_J predecessor of NBPending(q) has executed, and J.CCert reconstructs ReqSnapshot\_c(q)=S. Then q has exactly one legal A3 result J.NBDecision(q,S). On failure, A3 writes the status to NBResult(q), retires NBPending(q)/the one-cycle attempt, produces ReturnedAtCycle(q,t), and changes no channel history. On success, A3 is contracted with the corresponding committed transfer and writes the returned result. A3 does not perform ordinary outcome-independent control advance: such same-Horizon actions have already occurred in L1 under NoDependentUse, while every outcome-dependent continuation is an A1 action at a later Horizon. The result agrees with WC by C5 and no process-local script may choose or overwrite it. ■

### Lemma J.UnitEnable (local certificates imply complete-unit enablement)

After J.RegPhaseNF has executed all required L1 actions and formed the common L2 snapshots for a Horizon, the endpoint requests/attempts, stable payloads, channel state, participant state, and deferred-hole records satisfy the premises of each selected L3 complete unit. J.NBDecision supplies failed or successful A3 results from those snapshots; failure is an  $\epsilon$  result and success is the contracted unit. Buffered enablement follows from the common pre-update occupancy/head and non-fall-through E4, with J.BufSameCycle for legal dual updates; rendezvous requires simultaneous attributed endpoint requests/attempts and value agreement. Synchronization, E2 anchor, R2C, environment, and RESET units use their named certificates. ACert/CCert supply constituent facts while this lemma derives complete-unit enablement. ■

### Lemma J.UnitExt (one Horizon-batched locally certified extension)

Assume the matched cycle-entry invariant for the next nonempty Horizon bucket  $B_t$  of the J.ValidatedConstructionOrder and LocalGWRPackage. Execute the finite Dep\_J-minimal L1 A1/A2 predecessor set by J.LocalLift/J.PrepareCommute, carrying deferred holes through NoDependentUse work. Apply J.RegPhaseNF to obtain the canonical L1/L2/L3 arrangement, freeze requests, settle the explicit network, reconstruct all snapshots, and execute the bucket's failed A3  $\epsilon$  returns and complete A3/A4 units in commit-action rank order, using J.BufSameCycle where applicable. J.UnitEnable supplies each complete transition. No L1 action is executed after the L2 settle; dependent continuation is deferred to a later Horizon; all t-holes are empty at the exported post-boundary state. The resulting finite extension preserves replay, channel/atomic, attribution, reset, and canonical-history invariants and advances from the prior bucket ideal to the ideal containing all units in  $B_t$ . It uses no B2/B3 fairness. ■

### Theorem J.GWR-Comp (local certificates construct global witness realizability)

Fix a finite reachable RTL\_B prefix  $\tau_{\text{post}}$ , the theorem instance, stimulus, elaboration E, observation boundaries, witness w, and reset interpretation. Assume Core.RegSeq, LocalGWRPackage( $\tau_{\text{post}}$ , L), the Sys\_ref decomposition used by J.Frame-J.UnitExt, I.A0/WC, the source/RW premises, B-faithful E4 boundaries, and QualifiedSynchronousModel\_Sys\_ref(I). By Theorem Core.QSync-Ref, every reachable constructed Horizon therefore has the finite L2 SettlePoint required by J.RegPhaseNF/J.UnitExt. Then Dep\_J satisfies J.DepSound/J.HorizonOrderSafe/J.DepComplete; FR/J.DepRank prove J.DepAcyclic; J.CanonicalRank and the Horizon partition define the theorem-derived J.ValidatedConstructionOrder; J.RegPhaseNF proves its V6 phase property; and Horizon-batched J.UnitExt constructs W and a reachable  $\tau_{\text{ref}}$  satisfying GlobalWitnessRealizable( $\tau_{\text{post}}$ , W).

Proof. Generate the field-sensitive actions, deferred-hole records, Dep\_J, and FR metadata. J.DepSound/J.HorizonOrderSafe/J.DepComplete establish prerequisite coverage and commutation; J.DepRank/J.DepAcyclic derive the order. Partition the canonical order by Horizon. For each bucket, J.RegPhaseNF permutes unordered actions into the Core.Phase L1/L2/L3 order modulo J.HCanon, and J.UnitExt performs the finite phase-normal extension. J.CCert/ACert reconstruct common snapshots and updates; J.NBDecision derives shared outcomes; J.BufSameCycle handles legal buffered dual updates; J.Frame and LocalAgree preserve unrelated locations and coherence; ResetCert/J.RS handle resets. Outer induction over Horizons, with inner commit-action order within L3, yields a reachable chain covering every unit and no exported deferred holes. Assemble W from the records, order, phase-normal chain, snapshots, and invariants; G1–G3 and E1–E5 follow. ■

Non-circularity. LocalGWR supplies field-sensitive actions, deferred-hole/NoDependentUse records, snapshots, local dependencies, and Core.RegSeq evidence, but no global order, phase-normal execution, or extension. J.DepComplete/J.DepRank/J.DepAcyclic derive the order; J.RegPhaseNF derives the per-Horizon phase arrangement; J.NBDecision derives shared outcomes; and J.UnitExt/J.GWR-Comp prove

the reachable bucketed chain. The normal form is computed from graph-independent Horizon/stage/footprint facts and therefore does not assume the order it proves replayable.

### Proposition J.0 (Pairwise Composition Lemma)

For each process  $P$ , let  $A_{\text{ref},P}$  and  $A_{\text{post},P}$  be the annotated  $P$ -local projection records induced by  $\tau_{\text{ref},\text{system}}$  and  $\tau_{\text{post},\text{system}}$  under the chosen witness/annotation package. Their forgotten observable components are the ordinary projections  $\tau_{\text{ref},P}$  and  $\tau_{\text{post},P}$ , but the records also carry the issue annotations, dynamic message-call universe,  $\mu_Q$  matching, and request-attribution facts needed by E3/E5 and WC. Assume that for every process  $P$ , these annotated projections satisfy the per-process compatibility predicate  $A_{\text{ref},P} \approx_P A_{\text{post},P}$ . In readable prose below, this may be abbreviated as  $\tau_{\text{ref},P} \approx_P \tau_{\text{post},P}$  when the annotation package is fixed by context.

Thus Proposition J.0 is only a conditional composition result for a chosen compatible trace pair. It neither constructs a source execution nor proves J.GWR. Theorem J.GWR-Comp performs the local-to-global execution construction; Theorem J.1 then instantiates Proposition J.0 with that constructed pair. The restricted reverse clause continues to use the selected pair carried by `RTLConsistentRefPrefix`.

Assume every explicit abstract channel  $c$  in the chosen  $\text{Sys\_ref} / \text{RTL\_B}$  comparison has finite capacity  $B(c) \geq 0$ . Also assume, as a per-channel  $\text{Sys\_ref} / \text{RTL\_B}$  well-formedness condition, that both chosen traces satisfy the applicable E4 channel semantics at the chosen abstract boundaries. Thus, for  $B(c)=0$  channels, committed  $\text{Push}_c / \text{Pop}_c$  events obey the same-cycle rendezvous semantics; for  $B(c)>0$  channels, the committed  $\text{Push}_c / \text{Pop}_c$  history obeys the reset-empty, cycle-boundary, non-fall-through bounded-FIFO semantics with capacity  $B(c)$ , including FIFO order, no underflow, no overflow, no duplication, and no loss. Proposition J.0 does not derive these per-channel E4 facts; it assumes them as part of the definition of the compared well-formed  $\text{Sys\_ref} / \text{RTL\_B}$  traces.

No weak-fairness or channel-progress premise is needed for this conditional pairwise lifting lemma: Proposition J.0 does not construct a matching trace, but only lifts already-compatible per-process trace projections and already well-formed per-channel communication histories to the aggregate system trace. The finite selected-unit witness-construction premises are used later in Theorem J.1 when proving the finite execution-prefix existence result; weak fairness and channel progress are used only when a corresponding infinite fair execution is asserted. Then the aggregate traces are compatible:  $\tau_{\text{ref},\text{system}} \approx_{\text{sys}} \tau_{\text{post},\text{system}}$  under rules E1–E5 (given the R rules and B rules).

**Notation:** The order relation  $<_{\text{src}}$  used below is the dynamic source-induced order from Appendix G / R1–R4: it ranges over dynamic operation instances in the compared execution / witness, not over all syntactic call sites in the source text. Equivalently, when the process must be explicit, write  $<_P$ . Untaken branches and unexecuted loop iterations are not members of the witness-specific dynamic universe.

#### Proof

We prove each compatibility property. The per-channel E4 facts are not derived inside this proposition; they are part of the per-channel  $\text{Sys\_ref} / \text{RTL\_B}$  well-formedness hypothesis for the chosen traces.

#### E1 (Synchronization-order preservation):

Fix any process  $P$ . By the proposition hypothesis,  $\tau_{\text{ref},P} \approx_P \tau_{\text{post},P}$ , so the synchronization events of  $P$  occur in the same source order in both traces. Since this holds for every  $P$ , E1 holds for the system traces  $\tau_{\text{ref},\text{system}}$  and  $\tau_{\text{post},\text{system}}$ .

#### E2 (Signal-visibility preservation):

Fix any process  $P$ . If  $P$  is a sequential process, then by the proposition hypothesis (using the E2 anchoring relation embodied in  $\tau_{\text{ref},P} \approx_P \tau_{\text{post},P}$ ), every signal Read in  $P$  is anchored to  $P$ 's closest preceding synchronization event, and every signal Write in  $P$  is anchored to  $P$ 's closest succeeding synchronization event, in both  $\tau_{\text{ref},P}$  and  $\tau_{\text{post},P}$ .

If  $P$  is a combinational process, then R2C rather than  $\text{pred}_S/\text{succ}_S$  anchoring applies:  $P$ 's observable

signal Read/Write events are represented by the R2C fixed-point signal-observation unit for the relevant combinational process/network component, matched by that component's local R2C occurrence ordinal as specified in the R2C unit matching/index convention. By the combinational well-formedness condition and the compatibility hypothesis for the fixed stimulus / selected witness, the pre-HLS and post-HLS combinational networks reach corresponding unique  $\epsilon$ -fixed-point valuations for the matched R2C occurrence unit, so the matched Read/Write labels agree within that fixed-point unit. This local R2C-unit agreement is an explicit atomicity constraint; it does not require Sys\_ref and RTL\_B to use the same global observable cycle-bucket decomposition for unrelated independent events. Since the appropriate sequential or combinational signal-visibility condition holds for every process P, E2 holds for  $\tau_{\text{ref,system}}$  and  $\tau_{\text{post,system}}$ .

**E3 (Safe message issue ordering):**

Fix any process P. E3 is the post-side issue-order conjunct of the per-process compatibility relation  $\approx_P$ , not an additional ordering property derived from the Sys\_ref trace. Thus what must be checked here is that P's post-HLS execution satisfies E3 as stated in Appendix G: for any  $q1, q2 \in Q_{\text{exec},P}$  with  $q1 <_{\text{src}} q2$ , including ordinary distinct-interface message calls that appear in dynamic source sequence,  $\text{issue\_post}(q1) \leq \text{issue\_post}(q2)$ , and the issued set is prefix-closed under that source order. Same-cycle issue is permitted where allowed, and a later operation may be issued before an earlier operation commits under Policy L, but a source-later operation may not be issued before the source-earlier operation has issued. This post-side property is part of the local per-process hypothesis  $\tau_{\text{ref},P} \approx_P \tau_{\text{post},P}$ ; in the execution-level use of Proposition J.0 / Theorem J.1, the needed post-side witness/order fact is supplied by Appendix I / Implementation assumption I.A0. Since the E3 post-side conjunct holds for every P, the aggregate system pair satisfies E3.

**E4 (Per-channel channel semantics):**

E4 holds by the per-channel Sys\_ref / RTL\_B well-formedness hypothesis of Proposition J.0. That hypothesis states that, for every explicit abstract channel c in the chosen comparison, the Sys\_ref and RTL\_B traces already satisfy the applicable E4 channel semantics at the chosen abstract boundary: same-cycle rendezvous semantics when  $B(c)=0$ , and reset-empty, cycle-boundary, non-fall-through bounded-FIFO semantics when  $B(c)>0$ . Thus Proposition J.0 uses E4 as an assumed channel-realization condition for the chosen traces; it does not attempt to derive E4 from the per-process compatibility relation. Accordingly, Proposition J.0 should be read as a composition theorem under already-established process and channel conformance hypotheses. It does not prove that the implementation realizes the chosen abstract channel boundaries, nor that the back-annotated capacities  $B(c)$  satisfy B-faithfulness as defined in the Normative Formal Core. Those facts are separate RTL/tool-flow compliance obligations required before the theorem is applied to a concrete generated design.

**E5 (Messages cannot cross syncs):**

For every synchronization call s (in the source order used by R3 / pref\_S / suff\_S over  $Q_{\text{exec},P}$ ):

$\forall q \in \text{pref\_S}(s) : \text{issue\_post}(q) \leq \text{clk\_post}(s)$

$\forall q \in \text{suff\_S}(s) : \text{issue\_post}(q) > \text{clk\_post}(s)$

$\forall q \in \text{pref\_S}(s)$  with q a blocking Push/Pop :  $m(q)$  is defined and  $\text{clk\_post}(m(q)) \leq \text{clk\_post}(s)$ .

The third clause is the E5/Core.SyncFence completion fence. Same-cycle completion is allowed; on the post side, the "accounted for before interval advance" convention is proof/certificate attribution for the same-cycle batch, not a requirement for observable physical sub-cycle ordering in RTL. Failed non-blocking polls in  $Q_{\text{exec},P}$  are constrained at the issue level by E5 but contribute no committed  $\Sigma$ -event when they fail.

This property is guaranteed per-process by the proposition hypothesis on the chosen compatible pair (with Appendix I supplying the witnesses in the execution-level Post  $\rightarrow$  Ref use case) and requires no inter-process reasoning, so it lifts directly to the system level.

**Conclusion:**

All five compatibility properties hold at the system level for this chosen compatible trace pair. The composition is valid because:

- intra-process properties are preserved by the per-process compatibility hypothesis;
- inter-process communication respects the applicable E4 channel semantics by the per-channel Sys\_ref / RTL\_B well-formedness hypothesis; and
- no progress or fairness argument is needed inside Proposition J.0, because the lemma composes already-given traces.

The existence of the source trace used in the normal Post  $\rightarrow$  Ref application is supplied by Theorem J.GWR-Comp; a J.GWR-Direct package supplies the alternate entry route. Therefore  $\tau_{\text{ref},\text{system}} \approx_{\text{sys}} \tau_{\text{post},\text{system}}$ .  $\square$

**Remark 1.** When Sys\_ref = Sys\_B and all B(c)=0, Sys\_ref coincides with the rendezvous interpretation of Sys, so Proposition J.0 and Theorem J.1 both specialize to the rendezvous case. In the elaborated cases, the source-side proof target is Sys\_ref = Sys\_B^elab.

**Remark 2 (Reference-model ladder: raw Sys, abstract Sys\_B, and elaborated Sys\_B^elab).**

Definition Core.1 is the normative definition of Sys\_ref. This remark explains the model ladder, back-annotation intent, and circumstances selecting Sys\_B or Sys\_B^elab; it introduces no alternative reference-model definition.

The Matchlib library provides a capacity back-annotation feature that extracts finite capacities B(c) from the RTL realization of each message-passing channel and makes them available to the source-level model. To avoid ambiguity, this document distinguishes three source-side models:

- Sys: the raw source-level rendezvous interpretation of the user-written processes, i.e., the model obtained when every effective channel capacity is zero and no additional abstract realization structure is introduced.
- Sys\_B: the abstract bounded-FIFO source-level interpretation of the same user-written processes under E4, with finite capacities B(c) at the chosen abstract channel boundaries. When all B(c)=0 and no additional abstract realization structure is needed, Sys\_B coincides with Sys.
- Sys\_B^elab: the elaborated back-annotated source-level model used in the more complex cases discussed in Appendix B / Appendix Q, in which additional explicit abstract staged/bundle/join channels are introduced so that conjunctive/join dependencies, synchronization-boundary ramp-down / epoch-gating, occupancies, and enablement are represented at explicit abstract boundaries rather than as hidden predicates at the outer  $\Sigma$ -visible endpoints.

Formal proof target. Unless a statement explicitly says otherwise, the formal results of Appendices J–K target the prescribed source reference model Sys\_ref, defined as follows:

- (i) Sys\_ref = Sys\_B in the ordinary case where finite capacities B(c) at the chosen abstract boundaries suffice to capture the RTL realization; and
- (ii) Sys\_ref = Sys\_B^elab in the elaborated cases where additional explicit abstract staged/bundle/join channels or explicit sync-boundary / epoch-gate channels are required.

In particular, if an RTL pipeline with an explicit synchronization boundary ramps down by draining already accepted work and withholding producer-facing availability for capacity usable only after the synchronization call commits, then the prescribed source reference model is Sys\_B^elab. That withholding shall be represented as ordinary ChannelDisabled behavior at an explicit abstract boundary

of  $\text{Sys\_B}^{\text{elab}}$ . It shall not be modeled as an extra hidden disablement predicate on an outer bounded channel whose E4 occupancy test would otherwise make the transfer enabled.

For a simple pipelined loop with a single message-passing input and output,  $\text{Sys\_ref}$  is ordinarily just  $\text{Sys\_B}$ : the RTL pipeline’s internal staging capacity is accounted for entirely by the back-annotated effective capacity  $B(c)$  at the chosen  $\text{Sys\_B}$  channel boundary. Concretely,  $\text{Sys\_B}$  still has exactly one  $\Sigma$ -visible  $\text{Push\_c}(v)$  at the producer endpoint and one  $\Sigma$ -visible  $\text{Pop\_c}(v)$  at the consumer endpoint. With overlapped iterations, the post-HLS RTL may accept/read-ahead up to  $B(c)$  values into internal pipeline staging before older iterations produce their corresponding outputs; these accept/read-ahead transfers are modeled as internal microstate ( $\epsilon$ ) within the concrete realization contributing to  $B(c)$ . The RTL may realize this staging on either side of a particular module-local “Pop interface” handshake point; that handshake point need not coincide with the chosen  $\Sigma$ -visible Pop endpoint. By construction, the chosen  $\text{Sys\_B}$  boundary encloses all staging counted in  $B(c)$  between the  $\Sigma$ -visible endpoints, so movements within that enclosed realization (including stage-to-stage handoff) do not create additional  $\Sigma$ -visible committed Push/Pop events beyond the single boundary Push/Pop events.

In more complex cases such as HW pipelines with multiple message-passing inputs, the back-annotation mechanism may elaborate the source-side simulation into  $\text{Sys\_B}^{\text{elab}}$  by introducing additional explicit abstract realization structure. In those cases the formal target is  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$ , and  $B(\cdot)$ ,  $\text{occ\_}(t)$ , E4,  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled/ChannelDisabled}$ , and the Appendix K blocking/WFG machinery are interpreted over the explicit abstract channels of that elaborated model. See Appendix B and Appendix Q.

Operational note. In all three cases,  $\text{Sys\_ref}$  is a transition system with explicit process states, explicit abstract channel states, operation-attributive pending requests, and  $\epsilon$ -microstate as defined in Appendix G’s operational interpretation of  $\text{Sys\_ref}$ .  $\text{Sys\_B}$  and  $\text{Sys\_B}^{\text{elab}}$  differ in the channel/elaboration component of that transition system, not in the ordinary dynamic source order of the user-written process code. Source-level message-call instances that appear in sequence in a process remain ordered by  $<_{\text{src}}$ ; elaboration may introduce additional explicit proof-internal frontier items and abstract channels, but it does not make ordinary source-sequenced calls incomparable.

Note that in all cases the elaborated structure (staged/bundle/join channels, combinational couplers, and the projection map  $\Pi_{\text{elab}}$ ) introduced for a particular pipelined realization is local to that process’s side of the abstract channels at its boundaries. The complementary endpoint processes see only the abstract  $\Sigma$ -visible channels at the boundary and are unaffected by the elaboration choices.

For Lean-style formalization, the phrase “chosen  $\text{Sys\_ref}$ ” should be read as an explicit theorem parameter: either  $\text{Sys\_B}$  itself, or  $\text{Sys\_B}^{\text{elab}}(E)$  for a well-formed elaboration package  $E$  as defined in Appendix Q. Likewise, references to the Appendix I/L witness machinery assume that the actual WC premise has been discharged. The corresponding audit entry records whether WC was proved through NB-CF, an Appendix L snooping / witness-selector construction satisfying Lemma L.2, another direct witness construction, or complete mixed per-site coverage. NB-CF and Appendix L snooping are provenance routes for proving WC, not independent theorem premises alongside WC.

Side-condition audit before theorem application. Applying Theorem J.1, Corollary J.1, or Appendix K to a concrete design requires the side-condition discharge record in Appendix M.7a. The normal  $\text{Post} \rightarrow \text{Ref}$  route shall identify  $\text{Sys\_ref}$ , discharge WC, and provide Definition J.LocalGWR: complete J.PCert, J.CCert, J.ACert, environment/reset, construction-action inventory, typed footprint, request-snapshot, locally generated  $\text{Dep\_J}$ , LocalAgree, and coverage evidence, together with per-channel B-faithfulness.

J.CanonicalRank is derived by the theorem. A flow may instead record J.GWR-Direct together with its direct-provenance J.ValidatedConstructionOrder and independently checked V1–V5 obligations. When Corollary J.1 or Appendix K is used, the application must additionally discharge J.OfferReach, J.OfferConf, and J.KObsConf for the same theorem-produced or directly supplied W, source prefix, RTL prefix, H, O, and cutpoints. Liveness uses the additional K.Drain,  $K\epsilon$ , K.CV, A5-L, selected A5-L discharge route, progress, NB-Align, and L.Dir obligations as applicable. Ordinary synthesis does not automatically discharge any of these named conditions.

**Remark 3 (Practical rendezvous liveness screening under determinism).**

When B1 holds and the design’s control flow does not branch on empty/full observations of non-blocking interfaces (i.e., it depends only on the ordered history of observable Push/Pop/synchronization actions, not on “probe” outcomes (NB-CF, Appendix I.2)), the rendezvous specialization  $B(c)=0$  is often an effective practical screen for design-level deadlocks: it exercises the most constrained communication discipline and can expose true cyclic data-dependency deadlocks early. Because rendezvous is strictly more constrained than buffered/elaborated operation, it can also yield false positives: a deadlock observed under  $B(c)=0$  may disappear once finite buffering or explicit elaborated staged/bundle/join structure is present. However, bounded buffering can still introduce capacity-induced deadlocks (FIFO-depth artifacts) when FIFO depths are underestimated; these deadlocks are real for that modeled capacity configuration but may disappear once capacities are increased to the intended design point. Increasing buffer capacity (or applying dynamic buffer-growth strategies) is a standard way to distinguish depth-limited deadlocks from true cyclic dependency deadlocks. Accordingly, the formal results of Appendices J–K deliberately target the prescribed source reference model Sys\_ref of Remark 2, with the actual back-annotated capacities and explicit abstract boundaries required by the chosen realization, rather than relying on rendezvous alone.

**Remark 4 (When raw Sys may be used instead of the prescribed source reference model for safety-only checks).**

If verification is concerned only with safety properties over the chosen DUT-boundary projection  $\Sigma$  (denote it  $\Sigma_{\text{DUT}}$ ) (and not with deadlock/liveness), the prescribed source reference model Sys\_ref of Remark 2 remains the default reference model. This is because  $\Sigma_{\text{DUT}}$  includes committed channel-transfer events, and bounded buffering / explicit staged-bundle-join structure generally changes the set/timing of those committed Push\_c/Pop\_c events relative to the raw rendezvous case.

**Practical note (early bring-up):** Before accurate capacities  $B(c)$  are available (or before back-annotation/elaboration can be used), running the rendezvous specialization Sys against the intended testbench is still useful to validate functional intent and catch gross protocol/ordering bugs early; however, it remains a screening check, not a proof about the buffered/elaborated RTL-correspondent model.

**Important (soundness):** When any channel or explicit abstract staged/bundle/join boundary that can influence  $\Sigma_{\text{DUT}}$ -observable behavior has positive effective capacity or nontrivial enablement structure, the raw rendezvous model Sys is often a restriction of Sys\_ref / RTL\_B at  $\Sigma_{\text{DUT}}$  (it can admit fewer  $\Sigma_{\text{DUT}}$  behaviors). Therefore, using Sys in place of Sys\_ref is generally suitable only as a debug/screening check: a failure can be useful diagnostically, but a pass does not by itself prove  $\Sigma_{\text{DUT}}$ -safety of the buffered/elaborated implementation.

Sys may be used as a proof-equivalent substitute for Sys\_ref only when Sys and Sys\_ref are  $\Sigma_{\text{DUT}}$ -equivalent for the chosen  $\Sigma_{\text{DUT}}$  (i.e., their  $\Sigma_{\text{DUT}}$ -projected trace sets coincide under the intended stimulus). A simple sufficient condition is that every  $\Sigma_{\text{DUT}}$ -visible explicit abstract channel/boundary in Sys\_ref has effective capacity zero and that buffering/elaboration elsewhere is  $\Sigma_{\text{DUT}}$ -opaque (it cannot enable/disable/reorder  $\Sigma_{\text{DUT}}$ -events or change when  $\Sigma_{\text{DUT}}$ -events occur). In all other cases—i.e., if

any  $\Sigma_{\text{DUT}}$ -visible explicit abstract boundary has positive effective capacity, or if buffered/elaborated internal channels can affect when  $\Sigma_{\text{DUT}}$ -events occur—use the prescribed  $\text{Sys\_ref}$  of Remark 2.

**Formal soundness condition.** Let  $\text{Traces}_{\Sigma_{\text{DUT}}}(\text{M})$  denote the set of  $\Sigma_{\text{DUT}}$ -projected traces of model  $\text{M}$  under the intended fixed stimulus.  $\text{Sys}$  may replace  $\text{Sys\_ref}$  for proving a universal  $\Sigma_{\text{DUT}}$ -safety property  $\phi$  only if  $\text{Traces}_{\Sigma_{\text{DUT}}}(\text{Sys}) = \text{Traces}_{\Sigma_{\text{DUT}}}(\text{Sys\_ref})$ ; under that condition,  $\text{Sys} \models \phi$  iff  $\text{Sys\_ref} \models \phi$  (and otherwise  $\text{Sys}$  is only a diagnostic/screening check as stated above).

### Lemma J.RS (MatchedResetEpochSplice).

Assume RW1–RW5 and the fixed-stimulus interpretation. Let  $\text{H}$  be a matched Theorem J.1 prefix satisfying  $\text{Inv}(\text{H})$  whose next unit in the selected validated construction-order enumeration is a RESET unit  $\gamma_{\text{R}} = \text{RESET}(\text{S}_{\text{p}}, \text{S}_{\text{c}})$ , matched between  $\text{RTL}_{\text{B}}$  and  $\text{Sys\_ref}$  with the same scope. Then  $\text{Sys\_ref}$  extends from  $\sigma_{\text{ref}}[\text{H}]$  by its corresponding reset transition, and  $\text{Inv}(\text{H} \cup \{\gamma_{\text{R}}\})$  holds: (I1) holds because  $\gamma_{\text{R}}$  is matched with the same scope and the only required orders at the boundary are the epoch-boundary orders above; (I2) holds because, for every  $\text{P} \in \text{S}_{\text{p}}$ , the  $\text{P}$ -local replay state after  $\gamma_{\text{R}}$  is the deterministic epoch-initial state prescribed by RW3 — identical in both models by the reset-correspondence obligation — while for  $\text{P} \notin \text{S}_{\text{p}}$  the replay state is unchanged by RW5; RW2 guarantees that no request-attribution fact survives the boundary for  $\text{S}_{\text{p}}$ , so Lemma I.DR' applies epoch-locally with  $\gamma_{\text{R}}$  delimiting the  $\text{P}$ -local projection; and (I3) holds because channels in  $\text{S}_{\text{c}}$  restart with reset-empty histories per RW1 and the Core reset-empty FIFO convention — the  $\text{occ}_{\text{c}}(\text{t})$  definition is already measured from the most recent reset boundary — while channels outside  $\text{S}_{\text{c}}$  retain their histories by RW5. ■ Subsequent induction steps within the new epoch proceed exactly as in the initial epoch;  $\text{START}_{\text{P}}$  for  $\text{P} \in \text{S}_{\text{p}}$  is matched as the new epoch's first ordinary unit. Open dispositions that are not covered by RW1–RW5 — for example, an un-modeled partial reset of one endpoint of a live channel, retention of in-flight messages across the boundary, or cross-epoch request re-attribution — place the design outside Theorem J.1, Corollary J.1, and Appendix K until modeled explicitly. Under the Core.CycleResult interface,  $\gamma_{\text{R}}$  is the emitted resetUnits occurrence corresponding to that scoped reset transition; the induction index remains  $\gamma_{\text{R}}$  itself, not the enclosing cycle result. Thus the CycleResult refactor does not change the reset splice case or its epoch-local replay/channel-history obligations.

### Expanded proof of Theorem J.GWR-Comp and Theorem J.1

$\text{Post} \rightarrow \text{Ref}$ , compositional route. Fix  $\tau_{\text{post}} \in \text{Reach}_{\text{post}}$  and  $\text{LocalGWRPackage}(\tau_{\text{post}}, \text{L})$ . Generate the complete field-sensitive action and deferred-hole inventory,  $\text{Dep}_{\text{J}}$ ,  $\text{FR}$ , and  $\text{J.CanonicalRank}$ . Partition the complete units by their selected Horizon, retaining explicit atomic groups. Within each Horizon, stable proof keys order only otherwise commuting actions;  $\text{J.RegPhaseNF}$  later removes dependence on those incidental choices modulo  $\text{J.HCanon}$ .

Let  $\text{B}_1 \dots \text{B}_n$  be the nonempty Horizon buckets and  $\text{H}_{\text{i}}$  the finite required-prefix ideal containing all units through  $\text{B}_{\text{i}}$ ; within  $\text{B}_{\text{i}}$  retain the commit-action suborder only for the inner L3 pass. Construct by outer induction reachable cycle-entry and post-boundary states. The invariant is: (I1) exactly  $\text{H}_{\text{i}}$  has committed with matching units/values/identities and required order; (I2) process replay and every pending/deferred-hole record agree with the local certificates at cycle entry, while no deferred hole survives a post-boundary state; (I3) explicit boundary/channel/atomic state equals the CCert/ACert reconstruction; (I4) requests, witness choices, reset epochs, direct-input anchors, and environment history agree; and (I5) the constructed prefix through  $\text{H}_{\text{i}}$  satisfies  $\text{J.RegPhaseReach}$  and has the same CanonHist quotient as the selected RTL prefix through those units.

Base case. Use the matched initial state, reset-empty channel state, empty request inventory, fixed environment history, and LocalAgree. For later dynamic resets, ResetCert and Lemma J.RS re-establish the same invariant per reset epoch while framing components outside the reset scope.

Induction step. Let  $B_t$  be the next Horizon bucket. Execute all finite L1 A1/A2 predecessors, including actions carried past unresolved NB calls under NoDependentUse. Field-sensitive framing and J.DepComplete permit the required commutations; true dependencies remain ordered. J.RegPhaseNF then freezes the complete request set, performs the L2 settle, and orders all A3/A4 decisions/commits in L3. A failed NB writes its returned status and retires its hole; a successful NB is one of the bucket's complete units. No decision-dependent continuation remains in  $t$ , and no new request is issued after settle.

Every witness-significant failed or successful NB call is therefore evaluated from one common settled incident-boundary snapshot. The construction cannot choose failure with a matching rendezvous request present, success in an empty/full state that forbids it, or an outcome for the wrong dynamic call. WC identifies the selected result; J.CCert/J.NBDecision prove it unique. Outcome-independent same-Horizon work is already in L1; outcome-dependent work begins only in a later bucket.

J.UnitEnable proves every complete unit in the bucket enabled at the appropriate L3 point. The buffered case uses common pre-update occupancy/head and J.BufSameCycle; rendezvous uses simultaneous attributed requests/attempts and payload identity; synchronization/E2/R2C/environment/reset use their named certificates. Executing the inner commit-action order reaches the post-boundary state for  $H_{iUB_t}$ . The phase-normal segment has the same complete units, same post-boundary semantic state, and same J.HCanon quotient as any permitted same-Horizon topological ordering.

Lemma I.DR' updates participant replay state; J.CCert/ACert update exact channel/atomic state; J.Frame preserves nonparticipants; LocalAgree preserves coherence; and the all-holes-resolved invariant establishes the clean post-boundary cutpoint. Thus (I1)–(I5) hold for the next bucket. Finiteness follows from the local certificates/FPD, not fairness.

After the final Horizon bucket, the chain covers every unit of  $\tau_{\text{post}}$ . Assemble  $W$  from the local certificates, generated graph, derived bucketed order, phase-normal segments, snapshots, and invariants. Definition J.GWR and J.RegPhaseReach hold. The terminal source prefix and  $\tau_{\text{post}}$  have compatible annotated projections and well-formed explicit-channel histories, so Proposition J.0 proves Theorem J.1's compositional Post  $\rightarrow$  Ref clause.

Post  $\rightarrow$  Ref, direct route. If a validated  $W$  satisfying Definition J.GWR is supplied directly, take the reachable terminal source prefix from its chain and apply Proposition J.0 to the annotated pair. This route has the same semantic conclusion but bypasses the local-certificate derivation.

Normalization and cutpoints. Neither route appends a maximal  $\varepsilon$ -normalization suffix after each matched unit. Corollary J.1 separately applies B4/B5 to the selected terminal prefixes and then requires J.OfferReach, J.OfferConf, and J.KObsConf for the normalized cutpoints. The theorem-produced  $W$  is the authoritative source of the operational identities and history used by those obligations.

Ref  $\rightarrow$  Post clause. Let  $\tau_{\text{ref}} \in \text{Reach}_{\text{ref}}$  and choose  $\tau_{\text{post}}^* \in \text{Reach}_{\text{post}}$  and  $W$  with  $\text{RTLConsistentRefPrefix}(\tau_{\text{ref}}, \tau_{\text{post}}^*, W)$ . By definition, the selected pair already carries compatible annotated projections, Issue/Commit and request-attribution evidence, dynamic identities, WC/ $\mu_Q$  data, and E4 histories. Proposition J.0 therefore gives  $\tau_{\text{ref}, \text{system}} \approx_{\text{sys}} \tau_{\text{post}}^*, \text{system}$ ; take  $\tau_{\text{post}} = \tau_{\text{post}}^*$ . B1 supplies uniqueness only of the selected RTL  $\Sigma$  projection, not of the annotation package. This is not a construction from an arbitrary source run.  $\square$

### Lemma J.HCanon (Canonical replay-history quotient agreement)

Fix the same theorem instance  $I$ , elaboration  $E$ , witness  $w$ , reset interpretation, dynamic operation identities, and theorem-produced or directly validated J.GWR annotation package. Let  $\tau_{\text{post}}$  be a finite RTL<sub>B</sub> prefix and  $\tau_{\text{ref}}$  the Theorem J.1 source witness prefix selected for  $\tau_{\text{post}}$ . If  $\tau_{\text{ref}}$  and  $\tau_{\text{post}}$  satisfy the annotated observable-unit compatibility relation of Theorem J.1 under that fixed package, then their canonical enriched replay-history quotients are equal:

$\text{CanonHist}_{E,w}(\tau_{\text{ref}}) = \text{CanonHist}_{E,w}(\tau_{\text{post}})$ .

Proof. The compatibility relation matches the same atomic observable units with the same labels, values, dynamic identities, and reset scopes. The fixed witness  $w$  and  $\mu_Q/WC$  package identify the same witness-selected failed non-blocking, arbitration, signal, and  $\varepsilon$ -choice outcomes used by replay. Theorem J.1 preserves each process-local required order, each per-channel committed-transfer ordinal/FIFO order, every explicit atomic rendezvous or paired-synchronization grouping, every RESET marker, and every predecessor required by  $\prec_J$ . CanonHist retains exactly those fields. It discards only incidental source-versus-RTL clock-bucket placement and ordering among units that are independent under the retained constraints. Therefore both annotated prefixes map to the same typed quotient  $H$ . This lemma asserts equality of the canonical replay quotient, not equality of concrete cycle buckets or historical Issue timestamps. ■

### Definition J.RegPhaseReach (phase-correct reachability of the selected source witness)

For fixed  $I, E, w$  and a selected J.GWR source prefix  $\tau_{\text{ref}}$  ending at  $\sigma_{\text{ref}}$ ,  $\text{RegPhaseReach}_{I, E, w}(\tau_{\text{ref}}, \sigma_{\text{ref}})$  means that  $\tau_{\text{ref}}$  is Policy-L-legal under  $\text{Core.RegSeq}$  and admits the Horizon-bucket decomposition of  $\text{J.RegPhaseNF}$ : every cycle's process-owned A1/A2 actions occur in L1, one L2 settle fixes the requests/snapshots, all A3/A4 decisions/commits occur in L3, ReturnedAtCycle-dependent work occurs only in later Horizons, and no deferred NB hole remains at  $\sigma_{\text{ref}}$ . A J.GWR-Direct package must supply this checked property for its selected prefix.  $\text{RegPhaseReach}$  establishes the L0–L4 ordering/legality component used by Definition  $\text{Core.LAdm}$  through  $\text{Core.Phase}$ ; it does not establish nonwithdrawal,  $K\varepsilon$ , or the continuation-level  $\text{Core.Drain}/K.\text{Drain}$  discipline.

### Corollary J.RegPhaseReach-Comp (compositional derivation of phase-correct reachability)

Under the hypotheses of Theorem J.GWR-Comp, every constructed source prefix  $\tau_{\text{ref}}$  ending at  $\sigma_{\text{ref}}$  satisfies  $\text{RegPhaseReach}_{I, E, w}(\tau_{\text{ref}}, \sigma_{\text{ref}})$ . Proof. Lemma I.3 carries each enriched deferred NB hole through exactly the same-Horizon work certified by  $\text{NoDependentUse}$ ;  $\text{J.RegPhaseNF}$  normalizes each Horizon modulo  $\text{J.HCanon}$  into L1 process actions, one L2 settled snapshot, and L3 decisions/commits, using field-sensitive framing and  $\text{J.BufSameCycle}$ ; and Horizon-batched  $\text{J.UnitExt}$  preserves that normal form while constructing the reachable prefix. ReturnedAtCycle-dependent work is assigned only to later Horizons, and the exported-cutpoint invariant removes every unresolved deferred hole at  $\sigma_{\text{ref}}$ . ■

### Definition J.OfferReach (reachable source realization of a KObs record)

Fix  $I, E, w, H, O$ , and the particular  $\text{Sys}_{\text{ref}}$  witness prefix selected for the current Theorem J.1 / Corollary J.1 application.  $\text{OfferReach}_{I, E, w}(H, O; \tau_{\text{ref}}, \sigma_{\text{ref}})$  holds when  $\tau_{\text{ref}}$  is a finite reachable  $\text{Sys}_{\text{ref}}$  prefix under the same stimulus; its selected proof-relevant source normalization satisfies  $\text{NormRef}_I(\tau_{\text{ref}}, v_{\text{ref}}, \sigma_{\text{ref}})$  for some recorded  $v_{\text{ref}}$  and ends at  $\sigma_{\text{ref}} \in \text{KCutReach}_{\text{ref}}(I)$ ;  $H = H_{E, w}(\sigma_{\text{ref}})$ ;  $O = \text{ResOffers}_{E, w}(\sigma_{\text{ref}})$ ; the Issue/ReturnedAtCycle/Commit, E3/E5,  $\text{Core.IC}$ , reset, elaboration, and explicit-boundary rules hold; and no proof-internal deferred NB hole is exported at  $\sigma_{\text{ref}}$ .  $\text{OfferReach}$  remains an image/realizability predicate, not a  $\text{Core.LAdm}$  qualification.  $\text{OfferReach}$  is an image/realizability predicate. It is not implied by type-correctness, source-prefix closure, or endpoint cardinality of an arbitrary tuple  $\langle E, w, H, O \rangle$ . In Corollary J.1 it is discharged for the same normalized source witness selected from Theorem J.1, not for an unrelated existential source state. A source execution log, mechanized reachability proof, or tool/witness certificate may discharge it.

Historical Issue and ReturnedAtCycle data may be used internally to prove reachability, phase legality, and the current residual-offer set, but they are not KObs fields and are not compared as cross-model cycle timelines.  $\text{OfferReach}$  records the realized H/O cutpoint;  $\text{J.RegPhaseReach}$  records the theorem-

derived/direct L0–L4 phase property; and K.DrainReach records the remaining Appendix K continuation discipline. Their composed K.RLAdmReach is the stricter  $\text{Pref\_L}^{\text{adm}}(\text{I})$  qualification.

#### Definition K.DrainReach and K.RLAdmReach (composed Core.LAdm qualification)

Fix  $\text{I}, \text{E}, \text{w}, \text{H}, \text{O}, \tau_{\text{ref}}, \sigma_{\text{ref}}$  satisfying OfferReach.  $\text{DrainReach\_I}, \text{E}, \text{w}(\text{H}, \text{O}; \tau_{\text{ref}}, \sigma_{\text{ref}})$  holds when the exact prefix and instance binding satisfy request nonwithdrawal and  $\text{K}\epsilon$ ; discharge Core.Drain through K.Drain from every L2 fully-disabled-frontier SettlePoint reached by  $\tau_{\text{ref}}$ ; and discharge Core.SettleKBridge for the exact normalization/cutpoint path that ends at  $\sigma_{\text{ref}}$ , including the no-new-later-work, finite non-replenishing drain, sync-boundary, attribution, and  $\epsilon$ -inertness obligations.

$\text{RLAdmReach\_I}, \text{E}, \text{w}(\text{H}, \text{O}; \tau_{\text{ref}}, \sigma_{\text{ref}})$  is the conjunction of  $\text{J.RegPhaseReach\_I}, \text{E}, \text{w}(\tau_{\text{ref}}, \sigma_{\text{ref}})$  and  $\text{DrainReach\_I}, \text{E}, \text{w}(\text{H}, \text{O}; \tau_{\text{ref}}, \sigma_{\text{ref}})$ , with the same fixed I/E/w/stimulus/environment/arbitration/reset choices. Hence RLAdmReach proves  $\tau_{\text{ref}} \in \text{Pref\_L}^{\text{adm}}(\text{I})$ . On the compositional route the phase component is theorem-derived, while DrainReach remains an explicit checked bridge from implementation/source operational evidence; on the direct route both components are directly validated. Neither equal KObs nor OfferReach implies DrainReach. A K-facing CF-0 package shall carry/refer to the RegPhaseReach provenance and the DrainReach/Core.SettleKBridge proof for the exact cutpoint tuple.

#### Definition J.RelevantReq (KObs-relevant RTL endpoint request)

At an RTL K-facing cutpoint  $\rho_{\text{post}} \in \text{KCutReach\_post}(\text{I})$ , an asserted request level  $r$  is KObs-relevant for elaboration  $\text{E}$  when: (i)  $r$  is endpoint-owned and lies at an explicit abstract boundary selected by  $\text{E}$ ; and (ii) the presence or absence of  $r$  can affect the reconstruction of BoundaryReq, a rendezvous endpoint-request/co-presence fact, ChannelEnabled/ChannelDisabled, Front, WFG, WFG+, or a derived Dead\_K/Dead\_KS diagnostic predicate. Relevance is determined from the selected physical/abstract boundary and the reconstructed cutpoint interface, not from whether the flow happened to supply an attribution. Thus an asserted boundary request with missing attribution remains relevant and causes J.OfferConf to fail; it cannot be dropped from the abstraction by calling it unrecognized. K.Resp timers are not KObs fields and are validated separately by CF-S.

The dynamic operation/frontier/epoch attribution used below is supplied by the operational request-attribution component constructed by J.GWR-Comp or supplied by J.GWR-Direct, constrained by Appendix C's Issue/Commit linkage and Implementation Assumption I.A0. It is not inferred from the ready/valid wire level alone. A post-cycle request level proved by those annotations to belong only to an operation that committed in the selected cycle, with no next dynamic operation owning the continuing level, is not a residual offer. For a continuing assertion across a back-to-back request handoff, the annotation shall identify the next owning dynamic operation.

#### Obligation J.OfferConf (bidirectional RTL residual-offer conformance)

For an RTL K-facing cutpoint  $\rho_{\text{post}} \in \text{KCutReach\_post}(\text{I})$ , fixed  $\text{E}$  and  $\text{w}$ , and candidate residual-offer set  $\text{O}$ ,  $\text{OfferConf\_I}, \text{E}, \text{w}(\rho_{\text{post}}, \text{O})$  holds only if the following boundary conditions are satisfied at every explicit abstract boundary used by the theorem instance. All references to attribution use the theorem-produced or directly supplied operational package, Appendix C, and I.A0 as specified by Definition J.RelevantReq; no attribution is inferred solely from an asserted wire level.

(1) Offer-to-request completeness. For every  $o = (\text{P}, \text{x}, \text{q}, \text{c}, \text{dir}, \text{e}) \in \text{O}$ , the corresponding endpoint-owned RTL request at  $\text{c}$  and direction  $\text{dir}$  is asserted at  $\rho_{\text{post}}$ , is KObs-relevant, and is attributed by the selected J.GWR-Comp-produced operational package (or J.GWR-Direct package) to that same frontier item  $\text{x}$  and source operation  $\text{q}$  in reset epoch  $\text{e}$ , with the operation kind derived from  $\text{q}$ .

(2) Request-to-offer completeness and uniqueness. Every KObs-relevant endpoint-owned RTL request asserted at the selected RTL K-facing cutpoint is attributed by the selected J.GWR-Comp-produced operational package (or J.GWR-Direct package), Appendix C, and I.A0 to exactly one member of O, unless that package proves that the level belongs only to an operation that committed in the selected cycle and no next operation owns the continuing assertion. If the level remains high across a back-to-back request handoff, the attribution shall identify that next dynamic operation and the corresponding member of O. No relevant asserted request may be hidden, left unattributed, or omitted from O. At most one attributed outstanding source-level offer exists per explicit boundary and endpoint direction.

(3) Snapshot issue-order consistency. The issued-summary represented by the committed/replay history H, the WC witness information, and the current residual offers O is prefix-closed under the applicable dynamic source order and is consistent with E3, E5, reset epochs, and the selected elaboration. This is a cutpoint set invariant; OfferConf does not compare historical first-issue timestamps across models.

(4) Boundary and reset typing. Every x, q, c, direction, kind, owner, and epoch in O is supplied by E and agrees with the live RTL boundary attribution. Reset clears ending-epoch offers as required by RW2. Internal movement already represented as E4 occupancy is not also exposed as a second source-level offer.

(5) Payload scope. O need not contain pending Push payload values. Stable request payload is still required by Appendix C/Core.IC, and committed payload equality is checked by the value-labeled replay history. A flow that needs mid-flight data-wire correspondence may add it as a separate implementation obligation.

Audit-critical status of J.OfferConf(2). J.OfferConf is the authoritative RTL request-abstraction boundary for the KObs cutpoint result. In particular, request-to-offer completeness, no-hidden-request, explicit-boundary completeness, and attribution uniqueness are load-bearing obligations: omitting any asserted request that is KObs-relevant can remove a reconstructed wait-for edge or rendezvous-enablement fact and invalidates J.OfferConf, J.KObsConf, and the equal-KObs conclusion. B-faithfulness/E4 and canonical-history agreement remain separate independent obligations; satisfying them does not compensate for an incomplete request abstraction.

### Obligation J.KObsConf (composite KObs cutpoint conformance)

For every RTL K-facing cutpoint  $p\_post \in KCutReach\_post(I)$  used by Corollary J.1 or Appendix K, the flow shall select one exact package  $(\tau\_ref, \sigma\_ref, p\_post, I, E, w, H, O)$  such that: (a) the selected source and RTL prefixes use the same fixed instance, elaboration, witness, stimulus, reset-epoch interpretation, and observation boundaries; (b) Lemma J.HCanon yields H as their common canonical replay-history quotient, without requiring equal incidental cycle bucketings; (c)  $J.OfferReach\_I, E, w(H, O; \tau\_ref, \sigma\_ref)$  holds and therefore realizes O at the reachable source cutpoint; (d)  $J.OfferConf\_I, E, w(p\_post, O)$  holds and therefore identifies O bidirectionally and uniquely with the complete KObs-relevant RTL request inventory; and (e) the selected RTL abstract boundaries satisfy E4, B-faithfulness, reset correspondence, and  $\epsilon$ -opacity of the finite normalization. J.KObsConf is tied to the particular source witness chosen by Theorem J.1; an unrelated existential source cutpoint or an unrealizable residual-offer record does not discharge it.

Define the J-facing RTL observation certified by that package as  $KObsPost\_J(p\_post; E, w, H, O) := \langle E, w, H, O \rangle$ . J.OfferReach and Definition Core.18c give  $KObs\_E, w(\sigma\_ref) = \langle E, w, H, O \rangle$ . Hence J.KObsConf entails  $KObsPost\_J(p\_post; E, w, H, O) = KObs\_E, w(\sigma\_ref)$ . This typed equality is the sole cutpoint-level correspondence used by Appendix K. It is not equality of arbitrary RTL and source microstates, and it creates no independent clause-based state relation.

## J.2a KObs Reconstruction Theory

The following lemmas are source-side reconstruction results over reachable Core.18c KObs records. They are placed here because Corollary J.1 consumes them after J.OfferReach and J.OfferConf establish one common typed record across the selected source and RTL cutpoints.

### Lemma J.KObs-Proc (Process replay reconstruction).

Let  $\sigma \in \text{KCutReach\_ref}(I)$  under  $I, E, w$  and let  $K = \text{KObs\_E}, w(\sigma)$ . For every process  $P$ ,  $\text{ProcRec\_P}(K)$  equals the  $I.\text{ReplayObs}$  source slice of  $\sigma$  used by Lemmas I.DR/I.DR' and the Appendix J finite-ideal construction. Consequently  $\text{DoneRec\_P}(K)$ ,  $\text{RemainRec\_P}(K)$ ,  $\text{NextFrontRec\_P}(K)$ , all replay-derived next-item selection predicates, all values needed to determine future committed  $\Sigma$  labels, and  $P$ 's normal-completion status agree with their operational definitions at  $\sigma$ .  $\text{ProcRec}$  does not reconstruct  $\text{Pending\_P}$ ,  $\text{Front\_P}$ , or  $\text{BoundaryReq}$ ; those current offer facts are supplied by  $O$  and proved separately by Lemma J.KObs-Offers. Proof. Project  $H$  to  $P$ , include the  $WC$ /witness choices and  $\text{RESET}$  epoch markers, and apply deterministic replay I.DR/I.DR'; the replayed process state together with Definition Core.6 determines completed/remaining items and candidate  $\text{NextFront}$ .  $\square$

### Lemma J.KObs-Chan (Abstract channel reconstruction).

Let  $\sigma$  and  $K$  be as in Lemma J.KObs-Proc. For every explicit abstract channel  $c$  declared by  $E$ ,  $\text{ChanRec\_c}(K)$  equals the  $E4$  abstract channel state of  $\sigma$ . For  $B(c) > 0$ , occupancy and logical contents are uniquely obtained from reset/initial contents and the committed Push/Pop sequence and per-channel ordinals in  $H$ . For  $B(c) = 0$ , occupancy is identically zero and current rendezvous enablement additionally uses the endpoint offers in  $O$ . In  $\text{Sys\_B}^{\text{elab}}$ , the same statement holds at each explicit proof boundary, and  $\Pi_{\text{elab}}$  supplies the declared outer projection. Proof. Induction over the committed channel-history projection of  $H$ , using  $E4$ ,  $B$ -faithfulness,  $\text{RESET}/\text{RW1-RW3}$ , and  $\Pi_{\text{elab}}$  conservation.  $\square$

### Lemma J.KObs-Offers (Residual-offer, BoundaryReq, and frontier reconstruction).

Let  $\sigma$  and  $K = \langle E, w, H, O \rangle$  be as above. For every process  $P$  and blocking message-operation frontier item  $x$ ,  $x \in \text{Pending\_P}(\sigma)$  iff  $x \in \text{PendingRec\_P}(K)$ , and  $\text{BoundaryReq}(x, \sigma)$  iff  $\text{BoundaryReqRec}(x, K)$ . Hence  $\text{Front\_P}(\sigma) = \text{FrontRec\_P}(K)$ . For every point-to-point rendezvous channel,  $\text{Req\_prod}$  and  $\text{Req\_cons}$  at  $\sigma$  are exactly the existence of the corresponding producer-direction and consumer-direction offers in  $O$ . Proof.  $O$  was defined as the exact set of current operation- and boundary-attributed blocking endpoint requests.  $\text{Core.SyncFence}$  excludes a cross-interval survivor: after any synchronization commit, no blocking offer from the preceding interval remains outstanding. Therefore the  $P$ -owned blocking offers in  $O$  are exactly the current-interval items admitted by  $\text{Pending\_P}$ . Apply Definitions Core.14–Core.15 and take the  $\leq_P$ -minimal subset. The one-offer-per-endpoint-direction invariant makes the rendezvous projection functional.  $\square$

### Lemma J.KObs-WFG (Enablement and wait-for-graph reconstruction).

For every residual offer  $x$  at  $\sigma$ ,  $\text{ChannelEnabledRec}(x, K) = \text{ChannelEnabled}(x, \sigma)$ , and likewise for  $\text{ChannelDisabled}$ . Consequently  $\text{WFGRec}(K) = \text{WFG}(\sigma)$ . Proof. Use Lemma J.KObs-Chan for buffered occupancy/content state, Lemma J.KObs-Offers for rendezvous endpoint requests and attributed frontier membership, and Definitions Core.11–Core.13 together with Definition Core.WFG at the explicit boundary supplied by  $E$ . The no-hidden-cross-channel-disablement convention ensures that no additional enabling fact is required.  $\square$

## Corollary J.1 (K-observation Correspondence at the End of a Matched Observable Prefix)

Purpose.

This corollary is the bridge between execution-level trace compatibility and the K-facing cutpoint observation used directly by Appendix K. Its primary and authoritative cutpoint conclusion is equality of the compact KObs record, not equality of arbitrary source and RTL state. K-relevant frontier, request, channel, enablement, WFG, drain, and deadlock facts are derived from that equality by the named reconstruction functions; no clause expansion is required by the Appendix K transfer layer. Corollary J.1 establishes a reachable source realization of the common record but does not by itself establish membership of that witness in  $\text{Pref}_L^{\text{adm}}(I)$ . An Appendix K application supplies the separate  $\text{K.RLAdmReach}$  proof for the exact selected source prefix.

Statement. Assume the hypotheses of Theorem J.1 for  $\text{Sys\_ref}$  versus  $\text{RTL\_B}$  under the fixed stimulus, including B1 for the  $\text{RTL\_B} \Sigma$  projection. Let  $\tau_{\text{post}} \in \text{Reach\_post}$  carry the J.GWR package constructed by J.GWR-Comp or supplied by J.GWR-Direct. Select and record a finite RTL normalization  $v_{\text{post}}$  with  $\text{NormPost}_I(\tau_{\text{post}}, v_{\text{post}}, p_{\text{post}})$ , so  $p_{\text{post}} \in \text{KCutReach\_post}(I)$ . Select the matching Theorem J.1 source witness prefix and the proof-relevant source normalization recorded by J.OfferReach, ending at  $\sigma_{\text{ref}} \in \text{KCutReach\_ref}(I)$ . B4/B5 may be used to discharge existence/termination of these selected normalizations only under their stated premises; Corollary J.1 does not identify an L2 SettlePoint with either K-facing endpoint. Fix the same  $I$ ,  $E$ ,  $w$ ,  $H$ , and  $O$  throughout, and assume J.KObsConf for this exact package, namely: canonical-history agreement  $H$  from Lemma J.HCanon,  $\text{J.OfferReach}_I, E, w(H, O; \tau_{\text{ref}}, \sigma_{\text{ref}})$ ,  $\text{J.OfferConf}_I, E, w(p_{\text{post}}, O)$ , B-faithful/E4 interpretation at  $E$ 's explicit boundaries, reset correspondence, and  $\epsilon$ -opacity. Then:

- (1)  $\tau_{\text{ref}}, \text{system} \approx \tau_{\text{post}}, \text{system}$  under E1–E5; and
- (2)  $\text{KObsPost}_J(p_{\text{post}}; E, w, H, O) = \text{KObs}_E, w(\sigma_{\text{ref}})$ .

Derived K-facing consequence. Let  $K$  denote the common record in conclusion (2). Every typed reconstruction function in Definitions Core.18d–Core.18e has the same value on the selected post and source views. In particular,  $\text{ProcRec}$ ,  $\text{ChanRec}$ ,  $\text{NextFrontRec}$ ,  $\text{PendingRec}$ ,  $\text{BoundaryReqRec}$ ,  $\text{FrontRec}$ ,  $\text{ChannelEnabledRec/ChannelDisabledRec}$ ,  $\text{WFGRec}$ , and  $\text{DrainClosedNowRec}$  agree. These are functional consequences of conclusion (2), not additional implementation-side state-equality assumptions. Corollary J.1 makes no  $\text{Dead}_K$ ,  $\text{Dead}_K^W$ ,  $\text{Dead}_{KS}^W$ , or  $\text{K.RespFail}$  conclusion; Appendix K applies its own factoring results to the common record when those predicates are needed.

### Proof of Corollary J.1

Step 0 (Choose the K-facing normalized cutpoints). Use the recorded  $\text{NormPost}_I$  witness to obtain  $p_{\text{post}} \in \text{KCutReach\_post}(I)$ , and use the source normalization recorded by J.OfferReach to obtain  $\sigma_{\text{ref}} \in \text{KCutReach\_ref}(I)$ . Each endpoint is globally  $\epsilon$ -quiescent for its own model under Core.4 and has settled explicit-boundary enablement under Core.KCut. Neither endpoint is the L2 SettlePoint merely by definition. B4/B5 or another stated bounded-normalization argument discharges finiteness where applicable; uniqueness of the normalization is not required because all conclusions are relative to the selected package.

Step 1 (Obtain the matched replay history). Apply Theorem J.1 to the selected RTL prefix and its theorem-produced or directly supplied J.GWR package to obtain the matching  $\text{Sys\_ref}$  witness prefix under the same stimulus. Lemma J.HCanon then gives

$$\text{CanonHist}_E, w(\tau_{\text{ref}}) = \text{CanonHist}_E, w(\tau_{\text{post}}) = H.$$

Thus conclusion (1) holds, and the replay-history field of the two KObs records is equal even when their concrete cycle-bucket decompositions differ. The retained fields and the erased incidental ordering are exactly those stated in Lemma J.HCanon.

Step 2 (Establish the common residual offers). J.OfferReach identifies  $O$  exactly with  $\text{ResOffers}_E, w(\sigma_{\text{ref}})$  at the selected reachable source cutpoint. J.OfferConf identifies that same  $O$  bidirectionally and uniquely with the relevant asserted endpoint-owned RTL requests at  $p_{\text{post}}$ . The

reset-epoch, source-prefix-closure, boundary-cardinality, and no-hidden-request conditions ensure that  $O$  is a well-typed residual-offer component for both sides.

Step 3 (Conclude equal KObs). Both records have the same elaboration  $E$ , witness  $w$ , replay history  $H$ , and residual-offer set  $O$ . Therefore  $\text{KObsPost\_J}(\rho_{\text{post}}; E, w, H, O) = \langle E, w, H, O \rangle = \text{KObs\_E}, w(\sigma_{\text{ref}})$ , proving conclusion (2). B-faithfulness and E4 ensure that the post-side interpretation of the committed history at each explicit boundary is the same abstract channel interpretation used by ChanRec; finite normalization is  $\varepsilon$ -opaque by J.KObsConf.

Step 4 (Derive the K-facing facts). Lemmas J.KObs-Proc–J.KObs-WFG reconstruct the process slice, explicit abstract channel state, NextFront, residual endpoint requests, Pending/BoundaryReq attribution, enablement, Front, WFG, and DrainClosedNow from the common record  $K$ . These conclusions follow by functional extensionality of the named reconstruction functions; no broad state correspondence is assumed. Deadlock-, waiver-, environment-terminal-, and diagnostic-predicate factoring is deliberately deferred to Appendix K, which consumes this equal-KObs conclusion. The Policy-S K.Resp discharge is separately checked over  $\rho_{S^{\wedge}\text{det}}$  and is not a KObs cross-model predicate.  $\square$

## J.2b QSync-Specialized Certificate-Existence, Cycle Coverage, and Ideal Closure

The preceding results prove one matched RTL/source cutpoint package. The standard QSync scope permits two related instance-wide consequences from one-cycle induction. On the K-facing side, Core.QSync-Post supplies one canonical RTL KCut for each reachable K-origin, Core.QSync-Ref supplies the corresponding canonical source settling endpoint on the normal compositional route, and the richer correspondence data can be carried from one canonical cycle cutpoint to the next. On the safety side, the same per-cycle construction has a smaller exact-prefix projection over RTL prefixes ending at reachable K-origins. This subsection formalizes both routes. It does not make Core.QSync itself a source/RTL simulation theorem and does not assume J.TraceCertCov or K.CertCov. The cycle-aligned safety theorem below discharges J.TraceCertCov only on Definition J.CycleReach\_post; it does not manufacture route-complete J.TraceCertCov packages for arbitrary non-cycle-aligned finite prefixes. Instead, Lemma J.CycleIdealCover proves that every reachable finite outer observable history is a finite ideal of a reachable cycle-aligned history, and Corollary J.1-Safe-QSync uses J.IdealPrefixClosed to transfer safety to those ideals. During the qualified same-cycle settling relation, Core.QSync/QS2 and the L2 frame emit no  $\Sigma$ -visible commit. R2C intermediate delta-cycle evaluations are  $\varepsilon$ /stutter, and an emitted R2C fixed-point observation unit is atomic and may not be split.

### Definition J.CycleReach\_post (cycle-aligned RTL safety domain).

For a fixed theorem instance  $I$  in the standard QSync scope satisfying QualifiedSynchronousModel\_RTL\_B( $I$ ), define  $\text{CycleReach\_post}(I) := \{ \tau_{\text{post}} \in \text{Reach\_post}(I) \mid \tau_{\text{post}} \text{ ends at a reachable RTL K-origin} \}$ . The K-origin is the cycle-aligned proof point after the current registered/FSM/pipeline frame and process-owned registered request outputs have been established and before the qualified same-cycle settling relation. A reachable cycle-entry state is not excluded from being the final K-origin merely because it also satisfies DeclaredTerminal: when the Core.QSync registered/input/request frame is established, terminality removes the obligation for a later CycleResult but does not require an additional post-terminal clock transition. Thus a member of CycleReach\_post contains the selected observable history of the preceding completed real cycles and is the exact  $\tau_{\text{post}}$  carrier used by the safety projection below. The definition does not identify arbitrary intra-cycle required-prefix ideals with cycle-aligned prefixes.

### Definition J.QSyncTraceState (cycle-aligned safety correspondence state).

Fix the standard normal-compositional theorem instance  $I$  and its standing/static Theorem J.1 premises that do not themselves quantify a LocalGWR package over every reachable prefix. A tuple  $T = (\tau_{\text{post}};$

$\tau_{\text{ref}}, E, w, H$ ) satisfies  $\text{QSyncTraceState}_I(T)$  when all of the following hold. (TS1)  $\tau_{\text{post}} \in \text{CycleReach\_post}(I)$ . (TS2) The exact  $\tau_{\text{post}}$  carries the route-(a)-complete  $\text{Post} \rightarrow \text{Ref}$  evidence of Definition J.TraceCertCov: a J.LocalGWR package accepted by J.GWR-Comp, QualifiedSynchronousModel\_Sys\_ref(I)/Core.QSync-Ref supplies the finite L2 settling required by J.RegPhaseNF/J.UnitExt, all remaining applicable  $\text{Post} \rightarrow \text{Ref}$  premises of Theorem J.1 hold for the fixed instance, and J.GWR-Comp selects  $\tau_{\text{ref}} \in \text{Reach\_ref}$ . (TS3) For that same selected package and witness, the theorem-derived J.HCanon relation holds and  $H = \text{CanonHist\_E}, w(\tau_{\text{post}}) = \text{CanonHist\_E}, w(\tau_{\text{ref}})$ . Corollary J.RegPhaseReach-Comp may re-derive J.RegPhaseReach from the retained J.GWR-Comp evidence whenever a downstream use needs it; J.RegPhaseReach is therefore not an additional safety-side premise or required stored field. QSyncTraceState contains no J.OfferReach, J.OfferConf, J.KObsConf, K.DrainReach, or K.RLAdmReach requirement.

#### Definition J.QSyncTraceInd (base-and-step cycle-aligned safety-package induction).

QSyncTraceInd\_I holds when the following proof obligations are discharged over the fixed theorem instance. (TI0) Base. The initial/reset-established reachable RTL K-origin has an exact cycle-aligned prefix  $\tau_{\text{post}}^0 \in \text{CycleReach\_post}(I)$  and there exists a QSyncTraceState\_I tuple for that prefix. (TI1) One-cycle preservation. Assume QSyncTraceState\_I( $T_t$ ) for an exact prefix  $\tau_{\text{post}}^t$  ending at the cycle- $t$  K-origin, and let  $r_t$  be any reachable RTL CycleResult / next registered transition of the fixed instance whose successor has the next K-origin  $o_{\text{post}}^{(t+1)}$ . Let  $\tau_{\text{post}}^{(t+1)}$  be the exact reachable prefix obtained by extending  $\tau_{\text{post}}^t$  through that real cycle to the successor K-origin. The flow proves the safety-producing local extension facts for that transition: every newly completed observable unit and request-attribution change, including any emitted R2C fixed-point unit, satisfies the applicable J.LocalGWR/I.A0 obligations; J.GWR-Comp extends the selected source witness; Core.QSync-Ref supplies the required source settling; J.HCanon extends the canonical-history equality; and RESET units, if any, are handled through J.RS and RW1-RW5. These facts establish a QSyncTraceState\_I( $T_{(t+1)}$ ) tuple for the exact successor prefix. TI1 is discharged without consuming J.OfferReach, J.OfferConf, J.KObsConf, K.DrainReach, or K.RLAdmReach. Corollary J.RegPhaseReach-Comp remains available as a theorem-derived consequence of J.GWR-Comp rather than an extra TI1 input. (TI2) Exhaustive successor coverage. TI1 is proved for every reachable next-cycle RTL transition from a related cycle-aligned state, not merely the transition observed in one simulation. A deterministic implementation theorem may collapse that successor set; B1 alone does not license dropping unobserved internal successors. TI0/TI1 may be supplied by an inductive simulation/refinement invariant or a verified certificate-extraction procedure.

#### Definition J.QSyncCertState (canonical-cutpoint correspondence state).

Fix the standard normal-compositional theorem instance  $I$ , including the fixed source/RTL artifacts, Sys\_ref elaboration  $E$ , capacities, stimulus, environment/reset interpretation, witness  $w$ , and the standing/static Theorem J.1 premises that do not themselves quantify a LocalGWR package over every reachable prefix (including B1, E4/B-faithfulness, witness/replay machinery, and the qualified source/RTL structural semantics). No J.TraceCertCov or other prefix-package-totality premise is assumed. A tuple  $C = (\tau_{\text{post}}, v_{\text{post}}, \rho_{\text{post}}; \tau_{\text{ref}}, \sigma_{\text{ref}}, E, w, H, O)$  satisfies QSyncCertState\_I( $C$ ) when all of the following hold: (QC1)  $\tau_{\text{post}} \in \text{Reach\_post}$  ends at the reachable RTL K-origin for the represented cycle, and  $v_{\text{post}}$  is the canonical finite Core.QSync-Post normalization to  $\rho_{\text{post}} \in \text{KCutReach\_post}(I)$ ; (QC2) the per-prefix J.LocalGWR evidence for  $\tau_{\text{post}}$  is constructed/validated by the induction discharge, is accepted by J.GWR-Comp, and selects the exact Corollary J.1 source witness  $\tau_{\text{ref}}$  and source endpoint  $\sigma_{\text{ref}} \in \text{KCutReach\_ref}(I)$ , with source settling supplied by Core.QSync-Ref; (QC3) J.HCanon, J.OfferReach, J.OfferConf, and J.KObsConf hold for the exact common tuple  $(\rho_{\text{post}}, \tau_{\text{ref}}, \sigma_{\text{ref}}, E, w, H, O)$ ; (QC4) Corollary J.RegPhaseReach-Comp supplies J.RegPhaseReach for that

exact source witness; this is theorem-derived from the QC2 J.GWR-Comp machinery rather than an independent input obligation; and (QC5) K.DrainReach holds for that exact tuple, including Core.SettleKBridge for every newly or previously activated persisting blocked frontier, so K.RLAdmReach and  $\tau_{\text{ref}} \in \text{Pref\_L}^{\text{adm}}(I)$  follow. QSyncCertState is deliberately the semantic payload later bound by K.CertHdr; it contains no J.TraceCertCov assumption, no K.CertCov assumption, and no successor K.CertifiedCutpoint premise.

#### Definition J.QSyncCertInd (base-and-step certificate induction discharge).

QSyncCertInd\_I holds when the following proof obligations are discharged over the fixed theorem instance. (CI0) Base. The initial/reset-established reachable RTL K-origin has a canonical QSync-Post endpoint  $\rho_{\text{post}}^0$  and there exists a QSyncCertState\_I tuple for that endpoint. (CI1) One-cycle preservation. Assume QSyncCertState\_I( $C_t$ ) at a canonical RTL KCut for cycle  $t$ , and let  $r_t$  be any reachable RTL CycleResult / next registered transition of the fixed instance whose successor has the next K-origin  $o_{\text{post}}^{(t+1)}$ . The flow proves the local extension facts for that transition: the newly committed units and request-attribution changes satisfy the applicable J.LocalGWR/I.A0 obligations; J.GWR-Comp extends the selected source witness; Core.QSync-Post and Core.QSync-Ref provide the canonical next post/source settling endpoints; J.HCanon and J.OfferReach/J.OfferConf/J.KObsConf update H and O at those endpoints; Corollary J.RegPhaseReach-Comp supplies the next phase-correct source prefix; and the existing K.DrainReach proof is preserved while every newly activated drain/SettleKBridge obligation is discharged. RESET units, if any, are handled through J.RS and RW1–RW5 so the next tuple uses the correct reset epoch. These local facts establish a QSyncCertState\_I( $C_{(t+1)}$ ) tuple for the canonical successor endpoint. (CI2) Exhaustive successor coverage. CI1 is proved for every reachable next-cycle transition from a related canonical cutpoint, not merely the transition observed in one simulation. A deterministic implementation theorem may collapse that successor set; B1 alone does not license dropping unobserved internal successors. CI0/CI1 may be supplied by an inductive simulation/refinement invariant or a verified certificate-extraction procedure, but they may not cite K.CertCov, the successor K.CertifiedCutpoint fact, or J.CertExist-QSync itself.

#### Lemma J.QSyncCertInd-Trace (K-facing induction projects to the cycle-safety induction).

If J.QSyncCertInd\_I holds, then J.QSyncTraceInd\_I holds. Proof. Project each QSyncCertState tuple  $C$  to  $T$  by retaining the exact  $\tau_{\text{post}}$ , the selected  $\tau_{\text{ref}}$ ,  $E$ ,  $w$ , and  $H$ . QC1 gives TS1; QC2 gives TS2; and retain the J.HCanon component of QC3 to obtain TS3. The OfferReach/OfferConf/KObsConf remainder of QC3 and QC5 are omitted. QC4 need not be stored because Corollary J.RegPhaseReach-Comp re-derives it from the retained J.GWR-Comp evidence. CI0 and CI2 project directly. For CI1, the sub-obligations that produce QC2 and J.HCanon are the newly completed-unit/request-attribution LocalGWR/I.A0 checks, J.GWR-Comp witness extension, the applicable QSync settling facts, reset splice where needed, and J.HCanon itself. These do not consume J.OfferReach, J.OfferConf, J.KObsConf, K.DrainReach, or K.RLAdmReach as inputs: those facts are downstream K-facing outputs of the full CI1 step, and none is a hypothesis of J.GWR-Comp. Therefore the same CI1 proof restricted to the retained dependencies establishes TI1 from the projected state.  $\square$

#### Theorem J.TraceCertCov-QSyncCycle (QSync induction gives cycle-aligned safety package coverage).

Assume QualifiedSynchronousModel\_RTL\_B( $I$ ), QualifiedSynchronousModel\_Sys\_ref( $I$ ), the standing/static normal-compositional Theorem J.1 premises for the fixed instance other than per-prefix LocalGWR/TraceCertCov package totality, and J.QSyncTraceInd\_I. Then J.TraceCertCov\_I(CycleReach\_post( $I$ )) holds. Proof. Induct on the finite number of RTL real clock transitions from the initial/reset-established K-origin to the K-origin ending an arbitrary  $\tau_{\text{post}} \in$

CycleReach\_post(I). T10 supplies the route-complete exact-prefix state at the base. T11 extends that exact prefix and its selected J.LocalGWR/J.GWR-Comp witness through each reachable real cycle, while T12 makes the induction universal over the reachable RTL transition tree rather than simulation-path specific. Hence every  $\tau\_post \in \text{CycleReach\_post}(I)$  carries a QSyncTraceState tuple. TS2 is exactly route (a) of Definition J.TraceCertCov for that same  $\tau\_post$ , and TS3 retains the J.HCanon equality used by the safety transfer. Therefore TraceCertCov\_I(CycleReach\_post(I)) holds. No J.OfferReach, J.OfferConf, J.KObsConf, K.DrainReach, K.RLAdmReach, or K.CertCov premise is used.  $\square$

#### Corollary J.1-Safe-QSyncCycle (cycle-aligned source-side safety transfer).

Assume the hypotheses of Theorem J.TraceCertCov-QSyncCycle and the applicable source-side clause of Corollary J.1-Safe for a predicate  $\Phi$  on finite canonical  $\Sigma\_DUT$  histories satisfying J.IdealPrefixClosed\_I, $\Sigma\_DUT(\Phi)$  and J.HCanon-invariance. Then the corresponding conclusion of Corollary J.1-Safe holds for every  $\tau\_post \in \text{CycleReach\_post}(I)$ . Proof. The preceding theorem supplies J.TraceCertCov\_I(D) with  $D := \text{CycleReach\_post}(I)$ ; instantiate Corollary J.1-Safe with that D.  $\square$

#### Lemma J.CycleIdealCover (cycle-aligned histories cover finite observable ideals).

Assume QualifiedSynchronousModel\_RTL\_B(I). Let  $\tau\_post \in \text{Reach\_post}(I)$  be any reachable finite RTL\_B prefix in the standard QSync scope, and let  $H := \text{CanonHist\_}\Sigma\_DUT(\tau\_post)$ , interpreted in the Appendix-J theorem-prefix sense as the finite required-prefix ideal of complete selected outer observable units carried by that prefix. Then there exists  $\tau\_post^{cyc} \in \text{CycleReach\_post}(I)$  such that H is a finite ideal of CanonHist\_ $\Sigma\_DUT(\tau\_post^{cyc})$ . Proof. By Definition Core.3 and Definition J.Reach,  $\tau\_post$  inherits its cycle partition and current-cycle observable bucket from a legal reachable RTL\_B execution; the argument uses only that execution carrier and assumes no fair or infinite extension. If  $\tau\_post$  ends at the current K-origin or during the qualified same-cycle settling relation before any complete selected outer observable unit of the current cycle is added, QS2 makes that settling  $\Sigma$ -silent and the current K-origin has the same H. Otherwise  $\tau\_post$  carries a nonempty finite ideal  $H\_t$  of complete selected outer observable units from the current real-cycle bucket;  $H\_t$  may be the entire bucket when the prefix stops after all complete units but before the next registered update. By Lemma Core.RTLCycleCarrier, the enclosing legal cycle-t CycleResult exists in the same reachable execution carrier, and  $\tau\_post$  extends within that already-entered cycle through its completion to a successor prefix  $\tau\_post^{cyc}$  ending at the cycle-(t+1) reachable K-origin. If that successor cycle-entry state is also DeclaredTerminal, it may still serve as this final K-origin; no later CycleResult or fair extension is required. For every  $\gamma \in H\_t$  and every retained outer unit  $\gamma'$  with  $\gamma' \prec\_J \gamma$ , the clock-order argument used to prove acyclicity of  $\prec\_J$  in Theorem J.1 gives  $\text{clk}(\gamma') < \text{clk}(\gamma)$ , so  $\gamma'$  belongs to an earlier completed cycle and is already in H. Any selected same-cycle interaction that may not be split is represented as one explicitly atomic observable unit, including rendezvous transfers, paired synchronization, grouped E2 anchor units, and emitted R2C fixed-point units. When proof-internal RESET units are projected away from  $\Sigma\_DUT$ , projection removes them only as whole atomic units. If the selected outer history retains a reset marker, RW1-RW5 and the reset-epoch ordering are already part of theorem-prefix legality, so any required reset predecessor is already present in H. Therefore omitting independent same-cycle units cannot violate down-closure of H in the completed-cycle outer history. Hence H is a finite ideal of CanonHist\_ $\Sigma\_DUT(\tau\_post^{cyc})$ .  $\square$

#### Corollary J.1-Safe-QSync (standard QSync all-finite-prefix safety transfer).

Assume the hypotheses of Corollary J.1-Safe-QSyncCycle for a predicate  $\Phi$  on finite canonical  $\Sigma\_DUT$  histories satisfying J.IdealPrefixClosed\_I, $\Sigma\_DUT(\Phi)$  and J.HCanon-invariance. Then the corresponding source-side safety conclusion holds for every  $\tau\_post \in \text{Reach\_post}(I)$ . Proof. Fix arbitrary  $\tau\_post \in \text{Reach\_post}(I)$ . Lemma J.CycleIdealCover supplies  $\tau\_post^{cyc} \in \text{CycleReach\_post}(I)$  such that CanonHist\_ $\Sigma\_DUT(\tau\_post)$  is a finite ideal of CanonHist\_ $\Sigma\_DUT(\tau\_post^{cyc})$ . Corollary J.1-Safe-

QSyncCycle establishes  $\Phi$  for the covering cycle-aligned history.  $J.IdealPrefixClosed\_I, \Sigma\_DUT(\Phi)$  therefore gives  $\Phi$  for  $CanonHist\_I, \Sigma\_DUT(\tau\_post)$ , and  $J.HCanon$ -invariance makes the conclusion independent of the permitted canonical quotient representation. Since  $\tau\_post$  was arbitrary, the result holds for every reachable finite  $RTL\_B$  prefix.  $\square$

#### Scope note (ideal closure does not create per-prefix correspondence packages).

Lemma  $J.CycleIdealCover$  and Corollary  $J.1-Safe-QSync$  close the residual non-cycle-aligned safety case only for properties of finite canonical  $\Sigma\_DUT$  histories satisfying Definition  $J.IdealPrefixClosed$ . They do not assert  $J.TraceCertCov$  for every non-cycle-aligned  $\tau\_post$ , do not construct a separate matching  $Sys\_ref$  witness or  $J.GWR$  package for each such prefix, and do not extend raw latency/performance, properties that fail ideal-prefix closure, or non- $QSync$  claims. In particular, a cross-interface scoreboard whose correctness relation depends on an event that may be omitted by a  $\prec\_J$  ideal is not covered by the ideal-closure shortcut merely because it is informally called a safety check. Applications that need direct correspondence evidence for an arbitrary prefix continue to discharge Definition  $J.TraceCertCov$  on the selected broader domain.

#### Definition $J.QSyncCertCov$ (J-side total canonical-cutpoint package coverage).

$J.QSyncCertCov(I)$  holds when for every  $p\_post \in KCutReach\_post(I)$  there exists at least one tuple  $C$  satisfying  $QSyncCertState\_I(C)$  whose post endpoint is  $p\_post$ . This predicate records semantic package existence only;  $K.CertHdr$  binding and the downstream  $K.CertifiedCutpoint/K.CertCov$  packaging are performed in Appendix K.

#### Theorem $J.CertExist-QSync$ (canonical-cycle induction gives total J-side package existence).

Assume  $QualifiedSynchronousModel\_RTL\_B(I)$ ,  $QualifiedSynchronousModel\_Sys\_ref(I)$ , the standing/static normal-compositional Theorem  $J.1$  premises for the fixed instance other than per-prefix  $LocalGWR/TraceCertCov$  package totality, and  $J.QSyncCertInd\_I$ . Then  $J.QSyncCertCov(I)$  holds. Proof. Induct on the finite number of RTL real clock transitions from the initial/reset-established  $K$ -origin to a reachable  $K$ -origin.  $CI0$  establishes the first canonical cutpoint tuple.  $CI1$  constructs or extends the required per-prefix local package and preserves  $QSyncCertState$  across each reachable successor transition, while  $CI2$  makes the induction universal over the reachable RTL transition tree rather than simulation-path specific. Hence every canonical  $QSync$ -Post endpoint reached after finitely many cycles carries a  $QSyncCertState$  tuple. By Lemma  $Core.QSync-KCutCanonical$ , every  $p\_post \in KCutReach\_post(I)$  in the standard scope is exactly such a canonical endpoint, so the relation is total on  $KCutReach\_post(I)$ . The same induction is constructive when  $CI0/CI1$  are supplied as proof-producing base/step functions: it defines an on-demand extractor  $CertExtract\_QS$  that returns the corresponding  $QSyncCertState$  witness for any represented reachable  $KCut$ . No  $J.TraceCertCov$  premise, arbitrary normalization choice, or independent per-cutpoint package-existence axiom is used.  $\square$

### J.3 Causal Dependency and What System-Level Compatibility Guarantees (Informative)

The system-level result of Appendix J establishes compatibility of the aggregate observable behavior, i.e.  $\tau\_ref, system \approx \tau\_post, system$  over the alphabet  $\Sigma$  of channel operations, synchronization events, and signal Read/Write actions.

This compatibility is intentionally *observational*: it constrains what an external environment can distinguish at  $\Sigma$ , not internal RTL micro-timing.

As a consequence, Appendix J should be read as preserving explicitly required observable ordering, atomicity, and FIFO/synchronization semantics, not incidental latency. The causal-dependency relation

( $\rightarrow$ ) defined in the “Causal Dependency Between Processes” section remains useful for explaining information flow and for documenting which interactions a designer intends to be functionally significant. However,  $\rightarrow$  is not itself the strict induction order for Theorem J.1, and the theorem does not require the unquotiented dependency graph  $\rightarrow$  to be acyclic. The proof order is the required-order relation  $\prec_J$  over observable units.

Accordingly, the system-level theorem implies the following required-order preservation principle:

- If two observable units  $\gamma_1$  and  $\gamma_2$  are ordered by the required-order relation  $\gamma_1 \prec_J \gamma_2$ , then the matched Sys\_ref and RTL\_B executions preserve that required order. This follows because each elementary  $\prec_J$  generator is preserved by the corresponding rule: synchronization source order by E1; signal-anchor timestamping by E2; issue-side and request-attribution obligations by E3; channel FIFO legality and rendezvous atomicity by E4; immediate synchronization-boundary constraints by E5; and paired-barrier semantics by the explicit synchronization well-formedness conditions. In particular, for each explicit abstract channel  $c$ , the matched executions preserve the committed Push\_c/Pop\_c history: for  $B(c)=0$ , a committed transfer is one explicitly atomic rendezvous Push\_c(v)/Pop\_c(v) unit; for  $B(c)>0$ , the  $k$ -th committed Pop\_c observes the  $k$ -th earlier unmatched Push\_c according to reset-empty, FIFO-order, no-underflow/no-overflow, and non-fall-through E4 semantics.
- If two observable units are not ordered by  $\prec_J$  and are not members of the same explicitly atomic unit, then their relative cycle placement or interleaving is not constrained by the methodology, and differences are treated as incidental latency variation. A designer who needs a particular cross-process ordering to be preserved must express it through an explicit  $\Sigma$ -visible mechanism such as message-passing FIFO order, a rendezvous/handshake atomic unit, explicit synchronization, or signal-anchor semantics.

This perspective is what makes the “single testbench / no surprises” implication precise: for the fixed-stimulus execution sets covered by Theorem J.1, any testbench that observes only  $\Sigma$  and encodes its expectations through causal dependencies (channels, barriers, or explicit handshakes) cannot distinguish the matched Sys\_ref / RTL\_B execution pairs provided by the theorem.

Equivalently, RTL\_B introduces no new  $\Sigma$ -observable behavior relative to Sys\_ref under that fixed stimulus, and source-to-RTL transfer is asserted only for the annotated RTL-consistent source prefixes covered by Theorem J.1. (Here Sys\_ref is as fixed in Remark 2. The rendezvous specialization Sys with  $B(c)=0$  is covered only under the explicit conditions of Remark 4.)

## Appendix K – Preservation of Internal Wait-Cycle Freedom (Reachable-Cutpoint Deadlock-Freedom) for the Prescribed Source Reference Model

### K.1 Scope, Notation, and Definition of Internal Message-Passing Deadlock

#### Terminology alignment

Throughout Appendix K, when evaluating K-facing predicates at a cutpoint  $\sigma$  in  $K\text{CutReach\_ref}(I)$  or the corresponding certified RTL K-facing view, let  $t_\sigma$  denote the cycle of that cutpoint and use the cutpoint shorthand  $\text{occ\_c}(\sigma) \triangleq \text{occ\_c}(t_\sigma)$ . For any frontier item  $x$  on an explicit abstract channel  $c = \text{BoundaryOf\_P}(x)$  of Sys\_ref, with owning dynamic source-level message instance  $q = \text{MsgOf\_P}(x)$ , use  $\text{BoundaryReq}(x, \sigma)$ ,  $\text{ChannelEnabled}(x, \sigma)$ , and  $\text{ChannelDisabled}(x, \sigma)$  for the endpoint-owned request,

enablement, and disabledness facts at that specific Core.KCut evaluated boundary. In the ordinary unelaborated case, where  $q$  has a unique evaluated boundary and a unique frontier item,  $\text{BoundaryReq}(q, \sigma)$ ,  $\text{ChannelEnabled}(q, \sigma)$ , and  $\text{ChannelDisabled}(q, \sigma)$  may be used as shorthands. In  $\text{Sys\_B}^{\text{elab}}$ , however, the frontier-item form is authoritative because one source-level operation may own multiple proof-internal frontier items at different explicit boundaries. Thus “boundary request is asserted” means  $\text{BoundaryReq}(x, \sigma)$ , “channel-enabled” means  $\text{ChannelEnabled}(x, \sigma)$ , and “channel-disabled” means  $\text{ChannelDisabled}(x, \sigma)$ , all evaluated at the Core.KCut boundary fixed point of the selected cutpoint cycle and read through  $q = \text{MsgOf\_P}(x)$  and  $\text{BoundaryOf\_P}(x)$ . These predicates are not applied to already-committed  $\Sigma$ -event labels except through the operation/frontier item that contributes  $m(q)$  when it commits.

We work with a fixed design:

**System  $\text{Sys\_ref}$ :** The collection of pre-HLS processes obeying the R-rules and B-rules, interpreted under the prescribed source reference model of Definition Core.1 and Appendix J, Remark 2. In this appendix, capacities/occupancies/enablement predicates are interpreted at the chosen abstract boundaries of  $\text{Sys\_ref}$ ; in elaborated cases, this means the explicit abstract staged/bundle/join channels of that model. Each explicit abstract channel  $c$  of  $\text{Sys\_ref}$  has finite capacity  $B(c) \geq 0$ , chosen to match the corresponding RTL realization at that abstract boundary.

**Post-HLS implementation  $\text{RTL\_B}$ :** The hardware implementation of the same processes in which each chosen abstract channel  $c$  is realized with finite capacity  $B(c) \geq 0$ . In this appendix, we compare  $\text{Sys\_ref}$  (the prescribed source-side proof target) to  $\text{RTL\_B}$  (the micro-architectural realization). The proof does not require mapping buffered RTL states to a distinct raw rendezvous ( $B(c)=0$ ) state.

**Witness-selection and post-view convention.** Corollary J.1 supplies a selected reachable  $\text{Sys\_ref}$  cutpoint  $\sigma_{\text{ref}}$  and a common record  $K$  with  $\text{KObs\_E}, w(\sigma_{\text{ref}}) = \text{KObsPost\_J}(p_{\text{post}}; E, w, H, O)$  for the selected RTL K-facing cutpoint  $p_{\text{post}}$ . Appendix K does not compare arbitrary RTL microstate with source state. Define the K-facing post abstract view by applying the reconstruction functions of Definitions Core.18d–Core.18e to  $\text{KObsPost\_J}$ . Unless a raw RTL signal is expressly named as part of  $\text{J.OfferConf}$ , every post-side occurrence of  $\text{NextFront}$ ,  $\text{Pending}$ ,  $\text{BoundaryReq}$ , channel state,  $\text{ChannelEnabled/ChannelDisabled}$ ,  $\text{Front}$ ,  $\text{WFG}$ ,  $\text{DrainClosedNow}$ , or  $\text{Dead\_K}$  in Appendix K denotes that reconstructed view. If a reproducible source witness is desired for debugging, the Appendix L snooping wrapper may select one aligned to the RTL latency-sensitive events; uniqueness is not required by the theorem.

A global state  $\sigma$  of either system consists of:

1. For each process  $P$ : a control location (program point) and local state.
2. For each explicit abstract channel  $c$  of  $\text{Sys\_ref}$ : its abstract FIFO state at the chosen  $\text{Sys\_ref}$  boundary—its current occupancy  $\text{occ\_c}(t)$  and (for buffered channels) its ordered contents. Here  $\text{occ\_c}(t)$  counts all items that have been committed by  $\text{Push\_c}$  but have not yet been committed by  $\text{Pop\_c}$  at that chosen abstract boundary. In  $\text{RTL\_B}$ , this count includes items residing anywhere in the concrete sub-realization that contributes to that particular abstract channel’s back-annotated capacity  $B(c)$ . Thus internal movement within the concrete realization of one abstract channel  $c$  is  $\varepsilon$  and does not create additional committed  $\text{Push\_c/Pop\_c}$   $\Sigma$ -events beyond the boundary events counted by  $\text{occ\_c}$ ; however, transfers between distinct explicit abstract channels introduced in  $\text{Sys\_B}^{\text{elab}}$  are represented in  $\text{Sys\_ref}$  by those channels themselves and are not collapsed back into one outer  $\text{Sys\_B}$  boundary.

3. Any additional architectural state not representing buffered messages on any channel (e.g., computation pipeline registers, arbitration state, bookkeeping, etc.).

Appendix K uses the normative wait-for graph  $WFG(\sigma)$  of Definition Core.WFG, restricted there to internal message-passing channels. We write  $WFG\_ref$  for the graph reconstructed at a  $Sys\_ref$  cutpoint and  $WFG\_post$  for the corresponding K-facing graph at the selected  $RTL\_B$  cutpoint. All vertices, edge conditions, the edge-versus-blocked and multi-frontier clarifications, frontier-item attribution, complementary-endpoint ownership, buffered/rendezvous treatment, and exclusion of environment-facing channels are exactly those of Definition Core.WFG; this appendix adds the deadlock, closure, waiver, and preservation predicates but does not redefine WFG.

(No deassert-before-commit assumption.) Blocking  $Push\_c/Pop\_c$  requests are non-withdrawable within one reset epoch: once the request corresponding to a frontier item  $x \in \Omega\_msg, P$  is asserted through its owning message instance  $q = MsgOf\_P(x)$ , it is not deasserted before  $q$  commits, and  $Push\_c$  payload remains stable while asserted. If an affecting RESET occurs first, RW2 ends that dynamic instance at the epoch boundary. No other nonterminal transition may withdraw or reattribute it. This rules out “try-and-withdraw” behavior for blocking transfers within the WFG abstraction.

K-facing cutpoint convention. All WFG and Dead\_K reasoning is over Core.KCut states: the source state lies in  $KCutReach\_ref(l)$ , and the selected/certified RTL state lies in  $KCutReach\_post(l)$ . Core.4 quiescence means no internal  $\epsilon$ -step is enabled in the corresponding model at its current cycle/phase; it does not mean that no  $\Sigma$  action or environment action is enabled. Purely combinational ready/valid propagation is additionally required to be at its fixed point by Core.KCut. Every source-side L2 SettlePoint is a KCut by Lemma Core.SettleIsKCut, although a Dead\_K evaluation may occur later after Core.16-permitted drain has reached a drain-closed KCut. Core.SettleKBridge/K.DrainReach supplies that continuation relation. On  $RTL\_B$ , Core.KCutClosure\_post is the separate non-vacuity premise ensuring that each reachable K-facing cycle-evaluation origin reaches a finite KCut; with  $K\epsilon$ , Lemma Core.KCutObserve prevents a persistent relevant blocker from being hidden solely in pre-normalization RTL microstate.

(WFG-inert  $\epsilon$  premise.) Appendix K uses  $\epsilon$ -steps only after the WFG-relevant boundary-request / attribution cutpoint has been fixed. More precisely, consider an  $\epsilon$ -path  $\sigma \xrightarrow{\epsilon^*} \sigma'$  that contains no  $\Sigma$ -visible committed  $Push\_c/Pop\_c$  event, no synchronization event, no signal Read/Write event, and no source-level issue/attribution transition that creates, removes, or reassigns a pending blocking boundary request. Such  $\epsilon$ -steps are observationally silent with respect to the WFG abstraction: they do not change (i) the potentially-blocking frontier  $Front\_P(\sigma)$  for any process  $P$  (i.e., which guarded blocking  $Push/Pop$  frontier items are minimal and pending), nor (ii) the truth or attribution of any boundary requests relevant to those frontier items, nor (iii) the channel-side enabling status (as defined in K.1) of any  $x \in Front\_P(\sigma)$ , read through  $q = MsgOf\_P(x)$ . Equivalently, these WFG-inert  $\epsilon$ -steps do not change the channel state relevant to ChannelEnabled/ChannelDisabled (e.g.,  $occ\_c(t)$  for  $B(c)>0$  buffered channels), and they do not create/remove rendezvous enablement for a frontier transfer except via  $\Sigma$ -visible committed  $Push\_c/Pop\_c$  events.

This premise does not say that every non- $\Sigma$  microstep is WFG-inert. In particular, an  $\epsilon$ -normalization segment may include source-level issue/attribution steps that establish a persistent blocking request, such as moving from a cutpoint before a Pop has issued to a cutpoint where the Pop request is pending and BoundaryReq is true. Those steps may change  $Front\_P$ , BoundaryReq, or request attribution, and are therefore not the  $\epsilon$ -steps quantified over by the WFG-inert premise. Appendix K's WFG construction is applied only at the resulting Core.KCut cutpoints, after such request/attribution state has been fixed and the boundary

network has settled.

In particular (pipelined loops / overlapped iterations): when HLS introduces overlap that may initiate later-iteration message reads “early,” we assume the design obeys the Automatic flush (drain-only blocking discipline) defined in Appendix B. Under that discipline, once the process has reached a stable blocked frontier, overlapped execution does not introduce any later-iteration blocking Push/Pop frontier item into  $\text{Front\_P}(\sigma)$ , so transient internal pipeline activity that merely moves already-accepted work inside the concrete realization remains WFG-inert  $\epsilon$  and does not contribute wait-for edges. Overlapped execution in pipelined loops is modeled only as a potential retiming of ordinary  $\Sigma$ -visible boundary commits (Push\_c/Pop\_c) relative to the unpipelined source, and is accounted for by the back-annotated semantics at the chosen abstract boundary of  $\text{Sys\_ref}$  (E4). In particular, the channel facts relevant to ChannelEnabled/ChannelDisabled and WFG edges—e.g.,  $\text{occ\_c}(t)$  for  $B(c) > 0$ , and rendezvous enablement for  $B(c) = 0$ —change only on  $\Sigma$ -visible committed Push\_c/Pop\_c events at that boundary, or on explicit source-level issue/attribution steps before the WFG cutpoint is formed; internal movement within the concrete realization of a single abstract channel after that cutpoint is WFG-inert  $\epsilon$  and does not affect ChannelEnabled/ChannelDisabled or introduce/remove WFG edges.

Precondition ( $K\epsilon$ : WFG-inert  $\epsilon$  / no hidden micro-timing control decisions). All results in Appendix K are stated under the precondition  $K\epsilon$  that the narrowed WFG-inert  $\epsilon$  premise above holds for the design on the  $\epsilon$ -normalization segments leading to and between the  $\text{Core.KCut}$  points used for WFG reasoning. If an implementation intentionally uses internal micro-timing on such a segment to change  $\text{Front\_P}$  membership, boundary-request attribution, or ChannelEnabled/ChannelDisabled facts without a corresponding  $\Sigma$ -visible event or explicit source-level issue/attribution transition, then  $K\epsilon$  is violated unless that behavior is made  $\Sigma$ -observable (e.g., via explicit modeling/snooping) or is witness-selected under the Appendix I.2 framework.

A common instance is branching on the success/failure of a non-blocking poll in a way that changes the next  $\Sigma$ -visible candidate frontier **NextFront\_P( $\sigma$ )** (Definition Core.6; see Lemma J.KObs-Proc and Corollary J.1); in that case, either NB-CF must hold (Appendix I.2), or the poll outcomes must be witness-selected as stated there.

Non-blocking polling attempts and other purely internal actions do not contribute edges to the WFG; they are modeled as internal  $\epsilon$ -steps. When a non-blocking call succeeds, that success is already represented at the  $\Sigma$  level as the corresponding committed Push\_c/Pop\_c event, but it does not itself add a wait-for edge because it is not a blocking operation.

### **SyncChannels and barrier well-formedness.**

SyncChannel (barrier) operations are treated as pure synchronization events and are omitted from the Wait-For Graph, which tracks only blocking Push/Pop dependencies. We therefore interpret “deadlock” in this appendix as **internal message-passing deadlock** (i.e., a deadlocked set in which all processes are blocked on some internal Push/Pop dependency captured by the WFG). Any global stall at a barrier caused by mismatched or conditional participation (e.g., a process can permanently bypass the barrier or reach it a different number of times) is a specification-level error already present in the pre-HLS model; such stalls are not created by the HLS transformation and are excluded by an explicit barrier well-formedness assumption (A8 below), not by the internal message-passing deadlock premise. Under this assumption, omitting SyncChannels from the WFG is sound: a post-HLS deadlock implies the existence of a Push/Pop wait-cycle as characterized below.

Definition (Blocked on message passing). Using  $\text{Front\_P}(\sigma)$  as defined above, we say that  $P$  is blocked on message passing in state  $\sigma$  iff  $\text{Front\_P}(\sigma)$  is non-empty and, for every frontier item  $x \in \text{Front\_P}(\sigma)$ ,  $\text{BoundaryReq}(x, \sigma)$  holds and  $\text{ChannelDisabled}(x, \sigma)$  holds (equivalently,  $\text{BoundaryReq}(x, \sigma) \wedge \neg \text{ChannelEnabled}(x, \sigma)$ ), read operation-attributively through the owning message instance  $q = \text{MsgOf\_P}(x)$ . Note ( $\text{ALL disabled} \Rightarrow \text{stall}$ ). If any frontier item  $x \in \text{Front\_P}(\sigma)$  is channel-enabled at the chosen  $\Sigma$ -visible boundary, then  $P$  is not “blocked on message passing” for WFG purposes. Consequently, an implementation may not be physically stalled at the chosen  $\Sigma$  boundary by a cross-channel “join” predicate while some frontier item  $x \in \text{Front\_P}(\sigma)$  at that boundary remains channel-enabled; any such conjunctive/join dependency must be represented as an explicit staged/bundle/join boundary channel so that, when the process is physically stalled, the relevant frontier items on that explicit boundary are all channel-disabled.

Definition (Drain-closed cutpoint for deadlock extraction). Let  $D$  be a non-empty set of processes. A  $K$ -facing state  $\sigma \in \text{KCutReach\_M}(I)$  is drain-closed for  $D$  iff no already-issued non-frontier blocking Push/Pop request of any process  $P \in D$  is channel-enabled at  $\sigma$ . Equivalently, after the frontier members  $\text{Front\_P}(\sigma)$  are all disabled, there is no remaining already-issued downstream transfer of any process in  $D$  that can still commit as drain-only progress.

Here “already-issued downstream transfer” means an already-issued non-frontier blocking message-operation frontier item  $x \in \text{Pending\_P}(\sigma) \setminus \text{Front\_P}(\sigma)$ , with owning message instance  $q = \text{MsgOf\_P}(x)$ , interpreted at the chosen abstract  $\text{Sys\_ref}$  /  $\text{RTL\_B}$  boundary and within the current source interval, i.e., the interval between the adjacent synchronization calls that bracket  $\sigma$  in  $P$ ’s current execution;  $\text{Core.SyncFence}$  makes this restriction lossless for blocking requests: no issued-but-uncommitted blocking item from an earlier interval can survive the synchronization commit that established the current interval; hidden internal pipeline staging or later-iteration prefetch absorbed into  $B(c)$  is not a separate source-level operation for this definition unless it has been made an explicit abstract channel in  $\text{Sys\_B}^{\text{elab}}$ .

If such a non-frontier drain transfer is enabled,  $\sigma$  is not yet a stable deadlock cutpoint for  $D$ ; either that drain transfer must be allowed to commit finitely often until a drain-closed cutpoint is reached, or the resulting commit changes the WFG-relevant channel facts so that some frontier operation becomes enabled and the candidate deadlock disappears.

**Finite drain note.** Finiteness here is not derived from finite channel capacity alone. Definition  $\text{Core.16(6)}$ , as discharged by  $\text{K.Drain}$ , explicitly requires the downstream drain set to be finite and non-replenishing. Finite  $B$ -faithful storage/credit on every selected explicit boundary, finite  $\epsilon$ -depth  $D\_P$  for internally advancing work, and  $\text{Core.16(2)}$ ’s no-new-later-work rule are the standard evidence used to discharge that obligation. Once  $\text{Core.16(6)}/\text{K.Drain}$  is established, only work already accepted before the blocked condition can drain, and the number of  $D$ -owned downstream drain commits is finite.  $\text{Core.SettleKBridge}$  requires those explicit drain commits and intervening WFG-inert  $\epsilon$ -normalizations to be accounted for before a persisting blocked frontier is evaluated at a later drain-closed  $K$ -facing cutpoint. A certificate may use a coarse bound derived from the relevant in-flight occupancies, explicit finite credits, finite elaboration chains, and finite per-process  $\epsilon$ -depths; Theorem K.1 consumes only the proved finiteness/non-replenishment fact, not a particular numeric bound.

**Definition K.Dead (ordinary internal message-passing deadlock predicate).** For a  $K$ -facing cutpoint  $\sigma \in \text{KCutReach\_M}(I)$  of a source-side or lowered proof model  $M$  used by Appendix K,  $\text{Dead\_K}(\sigma)$  holds iff there exists a non-empty set of processes  $D$  such that  $\sigma$  is drain-closed for  $D$  and:

- Every process  $P$  in  $D$  is blocked on message passing, and in particular there exists at least one frontier item  $x \in \text{Front\_P}(\sigma)$  on an internal message-passing channel  $c$  (i.e., a channel included in the WFG), with

owning message instance  $q = \text{MsgOf\_P}(x)$ , such that  $\text{BoundaryReq}(x, \sigma)$  holds and  $\text{ChannelDisabled}(x, \sigma)$  holds.

- The vertices in  $D$ , together with the internal channels they access, form a strongly connected component in the WFG that has no outgoing edges (a closed cycle): processes in  $D$  can only wait on each other through internal WFG channels. (Members of  $D$  may additionally hold environment-facing frontier items, which are excluded from the WFG; see the Terminology note below.)

Intuitively,  $D$  is a set of processes that are waiting only on each other (via internal message-passing channels tracked by the WFG), with no remaining enabled drain-only transfer of any process in  $D$  that could change the WFG-relevant channel facts. Thus  $D$  cannot be unblocked by internal modeled message-passing progress represented in the closed WFG component, nor by already-issued downstream drain from within  $D$ . This is a reachable  $K$ -facing-state predicate, not a permanence claim: a member of  $D$  may also hold environment-facing frontier items (excluded from the WFG), and later environment/testbench activity under the fixed stimulus may dissolve such a state; the Appendix K results and the A5 premise quantify over reachability of these states, not over their permanence. On the RTL side Appendix K evaluates the corresponding reconstructed predicate  $\text{DeadKRec}/\text{DeadKRec}^W$  on the certified KObs view; it does not apply this operational source definition to an arbitrary RTL microstate. A process that is blocked solely on channels whose complementary endpoint is the external environment/testbench (excluded from the WFG) does not witness an internal message-passing deadlock in this appendix.

**Terminated-process convention.** For purposes of Appendix K, termination is interpreted per reset epoch. After a process  $P$  emits  $\text{FINISH\_P}$  for a given reset epoch, the infinite execution is padded only by process-local idle stutter for  $P$  for the remainder of that epoch. This padding creates no new  $P$ -owned  $\Sigma$ -events, no new  $\text{BoundaryReq}$ , no new  $\text{Front\_P}$  or  $\text{NextFront\_P}$  item, and no new outgoing WFG edge from  $P$ . This epoch-local padding does not preclude a later reset boundary affecting  $P$ . If such a reset occurs and  $P$  later exits reset, the subsequent  $\text{START\_P}$  is a new dynamic occurrence that begins a fresh reset epoch, and the earlier  $\text{FINISH\_P}$  has no ordering, anchoring, or WFG force across that reset boundary. Any later termination of  $P$  is represented by a distinct dynamic  $\text{FINISH\_P}$  occurrence. Thus a process blocked waiting for a complementary Push/Pop from a peer that has terminated in the current reset epoch is not, by itself, an internal message-passing deadlock under the  $\text{Dead\_K}$  predicate. It is instead outside the  $\text{Dead\_K}$  predicate unless the design includes an explicit modeled termination protocol that makes such waiting part of the internal channel protocol being analyzed.

**Terminology (adopted reading: reachable closed internal wait-cycle cutpoint).** The predicate just defined is, precisely, a reachable, drain-closed, closed internal wait-cycle cutpoint. It is deliberately weaker than a conventional (permanent) deadlock: a member of  $D$  may hold environment-facing frontier items excluded from the WFG, and — in a declared timed or reactive environment outside the B1-avail convention — a fair continuation in which the environment later enables such an item can leave the state. (Under the default B1-avail pull-based stimulus convention, an environment-facing input item that is disabled at a cutpoint is stimulus-exhausted and remains disabled forever, per Lemma K.EnvX-Auto in K.4b; the later-enablement possibility, and with it the input-side  $W.\text{Env}$  rescue class, arises only for declared timed/reactive environments.) Three consequences are adopted normatively throughout this appendix. (1) The property preserved by Theorem K.1 — absence of reachable states satisfying this predicate — is an invariance (safety-style) reachability property over admissible executions, not a temporal liveness property in the eventually/always sense; the historical word “liveness” is retained in titles for continuity but shall be read with this meaning. (2) The phrase “processes in  $D$  can only wait on each other” means, wherever it appears, “only wait on each other through internal WFG channels”; it does not exclude coexisting environment-facing waits. (3) The A5 premise is correspondingly conservative: it can be violated by a transient internal wait-cycle cutpoint even in a design whose fixed environment is guaranteed to release an environment-facing frontier item so that the system always

progresses; such a report is a true positive for the defined predicate and a deliberate conservatism with respect to permanent deadlock. Designs for which this conservatism is unacceptable may instead be assessed against the stronger persistent-deadlock predicate — every blocking frontier item of every member of  $D$  internal, with its complementary owner in  $D$ , and the state a trap under all admissible continuations — but the theorems of this appendix neither require nor establish results for that stronger predicate; adopting it would require re-proving Lemmas K.1a–K.2b and Theorem K.1 with an added permanence argument.

**Scope warning.** This exclusion is only a scope convention for Theorem K.1. It does not imply that waiting forever for a terminated peer is benign or acceptable design behavior. If termination, EOF, done, cancellation, or end-of-stream behavior is part of the intended protocol, that behavior must be modeled explicitly in  $\text{Sys\_ref}$  and  $\text{RTL\_B}$ , or verified separately outside the Appendix K internal message-passing-deadlock theorem.

**Our notion of the preserved property (historically labeled “liveness”) in this appendix is:** no reachable internal message-passing-deadlock state (reachable closed internal wait-cycle cutpoint) as defined above, under the fixed stimulus and fair scheduling. Formally this is an invariance (safety-style) reachability property over admissible executions — see the Terminology note above — and it does not assert that a reached deadlock state is permanent when environment-facing frontier items are present. (Starvation of a single process in an otherwise live system is ruled out separately by weak fairness.)

**Implication note (relation to persistent deadlock).** Within the closed internal wait-cycle class captured by  $\text{Dead\_K}$ , the reachable-cutpoint property is conservative with respect to permanence: it excludes even a transient reachable  $\text{Dead\_K}$  cutpoint, and therefore also excludes a persistent internal hang whose blocked condition contains such a drain-closed closed WFG component. This implication is intentionally limited to the  $\text{Dead\_K}$  structural class. The theorem does not establish the broader operational guarantee that every form of permanent internal-communication non-progress is impossible. In particular, starvation or perpetual under-supply by a complementary owner that remains live on unrelated work can leave downstream operations permanently unanswered without producing a closed  $\text{Dead\_K}$  component; Example K.CE-5/H2 is the canonical case. Waiting for a terminated peer and the other stalls excluded by the scope warnings above likewise require explicit protocol modeling or a separate argument. The stronger reachability form is retained as the normative  $\text{Dead\_K}$  predicate because, within its stated class, it is safety-shaped—an invariance over reachable states—which makes it checkable by inductive invariants and preservable through finite-prefix correspondence (Definition J.Reach, Corollary J.1). A theorem covering permanent non-progress outside that closed-wait-cycle class would instead require additional continuation/fairness and permanence reasoning and is not claimed here. Designs with intentional environment-rescued  $\text{Dead\_K}$  cutpoints are handled by the Waiver  $\text{W.Env}$  mechanism (K.2).

## K.2 Assumptions and Imported Results

**Standing Assumptions.** We assume throughout Appendix K:

**A1. R-rules and B-rules.** The prescribed source reference model  $\text{Sys\_ref}$  and the post-HLS implementation  $\text{RTL\_B}$  satisfy the process- and channel-level semantic rules defined earlier (R-rules and B-rules).

**A2. Finite Pipeline Depth (FPD) / B4 finite  $\epsilon$ -stutter.**

For each process  $P$  there is a finite bound  $D\_P$  such that, along any consecutive epsilon-only evolution in which  $P$  is internally advancing toward its next observable  $\Sigma$ -action or its next  $\Sigma$ -action is locally enabled, and  $P$  is not externally stalled by a peer/environment/back-pressure condition,  $P$  executes at most  $D\_P$  consecutive epsilon-steps before either (i) executing a  $\Sigma$ -action, or (ii) reaching a stable waiting state in

which no further epsilon-step is available until some external/peer/environment condition changes the relevant enabling predicates. This is not a bound on clock cycles spent waiting for external stimulus, peer action, or back-pressure relief; such waiting is governed by WF/B2 and B3. The system-level all-disabled quiescence bound is the separate B5 cycle-valued bound  $C_{sys}$ . A theorem instance may discharge B5 by supplying  $C_{sys}$  directly or by proving per-process cycle bounds  $C_P$  from  $D_P$  through the bucket-semantic charging lemma and taking  $C_{sys} = \max_P C_P$ . The identification  $C_{sys} = \max_P D_P$  is permitted only when that bridge is proved.

**A3. Weak Fairness (WF) (B2).** If an action remains continuously enabled (its guard remains true), the scheduler eventually selects it.

**A4. Channel Progress (B3 (Additional Semantic Assumptions (B-rules)); Appendix N).** For each channel  $c$  with capacity  $B(c)$ , assume the progress invariants of Appendix N hold. This is an obligation on the scheduler/environment (e.g., fairness of arbiters/back-pressure) and is not implied by the local R-rules alone.

A5. Primary selected-instance Policy-S bounded-response premise (A5-S). For the selected  $Sys\_K$  instance, let  $p\_S^{det}$  be the deterministic Policy-S execution selected by  $SDet$  after fixing capacities, elaboration, initial/reset state, stimulus, witness choices, environment behavior, arbitration, concrete pre-HLS scheduling semantics, Policy-S issue semantics,  $\epsilon$ -normalization, and  $K.Low$  lowering. A5-S requires: (i)  $K.CompleteS$  evidence that  $p\_S^{det}$  is maximal, fair, and complete; (ii)  $K.RespCov$  evidence defining a checkable proof disposition and finite bound for every static internal process-facing blocking site and relevant dynamic class and proving that every dynamically reachable instance belongs to a monitored or statically bounded class, with coverage-preserving waivers and the separate environment-site rule; and (iii)  $K.RespClean$  evidence that no finite prefix of the complete run witnesses a non-waived  $K.RespFail$ .  $K.RespFail$  is a finite-prefix predicate covering certified-bound expiration without process-facing commit/return under the deadline convention of Definition  $K.RespFail$ , a declared maximal finite terminal run with the monitored frontier still pending, or an affecting RW2 RESET that clears the monitored dynamic instance before response without an explicit waiver. A5-S is used by Corollary K.2c to derive  $A5L\_Sys\_K$  and then, through  $K.Low.A5$ , A5-L only when  $K.EnvMF$  and the shared  $K.InterruptionGate$  applicability hold for the candidate components, with  $K.ResetEpoch$  evaluated first and  $K.TerminalDisposition$  required only on  $K.ResetEpoch.Continue$ .

For A5, Policy S is the conservative serial-blocking interpretation of Appendix G. Within each source interval, a process does not issue a source-later blocking operation until every source-earlier blocking operation required by the serial discipline has committed in a strictly earlier cycle. The condition is process-facing: proof-internal staged/bundle/join/epoch-gate transfers do not satisfy it unless the elaboration identifies them as the commit that releases the source-level call. Channel-side E4 movement, FIFO drainage, independent-process execution, and explicit elaborated transfers remain concurrent.

Maximality, completeness, and fairness of  $p\_S^{det}$  are part of A5-S/ $K.CompleteS$ , not  $SDet$ .  $K.RespCov$  and  $K.RespClean$  are additional obligations checked over that complete artifact, not definitions of completeness itself. A finite simulation discharges A5-S only when it reaches a declared maximal terminal state and has no pending covered frontier, or when a complete proof establishes the required monitor result. A finite prefix of a still-running system is insufficient. For a nonterminating design, the selected run requires an invariant, sound model checking, recurrence/lasso evidence, or another complete argument that every covered response occurs within its finite bound and no reset-pending failure occurs.

Instance dependence. A5-S and  $SDet$  are not properties of the raw SystemC text and are not discharged by an arbitrary simulation. They are re-established whenever capacities, boundaries, elaboration, lowering, stimulus, witness assignment, environment/reset contract, arbitration rule, concrete simulator

scheduling semantics, Policy-S implementation, or normalization changes. Such a change defines a different selected theorem instance.

Practical discharge route (Corollary K.2c). Under SDet there is one selected Policy-S execution  $p\_S^{\text{det}}$  for the selected instance, with a committed history unique modulo finite WFG-inert  $\epsilon$ -stutter.  $K.\text{EnvMF}$  restricts the supported environment/frontier cases.  $K.\text{ResetEpoch}$  and  $K.\text{TerminalDisposition}$  instantiate the shared  $K.\text{InterruptionGate}$  schema in the normative nested order: reset first, then terminal only on  $K.\text{ResetEpoch}.\text{Continue}$ . A  $\text{DirectFail}$  branch yields the corresponding finite  $K.\text{RespFail}$  without invoking dominance; the two nested  $\text{Continue}$  branches select  $K.2b.E/R/P/Q$ .  $A5\text{-}S/CF\text{-}S$  separately supply  $K.\text{CompleteS}$ ,  $K.\text{RespCov}$ , and  $K.\text{RespClean}$  evidence. Lemma K.2b shows that any internal deadlock reached by a finite Policy-L prefix admissible under Definition  $\text{Core.LAdm}$  forces that same selected execution to produce a finite non-waived  $K.\text{RespFail}$  witness. Therefore a complete clean bounded-response certificate for  $p\_S^{\text{det}}$  is sufficient on that supported fragment. No alternate Policy-S schedulers or interleavings exist within the selected instance, and no confluence theorem is relevant. A source region whose liveness depends on favorable same-cycle or overlapped issue of multiple blocking operations is outside the scheduler-robust  $A5\text{-}L$  framework whenever an admissible serial-shaped Policy-L prefix reaches  $\text{Dead\_K}^W$ . In that case  $A5\text{-}L$  is false, not merely difficult to discharge: direct Policy-L verification, structural conditions, invariants, exploration, and tool certificates cannot prove it for the unchanged instance. The design must be rewritten or buffered, placed under explicit protocol/synchronization structure that removes the serial deadlock prefix, or verified under a different scheduler-specific theorem not supplied by this document.

$A5$  and Appendix K do not assert freedom from specification-level barrier errors, reset stalls, or external-environment stalls that do not contain the internal wait structure represented by the operative predicate.

**A6. Point-to-point channels (single-producer/single-consumer).** For each message-passing channel  $c$  (including buffered channels with  $B(c)>0$  and rendezvous channels with  $B(c)=0$ ), exactly one process performs all  $\text{Push}_c$  operations on  $c$  (the producer) and exactly one process performs all  $\text{Pop}_c$  operations on  $c$  (the consumer). Hence the complementary endpoint for any blocked  $\text{Push}_c/\text{Pop}_c$  is unique, and WFG edges are well-defined even for mixed WFGs that include both buffered and rendezvous channels.

If a design has a shared resource that would violate this uniqueness, it must be modeled via an explicit arbiter process plus point-to-point channels (as noted in “Normative Formal Core”  $B(c)$ ).

$A7$  (RTL determinism / witness selection / equal-KObs correspondence). Assume  $B1$  for  $\text{RTL\_B}$ .  $J.\text{GWR-Comp/Theorem J.1}$  and Corollary J.1 supply one selected source witness and equal KObs. For Appendix K,  $J.\text{RegPhaseReach}$  is derived from the revised registered, deferred-hole, phase-normal construction (or directly supplied), while  $K.\text{DrainReach}$  validates nonwithdrawal,  $K\epsilon$ , and every activated  $\text{Core.Drain/K.Drain}$  obligation for that exact prefix; together they are  $K.\text{RLAdmReach}$ .  $J.\text{OfferReach}$ ,  $J.\text{OfferConf}$ ,  $J.\text{KObsConf}$ ,  $\text{RegPhaseReach}$ ,  $\text{DrainReach}$ , and  $K.\text{RLAdmReach}$  must bind the same RTL/source cutpoints,  $H,O,E,w$ , instance, and stimulus.  $\text{Core.KCutClosure\_post}$  and  $K.\text{CertCov}$  are additionally required only for the universal reachable-cutpoint conclusion. In the standard  $\text{QSync}$  normal-compositional scope,  $\text{Core.QSync-Post}$  derives the closure premise and Lemma  $\text{Core.QSync-KCutCanonical}$  identifies the complete  $K\text{Cut}$  domain with canonical endpoints; if the flow proves  $J.\text{QSyncCertInd}$ , Theorem  $J.\text{CertExist-QSync}$  and Corollary  $K.\text{CertCov-QSync}$  derive  $K.\text{CertCov}$  by induction over those canonical cycle cutpoints. Otherwise  $K.\text{CertCov}$  remains an independently discharged coverage obligation.

When a deterministic, reproducible  $\text{Sys\_ref}$  witness is desired for simulation/debug, apply the Appendix L snooping wrapper to select a particular witness aligned with the latency-sensitive  $\epsilon$ -choices.  $B4$  and  $B5$  justify the selected  $\epsilon$ -normalized endpoints required by Corollary J.1. Every Appendix K transfer of a reconstructed frontier, request, channel, enablement, WFG, drain, or deadlock predicate uses the same

equal-KObs package. A discharge for an unrelated source cutpoint, different witness, different O, or noncanonical unmatched history is insufficient.

**Premise SDet** (deterministic selected Policy-S simulator). Fix the selected Sys\_K, initial/reset state, capacities/elaboration, stimulus and reactive environment, witness-selected  $\epsilon$  outcomes, deterministic arbitration, concrete SystemC/pre-HLS scheduler, Policy-S issue rule, and  $\epsilon$ -normalization. These define a functional Policy-S next-state semantics and select one run artifact  $p\_S^{\text{det}}$ . Every finite or infinite artifact generated from the same initial state by those exact semantics has a committed history prefix-comparable with that of  $p\_S^{\text{det}}$  modulo finite WFG-inert  $\epsilon$ -stutter; hence no two incompatible committed histories arise from the selected instance. WC and I.DR/I.DR' determine replay state/actions relative to the selected history; SDet excludes multiplicity caused by an unfixed scheduler, arbiter, environment, witness, issue, or normalization choice. SDet does not by itself assert maximality, fairness, completeness, response coverage, or response cleanliness, and a shorter finite artifact may be a proper prefix of the selected history. K.CompleteS and A5-S/CF-S separately establish maximality/fairness/completeness and K.RespCov/K.RespClean for  $p\_S^{\text{det}}$ . Changing any listed choice defines a different selected instance.

**A8. Barrier well-formedness (SyncChannels).** For each SyncChannel instance, the set of designated participant processes reaches the barrier the same number of times under the fixed stimulus/initial state; equivalently, no participant can permanently bypass a barrier call while another participant is waiting at that barrier. (Barrier stalls are therefore outside the behaviors considered in Appendix K's internal message-passing deadlock analysis.)

**A9. Single-transfer-per-cycle endpoint constraint (scalar channel interface).** The selected instance satisfies the authoritative Core single-transfer-per-cycle endpoint constraint; multi-lane/burst resources are modeled explicitly as specified there.

**A10. Reset-empty FIFO assumption.** The selected instance satisfies the authoritative Core reset-empty FIFO convention at every reset boundary in scope.

**A11. WFG-inert  $\epsilon$  (K $\epsilon$ ).** The design satisfies K $\epsilon$ .

No further numbered A-assumption is introduced beyond A1–A11; SDet is a separately named selected-simulator premise. A5-S is part of the selected-instance liveness obligation, not a conclusion derived from the local R-rules, WC alone, or the no-reverse rule.

**Drain-only blocked-frontier discharge for Appendix K (K.Drain).** K.Drain is the design/application evidence obligation that the selected theorem instance satisfies Definition Core.Drain. Its continuation content is the complete seven-clause discipline of Definition Core.16. The K.2b dominance/extraction bridge and Appendix K drain-normalization arguments require that discipline for every interval in which Policy L can leave source-later same-interval work outstanding behind a disabled earlier frontier. Pipelined or otherwise overlapped control discharges this through Appendix B automatic flush. Ordinary non-overlapped control may discharge it through a structural proof of the same discipline. In every case, the discharge must establish the blocked point, no new source-later dynamic-operation launch, request non-withdrawal, exclusion of non-frontier requests as WFG sources, drain-only downstream progress, finite non-replenishing B-faithful boundary storage or credit, and explicit sync-boundary / epoch-gate treatment for any withheld producer-facing capacity. This is the K.Drain application-specific discharge of the existing scheduling and channel assumptions, not an additional numbered A-assumption.

**Definition K.A5L** (model-indexed scheduler-robust bad-cutpoint exclusion). For a source proof model M used by Appendix K, write  $A5L\_M(I, W)$  for the A5-L formula interpreted in M: every finite Policy-L prefix admissible under M's Core.LAdm/Pref\_L<sup>adm</sup> domain, with the fixed theorem-instance stimulus, capacities, elaboration/lowering, environment, arbitration, reset interpretation, and witness-selected outcomes, excludes every reached  $\sigma \in K\text{CutReach\_M}(I)$  satisfying the corresponding Dead\_K<sup>W</sup>

predicate. The model index fixes the operation/frontier universe and explicit boundaries on which the predicate is evaluated; it is not an additional execution choice.

Premise A5-L (scheduler-robust liberal source liveness). A5-L is the abbreviation  $A5L\_Sys\_ref(I, W)$ . Thus, for every  $p\_L \in Pref\_L^{adm}(I)$  of the selected  $Sys\_ref$  instance, under the fixed stimulus, capacities  $B(c)$ , elaboration, and fixed witness-selected outcomes, no source  $K$ -facing cutpoint  $\sigma \in KCutReach\_ref(I)$  reached by  $p\_L$  satisfies  $Dead\_K^W$ . Theorem K.1A consumes this  $Sys\_ref$  form for one source prefix whose  $Pref\_L^{adm}(I)$  membership is supplied by  $K.RLAdmReach$ . Where synchronization waits or witness-selected non-NB-CF sites can participate in a potential wait cycle, the serial route first proves the corresponding  $A5L\_Sys\_K(I, W\_K)$  result on the normalized blocking instance  $Sys\_K$  and then transports it to A5-L by Lemma K.Low.A5.

Policy-S-primary architecture.  $SDet$  fixes one concrete deterministic Policy-S execution  $p\_S^{det}$ .  $K.EnvMF$  limits the serial reduction to  $StandardEnv/B1$ -avail when relevant  $Dead\_K$  components are multi-frontier, and to certified single-internal-frontier components for timed/reactive or otherwise nondefault environments.  $K.InterruptionGate$  supplies the common applicability branch discipline;  $K.ResetEpoch$  is evaluated first and  $K.TerminalDisposition$  only on  $K.ResetEpoch.Continue$ . Lemma K.2b compares a liberal counterexample with the selected execution and proves that a Policy-L internal deadlock forces a finite  $K.RespFail$  witness on  $p\_S^{det}$ . Corollary K.2c therefore reduces A5-L to the single complete A5-S check consisting of  $K.CompleteS$ , total  $K.RespCov$ , and  $K.RespClean$ . There is no family of Policy-S scheduler interleavings inside one selected instance and no confluence question.

$Dead\_KS$ ,  $EnvStop$ ,  $DoneStop$ ,  $K.EnvOrd$ , and  $K.Done$  are optional diagnostic classifications only. Normative organization of Appendix K. The spine is: the  $Dead\_K$  predicate;  $K.O$  frontier reconstruction;  $K.2$  WFG extensionality;  $K.2a$   $DrainClosedNow$  extensionality; the  $K.RLAdmReach$  qualification and  $K.CertifiedCutpoint$  interface; Theorem K.1A from A5-L for one certified cutpoint;  $Core.QSync$ - $Post/Core.QSync$ - $KCutCanonical$  plus  $J.QSyncCertInd/J.CertExist$ - $QSync$  and  $K.CertCov$ - $QSync$  for the standard universal package-totality route;  $Core.KCutClosure\_post$ ,  $K.CertCov$ , and Corollary K.1A-Total for the universal reachable-cutpoint conclusion;  $Sys\_K$  and the  $SDet$ -selected  $p\_S^{det}$ ;  $K.EnvMF$ ; the shared  $K.InterruptionGate$  schema with reset-first  $K.ResetEpoch$  and nested  $K.TerminalDisposition$  instantiations; deterministic serial-dominance K.2b; Corollary K.2c; A5-S on  $p\_S^{det}$ ; and Theorem K.1. Appendix K-P contains the operational companion proof. Structural and direct certificates remain alternate A5-L routes only when A5-L is true for the selected instance.

Waiver  $W.Env$  (declared environment-rescued cutpoints). The Appendix M.7a side-condition record may exclude a precisely identified cutpoint class when a declared environment-facing co-frontier is intended and proved to rescue it. The record shall identify the component class, rescuing item, environment assumption, and design justification. A waiver does not alter the default predicates and does not excuse an Example K.CE-style bypass without an accompanying no-bypass or equivalent argument.

Definition K. $DeadW$  and response waivers. Let  $W$  be the union of declared  $KObs$ -facing waived cutpoint classes.  $Dead\_K^W(\sigma)$  means  $Dead\_K(\sigma)$  and  $\sigma \notin W$ ; optional diagnostic  $Dead\_KS^W$  is defined analogously.  $KObs$ -facing waiver closure requires  $KObsClosed\_I, E, w(W)$  whenever  $W$  is non-empty. Separately, let  $W\_resp$  be the declared dynamic  $K.Resp$  monitor-waiver classes.  $RespFail^W$  means a  $K.RespFail$  whose monitor instance is not in  $W\_resp$ .  $K.RespCov$  requires coverage-preserving waiver closure: for every non-waived liberal  $Dead\_K$  component admitted to the K.2b premise, at least one permanently pending serial monitor extracted by K.2b.Q is outside  $W\_resp$ . A response waiver depending on hidden issue timing, reset disposition, or future environment behavior must carry the corresponding operational proof. Neither  $W$  nor  $W\_resp$  may be used as a broad label to hide an unclassified deadlock or end-of-test hang.

Scope of the preservation theorem. Theorem K.1A transfers a deadlock at one  $K.CertifiedCutpoint$  RTL cutpoint to a  $Core.LAdm$ -admissible Policy-L  $Sys\_ref$  deadlock through equal  $KObs$  and contradicts A5-L; it consumes neither  $SDet$  nor the serial-route premises and makes no package-totality claim. Corollary

K.1A-Total adds Core.KCutClosure\_post and K.CertCov and obtains the universal reachable-cutpoint conclusion. Theorem K.1 adds SDet, K.CV, NB-Align/K.Low where applicable, K.Resp, and the remaining serial conditions, obtains A5-L from A5-S for the SDet-selected execution  $p\_S^{\text{det}}$  via Corollary K.2c, and uses Core.KCutClosure\_post together with K.CertCov for the user-facing universal RTL conclusion. Within Lemma K.2b, K.InterruptionGate is instantiated first by K.ResetEpoch and, only on K.ResetEpoch.Continue, by K.TerminalDisposition. Either DirectFail branch bypasses the dominance construction; the two nested Continue branches select K.2b.E/R/P/Q. K.CompleteS supplies maximality/fairness/completeness, K.RespCov supplies total finite-bound coverage, and K.RespClean is the contradiction used by K.2c.

The no-reverse rule is not by itself a liveness theorem. Its practical force is captured by the deterministic Policy-S reduction: the selected simulator withholds source-later requests and serves as the conservative deadlock-stress execution. K.2b does not claim that  $p\_S^{\text{det}}$  reproduces the Policy-L trace or the same component; it states that any Policy-L internal deadlock forces a permanent covered process-facing serial frontier and therefore a finite K.RespFail witness.

Example: blocking rendezvous operations whose liveness depends on same-cycle/eager issue. Consider two rendezvous channels a and b with  $B(a)=B(b)=0$  and two processes; assume v and w are independent of any data/status returned by the preceding Pops:

```
P1: b.Pop(); a.Push(v)
P2: a.Pop(); b.Push(w)
```

Under Policy S's conservative serial-blocking issue discipline, each process issues only its first source-order blocking operation: P1 waits on Pop\_b and P2 waits on Pop\_a; neither process reaches the complementary Push. That two-Pop execution is not merely a Policy-S artifact—it is itself a Core.LAdm-admissible Policy-L prefix. After the L2 settled snapshot exposes both Pops disabled, Core.Drain/K.Drain forbids newly issuing the source-later Pushes, and the resulting drain-closed source K-facing cutpoint contains the closed internal rendezvous deadlock. Thus A5-L is false for the unchanged design, even though, when the selected instance admits the straight-line Core.ReachPrep case, a different eager Policy-L schedule may assert both distinct-interface requests and complete both rendezvous transfers in the same cycle; the RTL may likewise exhibit that behavior when its validated scheduling permits it. No direct Policy-L proof, structural proof, invariant, exploration, or alternate A5-L certificate can establish a false premise. The Appendix K scheduler-robust framework therefore cannot certify a design whose deadlock freedom depends on this favorable eager/overlapped issue; the design must be rewritten, buffered, explicitly synchronized, or verified under a different scheduler-specific theorem. Completeness boundary (normative): if some  $p \in \text{Pref\_L}^{\text{adm}}(I)$  reaches  $\text{Dead\_K}^W$ , then  $\neg A5-L(I)$ , regardless of whether another admissible schedule is live.

The Front/NextFront bridge used later in Appendix K is the source-local reconstruction result of Lemma K.0. Cross-model frontier, raw rendezvous-request, enablement, and WFG agreement are obtained by applying the same reconstruction functions to the equal KObs record of Corollary J.1. No asserted rendezvous request is identified with frontier membership merely because it is present in O.

#### Remark K.0-pre (Derived bridge, not an extra premise).

The asserted blocking-message slice of NextFront is not an additional theorem premise. Core.7a defines the candidate blocking slice, Core.10 and Core.IC supply operation-attributive request persistence, and Lemma K.0 derives that the asserted slice is exactly the pending blocking frontier. BoundaryReq remains undefined for non-message frontier items, and successful non-blocking frontier items do not create pending blocking WFG members.

Definition K.StimX. Under convention B1-avail, the next fixed input item is available as soon as earlier items on that environment channel have been consumed. An environment-facing Pop is stimulus-exhausted exactly when no item remains; an environment-facing Push is exhausted only under a declared permanent-stop contract, and never under the default always-ready output environment. Diagnostic Definition K.WFG+ and K.DeadKS. For debugging and failure triage only, extend the ordinary wait-for graph with two terminal-node classes. For each environment-facing boundary  $c$  whose next required interaction is permanently unavailable under Definition K.StimX, create  $\text{EnvStop}_c$  and add  $P \rightarrow \text{EnvStop}_c$  when blocked  $P$  has the corresponding disabled frontier. For each internal point-to-point boundary  $c$  whose unique complementary owner has normally completed before the demanded ordinal, create  $\text{DoneStop}_c$  and add  $P \rightarrow \text{DoneStop}_c$ . Ordinary internal dependencies remain process-to-process edges  $P \rightarrow Q$ . At a reachable drain-closed Policy-S K-facing cutpoint,  $\text{Dead}_K$  holds when the extended graph contains a nonempty closed blocked component containing at least one process-to-process edge;  $\text{Dead}_K$  is the terminal-free special case. These predicates remain KObs-reconstructible under their stated environment and waiver premises, but they are not complete for live endpoint under-supply and are not used to prove Corollary K.2c. They may classify why a K.Resp monitor fired and support familiar diagnostic reports for Examples K.CE through K.CE-4.

### K.2.1 KObs Deadlock Factoring and Extensionality

These results factor the Appendix K snapshot predicates through the common Core.18c KObs record. They transfer only cutpoint facts; Core.Drain/K.Drain, future issue legality, fairness, and bounded-response monitor obligations remain operational premises.

#### Definition K.EnvTerminalAdequate (environment-terminal reconstruction premise).

$\text{EnvTerminalAdequate}(I)$  holds when the selected instance records either  $\text{StandardEnv}$  or an  $\text{EnvReplayAdequate}$  certificate. Under  $\text{StandardEnv}$ , Definition K.StimX and the default always-ready output convention determine permanent next-interaction availability from the committed consumption history; Lemma K.EnvX-Auto supplies the corresponding environment-co-frontier fact used by the serial route. Under  $\text{EnvReplayAdequate}$ , the certificate supplies a typed reconstructed environment state  $\text{EnvStateRec}_I(H, w)$  and proves that the permanent-availability predicates used by K.WFG+ are determined by  $H$ ,  $w$ , and that state. This is an extensionality premise for optional diagnostics, not a claim that every well-typed KObs record is reachable.

#### Definition K.KObsDerived (K-specific reconstruction functions).

For a Core.18c record  $K = \langle E, w, H, O \rangle$ , define  $\text{DeadKRec}(K, I)$  by evaluating Definition K.Dead on the Core-reconstructed  $\text{FrontRec}$ ,  $\text{WFGRec}$ , process-completion, and  $\text{DrainClosedNowRec}$  facts. Define  $\text{DeadKRec}^W(K, I, W)$  by applying Definition K.DeadW to  $\text{DeadKRec}$  and the KObs-facing waiver set  $W$ . When  $\text{EnvTerminalAdequate}(I)$  holds, define  $\text{EnvStopRec}$ ,  $\text{DoneStopRec}$ ,  $\text{WFGPlusRec}$ ,  $\text{DeadKSRec}$ , and their KObs-closed waived variants by applying Diagnostic Definition K.WFG+ and K.DeadKS to the same Core-reconstructed process, frontier, request, channel, and committed-history facts together with the selected environment-reconstruction mode. These are Appendix K functions over a source-realizable Core record. No K.RespFail reconstruction is defined from  $K$  alone because monitor activation, age, bound class, and reset/terminal disposition state are operational data outside KObs.

#### Lemma K.KObs-Dead (DrainClosedNow and K-predicate factoring).

For every non-empty process set  $D$ ,  $\text{DrainClosedNowRec}(D, K)$  holds iff the operational cutpoint  $\sigma$  is drain-closed for  $D$  in the snapshot sense used by Appendix K. The unwaived ordinary predicate  $\text{Dead}_K(\sigma)$  is therefore determined by  $K$ . If  $\text{KObsClosed}_I(E, w(W))$  holds,  $\text{Dead}_K^W(\sigma)$  is determined by

$\text{DeadKRec}^W(K, I, W)$ . If  $\text{EnvTerminalAdequate}(I)$  holds, the optional diagnostic  $\text{WFG}+(\sigma)$ ,  $\text{EnvStop}$ ,  $\text{DoneStop}$ , and unwaived  $\text{Dead\_KS}(\sigma)$  are determined by  $\text{WFGPlusRec}(K, I)$  and  $\text{DeadKRec}(K, I)$ ; with  $\text{KObsClosed}$ , the diagnostic waived predicate is also determined. No  $\text{K.RespClean}$  or  $\text{K.RespFail}$  conclusion follows from  $\text{KObs}$ .

Proof. Lemmas  $\text{J.KObs-Proc}$ – $\text{J.KObs-WFG}$  reconstruct process completion, consumed ordinals, frontier membership, endpoint requests, channel enablement, and ordinary WFG edges. The optional terminal reconstructions use the stated environment adequacy premises.  $\text{KObsClosed}$  supplies waiver extensionality for cutpoint predicates. The snapshot drain clause is exactly the absence of an enabled non-frontier residual offer owned by  $D$ . No continuation-level claim about Definition  $\text{Core.16/DrainDiscipline}$  or response-monitor age follows from this factoring result.  $\square$

### Corollary K.KObs-Ext (KObs extensionality).

Let  $\sigma_1, \sigma_2 \in \text{KCutReach\_ref}(I)$  be source  $K$ -facing cutpoints of the same fixed instance with  $\text{KObs\_E}, w(\sigma_1) = \text{KObs\_E}, w(\sigma_2)$ . Then their reconstructed process slices, explicit abstract channel states,  $\text{NextFront}$  and  $\text{Front}$  sets, operation-attributive  $\text{BoundaryReq}$  values, channel-enable predicates, ordinary WFGs,  $\text{DrainClosedNow}$  predicates, and unwaived  $\text{Dead\_K}$  truth values are equal. If  $\text{KObsClosed\_I}, E, w(W)$  holds, their  $\text{Dead\_K}^W$  truth values are equal. If  $\text{EnvTerminalAdequate}(I)$  holds, their optional diagnostic  $\text{EnvStop/DoneStop}$  classifications,  $\text{WFGPlusRec}$  graphs, and unwaived  $\text{Dead\_KS}$  truth values are equal; with  $\text{KObsClosed}$ , their diagnostic  $\text{Dead\_KS}^W$  values are equal. The conclusion is independent of legal historical  $\text{Issue}$  timestamps not retained in  $O$ , and it makes no claim about  $\text{K.Resp}$  timer activation or age. Without waiver closure or environment-terminal adequacy, the extensionality conclusion is correspondingly limited. Proof. Functional extensionality of the named reconstruction definitions and Lemmas  $\text{J.KObs-Proc}$ – $\text{J.KObs-WFG}$  and  $\text{K.KObs-Dead}$ .  $\square$

Cross-model-use note. Corollary K.KObs-Ext is the source-side extensionality theorem for reachable records. Appendix J supplies  $\text{J.OfferReach}$  and  $\text{J.OfferConf}$  and proves through  $\text{J.KObsConf}$  that the selected post certificate and source cutpoint share one record. Lemma K.0, Corollary K.0a, Lemma K.2, Lemma K.2a, and Theorem K.1A state the required cross-model consequences directly from that equality.

## K.2.2 Lemma K.0 — Frontier Reconstruction from KObs

Purpose. Appendix K forms WFG edges from the minimal pending blocking frontier. The  $\text{KObs}$  layer already separates the candidate source frontier reconstructed from  $H$  from the currently asserted attributed offers reconstructed from  $O$ . This lemma identifies the operational source frontier with the corresponding pure reconstruction and states the cross-model consequence by equal-KObs extensionality.

Statement. Fix  $I, E, w$  and  $\sigma \in \text{KCutReach\_ref}(I)$ , and let  $K = \text{KObs\_E}, w(\sigma)$ . For process  $P$  define  $\text{BlockingMsgNextFrontRec\_P}(K) := \{ x \in \text{NextFrontRec\_P}(K) \mid x \in \Omega\_msg, P \text{ and } \text{MsgOf\_P}(x) \text{ is a blocking Push or Pop instance} \}$ .

Then

$\text{Front\_P}(\sigma) = \text{FrontRec\_P}(K) = \{ x \in \text{BlockingMsgNextFrontRec\_P}(K) \mid \text{BoundaryReqRec}(x, K) \}$ .

Here  $\text{NextFrontRec}$  is witness-relative over the fixed  $\Omega\_P$  and  $\leq\_P$  supplied by  $E$ ; it is a candidate frontier, while  $\text{BoundaryReqRec}$  selects the currently outstanding blocking offers.

Cross-model consequence. Let  $K\_post = \text{KObsPost\_J}(p\_post; E, w, H, O)$  and  $K\_ref = \text{KObs\_E}, w(\sigma\_ref)$  be the exact records selected by Corollary J.1. If  $K\_post = K\_ref$ , then for every process  $P$ ,  $\text{FrontRec\_P}(K\_post) = \text{FrontRec\_P}(K\_ref)$ , and  $\text{BoundaryReqRec}(x, K\_post) = \text{BoundaryReqRec}(x, K\_ref)$  for every typed blocking message-operation frontier item  $x$ . Thus the reconstructed post frontier is exactly

the operational source frontier at  $\sigma_{\text{ref}}$ . This consequence uses only the common record; it does not compare historical Issue times or raw RTL control state.

Mixed blocking/non-blocking clarification. Successful non-blocking transfers may contribute committed Push/Pop events to H, and failed polls may contribute WC-selected  $\varepsilon$  outcomes, but neither creates a pending blocking frontier member. A blocking endpoint may nevertheless wait on a complementary process whose source endpoint uses a non-blocking call; the non-blocking side can change occupancy or complete the complementary transfer, but does not itself become a WFG source unless it also owns a separate pending blocking frontier.

Proof. For a reachable source cutpoint, Lemma J.KObs-Offers identifies Pending and BoundaryReq with PendingRec and BoundaryReqRec and therefore identifies Front\_P with FrontRec\_P. The displayed BlockingMsgNextFrontRec characterization is the Core.7a and Definitions Core.14–Core.15 classification of the  $\leq_P$ -minimal pending blocking items; source-prefix issue order, synchronization anchoring, and the exclusion of failed polls ensure that no earlier unrealized nonblocking, signal, or synchronization item invalidates the candidate-frontier characterization. The cross-model conclusion follows by substituting  $K_{\text{post}} = K_{\text{ref}}$  into the pure reconstruction functions.  $\square$

Corollary K.0a (Rendezvous endpoint-request extensionality). For a point-to-point rendezvous channel  $c$  with  $B(c)=0$ , define  $\text{ReqProdRec}_c(K)$  and  $\text{ReqConsRec}_c(K)$  as the producer-direction and consumer-direction projections of  $O$ . If  $K_{\text{post}} = K_{\text{ref}}$  as above, then  $\text{ReqProdRec}_c(K_{\text{post}}) = \text{ReqProdRec}_c(K_{\text{ref}})$  and  $\text{ReqConsRec}_c(K_{\text{post}}) = \text{ReqConsRec}_c(K_{\text{ref}})$ . This raw endpoint-request equality is independent of frontier membership: an already-asserted non-frontier blocking offer remains represented in  $O$  and can participate in rendezvous enablement.

### K.2.3 Example K.0b — Front\_P vs. BlockingMsgNextFront\_P Example

The following example illustrates the distinction between Front\_P and BlockingMsgNextFront\_P. In this example all message-operation items under discussion are blocking Push/Pop operations, so BlockingMsgNextFront\_P is the relevant WFG-facing slice of the source-level NextFront\_P frontier over  $\Omega_P$ . The broader notation MsgNextFront\_P, when used elsewhere, denotes the general message-operation-item slice and may also include successful non-blocking transfer items; it is not the object used by the Front/NextFront bridge for WFG construction.

```
void P() {
  while (1) {
    wait();           // sync point
    int x = in.Pop(); // blocking Pop
    out.Push(x + 1);  // blocking Push
  }
}
```

Consider two source K-facing cutpoints for this same process:

**Cutpoint  $\sigma_0$ : just after wait() has committed, before in.Pop() has issued**

- $\text{BlockingMsgNextFront}_P(\sigma_0) = \{ \text{Pop}_{\text{in}} \}$
- $\text{BoundaryReq}(\text{Pop}_{\text{in}}, \sigma_0) = \text{false}$
- $\text{Front}_P(\sigma_0) = \emptyset$

**Cutpoint  $\sigma_1$ : in.Pop() has issued, rdy is asserted, but no data has arrived yet**

- $\text{BlockingMsgNextFront}_P(\sigma_1) = \{ \text{Pop}_{\text{in}} \}$
- $\text{BoundaryReq}(\text{Pop}_{\text{in}}, \sigma_1) = \text{true}$
- $\text{Front}_P(\sigma_1) = \{ \text{Pop}_{\text{in}} \}$

**So the difference is:**

- $\text{BlockingMsgNextFront}_P$  says which blocking message-operation items are next candidates in the

source-level frontier over  $\Omega_P$  for WFG/frontier classification.

·  $\text{Front}_P$  says which of those next blocking message-operation items are actually outstanding pending blockers right now.

### K.3 Lemma K.1a — Characterization of Internal Message-Passing Deadlock (Buffered and Rendezvous Cases)

Elaboration prerequisite. The channel boundary  $c$  used in this lemma is the chosen explicit abstract boundary of the selected source-side or lowered proof model  $M$  after applying any required elaboration from Definition Core.1 and Appendix J, Remark 2. In particular, synchronization-boundary ramp-down / epoch-gating is not treated as a hidden exception to E4 on an otherwise ordinary buffered channel. If such gating can withhold producer-facing availability while outer projected capacity would otherwise appear available, the relevant blocking point is represented by an explicit abstract channel of  $\text{Sys}_B^{\text{elab}}$ .

Statement: Let  $M$  be a source-side or lowered proof model used by Appendix K and let  $\sigma \in \text{KCutReach}_M(I)$ . Suppose there is a non-empty set of processes  $D$  witnessing  $\text{Dead}_K(\sigma)$  under Definition K.Dead. Then:

Every process  $P$  in  $D$  is blocked on a blocking channel operation (Push or Pop), not on an enabled internal  $\epsilon$ -step.

Buffered-channel case ( $B(c) > 0$ ):

- If  $P$  is blocked on a frontier item  $x \in \text{Front}_P(\sigma)$ , with owning message instance  $q = \text{MsgOf}_P(x)$ ,  $\text{kind}(q) = \text{Push}_c$ , and  $B(c) > 0$ , then  $\text{occ}_c(\sigma) = B(c)$  (channel is full at that cutpoint).
- If  $P$  is blocked on a frontier item  $x \in \text{Front}_P(\sigma)$ , with owning message instance  $q = \text{MsgOf}_P(x)$ ,  $\text{kind}(q) = \text{Pop}_c$ , and  $B(c) > 0$ , then  $\text{occ}_c(\sigma) = 0$  (channel is empty at that cutpoint).

Rendezvous case ( $B(c) = 0$ ):

- If  $P$  is blocked on a frontier item  $x \in \text{Front}_P(\sigma)$ , with owning message instance  $q = \text{MsgOf}_P(x)$ ,  $\text{kind}(q) = \text{Push}_c$ , and  $B(c) = 0$ , then the complementary endpoint is not simultaneously requesting the matching  $\text{Pop}_c$  in  $\sigma$  (i.e., the complementary  $\text{Pop}_c$  endpoint request is not true in that cycle, so rendezvous enabling for  $x$  is false).
- If  $P$  is blocked on a frontier item  $x \in \text{Front}_P(\sigma)$ , with owning message instance  $q = \text{MsgOf}_P(x)$ ,  $\text{kind}(q) = \text{Pop}_c$ , and  $B(c) = 0$ , then the complementary endpoint is not simultaneously requesting the matching  $\text{Push}_c$  in  $\sigma$ .

The processes in  $D$  form a closed strongly connected component in the WFG.

**Proof.** Definition K.Dead supplies, for every  $P \in D$ , a channel-disabled blocking message-passing frontier item and the closed strongly connected WFG component. Because  $\sigma \in \text{KCutReach}_M(I)$ , Definition Core.KCut together with Core.4 gives Quiescent $_M(\sigma)$ , so no enabled internal  $\epsilon$ -step remains at  $\sigma$ . For  $B(c) > 0$ , the full/empty conclusions are exactly Definition Core.12. For  $B(c) = 0$ , the missing-complementary-request conclusions are exactly Definition Core.13. The closed-SCC conclusion is the second defining clause of Definition K.Dead.  $\square$

### K.4 Lemma K.2 — WFG Extensionality under Equal KObs

Statement. Let  $p_{\text{post}} \in \text{KCutReach}_{\text{post}}(I)$  and  $\sigma_{\text{ref}} \in \text{KCutReach}_{\text{ref}}(I)$  be the selected K-facing cutpoints of one Corollary J.1 application, and write

$K_{\text{post}} := \text{KObsPost}_J(p_{\text{post}}; E, w, H, O)$ ,

$K_{\text{ref}} := \text{KObs}_E, w(\sigma_{\text{ref}})$ .

If  $K_{\text{post}} = K_{\text{ref}}$ , then

$\text{WFGRec}(K\_post) = \text{WFGRec}(K\_ref)$ .

Equivalently, the K-facing reconstructed post wait-for graph and the operational source wait-for graph have exactly the same vertices and directed edges. The statement applies uniformly to buffered and rendezvous boundaries and to  $\text{Sys\_B}^{\text{elab}}$  explicit staged, bundle, join, and epoch-gate boundaries supplied by E.

Proof.  $\text{WFGRec}$  is a typed pure function of K.  $\text{FrontRec}$  and  $\text{BoundaryReqRec}$  are reconstructed from O by Lemma K.0/J.KObs-Offers; buffered channel state is reconstructed from H and E by Lemma J.KObs-Chan; rendezvous complementary requests are direction projections of O by Corollary K.0a;  $\text{ChannelEnabledRec}/\text{ChannelDisabledRec}$  and the edge relation are then reconstructed by Lemma J.KObs-WFG. Substitution of  $K\_post = K\_ref$  therefore gives graph equality by functional congruence. No separate case split over full, empty, or rendezvous channels and no clause-by-clause state correspondence is required. Lemma K.1a remains the local semantic characterization of what a disabled frontier means at a concrete boundary, but is not needed to transfer the graph.  $\square$

#### K.4a Lemma K.2a — $\text{DrainClosedNow}$ Extensionality under Equal KObs

Statement. Let  $K\_post$  and  $K\_ref$  be the equal records of Lemma K.2, and let D be a non-empty process set. Then

$\text{DrainClosedNowRec}(D, K\_post) = \text{DrainClosedNowRec}(D, K\_ref)$ .

Consequently, if the reconstructed post cutpoint is drain-closed for D in the  $\text{DrainClosedNow}$  snapshot sense, the selected operational source cutpoint  $\sigma\_ref$  is drain-closed for D; the converse snapshot implication also holds.

Proof.  $\text{DrainClosedNowRec}(D, K)$  is the pure predicate that no residual offer owned by D and lying outside  $\text{FrontRec}$  is  $\text{ChannelEnabledRec}$ . All of its arguments are fields or reconstruction functions of K, so equality follows by functional congruence from  $K\_post = K\_ref$  and Lemmas J.KObs-Offers/J.KObs-WFG and K.KObs-Dead. This lemma transfers only the cutpoint predicate. It does not transfer, erase, or reconstruct the continuation-level  $\text{Core.Drain}/K.\text{Drain}$  discipline, which remains an operational premise for the executions used by Appendix K and Appendix K-P.  $\square$

#### K.4b Deterministic Serial-Source Dominance and the Primary Policy-S Route

Organization. The serial route normalizes the selected instance, applies  $\text{SDet}$  to select the Policy-S run artifact  $p\_S^{\text{det}}$  and to exclude any incompatible committed history under the same selected functional semantics, and uses separately supplied K.CompleteS evidence to establish that this artifact is maximal, fair, and complete. K.RespCov assigns every static internal blocking site and relevant dynamic class a proof disposition and proves that every actual persistent process-facing serial blocker is covered by a monitor or static bounded-response proof; K.RespClean proves that the complete selected run violates none of those obligations. Lemma K.2b proves that a Policy-L internal deadlock would force a finite K.RespFail witness on  $p\_S^{\text{det}}$ . The  $\text{Dead\_KS}/K.\text{EnvOrd}/K.\text{Done}$  owner-walk remains diagnostic machinery and is not needed for Corollary K.2c.

Definition K.BF (blocking-only fragment). Every wait relevant to the WFG is a persistent blocking Push/Pop at an explicit point-to-point boundary with finite B-faithful capacity; control, payloads, and later operation identities are deterministic functions of certified value/identity provenance, fixed stimulus, and fixed witness choices under K.CV, including Definition K.CVPred and K.CV-WF below; shared arbitration is remodeled as explicit point-to-point subchannels through an arbiter whose choices are deterministic or witness-fixed. Bounded local progress: computation between message operations terminates, so at every point each process either normally completes, reaches its next message operation, or is blocked at one; a process that diverges in pure computation is outside the fragment. Persistence/interruption scope: because every WFG-relevant wait in K.BF is an ordinary blocking

Push/Pop, the exhaustive Core.IC lifecycle applies once its request has issued: the request remains the same asserted, attributed, pending dynamic instance until process-facing commit or an affecting RW2 RESET; a DeclaredTerminal disposition may end the execution while it remains pending. Independent nonterminal cancellation, withdrawal, retry, or reattribution of an outstanding blocking request is not part of the G–K theorem model and requires a separately defined protocol extension outside this theorem stack. Lemma K.2b.Q uses K.BF to identify a permanently non-advancing process with a persistent process-facing frontier that K.RespCov must cover. No missing-supply owner trichotomy is assumed.

Lemma K.BF-Persist (frozen-frontier interruption completeness). Fix a K.BF instance, a candidate component D, and a selected WFG-relevant blocking frontier  $f_P$  whose attributed endpoint request is pending at a Core.LAdm-admissible prefix. Along any finite continuation prefix before a DeclaredTerminal state, if  $f_P$  has not process-facing committed and no RESET affecting P or its explicit boundary has occurred, then the request for  $f_P$  remains asserted, pending, and attributed to the same dynamic operation. In particular, no independent nonterminal cancellation, withdrawal, retry, reattribution, retirement, or replacement can dissolve a K.2b frozen component. Proof. Definition K.BF establishes that  $f_P$  is an ordinary blocking Push/Pop request. Core.IC(1)–(2) exhausts the ordinary blocking-instance lifecycle and gives persistence plus stable attribution until process-facing commit or an affecting RW2 RESET. A DeclaredTerminal disposition is handled separately by K.TerminalDisposition and may end the run with the request still pending. Therefore, absent process-facing commit, affecting RESET, or DeclaredTerminal, the selected request remains the same pending dynamic instance.  $\square$

Definition K.Low and normalized instance Sys\_K. Let  $\Lambda_W(\text{Sys\_ref})$  be the witness-indexed lowering that represents synchronization by explicit epoch-gate channels and every aligned non-NB-CF non-blocking dependency by a fresh blocking guard channel. The default guard has  $B(g)=1$  with ordinary non-fall-through E4 timing; the finite skew after supplier commit is WFG-inert and cannot remain at a drain-closed cutpoint. Define  $\text{Sys}_K \triangleq \Lambda_W(\text{Sys\_ref})$  where lowering is needed, and  $\text{Sys}_K \triangleq \text{Sys\_ref}$  otherwise. All serial-route predicates below are evaluated on  $\text{Sys}_K$ .

Lemma K.Low.C (normalization correspondence) — statement. Projection erasing guard/epoch transitions maps executions of  $\text{Sys}_K$  to executions of the original selected instance with the same external  $\Sigma$  history; witness-consistent executions lift to  $\text{Sys}_K$ ; and K-facing internal wait cutpoints correspond to literal blocking wait cutpoints of  $\text{Sys}_K$ . More precisely, the load-bearing cutpoint clause relates  $\text{KCutReach\_Sys}_K(l)$  to  $\text{KCutReach\_ref}(l)$  under the lowering projection. The local proof obligations are per lowering rule and are collected in Appendix K-P.

Definition K.OpKey (common cross-policy dynamic-operation key and source order). For each selected K.BF theorem instance I, let  $\text{OpKey}_I$  be a theorem-instance universe of typed process-local dynamic-operation occurrence keys under the fixed source code, stimulus, witness, elaboration/lowering, and reset-epoch schema. A key  $\kappa$  records at least its owner P, dynamic source-position/iteration occurrence identifier, operation kind/site, typed boundary/direction schema, and logical reset-epoch slot; the key exists independently of whether a particular admissible execution reaches that occurrence. Each admissible execution  $p$  has a partial realization map  $\text{Realize}_p : \text{OpKey}_I \rightarrow Q_{\text{exec}}, P(p)$ , and every realized execution-local operation  $q$  has a unique  $\text{Key}_p(q)=\kappa$ . Realization preserves kind/boundary/direction and the process-local source order. Define  $\kappa_1 <^{\text{K\_src}} \kappa_2$  on common keys by that source-induced order. The execution-internal relation  $<_{\text{src}}$  of Core.5 remains unchanged; whenever Appendix K compares an operation from Policy S with one from Policy L, notation such as  $g_P \leq_{\text{src}} f_P$  is a readability shorthand for  $\text{Key}(g_P) \leq^{\text{K\_src}} \text{Key}(f_P)$ , never a direct comparison of elements of two execution-local  $Q_{\text{exec}}$  universes. Likewise, cross-policy “same dynamic operation identity” means the same OpKey together with the typed fields/value facts determined by K.CV.

Lemma K.Low.A5 (lowered-to-source A5-L transport). Let  $\pi_K$  be the K.Low projection from  $\text{Sys}_K$  cutpoints to  $\text{Sys\_ref}$  cutpoints, and let  $W_K$  be the pullback of the selected source waiver class W along

$\pi_K$ ; any additional lowering-only waiver must be separately proved not to waive the lift of a non-waived  $\text{Sys\_ref}$  component. Under Lemma K.Low.C, including preservation of Policy-L admissibility on projection/lift and the cutpoint correspondence of part (iii),  $A5L\_Sys\_K(I, W_K)$  implies  $A5L\_Sys\_ref(I, W)$ , i.e. A5-L. Proof by contrapositive. A finite admissible  $\text{Sys\_ref}$  prefix reaching a non-waived  $\text{Dead\_K}$  cutpoint is a witness-consistent execution and therefore lifts by K.Low.C(ii) to an admissible  $\text{Sys\_K}$  prefix. An ordinary  $\text{Sys\_ref}$  internal message wait cycle is an instance of the extended wait-cycle class in K.Low.C(iii), so the lifted drain-closed K-facing state contains the corresponding literal blocking  $\text{Dead\_K}$  component in  $\text{Sys\_K}$ ; waiver pullback preserves non-waiver. This contradicts  $A5L\_Sys\_K(I, W_K)$ . If no lowering is required,  $\pi_K$  and the lemma are the identity case.  $\square$

Example K.CE (why ordinary Policy-S  $\text{Dead\_K}$  avoidance is not a sufficient serial check). Let  $P$  execute  $\text{Pop\_e}$ ;  $\text{Push\_c}(v0)$ , where  $e$  is an exhausted environment input and  $\text{Push\_c}$  is independent of  $\text{Pop\_e}$ . Let downstream processes use  $c$  and then form an internal wait cycle. Under  $\text{Core.ReachPrep}$ , the ordinary straight-line  $\text{Push\_c}$  may become reached and issue in the L1 window after  $\text{Pop\_e}$  has issued but before  $\text{Pop\_e}$  process-facing commits, because its reachability/payload is independent of the unresolved  $\text{Pop\_e}$  result; source order and issue-prefix closure are still preserved. Policy L can therefore reach the downstream internal cycle. Policy S withholds  $\text{Push\_c}$  and instead leaves covered process-facing frontiers pending along a chain ending at exhausted  $e$ . Thus absence of a Policy-S internal cycle alone does not imply A5-L; K.Resp detects the unanswered serial frontier without requiring a closed Policy-S WFG component.

Definition K.RespMon (process-facing bounded-response monitor). For a dynamic blocking source-operation instance  $q$  of process  $P$  in reset epoch  $e$ , let  $x$  be its E-typed process-facing frontier item or certified joint-frontier representative. The monitor triggers at the first selected Policy-S normalized cycle  $t$  in which  $q$  is the active source-minimal pending blocking frontier, its persistent request attribution is established, and  $q$  has not process-facing committed. The response is the commit/return that releases  $q$  from  $P$ 's source execution; proof-internal channel movement does not count unless  $E$  identifies it as that process-facing commit. The monitor carries a finite bound  $B\_resp(P, q, e, \text{state-class})$  measured in selected Policy-S clock cycles. Its age advances with the selected clock even while unrelated processes commit. For a certified multi-frontier source operation, the monitor is attached to operation return and the K-O2 composition lemma proves that permanent joint blockage leaves the monitor pending.

Definition K.RespFail, K.RespCov, and K.RespClean. K.RespFail is a predicate on finite Policy-S prefixes. A non-waived witness exists for a covered monitor when: (i) timeout—the monitor triggers with age zero in normalized cycle  $t$ ; a response through the boundary-action phase of cycle  $t + B\_resp$  is timely, and otherwise timeout is witnessed at the first normalized cutpoint after that phase; (ii) maximal-pending—the selected execution reaches a declared maximal finite terminal state while the monitor remains pending; or (iii) cancelled-pending—an affecting RESET governed by RW2 clears  $q$  before response. A temporarily silent/noncommitting cycle is not maximal:  $\text{Core.CompleteCycle}$  continues the clock and therefore continues K.Resp aging. An affecting reset is a failure unless the exact dynamic class is explicitly waived with the K.Resp coverage-preservation proof. Under the present G–K blocking-request model, case (iii) is the reset-epoch cancellation case; there is no independent nonterminal cancellation/withdrawal disposition for an outstanding blocking request. A modeled abort or end-of-test that ends the selected execution is represented by  $\text{DeclaredTerminal}$  and is therefore covered by maximal-pending when the monitor remains pending. K.RespCov is default-total over a declared coverage universe of static internal process-facing blocking sites and relevant dynamic classes. Each class has one disposition: a literal monitor, a validated static bounded-response proof, or an unreachability proof. Completeness requires every dynamically reachable instance to belong to a declared class; once reached, an unreachability disposition cannot apply. The user or checker need not first decide whether a site can become the persistent serial blocker; that fact motivates coverage but is

not its quantification domain. A production-count argument alone is not an exemption.  $K.RespClean$  states that no finite prefix of the complete  $p\_S^{\wedge det}$  witnesses a non-waived  $K.RespFail$ . Environment and waiver rule for  $K.RespCov$ . Every internal process-facing blocking site is covered by default. For  $StandardEnv/B1-avail$ , a source-earlier environment operation that committed in the frozen liberal prefix has the same consumption ordinal available to Policy S and, under persistent request,  $B2/B3$ , and  $Core.IssueFair$ , cannot be the permanent serial blocker; that theorem may discharge coverage for the corresponding dynamic instance. For a timed/reactive or otherwise nondefault environment,  $K.EnvMF$  first requires every relevant liberal  $Dead\_K$  component to be certified single-internal-frontier. Environment-facing operations that may block the selected Policy-S execution must then be monitored, proved available, or placed outside the serial route. A per-site monitor, availability proof, no-bypass argument, or waiver does not admit a timed/reactive multi-frontier liberal component; that combination is outside  $K.2b/K.2c$ . A response waiver is valid only if every non-waived liberal  $Dead\_K$  component extracted by  $K.2b$  maps to at least one non-waived covered serial monitor.

Complete-run and bound rule.  $K.CompleteS$  remains the semantic proof that  $p\_S^{\wedge det}$  is the complete selected artifact.  $K.RespClean$  is a separate property checked over it. For a terminating artifact, the checker validates every monitor through the declared maximal terminal state; absence of current commits in an otherwise nonterminal cycle does not make the artifact complete, because  $Core.CompleteCycle$  continues the selected clock and monitor ages. For an infinite artifact, a finite trace is insufficient: an invariant, exhaustive model check, recurrence/lasso proof, or equivalent certificate must establish that monitor ages never exceed their bounds. For theorem soundness, the bound need only be finite and compliance must be proved over the complete run; a separately justified worst-case-latency proof is required only to classify a timeout as a design-specification violation rather than as a failed certificate attempt. No  $K.Resp$  bound is transferred to RTL by Theorem K.1.

Diagnostic Definition  $K.Cone$ , Condition  $K.EnvOrd$ , and Condition  $K.Done$ .  $K.Cone$  and its may-follow closure may be used to explain environment-rooted or completion-rooted response failures and to prove that particular eager bypasses are impossible.  $K.EnvOrd$  retains the transitive, loop-aware dynamic no-bypass meaning described below.  $K.Done$  remains only a normal-completion audit: it can prove  $DoneStop$  unreachable by showing that every normally terminating endpoint produces or consumes enough operations before  $FINISH$ , or by proving that no process normally completes. The latter does not establish continuing service on every endpoint and does not address the H2 perpetual-under-supply case. Consequently  $K.EnvOrd/K.Done$  may simplify diagnostic  $Dead\_KS$  reports but never permit omission of  $K.Resp$  coverage and never discharge A5-L by themselves.

Example K.CE-2 (transitive starvation—diagnostic value and uniform  $K.Resp$  detection). Let  $e$  be an environment input channel with empty stimulus, and let  $c\_E, c\_V, a, b$  be internal point-to-point channels.  $E$ :  $Pop\_e$ ;  $Push\_cE$ .  $W$ :  $Pop\_cE$ ;  $Push\_cV(v0)$ , with  $v0$  independent of the  $Pop\_cE$  payload.  $V$ :  $Pop\_cV$ ;  $Push\_a$ ;  $Pop\_b$ .  $P2$ :  $Push\_b$ ;  $Pop\_a$ . The straight-line dependency-independent successors  $Push\_cE$  and  $Push\_cV$  use the same  $Core.ReachPrep$  permission as Example K.CE: they may become reached/issued in L1 before the earlier blockers commit while source order remains intact. An admissible Policy-L schedule can therefore feed the downstream internal cycle  $\{V, P2\}$ , so A5-L fails. Policy S withholds the feeding operations and leaves the relevant downstream/internal serial frontiers pending. Total  $K.RespCov$  therefore produces a finite timeout or maximal-pending failure without needing to walk the chain to  $EnvStop$ .  $K.Cone/K.EnvOrd$  may still diagnose the transitive environment root and explain which bypass admitted the liberal cycle, but that diagnosis is not part of the soundness proof.

Example K.CE-3 (loop-carried bypass—diagnostic value and uniform  $K.Resp$  detection). Let  $e$  and  $c\_E$  be as in Example K.CE-2, with  $E$ :  $Pop\_e$ ;  $Push\_cE(f(e))$  fully dependent, and let  $W$  be the pipelined loop  $W$ :  $loop \{ Push\_cV(v0); Pop\_cE \}$ , with  $v0$  independent of the  $Pop\_cE$  payload. Policy L can issue iteration  $i+1$ 's  $Push\_cV$  while iteration  $i$ 's  $Pop\_cE$  remains pending and feed the downstream cycle; Policy S

withholds the crossing and leaves a covered process-facing serial frontier unanswered. K.Resp therefore rejects the selected run by a finite response failure. The may-follow form of K.EnvOrd remains useful diagnostically because it identifies the loop-carried crossing that a merely textual source-later analysis misses.

Example K.CE-4 (completed-producer starvation—diagnostic value and uniform K.Resp detection). Let Z: Push\_c(w); end produce exactly one item on internal channel c and normally complete, and let Y: Pop\_c; Pop\_c; Push\_d(v0) demand two, with v0 independent; V: Pop\_d; Push\_a; Pop\_b and P2: Push\_b; Pop\_a. Policy L may issue Push\_d before Y's second Pop\_c settles and thereby reach the downstream internal cycle. Under Policy S, Push\_d is withheld and Y's second Pop\_c remains unanswered after Z completes. K.RespCov monitors that process-facing Pop and therefore yields a finite timeout or maximal-pending failure. The diagnostic WFG+ may classify the cause as DoneStop\_c, but K.2b no longer depends on completion being an exhaustive owner-walk case.

Example K.CE-5 / H2 (perpetual under-supplying owner). Let Z: Push\_c(1); while (1) { Push\_d1(0); Pop\_d2; }, W: while (1) { Pop\_d1; Push\_d2(0); }, X: Pop\_c; Pop\_c; Push\_g(v0) with v0 independent of the Pops, V: Pop\_g; Push\_a; Pop\_b, and P2: Push\_b; Pop\_a. Policy L may issue Push\_g before Pop\_c#2 settles and then reach the internal cycle {V,P2}. Under Policy S, Push\_g is withheld; Z and W may continue their d1/d2 ping-pong forever while X and the downstream chain remain permanently unanswered. No closed Dead\_KS component need exist because the owner Z stays live. K.RespCov nevertheless monitors the process-facing serial blockers, and their finite bounds are violated on the complete run. This example is the reason the owner trichotomy and terminal taxonomy are not soundness-critical in the revised route.

Definition K.JointFreeze (elaborated multi-frontier no-release condition). Fix a candidate Dead\_K component D at the K.2b.E activation cutpoint  $\sigma_L$ , the selected internal frozen operation  $f_P$  for each  $P \in D$ , and the fixed K.DomResetScope DomEpoch. This condition is required only on the StandardEnv/B1-avail branch when some process in D has more than one member of  $\text{Front}_P(\sigma_L)$ .

K.JointFreeze<sub>I</sub>(D,  $\sigma_L$ , { $f_P$ }) holds when the certificate proves the following over every finite Core.LAdm-legal continuation in the same DomEpoch while the selected  $f_P$  remain pending. (JF1) Cross-process launch/issue closure. For every  $Q \in D$  and every not-yet-issued blocking operation q of Q whose source-evaluation launch, new issue, attributed request, or elaboration-descendant request could enable, process-facing commit, remove, or otherwise release any selected frozen  $f_P$ , there exists a still-pending frontier item  $x \in \text{Front}_Q$  in the frozen component with  $x \leq_Q q$  and  $x \neq q$ . Hence, while  $\text{Front}_Q$  remains nonempty and fully channel-disabled so that Core.16 remains active, q may neither be newly launched past the blocked point nor become IssueEnabled. The certificate shall treat incomparable  $\text{Sys}_B^{\text{elab}}$  frontier items explicitly; incomparability is not equivalent to “source-later.” (JF2) Non-issue preservation. Every phase-legal action still permitted while D is frozen—including work outside D and finite drain/elaboration work that was already issued at  $\sigma_L$ —preserves the selected  $f_P$  and cannot create a new complementary request, grant, guard/join condition, or internal availability fact that releases D, except through a process-facing commit already excluded by the Dead\_K/drain hypotheses. JF2 may cite Core.Drain/K.Drain, point-to-point ownership, B-faithfulness, and the concrete elaboration semantics. For an ordinary non-elaborated or otherwise certified singleton frontier, no separate K.JointFreeze certificate is required; Lemma K.JointFreeze-Ord below supplies the corresponding no-new-release result. K.JointFreeze is a serial-route applicability condition preserving the selected frozen component; it is separate from the Core.16 comparison-advance condition consumed by K.2b.P0/K.2b.R. Lemma K.JointFreeze-Ord (ordinary/singleton no-release). Let D be a candidate Dead\_K component at the K.2b.E activation cutpoint  $\sigma_L$ . Suppose every process  $Q \in D$  has an active message frontier consisting solely of its selected frozen operation  $f_Q$ , and either that frontier is ordinary/non-elaborated or its singleton certificate also supplies the same relevant total-order/source-comparability property.

Assume further that  $D$  is component-closed and drain-closed, the A6 point-to-point evaluated-boundary ownership premise holds,  $\text{Core.16}$  is active on those frontiers, and  $\text{Core.Drain/K.Drain}$  holds. Then, while the selected  $f_Q$  remain pending in the fixed  $\text{DomEpoch}$ : (i) no not-yet-issued blocking operation of a member of  $D$  can be newly launched or issued so as to release  $D$ ; (ii) no phase-legal action owned by a process  $R \notin D$  can create the complementary request or boundary-availability fact needed to release a selected internal  $f_Q$ ; and (iii) every phase-legal already-issued drain action preserves the frozen component. Hence the full JF1/JF2 no-release conclusion used by Lemma K.2b.E holds without a separate  $\text{K.JointFreeze}$  certificate. Proof. Fix  $Q \in D$  and a distinct not-yet-issued blocking operation  $q$  of  $Q$  whose launch or issue could release a selected frozen frontier. Because  $\text{Front}_Q$  is the singleton  $\{f_Q\}$ , source-order prefix closure/E3 and the fact that  $f_Q$  is already the pending frontier exclude  $q$  being source-earlier than  $f_Q$ ; under the assumed ordinary/total-order comparability  $q$  is therefore source-later than  $f_Q$ . While  $f_Q$  remains channel-disabled,  $\text{Core.16(2)}$  forbids both the broader source-evaluation launch of  $q$  past the blocked point and its message  $\text{Issue}$ . Next fix  $R \notin D$  and suppose a phase-legal action of  $R$  could release some selected internal  $f_Q$ . The selected  $f_Q$  is evaluated at a point-to-point explicit boundary, so A6 gives it one complementary endpoint owner. Any new  $R$ -owned request or boundary-availability fact capable of making that blocked frontier  $\text{ChannelEnabled}$  would therefore make  $R$  that complement owner and induce an internal WFG edge  $Q \rightarrow R$ . Component closure of  $D$  forbids such an outgoing internal edge, so  $R$  must in fact belong to  $D$ , a contradiction; an  $R$ -owned action disjoint from that unique evaluated boundary cannot change  $f_Q$ 's E4 enablement. Finally, work already issued before the activation point is confined to the finite non-replenishing drain and, by component closure together with  $\text{Core.Drain/K.Drain}$  and nonwithdrawal, cannot create a new request or availability fact that releases or replenishes  $D$ . Thus no permitted continuation action dissolves the selected frozen component.  $\square$

**Definition K.CompareAdvanceLegal** (dominance-comparison  $\text{Core.16}$  advance compatibility). Fix a candidate component  $D$  at  $\sigma_L$ , its selected  $\{f_P\}$ , the certificate-bound  $\text{K.DomResetScope/DomEpoch}$ , and the K.2b.P least-missing/first-overtake comparison.  $\text{K.CompareAdvanceLegal}_I(D, \sigma_L, \{f_P\})$  holds when the following condition is valid for every process  $P$  in  $\text{K.DomResetScope}$  and every common operation key  $\kappa$  consumed by Lemma K.2b.P0 or Lemma K.2b.R on a finite comparison tuple  $(\pi_S, \pi_L)$  while the selected  $f_P$  remain pending and  $\text{DomEpoch}$  is fixed. Suppose the strengthened comparison invariant has already matched the source-earlier keys required by Policy  $S$ , the endpoint transfers earlier than  $\kappa$  in the fixed dominance order, and the  $\text{K.CVPred}$  comparison-availability step has supplied every required predecessor fact while independently preserving the frozen component. Then either  $\text{Core.16}$  is inactive for  $P$  at the applicable settled frontier, or no active  $x \in \text{Front}_P$  at that comparison prefix satisfies  $\text{Key}(\text{MsgOf}_P(x)) <^{\wedge K\_src} \kappa$ . Consequently  $\text{Core.16}$  forbids neither (a) the finite dependency-independent  $\text{Core.ReachPrep}$ /source-evaluation launch needed by Lemma K.2b.P0 to reach  $q_L := \text{Realize}(\kappa)$ , nor (b), once  $q_L$  is reached and unissued, its subsequent endpoint-request  $\text{Issue/assertion}$  used by Lemma K.2b.R. This condition is applied before the P0  $\text{Core.ReachPrep}$  extension and is retained through the later  $R$  request assertion. The same condition therefore discharges both distinct  $\text{Core.16(2)}$  obligations; issue legality is not inferred merely from successful reach, and reach legality is not inferred from later  $\text{IssueEnabled}$ . This condition applies whether  $P \in D$  or  $P \notin D$ . For an ordinary non-elaborated/singleton frontier it is theorem-derived by Lemma  $\text{K.CompareAdvanceLegal-Ord}$  below. If an explicit  $\text{Sys\_B}^{\wedge \text{elab}}$  package can give a K.2b.P0/K.2b.R-consumed owner a multi-frontier on a covered comparison prefix, the flow shall instead provide checked elaboration-specific evidence for this condition, for example by identifying  $\kappa$  with the represented frontier operation or by proving an appropriate joint-frontier order relation; incomparability alone is not evidence. Consistency with  $\text{K.JointFreeze}$  is mandatory: no same prefix/key may simultaneously be classified by JF1 as a releasing launch/issue that  $\text{Core.16}$  must suppress and by  $\text{K.CompareAdvanceLegal}$  as a comparison advance that

must remain legal. If both classifications apply, no joint serial-route certificate exists for that candidate and the candidate requires another A5-L discharge.

Lemma K.CompareAdvanceLegal-Ord (ordinary/singleton comparison advance legality). On a K.2b.P least-missing/first-overtake comparison tuple satisfying the strengthened transfer/provenance invariant, let  $\kappa$  be a blocking operation key owned by process P for which the selected Policy-S prefix requires the same source-earlier blocking-transfer prefix before  $\kappa$  may issue, and suppose P's applicable active message frontier is ordinary non-elaborated or otherwise certified singleton with the same relevant total-order/source-comparability property. Then the no-active-key-earlier conclusion used by K.CompareAdvanceLegal holds for  $\kappa$ , regardless of whether P belongs to D. In particular, the lemma discharges every ordinary/singleton K.2b.P0/K.2b.R comparison key and may be cited pointwise by the K.CVPred recursion for a lower-ranked blocking OpFact predecessor once that predecessor's own lower-ranked facts are available. Proof. If Core.16 is inactive, the claim is immediate. Otherwise suppose the unique active frontier item  $x$  were key-earlier than  $\kappa$ . In the ordinary case Core.5a/Core.OrdFrontSingleton, and in the certified-singleton extension its explicit total-order/source-comparability certificate, make  $x$ 's owning blocking operation source-earlier than  $\kappa$ . Policy S therefore requires that operation's process-facing transfer before  $\kappa$  may issue. By the least-missing/first-overtake comparison invariant, that earlier transfer is already present on the liberal side; E3/source-order prefix closure together with Core.IC then prevents the same earlier operation from remaining an uncommitted active frontier item. Contradiction. Hence no active Core.16 frontier item is key-earlier than  $\kappa$ . By Core.16(2), the finite dependency-independent preparation/evaluation that reaches Realize( $\kappa$ ) is not prohibited as a source-later launch, and after reach the request assertion is not prohibited as a source-later Issue. Thus the same ordinary-case argument discharges both halves of the comparison-advance condition whenever they are needed.  $\square$

Definition K.EnvMF (Policy-S environment/multi-frontier applicability). For every application of Lemma K.2b.E to a component D at cutpoint  $\sigma\_L$ , one of the following shall hold: (1) the selected instance uses StandardEnv/B1-avail and the default always-ready output environment; if any  $P \in D$  has a certified multi-frontier, K.JointFreeze\_I(D,  $\sigma\_L$ , {f\_P}) shall additionally hold for that component; or (2) the selected instance uses a timed/reactive or otherwise nondefault environment, and every  $P \in D$  has a certified single frontier at  $\sigma\_L$  consisting solely of the selected internal blocking item f\_P. Under clause (2), no environment-facing co-frontier, second internal frontier item, elaborated join/bundle alternative, or other multi-frontier item may coexist in Front\_P( $\sigma\_L$ ). Timed/reactive/nondefault multi-frontier components are outside Lemma K.2b.E, Lemma K.2b, Corollary K.2c, K.PSF, and the Policy-S form of Theorem K.1. StandardEnv multi-frontier components remain inside the route only when the explicit K.JointFreeze no-release obligation is discharged; deferring that fact solely to later mechanization is insufficient. K.EnvMF governs preservation of D; K.2b.P0/K.2b.R comparison-advance legality for all consumed owners is governed separately by K.CompareAdvanceLegal.

Lemma K.EnvX-Auto (environment co-frontiers). Under B1-avail and the default always-ready output environment, every environment-facing frontier item held by a blocked member of a Dead\_K component is stimulus-exhausted. Proof. Blockedness requires every frontier item to be channel-disabled. A remaining input item or ready output environment would enable the environment-facing item. Hence no separate environment-freeze premise is needed under StandardEnv. Scope emphasis (normative): B1-avail is a genuine environment-model restriction, not a bookkeeping convention—it asserts that stimulus availability is consumption-ordered and untimed. A testbench that delivers stimulus at declared clock times, or a nondefault output environment that may release back-pressure later, is outside B1-avail. For such an environment the Policy-S route is supported only through K.EnvMF's certified single-internal-frontier clause. Timed/reactive/nondefault multi-frontier cases require a non-serial A5-L proof or another theorem and are not admitted by a substitute exhaustion/no-rescue certificate in this revision.

Definition K.InterruptionGate (shared nested comparison-interruption schema). Fix a theorem instance  $I$ , a candidate liberal component  $D$ , its  $\text{Core.LAdm}$ -admissible counterexample prefix  $p_L$  ending at  $\sigma_L$ , and the selected frozen frontier operation  $f_P$  for each  $P \in D$ . A gate specification  $G$  supplies two gate-specific predicates. First,  $\text{Interrupt\_G}(I, D, \{f_P\}, \eta)$  is a prefix/state-local interruption predicate evaluated only on finite  $\text{Core.LAdm}$ -admissible extension prefixes  $\eta$  of  $p_L$  that are consistent with the fixed instance choices and throughout which every selected  $f_P$  remains pending; it may inspect the fixed contract,  $\eta$ 's final state, and the legality of the next proof-relevant transition/unit (with any executed real-cycle occurrence recorded in the enclosing  $\text{Core.CycleResult}$ ), but it shall not depend on a continuation already constructed by Lemma K.2b.E. Second,  $\text{DirectFail\_G}$  supplies an independently certified mapping  $P \in D$ ,  $f_P$ , a process-facing Policy-S serial frontier  $x_{P^S}$ , and  $g_P := \text{Op}(x_{P^S})$  with  $g_P \leq_{\text{src}} f_P$ , together with a finite selected Policy-S prefix on which a non-waived  $K.\text{Resp}$  obligation for  $x_{P^S}$  has already triggered and remains pending, plus gate-specific evidence that the corresponding interruption/disposition occurs and therefore witnesses  $K.\text{RespFail}$ . The direct mapping and disposition shall be established without K.2b.R, K.2b.P, or K.2b.Q Step 1. A  $K.\text{InterruptionGate\_G}$  discharge uses one of two branch forms: Continue, when every such frozen prefix  $\eta$  satisfies  $\neg \text{Interrupt\_G}(I, D, \{f_P\}, \eta)$ ; and DirectFail, when  $\text{DirectFail\_G}$  is established. If neither branch is established, the candidate is outside the Policy-S reduction and requires another A5-L discharge. Multiple gate instances are not flattened into one unordered enumeration: the theorem states their evaluation order explicitly. For Lemma K.2b the required order is  $K.\text{ResetEpoch}$  first and, only on  $K.\text{ResetEpoch}$ .Continue,  $K.\text{TerminalDisposition}$  second. Exhaustiveness for  $K.\text{BF}$ . For the  $K.\text{BF}$  serial-reduction fragment,  $\text{Core.IC}$  and Lemma K.BF-Persist establish that, apart from process-facing commit, the only modeled interruptions that can terminate or invalidate persistence of a selected frozen blocking instance are an affecting RW2 RESET or a DeclaredTerminal disposition ending the execution while the request remains pending. Accordingly, Reset and Terminal are the complete interruption-gate instantiations consumed by Lemma K.2b. Adding another gate is a theorem-model extension and requires an explicit operational definition plus re-proof of K.BF-Persist and the affected K.2b obligations; it is not a routine additional instantiation of this schema.

Definition K.DomResetScope (dominance reset scope). For a candidate  $D$ ,  $p_L$ ,  $\{f_P\}$ , fixed  $\text{Sys\_K}$  elaboration, fixed source/dynamic operation universe, and selected Policy-S comparison artifact, a  $K.\text{DomResetScope}$  certificate supplies finite sets  $S_{p^{\text{dom}}}$  of processes and  $S_{c^{\text{dom}}}$  of explicit abstract boundaries. The certificate is valid iff: (D1)  $D \subseteq S_{p^{\text{dom}}}$ ; (D2) for every  $c \in S_{c^{\text{dom}}}$ , both endpoint processes of  $c$  belong to  $S_{p^{\text{dom}}}$ ; (D3) every process-owned endpoint operation and every explicit-boundary transfer identity consumed by the K.2b.P0/K.2b.R/K.2b.P comparison for the fixed selected Policy-S artifact is owned by a process in  $S_{p^{\text{dom}}}$  and, when boundary-specific, by a boundary in  $S_{c^{\text{dom}}}$ ; and (D4)  $S_{p^{\text{dom}}}$  and  $S_{c^{\text{dom}}}$  are determined from the fixed elaboration, source/dynamic operation universe, and selected Policy-S comparison artifact and do not depend on the later liberal continuation constructed by Lemma K.2b.E. Define  $K.\text{DomResetScope\_I}(D, \{f_P\}) := (S_{p^{\text{dom}}}, S_{c^{\text{dom}}})$  for any validated certificate. Minimality is not required; a larger conservative scope is permitted. For any comparison prefix  $\eta$ ,  $\text{DomEpoch\_I}(\eta)$  is the vector of reset-epoch identifiers of the processes and explicit boundaries in this validated scope. A RESET is dominance-affecting exactly when its RW scope intersects this dominance scope. RW5 permits RESET occurrences disjoint from  $K.\text{DomResetScope}$ ; such resets may change epochs outside the dominance scope but cannot change any dynamic identity, boundary state, or request fact consumed by K.2b.P0/R/P.  $K.\text{ResetEpoch}$ .Continue therefore keeps  $\text{DomEpoch}$  fixed without requiring a global no-reset execution.

Definition K.ResetEpoch (reset instantiation of K.InterruptionGate). For the candidate  $D$ ,  $p_L$ , and  $\{f_P\}$ , let  $\text{Interrupt\_Reset}(I, D, \{f_P\}, \eta)$  hold exactly when the fixed reset contract permits, as the next proof-relevant transition/unit from  $\eta$ 's final state, a RESET whose scope intersects  $K.\text{DomResetScope\_I}(D, \{f_P\})$ . When that RESET executes within a real cycle, the occurrence is recorded

in the enclosing legal  $\text{Core.CycleResult.resetUnits}$ . Let  $\text{DirectFail\_Reset}$  be the independently mapped reset witness:  $P \in D$ , selected liberal  $f_P$ , process-facing Policy-S serial frontier  $x_{P^S}$  in the same pre-reset endpoint epoch,  $g_P := \text{Op}(x_{P^S})$  with  $g_P \leq_{\text{src}} f_P$ , an already-triggered non-waived  $K.\text{Resp}$  obligation that remains pending, and the first matched RESET affecting that monitored frontier whose emitted RESET unit causes RW2 to clear that exact dynamic instance, yielding cancelled-pending  $K.\text{RespFail}$ . Define  $K.\text{ResetEpoch} := K.\text{InterruptionGate\_Reset}$ . Thus  $K.\text{ResetEpoch.Continue}$  is the dominance-scope-reset-free comparison branch and  $K.\text{ResetEpoch.DirectFail}$  is the matched-reset direct-failure branch. The Continue predicate is tested before RW2 clearing can occur and is contract-wise rather than inferred from absence of an observed reset. A reset intersecting  $K.\text{DomResetScope}$  but not covered by  $\text{DirectFail\_Reset}$  makes the candidate ineligible for the Policy-S reduction and requires another A5-L discharge; a reset disjoint from  $K.\text{DomResetScope}$  is permitted and framed by RW5. All request-identity, endpoint-ordinal, nonwithdrawal, and persistence arguments on the Continue branch are scoped by the fixed  $\text{DomEpoch}$  rather than by a nonexistent single global reset epoch.

Definition  $K.\text{TerminalDisposition}$  (terminal instantiation of  $K.\text{InterruptionGate}$ ). This gate is instantiated only after  $K.\text{ResetEpoch.Continue}$  has been established for the candidate  $D$ . For the same  $p_L$  and  $\{f_P\}$ , let  $\text{Interrupt\_Terminal}(I, D, \{f_P\}, \eta)$  hold exactly when  $\eta$ 's final state satisfies  $\text{DeclaredTerminal\_Sys\_K}, I$ . Ordinary  $\text{Core.SilentCycle}$  progression does not satisfy this predicate. Let  $\text{DirectFail\_Terminal}$  be the independently mapped terminal witness:  $P \in D$ , selected liberal  $f_P$ , process-facing Policy-S serial frontier  $x_{P^S}$ ,  $g_P := \text{Op}(x_{P^S})$  with  $g_P \leq_{\text{src}} f_P$ , a finite selected Policy-S prefix on which a non-waived  $K.\text{Resp}$  obligation for  $x_{P^S}$  has already triggered and remains pending, and independently checked evidence that  $p_{S^{\text{det}}}$  reaches the specified declared maximal terminal state while that same monitor remains pending, yielding maximal-pending  $K.\text{RespFail}$ . Define  $K.\text{TerminalDisposition} := K.\text{InterruptionGate\_Terminal}$ . Thus  $K.\text{TerminalDisposition.Continue}$  is the terminal-free comparison branch and  $K.\text{TerminalDisposition.DirectFail}$  is the declared-terminal direct-failure branch. Modeled abort, watchdog, cycle-count, or end-of-test dispositions are handled here only when they satisfy  $\text{DeclaredTerminal}$ ; independent nonterminal cancellation/withdrawal is outside the present G–K model. The nesting is normative:  $K.\text{TerminalDisposition}$  is not evaluated on  $K.\text{ResetEpoch.DirectFail}$ .

Uniform instance-level gate discharge (normative sufficient condition). The candidate-relative  $K.\text{DomResetScope}/K.\text{ResetEpoch}/K.\text{TerminalDisposition}$  obligations above need not be enumerated one  $\text{Dead\_K}$  component at a time when the flow proves the following stronger facts for the fixed selected instance and selected Policy-S comparison artifact. Supply one finite conservative dominance scope  $U = (U_p, U_c)$ , fixed from the selected elaboration and source/dynamic operation universe before any liberal counterexample is chosen, such that: (U1)  $U$  contains every process that can participate in an Appendix-K blocking component and every process/boundary identity that  $K.2b.P0/K.2b.R/K.2b.P$  can consume; both endpoint processes of every boundary in  $U_c$  lie in  $U_p$ ; and  $U$  is independent of the later  $K.2b.E$  continuation. Thus  $U$  satisfies  $K.\text{DomResetScope}$  conditions D1–D4 whenever used for a candidate. The scope may conservatively contain all  $\text{Sys\_K}$  processes and all relevant explicit abstract message-passing boundaries; minimality is not required. (U2) No RESET permitted by the fixed reset contract has RW scope intersecting  $U$ . (U3) Either  $\text{DeclaredTerminal}$  is unreachable during the applicable comparison phase, or every reachable  $\text{DeclaredTerminal}$  source state has no pending  $K.\text{BF}$  WFG-relevant blocking frontier owned by  $U_p$  or represented at  $U_c$ . Under U1–U3, use  $U$  as  $K.\text{DomResetScope}$  for every candidate: U2 proves  $K.\text{ResetEpoch.Continue}$  uniformly, and because each frozen candidate retains a selected  $K.\text{BF}$  blocking frontier in  $U$ , U3 proves  $K.\text{TerminalDisposition.Continue}$  uniformly on the reset-Continue branch. A clean CF-S/ $K.\text{PSF}$  discharge may therefore cite this one instance-level record instead of enumerating candidate-specific gate records or  $\text{DirectFail}$  mappings. This is a sufficient, not necessary, discharge; if any clause cannot be proved, the existing candidate-specific nested-gate schema remains applicable without change.

Lemma K.2b.E (fair complete frozen extension). Let  $p_L \in \text{Sys}_K.\text{Pref}_L^{\text{adm}}(I)$  end at a source K-facing cutpoint  $\sigma_L \in \text{KCutReach}_{\text{Sys}_K}(I)$  satisfying  $\text{Dead}_K^{\text{AW}}$ , let  $D$  be a selected  $\text{Dead}_K$  component witnessing  $\text{Dead}_K^{\text{AW}}$  at  $\sigma_L$ , and let  $f_P$  be the component frontier operation selected for each  $P \in D$ . By Definition K.Dead,  $D$  is drain-closed at  $\sigma_L$ ; this fact is extracted from the deadlock hypothesis and is not an additional cutpoint premise. Assume  $\text{K.EnvMF}(I, D, \sigma_L)$ , with Lemma K.JointFreeze-Ord discharging every ordinary/singleton component and K.JointFreeze required on every admitted StandardEnv multi-frontier component, K.ResetEpoch.Continue, and—on that reset-free branch—K.TerminalDisposition.Continue for this application. Under Definition Core.LAdm, Core.CycleClosure, Core.Drain/K.Drain, B5, a Core.FairPick\_I, Policy L witness that constructively discharges the existing Core.IssueFair/B2/B3 behavioral obligations for the particular continuation, nonwithdrawal,  $\text{K}\epsilon$ , K.BF, point-to-point ownership, and the same fixed stimulus, witness, environment, reset, arbitration, termination contract, and normalization choices, there exists an infinite fair complete continuation  $\bar{p}_L \in \text{Exec}_L^{\text{adm}}(I)$  extending  $p_L$  in which every  $f_P$  remains pending while the  $\text{K.DomResetScope}$  DomEpoch remains fixed. Under StandardEnv, Lemma K.EnvX-Auto prevents later environment rescue of any co-frontier; for a StandardEnv multi-frontier component, K.JointFreeze additionally prevents a newly issued incomparable/internal operation or permitted elaboration action from releasing the selected frozen frontiers. Under a timed/reactive or otherwise nondefault environment, K.EnvMF excludes all co-frontiers by requiring each  $P \in D$  to have only the selected internal frontier. The continuation may schedule all enabled work outside the frozen component and may drain already-issued downstream work, but Lemma K.JointFreeze-Ord or K.JointFreeze, as applicable, together with Core.Drain/K.Drain prevent any newly introduced work from releasing the frozen frontiers. A disabled frozen request is not ChannelEnabled and therefore creates no B2 obligation to commit; Lemma K.BF-Persist applies the exhaustive Core.IC blocking lifecycle to that WFG-relevant blocker; and an operation excluded by the applicable K.JointFreeze-Ord/K.JointFreeze discharge is not IssueEnabled and therefore creates no Core.IssueFair obligation to issue. This lemma supplies only the reset-free, terminal-free continuation used by the dominance comparison. K.ResetEpoch.DirectFail and K.TerminalDisposition.DirectFail are direct K.RespFail branches of Lemma K.2b and do not invoke this extension.

Lemma K.2b (deterministic serial-source dominance with bounded-response extraction). Fix  $\text{Sys}_K$ , stimulus, capacities, elaboration, environment, witness, concrete Policy-S simulator/scheduler and arbitration semantics, Policy-S issue semantics,  $\epsilon$ -normalization, the reset and termination/end-of-test contracts, the K.InterruptionGate schema, and K.Resp. Assume SDet, A1-A4, B5, and the existing Policy-L Core.IssueFair/B2/B3 obligations in the constructive Core.FairPick form required to build the K.2b.E continuation; assume also A6-A11, K.CV, K.BF, point-to-point ownership, nonwithdrawal,  $\text{K}\epsilon$ , K.Drain, Core.CycleClosure, K.EnvMF, K.CompareAdvanceLegal for every candidate comparison (with ordinary/singleton cases discharged by Lemma K.CompareAdvanceLegal-Ord and explicit  $\text{Sys}_B^{\text{elab}}$  multi-frontier cases checked for every K.2b.P0/K.2b.R-consumed owner), K.ResetEpoch for every candidate component, K.TerminalDisposition on every K.ResetEpoch.Continue candidate, K.RespCov, the same fixed instance choices for the liberal prefix and its applicable continuation, and the applicable K.Low correspondence. Let  $p_{S^{\text{det}}}$  be the Policy-S execution selected by SDet, with separate K.CompleteS evidence establishing maximality, fairness, and completeness. If some Policy-L prefix  $p_L$  in  $\text{Sys}_K$ 's  $\text{Pref}_L^{\text{adm}}(I)$  domain reaches a source K-facing cutpoint whose  $K_L$  satisfies  $\text{DeadKRec}^{\text{AW}}$ , then  $p_{S^{\text{det}}}$  has a finite prefix containing a non-waived K.RespFail witness. The proof follows the gate order, not a flat case split. First evaluate K.ResetEpoch. Its DirectFail branch supplies the independently mapped pending Policy-S monitor and matched affecting RESET, so RW2 yields cancelled-pending directly. Only on K.ResetEpoch.Continue evaluate K.TerminalDisposition. Its DirectFail branch independently supplies the mapped pending monitor and DeclaredTerminal occurrence, yielding maximal-pending directly. Only when both nested gates take Continue does Lemma K.2b.E construct the

infinite fair frozen continuation; Core.FairPick/Core.FairCycleDovetail prove that the constructed continuation satisfies Core.IssueFair and B2/B3, K.2b.R and K.2b.P establish deterministic serial dominance on it, and K.2b.Q extracts the finite timeout/maximal-pending witness from the complete selected Policy-S artifact. If either required gate instance satisfies neither shared-schema branch, the candidate is outside the Policy-S reduction. The direct-failure evidence is independent of K.2b.R/P/Q Step 1, so no gate branch is proved by the dominance continuation it is used to select.

Proof status. Appendix K-P gives explicit paper proofs of K.Low.C, K.Low.A5, K.2b.P0, K.2b.P1, K.2b.R, K.2b.P, K.2b.E, and K.2b.Q and assembles them into Lemma K.2b. The remaining work is mechanization and tool/flow qualification rather than an omitted paper argument. The prerequisite mechanization now includes the per-lowering-rule cases of K.Low.C/K.Low.A5; the common K.OpKey realization/source-order layer; the K.2b.P0 occurrence/reachability lemma; the K.2b.P1 determinacy proof over Definition K.CVPred/K.CV-WF; and the constructive fair-cycle machinery of Core.FairPick/Core.FairCycleDovetail used by K.2b.E. K-O0 covers K.OpKey plus the K.CV fact-key/provenance certificate, comparison-availability closure, well-foundedness/rank proof, and K.2b.P1; K-O1 covers Core.IssueFair together with the Core.FairPick issue-selection/starvation-freedom discharge; K-O2 covers Lemma K.JointFreeze-Ord, mechanization of K.JointFreeze JF1/JF2 including launch/issue closure, and process-facing monitor-trigger composition; K-O3 covers deterministic serial progress dominance with the joint earlier-transfer/K.CV-predecessor invariant on the reset-free, terminal-free infinite-fair branch; K-O4 covers K.CompareAdvanceLegal, Lemma K.CompareAdvanceLegal-Ord, their consistency with K.JointFreeze, K.2b.P0/K.2b.R, and the Continue predicates of the two K.InterruptionGate instantiations; K-O5 covers default-total K.Resp coverage and waiver preservation; K-O6 validates the gate-independent DirectFail payload shape and its reset-specific cancelled-pending and terminal-specific maximal-pending instantiations; and K-O7 mechanizes the shared K.InterruptionGate anti-circularity/prefix-local rule, the nested ResetEpoch.Continue  $\rightarrow$  TerminalDisposition.Continue ordering, Core.CycleChoiceExt, the two-stage Core.FairCycleDovetail construction, its frozen-prefix invariant preservation, and dependent choice over the resulting typed successors. K-O8 remains optional diagnostic machinery. The former missing-supply-owner trichotomy, owner-walk closure, K.EnvOrd/K.Done collapse, and KObs drain-bag descent are not part of the primary soundness stack.

#### K.4c Corollary K.2c — The Deterministic Policy-S Execution Suffices

Definition K.RespSafe. For the selected  $p\_S^{\text{det}}$  and certified K.Resp configuration, RespSafe\_ $S^W$  means K.RespCov  $\wedge$  K.RespClean. Here K.RespClean means that no finite prefix of the complete run witnesses a non-waived K.RespFail. RespSafe\_ $S^W$  is an operational complete-run property, not a KObs cutpoint predicate and not an RTL latency statement.

Premise A5-S (primary Policy-S check). Let  $p\_S^{\text{det}}$  be the Policy-S execution selected by SDet. A5-S requires K.CompleteS evidence establishing maximality, fairness, and completeness of that run, K.RespCov establishing total finite-bound monitor coverage, and K.RespClean establishing RespSafe\_ $S^W$ . CF-S records these separate layers together with the shared K.InterruptionGate schema and either the candidate-specific K.ResetEpoch/K.TerminalDisposition evidence required by K.2b or one uniform instance-level U1-U3 record that proves all required Continue branches. A5-S alone is not an A5-L discharge outside those applicability conditions.

Corollary K.2c (deterministic Policy S is a sufficient bounded-response check). Under SDet, K.EnvMF, the shared K.InterruptionGate schema with a certificate-bound K.DomResetScope and K.ResetEpoch for every candidate component and K.TerminalDisposition on every K.ResetEpoch.Continue candidate, and Lemma K.2b's remaining premises, A5-S implies A5-L. The universal reset/terminal gate premises may be discharged either candidate-by-candidate or by the uniform instance-level U1-U3 sufficient condition above. More precisely, Lemma K.2b first proves A5L\_Sys\_K(I,W\_K) for the selected normalized blocking instance. If that lowered property were false, its finite Core.LAdm-admissible counterexample prefix

would satisfy  $K.EnvMF$  and one shared-schema branch of  $K.ResetEpoch$ .  $K.ResetEpoch.DirectFail$  independently supplies a finite non-waived cancelled-pending  $K.RespFail$  at the matched affecting  $RESET$ . On  $K.ResetEpoch.Continue$ , evaluate  $K.TerminalDisposition$ : its  $DirectFail$  branch independently supplies the mapped pending Policy-S response obligation and actual  $DeclaredTerminal$  disposition, yielding maximal-pending directly. Only under  $K.ResetEpoch.Continue$  followed by  $K.TerminalDisposition.Continue$  does Lemma K.2b.E supply the infinite fair frozen continuation required by K.2b, after which  $K.2b.P0/R/P/Q$  force a finite non-waived  $K.RespFail$  on the complete selected  $\rho\_S^{det}$ . Every branch contradicts  $K.RespClean$ , so  $A5L\_Sys\_K(I,W\_K)$  holds. Lemma K.Low.A5 then transports that result to  $A5L\_Sys\_ref(I,W)$ , which is A5-L. The corollary makes no claim for candidates that fail  $K.EnvMF$  or for which a required gate instantiation satisfies neither  $Continue$  nor  $DirectFail$ .  $\square$

**Definition K.CompleteS** and practical completeness rule. The  $SDet$ -selected  $\rho\_S^{det}$  is complete when it is maximal and fair for the selected deterministic simulator semantics under the  $Core.CompleteCycle$  projection of the selected  $Core.CycleResult$  semantics. A terminating run must reach a declared maximal terminal state; a cycle with no selected  $\Sigma$  commit is not terminal merely for that reason. A nonterminating run requires invariant, model-checking, recurrence/lasso, or equivalent completeness evidence.  $K.CompleteS$  identifies the entire artifact over which  $K.RespClean$  is checked; it does not itself imply any response bound. Changing capacities, elaboration, witness, stimulus, environment/reset contract, arbitration, simulator scheduler, Policy-S implementation, response-monitor configuration, waiver set, or normalization creates a different selected instance.

**Determinism and bound note.** Corollary K.2c concerns the selected artifact  $\rho\_S^{det}$  generated by the concrete Policy-S simulator semantics and certified complete by  $K.CompleteS$ . It neither assumes nor proves confluence among multiple Policy-S histories. A successful proof may use any finite response bounds if it proves complete-run compliance; increasing an unproved bound after a timeout changes the certificate attempt but not the functional design instance. A separately justified bound is needed to call the timeout a functional performance violation. WC and deterministic replay support reproducibility relative to the selected history;  $SDet$  fixes the remaining global choices that could change that history or the monitor outcomes.

#### K.4d Structural Sufficient Conditions for A5-L

The theorems of this subsection are alternate direct discharges of A5-L. They do not use Policy S, Lemma K.2b, or Corollary K.2c. They remain useful for nonterminating systems, designs outside the serial fragment, timed/reactive multi-frontier instances, or flows that prefer a compact structural certificate—but only when A5-L is actually true. They cannot certify the eager-dependent crossing example above, because that design has an admissible serial-shaped prefix that already falsifies A5-L.

**Definition K.Dep (full static dependency graph).**  $G\_dep$  is the directed graph over the processes of the selected instance with an edge  $P \rightarrow Q$  for every way  $P$  can, at some reachable cutpoint, hold a disabled blocking frontier item whose actual unique complement owner is  $Q$ : a data edge from the consumer to the producer of every internal channel; a capacity (back-pressure) edge from the producer to the consumer of every finite-capacity internal channel; both edges for every rendezvous channel; both directions between synchronization partners through their epoch-gate channels; and, for every arbiter process, edges in both directions between the arbiter and each client along the remodeled point-to-point sub-channels. Caution (normative):  $G\_dep$  shall be computed on this full relation. The data-flow topology alone is insufficient: a feed-forward data graph with finite capacities, synchronization, or arbitration can still contain dependency cycles through capacity, epoch-gate, or arbiter edges.

**Theorem K.Str-1 (acyclic dependency).** If  $G\_dep$  is acyclic, then A5-L holds for the selected instance. **Proof.** At every reachable source  $K$ -facing cutpoint under every admissible issue policy, every literal WFG edge is an instance of a  $G\_dep$  edge, by the E4 evaluation of  $ChannelDisabled$  at point-to-point boundaries and the WFG edge definition; a WFG cycle would therefore yield a  $G\_dep$  cycle. ■

**Theorem K.Str-2 (ranked dependency).** If there is a function  $r$  from processes (or from process–boundary pairs) to a well-ordered set such that every  $G\_dep$  edge realizable as a WFG edge at a reachable cutpoint strictly decreases  $r$ , then A5-L holds, since a WFG cycle would yield an infinite descent in  $r$ . Edges provably unrealizable at reachable cutpoints may be excluded from the obligation, with each exclusion proved as an invariant of the selected instance and recorded in the side-condition discharge record. ■

**Theorem K.Str-3 (credit / initial-token invariant).** If every cycle of  $G\_dep$  contains at least one edge that is unrealizable as a WFG edge at any reachable cutpoint — for instance, an edge of a channel with a proved occupancy invariant  $0 < occ\_c < B(c)$  whenever both endpoints are pending, as arranged by an initial token or a credit protocol — then A5-L holds. ■ Each such invariant is itself a proof obligation over the selected instance, including its capacities and elaboration, and shall be recorded in the side-condition discharge record.

#### K.4e Certificate and Discharge Formats (Normative)

A concrete theorem application uses one versioned certificate package. It may contain CF-0 cutpoint certificates and one A5-L discharge payload. A K-facing CF-0 carries or references the theorem-derived/direct  $J.RegPhaseReach$  provenance and the explicit  $K.DrainReach$  proof whose conjunction is  $K.RLAdmReach$  for the exact selected source witness. A universal  $K.1A\text{-Total}$  or  $K.1$  claim records the  $Core.QSync$  structural evidence and theorem-derived  $Core.KCutClosure\_post$  reference together with  $K.CertCov$ . In the standard normal-compositional  $QSync$  route, the package may cite  $J.QSyncCertInd$ ,  $J.CertExist\text{-}QSync$ , and  $K.CertCov\text{-}QSync$  as the instance-wide coverage derivation rather than enumerating an unrelated existence proof for every  $KCut$ . Policy-S certificates remain separate complete-run artifacts.

**Definition K.CertHdr** (common certificate header). The header binds every artifact to one theorem instance and records schema/model versions; source,  $Sys\_ref/Sys\_K$ , and RTL identities; elaboration, capacities, projections, ownership, and lowering; fixed stimulus, environment, reset, witness/WC provenance; the  $Core.CycleResult/Core.CycleClosure$  semantics identity where a complete source or Policy-S run is consumed, including full  $nextState$  environment/auxiliary typing and reset-unit emission; and, for the Policy-S route, the concrete deterministic simulator/scheduler, deterministic arbitration, Policy-S issue semantics,  $\epsilon$ -normalization identity,  $SDet$  evidence,  $K.Resp$  monitor schema, finite-bound function, reset/end-of-test disposition, the  $K.InterruptionGate$  schema/version, each candidate  $K.DomResetScope/DomEpoch$  definition and provenance,  $K.ResetEpoch$  evidence for candidate components,  $K.TerminalDisposition$  evidence on  $K.ResetEpoch.Continue$  candidates, and response-waiver policy. It also records cutpoint waivers, route, side-condition references,  $Core.QSync$  structural-evidence provenance and theorem references,  $K.CertCov$  coverage method when applicable, generator/checker versions, and the trusted computing base. Any QS/settling premise accepted solely from an unverified HLS/synthesis-tool report shall be named in the declared trusted base; independently checked concrete evidence records the checker/certificate identity instead. Different header identities denote different selected instances.

**Definition K.CertifiedCutpoint** (K-ready package for one reachable RTL cutpoint). Fix an RTL K-facing cutpoint  $p\_post \in KCutReach\_post(I)$ , a finite reachable  $RTL\_B$  prefix  $\tau\_post$ , and an explicitly recorded finite normalization witness  $v\_post$  satisfying  $NormPost\_I(\tau\_post, v\_post, p\_post)$ .

$CertifiedCutpoint\_I(\tau\_post, v\_post, p\_post)$  holds when one  $K.CertHdr$ -bound package records that triple and establishes: (1)  $J.LocalGWR/J.GWR\text{-}Comp$  or  $J.GWR\text{-}Direct$  for the selected prefix; (2) the exact Corollary J.1 tuple  $(\tau\_ref, \sigma\_ref, E, w, H, O)$ , with  $\sigma\_ref \in KCutReach\_ref(I)$ ; (3)  $J.HCanon$ ,  $J.OfferReach$ ,  $J.OfferConf$ , and  $J.KObsConf$ ; (4)  $J.RegPhaseReach$  for that exact  $\tau\_ref/\sigma\_ref$ , theorem-derived on the normal route or directly validated; (5)  $K.DrainReach$ , including  $Core.SettleKBridge$ , for the same tuple;

and hence (6)  $K.RLAdmReach$  and membership of  $\tau\_ref$  in  $Pref\_L^{adm}(I)$ . The package proves one prefix/normalization/ $K$ -cutpoint triple only and does not imply  $K.CertCov$ .

**Definition  $K.CertCov$**  ( $K$ -ready package coverage).  $K.CertCov(I)$  holds when every reachable RTL  $K$ -facing cutpoint  $p\_post \in KCutReach\_post(I)$  has at least one theorem-constructible or explicitly recorded reachable-prefix/normalization pair  $(\tau\_post, v\_post)$  satisfying  $NormPost\_I(\tau\_post, v\_post, p\_post)$  and a corresponding  $CertifiedCutpoint\_I(\tau\_post, v\_post, p\_post)$  package. A total proof-producing extractor or inductive existence theorem may represent this family without eagerly materializing one file record per cutpoint, provided the checker can derive or validate the package for any queried reachable cutpoint under the bound  $K.CertHdr$  instance. Under the standard  $Core.QSync$  route, the canonical same-cycle normalization/endpoint supplied by Theorem  $Core.QSync-Post$  may be used for this  $K$ -facing  $NormPost$  selection. This canonicalization does not replace Appendix I/J  $ReplayNormalizationConvention$  or its selected replay-normalization/congruence witnesses.  $K.CertCov$  does not require every arbitrary reachable RTL prefix to admit a normalization before the next  $\Sigma$  action; coverage is over the complete reachable  $K$ -facing domain on which  $DeadKRec^W$  is defined. Equivalently, the package-existence relation is total on  $KCutReach\_post(I)$ . For Corollary  $K.1A$ -Total and Theorem  $K.1$ ,  $K.CertCov$  is paired with  $Core.KCutClosure\_post(I)$ . In the standard  $QSync$  scope,  $Core.QSync-Post$  proves the closure/non-vacuity side, while Theorem  $J.CertExist-QSync$  plus Corollary  $K.CertCov-QSync$  below provide a generic inductive route to the package-totality side from  $J.QSyncCertInd$ . Thus  $K.CertCov$  remains the downstream semantic interface, but it need not be supplied as an unrelated per-cutpoint axiom or pre-enumerated package set when the  $QSync$  certificate-induction hypotheses are discharged.

#### Corollary $K.CertCov-QSync$ ( $QSync$ certificate existence discharges $K.CertCov$ ).

Assume the standard  $QSync$  normal-compositional scope and the hypotheses of Theorem  $J.CertExist-QSync$ . Then  $K.CertCov(I)$  holds. Proof.  $J.CertExist-QSync$  gives  $J.QSyncCertCov(I)$ , and on a proof-producing  $CI0/CI1$  discharge it supplies the on-demand extractor  $CertExtract\_QS$ . Fix any  $p\_post \in KCutReach\_post(I)$  and obtain its  $QSyncCertState$  tuple  $C$ .  $QC1$  supplies the canonical  $NormPost$  pair;  $QC2$  supplies  $J.LocalGWR/J.GWR-Comp$  and the exact Corollary  $J.1$  source tuple;  $QC3$  supplies  $J.HCanon/J.OfferReach/J.OfferConf/J.KObsConf$ ;  $QC4$  supplies  $J.RegPhaseReach$ ; and  $QC5$  supplies  $K.DrainReach$  and therefore  $K.RLAdmReach$ . Bind those already-proved semantic facts to the  $K.CertHdr$  for the fixed theorem instance. The resulting record satisfies Definition  $K.CertifiedCutpoint$  for  $p\_post$ . Because  $p\_post$  was arbitrary, Definition  $K.CertCov$  holds. The theorem/extractor reference may represent the family and materialize individual packages on demand;  $K.CertHdr$  adds version/provenance/trust accounting, not an additional semantic existence assumption.  $\square$

**CF-0** —  $KObs$  cutpoint correspondence certificate. This certificate supports Corollary  $J.1$  and, when augmented with item (8), Theorem  $K.1A$  for one cutpoint; it neither discharges  $A5-L$  nor proves  $K.CertCov$ . It records or commits to: (1)  $K.CertHdr$ ; (2) selected source/RTL prefixes and finite normalizations; (3) canonical  $H$ ; (4) typed residual offers  $O$ ; (5)  $J.HCanon/J.OfferReach$ ; (6) complete  $RelevantReqSet/ReqProjection$  equality, attribution, cardinality, prefix-closure, and no-hidden-request evidence for  $J.OfferConf$ ; (7)  $B$ -faithfulness/ $E4$ , reset, and  $\epsilon$ -opacity for  $J.KObsConf$ ; and (8), for Appendix  $K$ , the  $J.RegPhaseReach$  provenance plus  $K.DrainReach$  evidence whose checked conjunction is  $K.RLAdmReach$  for the exact tuple. The phase provenance includes  $Core.RegSeq$ , enriched deferred-hole/ $NoDependentUse$  records, field-sensitive footprints,  $J.RegPhaseNF$ , Horizon-batched  $J.UnitExt$ , and the no-unresolved-hole cutpoint invariant, or a direct equivalent. Derived  $Front/WFG/deadlock$  records are checker-reconstructed. Historical  $Issue/ReturnedAtCycle$  logs and  $K.Resp$  timers need not be duplicated, but referenced artifacts must validate their consumed obligations.

**CF-S** — Primary deterministic Policy-S bounded-response certificate. The payload records the header; the functional Policy-S simulator semantics, initial state, fixed witness/stimulus/environment, deterministic scheduler and arbitration, capacities, elaboration, lowering, normalization, response-

waiver policy,  $K.CVPred/K.CV-WF$  provenance, including the  $K.CVPred$  comparison-availability closure used by  $K.2b.P0/R/P$  and the explicit  $Core.16$  reach-legality disposition for every recursively materialized non-transfer predecessor, and the shared  $K.InterruptionGate$  schema with either the per-candidate certificate-bound  $K.DomResetScope/DomEpoch$  and ordered  $K.ResetEpoch/K.TerminalDisposition$  instantiations or one uniform instance-level gate record satisfying U1-U3 above; the selected run artifact  $p\_S^{det}$  as a sequence of  $Core.CycleResult$  records or an equivalent symbolic representation whose  $CompleteCycle$  projection is checkable;  $SDet$  evidence; fairness/maximality and  $B2/B3/B5/K.Drain/K\epsilon/nonwithdrawal$  evidence, including the exhaustive  $Core.IC$  lifecycle and Lemma  $K.BF-Persist$ ; and  $K.CompleteS$ . For a successful clean discharge,  $K.ResetEpoch.Continue$  is the normal prospective reset evidence, and only on that branch  $K.TerminalDisposition.Continue$  is the normal prospective terminal evidence; both are proved contract-wise over the applicable frozen-prefix classes rather than inferred from absence of observed failures. Either  $DirectFail$  branch may be proved conditionally for a hypothetical candidate or instantiated to classify a direct failure; a concrete  $DirectFail$  instantiation is itself a  $K.RespFail$  witness and cannot coexist with  $K.RespClean$  on a successful clean run. The generic direct-failure payload shape binds  $P, f\_P, x\_P^AS, g\_P$  with  $g\_P \leq_{src} f\_P$  and an already-triggered pending non-waived monitor independently of  $K.2b.R/P/Q$  Step 1. The  $ResetEpoch.DirectFail$  specialization additionally binds the common pre-reset epoch, matched affecting RESET in  $CycleResult.resetUnits$ , and RW2 clearing of that exact monitor, yielding cancelled-pending. The  $TerminalDisposition.DirectFail$  specialization instead binds the independently validated declared maximal terminal state while that monitor remains pending, yielding maximal-pending. CF-S also records timer progression through  $CycleResult.nextState$ , complete-run results,  $K.EnvMF$  applicability, including the  $K.JointFreeze$  certificate for every admitted  $StandardEnv$  multi-frontier candidate;  $K.CompareAdvanceLegal$  coverage for every  $K.2b.P0/K.2b.R$ -consumed owner/key, including the  $P0$  preparation/launch and  $R$  Issue/assertion halves and any required explicit  $Sys\_B^{elab}$  multi-frontier certificate and the  $JointFreeze/CompareAdvanceLegal$  consistency check; the  $K.CVPred$  recursive dispositions may cite Lemma  $K.CompareAdvanceLegal-Ord$  pointwise for qualifying ordinary/singleton blocking  $OpFact$  predecessors and must identify inert-observation or equivalent launch-legality evidence for the remaining predecessor reaches; and waiver preservation, and references the instance-bound  $Core.CycleClosure/Core.FairPick/B5$  discharge used to construct the Policy-L continuation in Lemma  $K.2b.E$ .  $Core.FairPick$  does not strengthen the behavioral  $Core.IssueFair/B2/B3$  property, but it is an additional constructive evidence form required because  $K.2b.E$  builds a Policy-L continuation rather than consuming one already supplied. Checker obligations include validating the  $K.CVPred$  provenance equations, comparison-availability closure, every recursive predecessor  $Core.16$  reach-legality disposition, and  $K.CV-WF/rank$  certificate used by  $K.2b.P0/P1/R/P$ ; validating the common gate schema and its  $K.BF$  exhaustiveness once and then either (i) validating each certificate-supplied  $K.DomResetScope$  against D1–D4 and its associated  $DomEpoch$ , each instantiation's  $Interrupt$  predicate and  $DirectFail$  specialization, or (ii) validating the single uniform instance-level gate record against U1-U3 and its claimed universal  $Continue$  discharge; in either route the required reset-first nesting and rejection of any clean candidate for which the applicable gate discharge fails remain enforced. For an ordinary non-elaborated singleton-frontier instance, the  $K.JointFreeze$  and main  $K.CompareAdvanceLegal$  fields are theorem references to  $K.JointFreeze-Ord/K.CompareAdvanceLegal-Ord$  rather than separate per-candidate frontier certificates.

Definition  $K.A5Cert$  ( $A5-L$  discharge package). The package consists of  $K.CertHdr$  plus exactly one declared route payload: CF-S; a proof reference for  $K.Str-1$ ; CF-1; CF-2; CF-3; CF-4; CF-5; or a direct Policy-L invariant/exploration/proof artifact. The package records the route and target property and shall expose enough data for an independent checker to validate the route without trusting a tool-emitted conclusion string. Policy-S route packages target  $K.RespClean$  and include dominance, coverage, finite-bound, waiver-preservation, and complete-run evidence; direct Policy-L packages target

Dead\_K<sup>W</sup>. Optional diagnostic environment-terminal records do not substitute for the response payload.

**CF-1 — Rank certificate.** A function  $r$  from processes (or process–boundary pairs) of the selected instance to a declared well-ordered set, together with the Definition K.Dep edge enumeration, such that every realizable edge strictly decreases  $r$  (Theorem K.Str-2). Checker obligations: the edge enumeration covers data, capacity back-pressure, rendezvous, epoch-gate, and arbiter edges of the selected instance including its elaboration and lowering; every non-excluded edge strictly decreases  $r$ ; every excluded edge carries a recorded invariant proof of unrealizability.

CF-2 — Credit / cycle-breaking certificate. For each cycle of the full dependency graph  $G_{\text{dep}}$ , provide an invariant—such as token presence, occupancy bounds, or a circulating-credit invariant—establishing that at least one edge of the cycle is unrealizable as a WFG edge at any reachable cutpoint (Theorem K.Str-3). Checker obligations: the dependency-cycle enumeration is complete; each invariant is inductive over the selected instance; and the proved edge exclusion applies at the explicit capacities and elaborated boundaries. Petri-net or siphon analyses may be used by a tool as one way to generate this certificate, but no Petri-net equivalence theorem is part of normative Appendix K.

CF-3 — Inductive-invariant certificate. Supply an abstraction map  $\alpha$  from the selected operational transition system into a finite abstraction  $A$ . For direct Policy-L verification, the observation component shall be KObs, or an explicitly proved quotient sufficient to reconstruct DeadKRec<sup>W</sup>. For the Policy-S response route,  $A$  shall additionally carry the monitor activation, dynamic attribution, age, bound-class, reset/terminal-disposition, and maximality state needed to reconstruct K.RespFail; KObs alone is intentionally insufficient. Supply an invariant  $\text{Inv}$  such that  $\alpha(\text{Init})$  is contained in  $\text{Inv}$ ,  $\text{Post}_A(\text{Inv})$  is contained in  $\text{Inv}$ , and  $\text{Inv}$  excludes the route’s bad event. Checker obligations:  $\alpha$  and  $\text{Post}_A$  soundly overapproximate the selected operational relation; every cutpoint or monitor fact is reconstructed from the appropriate observation/monitor components and ambient instance data; reset/environment/waiver distinctions are represented; witness-selected non-blocking outcomes are source-describable through NB-CF or K.Low; and all invariant inclusions are mechanically validated. Equality of KObs records alone shall not be treated as transition equivalence or as proof of future Issue legality, fairness, DrainDiscipline, or response timing.

CF-4 — Exploration certificate. Record an exhaustive or soundly bounded model-checking result over the selected Policy-L relation establishing unreachability of DeadKRec<sup>W</sup>, or over the selected Policy-S relation establishing absence of K.RespFail, with K.CertHdr, the explored transition-system identity, abstraction map, cutpoint normalization rule, monitor semantics, tool/version, bound or completeness justification, fairness interpretation, and coverage argument. The checker shall recompute Dead\_K from KObs for the direct route and shall recompute monitor outcomes from the augmented Policy-S state for the response route. A finite exploration depth or simulation prefix without a proof of completeness is supporting evidence only.

CF-5 — Tool-emitted certificate. Any CF-S or CF-1 through CF-4 emitted by the HLS tool or a companion tool over the selected instance may exploit the tool’s knowledge of the scheduled control graph, stage allocation, actual FIFO and skid-buffer capacities, request ownership, join and arbitration structure, automatic-flush implementation, and hidden storage or credits. It shall use K.CertHdr and the applicable CF-0/A5 payload schema and be validated by an independent checker whose trusted base is recorded. The checker shall validate structure, evidence references, and reconstructed K-facing predicates, not merely the certificate generator’s final claim. Labeling rule (normative): a certificate stated over the generated RTL itself, rather than over the selected Sys\_ref instance, is admissible evidence of deadlock-freedom for that design but bypasses the Appendix K preservation architecture — it is output verification, not a discharge of A5-L — and shall be labeled as such in the record.

Soundness restriction (normative). An arbitrary eventuality claim—such as ‘the arbiter eventually grants’—does not by itself discharge A5-L, because A5-L prohibits reaching the closed wait-cycle

cutpoint. The K.Resp route is different: Lemma K.2b proves that any such liberal cutpoint forces a permanent covered serial frontier and therefore a finite timeout/maximal-pending or matched-reset cancelled-pending event under a certified finite bound. Only response properties satisfying K.RespCov, the K.2b trigger mapping, complete-run checking, and waiver preservation may be used in Corollary K.2c. An unproved timeout threshold or a finite clean simulation is not a certificate.

#### K.4f The Policy-S Standard Fragment (K.PSF)

Definition K.PSF (Policy-S standard applicability fragment). A selected instance is in K.PSF when it is in Sys\_K form; channels are point-to-point with finite B-faithful capacities; SDet holds for the concrete Policy-S simulator; K.CV holds, including K.CVPred/K.CV-WF, and arbiters are deterministic and part of that simulator semantics; K.EnvMF holds, including K.JointFreeze for every admitted StandardEnv multi-frontier candidate; K.CompareAdvanceLegal holds for every candidate comparison, with ordinary/singleton main comparison cases theorem-derived and explicit Sys\_B^elab multi-frontier owners checked regardless of membership in D; K.CVPred recursively records the explicit Core.16 reach-legality disposition for every non-transfer predecessor materialization, using Lemma K.CompareAdvanceLegal-Ord pointwise for qualifying ordinary/singleton blocking OpFact predecessors and explicit reach evidence otherwise; the shared K.InterruptionGate schema is fixed and its universal candidate obligations are discharged either by the existing per-candidate K.DomResetScope/DomEpoch, K.ResetEpoch, and nested K.TerminalDisposition records or by the uniform instance-level U1-U3 sufficient condition above; requests are nonwithdrawable under the exhaustive Core.IC lifecycle, with Lemma K.BF-Persist supplying the frozen-frontier specialization for every WFG-relevant K.BF blocking request; Kε/K.Drain hold; Core.CycleClosure and B5 hold for the selected Policy-L semantics, and its Core.IssueFair/B2/B3 discharge is available in the constructive Core.FairPick form required by Lemma K.2b.E; and a finite-bound K.Resp configuration with total K.RespCov is supplied. Completeness, fairness, and K.RespClean of  $p\_S^{\text{det}}$  belong to A5-S/CF-S.

Theorem K.PSF-1. For a K.PSF instance, the SDet-selected Policy-S execution  $p\_S^{\text{det}}$  establishes A5L\_Sys\_K(I, W\_K), and hence A5-L by Lemma K.Low.A5, when CF-S supplies K.CompleteS and K.RespClean for the totally covered finite-bound monitor set. Proof: Lemma K.2b, Corollary K.2c, and K.Low.A5.  $\square$

Intent. K.PSF is the recommended standardized applicability target for new latency-insensitive designs. The general user-facing evidence is the conservative Policy-S execution checked by complete process-facing bounded-response obligations. For finite transactions and bounded hardware protocols, these are ordinary timeout/service assertions and are expected to be highly automatable. A correct design with fair but genuinely unbounded response gaps lies outside this serial fragment and must use a structural, direct Policy-L, invariant, exploration, or tool-certificate route. Dead\_KS/K.EnvOrd/K.Done remain optional diagnostic aids rather than primary proof premises. Ordinary-frontier simplification: for an ordinary non-elaborated instance whose relevant message frontiers are singleton, neither a separate K.JointFreeze certificate nor a separate K.CompareAdvanceLegal certificate is an additional per-candidate obligation; Lemmas K.JointFreeze-Ord and K.CompareAdvanceLegal-Ord derive those frontier facts from the standard source-order, Core.16, ownership, comparison-invariant, and drain premises. Explicit frontier certificates are needed only where relevant Sys\_B^elab multi-frontier structure prevents those ordinary lemmas from applying. In the common no-relevant-reset/clean-terminal case, the uniform instance-level gate discharge above replaces candidate-by-candidate reset/terminal gate enumeration without weakening the underlying universal premise.

## K.5 Theorems K.1A, K.1A-Total, and K.1 — Reachable-Cutpoint Deadlock-Freedom Preservation (“Liveness Preservation”)

Theorem K.1A (certified-cutpoint preservation from the scheduler-robust premise). Statement: Assume A1–A4 and A6–A11, WC/Core.RegSeq, the Appendix I/J correspondence prerequisites, and the point-to-point evaluated-boundary condition (the A6 ownership premise applied at every explicit boundary on which KObs/WFG facts are evaluated). Fix one  $p\_post \in \text{KCutReach\_post}(I)$  and one recorded reachable-prefix/normalization pair  $(\tau\_post, v\_post)$  with  $\text{NormPost\_I}(\tau\_post, v\_post, p\_post)$  whose package satisfies  $\text{CertifiedCutpoint\_I}(\tau\_post, v\_post, p\_post)$ . Thus Corollary J.1 yields equal KObs for  $p\_post$  and  $\sigma\_ref$ ; Corollary J.RegPhaseReach-Comp supplies J.RegPhaseReach for the exact source prefix on the normal compositional route (or the direct package supplies the equivalent fact); and K.DrainReach proves nonwithdrawal,  $K\epsilon$ , and Core.Drain/K.Drain. Their conjunction K.RLAdmReach places that same source prefix in  $\text{Pref\_L}^{\text{adm}}(I)$ . If W is non-empty, also assume  $\text{KObsClosed\_I,E,w}(W)$ . If A5-L holds for the selected Sys\_ref instance, then  $p\_post$  does not satisfy  $\text{DeadKRec}^W$ , equivalently the  $\text{Dead\_K}^W$  predicate of the selected theorem instance. With  $W = \emptyset$ , the fixed certified cutpoint is not an internal message-passing-deadlock cutpoint. The theorem consumes neither K.CV, NB-Align, nor Lemma K.2b, and it makes no K.CertCov or package-totality claim.

Proof (by contradiction).

1. Assume, for contradiction, that the fixed certified RTL K-facing cutpoint  $p\_post \in \text{KCutReach\_post}(I)$  has K-facing reconstructed observation  $K\_post$  satisfying  $\text{DeadKRec}^W(K\_post, I, W)$ . By Definitions K.Dead and K.DeadW and the reconstruction interface, this includes a non-empty blocked closed WFG component and DrainClosedNow for that component. The asserted RTL requests and channel facts used in this observation are covered by J.OfferConf and B-faithfulness/E4; no unrecorded raw RTL request may be omitted.
2. By  $\text{CertifiedCutpoint\_I}(\tau\_post, v\_post, p\_post)$ , apply the recorded Corollary J.1 instance to this same finite prefix and package. It supplies the reachable source K-facing cutpoint  $\sigma\_ref \in \text{KCutReach\_ref}(I)$  and  $K\_ref = \text{KObs\_E,w}(\sigma\_ref)$  such that  $K\_post = K\_ref$ . J.OfferReach ensures that  $\sigma\_ref$  is the endpoint of the selected reachable source witness rather than an arbitrary or unrealizable record.
3. By Lemma K.0, Lemma K.2, and Lemma K.2a — equivalently, by functional extensionality of the reconstruction functions —  $\text{DeadKRec}^W(K\_ref, I, W)$  has the same truth value as  $\text{DeadKRec}^W(K\_post, I, W)$ . When W is non-empty,  $\text{KObsClosed\_I,E,w}(W)$  makes the waiver predicate well-defined on the common observation, and Lemma K.KObs-Dead identifies the reconstructed source predicate with  $\text{Dead\_K}^W(\sigma\_ref)$ . Hence  $\text{Dead\_K}^W(\sigma\_ref)$  holds.
4. The same K.CertifiedCutpoint package supplies theorem-derived/direct J.RegPhaseReach and explicit K.DrainReach for the exact Corollary J.1 witness. The former proves Policy-L legality and L0–L4 phase ordering by construction; the latter proves nonwithdrawal,  $K\epsilon$ , and every activated Core.Drain/K.Drain clause. Therefore K.RLAdmReach establishes  $\tau\_ref \in \text{Pref\_L}^{\text{adm}}(I)$ . No phase inference is made from equal KObs, and no drain inference is made from J.RegPhaseNF alone. Thus  $\sigma\_ref$  is the reachable Core.LAdm-admissible cutpoint prohibited by A5-L.
5. This contradicts A5-L. Therefore the fixed certified cutpoint  $p\_post$  cannot satisfy  $\text{Dead\_K}^W$ .

□

Mechanization note. The proof core is Corollary J.1, equality of typed KObs, pure reconstruction of  $\text{Dead\_K}^W$ , theorem-derived/direct J.RegPhaseReach, and explicit K.DrainReach for the exact witness. Corollary J.RegPhaseReach-Comp proves J.RegPhaseReach upstream from Core.RegSeq, enriched holes, field-sensitive footprints, J.RegPhaseNF, and bucketed J.UnitExt; Theorem K.1A consumes only the composed K.RLAdmReach fact. K.Drain continuation evidence is not derived from KObs or phase normality. Package totality remains outside this theorem.

Corollary K.1A-Total (universal reachable-cutpoint preservation under K.CertCov and KCut closure). Assume the premises of Theorem K.1A hold uniformly for the selected instance, assume  $\text{Core.KCutClosure\_post}(I)$ , and assume  $\text{K.CertCov}(I)$ . Then no  $\rho\_post \in \text{KCutReach\_post}(I)$  satisfies  $\text{Dead\_K}^W$ . With  $W = \emptyset$ ,  $\text{RTL\_B}$  is internally message-passing-deadlock-free over the complete reachable K-facing cutpoint domain, and by Lemma  $\text{Core.KCutObserve}$  a persistent K-relevant blocked configuration cannot evade that domain solely through endless pre-normalization  $\epsilon$  activity. Standard QSync instantiation. In the standard theorem scope,  $\text{Core.QSync-Post}$  supplies  $\text{Core.KCutClosure\_post}$ ; if  $\text{J.QSyncCertInd}$  is discharged, Theorem  $\text{J.CertExist-QSync}$  and Corollary  $\text{K.CertCov-QSync}$  additionally supply  $\text{K.CertCov}$ . Thus both universal-domain obligations are theorem-derived from checked structural plus base/step semantic-compliance evidence rather than from arbitrary normalization or sampled cutpoints. Proof.  $\text{Core.KCutClosure\_post}$  supplies a finite KCut for every reachable RTL K-facing cycle-evaluation origin;  $\text{K}\epsilon$  preserves any blocker that persists through that normalization. Any alleged prohibited KCut has a  $\text{K.CertifiedCutpoint}$  package by  $\text{K.CertCov}$  and is therefore excluded by Theorem K.1A.  $\square$

Theorem K.1 (user-facing deterministic Policy-S preservation under cutpoint closure and certificate coverage). Assume Theorem K.1A's per-cutpoint premises uniformly,  $\text{Core.KCutClosure\_post}(I)$ ,  $\text{K.CertCov}$ ,  $\text{SDet}$ ,  $\text{K.CV}$ ,  $\text{NB-Align/K.Low}$  where applicable, point-to-point boundaries, nonwithdrawal,  $\text{K}\epsilon$ ,  $\text{K.Drain}$ ,  $\text{K.EnvMF}$ , the shared  $\text{K.InterruptionGate}$  schema with  $\text{K.ResetEpoch}$  and nested  $\text{K.TerminalDisposition}$  applicability, and  $\text{K.RespCov}$ . If A5-S holds for the  $\text{SDet}$ -selected execution  $\rho\_S^{\text{det}}$ —supplying  $\text{K.CompleteS}$  and  $\text{K.RespClean}$ —then  $\text{RTL\_B}$  cannot reach a persistent K-relevant internal message-passing blocking configuration whose normalized K-facing observation satisfies  $\text{Dead\_K}^W$ . Equivalently, no reachable RTL KCut satisfies  $\text{Dead\_K}^W$ , and KCut closure prevents persistent blockers from remaining forever outside that observation domain. Proof: K.2c gives A5-L. In the standard QSync normal-compositional scope,  $\text{Core.QSync-Post}$  derives  $\text{Core.KCutClosure\_post}$  and, when  $\text{J.QSyncCertInd}$  is discharged,  $\text{J.CertExist-QSync}$  plus  $\text{K.CertCov-QSync}$  derive  $\text{K.CertCov}$ ; Corollary K.1A-Total then gives the universal RTL conclusion. A flow that does not use that induction route may instead establish  $\text{K.CertCov}$  by one of the alternate methods in M.7a. Without  $\text{K.CertCov}$ , the same A5-L discharge supports only the certified-cutpoint conclusion of Theorem K.1A; without  $\text{KCutClosure\_post}$ , no universal claim is made that persistent pre-normalization RTL blocking states become observable as KCuts. The Policy-S form makes no claim for timed/reactive/nondefault multi-frontier components or for candidates for which a required nested gate instantiation satisfies neither Continue nor DirectFail.  $\square$

Implementation status. The operational Policy-S check is specified in terms of conventional pre-HLS simulation and bounded-response assertions and is expected to be highly automatable for finite transactions and bounded protocols. Treating a clean run as a verified discharge of A5-L retains proposed-reduction status until the K-P proof obligations and required certificate interfaces are mechanized and qualified. For the universal RTL conclusion, the paper-level standard QSync route is now complete at the theorem-interface level:  $\text{Core.QSync-Post}$  derives  $\text{Core.KCutClosure\_post}$ , and  $\text{J.CertExist-QSync/K.CertCov-QSync}$  derive  $\text{K.CertCov}$  from an independently checked  $\text{J.QSyncCertInd}$  base-and-step invariant covering every reachable RTL successor. This removes the former generic package-existence gap but does not make coverage automatic: a concrete flow must still prove or independently check QSync and the QSyncCertInd base/step obligations (or use an alternate  $\text{K.CertCov}$  route), and no verified generic certificate generator is supplied here.

Discharge routes for A5-L (normative summary). Primary: an  $\text{SDet}$ -qualified CF-S certificate and K.2c from A5-S for the  $\text{SDet}$ -selected  $\rho\_S^{\text{det}}$ , including  $\text{K.CompleteS}$ ,  $\text{K.RespCov}$ , and  $\text{K.RespClean}$ . Alternatives are structural, invariant, exploration, CF-5, or direct Policy-L artifacts. The header records the selected normalized instance and, when Policy S is used, the deterministic simulator/scheduler, arbitration, Policy-S issue semantics, normalization identity,  $\text{SDet}$ , response-monitor schema and finite bounds, completeness/fairness, waivers, and trust boundary.

## Appendix K-P – Proofs Companion for the Serial-Reduction Route

Status. This companion collects the deterministic serial-reduction proof.  $\text{SDet}$  fixes the selected functional Policy-S semantics and one run artifact  $\rho_{\text{S}}^{\text{det}}$  at the outset;  $\text{K.CompleteS}$  separately establishes that this artifact is maximal, fair, and complete. Below write  $\rho_{\text{S}} := \rho_{\text{S}}^{\text{det}}$ . The soundness-critical proof retains  $\text{K.Low.C}$ , identity/dataflow lemmas, serial-prefix reachability, progress dominance, joint frontier freeze, and finite bounded-response extraction. The former missing-supply owner trichotomy, stuck-owner walk,  $\text{EnvStop/DoneStop}$  closure, and  $\text{K.EnvOrd/K.Done}$  collapse are retired from the trusted route and remain at most diagnostic. Optional serial replay is unchanged.

K-P KObs interface convention. For every source K-facing cutpoint  $\sigma \in \text{KCutReach\_Sys\_K(I)}$  used below, write  $\text{K}_{\sigma} := \text{KObs\_E,w}(\sigma) = \langle \text{E,w,H}_{\sigma}, \text{O}_{\sigma} \rangle$ , where  $\text{O}_{\sigma} = \text{ResOffers\_E,w}(\sigma)$ . For  $\text{K} = \langle \text{E,w,H,O} \rangle$ , use  $\text{owner(o)}$ ,  $\text{item(o)}$ ,  $\text{op(o)}$ ,  $\text{boundary(o)}$ ,  $\text{dir(o)}$ , and  $\text{epoch(o)}$  for the projections of the attributed residual-offer tuple  $\text{o} = \langle \text{P,x,q,c,dir,e} \rangle$  of Definition Core.18b. Write  $\text{Proc(I)}$  for the finite process set of the selected instance. For a process set  $\text{D}$  define  $\text{Offers\_D(K)} := \{\text{o} \in \text{O} \mid \text{owner(o)} \in \text{D}\}$ ;  $\text{FrontOffers\_D(K)} := \{\text{o} \in \text{Offers\_D(K)} \mid \text{item(o)} \in \text{FrontRec\_owner(o)(K)}\}$ ; and  $\text{DrainBag\_D(K)}$  as the finite multiset of offers  $\text{o} \in \text{Offers\_D(K)}$  whose item lies outside  $\text{FrontRec\_owner(o)(K)}$ .  $\text{DrainCandidates\_D(K)}$  is the submultiset of  $\text{DrainBag\_D(K)}$  whose item is  $\text{ChannelEnabledRec}$  at  $\text{K}$ .  $\text{CmtRec\_P(K)}$ ,  $\text{PushOrdRec\_c(K)}$ , and  $\text{PopOrdRec\_c(K)}$  denote the committed-identity and per-endpoint ordinal projections of  $\text{H}$ . These names are pure snapshot selectors; at an operational cutpoint  $\text{K}_{\sigma}$  they agree with the corresponding  $\text{Sys\_K}$  source facts by Lemmas  $\text{J.KObs-Proc-J.KObs-WFG}$  and  $\text{K.KObs-Dead}$ .

**Operational and elaboration boundary.** The graph and residual-offer arguments in this appendix range over the full E-typed frontier items and tuples, including explicit  $\text{Sys\_B}^{\text{elab}}$  staged, bundle, join, guard, and epoch-gate boundaries. When the proof uses an operation symbol  $\text{f\_P}$ ,  $\text{g\_P}$ , or  $\text{q}$ , it abbreviates  $\text{MsgOf\_P(x)}$  only after the corresponding typed item  $\text{x}$  and boundary have been fixed; the item identity, boundary, direction, epoch, and owner remain part of the formal object. KObs deliberately omits historical Issue times and does not determine future issue legality.  $\text{Core.Drain/K.Drain}$  remains the continuation-level  $\text{DrainDiscipline}$  used to forbid replenishment past a blocked frontier.

$\text{DrainClosedNowRec}$  and  $\text{DrainCandidates}$  are only cutpoint predicates. Channel-side E4 movement inside the selected elaboration is handled by B4/B5  $\epsilon$ -normalization and is not duplicated as a source residual offer.

Lemma  $\text{K.Low.C}$  (normalization correspondence) — restated with proof. For the same fixed stimulus and witness choices: (i) every admissible execution of  $\Lambda_{\text{W}}(\text{Sys\_ref})$  projects, by erasing guard-channel and epoch-gate transitions, to an admissible execution of  $\text{Sys\_ref}$  with the same issue-policy class ( $\text{S}$  or  $\text{L}$ ), the same committed message/synchronization/observation history, and the same  $\Sigma$  projection; (ii) every admissible execution of  $\text{Sys\_ref}$  in which each witness-selected site takes its  $\text{W}$ -selected outcome lifts to an admissible execution of  $\Lambda_{\text{W}}(\text{Sys\_ref})$  in which each guard Pop commits within a finite WFG-inert  $\epsilon$ -delay of the commit of its modeled obstruction — in the default  $\text{B(g\_s)} = 1$  form, in the cycle after that commit once the guard request is asserted; exactly at that commit only in a declared atomic rendezvous variant; (iii) a reachable source K-facing cutpoint  $\sigma_{\text{K}} \in \text{KCutReach\_Sys\_K(I)}$  of  $\Lambda_{\text{W}}(\text{Sys\_ref}) = \text{Sys\_K}$  satisfies  $\text{Dead\_K}$  if and only if its lowering projection  $\sigma_{\text{ref}}$  lies in  $\text{KCutReach\_ref(I)}$  and contains a closed internal wait cycle in the extended sense — a non-empty process set closed under waiting through internal message frontiers, infeasible NB-aligned selected outcomes, and pending explicit synchronizations. Proof. Parts (i) and (ii) are per-rule stuttering simulations: the added transitions touch only the fresh channels; by NB-Align and  $\text{K.CV}$ ,  $\text{W}$ 's selected outcome at each lowered site is a deterministic function of the commit of its modeled obstruction, so the guarded process's post-guard behavior is identical in both models; clause (7) already requires synchronization refusal to be represented as  $\text{ChannelDisabled}$  at an explicit boundary, so the synchronization/epoch-gate lowering rule is representational rather than semantic; point-to-point ownership and E3 prefix closure identify

the unique  $g_s$  producer;  $K\epsilon$  keeps the added microsteps WFG-inert. The guard skew of the default  $B(g_s) = 1$  form is absorbed by the same stuttering simulations: during the skew the guard Pop is channel-enabled with its unique producer already committed, so it is WFG-inert and is a drain candidate, and therefore no drain-closed KCut analyzed by part (iii) contains a half-completed guard; an exactly-at reading would be inconsistent with the non-fall-through E4 semantics for positive-capacity channels. For (iii), left to right: a member blocked at a guard or epoch-gate frontier projects to a process whose  $W$ -selected outcome or synchronization is infeasible, and the unique complement owner of that frontier projects to the modeled supplier, so a lowered literal WFG cycle projects to a closed wait cycle in the extended sense; right to left: each frozen or waiting node of an extended cycle is, in the lowered instance, message-blocked at its guard or epoch-gate frontier with its complement owner the image of its modeled supplier, so the extended cycle is a literal all-message-blocked  $Dead_K$  component in  $\Lambda_W(Sys\_ref)$ . ■ Mechanization note. Lemma K.Low.C is the single successor to the former obstruction-pointer conversion obligation. It shall be discharged per lowering rule — one case lemma for the epoch-gate rule and one for the guard rule — as a local correspondence at each lowered site, not as a global argument over terminal cutpoints. Part (iii) is the load-bearing claim; a development that mechanizes only the blocking-only Lemma K.2b without K.Low.C(iii) covers only designs natively in the blocking-only fragment.

**Definition K.Id (committed-identity state and serial frontier).** For an execution prefix  $p$  and process  $P$ ,  $Cmt\_P(p)$  is the set of dynamic source-operation identities of  $P$  that have process-facing committed in  $p$ , and  $Push\_c(p)$ ,  $Pop\_c(p)$  are the per-endpoint commit ordinals at channel  $c$ . At a terminal cutpoint  $\sigma$  with  $K_\sigma = \langle E, w, H_\sigma, O_\sigma \rangle$ , write  $CmtRec\_P(K_\sigma)$ ,  $PushOrdRec\_c(K_\sigma)$ , and  $PopOrdRec\_c(K_\sigma)$  for the same facts reconstructed from  $H_\sigma$ . Because operations on a single message-passing interface issue and commit in source order (R4/E3 with FIFO transfer order and point-to-point ownership), the per-channel per-endpoint commit ordinal of any fixed dynamic operation is the same in every admissible execution in which it commits; per-channel counts therefore identify committed transfers by identity through Lemma K.2b.P1. Across distinct interfaces, however, Policy L may commit a source-later operation while a source-earlier operation remains uncommitted, so  $Cmt\_P(p_L)$  need not be a source-order prefix, and a scalar commit count identifies neither the committed set nor the frontier under Policy L. Under Policy S,  $Cmt\_P(p_S)$  is always a source-order prefix of  $P$ 's dynamic operation sequence; in the blocking-only fragment every commit advances the frontier, and  $SerFront\_P(p_S)$  —  $P$ 's serial frontier — is its source-minimal uncommitted blocking operation, with every source-earlier operation committed. All cross-policy progress comparisons identify dynamic operations through Definition K.OpKey and remain statements over operational prefixes, keyed committed-identity sets, and per-channel ordinals; the reconstructed KObs projections are used only when a terminal cutpoint is named. Scalar counts are consumed only as inequalities (Lemma K.2b.P). The earlier identification of a process's liberal frontier as operation  $prog\_P + 1$  was unsound and remains retired.

Definition K.CVPred and K.CV-WF (well-founded value/identity provenance). Fix a K.BF theorem instance  $I$ , its common OpKey\_I universe of Definition K.OpKey, and its K.CV discharge. Let CVFact\_I be the common semantic fact-key universe used by the K.CV proof. It contains at least OpFact( $\kappa$ ) for  $\kappa \in OpKey\_I$ , whose deterministic value is an optional typed operation descriptor: none when the fixed predecessor/control facts exclude that keyed occurrence, and some(desc) when  $\kappa$  is selected, with desc recording operation kind/site identity, typed boundary/direction, and any Push payload; ObsFact( $u$ ), recording the value or status of each explicitly modeled observation that K.CV permits to influence a later operation; and XferFact( $c, n$ ), the distinguished ObsFact for the payload of the  $n$ -th committed transfer at explicit boundary  $c$ . Fixed constants, fixed stimulus values, and witness-fixed choices are roots rather than recursive facts. For an admissible execution  $p$ , Occurs\_ $p$ ( $\kappa$ ) means Realize\_ $p$ ( $\kappa$ ) is defined; whenever Occurs\_ $p$ ( $\kappa$ ), OpFact( $\kappa$ )=some(desc) for that realization. The K.CV certificate supplies, for every non-root fact  $b$ , a finite predecessor set Pred\_CV( $b$ ) and a deterministic evaluator Eval\_ $b$ .

Occurrence closure is explicit: whenever a fact  $b$  occurs in an admissible execution, every  $a \in \text{Pred\_CV}(b)$  also occurs in that execution; and whenever the predecessor facts are available,  $\text{Eval\_b}$  determines the same fact value from the fixed roots and those keyed predecessor values. For  $\text{OpFact}(\kappa)$ , this evaluator includes the control/branch/iteration selection needed to determine none versus some(desc), not merely the fields of an already-assumed occurrence. The certificate shall validate this provenance against  $\text{Core.RegSeq/Core.ReachPrep}$  and the applicable explicit causal mechanisms; for  $\text{XferFact}(c,n)$ , it also validates the producer  $\text{OpKey}$  attribution and ordinal relation using point-to-point ownership,  $\text{R4/E3}$ ,  $\text{E4/FIFO}$  or rendezvous semantics, and the fixed elaboration. Define  $a \prec_{\text{CV}} b$  iff  $a \in \text{Pred\_CV}(b)$ . The same  $\text{OpKey/CVFact}$  keys, predecessor sets, and evaluators are bound to the fixed source, elaboration, stimulus, witness, and reset schema across the admissible executions compared by Lemmas  $\text{K.2b.P0/P1}$ ; they are not reconstructed from incidental commit timing. The  $\text{K.CV}$  discharge shall additionally supply or derive  $\text{K.CV-WF: WellFounded}(\prec_{\text{CV}})$ . A checked  $\text{CVRank}$  into the natural numbers or another well-founded lexicographic order, with  $a \prec_{\text{CV}} b$  implying  $\text{CVRank}(a) < \text{CVRank}(b)$ , is a sufficient certificate form. No  $\prec_{\text{CV}}$  edge is introduced merely because two requests mutually enable a rendezvous, share a cycle, or are related by the broad Appendix-G causal-dependency graph  $\rightarrow$ . Thus a legal cycle in  $\rightarrow$  does not imply a cycle in  $\prec_{\text{CV}}$ . Conversely, an actual circular value/control provenance for which no  $\text{K.CV-WF}$  certificate exists is outside  $\text{K.CV}$  and therefore outside the  $\text{K.2b/K.2c}$  serial-reduction route.

Comparison-availability closure (normative  $\text{K.CVPred}$  serial-route clause). For every blocking  $\text{OpFact}(\kappa)$  consumed by Lemma  $\text{K.2b.P0}$ ,  $\text{K.2b.R}$ , or  $\text{K.2b.P}$ , the  $\text{K.CV}$  certificate shall additionally establish cross-policy predecessor availability for the least-missing/first-overtake comparison. Fix  $\text{Policy-S}$  and  $\text{Policy-L}$  comparison prefixes in one  $\text{K.DomResetScope DomEpoch}$ , with  $\kappa$  reached on the  $\text{Policy-S}$  side. Once the  $\text{Policy-L}$  side contains, with the common keyed values, every source-earlier blocking operation that  $\text{Policy S}$  requires to have committed before  $\kappa$  may issue and every dominance-domain endpoint-transfer fact that must precede  $\kappa$  in the selected serial prefix, and no still-frozen  $\text{Core.16}$  frontier of  $\kappa$ 's owning process that is key-earlier than  $\kappa$  prohibits the owner-local preparation, there exists a finite phase-legal  $\text{Policy-L}$  extension that preserves the frozen component and fixed  $\text{DomEpoch}$ , introduces no unmatched dominance-domain endpoint transfer, and contains every  $a \in \text{Pred\_CV}(\text{OpFact}(\kappa))$  that occurs in the  $\text{Policy-S}$  prefix with the same keyed value. An  $\text{XferFact}$  predecessor may be discharged only by showing that it is already one of the matched earlier serial transfers. A non-transfer  $\text{OpFact/ObsFact}$  predecessor is discharged recursively along the checked  $\text{CVRank}$  using its certified  $\text{Core.ReachPrep/observation reach}$  rule; after occurrence is established, Lemma  $\text{K.2b.P1}$  supplies value/identity equality. Recursive  $\text{Core.16 reach-legality}$  disposition (normative). Before any recursive non-transfer predecessor-reach extension is accepted, the  $\text{K.CVPred}$  certificate shall record one of the following dispositions for that extension. (a) If the predecessor is a blocking  $\text{OpFact}(\kappa_a)$  whose materialization performs source preparation/evaluation or another new launch, the comparison shall establish the same no-active-key-earlier condition used by  $\text{K.CompareAdvanceLegal}$  after the lower-ranked predecessor facts of  $\kappa_a$  are available. For an ordinary/non-elaborated or otherwise comparability-qualified singleton frontier, Lemma  $\text{K.CompareAdvanceLegal-Ord}$  may be cited pointwise only when, at the recursive comparison prefix, every blocking transfer that the selected  $\text{Policy-S}$  semantics requires to precede  $\kappa_a$ 's Issue is already present in the matched liberal transfer prefix with the required keyed identity/ordinal; none of those prerequisite transfers may be created by the predecessor-materialization extension itself. If that qualification test is not established, the pointwise lemma citation is unavailable and the certificate must instead supply independent owner/frontier  $\text{Core.16}$  legality for the predecessor reach or reject the candidate. An explicit  $\text{Sys\_B}^{\text{elab}}$  multi-frontier requires checked elaboration/order evidence. Only the reach/launch half is consumed when  $\kappa_a$  is materialized solely as a provenance predecessor. (b) If an  $\text{ObsFact}(u)$  reach rule consists only of observation, settling, or other actions that launch no new source-later work and issue no new message request, the reach rule shall certify that extension  $\text{Core.16-inert}$ .

(c) If an ObsFact or other non-transfer predecessor reach rule requires source-control advance or another launch not represented by a blocking OpFact key covered by clause (a), its reach certificate shall supply the equivalent owner/frontier proof that no active Core.16 frontier is source/key-earlier than each launched operation. The adjective “phase-legal” is not a substitute for one of these explicit dispositions. If no applicable disposition can be proved, comparison availability is false for  $\kappa$  and the candidate is outside the K.CV/K.2b serial route. The recursion is well-founded because CVRank strictly decreases on Pred\_CV edges. This clause supplies occurrence/reachability only; K.2b.P1 remains conditional on occurrence and does not by itself establish predecessor availability. The extension-preservation requirement is global over the frozen component, not merely owner-local: if a required non-transfer predecessor is produced by another process in D and its certified reach/observation rule would have to advance or release that process’s frozen frontier, issue work prohibited by Core.16/K.JointFreeze, change DomEpoch, or otherwise violate preservation of D, then the comparison-availability clause is false for that  $\kappa$  and the candidate is outside the K.CV/K.2b serial route. The owner-local Core.16 caveat therefore does not authorize materialization through another frozen process. K.CVPred owns this recursive predecessor-availability obligation; K.CompareAdvanceLegal remains the named coverage condition for the main K.2b.P0/K.2b.R comparison key, while Lemma K.CompareAdvanceLegal-Ord or equivalent elaboration-specific no-key-earlier evidence may be reused pointwise inside the K.CVPred recursion as specified above.

Lemma K.2b.P1 (keyed dataflow determinacy) — restated with proof. In the blocking-only fragment, for the fixed instance, stimulus, witness choices, and OpKey\_I universe: for every  $\kappa \in \text{OpKey\_I}$ , any two admissible executions—of either policy—in which Occurs\_p( $\kappa$ ) holds realize  $\kappa$  with the same operation kind/site identity, typed boundary/direction, and any K.CV-determined Push payload; and for every channel c and ordinal n, the payload of the n-th committed transfer at c is the same in every admissible execution in which that transfer fact occurs. Proof. Strengthen the induction claim to all facts in CVFact\_I and perform well-founded induction on  $\leq_{\text{CV}}$  using K.CV-WF; this induction does not use the broad causal-dependency graph  $\rightarrow$ . Fix a fact b and assume the execution-independent equality claim for every  $a \leq_{\text{CV}} b$ . By provenance adequacy in Definition K.CVPred, every admissible execution in which b occurs evaluates b with the same deterministic Eval\_b from the same fixed roots and the same keyed predecessor facts Pred\_CV(b). Occurrence closure supplies an occurrence of every  $a \in \text{Pred\_CV}(b)$  in each such execution; each predecessor has smaller  $\leq_{\text{CV}}$  rank and therefore has the same value/identity by the induction hypothesis. The fixed roots are identical by the theorem-instance stimulus/witness binding. Hence b has the same value/identity in every execution in which it occurs. For OpFact( $\kappa$ ), this yields the same typed realization fields and any Push payload for the common operation key  $\kappa$ ; for XferFact(c,n), the certified producer-key attribution plus point-to-point ownership, R4/E3, and E4/FIFO or rendezvous ordinal semantics make the n-th transfer payload the deterministic value of its lower-ranked producer/provenance facts. Other K.CV observation kinds are handled by their explicitly modeled deterministic Eval\_b equations. Same-cycle rendezvous request coincidence is an enablement condition only and is not a  $\leq_{\text{CV}}$  value predecessor, so it creates no induction cycle.  $\square$  This lemma is deliberately conditional on occurrence; existence/reachability of the corresponding operation in the other execution is the separate content of Lemma K.2b.P0. At any named terminal cutpoint, the determined committed keys, payload ordinals, and process-local replay facts are exactly the corresponding projections of H in its KObs record; this observation does not replace the operational induction.

Lemma K.2b.P0 (keyed next-operation occurrence and reachability) — Statement. Fix a common key  $\kappa \in \text{OpKey\_I}$  owned by process P that is reached in the selected Policy-S execution; fix the candidate component D, its K.2b.E activation cutpoint  $\sigma_{\text{L}}$  and selected  $\{f_{\text{P}}\}$ ; and fix Policy-S/Policy-L comparison prefixes  $\pi_{\text{S}}$  and  $\pi_{\text{L}}$  within the same K.DomResetScope DomEpoch for that candidate. By the K.CVPred comparison-availability clause, first extend  $\pi_{\text{L}}$  if necessary by a finite phase-legal segment that

preserves the fixed DomEpoch and frozen component and adds no unmatched dominance-domain endpoint transfer, obtaining a comparison prefix, renamed  $\pi\_L$  below, in which every K.CV predecessor fact required by OpFact( $\kappa$ ) is present with the same keyed value as in  $\pi\_S$ . Using that extended comparison prefix, assume every source-earlier key that Policy S requires to complete before  $\kappa$  may issue has committed as required. Before taking any further Core.ReachPrep/source-evaluation step to reach  $\kappa$ , assume the applicable K.CompareAdvanceLegal\_I( $D, \sigma\_L, \{f\_P\}$ ) coverage for  $\kappa$  on this comparison tuple; for ordinary/singleton frontiers Lemma K.CompareAdvanceLegal-Ord supplies it, while any explicit Sys\_B^elab multi-frontier owner consumed by K.2b.P0 requires checked elaboration-specific evidence regardless of whether  $P \in D$ . Then either  $\kappa$  is already realized/reached in  $\pi\_L$  or  $\pi\_L$  has a finite phase-legal Policy-L extension, within the same DomEpoch, that reaches  $q\_L := \text{Realize}(\kappa)$ . The realized  $q\_L$  has the operation kind/site, typed boundary/direction, and payload determined by OpFact( $\kappa$ ). This lemma establishes occurrence/reachability only; request assertion and persistence are supplied by K.2b.R.

Proof. Lemma K.2b.P1 alone cannot prove this conclusion because P1 is conditional on occurrence. Here the K.CVPred comparison-availability clause makes every input of Eval\_OpFact( $\kappa$ ) available in the liberal execution. Because the selected Policy-S execution reaches  $\kappa$ , its evaluator value is some(desc); determinism of the common K.CV evaluator therefore gives the same some(desc) value from the equal keyed predecessors on the liberal side, establishing that the same keyed occurrence is control-selected rather than an untaken-branch artifact. Definition K.OpKey supplies that potential process-local occurrence independently of execution, and realization preserves source order. The K.CVPred predecessor-materialization extension has already been certified separately to preserve the frozen component. Before advancing the liberal preparation cursor for  $\kappa$  itself, apply K.CompareAdvanceLegal. If Core.16 is active, its conclusion excludes any active frontier key-earlier than  $\kappa$ ; by the broader no-new-launch clause of Core.16(2), the finite dependency-independent source-evaluation/preparation needed to reach  $\kappa$  is therefore not prohibited as work launched past a blocked point. Core.ReachPrep together with K.BF bounded local progress may then advance P through that finite local computation, and Core.CycleChoiceExt supplies a typed completion whenever the finite reach segment crosses a source-cycle boundary. Thus  $\kappa$  has the claimed finite liberal realization/reach extension, with the typed fields/value fixed by OpFact( $\kappa$ ) and K.CV/P1. The subsequent request Issue is not part of this lemma; K.2b.R consumes the same K.CompareAdvanceLegal fact for that second Core.16(2) obligation.  $\square$

Lemma K.2b.R (serial-prefix request reachability) — Statement. Under K.ResetEpoch.Continue and K.TerminalDisposition.Continue, let D be the frozen component associated with the infinite fair continuation  $\bar{p}\_L$  of Lemma K.2b.E, and let  $q\_S$  be a blocking endpoint operation of process P reached by the selected Policy-S execution, with common key  $\kappa := \text{Key}(q\_S)$ . By the definition of K.DomResetScope, P and  $q\_S$ 's explicit boundary belong to the dominance scope whenever  $\kappa$  is consumed by this comparison; let  $e\_{\{P, \kappa\}}$  denote the corresponding endpoint/boundary reset epoch within the fixed DomEpoch. Fix Policy-S and Policy-L comparison prefixes  $\pi\_S$  and  $\pi\_L$  immediately before a first alleged Policy-S overtake, with  $\pi\_L$  a prefix of  $\bar{p}\_L$ . Assume every source-earlier key that Policy S requires before  $\kappa$ , every endpoint transfer earlier than  $\kappa$  in the fixed serial commit order and K.2b.P dominance domain, and every K.CV predecessor fact required by OpFact( $\kappa$ ) are present in  $\pi\_L$  with the keyed identities/values supplied by the strengthened K.2b.P transfer/provenance invariant; its K.CV predecessor portion is obtained from the K.CVPred comparison-availability closure. Assume also the applicable K.CompareAdvanceLegal\_I( $D, \sigma\_L, \{f\_P\}$ ) coverage for this comparison tuple; for ordinary/singleton frontiers Lemma K.CompareAdvanceLegal-Ord supplies it, while any explicit Sys\_B^elab multi-frontier owner consumed by K.2b.P0/K.2b.R requires checked elaboration-specific evidence regardless of whether  $P \in D$ . If  $\kappa$ 's liberal realization has not committed in  $\pi\_L$ , then  $\bar{p}\_L$  has a finite extension of  $\pi\_L$ , with the same DomEpoch and endpoint epoch, in which the attributed endpoint request for  $q\_L := \text{Realize}(\kappa)$  is asserted. On this K.BF reset-free, terminal-free comparison branch, once

asserted, Lemma K.BF-Persist retains that request as the same pending keyed dynamic request until  $q_L$  commits. No reachability or nonwithdrawal claim crosses a dominance-affecting RESET or a declared-terminal branch;  $K.ResetEpoch.DirectFail$  and  $K.TerminalDisposition.DirectFail$  are handled directly through  $K.RespFail$ .

Proof. Under Policy S, every source-earlier blocking operation of  $P$  required by the serial discipline has committed before  $q_L$ 's request may issue. By the comparison hypothesis, those committed operation keys, returned values, earlier endpoint-transfer facts, and the remaining  $K.CV$  predecessors of  $OpFact(\kappa)$  also occur with the same keyed values in the liberal prefix. Lemma K.2b.P0 therefore supplies the corresponding liberal realization  $q_L := Realize(\kappa)$ , either already reached or after a finite phase-legal local extension; Lemma K.2b.P1/ $K.CV$  supplies its common typed identity fields and payload. Because  $P$  and  $q_L$ 's explicit boundary belong to  $K.DomResetScope$  and this comparison remains within the fixed  $DomEpoch$ ,  $q_L$  is in the corresponding endpoint/boundary reset epoch  $e_{\{P,\kappa\}}$ . If  $\pi_L$  has already issued  $q_L$ , nonwithdrawal gives the conclusion. Otherwise reuse  $K.CompareAdvanceLegal$ , whose reach/launch half was already consumed by  $K.2b.P0$  and whose Issue/assertion half applies now. If  $P$ 's applicable active message frontier is ordinary/non-elaborated or otherwise certified singleton, Lemma K.CompareAdvanceLegal-Ord derives the required Core.16 compatibility from the least-missing/first-overtake invariant, E3/source-order prefix closure, and the fact that any genuinely source-earlier required blocking transfer is already matched on the liberal side. This argument does not depend on  $P \in D$ . If an explicit  $Sys\_B^{elab}$  package gives  $P$  a multi-frontier on the comparison prefix, the explicit  $K.CompareAdvanceLegal$  certificate supplies the corresponding result, again regardless of membership in  $D$ . Thus, whenever Core.16 is active for  $P$ , no active frontier item is key-earlier than  $\kappa$ , and the specific no-new-Issue clause of Core.16(2) does not suppress  $q_L$ . This is the second half of the same comparison-advance fact that justified  $P0$ 's preparation launch. The former shortcut " $P \notin D$ , so no freeze restriction applies" is not used: a process outside  $D$  may itself acquire a fully disabled frontier and is covered by the same comparison-advance interface. Hence  $q_L$  is either already asserted or remains legally IssueEnabled under Core.16 at every applicable frozen comparison prefix. Policy L admits the issue because it admits every Policy-S issue and assertion of a persistent blocking request does not require the channel transfer itself already to be enabled. At every finite comparison prefix on which the relevant frozen requests remain pending,  $P$  and  $q_L$ 's explicit boundary are members of  $K.DomResetScope$  by the comparison-domain definition, so  $K.ResetEpoch.Continue$  forbids a RESET affecting either as the next transition and preserves the same  $DomEpoch$  and  $e_{\{P,\kappa\}}$ ;  $K.TerminalDisposition.Continue$  makes the corresponding frozen-prefix final state nonterminal. Because  $q_L$  remains legally IssueEnabled at every applicable L1 opportunity, Core.IssueFair therefore schedules the assertion after finitely many such opportunities; the infinite fair continuation cannot postpone it forever. Lemma K.BF-Persist together with nonwithdrawal then retains the attributed request as the same pending keyed dynamic instance until  $q_L$  commits: the exhaustive Core.IC lifecycle admits no independent nonterminal withdrawal/cancellation transition,  $K.ResetEpoch.Continue$  excludes an affecting RESET on the comparison prefix, and  $K.TerminalDisposition.Continue$  excludes a declared-terminal escape. RW2 would end the realized dynamic instance at an affecting RESET, which is why the conclusion is expressly  $DomEpoch$ - and endpoint-epoch-local and  $K.ResetEpoch.DirectFail$  is handled separately.  $\square$

**Lemma K.2b.P (deterministic serial progress dominance) — Statement.** Under  $K.ResetEpoch.Continue$  and  $K.TerminalDisposition.Continue$ , fix a Policy-L prefix and the infinite fair reset-free, terminal-free continuation  $\bar{p}_L$  supplied by Lemma K.2b.E, and let  $p\_S^{det}$  be the Policy-S execution selected by  $SDet$ , with separate  $K.CompleteS$  evidence establishing maximality, fairness, and completeness. Every endpoint-transfer identity and payload committed in  $p\_S^{det}$  whose owning endpoints/boundary belong to  $K.DomResetScope$  for this candidate, while the comparison remains in the fixed  $DomEpoch$ , also occurs in  $\bar{p}_L$ . The conclusion is transfer inclusion over the dominance comparison footprint; it does

not assert identical cycle timing, the same global interleaving, or cross-epoch inclusion for reset activity framed outside that scope.

**Proof.** Order the endpoint-transfer commits of  $p\_S^{\text{det}}$  in the  $K.\text{DomResetScope}$  dominance domain and fixed  $\text{DomEpoch}$  by operational transition index, using the fixed normalization and boundary-action order to break any same-cycle ties. The tie-break orders identities only for the least-missing-transfer induction: every occupancy and enablement fact below is evaluated from the common begin-of-cycle state, so a same-cycle Push cannot supply a Pop and a same-cycle Pop cannot create capacity for a Push under non-fall-through E4. Suppose the conclusion is false and choose the least serial transfer  $T$  absent from the complete liberal continuation. Let  $c$  be the explicit E-typed abstract boundary named by  $T$  in  $\text{Sys}_K$ —including any staged, bundle, join, guard, or epoch-gate boundary introduced by elaboration or lowering—and let  $n$  be  $T$ 's ordinal in the applicable endpoint direction and reset epoch. Every earlier serial transfer occurs in  $\bar{p}_L$  by minimality. Strengthen the least-missing-transfer induction with a provenance-availability invariant: for each endpoint key  $\kappa$  needed by  $T$ , every fact in  $\text{Pred\_CV}(\text{OpFact}(\kappa))$  that occurs in the Policy-S prefix is available with the same keyed value in a liberal comparison prefix before any unmatched dominance-domain endpoint transfer. The  $K.\text{CVPred}$  comparison-availability closure establishes this by well-founded recursion on  $\text{CVRank}$ .  $\text{XferFact}$  predecessors are certified to be strictly earlier required serial transfers and are therefore present by minimality; lower-ranked non-transfer  $\text{OpFact}/\text{ObsFact}$  predecessors are materialized by their certified finite  $\text{Core.ReachPrep}/\text{observation-reach}$  extensions without disturbing the frozen component, fixed  $\text{DomEpoch}$ , or matched transfer prefix, and  $K.2b.P1$  fixes their values after occurrence. Thus the full predecessor-agreement hypotheses of  $K.2b.P0$  and  $K.2b.R$  hold for the endpoint keys of  $T$ ; the separately assumed  $K.\text{CompareAdvanceLegal}$  coverage supplies their  $\text{Core.16}$  preparation/launch and Issue/assertion side condition for each consumed owner/key. Lemma  $K.2b.P0$  supplies the corresponding liberal endpoint occurrence/reachability, Lemma  $K.2b.P1/K.CV$  gives the same dynamic endpoint identity and payload in both executions, and Lemma  $K.2b.R$  supplies the persistent liberal request for every endpoint needed by  $T$  unless  $T$  has already committed, which would contradict the choice of  $T$ .

**Buffered Pop case.** Assume  $B(c) > 0$  and  $T$  is the  $n$ -th  $\text{Pop}_c$  transfer. Under the non-fall-through E4 semantics, the FIFO is nonempty before that Pop commits, so at least  $n$   $\text{Push}_c$  transfers have already committed before  $T$ . Those Push transfers precede  $T$  and therefore occur in  $\bar{p}_L$ . If the  $n$ -th  $\text{Pop}_c$  has already occurred there,  $T$  is not missing. Otherwise same-interface order gives  $\text{PopOrd}_c < n$ , while  $\text{PushOrd}_c \geq n$ ; hence  $\text{occ}_c = \text{PushOrd}_c - \text{PopOrd}_c > 0$ . The persistent Pop request supplied by  $K.2b.R$  is therefore  $\text{ChannelEnabled}$ . Because  $\bar{p}_L$  is an infinite fair continuation and the persistent request is continuously  $\text{ChannelEnabled}$ , B2/B3 force the transfer. Thus  $T$  occurs, a contradiction.

**Buffered Push case.** Assume  $B(c) > 0$  and  $T$  is the  $n$ -th  $\text{Push}_c$  transfer. Immediately before its serial commit there are  $n-1$  prior  $\text{Push}_c$  commits, and E4 enablement gives  $(n-1) - \text{PopOrd}_c < B(c)$ . Therefore at least  $m = \max(0, n - B(c))$   $\text{Pop}_c$  transfers precede  $T$ , and minimality places all of them in  $\bar{p}_L$ . If the  $n$ -th  $\text{Push}_c$  is absent from the liberal continuation, same-interface order gives  $\text{PushOrd}_c \leq n-1$ . Consequently  $\text{occ}_c = \text{PushOrd}_c - \text{PopOrd}_c \leq (n-1) - m \leq B(c) - 1$ , so the channel is nonfull. The persistent Push request is  $\text{ChannelEnabled}$  throughout the relevant suffix of the infinite fair continuation, so B2/B3 force its commit. Again  $T$  occurs, a contradiction.

**Rendezvous case.** Assume  $B(c) = 0$ . Then  $T$  is the atomic  $n$ -th  $\text{Push}_c/\text{Pop}_c$  rendezvous transfer. If  $T$  is absent, neither endpoint can have committed it separately. Apply Lemma  $K.2b.R$  to both endpoint operations. After the later of the two finite assertion points, both attributed requests are present and remain present by nonwithdrawal. E4 therefore makes the rendezvous transfer  $\text{ChannelEnabled}$  with the payload fixed by  $K.2b.P1/K.CV$ . Because  $\bar{p}_L$  is infinite fair and both persistent endpoint requests keep the rendezvous  $\text{ChannelEnabled}$ , B2/B3 force the atomic commit. Hence  $T$  occurs, a contradiction.

Conclusion. Every possible type of least missing transfer yields a contradiction. Therefore every endpoint transfer committed by  $\rho\_S^{\text{det}}$  in the  $K.\text{DomResetScope}$  dominance domain while  $\text{DomEpoch}$  is fixed occurs, with the same dynamic identity, channel ordinal, and payload, in  $\bar{\rho}\_L$ . The proof establishes only this one-way dominance inclusion;  $\bar{\rho}\_L$  may contain additional liberal transfers and may order independent transfers differently.  $\square$

**Composition with Lemma K.2b.E.** Lemma K.2b.P is a conditional contradiction argument on the reset-free  $K.\text{ResetEpoch}.\text{Continue}$ , terminal-free  $K.\text{TerminalDisposition}.\text{Continue}$  branch: if a transfer committed by the selected Policy-S execution were absent from the frozen liberal continuation, the preceding cases would force that transfer to occur there. Lemma K.2b.E keeps each selected frozen transfer pending and therefore absent. Consequently, K.2b.P implies that the selected Policy-S execution cannot commit that frozen transfer, or overtake it with a source-later transfer. This is the implication used in Lemma K.2b.Q, Step 1; the two lemmas are complementary rather than contradictory.

Lemma K.2b.E (fair complete frozen extension)—Proof. At every finite prefix constructed below while the selected  $f\_P$  remain pending,  $K.\text{ResetEpoch}.\text{Continue}$  forbids any RESET whose scope intersects  $K.\text{DomResetScope}$  as the next transition; RW2 therefore never clears the selected requests on this branch.  $K.\text{TerminalDisposition}.\text{Continue}$  makes every such frozen-prefix final state nonterminal, so the construction has no finite declared-terminal escape. Lemma K.BF-Persist supplies the remaining interruption-completeness fact by applying the exhaustive Core.IC blocking lifecycle to the selected WFG-relevant K.BF blockers. Thus every persistence and dynamic-identity claim below remains within the fixed  $\text{DomEpoch}$  for  $K.\text{DomResetScope}$  and the same selected frozen request instances; RW5-framed resets disjoint from that scope remain permitted. Start from the finite Core.LAdm-admissible prefix  $\rho\_L$  and its drain-closed component  $D$ , retaining the persistent requests  $f\_P$  already present at  $\sigma\_L$ . First discharge  $K.\text{EnvMF}$ . Under  $\text{StandardEnv}/B1\text{-avail}$ , Lemma K.EnvX-Auto makes every disabled environment-facing co-frontier permanently unavailable. Under a timed/reactive or otherwise nondefault environment,  $K.\text{EnvMF}$  certifies that each  $P \in D$  has exactly one frontier item, namely the selected internal  $f\_P$ , so no environment-facing or other co-frontier can later become enabled, deactivate Core.16, and release the component. In an ordinary/single-frontier component, invoke Lemma K.JointFreeze-Ord to obtain the same no-release result, including both the broader no-new-launch and the specific no-new-Issue consequences of Core.16(2). In an admitted StandardEnv multi-frontier component, that conclusion is supplied instead by K.JointFreeze: JF1 covers cross-process releasing issues even when the elaborated frontier contains incomparable members, and JF2 covers permitted drain/elaboration actions. Hence no releasing operation is an FP1 issue candidate and no permitted completion action can dissolve  $D$  merely through the multi-frontier incomparability that Core.16 alone does not order. Already-issued source-later work is finite at the cutpoint, may drain under the selected progress semantics, and cannot replenish the frozen set. Each frozen  $f\_P$  itself remains ChannelDisabled, so it is never an FP2/B2 commit candidate. Define Inv for a finite extension prefix to mean: the same  $D$  and selected  $f\_P$  remain pending and attributed to the same dynamic instances; the  $K.\text{DomResetScope} \text{ DomEpoch}$  is unchanged; Core.16 remains active wherever required; no prohibited replenishing source-later work has been launched; and the prefix remains Core.LAdm-legal with the fixed witness/environment/arbitration choices. The initial prefix satisfies Inv. Assume a constructed prefix satisfies Inv. The Core.FairPick witness supplied as the constructive discharge of the existing Core.IssueFair/B2/B3 obligations selects only phase-legal nonfrozen issue and boundary work; K.JointFreeze-Ord or K.JointFreeze, as applicable, exclude any releasing new launch/issue, K.EnvMF excludes an environment co-frontier escape, and Lemma K.BF-Persist plus nonwithdrawal retain each frozen request.  $K.\text{ResetEpoch}.\text{Continue}$  excludes an affecting RESET at every intermediate frozen prefix, and  $K.\text{TerminalDisposition}.\text{Continue}$  makes the resulting frozen-prefix state nonterminal. Therefore the two-stage construction of Lemma Core.FairCycleDovetail applies: its first Core.CycleChoiceExt call obtains an actual finite L2 settle path after the selected L1 issue segment; BoundaryPick is evaluated

only at that settled state; its second  $\text{Core.CycleChoiceExt}$  call produces a typed  $\text{CycleResult}$  containing both selected fair segments. Every additional action in the completion is phase-legal and, by  $\text{Core.Drain/K.Drain}$  together with Lemma  $\text{K.JointFreeze-Ord}$  on the ordinary/singleton branch or  $\text{K.JointFreeze}$  on the admitted  $\text{StandardEnv}$  multi-frontier branch, cannot replenish or release  $D$ . Hence the successor prefix again satisfies  $\text{Inv}$ . Iterate these typed  $\text{Inv}$ -preserving  $\text{CycleResult}$  successors. Because every  $\text{Inv}$  state is nonterminal on the nested  $\text{Continue}$  branch, dependent choice now yields an infinite  $\text{CompleteCycle}$  continuation  $\bar{p}_L$ .  $\text{FP4}$  of  $\text{Core.FairPick}$  proves  $\text{Core.IssueFair}$  and  $\text{B2}$  on  $\bar{p}_L$ , and  $\text{FP5}$  supplies the  $\text{B3}$  channel-progress cases; the separately assumed  $\text{B5}$  closure property applies to this operational continuation. If a selected cycle has no  $\Sigma$  commit, its typed completion is a  $\text{Core.SilentCycle}$  rather than an  $\epsilon/\text{identity}$  step, and the  $\text{EnvStep}$  component of  $\text{CycleResult}$  advances the time-aware environment. Thus  $\bar{p}_L \in \text{Exec}_L^{\text{adm}}(\text{I})$ , extends  $p_L$ , and preserves every  $f_P$  pending with the same dominance-scope  $\text{DomEpoch}$ .  $\text{K.ResetEpoch.DirectFail}$  and  $\text{K.TerminalDisposition.DirectFail}$  do not invoke this construction; each is discharged by its independently certified direct  $\text{K.RespFail}$  branch.  $\square$

Lemma  $\text{K.2b.Q}$  (deterministic bounded-response extraction)—Statement. Assume  $\text{SDet}$ , separate  $\text{K.CompleteS}$  evidence for  $p_S^{\text{det}}$ ,  $\text{K.RespCov}$ ,  $\text{K.ResetEpoch.Continue}$ ,  $\text{K.TerminalDisposition.Continue}$ , Lemma  $\text{K.2b.E}$ , and  $\text{K.2b}$ 's remaining premises. Let a  $\text{Policy-L}$  prefix reach  $\sigma_L$  with  $\text{DeadKRec}^W(\text{K}_L)$ . Then the  $\text{SDet}$ -selected execution  $p_S^{\text{det}}$  has a finite prefix containing a non-waived  $\text{K.RespFail}$  witness. No conclusion about  $\text{Dead}_K$ , an owner terminal, or a closed serial  $\text{WFG}$  component is required.

Proof, Step 1 (joint frontier freeze and permanent serial blockage). Let  $D$  be the non-waived  $\text{Dead}_K^W$  component reconstructed at  $\text{K}_L$ . In the single-frontier case, for each  $P \in D$  let  $x_{P^L}$  be its unique internal item in  $\text{FrontRec}_P(\text{K}_L)$  and  $f_P := \text{MsgOf}_P(x_{P^L})$ . Exact residual offers and source-prefix closure show that every process-facing blocking  $\text{Push/Pop}$  required by the  $\text{Sys}_K/\text{K.BF}$  serial prefix and source-earlier than  $f_P$  has committed by  $\sigma_L$ ; failed non-blocking polls are handled through their  $\text{WC}$ -selected return outcomes and are not claimed to have committed.  $\text{K.EnvMF}$  supplies either  $\text{StandardEnv/B1-avail}$ , with  $\text{K.JointFreeze}_I(D, \sigma_L, \{f_P\})$  for every admitted multi-frontier component, or, for a timed/reactive/nondefault environment, reduces this step to the single-internal-frontier case. Invoke Lemma  $\text{K.2b.E}$  to obtain the infinite fair complete  $\text{Core.LAdm}$  continuation in which all  $f_P$  remain pending.  $\text{K.ResetEpoch.Continue}$  applies at each finite frozen prefix and therefore prevents any relevant  $\text{RW2}$  reset from occurring as the next transition, so those requests are never cleared or re-identified during the comparison. Lemma  $\text{K.2b.E}$  has already established the required joint freeze for this same  $D$ . Ordinary/single-frontier cases use Lemma  $\text{K.JointFreeze-Ord}$ ; admitted  $\text{StandardEnv}$  multi-frontier cases use  $\text{K.JointFreeze JF1/JF2}$  to exclude releasing issues and release through permitted drain/elaboration actions.  $\text{K.2b.Q}$  does not re-derive that fact from an unqualified source-later argument. Lemmas  $\text{K.2b.R}$  and  $\text{K.2b.P}$  then prevent  $p_S$  from overtaking this continuation. Hence each  $P \in D$  eventually becomes permanently non-advancing at a process-facing  $\text{Policy-S}$  serial frontier  $x_{P^S}$  with operation  $g_P$  satisfying  $\text{Key}(g_P) \leq^{\text{K\_src}} \text{Key}(f_P)$  (the Appendix-K shorthand  $g_P \leq_{\text{src}} f_P$ ), and that frontier never process-facing commits.  $\text{K.BF}$  excludes an intervening pure-computation divergence. For an admitted  $\text{StandardEnv}$  multi-frontier component, the paper-level  $\text{K.JointFreeze}$  certificate is the soundness premise and  $\text{K-O2}$  mechanizes its  $\text{JF1/JF2}$  clauses; timed/reactive/nondefault multi-frontier components are outside the lemma.

Proof, Step 2 (coverage and non-waiver). By the default-total definition of  $\text{K.RespCov}$ , every dynamically reachable internal  $x_{P^S}$  produced by Step 1 already has a checkable proof disposition; no modal selection argument about whether its static site can become a serial blocker is permitted. A validated unreachability disposition cannot apply to the reached dynamic instance, and a validated static bounded-response disposition supplies the same finite-response fact as a literal monitor. If  $x_{P^S}$  is environment-facing and the instance uses  $\text{StandardEnv/B1-avail}$ , then  $g_P <_{\text{src}} f_P$  committed in the liberal prefix; its fixed consumption ordinal is therefore available in  $\text{Policy S}$ , and persistent request plus

Core.IssueFair and B2/B3 would force its commit, contradicting permanent blockage. Thus such an environment frontier cannot be the Step 1 blocker. For a timed/reactive or otherwise nondefault environment, K.EnvMF has already excluded every multi-frontier liberal component; any environment-facing serial blocker arising on  $p\_S^{\text{det}}$  requires an explicit K.Resp monitor or availability proof. Coverage-preserving waiver closure ensures that the original non-waived component D cannot map solely to waived dispositions. Therefore at least one non-waived covered response obligation triggers at a finite Policy-S cycle t and remains unsatisfied thereafter unless the process-facing response occurs; any reset or run-ending disposition is classified separately by Definition K.RespFail and the applicable reset/terminal branch.

Proof, Step 3 (finite response-failure witness). Let B be the finite bound of the monitor selected in Step 2. If  $p\_S$  continues through the boundary-action phase of cycle t+B without response, the first normalized cutpoint after that phase witnesses timeout. If  $p\_S$  reaches a declared maximal finite terminal state before that post-phase cutpoint while the monitor remains pending, the terminal prefix witnesses maximal-pending. On this reset-free branch these cases exhaust the complete artifact supplied by K.CompleteS: a nonterminating complete run eventually reaches the finite timeout bound, while a terminating complete run ends only at DeclaredTerminal. The matched-reset cancelled-pending case is the separate K.ResetEpoch.DirectFail branch handled directly in Lemma K.2b. Therefore  $p\_S$  contains a finite non-waived K.RespFail.  $\square$

Diagnostic note. The retired owner walk may still be run after K.RespFail to classify a likely root as an internal cycle, EnvStop, DoneStop, or another owner-chain condition. Example K.CE-5/H2 shows why that taxonomy cannot be used as an exhaustive proof premise: a complementary owner can remain live forever on unrelated channels while permanently under-supplying the demanded endpoint. Failure to classify a response failure affects diagnostics only, not Corollary K.2c.

Proof of Lemma K.2b. Apply K.Low.C if lowering is needed, then instantiate the shared K.InterruptionGate in the normative order. First evaluate K.ResetEpoch. On K.ResetEpoch.DirectFail, use the independently certified  $P \in D$ , selected liberal frontier  $f\_P$ , Policy-S serial frontier  $x\_P^{\text{AS}}$ , and  $g\_P := \text{Op}(x\_P^{\text{AS}})$  with  $g\_P \leq_{\text{src}} f\_P$  in the same pre-reset epoch. The associated non-waived K.Resp monitor has already triggered and remains pending before the first matched affecting RESET; RW2 clears that exact monitored dynamic instance at the RESET unit, so K.RespFail supplies cancelled-pending immediately. This branch does not invoke K.2b.Q Step 1. Only on K.ResetEpoch.Continue evaluate K.TerminalDisposition. On K.TerminalDisposition.DirectFail, the independently certified mapping supplies the same D-to-serial-frontier data, a finite prefix of  $p\_S^{\text{det}}$  with the mapped non-waived monitor already triggered and pending, and independently checked evidence that  $p\_S^{\text{det}}$  reaches the specified declared maximal terminal state while that monitor remains pending; K.RespFail therefore supplies maximal-pending directly. Only on K.TerminalDisposition.Continue invoke Lemma K.2b.E to obtain the infinite fair frozen continuation and then K.2b.R/P/Q to derive the response failure. Thus each DirectFail branch is discharged independently, and the dominance continuation is constructed only after both nested Continue checks succeed.  $\square$

**Lemma K.Ser (conditional serial trace replay). Statement.** Fix the selected instance Sys\_K, the fixed stimulus, and the fixed witness choices, and let  $p\_L$  be a finite Policy-L prefix admissible under Definition Core.LAdm ending at a source K-facing cutpoint  $\sigma\_L \in \text{KCutReach\_Sys\_K}(I)$  with  $K\_L = \langle E, w, H\_L, O\_L \rangle$ . Suppose: (a) commit-prefix condition — for every process P,  $\text{CmtRec\_P}(K\_L)$  is a source-order prefix of P's dynamic operation sequence; (b) serial-precedence acyclicity — the union of the causal dependencies of  $p\_L$  (per-channel transfer order and capacity dependencies, rendezvous atomic pairing, epoch-gate and guard dependencies, and process-local source order) with the Policy-S precedence constraints, requiring each source-earlier blocking operation of a process to commit strictly before a source-later one issues, is acyclic; and (c) terminal offer admissibility — every  $o \in O\_L$  is attributed to its owner's source-minimal uncommitted blocking item, equivalently  $\text{DrainBag\_Proc}(I)(K\_L)$  is empty, so

$\sigma\_L$  contains no source-later asserted request that Policy S would have withheld. Then there exists a finite admissible Policy-S execution of the same instance, stimulus, and witness with the same per-channel committed transfer sequences and payloads ordinal for ordinal, the same committed-identity sets and canonical process-local replay history  $H\_L$ , and therefore the same ProcRec, ChanRec, NextFrontRec, and source-relevant terminal values as  $K\_L$ . If a separate terminal-offer replay check also establishes the same residual-offer set  $O\_L$ , the two terminal cutpoints have equal KObs; equal O is not inferred from the committed history alone. Proof sketch. Linearize the finite acyclic combined precedence relation and replay commits in that order, inserting idle cycles to satisfy the strictly-earlier-cycle serial condition. The induction invariant — equal per-channel counts and payloads, equal process-local committed histories, equal channel state — is preserved at each step: source-earlier blockers have committed by the added precedence constraints, so each issue is Policy-S-legal; channel enabling conditions transfer by the equal-channel-state invariant; Lemma K.2b.P1 supplies operation-identity and payload agreement; and I.DR supplies per-process state agreement from equal replay histories. The optional equal-O conclusion requires one additional finite check that the terminal serial requests are asserted at the same typed boundaries and epochs. Mechanization note: a finite structural induction with no fairness or limit reasoning; mechanization is required before the lemma is cited as certified.

**Engineering note (serial replay of recorded traces; scope of Lemma K.Ser).** Lemma K.Ser characterizes when a recorded liberal or RTL-matched execution can be replayed under the conservative serial scheduler with identical committed histories — and, with the additional terminal-offer replay check, identical KObs. KObs alone is not a sufficient replay certificate. The recorded artifact must additionally contain the historical dynamic operation identities, Issue events, persistent-request intervals and attribution, failed non-blocking outcomes as witness selections, elaborated-boundary transfers, and arbiter selections needed to justify the acyclic operational replay; a committed-event history determines neither those issuance obligations nor the intermediate enablement facts.  $O\_L$  records only requests still outstanding at the terminal cutpoint. Conditions (a) and (c) fail exactly where the recorded run exploited eager issue; Examples K.CE and K.CE-2 are minimal failures, and their liberal histories are reproducible by no Policy-S execution. Lemma K.Ser is therefore deliberately not a substitute for Lemma K.2b: the non-serializable case is where the dominance content of the serial-reduction route resides. Mechanization obligations. K-O0: mechanize Definition K.OpKey, its partial per-execution realization maps and key-level  $<^K\_src$  order; mechanize Definition K.CVPred on the common CVFact\_I key universe, validate predecessor occurrence closure, the per-fact deterministic provenance equations, the serial-route comparison-availability closure, and the explicit recursive Core.16 reach-legality disposition for each non-transfer predecessor (pointwise ordinary/singleton blocking-OpFact proof only after the prerequisite-transfer qualification test, Core.16-inert observation reach, or equivalent owner/frontier launch proof), establish K.CV-WF (or an equivalent checked CVRank whose strict decrease proves it), and mechanize Lemma K.2b.P1 by well-founded induction on  $<\_CV$ ; do not use the possibly cyclic Appendix-G causal-dependency graph  $\rightarrow$  as that induction order. K-O1: mechanize Definition Core.IssueFair and the Core.16 compatibility fact. K-O2: mechanize Lemma K.JointFreeze-Ord and Definition K.JointFreeze, including JF1 coverage of every releasing launch/issue across incomparable  $Sys\_B^{\text{elab}}$  frontier members, JF2 preservation under permitted drain/elaboration actions, and process-facing monitor-trigger composition. K-O3: prove Lemma K.2b.P over the K.DomResetScope dominance domain with fixed DomEpoch on the reset-free, terminal-free infinite-fair branch, using the strengthened induction invariant that carries both all earlier serial endpoint transfers and every lower-ranked K.CV predecessor fact required to invoke K.2b.P0/K.2b.R; prove that each comparison-availability extension preserves the frozen component, introduces no unmatched dominance-domain endpoint transfer, and carries one of the required recursive Core.16 reach-legality dispositions rather than relying on “phase-legal” as an unanalyzed assertion. K-O4: mechanize Definition K.CompareAdvanceLegal and Lemma K.CompareAdvanceLegal-Ord; prove coverage for every K.2b.P0/K.2b.R-consumed owner/key in

$K.DomResetScope$ , including  $P \notin D$  and explicit  $Sys\_B^{elab}$  multi-frontier owners, and expose  $K.CompareAdvanceLegal-Ord$  as the reusable pointwise ordinary/singleton proof for qualifying blocking  $OpFact$  predecessor reaches required by  $K-O0$ , where qualification requires every Policy-S-required source-earlier blocking transfer for the predecessor key to be already present in the matched liberal transfer prefix before predecessor materialization and not created by that materialization extension; prove that the same condition discharges both  $Core.16(2)$  preparation/launch and Issue/assertion legality; enforce the  $JointFreeze/CompareAdvanceLegal$  consistency rejection rule; then validate the certificate-supplied finite  $S\_p^{dom}/S\_c^{dom}$  against  $K.DomResetScope$  conditions D1–D4, derive  $DomEpoch$ , mechanize Lemma  $K.2b.PO$ 's keyed occurrence/reachability result, and then prove Lemma  $K.2b.R$  using the shared  $K.InterruptionGate.Continue$  rule: instantiate  $Interrupt\_Reset$  as prefix-local legality of a RESET intersecting  $K.DomResetScope$  as the next proof-relevant transition/unit, with any executed occurrence recorded in the enclosing  $CycleResult.resetUnits$  and  $Interrupt\_Terminal$  as  $DeclaredTerminal$  of the frozen-prefix final state; prove both predicates prefix-locally, show every  $K.2b.PO/R/P$ -consumed endpoint lies in  $K.DomResetScope$ , and preserve  $DomEpoch$ -local request attribution through Lemma  $K.BF-Persist$ .  $K-O5$ : mechanize default-total  $K.RespCov$  and waiver closure.  $K-O6$ : mechanize the common  $K.InterruptionGate.DirectFail$  payload shape independently of  $K.2b.PO/R/P/Q$  Step 1, then prove the reset specialization yields cancelled-pending through matched RESET/RW2 and the terminal specialization yields maximal-pending through  $DeclaredTerminal$ .  $K-O7$ : mechanize the schema's  $K.BF$  exhaustiveness fact, anti-circularity rule, and the normative nesting  $K.ResetEpoch.Continue \rightarrow K.TerminalDisposition$ ; prove the  $Core.CycleClosure C2$  partial-cycle extension interface and Lemma  $Core.CycleChoiceExt$  for the selected source semantics; mechanize Definition  $Core.FairPick$ , including local obligation finiteness/keying, the actual fixed-scheduler/arbitration starvation proof,  $SyncFence$  prerequisite segments, and B3 exposure; mechanize Lemma  $Core.FairCycleDovetail$  with its two  $C2/CycleChoiceExt$  applications ( $L1$  selection  $\rightarrow$  provisional typed settle prefix  $\rightarrow$   $L3$  selection  $\rightarrow$  final typed  $CycleResult$ ) and the  $Inv$ -preservation argument used by  $K.2b.E$ ; then use dependent choice only over those already-typed successors and derive  $Core.IssueFair/B2/B3$  from  $FP4/FP5$ . Either  $DirectFail$  branch exits without invoking that construction. The supported environment branches remain those of  $K.EnvMF$ .  $K-O8$  is needed only for the optional diagnostic drain-normalization/ $Dead\_KS$  construction. These operational results shall be proved from  $Core.ReachPrep$ ,  $Core.CycleResult/Core.CompleteCycle/Core.CycleClosure/Core.CycleChoiceExt$ ,  $Core.16$ , Appendix C Issue/Commit linkage, RW2, nonwithdrawal,  $K\epsilon$ ,  $K.OpKey$ ,  $K.CV$  including  $K.CVPred/K.CV-WF$ , B1–B5,  $Core.IssueFair$ ,  $K.BF$ , point-to-point E4 ownership,  $K.EnvMF$ ,  $K.CompareAdvanceLegal$ ,  $K.InterruptionGate$  with  $K.DomResetScope$  and its two exhaustive  $K.BF$  instantiations,  $K.Low.C/K.Low.A5$ , and  $K.Resp$ —not from finite state-space reasoning.  $KObs$  selector facts do not replace operational monitor premises.

## Appendix L – Snooping Introduction, Implementation and Concerns

### Snooping Introduction

If designs are completely latency insensitive, verification of the pre-HLS versus post-HLS models is straightforward. However, if the design contains some components such as arbiters which have latency-sensitive behavior, and if that latency-sensitive behavior is in some cases externally visible to the DUT, then verification becomes somewhat more complex. Techniques can be applied to simplify verification.

Consider a design which has an arbiter like Matchlib toolkit example 09\*. The arbiter uses non-blocking PopNB() on its inputs, and blocking Push() on its output to emit the winner of the arbitration. Since the latency between the pre-HLS and post-HLS designs will differ, the order of transactions presented to the arbiter in the two scenarios will differ, and thus the order of the winners will differ. This may result in verification mismatches between the two models if the order of the winners is externally visible to the DUT.

To force the pre-HLS and post-HLS simulations to match, we can run the two designs side-by-side. We can snoop the inputs to the arbiter in the post-HLS model, and only allow the inputs to the pre-HLS arbiter to proceed when the corresponding inputs are seen in the post-HLS model. This will force the order of the inputs to be equivalent between the pre-HLS and post-HLS models, and thus both arbiters will pick the same winners. When this technique is used, the pre-HLS simulation will be throttled by the post-HLS simulation, but the overall verification will still work properly.

Another related example involves interrupt request signals feeding into an interrupt controller within a CPU model. Typically, each interrupt request signal is a single bit `sc_signal<>`, indicating that an interrupt request is pending. If the requests originate from accelerator blocks that are being synthesized through HLS, then because of the latency differences between the pre-HLS and post-HLS models, the order of interrupt requests arriving at the interrupt controller will differ between the two models, and this will likely result in verification mismatches if the differences are externally visible to the DUT. To force the order of the requests to match, we snoop the requests arriving at the controller in the post-HLS model, and only then allow the requests to be seen in the pre-HLS model.

### Simplified Snooping Approaches

In the snooping example directly above in which the arbiter is snooped, the entire post-HLS RTL DUT is simulated alongside the pre-HLS model to force the arbiter requests to be aligned in time. This is the most general case and assumes the input delays to the arbiter are difficult to determine via analysis.

Sometimes there are simpler cases where the input delays to the arbiter in the RTL DUT are easier to determine. For example, each input to the arbiter might arrive a fixed number of clock cycles after one of the primary inputs to the RTL DUT is pushed by the testbench. In this case, a much simpler model can be used to align the pre-HLS model with the post-HLS model. We can simply monitor the primary inputs to the pre-HLS model and then apply the fixed cycle delays to the pre-HLS arbiter inputs.

The two cases above illustrate two ends of a spectrum of possible approaches for extracting the delays from the RTL DUT. Between these two points there exist other possible approaches.

## Snooping Implementation

To enable perfect matching between the pre-HLS and post-HLS system simulations, latency-sensitive global signals and latency-sensitive non-blocking message-passing operations need to be synchronized between the two simulations if their latency differences are externally visible to the DUT.

As explained earlier in this appendix, in the general case the entire post-HLS DUT RTL must be run alongside the pre-HLS system to achieve alignment. Examples of signals that need to be synchronized include global interrupt request signals (as discussed earlier in this appendix) and rdy/vld signal pairs for non-blocking operations on message-passing channels. All such synchronization signals and their corresponding handshake signals must be level-stable:

- A latency-sensitive signal  $s$  is **level-stable** if, once  $s$  becomes 1 in cycle  $t$ , it must remain 1 until the corresponding handshake signal  $h$  is 1 in some cycle  $t' \geq t$ .

Once the set of signals that need to be aligned between the two simulations is identified, the general rule to implement snooping is simple:

- For each polarity-normalized Boolean commit-enabling snoop signal (for example, the vld/rdy handshake signals of snooped non-blocking operations, or level-stable request lines), make all readers in the pre-HLS simulation see the logical AND of the value driven in the pre-HLS simulation and the recorded post-HLS observation of the same dynamic instance ordinal — captured at the post-HLS-side event and retired at the pre-HLS-side handshake of that ordinal, per Wrapper Realization Requirements W1–W4 below. Payload/data values are never ANDed or substituted; they are checked for equality at the matched ordinal (Requirement W4). The post-HLS simulation runs free: its readers see only its own driven values, and the wrapper observes but never gates, delays, or modifies any post-HLS-side signal.

Often the post-HLS side will not run ahead of the pre-HLS side at the snoop points, because HLS typically only adds (rather than subtracts) latency within each process. This tendency is a heuristic, not an invariant: HLS can also subtract latency, for example by removing conservative waits present in the source, by improving loop initiation intervals, or by realizing channels with fall-through timing. The one-sided rule above is therefore normative, and a symmetric variant — driving the logical AND of the two simulations' nets into the readers on both sides, so that the pre-HLS side can gate the post-HLS side — must not be used, even as a fallback when the post-HLS side runs ahead. Under B1 the post-HLS  $\Sigma$ -trace under fixed stimulus is unique, so there is no post-HLS nondeterminism for a wrapper to select among: gating a post-HLS-side reader (in particular an internal net such as an arbiter input or an interrupt request line) does not select an admissible RTL\_B behavior; it produces a different circuit RTL\_B' with different internal latencies, and any result obtained about RTL\_B' does not transfer to the RTL\_B that ships. Gating the pre-HLS side, by contrast, is witness selection over an intentionally under-specified specification (see Snooping Concerns below) and is always permitted.

Two situations must be distinguished when the post-HLS side does run ahead at a snoop point. (a) Benign cycle skew: the RTL asserts a snooped level-stable signal some cycles before the pre-HLS side does, but the order of  $\Sigma$ -visible events at the snoop points is unchanged. Because the compatibility

results of Appendices G–K are stated over  $\Sigma$ -event order and values, with finite  $\epsilon$ -stutter tolerated (B4/B5), a matching pre-HLS witness still exists. The wrapper handles this case by recording the RTL assertion (level-stability makes the capture clean; the recorded assertion is held until the pre-HLS-side handshake of the same ordinal, per Requirement W1 below) and letting the pre-HLS side catch up, arbitrarily behind in host time. Cycle skew is not an error, and a cycle-locked per-cycle “perfect match” comparison must not be used as a validity gate, since it would misclassify a mere HLS speedup as a failure. (b) Divergence: no admissible pre-HLS execution can consume or produce the RTL’s event sequence — its ordinal order, kinds, and payloads — at the chosen boundary; the witness required by WC does not exist. This is the genuine failure mode; it violates Condition L.Dir below, must cause the wrapper to abort loudly, and is triaged as described under Snooping Concerns. Comparison of the pre-HLS-side AND against the value driven on the post-HLS side remains a mandatory monitor (Requirement W4 below), but its role is diagnostic: a per-cycle mismatch reports skew, while the L.Dir violation itself is defined at the event level — ordinal order, kind, and payload — and is detected by an order/kind/payload check together with a watchdog/timeout that converts witness non-existence, or an inconclusive failure to realize the witness within an unproved bound (Requirement W4), into a reported failure rather than a silent gate or an indefinite hang.

## Wrapper Realization Requirements (Recorded-Observation AND)

The AND rule above is stated over a recorded observation, not over the live post-HLS net. A naive realization in which the pre-HLS reader sees the AND of the two currently driven values is a cycle-window mechanism: it delivers a snooped event to the pre-HLS side only if the post-HLS assertion is still asserted when the pre-HLS side is ready to commit. Because the post-HLS side runs free, its assertion window ends at its own handshake, which is unrelated to whether the pre-HLS side has caught up; level-stability guarantees that the signal remains asserted until the post-HLS-side handshake, not until the pre-HLS-side handshake. The live-AND realization is therefore faithful exactly when, for every dynamic instance ordinal  $k$  at every snoop point, the pre-HLS-side commit of instance  $k$  falls inside the post-HLS-side assertion window of instance  $k$ . This per-ordinal window-overlap condition is strictly stronger than Condition L.Dir, and it is an empirical per-run property: it shall be checked, not assumed.

Where the window-overlap condition fails — including on runs that are L.Dir-clean, i.e., runs exhibiting only benign cycle skew — the live AND fails in three ways. First, it drops events: the post-HLS assertion retires before the pre-HLS side arrives, the AND never rises for the pre-HLS side, and the event is silently lost (for example, a lost interrupt downstream, or a pre-HLS process waiting forever on a request that never becomes visible), converting an L.Dir-clean run into an apparent failure or value divergence — precisely the misclassification this appendix forbids. Second, it aliases ordinals: if the post-HLS side runs two or more transactions ahead on the same signal, the pre-HLS side’s  $k$ -th commit can be enabled by the post-HLS side’s  $(k+1)$ -th assertion, and pulse-type signals held across two post-HLS instances can merge into one pre-HLS observation, or drop. Dropped and aliased events violate the no-add/remove/reorder premise of Lemma L.2 and break the per-instance  $\mu_Q$  matching supplied by WC: the harness is then sampling a wire, not replaying the witness. Third, it detects nothing: genuine order divergence (a real L.Dir violation) manifests as a hang indistinguishable from a dropped-event hang, so the triage of Snooping Concerns cannot begin; moreover, the harness-induced infinite wait appears as a wait-for dependency in any Appendix K analysis — a phantom deadlock manufactured by the harness rather than by the design, and outside the finite-delay tolerance that B2 fairness extends to the wrapper on L.Dir-clean runs.

**W1 (Recorded per-ordinal observation).** For each snoop point, the wrapper shall capture each post-HLS snooped event — the committed transfer for message-passing snoop points, the assertion edge for level-stable global signals such as interrupt request lines — indexed by its dynamic instance ordinal, and shall hold the captured observation for the pre-HLS side until the pre-HLS-side handshake of the same ordinal, at which point the observation is retired. For a single-outstanding interface this reduces to one sticky flop, set by the post-HLS event and cleared by the pre-HLS-side handshake; in the general case it is a small per-snoop-point event FIFO. The AND shall consume the captured observation, never the raw net; level-stability is what makes the capture clean, and pulse-like signals shall be queued as distinct edge events so that dynamic instances are neither merged nor dropped.

**W2 (Gate only the commit-enabling mirror signal).** Per operation kind, the wrapper shall gate only the mirror signal that enables the pre-HLS-side commit: the observed vld for a pre-HLS-side Pop, and the observed rdy for a pre-HLS-side Push. The endpoint-owned request driven by the pre-HLS side itself is never modified, consistent with the Appendix C non-withdrawal discipline.

**W3 (Registered sampling point).** In a shared-clock co-simulation, the post-HLS observation shall be registered by at least one cycle before it enters the AND, so that delta-cycle ordering races between the two simulations cannot affect the observed value. This is safe by construction: it adds only pre-HLS-side latency, which is witness selection.

**W4 (Monitor and watchdog; Lemma L.5(iv)).** At each pre-HLS-side commit at a snoop point, the wrapper shall compare the committed event against the recorded post-HLS event of the same ordinal, in order, kind, and payload. Payloads must match (K.CV); the wrapper shall never substitute a post-HLS payload into the pre-HLS side, since substitution relabels  $\Sigma$ -events and masks value divergence. A pre-HLS-side snoop commit with no recorded post-HLS counterpart, or an order/kind/payload mismatch at a matched ordinal, shall abort loudly as an L.Dir violation. A recorded event outstanding beyond a configured watchdog bound, or a pre-HLS-side request that remains unmatched beyond that bound, shall likewise abort loudly, but as an L.Dir/WC discharge failure: if the bound has been proved sufficient for the selected instance, the abort may be classified as an L.Dir violation; otherwise it shall be triaged as an inconclusive witness-realization failure — the compared run remains outside the discharged theorem scope, but the abort shall not be converted into a claimed design deadlock or a claimed proven violation. In all cases the abort converts witness non-existence, or failure to realize the witness within the bound, into a reported failure rather than a silent gate or an indefinite hang. Per-cycle skew shall be logged separately as diagnostics and shall not be treated as a failure.

**W5 (Coverage remains a separate obligation).** Requirements W1–W4 concern the fidelity of the replay mechanism at each snoop point; they do not discharge the first premise of Lemma L.2. If the snooped set misses an  $\epsilon$ -modeled choice that can affect  $\Sigma$ -visible control flow, frontier membership, request attribution, guard values, or payload values, the replayed witness is underdetermined regardless of the recording discipline, and the resulting failure shall be triaged as harness incompleteness under Snooping Concerns.

**Remark (relation to the naive live AND).** The live-AND and recorded-observation realizations coincide on single-outstanding snoop points at which the pre-HLS consumer commits within every post-HLS assertion window — for example, a non-blocking polling consumer that polls every cycle while the pre-HLS side remains at-or-ahead in cycle time at that snoop point throughout the run. Because W4 requires the W1 recording in any case as its comparison reference, gating with the live net offers no economy;

the recorded observation is therefore the normative gating term, and the live net may be used only as an additional diagnostic tap.

**Shared reactive testbenches.** Once the pre-HLS side is permitted to lag by more than a cycle, a single shared reactive testbench instance is problematic: its reactions are driven by whichever model it is wired to, so the two models no longer receive the same stimulus, contrary to the fixed-stimulus setting of the formal results. The flow shall therefore either replicate the stimulus/testbench per model, or structure the shared testbench as a function of the DUT-visible trace so that the wrapper's recorder can replay its reactions to the lagging side under the same recorded-observation discipline of W1–W4.

**Definition L.1 (Witness selector / Snooping).**

Fix an initial state, fixed external stimulus, and an RTL\_B execution prefix  $\tau_{\text{post}}$ . A witness selector  $W$  for a source process  $P$  is a partial strategy that, at each  $\epsilon$ -quiescent source cutpoint  $\sigma_{\text{ref}}$  and each source-level choice point whose outcome is  $\epsilon$ -modeled but may affect later  $\Sigma$ -visible behavior, returns the outcome selected by the matched RTL\_B execution.

Such choice points include failed/successful outcomes of non-blocking PushNB/PopNB calls when the outcome affects later control flow, frontier membership, request attribution, or payload/guard values; and any other  $\epsilon$ -only implementation choice that can affect the next  $\Sigma$ -visible frontier.

**Cadence-sensitive non-blocking polling.** If a source process computes externally visible behavior from the number or cadence of failed PushNB/PopNB polls — for example, retry counters, timeout counters, or arbitration policies whose winner depends on how long an input remained empty/full — then that polling logic is latency-sensitive in the sense of Appendix D. Such a process is not covered by the ordinary Appendix I/J latency-insensitive source-core argument merely by treating the failed polls as hidden  $\epsilon$ -steps. To bring such a design into scope, the cadence-sensitive polling logic shall be isolated as a latency-sensitive local protocol block, or shall be aligned by an Appendix L snooping/witness construction that makes the selected source execution an admissible execution of the same isolated protocol behavior observed in RTL\_B. The observable interface to the rest of the system shall then be the chosen boundary of the isolated block, not the internal failed-poll cadence itself.

The selected outcome must be causally available from the matched RTL\_B prefix and the fixed global  $\Sigma$ -causal history.  $W$  is not allowed to depend on future RTL\_B events beyond the matched prefix needed to identify the current source choice. For cadence-sensitive polling logic, the witness must select an admissible execution of the isolated latency-sensitive block; it may not manufacture a retry/timeout outcome that is inconsistent with the RTL\_B behavior being used as the witness, the fixed stimulus, or the chosen observable boundary of that block.

**Lemma L.2 (Snooping establishes WC).**

Let  $\tau_{\text{post}}$  be an RTL\_B execution under fixed stimulus, and let  $W$  be the witness selector induced by the Appendix L snooping wrapper for the corresponding source execution. Assume:

1. the snooped signals / outcomes include every  $\epsilon$ -modeled source choice whose result can affect the next  $\Sigma$ -visible control flow, frontier, request-attribution, guard value, or payload value;
2. the snooping wrapper does not add, remove, reorder, or relabel any  $\Sigma$ -visible event except by selecting  $\epsilon$ -only source outcomes at the chosen observable boundary;
3. each selected  $\epsilon$ -only outcome is consistent with the matched RTL\_B prefix and the fixed global  $\Sigma$ -causal history;
4. failed non-blocking polls remain  $\epsilon$ -events and successful non-blocking transfers produce exactly the corresponding committed Push\_c / Pop\_c  $\Sigma$ -event; and

5. if the source contains cadence-sensitive non-blocking polling whose externally visible behavior depends on the number or cadence of failed polls, that polling logic has been isolated or snoop-aligned as a latency-sensitive local protocol block as described in Appendix D and Definition L.1. The WC witness is then applied to the chosen observable boundary of that block, not to an unconstrained hidden failed-poll cadence inside the ordinary latency-insensitive source core.

Then the source execution selected by  $W$  satisfies the witness-compatibility premise WC used in Appendix I.2 and Appendix J.

**Corollary L.3 (NB-CF identity witness).**

If NB-CF holds for process  $P$ , then failed non-blocking poll outcomes do not affect the next  $\Sigma$ -visible control flow, frontier membership, request attribution, guard values, or payload values. Therefore the witness selector  $W$  may be taken to be the identity / arbitrary selector on failed non-blocking outcomes, and WC holds without additional snooping for those polls.

Corollary L.4 (Cadence-sensitive polling is handled by isolation).

If NB-CF does not hold because the source observes the number or cadence of failed PushNB/PopNB polls — for example, retry counters, timeout counters, or arbitration policies whose later  $\Sigma$ -visible behavior depends on how long a channel remained empty/full — then the design is latency-sensitive at that polling site. Such a site must be isolated or snoop-aligned under Appendix D / Appendix L before applying the ordinary Appendix I/J source-core proof. If no such isolation, snooping, or equivalent explicit timing model is supplied, the process is outside the scope of the Appendix I/J trace-compatibility theorem for the chosen observable boundary.

**Condition L.Dir (One-directional alignment / RTL-side non-interference).**

Fix the free-running RTL\_B execution  $\tau_{\text{post}}$  under the fixed stimulus (B1) and the Appendix L wrapper for a chosen set of snoop points. L.Dir holds for a compared run iff the source-side snoop sequence selected by the witness selector  $W$  of Definition L.1 is exactly an ordinal-preserving consumption of the free-running RTL\_B snoop sequence at the chosen boundary: (i) each pre-HLS-side snoop commit consumes the next recorded RTL\_B event of the same snoop point, with matching kind and payload; (ii) no pre-HLS-side snoop commit occurs without such a next recorded RTL\_B event; and (iii) every recorded RTL\_B snoop event within the compared run is eventually either consumed by the source witness or reported outstanding according to the proved or declared comparison bound of Requirement W4 above. Throughout, the alignment never gates, delays, or modifies any RTL\_B-side signal or event (RTL-side non-interference): the wrapper never drives a post-HLS-side reader with a value different from the value the free-running RTL\_B drives. Cycle-level lag of the pre-HLS side (benign skew) is permitted under L.Dir provided the ordinal-preserving consumption above is maintained via recorded/buffered replay of the level-stable assertions; only order, kind, or payload divergence — non-existence of the WC witness at the chosen boundary — violates L.Dir. L.Dir is a named side condition with its own entry in the side-condition discharge record (Normative Formal Core). A run that violates L.Dir provides no theorem-backed guarantee. The harness shall abort loudly on any detected divergence and on any exceeded Requirement W4 watchdog bound, with the classification of bound-exceeded aborts governed by Requirement W4, and every abort shall be triaged as described under Snooping Concerns below.

**Lemma L.5 (Online realization of  $W$  / harness soundness).**

The equivalence and liveness results of Appendices I–K are stated over the free-running RTL\_B execution  $\tau_{\text{post}}$  under fixed stimulus (B1), together with the source-side witness selector  $W$  of Definition L.1. This statement is one-directional by construction: no premise of those results requires RTL\_B to be stallable by the source, and  $\tau_{\text{post}}$  is a fixed object rather than a participant that can be gated. Separately, suppose the co-simulation wrapper (i) observes but never gates, delays, or modifies any post-HLS-side

signal or event; (ii) records snooped level-stable assertions and replays them to the pre-HLS side in RTL order, permitting arbitrary finite host-time skew (Wrapper Realization Requirements W1–W3); (iii) satisfies the premises of Lemma L.2 (see Requirement W5); and (iv) monitors Condition L.Dir and aborts loudly upon violation (Requirement W4). Then on every run on which the wrapper does not abort, the wrapper computes  $W$  online, the selected source execution satisfies the WC premise used by Appendices I–J (by Lemma L.2), and the results proved over free-running  $RTL\_B + W$  apply to that run. Bidirectional stalling therefore never enters any theorem whose conclusion is transferred to the shipping RTL: the post-stalls-pre direction is witness selection over the intentionally under-specified source, and the pre-stalls-post direction is excluded by the monitored Condition L.Dir rather than assumed, endorsed, or repaired by gating the RTL.

## Snooping Concerns

The snooping technique is used to align pre-HLS and post-HLS latency-sensitive global signals and latency-sensitive non-blocking message-passing operations in cases where their latency differences are externally visible at the DUT boundary. This includes cadence-sensitive polling blocks, such as retry/timeout or polling-arbiter logic, when their externally visible behavior depends on the number or cadence of failed non-blocking polls. When such a wrapper is applied, the pre-HLS model is throttled to follow the latency choices made by the post-HLS RTL, and the resulting traces at the observable interfaces are forced to agree. Snooping does not make failed polls into committed message-transfer events; failed polls remain  $\epsilon$ -steps. Rather, snooping selects an admissible execution of the latency-sensitive local protocol block, while the rest of the system is compared at the block’s chosen observable boundary.

Readers with a formal background may be concerned by this technique, because it appears to use the post-HLS RTL implementation to modify the behavior of the pre-HLS specification. From a formal point of view, the important observation is that the pre-HLS model is intentionally under-specified with respect to micro-timing/latency: subject to the R-rules, B-rules, weak fairness, and the explicit  $\Sigma$ -visible ordering, FIFO, signal-anchor, synchronization, and atomicity constraints used by  $\approx$ , it can admit multiple  $\epsilon$ -/latency-different executions that respect the same required-order constraints over the chosen observable units. This is compatible with B1, which asserts that the post-HLS  $\Sigma$ -projected trace is unique under a fixed stimulus; the pre-HLS side may still have multiple admissible realizations that (when projected to  $\Sigma$ ) match that unique RTL  $\Sigma$ -trace.

Snooping does not change what executions are allowed by the pre-HLS specification. Instead, it selects—purely for verification purposes—one admissible pre-HLS execution whose latency-sensitive events align with those observed in the RTL. In other words, the implementation is used to pick a witness execution of the specification, not to redefine the specification.

**Triage of L.Dir violations.** A violation of Condition L.Dir (order, kind, or payload divergence at a snoop point, including a pre-HLS-side snoop commit with no recorded  $RTL\_B$  counterpart) is a relational failure: the compared run is outside the discharged theorem scope, and no theorem-backed claim attaches to it. It shall not be defined a priori as a bug in the pre-HLS model, because witness non-existence has three possible root causes, and defining the event as a model bug by fiat forecloses detection of the third: (1) an ill-formed pre-HLS model — for example, reliance on the ordering of causally independent events, cadence-sensitive polling that was not isolated per Corollary L.4, or a snooped signal that is not level-stable — in which case the failure is of the same character as a standalone pre-HLS test failure and the

model is outside the well-formedness envelope of the stress-testing section below; (2) an incomplete snooping harness — the snooped set misses an  $\varepsilon$ -modeled choice covered by the first premise of Lemma L.2, so the replayed witness is underdetermined; or (3) a non-compliant tool/RTL — the RTL exhibits behavior outside the admissible envelope (for example, an R/B-rule or B-faithfulness violation), in which case the Post  $\rightarrow$  Ref result itself fails and the divergence is evidence of a synthesis defect. In a flow in which the harness has been certified against Lemma L.2 and the tool is trusted, the pre-HLS model is the default suspect and the failure should be filed against it first; the formal status of the run, however, is “L.Dir/WC undischarged — outside theorem scope,” not “pre-HLS model incorrect,” until triage assigns the root cause. A watchdog abort under a bound that has not been proved sufficient for the selected instance is triaged in the same way but is recorded as an inconclusive witness-realization failure per Requirement W4, not as a proven L.Dir violation. Benign cycle skew (unchanged  $\Sigma$ -event order with the RTL merely earlier in cycle time) is not an L.Dir violation and shall not be classified as a model defect. Experienced SoC architects tend to view this slightly differently, in terms of design tradeoffs and verification cost.

First, it is usually possible at the architectural level to avoid latency-sensitive behaviors entirely, for example by insisting that all communication be latency-insensitive and fully synchronized. However, the quality-of-results (QOR) costs of such designs may be unacceptable. Introducing latency-sensitive behavior—non-blocking arbiters, latency-sensitive global interrupt signals, and so on—is a deliberate choice made by the designer to meet performance, power, or area goals.

Second, in many practical systems, latency-sensitive behavior is not functionally observable at the DUT boundary under the trace-compatibility predicate  $\approx$  with  $\Sigma$  instantiated to include only the DUT-boundary observables (Appendix G; Normative Formal Core above). As an example, consider a DUT that contains a single-port RAM used to exchange data between two internal processes. An arbiter is required to control access to that RAM port, and the arbiter uses latency-sensitive non-blocking Pop operations on its request channels. During HLS, internal latencies will change, so the order in which the arbiter sees pending requests and chooses winners may differ between the pre-HLS and post-HLS models.

In this example:

- The individual RAM read/write operations and the internal arbitration decisions are not directly visible at the DUT interface.
- Only the higher-level IO of the two processes (for example, their message-passing interfaces to the rest of the system) is externally visible.

One might initially conclude that it is necessary to snoop the arbiter’s inputs to make the pre-HLS and post-HLS traces match. However, the modeling rules impose two additional constraints:

1. **Weak fairness for the arbiter.** The arbiter must satisfy the weak fairness assumptions, so that any request that remains pending is eventually granted in both the pre-HLS and post-HLS models.
2. **Causal-dependency discipline on shared memory.** The system must make any functionally significant shared-memory dependence explicit, either through synchronization that orders the relevant accesses or through declared array-access mapping / memory-serialization semantics. This discipline ensures that sufficient synchronization or declared serialization exists between the processes so that there are no read/write races through the RAM in either model.

Under these conditions, the different arbitration orders merely change the relative timing of internal operations and of externally visible actions that are not ordered by the explicit required  $\Sigma$ -constraints. They do not change the FIFO, synchronization, signal-anchor, atomicity, or other required ordering constraints at the DUT boundary. In the terminology of Appendices G and J, the differences are incidental variations in latency rather than changes to the required-order relation  $\leq_J$  over the chosen observable units, and therefore they are not considered observable differences in behavior at the DUT boundary. **In such cases, snooping is not required.** (See also Note 1 below).

(More precisely: the trace-compatibility predicate  $\approx$  is parameterized by the chosen observable alphabet  $\Sigma$ . If internal functionality (such as the above RAM arbiter) uses channels/signals that are not designated observable in the chosen instantiation of  $\Sigma$  (see Normative Formal Core), then mismatches on those internal actions are outside  $\approx$  by construction: they are not in  $\Sigma$ . If those internal channels/signals are designated observable (e.g., by snooping for debug, or by expanding  $\Sigma$  for Appendix K deadlock analysis), then they are  $\Sigma$ -events and must match under  $\approx$ . In addition, even when B1 holds (unique post-HLS  $\Sigma$ -trace under fixed stimulus), the pre-HLS model may still admit multiple  $\varepsilon$ -/latency-different executions consistent with the same required  $\Sigma$ -ordering, FIFO, signal-anchor, synchronization, and atomicity constraints; Theorem J.1 captures the intended quantification (“existence of a matching witness execution prefix”) rather than requiring a single pre-chosen trace.)

Experienced SoC architects therefore try to avoid designs in which latency-sensitive internal behaviors leak directly into the observable behavior of the system under test, since this both complicates verification and makes RTL behavior less predictable. When they do introduce such behaviors—examples include non-blocking arbiters whose winners affect externally visible ordering, global interrupt request signals at the DUT boundary—they typically know exactly where those latency-sensitive interfaces are and can isolate them.

A useful analogy is a subsystem whose clock frequency is adjusted dynamically based on on-die temperature. As temperature changes, the externally visible behavior of the SoC can change (for example, in terms of throughput or timing of events), and designers may choose such a temperature-dependent scheme to meet overall system requirements. To verify such a system, we may wish to compare a concrete RTL trace captured at a specific temperature profile with a higher-level reference model. To do so, the reference model must be driven with the **same** temperature evolution as the RTL saw. If that temperature profile is not otherwise under direct control, the simplest way to obtain it is to “snoop” the temperature as observed in the RTL trace and replay it into the reference model. In this analogy, temperature is an external parameter of the environment rather than a fundamental part of the functional specification, but it still influences observable behavior. Snooping simply provides a mechanism to recover that parameter from the implementation when it cannot be easily prescribed a priori. Similarly, snooping of latency-sensitive IO provides a mechanism to supply implementation-specific latency choices — subject to the fairness assumptions and the explicit required  $\Sigma$ -ordering, FIFO, signal-anchor, synchronization, and atomicity constraints — back to the pre-HLS specification so that the two models can be compared under a common environment.

**Note 1 (Unobservable non-blocking micro-timing).**

**Safety / functional equivalence.** The trace-compatibility predicate  $\approx$  is parameterized by the chosen observable alphabet  $\Sigma$  (Appendix G). If, for safety-only checking, we instantiate  $\Sigma$  to include only DUT-boundary observables ( $\Sigma_{\text{DUT}}$ ) and we assume that any latency-sensitive internal behavior is not visible at that boundary under  $\approx$ —i.e., it may change only internal micro-timing, but it does not change the values or presence of  $\Sigma_{\text{DUT}}$  events, and does not change the ordering of causally *dependent*  $\Sigma_{\text{DUT}}$

events—then snooping of those internal latency-sensitive interfaces is not required for proving  $\Sigma_{\text{DUT}}$ -safety. Any mismatches on actions outside  $\Sigma_{\text{DUT}}$  are outside  $\approx$  by construction. When source-reference safety is proved universally, or when one complete pre-HLS run satisfies Definition J.RefProjCover, Corollary J.1-Safe combines that result with Theorem J.1 Post  $\rightarrow$  Ref to obtain RTL\_B safety at  $\Sigma_{\text{DUT}}$ . This common-case transfer does not use the restricted Ref  $\rightarrow$  Post clause and does not require snooping.

**Liveness / deadlock analysis.** For message-passing deadlock analysis (Appendix K WFG), any latency-sensitive interface whose  $\varepsilon$ -level choices can affect whether a blocking Push/Pop is enabled, or can create/remove a wait-for dependency without an intervening committed  $\Sigma$ -event, must be accounted for. Practically, the default is to snoop and replay every such latency-sensitive (typically non-blocking) arbitration/probe interface in the DUT. Snooping can be avoided only if the design is certified “WFG-inert” with respect to that interface: it cannot change WFG edges except via committed Push\_c/Pop\_c (or explicit synchronization) events.

## Stress Testing Pre-HLS Models for Latency Robustness

The formal trace-compatibility results in Appendices G–K establish that, under the scheduling rules and fairness assumptions, the post-HLS RTL is trace-compatible with the pre-HLS/source-reference model at all observable interfaces.

These results implicitly assume that the pre-HLS specification is *well-formed*: it must not rely on incidental, implementation-specific timing alignments for functional correctness. Designs that do rely on such incidental alignments fall outside the formal guarantees.

Designs that use non-blocking message-passing operations (e.g., PushNB/PopNB, ac\_channel nb\_read/nb\_write) or latency-sensitive global signals are particularly vulnerable to this kind of fragility. For such designs, it is strongly recommended to subject the pre-HLS model to aggressive *latency stress tests* in which message and handshake latencies are varied across executions. The intent is to validate that the design behaves correctly across the range of weakly fair schedules permitted by the modeling rules, not just under one convenient execution order.

More concretely, verification should check that:

- **Causal dependencies are explicit.** All functionally significant causal-dependency relationships are enforced via message-passing, SyncChannel/ac\_sync operations, explicit signal handshakes/anchors, declared memory-serialization constraints, or Appendix L snooping/witness-selection boundaries, rather than by relying on the incidental execution order of concurrent processes. In the terminology of Appendix G, the design shall not depend on the ordering of causally independent events for correctness.
- **Weak fairness does not hide latent deadlocks or starvation.** The design continues to make progress under adversarial but *weakly fair* scheduling—i.e., enabled actions may be delayed arbitrarily long but not forever—consistent with the B2/B3 obligations on the scheduler and environment summarized in Appendix N.

If the pre-HLS model fails under such randomized-latency stress scenarios, then the specification itself is outside the formal model of this document: it violates the design rule that well-formed systems must not rely on the relative ordering of causally independent events. In that situation, the trace-compatibility theorems still apply to well-formed traces, but they no longer guarantee that the “golden” specification is robust under the full range of latency variations allowed by the environment.

The Matchlib library provides practical mechanisms to automate these stress tests in pre-HLS simulation, including:

- **Random stall injection on message-passing channels**, to simulate variable communication delays while preserving rendezvous semantics.
- **Latency and capacity back-annotation**, to model specific per-channel buffer depths and delays, including the capacities  $B(c)$  chosen during HLS.

Worked examples of this methodology are provided in Catapult Matchlib examples 60\* and 72\*, which illustrate how to configure randomized stalls and back-annotated latencies when validating that a design remains well-formed and latency-robust.

## Appendix M – Formal Results Overview

### M.1 Purpose and status

This appendix gives a readable overview of the formal results and the relationship among their principal components. It is informative: the definitions, premises, theorem statements, proof obligations, applicability restrictions, and coverage requirements in Appendices G-L, Appendix N, and the Normative Formal Core remain authoritative. The overview intentionally uses formal names where they help a reviewer locate the controlling result, but it does not reproduce the complete proof API.

The document proves two related but distinct kinds of result. The safety layer establishes ordered observable trace compatibility between the post-HLS implementation model and a prescribed source reference model. The liveness layer excludes a specified prohibited internal wait structure under stronger and more global conditions. Neither layer independently certifies that a particular source design, HLS invocation, generated RTL instance, environment, or back-annotation satisfies the premises required to instantiate the result.

### M.2 Models and observation boundary

The implementation-side model is `RTL_B`. The source-side theorem target is the selected `Sys_ref` under the fixed stimulus and the recorded environment, reset, arbitration, witness, and abstraction choices. In the ordinary bounded-channel case, `Sys_ref` is `Sys_B` with capacities `B(c)` faithful to the selected explicit abstract channel boundaries. The rendezvous case is the special case `B(c)=0` at the relevant ordinary boundaries. When the implementation contains staged, bundled, joined, coupled, or synchronization-gated realization structure that cannot be represented faithfully by one ordinary bounded FIFO, `Sys_ref` is `Sys_B^elab` for a well-formed elaboration package as described in Appendix Q.

The safety relation is defined over a selected observable alphabet and an ordered trace-compatibility relation rather than over equality of internal states or equality of cycle-by-cycle execution. Observable units include the committed channel transfers, synchronization events, and anchored signal reads and writes selected by the formal model. Independent units may commute or be grouped into cycles differently when the required ordering, value, atomicity, and observation-boundary conditions are preserved. Proof-internal elaboration events and epsilon steps are hidden or projected as specified by the applicable abstraction map.

Shared-memory communication between processes is covered by the proved safety result only under the J.Mem mapping-layer lowering condition stated in the Normative Formal Core. That condition makes memory-carried dependencies visible through operations already represented in the proof alphabet. A direct global memory-state extension is discussed as a possible future extension but is not part of the present theorem.

### M.3 Principal safety result: Post -> Ref

The ordinary safety direction is Post -> Ref. For a covered reachable `RTL_B` prefix, the flow supplies Definition J.LocalGWR or, alternatively, a directly validated J.GWR package with its required provenance and checks. J.LocalGWR organizes design-specific evidence into process-local, explicit-channel, named atomic-interaction, environment/reset, construction-action, footprint, request-snapshot, generated-dependency, local-agreement, and coverage components, together with the applicable channel-capacity faithfulness obligations.

Theorem J.GWR-Comp is the generic local-to-global construction. It generates the dependency structure, proves the required acyclicity and canonical construction order, combines the local preparation scripts, settles shared non-blocking decisions under the registered phase discipline, and

produces one reachable globally witness-realizable Sys\_ref package. Proposition J.0 then composes the selected source/RTL pair to establish the ordered observable compatibility required by Theorem J.1. The design-specific premises are primarily local even though the conclusion is system-wide; the global gluing argument is proved once rather than supplied as an unexplained replay premise for each design. This locality characterization concerns the finite-prefix J.1 safety construction. In the standard QSync normal-compositional scope, J.2b now exposes a smaller instance-wide safety projection: J.QSyncTraceInd carries only the exact cycle-aligned J.LocalGWR/J.GWR-Comp and J.HCanon facts and Theorem J.TraceCertCov-QSyncCycle derives J.TraceCertCov on J.CycleReach\_post. The richer J.QSyncCertInd used by the universal Appendix K route implies J.QSyncTraceInd but additionally carries the Offer/KObs/drain facts required downstream by K.

The practical safety interpretation is that a covered RTL\_B behavior cannot introduce a new canonical observable safety behavior that is absent from the selected source reference model. For source-side safety signoff, Corollary J.1-Safe transfers a J.HCanon-invariant safety predicate from the covered Sys\_ref prefixes to the covered RTL\_B prefixes. When one complete pre-HLS execution is used as the source-side verification artifact, Definition J.RefProjCover is the additional coverage fact showing that its observable prefixes represent all covered source-reference observable prefixes relevant to the claimed safety property. In the standard QSync route, J.1-Safe-QSyncCycle removes the need to posit an unrelated route-complete package separately for every completed-cycle RTL prefix once J.QSyncTraceInd has been discharged. Lemma J.CycleIdealCover then shows that every reachable finite canonical  $\Sigma_{\text{DUT}}$  history is a finite ideal of one of those covered cycle-aligned histories, and Corollary J.1-Safe-QSync extends any safety predicate satisfying Definition J.IdealPrefixClosed to all reachable finite RTL\_B prefixes. This ideal-closure step does not assert per-prefix J.TraceCertCov; general J.TraceCertCov remains the interface when a claim requires direct correspondence packages for non-cycle-aligned prefixes or uses a broader non-QSync coverage route.

#### M.4 Restricted Ref -> Post use and witness selection

The converse direction is intentionally not an unrestricted theorem saying that every raw source simulation execution is reproduced by RTL\_B. HLS may change latency, cycle grouping, arbitration timing, non-blocking outcomes, and other witness-sensitive choices while remaining within the scheduling contract. Accordingly, direct Ref -> Post transfer is limited to annotated RTL-consistent source prefixes or to source executions selected by the Appendix L snooping and witness-selection mechanism.

This asymmetry is important for interpreting the methodology. The primary safety argument does not require a claim that one arbitrary pre-HLS run is cycle-for-cycle or choice-for-choice identical to the implementation. Instead, it shows that each covered implementation behavior has a compatible source-reference explanation. A particular source run can be used directly in the reverse direction only when the additional annotation or witness-selection conditions establish that it is the run corresponding to the implementation behavior under review.

#### M.5 Cutpoint correspondence used by liveness

The liveness proof begins from a stronger cutpoint interface than ordinary trace compatibility. Corollary J.1 and CF-0/J.KObsConf connect one selected reachable RTL\_B cutpoint to one reachable Sys\_ref cutpoint through an equal KObs record. KObs contains the canonical replay history and the complete attributed residual-offer information needed to reconstruct the relevant frontier and wait-for structure. It is an observation record for the bad-state predicate, not by itself a complete operational state for future Policy-L or Policy-S execution.

For the exact source witness selected by that correspondence, `J.RegPhaseReach` supplies theorem-derived L0-L4 phase-correct reachability. `K.DrainReach` separately supplies the nonwithdrawal, epsilon-normalization, and activated `Core.Drain/K.Drain` evidence. Their checked conjunction is `K.RLAdmReach`, which places the witness in the admissibility domain consumed by the Appendix K theorem. Keeping these components separate prevents the trace correspondence alone from silently assuming the drain and progress properties needed by the liveness argument.

## M.6 Deadlock-preservation result

**Primary user workflow.** The primary user-facing discharge route is deterministic Policy-S serial reduction. For one fixed selected `Sys_K` instance, including capacities, elaboration, initial/reset state, stimulus, witness choices, environment behavior, arbitration, concrete pre-HLS scheduler, Policy-S issue semantics, lowering, and normalization, `SDet` selects one Policy-S execution artifact  $p\_S^{\text{det}}$ . Policy S is conservative: within each source interval, a process does not issue a source-later blocking operation until every required source-earlier blocking operation has committed in a strictly earlier cycle. The flow supplies total finite-bound `K.RespCov` for every relevant internal process-facing blocking site or dynamic class. For a terminating selected instance, the design user runs this instrumented model to a declared maximal terminal state; the complete artifact must contain no covered timeout, no covered frontier pending at termination, and no non-waived reset-pending failure.

**Why the run suffices.** `A5-S/CF-S` combines three separately checked layers for the selected execution: `K.CompleteS` establishes maximality, fairness, and completeness; `K.RespCov` establishes total finite-bound response coverage; and `K.RespClean` establishes that the complete run contains no non-waived `K.RespFail` witness. Lemma K.2b shows that a prohibited deadlock reached by an admissible Policy-L prefix would force that same selected Policy-S execution to produce a finite response failure. Corollary K.2c therefore derives `A5-L` when `SDet`, `K.EnvMF`, `K.ResetEpoch`, `K.TerminalDisposition`, and the remaining reduction premises hold. This replaces exploration of alternative Policy-S scheduler interleavings with one selected complete artifact; no confluence theorem is required. For a terminating selected instance, the complete-run response check itself does not require model checking or a hand-written liveness invariant, although the flow must still qualify the selected semantics, finite bounds, applicability, coverage, fairness, and other stated premises. Separately, the proof of Lemma K.2b.E requires an instance-bound `Core.FairPick` constructive witness for the fixed Policy-L semantics used to build the hypothetical frozen continuation. That witness is not supplied by the observed Policy-S run or by `K.CompleteS`; it is a flow/checker obligation that refines the same behavioral `Core.IssueFair/B2/B3` requirements into prefix-local selectors and starvation proofs.

What transfers to RTL. Theorem K.1A is first a certified-cutpoint theorem. At a reachable K-relevant `RTL_B` cutpoint carrying a complete `K.CertifiedCutpoint` package, the equal-KObs source witness, `K.RLAdmReach`, `A5-L`, and the applicable point-to-point, channel, structural, waiver, environment, progress, and fairness premises exclude the prohibited closed internal wait structure. A universal statement additionally requires `Core.KCutClosure_post` and `K.CertCov`. In the standard `QSync` normal-compositional route, `Core.QSync-Post` derives closure and canonical normalization; Lemma `Core.QSync-KCutCanonical` shows that every reachable `KCut` is one of those canonical endpoints; and a checked `J.QSyncCertInd` base/step invariant feeds Theorem `J.CertExist-QSync` and Corollary `K.CertCov-QSync` to derive total certificate coverage over that domain. Corollary K.1A-Total then combines both with the per-cutpoint theorem. Thus one clean Policy-S execution discharges the source-side `A5-L` premise through K.2c; it does not by itself prove the implementation-side `QSync` or `QSyncCertInd` obligations, and its finite response bounds are not transferred as RTL latency bounds. Alternative `K.CertCov` routes remain available where the `QSync` induction route is not used.

Deadlock observation scope. Appendix K uses a wider proof-internal observation boundary than a DUT-boundary-only safety check. For Appendix K,  $\Sigma$  includes every explicit abstract message-passing channel whose endpoint transfers can contribute a WFG edge, including proof-internal  $\text{Sys\_B}^{\text{elab}}$  boundaries where applicable. Each such boundary therefore carries the applicable E4/B-faithfulness and J.OfferReach/J.OfferConf/J.KObsConf evidence used by the KObs bridge. Internal RTL staging already represented within B(c) remains  $\varepsilon$ -level behavior and need not be exposed as a separate channel.

Scope, alternatives, and status. Timed, reactive, or otherwise nondefault environment behavior in a relevant multi-frontier component is outside the serial route unless the certified single-internal-frontier condition and the other K.EnvMF conditions hold. Under StandardEnv/B1-avail, ordinary/singleton candidates use theorem-derived Lemma K.JointFreeze-Ord, while an admitted multi-frontier candidate additionally requires K.JointFreeze\_I(D,  $\sigma_L$ , {f\_P}); failure of that certificate likewise puts the candidate outside the serial-reduction route and requires another A5-L discharge. For an ordinary non-elaborated instance whose relevant message frontiers are singleton, this combines with Lemma K.CompareAdvanceLegal-Ord to eliminate both special frontier certificates as per-candidate evidence: the flow cites the two ordinary lemmas instead of supplying K.JointFreeze or explicit K.CompareAdvanceLegal certificate payloads. Independently, every K.2b.P0/K.2b.R-consumed owner/key must satisfy K.CompareAdvanceLegal before P0 advances preparation and through R's later request assertion. Ordinary/singleton owners use Lemma K.CompareAdvanceLegal-Ord; any explicit  $\text{Sys\_B}^{\text{elab}}$  multi-frontier owner on a covered comparison prefix requires checked elaboration-specific evidence even when that process lies outside D. K.CVPred predecessor recursion separately records Core.16 legality for each intermediate non-transfer predecessor reach: qualifying ordinary/singleton blocking OpFact predecessors may reuse K.CompareAdvanceLegal-Ord pointwise, pure observation/settling reaches are certified Core.16-inert, and any other launch-carrying reach rule supplies equivalent owner/frontier evidence. Dynamic-reset instances additionally require K.ResetEpoch; every candidate on K.ResetEpoch.Continue additionally requires K.TerminalDisposition. For every candidate liberal component, the flow first binds a validated K.DomResetScope/DomEpoch and establishes K.ResetEpoch. On K.ResetEpoch.Continue, the comparison continuation must remain free of any RESET whose RW scope intersects the candidate's K.DomResetScope; RESETs disjoint from that scope remain permitted under RW5. Alternatively, K.ResetEpoch.DirectFail must independently map a selected liberal frontier to an already-triggered, non-waived process-facing Policy-S response monitor that the matched RESET clears under RW2. A RESET that intersects K.DomResetScope but does not satisfy the independently mapped K.ResetEpoch.DirectFail conditions satisfies neither serial-route branch; that candidate is outside the serial-reduction route and requires another A5-L discharge. On the K.ResetEpoch.Continue branch, K.TerminalDisposition likewise requires either a terminal-free frozen comparison or an independently mapped Policy-S declared-terminal maximal-pending K.RespFail; otherwise the cutpoint is outside the serial route. A source region whose deadlock freedom genuinely depends on favorable same-cycle or overlapped issue is outside the scheduler-robust A5-L framework when an admissible serial-shaped prefix reaches Dead\_K^W. A correct system with fair but genuinely unbounded response gaps also lies outside the finite-bound Policy-S fragment. Structural, direct Policy-L, invariant, exploration, and tool-certificate methods remain alternative A5-L routes when A5-L is true. Dead\_KS and the related owner/terminal classifications are optional diagnostic aids, not primary proof premises. Operationally, the route is expressed using conventional source simulation and timeout/service assertions and is expected to be highly automatable for finite transactions and bounded protocols. Accordingly, one complete clean run is the principal dynamic design-user artifact rather than the entire proof package. Its use as a verified theorem discharge retains proposed-reduction status until the K-P obligations and certificate interfaces are mechanized and qualified; a universal conclusion separately requires checked Core.KCutClosure\_post and K.CertCov discharges.

Common-case interruption-gate discharge. The reset/terminal applicability conditions do not require enumeration of hypothetical liberal deadlock components when the stronger uniform instance-level condition of Appendix K is proved. One conservative scope may be validated once; if no permitted reset intersects it and declared termination cannot occur while a relevant blocking frontier in that scope remains pending, all required `K.ResetEpoch.Continue` and nested `K.TerminalDisposition.Continue` facts follow uniformly. If those stronger instance-level facts are unavailable, the existing candidate-specific nested-gate route remains required.

## M.7 Implementation qualification and evidence

The formal proofs specify a contract between source modeling, implementation behavior, and validation evidence. They do not prove that ordinary synthesis automatically discharges the named conditions. A concrete theorem application must identify the selected `Sys_ref` and abstraction boundaries; establish source well-formedness and witness conditions; provide the local process, channel, atomic-interaction, environment, reset, footprint, request-snapshot, agreement, and coverage evidence required by the safety construction; and establish faithful realization of every selected channel capacity or elaborated boundary.

For liveness, the application must additionally establish the exact cutpoint correspondence, phase and drain bridge, `A5-L` and its chosen discharge route, fairness and progress assumptions, nonwithdrawal, applicable waiver rules, environment/frontier scope, point-to-point ownership or explicit arbiter modeling, and universal certificate coverage when the total conclusion is claimed. Tool documentation, generated certificates, static checks, formal proofs, simulation monitors, randomized-latency stress tests, and targeted RTL/interface checks may all contribute evidence, but the evidence must be sufficient for the named premise it is used to discharge.

At the architecture level, the proof separates operational semantics, safety/replay construction, `KObs` cutpoint correspondence, and complete-run `Policy-S` evidence. This separation prevents trace compatibility, cutpoint reconstruction, and source-side liveness evidence from being treated as interchangeable assumptions. M.7b gives a standard `QSync` packaging view over this separation so that a tool or user-facing flow may expose a small number of top-level certificate bundles while M.7a remains the definitive ledger of the individual obligations.

**Division of verification responsibilities.** The following table separates the principal dynamic design-user activity from per-instance flow evidence, reusable tool qualification, and framework work that remains to be completed. Ownership may be shared in a particular implementation; the table identifies the expected primary source of evidence.

| Primary source of evidence            | Principal items                                                                                                                                                                                            | Current interpretation                                                                                                                                                                             |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Design user                           | Run the selected instrumented <code>Policy-S</code> model to its declared maximal terminal state for a terminating instance; review response failures, declared waivers, and design/environment contracts. | This complete execution is the principal dynamic user artifact. It supplies <code>K.RespClean</code> only when bound to the qualified selected instance and accompanied by the remaining evidence. |
| Per-instance tool / verification flow | Select and bind <code>Sys_ref/Sys_K</code> , capacities, elaboration, stimulus, witness, reset and environment contracts, and discharge the applicable per-instance side                                   | These obligations are not discharged merely by a clean observed run. They may be generated or checked automatically, but remain theorem                                                            |

| Primary source of evidence                       | Principal items                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Current interpretation                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | conditions recorded in M.7a, including K.PSF/K.EnvMF/K.ResetEpoch applicability, K.JointFreeze where StandardEnv multi-frontier candidates are admitted, K.CompleteS, total K.RespCov, B-faithfulness, K.Drain/K $\epsilon$ , Core.FairPick when K.2b.E constructs a continuation, J.TraceCertCov for safety coverage, and J.QSyncCertInd/K.CertCov when the corresponding universal K-facing claim is made.                                                                                           | premises for the selected design instance.                                                                                                                                                                                                                                                                                                                                                                                                        |
| Reusable tool qualification                      | Qualify the concrete Policy-S simulator semantics and deterministic scheduling/arbitration used by SDet; validate standard lowering/elaboration schemes, certificate schemas, generators, and independent checkers.                                                                                                                                                                                                                                                                                    | Much of this qualification can be reusable across designs. SDet and the selected certificate binding must nevertheless be re-established whenever capacities, elaboration, lowering, arbitration, normalization, environment/reset behavior, or another history-affecting choice changes.                                                                                                                                                         |
| Framework / mechanization work not yet completed | Mechanize the Appendix K-P obligations, including K.JointFreeze/K-O2 and K-O3/K-O4 for freeze preservation, progress dominance, and request reachability; mechanize and qualify Core.FairPick/Core.FairCycleDovetail for proof-built continuations; qualify the certificate interfaces; and mechanize/qualify the J.QSyncCertInd $\rightarrow$ J.CertExist-QSync $\rightarrow$ K.CertCov-QSync theorem route (or an alternate verified package-existence route) for concrete universal-coverage flows. | The paper now supplies the generic QSync package-existence theorem route and the corresponding paper-level soundness arguments for the current formal interfaces. The remaining work is mechanization, independent checking/tool qualification, and concrete discharge of the design-wide refinement/fairness/freeze obligations; the reduction and certificate stack therefore retain proposed, not fully mechanized and flow-qualified, status. |

**Interpretation of the discharge ledger.** M.7a is an explicit accounting of the conditions required for the selected formal claim; it is not a list of separate manual tasks imposed on the design user. Many entries formalize engineering conditions or verification assumptions that were largely already present in either an HLS-generated or manually written RTL flow but might otherwise remain unnamed or implicit. The ledger makes those dependencies visible, assigns responsibility for establishing them, and permits them to be checked, reused, packaged, or qualified systematically. Depending on the selected theorem and instance, an entry may be inapplicable, satisfied by a simple design restriction, theorem-derived, discharged through reusable tool or library qualification, or established automatically by the per-instance flow. The methodology may require more explicit evidence and accounting than an informal

flow, but that additional explicitness should not be confused with newly creating the underlying engineering burden.

### M.7a Side-condition discharge record

Before applying Theorem J.1, Corollary J.1 cutpoint transfer, or the Appendix K liveness-preservation theorem to a concrete design, the verification flow shall maintain an explicit side-condition discharge record. The record need not use one physical file, but every referenced artifact shall be versioned and bound to one selected theorem instance through the common certificate header of Definition K.CertHdr. For each theorem premise or separately named qualification obligation identified below, the record shall state whether the item is inapplicable to the selected theorem or flow-qualification scope, satisfied by a simple syntactic/design restriction, discharged by tool/RTL evidence, or discharged by a source/RTL witness construction. A premise or qualification condition required by the selected theorem or flow-qualification scope may not be marked inapplicable. Remark Core.Det uses this same evidence axis; derived results such as I.DR or theorem-derived Core.KCutClosure\_post are not converted into independent assumptions merely because their proof artifacts are recorded. The named audit entries are Core.RegSeq registered sequential semantics; Core.ReachPrep overlap-aware source reach/preparation; ClockScope single-global-clock theorem scope; Core.QSync qualified synchronous same-cycle settling; Core.CycleClosure full-state source cycle qualification/closure; Core.FairPick constructive fairness witness for proof-built source continuations; Core.IssueFair source-issue weak fairness; B2/B3 scheduler/channel progress; B5/B6-Terminal quiescence-closure/terminal-idle adequacy; B1 observable RTL projection determinism; B4/FPD finite internal stutter/bounded local source progress; A6 point-to-point ownership/explicit arbitration modeling; A8/A9/A10 barrier/cardinality/reset-empty well-formedness; R/E scheduling-contract conformance (R1–R5/E1–E5, including R2C where applicable); Core.IC request-lifecycle/Issue–Commit conformance; Core.SyncFence synchronization-completion conformance; K.CertPkg certificate-package integrity; WC witness-compatible  $\epsilon$  choices/non-blocking outcomes; J.LocalGWR local witness-construction certificates and optional J.GWR-Direct; Sys\_B^elab; B-faithfulness; J.QSyncTraceInd cycle-aligned safety package induction; J.TraceCertCov Post  $\rightarrow$  Ref safety certificate coverage; J.OfferReach/J.OfferConf/J.KObsConf KObs cutpoint correspondence; K.RLAdmReach admissibility of the exact selected source witness; Core.KCutClosure\_post theorem-derived K-facing cutpoint existence/non-vacuity for universal RTL conclusions; J.QSyncCertInd canonical-cycle correspondence/package induction for the standard QSync coverage route; K.CertCov total K-ready package coverage for universal reachable-cutpoint conclusions; K.Drain; K.BF; K $\epsilon$ ; SDet deterministic selected Policy-S simulator; A5-S selected-instance serial liveness; A5-L scheduler-robust liberal liveness; K.CV value determinism/provenance; NB-Align witness-selected non-blocking alignment; K.Low witness-indexed blocking lowering/correspondence; K.EnvMF together with K.JointFreeze-Ord/K.JointFreeze for frozen-component preservation; K.CompareAdvanceLegal dominance-comparison Core.16 launch/issue compatibility for every K.2b.P0/K.2b.R-consumed owner/key; K.InterruptionGate with K.ResetEpoch/K.TerminalDisposition; K.Resp validated bounded-response coverage; optional K.EnvOrd/K.Done diagnostics; W.Env waivers; L.Dir one-directional alignment/RTL-side non-interference; Core.MemSyncOrder array/synchronization commit-order conformance; J.Mem shared-memory lowering; and RW reset well-formedness (RW1–RW5), as applicable to the selected theorem scope. M.7b below provides a standard logical packaging of these entries for QSync flows; it does not alter this discharge record or change the trigger, discharge, or evidence requirements of any entry.

R/E scheduling-contract conformance — R1–R5/E1–E5, including R2C where applicable.

Trigger: Required for every theorem or flow-qualification scope that invokes the Appendix G–K scheduling contract. Apply the applicable R1–R5 source rules to the selected Sys\_ref behavior and the applicable E1–E5 implementation/compatibility rules to the compared RTL\_B behavior at the selected

abstraction and observation boundaries. The R2C fixed-point semantics and matching obligations are required whenever combinational processes or explicitly modeled combinational networks are in scope. Discharge: Bind the concrete source/tool/RTL instance to the applicable R/E rules. Establish the required source dynamic-operation/source-order and synchronization/signal-anchor legality; post-HLS synchronization-order preservation; sequential signal visibility/anchoring or R2C fixed-point observation and matching; prefix-closed message issue order; E4 explicit-boundary/channel-capacity legality; and E5 synchronization-boundary issue/completion ordering. The record may cite constitutive Sys\_ref semantics where a source rule is built into the normative operational model, but corresponding RTL\_B uses shall be backed by tool/RTL/checker evidence or a proved refinement/conformance result. This roll-up does not replace separately named entries such as Core.RegSeq, Core.IC, Core.SyncFence, B-faithfulness, Core.MemSyncOrder, J.Mem, or RW; those remain separately discharged when triggered. Evidence examples: source well-formedness/lint and simulator-semantics qualification; HLS schedule and issue-order certificates; RTL/interface assertions or formal checks for synchronization and signal anchoring; an R2C combinational fixed-point/unique-settling certificate; and explicit-boundary E4/capacity conformance checks.

Core.RegSeq — registered sequential-interface scope and conformance.

Trigger: Required for every Appendix I/J/K theorem instance containing a sequential process whose message requests, Push payloads, or continuation can depend on a message result.

Discharge: Classify each process as R2C combinational or Core.RegSeq sequential. For each sequential process, prove or validate that process-driven request and Push-data outputs for Horizon t are determined from registered state and operands stable before the L2 settle point; that direct inputs satisfy their E2/direct-input stability contract; and that every use of a cycle-t returned data/status result has strictly later Horizon. Tool-side Cat# or equivalent schedule/RTL evidence and source-side simulation, lint, wrapper, or Appendix L evidence shall identify and reject, isolate, or explicitly remodel zero-time NB-result cascades and retry loops. Evidence examples include schedule reports, control-step allocation, generated RTL cone/register checks, source lint, and cycle-accurate transactor or snooping artifacts.

Core.ReachPrep — overlap-aware source reach/preparation.

Trigger: Required whenever the selected Sys\_ref/Policy-L theorem instance permits a source-later dynamic operation to become reached before an earlier same-interval blocking operation process-facing commits, including the straight-line eager cases used by Examples K.CE/K.CE-2, the Appendix K A5-L blocking-rendezvous example, and pipelined/overlapped cases.

Discharge: Identify the source/tool rule or certificate that validates each permitted finite dependency-independent L1 preparation segment under Definition Core.ReachPrep, including dynamic source order, no synchronization crossing, Core.RegSeq/NoDependentUse dependencies, and the Core.16 post-settle prohibition. For every RTL\_B prefix included in a Post → Ref coverage claim, the record shall verify that the selected Sys\_ref ReachPrep semantics admit every dependency-independent preparation/reach segment required by the accepted J.LocalGWR or J.GWR construction for that prefix; the source semantics may be more permissive than the observed RTL schedule but shall not exclude preparation needed to realize a covered RTL witness. The record shall state explicitly whether ordinary straight-line q0; q1 eager reach is admitted. If it is not admitted for an instance, examples or proof arguments that rely on that straight-line liberal behavior may not be used unchanged.

Evidence examples: a schedule/preparation certificate, source dependency/footprint check, a qualified eager-source simulator semantics, or a tool-generated mapping from RTL issue stages to legal Core.ReachPrep segments.

ClockScope — single-global-clock theorem scope.

Trigger: Required for every theorem instance using the formal results of Appendices G–K.

Discharge: Confirm that the selected theorem instance is interpreted under one global clock/cycle relation for every process and explicit abstract channel to which the G–K rules are applied. A larger multi-clock design may be partitioned into synchronous islands, but any clock-domain crossing used by the selected theorem instance must be excluded from the single-clock comparison or represented by a separately specified and verified CDC/refinement boundary under an explicit multi-clock semantic layer. In the absence of that layer, the G–K theorems shall not be applied across clock domains. Evidence examples include a clock-domain/elaboration manifest, structural CDC partition report, or an equivalent checked instance-scope declaration. Core.QSync — qualified synchronous same-cycle settling scope and conformance.

Trigger: Source-side Core.QSync is required on the normal J.LocalGWR → J.GWR-Comp → finite-prefix Theorem J.1 route, because J.RegPhaseNF/J.UnitExt form a finite L2-settled source snapshot for every constructed Horizon; it is also required whenever the application otherwise invokes source Core.Settle/NormRef/KCut settling. A directly validated J.GWR package may instead carry independently checked finite L2-settled segments for its Horizons and need not separately discharge global source-side Core.QSync solely for Theorem J.1. Post-side Core.QSync is required whenever the application invokes post KCut/NormPost settling, including Corollary J.1 cutpoint transfer and every Appendix K application. Any R2C combinational-process fixed-point obligation remains independently applicable.

Discharge: Establish the hypotheses of Definition Core.QSync for the concrete instance. Use the ClockScope entry as the concrete audit evidence for QS1. For the source side, show the Core.Settle L2 frame, stable sequential drives under Core.RegSeq/direct-input contracts, inclusion of R2C combinational components in the settle network, QS3 relation-to-network realization/adequacy for every phase-legal same-cycle settling step, finite acyclic deterministic cycle-local settling, and explicit-boundary completeness. For Sys\_B^elab(E), cite the Appendix Q qualified-settling/template evidence required by Q.1(i) for the elaboration-owned subnetwork and additionally establish template/design settling composition for the complete source network. Every dependency edge crossing the design/template partition shall be included in the whole-network analysis, the combined transition relation shall satisfy the QS3 realization/adequacy condition, and the combined network shall satisfy QS2–QS5; in particular, no combinational settling cycle may be introduced across the partition. A sufficient mechanically checkable QS4 discharge is a combined dependency rank or topological order in which every design-local, template-local, and cross-partition settling edge advances the order. For RTL\_B, establish the analogous K-origin frame, stable registered drives/cycle-t inputs, QS2a origin recoverability for every reachable post KCut in the standard K-facing domain, QS3 relation-to-network realization/adequacy for the complete K-facing same-cycle settling relation, finite acyclic deterministic combinational/handshake settling, and complete selected-boundary mapping/B-faithfulness. The QS2a evidence shall identify the designated same-cycle K-origin/frame and show that the reachable suffix to the KCut contains only the qualified settling relation before any later registered update; quiescence alone is not accepted as origin recovery. The checker shall validate enough of these structural facts to invoke Theorems Core.QSync-Ref and Core.QSync-Post.

Evidence examples: a syntactic/design restriction plus source/elaboration template proof; an independently checked combinational-dependency/clock/register-frame certificate; a combined dependency-rank or template/design interface-compatibility certificate covering cross-partition edges; structural RTL/netlist analysis; a qualified template-library theorem; or equivalent tool/RTL evidence. Trust accounting follows K.CertHdr: an unverified tool report used as an axiom enlarges the declared trusted base, while an independent check of the concrete artifact does not require trusting the producing HLS tool for that fact.

Scope boundary: Core.QSync is same-cycle settling evidence only. It does not discharge Core.CycleClosure full-state/source-cycle/environment definedness, B2/B3 cross-cycle progress, or the terminal Core.EnvTerminal premise. No generic RTL cross-cycle totality premise is required by the finite-prefix J.1 or K-facing cutpoint-closure results.

Core.CycleClosure — full-state source cycle qualification and closure.

Trigger: Required when a theorem constructs a complete/fair source continuation through Core.LAdm, notably Lemma K.2b.E/K-O7. It is not required by finite-prefix Theorem J.1 or by RTL K-facing cutpoint closure.

Discharge: Bind Definition Core.CycleResult and its Core.CompleteCycle projection to the selected source/environment semantics; prove the Core.CycleClosure record: full-state CycleResult typing, nextState clock/environment/auxiliary advancement, exact scoped RESET emission under RW1-RW5, nonterminal source-cycle existence, finite L1 preparation, finite L2 settling (Core.QSync-Ref or an allowed equivalent route), compatible EnvStep definedness, and the C2 partial-cycle extension property needed by Lemma Core.CycleChoiceExt: every finite phase-legal selected issue/boundary segment used by a continuation proof must admit completion to a typed CycleResult. Verify that source-side SilentCycle is a CycleResult with no selected  $\Sigma$  commit rather than an  $\epsilon$ /identity step, and that any proof-internal RESET remains present in resetUnits. For Lemma K.2b.E, C2 is used only through the two-stage Core.FairCycleDovetail construction; CycleClosure itself does not discharge fairness or legalize an L3 action chosen before the actual L2 settled state is known.

Evidence examples: mechanized source cycle semantics with a cycle-completion lemma; source FPD/preparation plus QSync certificate and checked partial-cycle extension; environment state-machine totality proof; or a qualified source-model theorem whose concrete artifact is independently checked.

Core.FairPick — constructive fairness witness for proof-built source continuations.

Trigger: Required when Lemma K.2b.E or another theorem constructs a complete/fair Policy-L execution from a finite prefix rather than consuming an already supplied execution. It is not separately required merely to state Core.IssueFair/B2/B3 of an existing run.

Discharge: Provide the FP1–FP5 IssuePick/BoundaryPick witness, local obligation finiteness/keying, two-stage settle-then-boundary applicability, and starvation proof for the particular fixed theorem instance. Distinguish proof-selectable Policy-L issue latitude from choices fixed by Core.LLegal/I. A round-robin/oldest-enabled selector may be introduced only for genuinely proof-selectable issue choices; any fixed scheduler component, arbitration/grant, environment, witness, reset, or other bound choice must be followed and proved fair enough for FP4/FP5. This is stronger constructive evidence than an abstract fairness assertion, but it does not strengthen the behavioral fairness property being established.

Evidence examples: mechanized dovetailing over proof-selectable Policy-L issue choices plus a rank proof for fixed arbiters; a qualified source operational-semantics theorem returning prefix-local fair selectors; or an independently checked scheduler/arbiter certificate exposing the required selection and starvation ranks.

Core.IssueFair — weak fairness of source issue.

Trigger: Required for complete Core.LAdm executions consumed by Lemma K.2b.E/K.2b/K.2b.R and wherever the Policy-S route invokes issue fairness.

Discharge: For each relevant reached operation, prove that if its issue transition remains legal at every applicable L1 opportunity within the reset epoch, it is eventually issued. The legality predicate must include Core.16; an operation blocked as source-later is not IssueEnabled and therefore creates no IssueFair obligation. Intermittently legal operations do not trigger this weak-fairness antecedent. When the theorem constructs the complete run itself, notably K.2b.E,

the separate `Core.FairPick` entry supplies the required prefix-constructive issue-selection witness; abstract fairness of some already-given execution is insufficient.

Evidence examples: simulator scheduler proof, source operational-semantics theorem, fairness assertion over issue-enable state, or mechanized dovetailing argument.

**B2/B3** — boundary-action and channel progress.

Trigger: Required whenever a complete/fair continuation is used by Lemma K.2b.E, the K.2b dominance/response arguments, K.CompleteS, or another theorem that explicitly requires such a run. Appendix I.5 and the finite-prefix J.1 correspondence do not require B2/B3 merely to construct finite replay prefixes.

Discharge: Record the concrete scheduler/arbitrator and channel evidence establishing B2 weak fairness for continuously `ChannelEnabled Push/Pop/Sync` actions and the B3 `P_push/P_pop` channel-progress obligations on every relevant explicit abstract channel. This entry is distinct from `Core.IssueFair`, which concerns request assertion before a boundary action is enabled. When K.2b.E constructs the continuation, the separate `Core.FairPick` entry supplies the boundary-selection/B3-exposure witness and binds it to the actual fixed scheduler/arbitrator; a proof may not replace a fixed unfair arbitration choice with an invented fair one.

Evidence examples: fair-arbitrator RTL proof, ready/valid liveness assertions, channel-library qualification, or a mechanized scheduler invariant.

**B5/B6-Terminal** — quiescence closure and terminal idle adequacy.

Trigger: B5 is required where its bounded quiescence/normalization role is invoked. The B6 terminal-idle branch is required only when an infinite execution is extended by idle ticks after a permanently idle state.

Discharge: Prove the B5 closure premise for the selected bucket semantics and, for any B6 suffix, prove `Core.TerminalIdle`: no future registered/internal machine transition and no permitted future environment evolution can change non-clock state or re-enable work. `Core.EnvTerminal` supplies only the environment component; `Core.CycleClosure`'s environment-definedness clause or `Core.EnvTerminal` alone is not sufficient. `StandardEnv` stimulus exhaustion plus a machine-side stable-terminal/permanent-idle proof is one sufficient route. A currently idle but nonterminal state must instead continue through `Core.SilentCycle`.

Evidence examples: bounded quiescence proof plus stimulus-exhaustion and stable-machine certificate, explicit terminal state with no registered successor effects, or a model-checked permanent-idle invariant.

**B1** — observable RTL projection determinism. Trigger: Required wherever B1 is a theorem premise, including the finite-prefix J.1 correspondence and Appendix K routes that cite RTL observable determinism. Discharge: Under the fixed initial state and stimulus, prove uniqueness of the `RTL_B` observable  $\Sigma$  projection required by Definition B1. Internal  $\epsilon$  paths, issue annotations, request attribution, and next-cycle totality need not be unique; those stronger properties are discharged separately. Evidence examples: deterministic RTL semantics/refinement proof, qualified simulation theorem, or formal trace-projection uniqueness check.

**B4/FPD** — finite internal stutter / bounded local source progress. Trigger: Required wherever Lemma I.1, finite replay/preparation normalization, B5 charging, or the generic source-cycle finiteness argument relies on FPD/B4. Discharge: For each relevant process P, provide the finite `D_P` bound (or an equivalent well-founded measure) showing that consecutive internally advancing  $\epsilon$ -work reaches the next  $\Sigma$  action or a stable external/peer wait in finitely many steps. A certified Lemma I.3 preparation segment may cite its local certificate; generic `Core.CycleClosure` use requires the corresponding uncapped per-cycle source-control finiteness evidence for its source-cycle-existence clause. Evidence examples: finite pipeline/FSM-depth

proof, source/control graph ranking, simulator semantics theorem, or tool-generated progress certificate.

A6 — point-to-point ownership / explicit arbitration modeling. Trigger: Required wherever Appendix K constructs WFG edges, complementary endpoint ownership, or K.2b point-to-point dominance obligations. Discharge: Show that every explicit message boundary used by the theorem has one producer and one consumer, or remodel shared resources through an explicit arbiter plus point-to-point subchannels whose choices are included in the selected instance. Evidence examples: elaboration/ownership manifest, structural channel check, arbiter-lowering certificate, or source/RTL connectivity proof.

A8/A9/A10 — barrier, endpoint-cardinality, and reset-empty well-formedness. Trigger: A8 is required whenever SyncChannels/barriers are omitted from the internal Push/Pop WFG; A9 is required for scalar per-cycle endpoint semantics; A10 is required for buffered channels at initial/reset boundaries. Discharge: For A8 prove matched barrier participation/no permanent bypass under the fixed stimulus; for A9 prove at most one Push and at most one Pop commit per channel per cycle unless an explicit multi-lane model is used; for A10 prove logical buffered occupancy is zero at each selected reset boundary or explicitly model any preload. Evidence examples: source/elaboration checks, barrier-count/participation proof, interface cardinality assertions, and reset-state/channel initialization checks.

Core.IC — request lifecycle / Issue-Commit conformance.

Trigger: Required on RTL\_B whenever the selected theorem or certificate uses implementation-side Issue/BoundaryReq/Commit attribution or blocking-request persistence, including the applicable E3/E5 trace obligations, J.OfferConf/J.KObsConf request abstraction, K.Drain, or K.BF. On Sys\_ref, Core.IC is constitutive operational legality and is not a separate premise to be assumed or independently discharged.

Discharge: For every admitted explicit boundary and endpoint direction in the selected scope, establish the applicable Core.IC lifecycle and attribution rules: an attributed blocking request, once issued, remains asserted with stable Push payload until its process-facing commit or an affecting RW2 RESET; no hidden nonterminal cancellation, withdrawal, retry, replacement, or reattribution is permitted; continuing/back-to-back request levels are assigned to the correct dynamic operation; the scalar same-endpoint serialization rule is respected; and synchronization-boundary attribution is consistent with Core.SyncFence. Evidence examples include an RTL/interface assertion or monitor, structural/control-state analysis, a formal refinement check, or a checked request-attribution certificate bound to the selected explicit-boundary mapping. A protocol requiring an additional nonterminal request disposition is outside the present Core.IC model unless that operational extension and its correspondence obligations are separately defined and proved.

Core.SyncFence — synchronization-completion conformance.

Trigger: Required whenever a synchronization call  $s$  selected into  $S$  can have a blocking Push/Pop  $q$  in  $\text{pref}_S(s)$  on a side of the comparison. The obligation is vacuous for a synchronization boundary with no such blocking predecessor.

Discharge: Establish the applicable R3/E5/Core.SyncFence condition that every such  $q$  process-facing commits no later than  $s$  and that no issued-but-uncommitted  $q$  survives the synchronization into the succeeding source interval. For source operational semantics, a same-cycle  $q/s$  completion is represented by the selected L3 boundary-action batch with  $q$  accounted for before the interval update. For RTL\_B, no physical sub-cycle ordering is required: the implementation evidence must establish the same commit cycle and the certificate/annotation convention must attribute the predecessor completion before advancing the formal source-interval bookkeeping. In Sys\_B^elab, bind  $q$  to the designated process-facing completion event

rather than an arbitrary internal staging transfer. Evidence examples include schedule/control-state evidence, an RTL/interface monitor or assertion, a structural/formal refinement check, or a checked process-facing completion map in the elaboration certificate.

K.CertPkg — certificate identity, schema, and trust boundary.

Trigger: Required whenever a generated or manually assembled certificate package is used to discharge Corollary J.1, Theorem K.1A, Corollary K.1A-Total, Theorem K.1, A5-S/A5-L, or CF-1 through CF-5.

Discharge: Provide the common K.CertHdr fields: certificate-schema and document/theorem version; a digest or unambiguous identifier for the selected source, Sys\_ref/Sys\_K, and RTL artifact; E, B\_E, explicit boundaries, projections, ownership, and K.Low lowering where applicable; fixed stimulus, environment, reset policy, witness w, and WC provenance; KObs-facing waiver set W and closure evidence; the selected A5-L discharge route; the Core.QSync structural-evidence provenance and Core.QSync-Ref/Core.QSync-Post theorem references used to derive standard K-facing settling/closure, together with the K.CertCov coverage route when a universal reachable-cutpoint conclusion is claimed; and, for the Policy-S route, the concrete deterministic simulator/scheduler, deterministic arbitration, Policy-S issue semantics,  $\epsilon$ -normalization identity, SDet evidence, K.Resp monitor schema, finite bounds, reset/end-of-test disposition, response waivers, and coverage-preservation evidence; plus the theorem scope, clock-scope, Core.IC, and Core.SyncFence side-condition references where applicable, other side-condition references, generator/checker versions, and declared trusted base. Every subordinate KObs or A5 certificate shall cite the same header identity. Derived K-facing predicates and K.Resp outcomes shall be recomputed by the checker from their respective artifacts rather than accepted as authoritative fields. A QSync property accepted only from an unverified tool report is part of the declared trusted base; a property independently checked from the bound artifact is recorded as checked evidence instead.

Evidence examples: A signed or hashed manifest binding all artifacts, an independently checked structural QSync certificate, qualified checker/tool versions, and reproducible source/RTL identities.

- WC — witness-compatible  $\epsilon$  choices / non-blocking outcomes.

Trigger: Required when failed non-blocking polls or other  $\epsilon$ -modeled choices can affect later  $\Sigma$ -visible control flow, frontier membership, request attribution, guard values, payload values, or other later  $\Sigma$ -visible choices.

Discharge: Provide evidence that proves the actual WC premise used by Appendices I–J. The audit entry shall identify the provenance route by which WC was established:

- NB-CF route: show that failed PushNB/PopNB outcomes do not affect later  $\Sigma$ -visible control flow, frontier membership, request attribution, guard values, payload values, or other later  $\Sigma$ -visible choices. Under this route, the failed-poll witness may be the identity or arbitrary selector, subject to the applicable latency-sensitive local-protocol conditions.
- Appendix L snooping route: provide a witness selector W satisfying Lemma L.2. Show that the snooped outcomes cover every  $\epsilon$ -modeled source choice that can affect later  $\Sigma$ -visible behavior, that the wrapper changes only  $\epsilon$ -choice selection, and that it does not add, remove, reorder, or relabel  $\Sigma$ -visible events.
- Direct witness-construction route: otherwise construct and prove the  $\mu_Q$  / WC witness used by Appendix I/J.
- Mixed per-site route: partition the relevant non-blocking or  $\epsilon$ -choice sites among the applicable NB-CF, snooping, or direct-construction routes, and show that the partition covers every relevant site.

NB-CF and Appendix L snooping are therefore provenance routes for proving WC, not separate theorem premises that replace WC. When useful for review or automation, the WC audit entry may include optional per-site metadata identifying each relevant call site or  $\epsilon$ -choice site, its selected provenance route, and the corresponding source restriction, tool/RTL evidence, wrapper evidence, or witness construction.

Evidence examples: Source inspection or a static control/dataflow check establishing NB-CF; a generated WC witness or certificate; a wrapper proof or monitor specification satisfying Lemma L.2; a co-simulation/snooping harness, subject to the L.Dir entry below; or a complete per-site coverage report for a mixed discharge.

- **Sys\_B<sup>elab</sup>** — elaborated source reference model.  
 Trigger: Required when ordinary Sys\_B capacities at the outer abstract boundaries are not sufficient to represent the RTL realization as ordinary E4 bounded-FIFO/rendezvous behavior; examples include multi-input/join couplers, staged/bundle/join channels, and sync-boundary / epoch-gate capacity withdrawal.  
 Discharge: Provide a well-formed elaboration package E defining the explicit abstract channels, capacities, projection  $\Pi_{elab}$ , and E4 ChannelEnabled/ChannelDisabled behavior used by Sys\_B<sup>elab</sup>(E). The package shall additionally satisfy Definition Q.1(i): its cycle-local elaboration/ready-valid/guard/join/coupler/epoch-gate subnetwork must satisfy the applicable Core.Settle frame and template-side finite-acyclic deterministic-settling/boundary-completeness conditions. That elaboration-side result may be proved once for each qualified elaboration template and then bound to the concrete E instance. It does not by itself establish whole-source QualifiedSynchronousModel\_Sys\_ref(I); the Core.QSync discharge above additionally combines the template evidence with the design-owned Core.RegSeq/direct-input/R2C/non-template-boundary facts and the checked template/design settling-composition condition, including complete-network QS4.  
 Evidence examples: Tool-generated elaboration plus an independently checkable template certificate, structural back-annotation report, Appendix Q-style construction, qualified elaboration-library theorem, or manual elaboration justified against the RTL channel structure; for the complete source-QSync discharge, also provide the applicable combined dependency/interface certificate required by the Core.QSync entry.
- **B-faithfulness** — faithfulness of each back-annotated capacity B(c) and chosen explicit abstract channel boundary.  
 Trigger: Required for every back-annotated explicit abstract channel c whose B(c), occupancy occ\_c(t), E4 ChannelEnabled/ChannelDisabled behavior, BoundaryReq facts, or Appendix K blocking/WFG behavior is used by the selected theorem instance. For such a channel, B-faithfulness is a required per-channel obligation and shall not be marked inapplicable.  
 Discharge: Show all of the following for the chosen abstract boundary of c:  
 Boundary completeness — every  $\Sigma$ -visible transfer of c in RTL\_B crosses the chosen abstract boundary as the corresponding committed Push\_c or Pop\_c, with no hidden side entrance or exit.  
 Exact boundary behavior — for every relevant execution state and boundary action, the concrete realization permits the action exactly when the E4 abstract channel permits it. In particular, realized unpopped occupancy is bounded by B(c), and the realization contains no hidden storage or credit beyond the capacity represented by B(c).  
 No hidden enablement or disablement — no unmodeled grant, arbitration, throttling, or backpressure condition changes whether a boundary action is permitted. Any such condition

that affects boundary behavior shall instead be represented by explicit  $\text{Sys\_B}^{\text{elab}}$  boundary structure.

$\epsilon$ -opacity of internal movement — movement of data within the realized channel between chosen boundary commits is internal  $\epsilon$ -behavior and creates no additional  $\Sigma$ -visible Push/Pop event or Appendix K WFG blocking cause.

Evidence examples: A trusted back-annotation algorithm; a per-channel capacity and boundary certificate; a structural RTL check; an RTL/interface monitor; a formal refinement check comparing the realized boundary behavior with E4; or equivalent tool/RTL-flow evidence establishing the complete B-faithfulness obligation.

- **J.QSyncTraceInd** — cycle-aligned safety package induction for the standard QSync route.  
 Trigger: Required when  $\text{J.TraceCertCov\_I}(\text{CycleReach\_post(I)})$  is discharged through Theorem  $\text{J.TraceCertCov-QSyncCycle}$  for either Corollary J.1-Safe-QSyncCycle or the all-finite-prefix Corollary J.1-Safe-QSync. It is not required for a one-prefix Theorem J.1 application, for a direct J.GWR route, or for a broader safety domain that is discharged independently through  $\text{J.TraceCertCov}$ . If  $\text{J.QSyncCertInd}$  is already discharged for the standard Appendix K route, Lemma  $\text{J.QSyncCertInd-Trace}$  supplies  $\text{J.QSyncTraceInd}$  and no second induction proof is required.  
 Discharge: Prove TI0 for the initial/reset-established exact RTL K-origin prefix and TI1 for every reachable one-cycle RTL successor, extending the exact  $\text{J.LocalGWR/J.GWR-Comp}$  source witness and  $\text{J.HCanon}$  history through the new complete observable units/request-attribution changes with source-side  $\text{Core.QSync-Ref}$  settling and  $\text{J.RS/RW}$  reset handling where applicable. TI1 shall not rely on  $\text{J.OfferReach}$ ,  $\text{J.OfferConf}$ ,  $\text{J.KObsConf}$ ,  $\text{K.DrainReach}$ , or  $\text{K.RLAdmReach}$ ; Corollary  $\text{J.RegPhaseReach-Comp}$  is theorem-derived from the retained  $\text{J.GWR-Comp}$  evidence if needed. TI2 requires exhaustive successor coverage; a single simulation path is insufficient unless an independent determinism theorem collapses the reachable successor set.  
 Evidence examples: a mechanized safety projection of the source/RTL cycle refinement relation; a checked cycle-step LocalGWR certificate generator plus induction theorem; bounded/exhaustive cycle-step proof; or projection of an already qualified  $\text{J.QSyncCertInd}$  proof.
- **J.TraceCertCov** — Post  $\rightarrow$  Ref certificate coverage for safety claims.  
 Trigger: Required whenever Corollary J.1-Safe is applied directly over a coverage domain D, and as the cycle-aligned package interface discharged by Theorem  $\text{J.TraceCertCov-QSyncCycle}$  in the standard QSync route.  
 Discharge: Identify D and prove that every  $\tau\_post \in D$  carries one route-complete Post  $\rightarrow$  Ref package. On the normal compositional route, provide the applicable  $\text{J.LocalGWR}$  evidence, source-side  $\text{QualifiedSynchronousModel\_Sys\_ref(I)}/\text{Core.QSync-Ref}$  settling evidence required by  $\text{J.GWR-Comp}$ , and all remaining applicable Post  $\rightarrow$  Ref premises of Theorem J.1. On the direct route, provide a directly validated  $\text{J.GWR}$  package whose checked reachable phase-normal chain contains a finite L2-settled segment for every represented Horizon, together with all remaining applicable Post  $\rightarrow$  Ref premises. Bind all evidence to the same theorem instance, stimulus, observation boundary, and selected RTL\_B prefix. For a direct universal RTL safety application of Corollary J.1-Safe, D shall contain every reachable RTL\_B prefix relevant to  $\Phi$ . This safety-side coverage fact does not imply  $\text{K.CertCov}$ . In the standard normal-compositional QSync route, it is sufficient for prefix-closed finite-history safety to take  $D = \text{CycleReach\_post(I)}$ , discharge  $\text{J.TraceCertCov\_I}(D)$  from  $\text{J.QSyncTraceInd}$  by Theorem  $\text{J.TraceCertCov-QSyncCycle}$ , and then invoke Lemma  $\text{J.CycleIdealCover}$  / Corollary J.1-Safe-QSync. No separate  $\text{J.TraceCertCov}$  package is thereby asserted or required for each non-cycle-aligned ideal. Lemma  $\text{J.QSyncCertInd-Trace}$  permits reuse of an already discharged K-facing  $\text{J.QSyncCertInd}$  proof.

- J.OfferReach / J.OfferConf / J.KObsConf — KObs cutpoint correspondence. Trigger: Required whenever Corollary J.1 cutpoint correspondence is used, including every Appendix K application that transfers Pending\_P, BoundaryReq, rendezvous endpoint requests, ChannelEnabled/ChannelDisabled, Front\_P, WFG, or derived deadlock facts between RTL\_B and Sys\_ref. Appendix R only summarizes this transfer and adds no separate obligation. The evidence shall be packaged as, or mapped unambiguously onto, the CF-0 KObs cutpoint certificate of K.4e. J.OfferReach discharge: For the same selected Theorem J.1 source witness, witness  $w$ , and  $\varepsilon$ -normalization used by the application, exhibit  $H$  and  $O$  and show that the selected source K-facing cutpoint  $\sigma_{\text{ref}} \in \text{KCutReach\_ref}(I)$  satisfies  $H = H_{E,w}(\sigma_{\text{ref}})$  and  $O = \text{ResOffers}_{E,w}(\sigma_{\text{ref}})$ , with all operational Issue/Commit, E3/E5, reset-epoch, elaboration, and source-transition rules respected. This is an image/realizability obligation; well-typedness of  $\langle E, w, H, O \rangle$  alone is insufficient. The complete Issue history may remain in the referenced source witness rather than being duplicated in the CF-0 record. J.OfferReach does not by itself establish Definition Core.LAdm admissibility; that K-specific fact is K.RLAdmReach below. J.HCanon discharge: Establish the hypotheses of Lemma J.HCanon for the selected source/RTL annotated prefixes and fixed theorem-produced or directly validated J.GWR package; record that both prefixes produce the same canonical enriched replay-history quotient  $H$ . J.OfferConf discharge: At the selected RTL K-facing cutpoint  $p_{\text{post}} \in \text{KCutReach\_post}(I)$ , enumerate every KObs-relevant asserted endpoint-owned request at the explicit boundaries of  $E$  and prove bidirectional, unique attribution between that physical request set and  $O$  using the operational-attribution component produced by J.GWR-Comp (or supplied by J.GWR-Direct), Appendix C, and I.A0. Record the physical RelevantReqSet\_ $E(p_{\text{post}})$ , ReqProjection\_ $E(O)$ , their equality, request-to-offer completeness, no-hidden-request, boundary completeness, issued-summary prefix closure, reset-epoch typing, and endpoint cardinality. A monitor or certificate that reports only requests it can already attribute is insufficient unless it separately proves that no other KObs-relevant asserted request exists. J.KObsConf discharge: For that same selected tuple  $(p_{\text{post}}, \tau_{\text{ref}}, \sigma_{\text{ref}}, E, w, H, O)$ , additionally establish B-faithful/E4 interpretation at every explicit boundary, reset alignment, and  $\varepsilon$ -opacity of normalization. J.KObsConf is the composite application wrapper; it no longer requires independent proof of a broad source-value/frontier/channel/enablement slice, because those facts are reconstructed from equal KObs by Lemmas J.KObs-Proc–J.KObs-WFG. Audit priority: J.OfferConf is the authoritative RTL request-abstraction boundary. Its completeness and no-hidden-request directions shall receive the same explicit review and certificate coverage previously assigned to the request/pending clauses of the broad J.KObsConf relation. Evidence examples: A CF-0 tool-generated paired KObs/cutpoint certificate; a source execution log proving OfferReach; an RTL/interface request-attribution monitor plus boundary-completeness proof proving OfferConf; structural back-annotation and B-faithfulness evidence; a co-simulation witness log; a bounded proof over generated RTL; or an equivalent combined certificate.

K.RLAdmReach — composed Core.LAdm qualification of the exact selected source witness. Trigger: Required whenever Theorem K.1A, Corollary K.1A-Total, or Theorem K.1 uses the source witness selected by Corollary J.1. It is not required for an ordinary safety-only Corollary J.1 application.

Discharge: Bind the exact tuple  $(p_{\text{post}}, \tau_{\text{ref}}, \sigma_{\text{ref}}, E, w, H, O)$  used by J.OfferReach/J.OfferConf/J.KObsConf. On the normal compositional route, reference Corollary J.RegPhaseReach-Comp and its theorem-derived J.RegPhaseReach fact obtained from Core.RegSeq, the enriched Lemma I.3 records, J.RegPhaseNF, and Horizon-batched J.UnitExt; on the direct route, supply an equivalent independently checked phase-normal witness. Separately supply K.DrainReach for the same prefix, proving nonwithdrawal,  $K\varepsilon$ , Core.Drain/K.Drain from

every L2 fully-disabled-frontier activation point, and Core.SettleKBridge from each persisting activation through any intervening drain to the exact selected K-facing cutpoint, together with the common instance/stimulus/witness binding. The checker combines J.RegPhaseReach and K.DrainReach to establish  $\tau_{\text{ref}} \in \text{Pref\_L}^{\text{adm}}(\text{I})$ . KObs equality, OfferReach alone, a bare “admissible=true” flag, or ambient premise names do not discharge this obligation.

Core.KCutClosure\_post — K-facing cutpoint existence / non-vacuity (derived in the standard QSync scope).

Trigger: Required by Corollary K.1A-Total and the universal user-facing Theorem K.1 conclusion. It is not required to apply Theorem K.1A to one already selected certified KCut.

Standard discharge: Cite the accepted Core.QSync audit entry for RTL\_B and Theorem

Core.QSync-Post. No independent per-instance semantic normalization argument is required beyond the structural QSync evidence: the theorem proves that every reachable RTL K-facing cycle-evaluation origin admits one finite same-cycle normalization to a unique  $p_{\text{post}} \in \text{KCutReach\_post}(\text{I})$  before a later cycle-advancing registered update. With Kε, Lemma Core.KCutObserve supplies the persistent-blocker observation consequence. Record the theorem/certificate identity in K.CertHdr.

Scope note (intentional standard-scope narrowing): Core.KCutClosure remains the downstream semantic interface consumed by Appendix K. The standard G–K scope now requires the finite-acyclic QSync settling discipline and therefore withdraws the former independent standard-scope Core.KCutClosure\_post discharge alternatives based solely on an arbitrary inductive RTL invariant or checked normalization certificate. A non-QSync design is not asserted incorrect, but it is outside the standard user-facing theorem scope unless an explicitly defined and separately qualified extension proves the same Core.KCutClosure interface under its alternative settling semantics.

J.QSyncCertInd — canonical-cycle correspondence/package induction for the standard QSync coverage route.

Trigger: Required only when K.CertCov is discharged through Theorem J.CertExist-QSync / Corollary K.CertCov-QSync. It is not required for one-cutpoint Theorem K.1A or when an alternate independently checked K.CertCov route is used. When it is discharged, Lemma J.QSyncCertInd-Trace also supplies J.QSyncTraceInd, so the same proof automatically supports the standard QSync safety route, including the J.CycleIdealCover / J.1-Safe-QSync ideal-closure step for finite-history properties satisfying J.IdealPrefixClosed.

Discharge: Prove CI0 at the initial/reset-established canonical QSync cutpoint and CI1 for every reachable one-cycle RTL successor from a related canonical cutpoint. The step proof shall extend the normal J.LocalGWR/J.GWR-Comp witness, update HCanon/OfferReach/OfferConf/KObsConf, obtain the canonical next post/source settling endpoints from QSync-Post/QSync-Ref, preserve or extend J.RegPhaseReach and K.DrainReach/Core.SettleKBridge, and handle RESET units through J.RS/RW1–RW5. CI2 requires exhaustive successor coverage: a single observed simulation path is insufficient unless a separately proved determinism theorem collapses the reachable successor set. The proof may be an inductive simulation/refinement invariant or a verified extraction relation, but it may not assume K.CertCov or a successor K.CertifiedCutpoint. Evidence examples: a mechanized inductive source/RTL relation whose invariant carries the QSyncCertState fields; a checked cycle-step certificate generator plus induction theorem; a bounded/exhaustive proof covering every reachable next-cycle transition; or a qualified tool-specific refinement theorem.

K.CertCov — total K-ready package coverage for universal reachable-cutpoint conclusions.

Trigger: Required whenever Corollary K.1A-Total or the universal user-facing Theorem K.1 conclusion is claimed, together with Core.KCutClosure\_post. It is not required to apply Theorem

K.1A to one already certified cutpoint.

Standard QSync discharge: Establish the Core.QSync audit entries for RTL\_B and Sys\_ref and prove J.QSyncCertInd. Theorem J.CertExist-QSync then proves J.QSyncCertCov over the complete canonical KCut domain, using Lemma Core.QSync-KCutCanonical; Corollary K.CertCov-QSync binds those semantic tuples to K.CertHdr records and derives K.CertCov. This is the preferred generic package-existence route in the standard scope.

Alternate discharge: A proved total certificate-extraction function on KCutReach\_post(I), another inductive simulation/refinement relation whose invariant contains the complete package data, exhaustive finite-state exploration, a verified certificate generator, or a separately qualified tool-specific package-existence theorem may establish the same K.CertCov interface.

Scope and anti-shortcut rule: Coverage is required exactly on the reachable Core.KCut RTL domain on which the K-facing deadlock predicate is defined; arbitrary transient RTL microstates and arbitrary pre-normalization prefixes need not carry CF-0 packages. Merely supplying a package for one observed run, a finite set of sampled K-facing cutpoints, or a clean source simulation does not establish K.CertCov. Core.KCutClosure\_post separately proves that a persistent K-relevant blocking configuration cannot remain hidden indefinitely in excluded transient states.

- K.Drain — drain-only blocked-frontier discipline and settle-to-KCut bridge. Trigger: Required for Appendix K whenever the selected liberal Sys\_ref or RTL\_B execution may have source-later same-interval blocking work already issued while an earlier minimal blocking frontier remains uncommitted. This includes pipelined-loop overlap, other explicitly modeled eager/overlapped issue under Policy L, and ordinary non-overlapped control when Lemma K.2b or drain normalization relies on the absence of newly launched work past the blocked point. Discharge: Show the full Definition Core.16 discipline—blocked-frontier identification, no new later work, no request withdrawal, drain-only downstream progress, finite explicit boundary storage/credit, and any required sync-boundary capacity withdrawal in Sys\_B^elab—and discharge Core.SettleKBridge from every persisting L2 activation to the selected later drain-closed KCut, with Kε/WFG-inert normalization and every intervening drain commit represented explicitly. Ordinary source operations on distinct interfaces remain source-ordered; any multi-frontier behavior used by Appendix K must be represented by explicit source/elaboration structure rather than by implicit incomparability. Under Definition Core.LAdm this discipline and bridge are part of Appendix K admissibility once the settled-frontier trigger occurs; they are not part of the basic Policy-L issue rule before that trigger.

K.BF — blocking-only serial fragment. Trigger: Required by Lemma K.2b, Corollary K.2c, K.PSF, and the Policy-S route whenever the bounded-response reduction is used. Discharge: Show that every WFG-relevant wait is a persistent blocking Push/Pop on a finite B-faithful point-to-point boundary; control, payload, and later operation identity are deterministic under K.CV/fixed witness choices; shared arbitration is explicit and deterministic or witness-fixed; and local computation between message operations is bounded so pure-computation divergence cannot replace the process-facing serial frontier used by K.2b.Q. Because those waits are ordinary blocking requests, Core.IC supplies the exhaustive lifecycle: once issued, the request remains the same pending dynamic instance until process-facing commit or an affecting RW2 RESET, while DeclaredTerminal may end the execution with it still pending. Lemma K.BF-Persist packages that fact for the frozen-frontier proof. A protocol with independent nonterminal cancellation, withdrawal, retry, or reattribution of an outstanding blocking request is outside the G–K theorem model and requires a separately defined operational extension before any A5-L route in this theorem stack can be used. Evidence examples: source/control-flow qualification,

explicit arbiter lowering, finite local-progress proof, per-boundary B-faithfulness certificate, and conformance of every admitted blocking interface to Core.IC/RW2.

- $K\epsilon$  — WFG-inert  $\epsilon$  / no hidden micro-timing control decisions.  
Trigger: Required for Appendix K liveness/deadlock preservation when  $\epsilon$ -steps occur between  $\Sigma$ -visible cutpoints.  
Discharge: Show that  $\epsilon$ -only steps do not create, remove, or redirect the relevant Appendix K WFG blocking causes except through the explicit  $\Sigma$ -visible or E4-modeled channel state already represented in Sys\_ref.  
Evidence examples: RTL structural argument, scheduler proof, bounded pipeline/drain argument, formal check over the WFG abstraction, or tool certificate.
- SDet — deterministic selected Policy-S simulator. Trigger: Required for K.2b, K.2c, K.1, K.PSF-1, or CF-S. After fixing Sys\_K, capacities/elaboration, initial/reset state, stimulus, environment function, witness-selected  $\epsilon$  outcomes, deterministic arbitration, concrete SystemC/pre-HLS scheduling, Policy-S issue semantics, and  $\epsilon$ -normalization, SDet defines a functional Policy-S next-state semantics and selects one run artifact  $p\_S^{\text{det}}$ . For every finite or infinite run artifact  $p$  generated from the same initial state under exactly those selected semantics, the committed histories of  $p$  and  $p\_S^{\text{det}}$  are prefix-comparable modulo finite WFG-inert  $\epsilon$ -stutter; equivalently, the selected semantics generate no second incompatible committed history. A finite artifact may therefore be a proper prefix of  $p\_S^{\text{det}}$  without contradicting SDet. WC and deterministic replay give reproducibility relative to the selected history; SDet is the separate global premise fixing every remaining history-affecting choice. Changing any such choice creates a different theorem instance. Discharge by a simulator-semantics proof, deterministic qualification, schedule/arbitration certificate, or mechanized functional-determinism and history-comparability theorem. SDet does not assert that  $p\_S^{\text{det}}$  is maximal, fair, complete, or bounded-response clean. Those facts are separate K.CompleteS and A5-S/CF-S obligations.
- A5 — primary selected-instance Policy-S bounded-response discharge (A5-S). Trigger: Required whenever Theorem K.1 is invoked; Theorem K.1A and Corollary K.1A-Total instead consume A5-L. A5-S is a property of the execution  $p\_S^{\text{det}}$  selected by SDet for the selected normalized instance Sys\_K, including capacities, elaboration, fixed stimulus, witness choices, deterministic simulator/scheduler and arbitration semantics, environment/reset contracts, K.Low lowering, and the K.Resp configuration. It combines three obligations for that exact run: (1) K.CompleteS evidence establishing maximality, fairness, and completeness over the CompleteCycle projection of the selected CycleResult semantics; (2) K.RespCov evidence defining a checkable disposition for every static internal process-facing blocking site and relevant dynamic class and proving that every dynamically reachable instance belongs to a monitored or statically bounded class, together with finite bounds and waiver-preservation, while environment-facing sites are handled by the separate StandardEnv/monitor/exclusion rule; and (3) K.RespClean evidence proving that the complete run contains no non-waived K.RespFail witness. Discharge: Supply CF-S identifying  $p\_S^{\text{det}}$  and carrying all three obligations. A finite prefix of a nonterminating execution is supporting evidence only unless accompanied by a complete invariant, model-checking, recurrence/lasso, or equivalent argument that all monitor ages remain bounded. K.Resp evidence is source-side A5-L-discharge evidence; it does not establish an RTL response-time bound.
- A5-L — scheduler-robust liberal source liveness. Statement: A5-L abbreviates Definition K.A5L on Sys\_ref: no finite Policy-L prefix admissible under Definition Core.LAdm for the selected Sys\_ref instance reaches a source K-facing cutpoint  $\sigma \in \text{KCutReach\_ref}(I)$  satisfying  $\text{Dead\_K}^W$ . A5-L is consumed by Theorem K.1A for one certified cutpoint, by Corollary K.1A-Total together with Core.KCutClosure\_post and K.CertCov for the universal RTL conclusion, and by Theorem K.1

after the selected A5-S route discharges it. Discharge: Primarily by an SDet-qualified CF-S certificate proving the model-indexed lowered property  $A5L\_Sys\_K(I, W\_K)$ , Corollary K.2c, and Lemma K.Low.A5 transporting that result to  $Sys\_ref$ ; alternatively by K.Str-1/K.Str-2/K.Str-3, direct Policy-L verification, an inductive invariant, exploration, or CF-5 directly on  $Sys\_ref$ . These alternatives do not weaken A5-L: if any admissible  $Sys\_ref$  Policy-L prefix—including a Policy-S-shaped prefix—reaches  $Dead\_K^W$ , A5-L is false and no alternate discharge can certify the unchanged instance. The Policy-S package records the selected normalized instance, capacities, elaboration, lowering/projection, witness, stimulus, environment/reset model, concrete simulator/scheduler, arbitration, Policy-S issue semantics, normalization identity, SDet evidence, K.EnvMF, K.ResetEpoch, K.TerminalDisposition, K.CompleteS, K.RespCov, K.RespClean, response bounds, monitor waivers,  $W\_K$  pullback, and the trust boundary.

- K.CV — design-level value determinism.

Serial-route provenance trigger (K.OpKey/K.CVPred/K.CV-WF): Required whenever Lemma K.2b.P0 or K.2b.P1 is consumed, including every use of Lemma K.2b, Corollary K.2c, K.PSF, CF-S, or the Policy-S form of Theorem K.1, regardless of whether the design has cross-process observations beyond blocking Pop values.

Additional K.CV observation-check trigger: Required when a process can branch on, compute payloads from, or select later operations based on cross-process observations beyond ordinary blocking Pop transfer values — anchored signal reads, direct-input(-sync) reads, shared/external-memory reads through the array-access mapping layer, emitted R2C fixed-point observation units, or synchronization outcomes.

Discharge: For the serial-route provenance trigger, supply or derive Definition K.OpKey plus the Definition K.CVPred certificate: one theorem-instance  $OpKey\_I$  universe with partial per-execution realization maps and a checked key-level source order, one common  $CVFact\_I$  key universe, finite  $Pred\_CV(b)$  sets, explicit predecessor occurrence closure, deterministic  $Eval\_b$  equations adequate for every admissible execution in the comparison scope, and for each  $OpFact(\kappa)$  a checked none/some(desc) evaluator that includes control/branch occurrence selection rather than assuming occurrence, plus K.CV-WF:  $WellFounded(<\_CV)$ , or an equivalent checked  $CVRank$  whose strict decrease proves it. The same K.CVPred certificate shall also discharge the serial-route comparison-availability closure: for every blocking  $OpFact(\kappa)$  consumed by K.2b.P0/R/P, once the liberal comparison has matched the source-earlier blocking operations and earlier serial-transfer facts required before  $\kappa$  may issue, all  $Pred\_CV(OpFact(\kappa))$  facts must be available with the same keyed values after a finite phase-legal extension that preserves the frozen component/DomEpoch and introduces no unmatched dominance-domain endpoint transfer; transfer predecessors must already be among the matched earlier transfers, while lower-ranked non-transfer predecessors are discharged recursively by  $CVRank$  and their certified reach rules. Each recursive non-transfer predecessor reach shall also carry an explicit Core.16 disposition: a qualifying ordinary/singleton blocking  $OpFact$  predecessor may cite Lemma K.CompareAdvanceLegal-Ord pointwise after its lower-ranked facts are available only if every Policy-S-required source-earlier blocking transfer for that predecessor is already in the matched liberal transfer prefix with the required keyed identity/ordinal before materialization, and the materialization extension itself supplies none of those prerequisites; a pure observation/settling reach shall be certified Core.16-inert; and any other reach that launches source work shall provide equivalent owner/frontier no-key-earlier evidence. “Phase-legal” alone is not an accepted discharge. The  $<\_CV$  relation shall contain actual value/control provenance only; rendezvous mutual enablement, same-cycle coincidence, and the broad Appendix-G causal-dependency graph  $\rightarrow$  do not become predecessor edges merely by being causal/temporal neighbors. A design with circular value/control provenance that cannot satisfy

K.CV-WF is outside the K.2b/K.2c serial route. For the additional observation-check trigger, also show that every such dependency is carried by an explicitly modeled causal mechanism and that the observed value is a deterministic function of the certified value/identity provenance feeding that mechanism, not of incidental schedule timing. These additional design checks include RULE 1 / RULE 2 anchored signal IO, proper handshakes or direct-input contracts on cross-process signals, mapping-layer discipline on shared memories, and R2C well-formedness; latency-sensitive sites are isolated or witness-aligned per Appendices D and L.

Evidence examples: a source/tool-generated value-provenance graph with a checked ranking function; a mechanized well-foundedness proof over typed dynamic operation/observation keys; source inspection or lint establishing RULE 1 / RULE 2 and handshake conformance where the additional observation-check trigger applies; a coverage-justified Appendix L latency-robustness stress-testing result with provenance/rank evidence; a refinement argument; or an Appendix D isolation argument. Randomized stress testing by itself is supporting evidence, not a complete K.CV discharge, unless accompanied by a coverage/refinement argument and the required K.OpKey/K.CVPred/K.CV-WF evidence. This K.CV evidence is necessary but not sufficient for the separate NB-Align condition (next entry), which is discharged independently.

- NB-Align — witness-selected non-blocking alignment.

Statement: NB-Align is the scope restriction stated normatively in the Lemma K.2b Statement (Appendix K), which is its authoritative wording: every witness-selected non-NB-CF non-blocking site that can participate in a potential wait cycle must satisfy the Appendix L alignment conditions that convert its selected outcome dependency into an explicitly modeled message/synchronization obstruction. NB-Align is consumed as the applicability condition of the Definition K.Low witness-indexed blocking lowering: an aligned site is lowered to a blocking guard on an explicit point-to-point supplier channel, and a witness schedule containing a non-aligned such site admits no lowering and is outside the Lemma K.2b / Corollary K.2c / Theorem K.1 scope. Theorem K.1A does not consume NB-Align. Trigger: Required whenever Lemma K.2b or Corollary K.2c is invoked (including through Theorem K.1) and the design contains witness-selected non-blocking outcomes that are not NB-CF. Discharge: Show that each witness-selected non-blocking site is NB-CF, is Appendix L-aligned (so that Definition K.Low applies to it), or is structurally excluded from every potential wait cycle of the selected instance.

Evidence examples: An NB-CF classification per Appendix L; a per-site Appendix L alignment argument identifying the modeled supplier used by the Definition K.Low guard channel; or a structural argument that no witness-selected site can participate in a wait cycle of the selected instance. NB-Align is distinct from, and not implied by, K.CV.

- K.Low — witness-indexed blocking lowering, cutpoint correspondence, and A5 transport.

Trigger: Required whenever the serial-reduction route (Lemma K.2b / Corollary K.2c / Theorem K.1) is applied to a design containing explicit synchronization waits or NB-Aligned witness-selected non-blocking sites in potential wait cycles.

Discharge: Exhibit the lowered instance  $\Lambda\_W(\text{Sys\_ref})$  of Definition K.Low — epoch-gate representation of synchronization per Definition Core.16 clause (7), and a blocking guard channel per aligned site with its modeled supplier as unique producer — and discharge the Lemma K.Low.C correspondence obligations (i)–(iii) for the lowering rules used. Define the waiver pullback  $W\_K$  and discharge Lemma K.Low.A5, including Policy-L admissibility preservation and the no-waive-on-lift condition, so that  $A5L\_Sys\_K(I, W\_K)$  proved by the Policy-S route transports to the  $\text{Sys\_ref}$  A5-L consumed by Theorem K.1A. Evidence examples: The elaboration record for  $\Lambda\_W(\text{Sys\_ref})$ ; per-site guard-channel construction tied to the Appendix L alignment; a mechanized or reviewed K.Low.C case proof per lowering rule; and a checked

K.Low.A5 projection/waiver-transport proof. Guard channels carry the corrected default  $B(g_s) = 1$  skewed timing, or a declared atomic rendezvous variant, under Definition K.Low.

- K.EnvMF — Policy-S environment/multi-frontier applicability. Trigger: Required for Lemma K.2b.E, Lemma K.2b, Corollary K.2c, Theorem K.PSF-1, Theorem K.1, and CF-S. Discharge: Establish either (a) StandardEnv/B1-avail with the default always-ready output environment; ordinary/non-elaborated components, or certified singleton components carrying the required total-order/source-comparability evidence, cite Lemma K.JointFreeze-Ord, while every admitted certified multi-frontier component additionally supplies Definition K.JointFreeze, proving JF1 cross-process launch/issue closure across incomparable  $Sys\_B^{elab}$  frontier items and JF2 no-release under permitted drain/elaboration actions; or (b) a timed/reactive or otherwise nondefault environment together with a certified single frontier for every process in every candidate Dead\_K component at the K.2b.E activation cutpoint, that frontier consisting solely of the selected internal blocking item. A timed/reactive/nondefault instance with any relevant multi-frontier—including an environment-facing co-frontier—is outside the Policy-S serial-reduction route. It may use Theorem K.1A only with a non-serial proof of A5-L, or a different theorem outside this document. A StandardEnv multi-frontier component for which K.JointFreeze cannot be discharged is likewise outside K.2b/K.2c. This revision deliberately does not admit a substitute future-environment no-rescue certificate for the excluded timed/reactive combination. K.EnvMF together with K.JointFreeze-Ord/K.JointFreeze preserves the frozen component; K.CompareAdvanceLegal is the separate audit entry for K.2b.P0/K.2b.R launch-and-issue compatibility.
- K.CompareAdvanceLegal — dominance-comparison Core.16 launch-and-issue compatibility. Trigger: Required by Lemma K.2b.P0/K.2b.R/K.2b and therefore by K.2c/K.PSF/CF-S for every candidate comparison. Discharge: Cover every process/key in K.DomResetScope that K.2b.P0 or K.2b.R may consume, whether or not the owner belongs to D. For an ordinary non-elaborated active frontier, or a certified singleton frontier carrying the same relevant total-order/source-comparability evidence, cite Lemma K.CompareAdvanceLegal-Ord and the checked least-missing/first-overtake transfer/provenance invariant. If a  $Sys\_B^{elab}$  package can present a multi-frontier for a consumed owner on a covered comparison prefix, supply an explicit elaboration/order certificate proving that no active frontier item is key-earlier than  $\kappa$  before the P0 preparation/reach extension and through the later R request assertion; incomparability alone is insufficient. The record shall state that this one condition discharges both distinct Core.16(2) obligations: the broader no-new-launch rule for P0's Core.ReachPrep/source-evaluation advance and the specific no-new-Issue rule for R's request assertion. The checker shall also enforce the consistency rule with K.JointFreeze: the same operation/prefix may not be both JF1-suppressed as a releasing launch/issue and certified here as a required legal comparison advance. A conflict or an uncovered elaborated multi-frontier owner puts that candidate outside the Policy-S reduction and requires another A5-L discharge. Reuse inside K.CVPred: when a lower-ranked predecessor is itself a blocking OpFact in the same ordinary/singleton comparison class, the predecessor-reach disposition may cite Lemma K.CompareAdvanceLegal-Ord pointwise only after the recursive qualification test confirms that every Policy-S-required source-earlier blocking transfer for that predecessor is already in the matched liberal prefix before materialization and is not created by the materialization extension; non-operation observations and other launch forms remain governed by the explicit K.CVPred recursive-disposition rules rather than being silently folded into this entry.
- K.InterruptionGate / K.ResetEpoch / K.TerminalDisposition — nested comparison-interruption applicability. Trigger: Required whenever Lemma K.2b, Corollary K.2c, K.PSF, CF-S, or the Policy-S form of Theorem K.1 uses the serial reduction. Discharge the shared schema once, including its

K.BF exhaustiveness rule: every gate instance has a prefix-local Continue predicate and an independently mapped DirectFail payload; Continue may depend only on the fixed contract and frozen prefix, DirectFail must bind  $P$ ,  $f_P$ ,  $x_P^A$ ,  $g_P$  with  $g_P \leq_{src} f_P$  and an already-triggered pending covered monitor independently of K.2b.R/P/Q Step 1, and a candidate satisfying neither branch is outside the route. For each candidate, bind finite  $S_p^{dom}/S_c^{dom}$ , validate K.DomResetScope conditions D1–D4—including D membership, both-endpoint boundary closure, coverage of every K.2b.R/P-consumed endpoint/boundary identity, and independence from the later K.2b.E continuation—and derive DomEpoch. Instantiate reset first.

K.ResetEpoch.Continue proves that no RESET intersecting K.DomResetScope is legal as the next proof-relevant transition/unit while the selected frozen requests remain pending and DomEpoch is fixed. Reset accounting for this gate is cycle-level rather than branch-local: any RESET occurrence that executes within the relevant real cycle, whether RW5-disjoint or the affecting RESET matched by DirectFail, must appear in the enclosing CycleResult.resetUnits. K.ResetEpoch.DirectFail binds the matched affecting RESET recorded there and exact RW2 clearing, yielding cancelled-pending. Only on K.ResetEpoch.Continue instantiate terminal.

K.TerminalDisposition.Continue proves every applicable frozen-prefix final state is non-DeclaredTerminal; K.TerminalDisposition.DirectFail independently binds the declared maximal terminal state while the same mapped monitor remains pending, yielding maximal-pending. For a successful clean CF-S discharge the two nested Continue branches are the normal prospective evidence; concrete DirectFail witnesses are themselves K.RespFail classifications and are not required on every clean run. Under the per-candidate route, the audit record shall name the K.InterruptionGate schema version, the K.BF exhaustiveness citation, each K.DomResetScope certificate, D1–D4 validation/provenance, and DomEpoch artifact, the reset-first nesting, each Interrupt predicate, and each invoked DirectFail specialization. Uniform instance-level alternative: instead of the per-candidate records just described, a clean CF-S/K.PSF application may supply one fixed conservative scope  $U$  satisfying U1–U3 of the normative uniform discharge after Definition K.TerminalDisposition. The checker validates  $U$  once and treats the resulting K.ResetEpoch.Continue and K.TerminalDisposition.Continue facts as universal over K.2b candidates. This alternative may not be used if any U1–U3 clause is unproved.

- K.Resp — validated process-facing bounded-response coverage. Trigger: Required for Lemma K.2b, Corollary K.2c, Theorem K.1, K.PSF-1, and CF-S. K.RespCov is default-total over a declared coverage universe of static internal process-facing blocking sites and relevant dynamic classes. Each class has one checkable disposition: (a) a monitor with an unambiguous trigger at the first normalized cycle in which the source operation is the active minimal pending frontier and its persistent boundary request is established, the process-facing commit/return as response, a finite Policy-S-cycle bound, and reset/end-of-test handling; (b) a validated static bounded-response proof; or (c) a proof that the class is unreachable in the selected instance. Completeness requires every dynamically reachable instance to belong to a declared class; once an instance is reached, disposition (c) cannot apply. A production-count argument alone is not an exemption, because the complementary owner may itself be blocked. Under StandardEnv, a source-earlier environment interaction that committed in the liberal witness may be proved incapable of remaining the serial blocker by replay availability and fairness. Under a timed/reactive or otherwise nondefault environment, K.EnvMF first requires the relevant liberal Dead\_K components to be single-internal-frontier; environment-facing sites that may block the selected Policy-S execution then require monitoring, a specific availability proof, or exclusion from the serial route. No monitor or no-bypass proof admits a timed/reactive multi-frontier liberal component. K.RespFail is a finite-prefix predicate/witness: certified-bound expiration without response under Definition K.RespFail's deadline convention, a declared-terminal

maximal finite selected run with the frontier still pending, or an affecting RW2 RESET that clears the monitored dynamic instance before response unless explicitly waived. A terminating abort/end-of-test disposition is represented by DeclaredTerminal and is covered by the maximal-pending case. K.RespClean proves absence of non-waived K.RespFail witnesses over the complete  $p\_S^{\text{det}}$ . A successful complete-run proof may use any finite bound; a bound violation under an unproved design-latency bound makes that certificate attempt inconclusive rather than proving a design bug.

- K.EnvOrd / K.Done — optional diagnostic starvation classification. These conditions are no longer part of the soundness-critical Policy-S discharge and cannot substitute for K.RespCov/K.RespClean. They may be used to classify a reported K.RespFail through the historical WFG+/EnvStop/DoneStop owner-walk machinery or to simplify diagnostic reporting from Dead\_K to Dead\_K. K.EnvOrd builds the finite may-follow starvation cone and proves the corresponding dynamic no-bypass property. K.Done audits normal completion sufficiently to eliminate DoneStop terminals; the alternative that every process is perpetual eliminates normal completion only and says nothing about live endpoint under-supply. In particular, neither K.Done nor process perpetuity permits omission of a K.Resp monitor. The K.CE-2/K.CE-3/K.CE-4 and H2 examples remain useful diagnostics, but no Corollary K.2c conclusion depends on completeness of this terminal taxonomy.
- W.Env — environment-rescue and response-monitor waiver. Trigger: Optional; used only for a specifically declared cutpoint or dynamic monitor class whose environment/reset/end-of-test behavior is intentional and proved compatible with the theorem scope. For KObs predicates, record the component class, rescuing item, environment contract, and KObs-closure justification. For K.Resp, record the exact dynamic frontier class, the reason timeout/reset interruption is permitted, and a coverage-preservation proof that a non-waived Policy-L deadlock component cannot map only to waived serial monitors. A broad end-of-test or reset waiver is insufficient. Waivers are removed explicitly from Dead\_K^W and optional diagnostic Dead\_KS^W, and from K.RespFail^W, and propagate to Theorems K.1A/K.1. A waiver depending on hidden issue timing or unconstrained future behavior must carry the corresponding operational proof; a label alone does not discharge the obligation.
- L.Dir — one-directional alignment / RTL-side non-interference.  
Statement: L.Dir is the scope restriction stated normatively in Condition L.Dir (Appendix L), which is its authoritative wording: the source-side snoop sequence selected by W is exactly an ordinal-preserving consumption of the free-running RTL\_B snoop sequence at the chosen boundary — each pre-HLS-side snoop commit consumes the next recorded RTL\_B event of the same snoop point with matching kind and payload, no pre-HLS-side snoop commit occurs without such a next recorded RTL\_B event, every recorded RTL\_B snoop event within the compared run is eventually consumed or reported outstanding per the Requirement W4 bound, and no RTL\_B-side signal or event is gated, delayed, or modified. Benign cycle skew is permitted; only order, kind, or payload divergence — non-existence of the WC witness at the chosen boundary — violates the condition. A run violating L.Dir is outside the Appendix I–K theorem scope for that comparison and carries no theorem-backed guarantee.  
Trigger: Required whenever WC is discharged through an Appendix L snooping / witness-selector co-simulation harness (Lemma L.2 / Lemma L.5).  
Discharge: Provide a per-run monitor per Lemma L.5(iv) — realized by a wrapper conforming to the Appendix L Wrapper Realization Requirements W1–W5 — that aborts loudly on violation, together with the run’s clean-monitor result and a statement of whether the Requirement W4 watchdog bound is proved sufficient for the selected instance (bound-exceeded aborts under an unproved bound are inconclusive witness-realization failures, not proven violations); or a

static/structural argument — for example, latency back-annotation or delay padding of the pre-HLS model, or latency-interval analysis at each snoop point in the spirit of the simplified snooping approaches — showing that the post-HLS side cannot run ahead in event order; or a model-checking proof that order divergence is unreachable for the selected instance.

Evidence examples: A harness monitor log showing no L.Dir abort, with cycle-skew statistics reported separately as diagnostics; a static latency analysis at the simplified-snooping end of the Appendix L spectrum; or a wrapper certification against Lemma L.2 together with a watchdog configuration that converts witness non-existence into a reported failure rather than a hang. L.Dir violations are triaged per the Snooping Concerns section of Appendix L: the pre-HLS model is the default suspect in a certified-harness, trusted-tool flow, but harness incompleteness and tool non-compliance must not be excluded a priori.

- J.LocalGWR — local certificates for the compositional J.GWR construction.

Trigger: Required whenever Theorem J.1's normal Post  $\rightarrow$  Ref route, Corollary J.1, or any result consuming their witnesses is applied, unless the optional J.GWR-Direct route below is used.

Discharge: For each compared RTL\_B prefix, provide the package of Definition J.LocalGWR. The package shall cover every process with J.PCert; every explicit abstract channel with J.CCert and B-faithfulness; every rendezvous, paired synchronization, R2C, anchored synchronization/signal, RESET, and other explicitly atomic unit with J.ACert; and the applicable J.EnvCert and J.ResetCert obligations. It shall also provide the finite typed construction-action inventory, per-action ReadSet/WriteSet, ownership facts, and locally derived epoch/horizon/stage/ordinal metadata for J.FoundationRank, request-snapshot keys for every witness-significant non-blocking call, the locally generated semantic D1–D4 edges and D5 construction-stratification edges required by Definition J.DepJ, including J.HorizonOrderSafe commutation evidence for every D5-only pair, and J.LocalAgree for every duplicated endpoint, identity, payload, reset-epoch, request-attribution, boundary, footprint, and snapshot field. The package shall prove complete coverage and the premises needed for the case-by-case J.DepRank proof: every relevant process, explicit boundary, construction action, non-blocking decision, and atomic unit in the selected prefix is represented exactly once or by a uniquely identified shared record. J.CanonicalRank is reconstructed from the checked dependency graph and is not a separately supplied certificate field.

The discharge may not contain a renamed global realizability premise. In particular, it may not assume that the local certificates admit a realizable merge, that an arbitrary phase/rank already gives a legal replay order, that the next complete unit has a global extension, or that a matching Sys\_ref execution already exists. Dependency edges must arise only from the enumerated local rules of Definition J.DepJ, and non-commuting action pairs must be covered by Lemma J.DepComplete or by a component-local footprint/commutation certificate. Global well-foundedness, coexistence, and extension are conclusions of Lemma J.DepRank, Lemma J.DepAcyclic, J.PrepareCommute, J.UnitExt, and Theorem J.GWR-Comp.

Evidence examples: per-process schedule/replay scripts with typed action footprints; per-channel boundary, occupancy, same-cycle transfer, and request-snapshot certificates; static or generated atomic-unit identity records; locally justified Dep\_J edge records; source/RTL structural checks; component-local footprint/commutation certificates for custom Sys\_B^elab elements; a translation-validation package whose individual records map to J.PCert/J.CCert/J.ACert; or equivalent independently checkable evidence.

J.GWR-Direct — optional direct global witness-realizability route.

Trigger: Optional alternate route when a flow does not use J.LocalGWR/J.GWR-Comp for a selected prefix.

Discharge: Supply a directly validated package satisfying Definition J.GWR, including the complete reachable Sys\_ref witness chain, all operational annotations, and a direct-provenance J.ValidatedConstructionOrder with an independently checked well-founded predecessor relation, complete unit enumeration, required-order refinement, operational-prerequisite coverage, and ideal-prefix proofs. This route may be implemented by translation validation, bounded symbolic replay, a validated global trace certificate, or a previously proved tool-specific theorem. It is not required when J.LocalGWR and Theorem J.GWR-Comp are used, and it may not cite the theorem-derived J.CanonicalRank without also supplying the corresponding LocalGWR package.

Historical Issue timestamps may remain inside either route’s operational witness evidence, but they are not fields of KObs and are not independently compared across cutpoints.

- Core.MemSyncOrder — array/synchronization commit-order conformance.

Trigger: Required whenever the selected design/flow instance contains array or memory accesses in a process that are source-ordered with synchronization calls selected into S, whether the storage is process-internal or externally mapped.

Discharge: Check the main-text “Scheduling of Array Accesses” commit-time fence at the schedule/mapping-layer level: mapped storage commits for accesses before s occur no later than s’s commit, and mapped storage commits for accesses after s occur strictly later. Any allowed cache/merge/split/reorder transformation shall carry mapping-layer evidence that preserves the fence. This is a scheduling-contract qualification check recorded by M.7a, not an Appendix I replay or Theorem J.1 semantic premise, and it does not add memory events to  $\Sigma$ . For a J.1 application with cross-process shared-memory value flow, additionally discharge J.Mem.

Evidence examples: HLS schedule/control-step reports; memory-latency and mapping metadata; generated-RTL memory-enable/address/commit assertions; or a mapping-layer proof/checker.

- J.Mem — shared-memory lowering.

Trigger: Required whenever the compared design contains a cross-process shared-memory dependency that can affect control flow, payloads, or operation identity/order.

Discharge: Show every such dependency is lowered by the array-access mapping layer onto covered IO operations (explicit point-to-point E4 channels and/or  $\Sigma$ -visible anchored signal IO), or supply the memory-state invariant extension described in the informative note in Normative Formal Core.

Evidence examples: use of pre-verified shared-memory library classes presenting channel/signal boundaries; mapping-layer configuration review; structural check that no un-lowered cross-process array access exists.

- RW — reset well-formedness (RW1–RW5).

Trigger: Required whenever dynamic reset can occur in the compared executions.

Discharge: Show every dynamic reset occurrence satisfies RW1–RW5 so that Lemma J.RS (MatchedResetEpochSplice) applies at each RESET boundary unit.

Evidence examples: reset-tree/scope review; RTL check that both endpoints of each live channel reset together or that an explicit modeled flush/abort protocol is used; confirmation that outstanding requests clear and affected channels re-initialize reset-empty.

This record is not itself an additional semantic assumption beyond the named premises used in Appendices I–K. It also records separately named qualification obligations, notably Core.MemSyncOrder, without converting them into Appendix I or Theorem J.1 premises. It makes the required single-global-clock scope check and the discharge of Core.IC, Core.SyncFence, Core.ReachPrep, Core.CycleClosure, Core.IssueFair, B1, B2/B3, B4/FPD, B5/B6-Terminal, A6, A8/A9/A10, R/E scheduling-contract conformance, WC, J.LocalGWR (or J.GWR-Direct), elaboration, B-faithfulness, J.OfferReach, J.OfferConf,

J.KObsConf, Core.MemSyncOrder, J.Mem, RW, K.Drain, K $\epsilon$ , K.BF, SDet, A5-L, and the selected A5-L discharge route—including K.CompleteS, K.RespCov, K.RespClean, and the shared K.InterruptionGate schema with K.ResetEpoch/nested K.TerminalDisposition when the deterministic Policy-S route is used—and K.CV explicit for the selected theorem or flow-qualification scope.

## M.7b Standard QSync certificate profile

**Normative packaging status.** For the standard normal-compositional QSync flow, the detailed M.7a discharges may be organized and exposed through the four logical certificate bundles below. These bundles are packaging interfaces only: they do not introduce a new semantic assumption and do not replace, merge, weaken, or make optional any theorem premise, qualification obligation, coverage condition, or evidence requirement defined elsewhere in this document. M.7a remains the definitive per-instance discharge ledger. If an item is required by the selected theorem or flow-qualification scope, it shall still be discharged as specified in M.7a; if an item is theorem-derived, the bundle may reference the applicable theorem and its checked inputs rather than duplicate the derived fact as an independent premise.

**Common binding and physical representation.** The four bundle names in this profile are logical interface names, not new theorem predicates. A conforming flow may represent a bundle in one physical file, several files, a proof object, a database record, or a set of versioned references. Every artifact used in one theorem claim shall map unambiguously to the M.7a obligations it discharges and shall be bound to the same selected theorem instance through K.CertHdr/K.CertPkg, including the source, Sys\_ref/Sys\_K and RTL identities, capacities/elaboration, observation boundaries, stimulus, environment, reset, witness/waiver provenance, and applicable tool/checker versions. A stronger checked bundle may cite theorem-derived projections instead of requiring a second independently generated certificate.

**1. Instance and QSync qualification bundle.** This bundle fixes the theorem instance and carries or references the primarily structural, scheduling-contract, and implementation-mapping qualifications used across J and K. In the standard QSync profile it includes the applicable ClockScope and source/RTL Core.QSync QS1–QS5 evidence, including RTL QS2a KCut-origin recoverability when the K-facing domain is used; Core.RegSeq and Core.ReachPrep where triggered; Core.IC and Core.SyncFence where triggered; the applicable M.7a R/E scheduling-contract conformance (R1–R5/E1–E5, including R2C where applicable); selected abstract boundaries, capacities, B-faithfulness and Core.ElabProj; and the applicable Core.MemSyncOrder, J.Mem, RW, direct-input, and elaboration/template-composition qualifications recorded by M.7a. It may cite Core.QSync-Ref/Core.QSync-Post and Core.QSync-KCutCanonical when their hypotheses have been checked. This bundle establishes that the selected source/RTL instance and its cycle/boundary abstraction are qualified; it does not by itself prove J trace correspondence, KObs equality, K.CertCov, or A5-L.

**2. J correspondence/refinement bundle.** This bundle carries the source/RTL witness-construction and correspondence evidence for the selected coverage profile. Its local/global core is J.LocalGWR with J.GWR-Comp, or a permitted J.GWR-Direct package, together with the applicable WC discharge, the exact selected witness, and the J.HCanon history relation. For the standard QSync safety profile, it carries or references J.QSyncTraceInd; Theorem J.TraceCertCov-QSyncCycle then supplies J.TraceCertCov on CycleReach\_post, Corollary J.1-Safe-QSyncCycle supplies cycle-aligned safety, and Lemma J.CycleIdealCover / Corollary J.1-Safe-QSync extend ideal-prefix-closed safety to all reachable finite outer observable prefixes without adding per-prefix certificate packages. For the richer K-facing profile, the bundle carries J.QSyncCertInd/QSyncCertState, including the exact J.OfferReach/J.OfferConf/J.KObsConf correspondence, theorem-derived J.RegPhaseReach provenance, and the exact K.DrainReach/Core.SettleKBridge field required by QC5. Carrying K.DrainReach here is a package-location choice for the exact cutpoint tuple and does not reclassify the underlying

K.Drain/K $\epsilon$ /nonwithdrawal obligations as Appendix-J safety premises. Lemma J.QSyncCertInd-Trace projects the richer K-facing induction to J.QSyncTraceInd, so a flow that has the K-facing bundle shall not require a second independent safety-induction certificate.

**3. K applicability and provenance bundle.** This bundle carries the route-applicability and proof-construction qualifications that make Appendix K sound beyond the J correspondence itself. For the primary Policy-S route it includes the applicable K.BF/Core.IC persistence and A6 point-to-point ownership facts; K.OpKey/K.CVPred/K.CV-WF provenance, including comparison-availability and recursive Core.16 reach-legality dispositions; K.Drain/K $\epsilon$ /nonwithdrawal qualifications used to justify the exact K.DrainReach evidence; K.EnvMF and, only where required, K.JointFreeze; K.CompareAdvanceLegal; Core.CycleClosure, Core.FairPick/Core.FairCycleDovetail, Core.IssueFair, B2/B3 and B5 as consumed by the proof-built continuation; the K.InterruptionGate schema with K.DomResetScope and either the uniform U1–U3 discharge or the required candidate-specific K.ResetEpoch/nested K.TerminalDisposition records; and applicable NB-Align, K.Low/K.Low.A5, waiver, and related M.7a conditions. Ordinary non-elaborated or otherwise qualified singleton-frontier cases may cite K.JointFreeze-Ord and K.CompareAdvanceLegal-Ord instead of supplying separate frontier certificates where the normative lemmas apply. This bundle establishes applicability of the K reduction; it does not substitute for the selected complete Policy-S response result.

**4. Policy-S response bundle.** For the primary bounded-response discharge of A5-L, this bundle binds the SDet-selected Policy-S simulator/run artifact and the response-monitor semantics to the same theorem instance, and carries or references K.CompleteS, total finite-bound K.RespCov, K.RespClean, the applicable response-waiver configuration, and the CF-S/A5-S result. It references rather than duplicates the provenance, lowering, fairness, environment, reset/terminal, and other route-applicability evidence owned by the preceding bundles. With the K applicability/provenance bundle, Corollary K.2c uses this evidence to establish the required A5-L result on the supported fragment. If A5-L is instead discharged by K.Str-1, CF-1 through CF-5, a direct Policy-L invariant/exploration proof, or another permitted route, the corresponding K.A5Cert route payload replaces this Policy-S response bundle for that theorem application; the other J/K correspondence and implementation-qualification obligations remain governed by their own triggers.

**Standard profile composition.** For a standard QSync safety claim satisfying J.IdealPrefixClosed, the profile consists of the instance/QSync qualification bundle plus the safety form of the J correspondence/refinement bundle; the K applicability/provenance and Policy-S response bundles are not safety premises. J.QSyncTraceInd establishes the cycle-aligned packages, and J.CycleIdealCover/J.1-Safe-QSync supply all-finite-prefix ideal closure without adding a new certificate obligation. If the richer J.QSyncCertInd K-facing bundle is already available, J.QSyncCertInd-Trace supplies the safety induction. For one-cutpoint Theorem K.1A, the flow uses the applicable instance qualification, the exact K-facing J/cutpoint package, the triggered K applicability evidence, and any valid A5-L discharge; universal QSync package induction and K.CertCov are not required merely for that one cutpoint. For the universal user-facing Theorem K.1 through the primary Policy-S route, all four bundles are used: checked QSync structure gives the standard KCut closure/canonical-domain results; J.QSyncCertInd together with J.CertExist-QSync and K.CertCov-QSync derives K.CertCov; and the K applicability plus Policy-S response bundles discharge A5-L through K.2b/K.2c before K.1A-Total is applied.

**Independent-checker and API boundary.** This profile intentionally does not prescribe a concrete file format, serialization, command-line interface, database schema, or certificate transport. Those are tool-flow API choices. Whatever representation is chosen, an independent checker shall validate the referenced subordinate evidence and reconstruct theorem-derived or KObs-derived facts where the existing certificate definitions require that behavior; a producer's final pass/fail Boolean is not a substitute for the M.7a discharges. A user-facing implementation may therefore report four top-level

bundle statuses and provide drill-down to the named M.7a obligations without exposing every formal predicate as a separate top-level user action.

## M.8 Practical interpretation

The architecture provides two different forms of scalability. Safety scales by keeping most design-specific evidence local to processes, explicit channel boundaries, and named atomic interactions, while using one generic theorem to construct the global replay witness. Liveness scales by separating the system-wide source premise A5-L from the available methods for proving it and by making the remaining progress, environment, drain, and cutpoint-coverage obligations explicit.

For a qualified HLS flow, the intended engineering result is that the pre-HLS SystemC model remains the primary place for system-level functional reasoning, coverage, and debug, while RTL work is concentrated on showing that the implementation conforms to the scheduling contract and its selected abstraction. The approach does not eliminate implementation verification; it changes its purpose from repeating the full functional test campaign at RTL to establishing the evidence needed to justify transfer from the source-reference verification result.

The same proof architecture is not inherently limited to HLS-generated RTL. Hand-written RTL, modified generated RTL, or RTL produced by another generation flow can use the same results when the same process, channel, observation-boundary, reset, environment, progress, and implementation-side obligations are established. The user-visible and tool-neutral form of the contract is also what makes the document suitable as a starting point for standardization discussions.

## Appendix N – Scheduler and Environment Fairness Assumptions

The formal results in Appendices I–K rely on several semantic assumptions about schedulers and environments. The B-rule table in “Normative Formal Core for Appendices G–K” gives the proof-level names B1–B6. This appendix restates the boundary-action fairness/progress rules in operational ready/valid form and separately restates Core.IssueFair for source request assertion. In particular, Appendix N’s “B3 Channel Progress Invariants” are exactly the P\_push(c) and P\_pop(c) obligations for each explicit abstract channel c; they are not additional assumptions beyond B3.

- **B2 (Weak fairness of the scheduler).** If the enabling predicate for an action (Push, Pop, Sync) remains continuously true from some cycle onward, the scheduler must eventually select that action within a finite, but unspecified, number of cycles.

Core.IssueFair (source issue; separate from B2). If a reached source operation remains legally IssueEnabled at every applicable L1 opportunity from some cycle onward within the same reset epoch, it is eventually issued. This is weak fairness over request assertion, not over commit. Core.16 is part of issue legality, so a source-later operation prohibited by an active blocked-frontier discipline is not IssueEnabled and imposes no IssueFair obligation. B2 begins after an applicable Push/Pop/Sync boundary action is ChannelEnabled. Core.LAdm/K.2b consume Core.IssueFair as a premise of complete admissible continuations; finite-prefix Appendix J does not require a generic issue-fair extension theorem. Constructive use in Lemma K.2b.E is through Definition Core.FairPick and Lemma Core.FairCycleDovetail. This does not strengthen the behavioral IssueFair/B2/B3 property, but it does require a stronger evidence form: prefix-local selector/starvation proofs for the particular constructed execution. Proof-selectable Policy-L issue latitude may be dovetailed; instance-fixed scheduler/arbitrer/environment/witness choices must be followed and justified, and the selected L3 action is chosen only after an actual typed L2 settle prefix has been obtained.

- **B3 Channel Progress Invariants (ready/valid form).**

For every channel  $c$  with capacity  $B(c)$ , interpret “continuously enabled” in terms of continuous endpoint request, not as an immediate commit.

**(No deassert-before-commit assumption.)** For blocking Push<sub>c</sub>/Pop<sub>c</sub> transfers, these requests are non-withdrawable during the outstanding transfer attempt within one reset epoch: once asserted for a transfer attempt, they are not deasserted before the corresponding commit, and Push<sub>c</sub> payload remains stable while vld<sub>c</sub> is asserted for that outstanding transfer. If an affecting RESET occurs first, RW2 ends that dynamic instance at the epoch boundary. No other nonterminal transition may withdraw or reattribute it. After the corresponding commit, the endpoint may deassert or may present a different request/payload for a later transfer.

Let vld<sub>c</sub> denote the producer’s persistent request to transfer (e.g., valid=1 with a stable payload for a Push<sub>c</sub>), and let rdy<sub>c</sub> denote the consumer’s persistent request to transfer (e.g., ready=1 for a Pop<sub>c</sub>). In the definitions below, “remains asserted continuously” is scoped to the outstanding interval from  $t_0$  until the first matching commit, or until an affecting RESET ends the dynamic instance under RW2. If neither occurs and the execution remains nonterminal in that reset epoch, then the request must remain asserted for all later cycles.

- **P<sub>push</sub>(c):** If vld<sub>c</sub> is asserted at cycle  $t_0$  while the channel is full ( $occ_c = B(c)$ ), and both endpoint requests continue until the matching discharge event — i.e., vld<sub>c</sub> remains asserted with stable payload and the complementary Pop<sub>c</sub> endpoint request remains asserted with BoundaryReq(Pop<sub>c</sub>, $t$ ) true until the first Pop<sub>c</sub> commit on  $c$  — then some Pop<sub>c</sub> commit must eventually occur, discharging the full condition. Equivalently: when both endpoints continuously request the matching transfer until discharge, the system cannot remain stuck forever in the full state without a Pop<sub>c</sub> commit.

- **P<sub>pop</sub>(c):** If rdy<sub>c</sub> is asserted at cycle  $t_0$  while the channel is empty ( $occ_c = 0$ ), and both endpoint requests continue until the matching discharge event — i.e., rdy<sub>c</sub> remains asserted and the complementary Push<sub>c</sub> endpoint request remains asserted with stable payload and BoundaryReq(Push<sub>c</sub>, $t$ ) true until the first Push<sub>c</sub> commit on  $c$  — then some Push<sub>c</sub> commit must eventually occur, discharging the empty condition. Equivalently: when both endpoints continuously request the matching transfer until discharge, the system cannot remain stuck forever in the empty state without a Push<sub>c</sub> commit.

For  $B(c)=0$  rendezvous channels: if both endpoint requests are asserted at cycle  $t_0$  and both remain asserted until the matching rendezvous discharge event, then the rendezvous completion eventually occurs. If no rendezvous completion occurs, both requests must remain asserted for all later cycles.

These obligations rule out “permanent back-pressure loops” that are logically independent of the local R-rules. They do not require an endpoint request or Push payload to remain asserted forever after the matching transfer has already committed.

**Clarification (scope of the B3 conclusion).**

For  $B(c)>0$ , P<sub>push</sub>(c) concludes that a complementary Pop<sub>c</sub> eventually commits and P<sub>pop</sub>(c) concludes that a complementary Push<sub>c</sub> eventually commits. Neither buffered clause directly states that the original blocked endpoint operation commits. Once the complementary discharge changes occupancy, E4 determines whether the original persistent request is ChannelEnabled, and B2 supplies eventual selection/commit if that enablement remains continuous. For  $B(c)=0$ , the matching discharge is the original same-cycle rendezvous completion.

- **B5 (System quiescence closure).** If at some cycle no observable actions are enabled in any process, and no external or environmental change during the closure interval changes the relevant enabling predicates, then within a bounded number of cycles the system reaches an  $\epsilon$ -fixed point. Thereafter, while no external or environmental change re-enables observable or internal work, the system executes no additional  $\epsilon$ -steps. This prevents a dead, all-disabled state from being hidden behind an infinite tail of internal activity, without requiring the system to remain quiescent after a later reset, input change, externally supplied message, or other environmental event re-enables work.

**B6 (terminal idle only).** B6 may provide an indefinite idle-clock suffix only when `Core.TerminalIdle` holds: the system is at a genuine B5/global fixed point, future registered/internal machine evolution cannot re-enable work, and `Core.EnvTerminal` establishes the same no-future-enabling fact for the environment. A time-aware environment can be perfectly total while delivering an input at a later cycle, so `Core.CycleClosure`'s environment-definedness clause does not imply `Core.EnvTerminal`; likewise environment terminality alone does not prove that an internal timer/FSM cannot advance. Temporary cycles with no committed  $\Sigma$  action are ordinary `Core.SilentCycle` transitions, not B6 terminal stutter. These B-rules are **assumptions on the execution environment**, not guarantees automatically provided by the HLS tool. In practice they place concrete requirements on arbiters, back-pressure, and external masters that interact with the post-HLS RTL.

### Priority arbiter example: fair vs unfair priority

Consider a simple two-input priority arbiter inside the post-HLS RTL. Each input is driven by a message-passing channel; the arbiter issues `Pop` operations to select which request to serve in each cycle.

- Input 0: high-priority channel `c_high`
- Input 1: low-priority channel `c_low`

Both channels may have pending requests at the same time.

#### 1. Fair priority arbiter (satisfies B2 / B3).

A fair implementation might still give `c_high` strict priority in *individual* decisions, but it ensures that a continuously pending `c_low` request cannot starve. In RTL terms, this can be realized by, for example:

- A **round-robin or rotating-priority** arbiter that periodically moves the “highest” priority to the next requester, or
- A **bounded-burst priority** scheme in which `c_high` may win at most  $N$  consecutive cycles while `c_low` is requesting; after that, the next grant is forced to `c_low`.

Under these implementations:

- If `Pop_low`'s `ChannelEnabled` predicate remains true indefinitely (the low-priority channel has a pending request and the relevant abstract boundary availability predicate holds), the scheduler/arbiter must eventually select `Pop_low`. The selected `Pop_low` then commits in the selected cycle. This satisfies B2 for that action.
- If the low-priority FIFO is full and keeps requesting service, the system ensures that some complementary `Pop_low` eventually commits, discharging data and freeing space. This is consistent with `P_push/P_pop` in B3; it is a channel-progress obligation that may be needed before a blocked producer's `Push` becomes `ChannelEnabled`.

Intuitively, the arbiter is still “priority-based,” but its micro-architecture ensures that every continuously enabled requester is eventually selected, and that full/empty channel states covered by B3 cannot persist forever under the stated persistent-request premises.

#### 2. Unfair priority arbiter (violates B2 / B3).

A more naive design might implement:

```
if (req_high) grant_high();  
else if (req_low) grant_low();
```

combined with an environment in which req\_high can remain asserted indefinitely. In this case, even if req\_low is also continuously asserted:

- Pop\_low's ChannelEnabled predicate may be true forever, but Pop\_low is never selected by the scheduler/arbiter.
- Low-priority requests can starve indefinitely, even though they are continuously eligible for selection.

This behavior violates B2, because B2 requires eventual selection of any continuously enabled action. If c\_low's FIFO is full and the only way to make space is to service it, permanent starvation can also violate B3's progress expectations for that channel, because the full condition may never be discharged under the stated persistent-request premises.

In such a design, the safety properties E1–E5 may still hold (trace-compatibility on actions that do occur), but the liveness/results that depend on B2/B3—especially the starvation-vs-deadlock arguments in Appendix K—no longer apply.

## Patterns that typically satisfy the fairness assumptions

The following design patterns normally satisfy B2 and are compatible with B3/B5, provided the rest of the system is well-behaved:

- **Round-robin or rotating-priority arbiters.** Any requester that remains enabled will eventually be at the front of the rotation and will be granted.
- **Age-based or credit-based arbiters.** Requesters accumulate age or credits while waiting; the arbiter prefers older or more-starved requests, guaranteeing that long-pending actions eventually win.
- **Bounded-burst fixed priority.** Fixed priority is combined with a counter that limits how long a higher-priority requester can dominate while lower-priority requesters are pending. After the burst limit is reached, lower-priority requests are forced to win until the system is “caught up.”
- **Back-pressure with bounded stall.** When ready/valid or similar handshake signals are used, the design (or its environment) must enforce that ready cannot remain low forever while valid remains high, and vice versa. For example:
  - Downstream consumers are required (by specification) to service queues at least once every N cycles when data is present.
  - External bus masters are configured such that they cannot indefinitely defer reading from full DUT FIFOs that are logically part of the interface contract.
- **Clock-gating and power-management schemes that respect B5.** Once no observable actions are enabled, the design either:
  - Quietly stabilizes (no further internal toggling), or
  - Explicitly enters a low-power state from which it only wakes when some observable action again becomes enabled.

In each case, the intent is the same: continuous ChannelEnabled status implies eventual scheduler/arbiter selection and commit under B2; persistent full/empty/rendezvous-blocked request premises imply eventual discharge under B3; and once nothing is enabled, the system converges rather than oscillating internally.

## Patterns that tend to violate the fairness assumptions

Conversely, the following patterns are likely to **break B2/B3/B5** and therefore fall outside the scope of the liveness arguments in this document:

- **Pure fixed-priority arbitration with unbounded high-priority traffic.** A static priority tree with no rotation or aging, combined with workloads in which a high-priority master can keep its request asserted indefinitely, can starve lower-priority channels forever.
- **“Best-effort” external masters with no progress guarantee.** For example:
  - A software driver that reads from a DUT output FIFO only when it happens to poll, and that may be pre-empted indefinitely by higher-priority threads.
  - An external bus or DMA engine that is architecturally allowed to ignore some requesters forever if higher-priority traffic remains heavy.  
Such environments can violate **P\_push/P\_pop** by allowing FIFOs to remain full or empty indefinitely while the DUT is continuously requesting progress.
- **Circular back-pressure loops with no escape.** Two or more processes form a cycle in which each is waiting for the other’s channel to change state, and no other process can break the cycle. Without an explicit design-level guarantee that some process in the loop will eventually act (for example, by dropping priority or issuing a compensating Pop/Push), **B3** is not satisfied.
- **Internal oscillation without quiescence.** A scheduler or control FSM that can toggle internal  $\epsilon$ -level state forever, even after all observable actions are disabled, violates **B5**. While such designs are uncommon in practice, patterns involving mis-configured clock gating, asynchronous feedback, or “keep-alive” timers that never expire can have this effect.

In all these cases, the trace-compatibility theorems **still describe what happens when actions do occur**, but the liveness claims that depend on B2/B3/B5 (for example, “a continuously enabled channel will not starve”) are no longer guaranteed. Designers and verification engineers should therefore either:

- Architect their arbiters and environments to satisfy these B-rules, or
- Treat the liveness guarantees in Appendix K as *out of scope* for those particular interfaces, and rely only on the safety-oriented parts of the model.

## Appendix O – Support for Multiple Clock Domains (Informative Sketch)

This appendix is an informative sketch of how the single-clock formalism of Appendices G–K could be organized in a multi-clock design. The formal results of Appendices G–K are stated for a single global clock unless an explicit multi-clock semantic layer is added.

Within a single clock domain  $d$ , one may treat that domain as a synchronous island with a local cycle counter  $\text{clk}_d$  and apply the existing R- and E-rules to processes wholly inside that domain, interpreting  $\text{clk}(\cdot)$  as  $\text{clk}_d(\cdot)$  for local synchronization, signal I/O, and message-passing events. Inter-domain communication should be represented by explicit clock-domain-crossing components, such as CDC FIFOs, whose source-side and destination-side commit semantics are specified at their respective clock-domain boundaries.

A full formal lift requires additional definitions not supplied by Appendices G–K themselves: a global event order or partial order relating local-domain cycles, precise CDC commit semantics, occupancy sampling rules across source and destination clocks, synchronizer fairness/metastability abstraction assumptions, and progress obligations for CDC components. Once those additional definitions are supplied, the intended proof strategy is to treat each synchronous island using the existing single-

domain rules and to treat each CDC component as an explicit channel/refinement boundary with its own stated E4-style legality and progress obligations. Without those additional definitions, Appendix O should not be read as claiming that the single-clock proofs apply unchanged to arbitrary multi-clock systems.

## Appendix P – AI Discussion of Alternative Formal Frameworks

The following links provide an AI discussion of alternative formal frameworks compared to the one presented in this document:

[https://drive.google.com/file/d/1ccU-eRW7H2ggh-AZyKnCOzO-KLPejob\\_/view?usp=sharing](https://drive.google.com/file/d/1ccU-eRW7H2ggh-AZyKnCOzO-KLPejob_/view?usp=sharing)

<https://chatgpt.com/share/69457119-ec30-8006-8684-9a2a8b19f96f>

<https://drive.google.com/file/d/1R3-46DtZLsBn1hdTbLEXyC0FVB50Gymz/view?usp=sharing>

## Appendix Q – Multiple Input Pipelines Back-Annotation

Appendix J Remark 2 discusses back-annotation of pipelined loops with a single message passing input. In more complex cases such as HW pipelines with multiple message-passing inputs, the back-annotation mechanism may elaborate the Sys\_B (source-level) simulation by introducing additional internal realization structure. In this document, we adopt the following simplified modeling discipline:

1. Terminology (elaboration and channel accounting). In an elaborated realization, the quantities  $B(\cdot)$ ,  $occ_{\cdot}(t)$ , and the E4 availability predicate are interpreted over the explicit abstract channels present in the elaborated model (including any introduced staged/bundle channels). When we continue to write a source-level channel name  $c$  at the outer DUT/testbench observation boundary, we mean the chosen outer abstract boundary channel associated with  $c$  in that elaboration; downstream staging that is reachable only through a join (and therefore may be withheld when “join not met”) is represented by distinct internal channels with their own  $B(\cdot)$  and  $occ_{\cdot}(t)$ , rather than as independent “slack” for  $c$  in isolation.
2. All buffering / capacity effects are represented explicitly as finite-capacity message channels (annotated with bounded capacity) that may appear upstream and/or downstream of any internal glue logic; and
3. Any internal “coupler” used to enforce mutual-stall between multiple inputs is purely combinational (stateless) handshake logic.

Observation-boundary convention for Appendix Q. In elaborated cases, distinguish the outer projected observation boundary from the proof-internal abstract boundaries. The outer  $\Sigma_{DUT}$  / testbench projection records only the chosen outer Push\_c / Pop\_c history after applying  $\Pi_{elab}$ . By contrast, Appendix K WFG reasoning is carried out over the explicit abstract channels of  $Sys\_ref = Sys\_B^{elab}$ , including introduced staged/bundle/join channels. Thus an introduced internal channel may be hidden

by  $\Pi_{\text{elab}}$  from the outer DUT-visible trace while still being an explicit proof-internal boundary with its own  $B(\cdot)$ ,  $\text{occ}_{\cdot}(t)$ ,  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled}/\text{ChannelDisabled}$ ,  $\text{Front\_P}$ , and  $\text{WFG}$  facts.

Concretely, a combinational coupler is a stateless component that (a) reads the upstream channels' valid and data signals and computes/drives the downstream channels' valid and data signals, and (b) reads downstream ready signals and upstream valid signals and then computes/drives upstream ready signals. The coupler contains no internal state and has no separate credit/capacity state; any shared pipeline capacity, per-input staging, skid/elastic buffering, or credit-like effects are modeled only by the explicit finite-capacity channels inserted by the back-annotation realization.

**Example (two inputs, staged Pops,  $II=1$ , latency=4):**

Consider a 4-stage RTL pipeline that (i) performs  $\text{Pop\_c1}$  in stage 1, (ii) performs  $\text{Pop\_c2}$  in stage 2, and (iii) performs  $\text{Push\_cout}$  in stage 4, with  $II=1$ . To model the pipeline's internal staging and the stage-2 input coupling in the source-level  $\text{Sys\_B}^{\text{elab}}$  simulation while keeping the same outer projected  $\Sigma$  boundary, the back-annotation elaboration may introduce the following internal realization structure:

- An explicit bounded **channel F1 of capacity 1** on the c1 path to represent stage-1 in-flight storage of values that have been accepted from c1 but have not yet reached the stage-2 join point.
- An explicit internal **rendezvous channel F2** on the c2 path at the stage-2 join boundary, with  $B(F2)=0$ . F2 represents the c2-side handoff into the stage-2 join. It is not additional c2-side buffering or slack, and it does not permit a c2 value to be accepted independently before the join is ready. Its purpose is to make the c2-side join blocking point an explicit proof-internal abstract boundary of  $\text{Sys\_B}^{\text{elab}}$ , so that any mutual-stall condition involving c1/F1 and c2/F2 is represented as ordinary  $\text{BoundaryReq}$  /  $\text{ChannelEnabled}$  /  $\text{ChannelDisabled}$  facts on explicit internal channels rather than as a hidden cross-channel disablement predicate on the outer projected c2 boundary.
- A **purely combinational coupler/join** at the stage-2 boundary that enforces mutual stall: it allows the joined transfer to proceed only when (a) a token is available from F1, (b) the c2-side rendezvous over F2 is requested/enabled, and (c) downstream staging has space; it then presents the paired value(s) to the downstream staging. The join's possible refusal to proceed is therefore visible to Appendix K through the explicit internal boundaries F1, F2, and F34, even though those internal events are projected away from the outer DUT/testbench-visible  $\Sigma_{\text{DUT}}$  trace by  $\Pi_{\text{elab}}$ .
- An explicit bounded **channel F34 of capacity 2** after the coupler to represent the in-flight capacity of stages 3–4 for the joined/paired work.

This example illustrates why multi-input pipelines cannot always be captured by a single per-channel capacity number in isolation: c1 and c2 do not have independent "slack" because acceptance across the stage-2 join boundary is coupled. In this particular example, the c1 side has one stored stage-1 token available through F1, the c2 side has no independent pre-join storage and is therefore modeled by the rendezvous channel F2 with  $B(F2)=0$ , and the joined/paired work has two downstream storage slots through F34.

All storage-bearing buffering/capacity resides in the explicit bounded channels F1 and F34, and in any additional positive-capacity stages the realization uses. The explicit rendezvous channel F2 has  $B(F2)=0$  and contributes no storage; it is an explicit proof-internal join boundary used to type the mutual-stall condition. The coupler itself remains stateless combinational handshake logic.

This elaboration is an implementation/refinement of the abstract bounded-FIFO channels at the chosen outer DUT/testbench observation boundary; it does not change the outer interface  $\Sigma_{\text{DUT}}$  or E4's per-channel FIFO meaning at that outer boundary. It may, however, introduce additional proof-internal  $\Sigma$  boundaries in  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}$  for Appendix K blocking/WFG reasoning. Concretely, any back-annotation elaboration (including extra FIFO stages and/or combinational couplers) must satisfy:

- No new outer DUT-boundary observables: It introduces no new  $\Sigma_{\text{DUT}}$ -observable actions at the chosen DUT/testbench boundary, and it does not change the labeling of existing outer  $\Sigma_{\text{DUT}}$  events ( $\text{Push\_c}(v)$ ,  $\text{Pop\_c}(v)$ , etc.).
- Boundary/E4 preservation at the outer projection: At the chosen outer DUT/testbench boundary, each outer  $\Sigma_{\text{DUT}}$ -visible committed  $\text{Push\_c}/\text{Pop\_c}$  event still denotes a committed transfer on the same abstract channel  $c$ , with per-channel FIFO order and the same E4 capacity/enablement interpretation under the annotated  $B(c)$  and  $\text{occ\_c}(t)$ . Any additional internal channels introduced by the elaboration (e.g., staged/bundle/join channels) are separate proof-internal abstract channels in  $\text{Sys\_B}^{\text{elab}}$  with their own  $B(\cdot)$ ,  $\text{occ\_c}(t)$ , and E4 availability predicates; if they participate in Appendix K, their committed transfers are proof-internal  $\Sigma$ -events even when  $\Pi_{\text{elab}}$  hides them from  $\Sigma_{\text{DUT}}$ .
- No double-counting of internal staging: internal transfers within the concrete realization of a single explicit abstract channel that contributes to that channel's effective capacity (extra FIFO stages, skid/elastic buffering, FIFO  $\rightarrow$  pipeline-stage handoffs, and value propagation through purely combinational couplers that does not cross an explicit abstract channel boundary) are  $\epsilon$ -steps and do not create additional committed  $\text{Push\_c}/\text{Pop\_c}$  events at that abstract boundary. By contrast, a transfer across an explicit staged/bundle/join channel introduced in  $\text{Sys\_B}^{\text{elab}}$  is a committed  $\text{Push}/\text{Pop}$  event at that proof-internal abstract boundary. Such an event may be projected away from the outer DUT/testbench-visible  $\Sigma_{\text{DUT}}$  trace by  $\Pi_{\text{elab}}$ , but it is not  $\epsilon$  for the Appendix K proof-internal WFG boundary.
- Conservation / projection of internal staging (normative): For each outer  $\Sigma_{\text{DUT}}$ -visible abstract channel  $c$  at the chosen DUT/testbench boundary, the elaborated realization shall admit a fixed projection/accounting map from the occupancies of the explicit internal channels introduced by the elaboration (including staged/bundle/join channels, as applicable) to the outer boundary occupancy  $\text{occ\_c}(t)$ . This projection/accounting map is part of the realization and witnesses that the proof-internal occupancy bookkeeping is only a refinement of the outer boundary model: (i) every  $\epsilon$ -step internal transfer and every committed transfer across an introduced proof-internal abstract channel preserves the projected outer boundary occupancy for each outer  $\Sigma_{\text{DUT}}$ -visible channel, except when it corresponds under  $\Pi_{\text{elab}}$  to an outer boundary commit, and (ii) each outer  $\Sigma_{\text{DUT}}$ -visible committed  $\text{Push\_c}/\text{Pop\_c}$  updates the projected outer boundary occupancy exactly as required by E4 (equivalently, exactly as the outer boundary  $\text{occ\_c}(t)$  defined from  $\Sigma_{\text{DUT}}$ -visible boundary commits). For bundled/joined work items, the projection may assign one proof-internal token to multiple outer  $\Sigma_{\text{DUT}}$ -visible channels as specified by the realization. Therefore, the internal capacities/occupancies  $B(\cdot)$ ,  $\text{occ\_c}(t)$  do not introduce an additional independent outer  $\Sigma_{\text{DUT}}$ -visible state; they refine the same bounded-FIFO accounting already represented at the chosen outer boundary, while still serving as proof-internal  $\Sigma$  state for Appendix K when those internal channels are explicit abstract boundaries.
- **Multi-input pipelines / combinational couplers (mutual stall + explicit capacity):**  
To preserve per-channel E4 at the  $\Sigma$  boundary, any downstream staging that is reachable only through a join (and therefore may be withheld when "join not met") shall be modeled as distinct internal finite-capacity channels with their own  $B(\cdot)$  and  $\text{occ\_c}(t)$ , rather than being treated as additional independent "slack" for an upstream  $\Sigma$ -visible channel in isolation; the back-annotation elaboration may use combinational couplers to enforce the intended mutual-stall behavior when some required inputs are unavailable. Any "shared capacity" or "distributed staging" effects needed for such a pipeline shall be represented only by explicit finite-capacity channels in the realization (e.g., bounded staging FIFOs,

bounded slot/credit channels, bounded bundle channels), not by coupler state. The coupler itself remains stateless combinational handshake logic and does not redefine the per-channel E4 capacity/availability predicates at the chosen observable boundary.

Projection/accounting invariant for  $\text{Sys\_B}^{\text{elab}}$  (normative implementation of Definition Core.ElabProj). Any elaborated reference model  $\text{Sys\_B}^{\text{elab}}$  used for multi-input / coupler cases shall come with a projection map  $\Pi_{\text{elab}}$  from the explicit staged/bundle/join channels of the elaboration back to the chosen outer  $\Sigma$ -visible channel boundaries. This projection map has these roles:

- **Observable-boundary preservation.** When the internal staged/bundle/join events introduced by elaboration are hidden or projected away, the remaining outer-boundary  $\text{Push\_c}(v)$  /  $\text{Pop\_c}(v)$  history is the same boundary history that the non-elaborated model would expose at the chosen  $\Sigma$ -visible endpoints. Elaboration may add explicit internal abstract channels for proof and WFG purposes, but it may not create extra outer-boundary  $\text{Push\_c}$  /  $\text{Pop\_c}$  events, drop outer-boundary events, duplicate them, or change their payload labels.
- **Capacity/occupancy accounting.** For each outer channel  $c$  represented by an elaborated subnetwork, the outer-boundary availability facts used by E4 are preserved by the aggregate accounting of the explicit internal channels in the preimage/fiber  $\Pi_{\text{elab}}^{-1}(c) = \{d : \Pi_{\text{elab}}(d) = c\}$ . Internal staged/bundle/join occupancies may refine where storage and blocking occur, but after projection they must implement the same externally visible capacity/acceptance behavior at the chosen boundary. Conversely, any cross-channel dependency or join condition that can block progress must appear as ordinary  $\text{BoundaryReq}$  /  $\text{ChannelEnabled}$  /  $\text{ChannelDisabled}$  facts on some explicit internal abstract channel of  $\text{Sys\_B}^{\text{elab}}$ , not as a hidden extra predicate on an otherwise E4-enabled outer-boundary operation.

All examples and constraints in Appendix Q are to be read relative to this projection/accounting invariant.

**Explicit join boundary requirement (normative):** The mutual-stall point shall be an explicit staged/bundle/join boundary inside the elaborated realization (i.e., the endpoint of an explicit finite-capacity message channel). “Join not met” shall therefore be expressed as ordinary  $\text{ChannelDisabled}$  at that explicit internal boundary (because that boundary’s own E4 availability predicate fails at the settled  $\text{BoundaryEvalPoint}$ ). The realization must not force a  $\Sigma$ -visible operation on an abstract channel  $c$  at the chosen boundary to be  $\text{ChannelDisabled}$  solely due to the state of some other channel; any cross-channel dependency shall be expressed only via the explicit staged/bundle channels introduced by the elaboration. Operationally, the coupler is invisible at the outer DUT/testbench observation boundary: it only propagates ready/valid between the explicit staged/bundle channels inside the elaborated realization, and it does not change the projected outer  $\text{occ\_c}(t)$ , which is defined solely by projected outer-boundary commits under  $\Pi_{\text{elab}}$ . This invisibility is only with respect to the outer projection; it does not hide any explicit staged/bundle/join channel from Appendix K’s proof-internal WFG boundary.

- **WFG compatibility (Appendix K):** The realization must not introduce a new blocking point that is invisible to the proof-internal abstract boundary of  $\text{Sys\_B}^{\text{elab}}$ . In particular, if the consuming process cannot advance the pending blocking frontier  $\text{Front\_P}(\sigma)$  (Appendix K.1), then at least one blocking frontier operation ( $\text{Push/Pop}$ ) at either the chosen outer boundary or an introduced explicit staged/bundle/join boundary is asserted with a true boundary request and is channel-disabled by the  $\text{ChannelEnabled/ChannelDisabled}$  test; the process is therefore

“blocked on message passing” exactly as defined in K.1. Because couplers are purely combinational, they have no internal state transitions; any changes in readiness/validity induced by the coupler are purely a function of (i) downstream readiness and (ii) the same channel occupancies /  $\Sigma$ -visible committed transfers already accounted for by E4 and Appendix K, with “downstream readiness” interpreted in the  $\varepsilon$ -quiescent ( $\varepsilon$ -normalized) state used for WFG evaluation.

Pure internal data movements that do not change any proof-internal BoundaryReq, ChannelEnabled/ChannelDisabled, Pending\_P, or Front\_P fact are  $\varepsilon$ -steps and are WFG-inert under the same pending-blocking-frontier / Front\_P( $\sigma$ ) discipline used elsewhere in Appendix K. By contrast, a staged/bundle/join channel boundary that can block progress is not hidden from Appendix K; it is part of the explicit Sys\_B<sup>elab</sup> proof boundary, even if it is projected away from the outer  $\Sigma$ \_DUT trace by  $\Pi$ \_elab.

- **Combinational well-formedness:** The network of combinational couplers and ready/valid connections shall be combinational acyclic (no pure combinational ready/valid loops). Any required feedback that would otherwise create a combinational loop must pass through an explicit finite-capacity channel stage (and is therefore represented in the back-annotated realization and its effective capacities).

Accordingly, the formal development need not depend on the particular back-annotation construction, provided it yields an explicit source-side reference model Sys\_ref = Sys\_B<sup>elab</sup> (Definition Core.1 and Appendix J, Remark 2) equipped with a projection/accounting map  $\Pi$ \_elab satisfying the invariant above. The elaborated model must respect E4, BoundaryReq, ChannelEnabled/ChannelDisabled, and the Appendix K blocking/WFG semantics at its explicit proof-internal abstract boundaries, while its projection back to the chosen outer  $\Sigma$ \_DUT-visible boundaries must preserve the externally visible Push\_c / Pop\_c history and aggregate capacity/occupancy behavior. In these cases the proofs do not target the raw outer-boundary Sys or the non-elaborated outer-boundary Sys\_B directly; they target the proof-internal explicit abstract channels of Sys\_B<sup>elab</sup> together with their  $\Pi$ \_elab projection back to the outer DUT/testbench-visible boundary.

### Lean-oriented clarification / proof-parameter convention

The formal theorems do not require a unique algorithm that constructs Sys\_B<sup>elab</sup> from RTL\_B. Instead, the chosen comparison boundary Sys\_ref is part of the theorem instance. For ordinary bounded-FIFO cases, Sys\_ref may be Sys\_B. For cases involving proof-relevant internal staging, multi-input joins, combinational couplers, sync-boundary epoch gates, or shared-capacity effects, Sys\_ref shall be supplied as Sys\_B<sup>elab</sup>(E) for some well-formed elaboration package E.

Thus elaboration is not an extra semantic freedom inside the proof. It is an explicit input to the proof instance, and all Appendix I–K definitions are interpreted over that chosen Sys\_ref.

#### Definition Q.1 (Elaboration package for Sys\_B<sup>elab</sup>).

For a chosen outer source reference boundary Sys\_B and outer observable alphabet  $\Sigma$ \_DUT, an elaboration package E consists of:

1. an elaborated source reference model Sys\_B<sup>elab</sup>(E);
2. a finite set of explicit abstract channels Chan\_elab(E), each with finite capacity B\_E(d)  $\geq 0$  and cycle-boundary occupancy occ\_d(t);
3. a distinguished subset of outer-boundary channels Chan\_outer and a projection/accounting map

$\Pi_{\text{elab}} : \text{Chan\_elab}(E) \rightarrow \text{Chan\_outer} \cup \{\perp\};$

4. a trace projection map  $\pi_{\Sigma, E}$  from proof-internal  $\Sigma_{\text{elab}}$  traces to outer  $\Sigma_{\text{DUT}}$  traces; and
5. a state-accounting map  $A_E$  that computes each outer  $\text{occ}_c(t)$  from the occupancies and token/accounting state of the elaborated channels in  $\Pi_{\text{elab}}^{-1}(c)$ .

The package is well formed only if the following obligations hold:

(a) Outer-trace preservation.  $\pi_{\Sigma, E}$  neither creates, drops, duplicates, nor relabels any outer-boundary  $\text{Push}_c(v) / \text{Pop}_c(v)$  event.

(b) Occupancy preservation. For every outer channel  $c$  and every  $\varepsilon$ -quiescent cycle-boundary state  $\sigma$  at time  $t$ ,

$$\text{occ}_c(\sigma) = A_E(c, \sigma),$$

and projected  $\text{Push}_c / \text{Pop}_c$  commits update  $A_E$  exactly as required by E4 for the outer channel  $c$ .

(c) Internal refinement. Any internal staged/bundle/join transfer either is projected away by  $\pi_{\Sigma, E}$  or projects to a unique corresponding outer-boundary commit. Projected-away internal transfers preserve all outer occupancies  $A_E(c, \sigma)$ .

(d) No hidden cross-channel disablement. If progress can be blocked by a join, mutual-stall condition, sync-boundary gate, or shared-capacity condition, then that blocking condition appears as ordinary  $\text{BoundaryReq} / \text{ChannelEnabled} / \text{ChannelDisabled}$  status at some explicit abstract channel  $d \in \text{Chan\_elab}(E)$ . It is not an additional hidden predicate on an otherwise E4-enabled outer-boundary operation.

(e) WFG visibility. Every proof-relevant blocking point introduced by the elaboration is represented at an explicit abstract boundary of  $\text{Sys\_B}^{\text{elab}}(E)$ , and therefore may be used by Appendix K's  $\text{Front\_P}$ ,  $\text{ChannelDisabled}$ , and WFG definitions, even if its events are projected away from the outer  $\Sigma_{\text{DUT}}$  trace.

(f) Combinational well-formedness. Any combinational coupler logic in the elaboration is stateless and acyclic after  $\varepsilon$ -settling; any feedback or storage needed for correctness is represented by an explicit finite-capacity abstract channel.

(g) Point-to-point elaborated channels. Every explicit abstract channel in  $\text{Chan\_elab}(E)$  at which  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled}/\text{ChannelDisabled}$ ,  $\text{Front\_P}$ , or Appendix K WFG facts are evaluated has exactly one producer process and exactly one consumer process at that boundary of evaluation. A join, bundle, or shared-capacity structure that would multiplex several producers or several consumers at an evaluated boundary shall be remodeled with explicit per-endpoint point-to-point sub-channels (for example, via an explicit arbiter process), as required by the point-to-point premise of Lemma K.2b.

(h) Frontier-item declaration and boundary attribution. The elaboration package declares every proof-internal frontier item introduced by  $\text{Sys\_B}^{\text{elab}}(E)$ , including its owning process, its membership in  $\Omega_P / \Omega_{\text{msg}, P}$  when applicable, its owning source-level message instance  $\text{MsgOf}_P(x)$ , its evaluated abstract boundary  $\text{BoundaryOf}_P(x) \in \text{Chan\_elab}(E)$ , where  $P$  is the owning process of  $x$ , and the partial order  $\leq_P$  among ordinary source items and introduced elaborated frontier items. For ordinary unelaborated

message calls there is a unique frontier item and a unique evaluated boundary. For elaborated calls, one source-level message instance  $q$  may own multiple frontier items at different explicit abstract boundaries; in that case  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled}$ ,  $\text{ChannelDisabled}$ ,  $\text{Pending\_P}$ , and  $\text{Front\_P}$  are interpreted for the pair  $(q, \text{BoundaryOf\_P}(x))$  represented by the frontier item  $x$ , not for  $q$  alone. Thus  $\text{BoundaryReq}(q, \sigma)$  remains only a shorthand for the ordinary unique-boundary case, while  $\text{BoundaryReq}(x, \sigma)$  means the endpoint-owned request for  $\text{MsgOf\_P}(x)$  at  $\text{BoundaryOf\_P}(x)$ . The package also states how these introduced items project through  $\pi_{\Sigma, E} / \Pi_{\text{elab}}$  back to outer  $\Sigma_{\text{DUT}}$  events and occupancy accounting.

(i) Qualified same-cycle settling. The elaborated source model as a whole shall satisfy the  $\text{Core.Settle}$  L2 frame and the source-side hypotheses of Definition  $\text{Core.QSync}$ . The elaboration package is responsible for the settling facts contributed by the staged/bundle/join/guard/coupler/epoch-gate structure. For that elaboration-owned cycle-local subnetwork, every applicable  $\rightarrow_{\varepsilon, L2}$  step shall respect the L2 frame—preserving the cycle index, registered process/FSM/pipeline state, registered explicit-channel storage/occupancy/credit state, and cycle- $t$  exogenous inputs, with no explicit-boundary  $\Sigma$  commit or next-cycle registered update—and, after cutting at the state-bearing boundaries, the subnetwork shall be finite, deterministic, combinational acyclic, and complete for the boundary predicates it owns. The package shall provide or cite a compositional/template proof of these template-side facts; that proof may be qualified once for each legal elaboration template and then bound to the concrete  $E$  instance.

Whole-source composition requirement. The template-side facts alone do not imply  $\text{QualifiedSynchronousModel\_M}(I)$  for  $M = \text{Sys\_B}^{\text{elab}}(E)$ . The selected theorem instance shall additionally establish the applicable design-owned source-settling facts, including stable sequential drives under  $\text{Core.RegSeq}$ /direct-input contracts, inclusion and deterministic evaluation of R2C combinational components, the applicable non-template L2-frame facts, and completeness at non-template selected boundaries. It shall also establish a checked template/design settling-composition condition: every dependency edge crossing the design/template partition is included in the complete same-cycle dependency analysis, and connecting the two subnetworks introduces no violation of QS2–QS5. In particular, the dependency graph of the complete combined same-cycle settling network shall satisfy QS4. A sufficient discharge is a checked combined dependency rank or topological order in which every design-local, template-local, and cross-partition settling edge advances the order. The template-side, design-side, and composition evidence together establish the corresponding  $\text{QualifiedSynchronousModel\_M}(I)$  fact for  $M = \text{Sys\_B}^{\text{elab}}(E)$ , so Theorem  $\text{Core.QSync-Ref}$  supplies existence and uniqueness of the source  $\text{SettlePoint/KCut}$ . This obligation makes the former implicit source-settling termination premise explicit; Definition  $\text{Core.Settle}$  by itself does not assume a fixed point exists.

When  $\text{Sys\_ref} = \text{Sys\_B}^{\text{elab}}(E)$ , all references to  $B(\cdot)$ ,  $\text{occ\_}$ ,  $\text{BoundaryReq}$ ,  $\text{ChannelEnabled}$ / $\text{ChannelDisabled}$ ,  $\text{Front\_P}$ , and the Appendix K WFG are interpreted over the explicit channels of  $\text{Chan\_elab}(E)$ , with outer observations recovered only through  $\pi_{\Sigma, E}$  and  $\Pi_{\text{elab}}$ .

**Notation convention for  $A\_E$ .** This document writes the accounting function in channel-first form  $A\_E(c, \sigma)$ . A mechanization may equivalently use the state-first form  $A\_E(\sigma, c)$  so that state-indexed maps compose naturally with elaborated-state projections. The two spellings denote the same accounting function with the argument order reversed; no semantic distinction is intended.

## Locality of elaboration

The elaboration package  $E$  is constructed locally, per process and per channel boundary. Each elaboration pattern in this appendix (multi-input pipelined-loop staging, sync-boundary epoch gating, R2C combinational coupler, etc.) is determined by the local channel structure of the process at the

boundary in question — its number of input channels, its sync-boundary discipline, its pipelining configuration — and does not depend on the structure or elaboration of any other process. Independently chosen per-process elaborations compose into a system-level  $\text{Sys\_B}^{\text{elab}}(E)$  without further global modeling.  $\Pi_{\text{elab}}$  then projects the introduced internal channels back to the outer  $\Sigma_{\text{DUT}}$  alphabet on a per-channel basis. In this sense, the elaboration step preserves the per-process compositionality stated in the introduction: it refines each process's abstract boundary, it does not introduce cross-process modeling obligations.

## Appendix R – Compact Derived Formal Core and Mechanization View

### R.1 Purpose and normative status

This appendix is a derived companion to the normative theorem stack. It introduces no definition, premise, execution domain, reconstruction function, or theorem interface consumed by an earlier section. The normative dependency order is: the Normative Formal Core; Appendix I process-local replay and preparation; Appendix J system construction, KObs reconstruction, and equal-KObs correspondence; and Appendix K wait-for/deadlock factoring and preservation. This appendix may be used as a compact review or mechanization map, but citations needed to instantiate a theorem shall point to those normative locations.

Deletion-safety rule. Removing this appendix shall not make any earlier formula, proof obligation, certificate field, or theorem statement undefined. A future edit that introduces an upstream citation to an Appendix R item is therefore an architectural regression.

### R.2 One-way dependency structure

Core  $\rightarrow$  I  $\rightarrow$  J  $\rightarrow$  K  $\rightarrow$  R.

The Core fixes  $\text{Sys\_ref}$ , explicit abstract channels and elaboration,  $\Sigma/\epsilon$ , global Core.4 quiescence, L2 Core.Settle points and Core.QSync, the Core.KCut comparison domain, Core.ReachPrep, dynamic frontier items and source order, Commit/Issue/BoundaryReq, Core.IC/Core.SyncFence, Core.IssueFair, ChannelEnabled/ChannelDisabled, Pending and Front, Core.7a BlockingMsgNextFront, Core.ElabProj, Core.WFG, Core.RegSeq, Core.LLegal, Core.Phase, Core.PolicyL/Core.PolicyS, Core.16/Core.Drain discipline, Core.CycleState/Core.CycleResult/Core.CompleteCycle/Core.SilentCycle, DeclaredTerminal/Core.TerminalIdle, Core.EnvStep/Core.EnvTerminal, Core.CycleClosure, Core.FairPick/Core.FairCycleDovetail, Core.LAdm, and the Core.18c KObs datatype and pure reconstruction functions. Appendix I proves process-local deterministic replay and finite preparation. Appendix J composes the local evidence, proves the J.KObs reconstruction lemmas, and establishes Corollary J.1 equal-KObs correspondence. Appendix K factors the deadlock predicates through that record and adds the operational liveness, drain, fairness, response, terminal/reset applicability, and certificate-coverage obligations. Appendix R only summarizes the completed chain.

#### R.2a K-observation architecture

The formal development has three explicit layers. First, the operational semantics define Core.RegSeq, Core.ReachPrep, ReturnedAtCycle, Policy L/S, channel commits, Core.IssueFair, Core.Settle/Core.QSync, Core.CycleState/Core.CycleResult/Core.CompleteCycle/Core.SilentCycle, DeclaredTerminal/Core.TerminalIdle, environment/cycle closure, the constructive Core.FairPick/Core.FairCycleDovetail fair-continuation interface, and Core.Drain/K.Drain. Second, Appendix I records enriched deferred NB holes and field-sensitive local preparation; Appendix J derives the dependency order, proves J.RegPhaseNF, performs Horizon-batched J.UnitExt, constructs

J.GWR/J.RegPhaseReach, and proves Theorem J.1/J.HCanon. Third, the K cutpoint layer combines H and O in KObs. J.KObsConf transfers one selected cutpoint; K.DrainReach supplies the remaining nonwithdrawal/ $K\epsilon$ /Core.16 bridge, and K.RLAdmReach is its conjunction with J.RegPhaseReach. K.CertifiedCutpoint and Theorem K.1A operate per cutpoint; Core.QSync-Post derives Core.KCutClosure\_post from the qualified synchronous structural evidence, and K.CertCov/K.1A-Total supply the separate certificate-totality/universal-preservation layer. The Policy-S reduction then adds K.EnvMF, K.DomResetScope, the shared K.InterruptionGate schema with K.ResetEpoch and nested K.TerminalDisposition instantiations, K.BF, K.Resp, and the K.2b/K.2c operational route.

(1) Operational Issue is retained. Issue transitions and the Issue/Commit linkage remain part of the operational semantics and continue to govern R4/E3/E5, Policy L, Policy S, Core.IC, Core.Drain/K.Drain, the applicable progress and fairness premises including K-O1, Lemmas K.2b.R/K.2b.P/K.2b.Q, and the K.Resp monitor trigger. KObs removes historical issue timestamps only from the cross-model cutpoint-observation and certificate interface; it does not redefine Issue(q), identify Issue(q) with Commit(q), or permit an operational proof to ignore whether work was already issued before a block.

(2) KObs is an observation-layer quotient, not a normalized execution. The factoring theorems prove that K-relevant cutpoint predicates depend only on replay history plus the current attributed residual offers. They do not construct a replacement execution in which committed operations issue in their commit cycles, and no issue-erasure normal form is assumed.

(3) Residual offers are typed over frontier items and explicit boundaries. A residual offer identifies the owning process P, frontier item  $x \in \Omega\_msg, P$ , owning dynamic message-call instance  $q = \text{MsgOf\_P}(x)$ , evaluated explicit abstract boundary  $c = \text{BoundaryOf\_P}(x)$ , endpoint direction, and reset epoch. Its operation kind is the explicitly derived field  $\text{kind}(o) := \text{kind}(q)$ , with well-typedness requiring producer-valid exactly for a Push instance and consumer-ready exactly for a Pop instance. A bare set of source calls q is insufficient in  $\text{Sys\_B}^{\text{elab}}$ , where one source-level call may own several proof-internal frontier items at different explicit boundaries.

(4) The full elaboration package E is a parameter of KObs. E includes the selected Sys\_ref form, explicit abstract channels, capacities B(c),  $\Pi\_elab$ , ownership, boundary and projection maps, frontier-item universes and partial orders, and any bundle/join/epoch-gate structure. KObs is not parameterized by B alone.

(5) Cross-model residual-offer conformance is the bidirectional boundary fact J.OfferConf of Appendix J: every abstract residual offer has its attributed RTL boundary request asserted, and every relevant asserted RTL boundary request is attributed to exactly one stated residual offer. The record enforces at most one attributed outstanding source-level offer per endpoint direction at each explicit abstract boundary. This prevents a proof-internal realization transfer already counted in channel occupancy from also being represented as a second source-level offer.

(6) Residual-offer reachability remains separate from boundary conformance. J.OfferReach establishes that a proposed KObs record is in the image of an admissible reachable Sys\_ref cutpoint with the same annotated replay history and witness w. J.OfferConf establishes agreement between the record's residual offers and the RTL requests at the selected cutpoint. J.KObsConf requires both; OfferConf alone cannot manufacture an unreachable source offer set, and I.ReplayObs/I.DR replay matching alone cannot determine I.CutpointOfferObs or which uncommitted requests are presently asserted.

(7) Snapshot drain closure is distinct from the transition discipline. DrainClosedNow(D,KObs) means that, at the selected Core.KCut K-facing cutpoint, no already-offered non-frontier blocking request owned by a process in D is channel-enabled. It is a function of the cutpoint observation. DrainDiscipline denotes the continuation-level no-new-later-work, non-withdrawal, finite non-replenishing drain, and sync-boundary conditions of Core.Drain/K.Drain; it remains an operational premise.

(8) The minimal K-specific residual-offer abstraction carries no pending Push payload. Payload stability while a blocking Push request is asserted remains required by Core.IC and Appendix C, and the

committed payload remains checked by the value-labeled replay/trace results. A proof or checker needing mid-flight data-wire correspondence may add that separate implementation obligation without enlarging the deadlock observation itself.

(9) Elaboration and issue policy remain separate. Pipeline storage, bundle/join structure, and internal stage movement are represented by  $E$  and the explicit  $\text{Sys\_B}^{\text{elab}}$  channels. Policy  $L/S$  governs issue of the actual source-level operation instances and their elaboration-provided frontier items.  $\text{KObs}$  records current source-attributed offers but does not turn intra-realization movement into additional source calls or double-count capacity already represented by  $E4$  occupancy.

(10)  $\text{KObs}$ -facing waiver classes are observationally closed. For any theorem, certificate, or checker that claims a waiver-parameterized cutpoint predicate is determined by  $\text{KObs}$ , write  $\text{KObsClosed\_I,E,w}(W)$  for the requirement that, for all  $\sigma_1, \sigma_2 \in \text{KCutReach\_ref}(I)$  of the same fixed instance,  $\text{KObs\_E,w}(\sigma_1) = \text{KObs\_E,w}(\sigma_2)$  implies  $(\sigma_1 \in W \text{ iff } \sigma_2 \in W)$ .  $\text{K.Resp}$  monitor waivers are a separate complete-run certificate interface: they are keyed by the dynamic process-facing frontier instance and must satisfy  $\text{K.RespCov}$ 's coverage-preservation rule. A waiver class that depends on historical Issue timing, implementation microstate, monitor age, or any other fact not reconstructible from  $\text{KObs}$  cannot be used in  $\text{KObs}$  extensionality unless a separate proof establishes  $\text{KObsClosed\_I,E,w}(W)$ .

(11) The primary serial route and diagnostic terminal reconstruction are separate instance data. The ambient theorem instance records the  $\text{K.Resp}$  monitor schema, finite-bound function, reset/end-of-test disposition, and response-waiver policy used by  $\text{CF-S}$ . Those monitor fields are not part of  $\text{KObs}$  and are not transferred to RTL as latency claims. Separately, an instance may request diagnostic reconstruction of  $\text{WFG+}$ ,  $\text{EnvStop}$ ,  $\text{DoneStop}$ ,  $\text{Dead\_KS}$ ,  $\text{K.EnvOrd}$ , or  $\text{K.Done}$ . For  $\text{EnvStop}$  diagnostics, the instance records either the standard  $\text{B1-avail}$ , pull-based, consumption-ordered input model with default always-ready outputs, or an explicit  $\text{EnvReplayAdequate}$  certificate showing that permanent next-interaction availability is a function of  $H$  and  $w$ . Without one of these environment conditions,  $\text{KObs}$  still factors the ordinary internal  $\text{Dead\_K}$  predicate but makes no extensionality claim for  $\text{EnvStop}$ -dependent diagnostic predicates.

(12) The certificate boundary is layered and explicit. A proof-backed application records: (a)  $\text{K.CertHdr}$ , including  $\text{Core.RegSeq/Cat\#}$  and source-simulation conformance plus the  $\text{Core.QSync}$  structural-evidence provenance; (b)  $\text{CF-O}$  with  $H, O, J.H\text{Canon}/J.O\text{fferReach}/J.O\text{fferConf}/J.K\text{ObsConf}$  and, for Appendix K,  $J.RegPhaseReach$  provenance plus  $\text{K.DrainReach}$ ; (c) the selected A5 payload; and (d) the  $\text{Core.QSync-Post}$  theorem reference establishing  $\text{Core.KCutClosure\_post}$  plus  $\text{K.CertCov}$  for a universal claim.  $J.RegPhaseReach$  is theorem-derived on the compositional route from enriched holes, field-sensitive footprints,  $J.RegPhaseNF$ , and Horizon-batched  $J.UnitExt$ ;  $\text{DrainReach}$  remains independently checked. Deferred-hole records are proof-internal and empty at exported cutpoints, so they do not enlarge  $\text{KObs}$ . Derived  $\text{Front/WFG/deadlock}$  values remain checker-reconstructed.

### R.3 Compact theorem-instance package

A theorem instance fixes the design, source and RTL identities,  $\text{Sys\_ref}$  form, elaboration package  $E$  and  $\Pi_{\text{elab}}$ , explicit abstract boundaries and capacities, initial/reset state, stimulus and environment model, witness choices, arbitration and  $\epsilon$ -normalization conventions, observation boundary, waiver classes, and—when the  $\text{Policy-S}$  route is used—the concrete deterministic simulator and  $\text{K.Resp}$  monitor configuration. These bindings are supplied by the  $\text{Core}$  and the certificate interfaces in  $J$  and  $K$  rather than by this appendix.

At a selected source  $K$ -facing cutpoint  $\sigma_{\text{ref}} \in \text{KCutReach\_ref}(I)$ ,  $\text{Core.18c}$  records  $\text{KObs\_E,w}(\sigma_{\text{ref}}) = \langle E, w, H, O \rangle$ .  $J.O\text{fferReach}$  proves that the record is source-realizable,  $J.O\text{fferConf}$  proves the attributed RTL request abstraction, and  $J.K\text{ObsConf}$  gives the same typed record to the selected RTL cutpoint.  $J.K\text{Obs-Proc}$ ,  $J.K\text{Obs-Chan}$ ,  $J.K\text{Obs-Offers}$ , and  $J.K\text{Obs-WFG}$  reconstruct the process, channel,

frontier, request, enablement, and WFG facts.  $K.KObs\text{-}Dead$  and  $K.KObs\text{-}Ext$  factor the relevant deadlock and waiver-dependent snapshot predicates.

#### R.4 Compact safety and cutpoint bridge

Derived summary R-C1. Under the hypotheses of Theorem J.1 and Corollary J.1, every covered reachable  $RTL\_B$  prefix has a compatible reachable  $Sys\_ref$  witness under E1–E5, and the selected  $\varepsilon$ -normalized  $RTL/source$  cutpoints share one  $Core.18c$   $KObs$  record. Equal reconstructed  $K$ -facing facts are functional consequences of that typed equality; they are not independent state-equality assumptions and do not transfer response-monitor age or continuation-level drain discipline.

#### R.5 Compact liveness bridge

Derived summary R-C2. For the exact source witness selected by Corollary J.1,  $J.RegPhaseReach$  supplies the  $Core.Phase$  L0–L4 phase-correct reachability component.  $K.DrainReach$  separately supplies nonwithdrawal,  $K\varepsilon$ , and every activated  $Core.Drain/K.Drain$  obligation. Their conjunction  $K.RLAdmReach$  places that exact prefix in  $Pref\_L^{adm}(I)$ . If A5-L excludes  $Dead\_K^W$  throughout that admissible domain, Theorem K.1A excludes the corresponding reconstructed deadlock at the certified  $RTL$  cutpoint;  $Core.KCutClosure\_post$  and  $K.CertCov$  are additionally required for the universal reachable-cutpoint conclusion.

For the principal serial-reduction route,  $SDet$  selects one concrete  $Policy\text{-}S$  execution and  $CF\text{-}S$  supplies complete bounded-response coverage and cleanliness under  $K.EnvMF$ , a certificate-bound  $K.DomResetScope/DomEpoch$ , and the shared  $K.InterruptionGate$  schema.  $K.ResetEpoch$  is instantiated for every candidate component; only on  $K.ResetEpoch.Continue$  is  $K.TerminalDisposition$  instantiated. Their  $DirectFail$  branches are independently mapped finite  $K.RespFail$  witnesses, while the nested  $Continue$  branches select the  $K.2b.E/R/P/Q$  dominance route.  $K.OpKey$  and  $K.CV$  including  $K.CVPred/K.CV\text{-}WF$  and Lemmas  $K.2b.P0/P1$ ,  $K.BF$  including Lemma  $K.BF\text{-}Persist$  and the exhaustive  $Core.IC$  lifecycle plus the gate-level exhaustiveness citation, lowering,  $Core.CycleClosure$  plus the  $Core.FairPick$  constructive discharge of  $Core.IssueFair/B2/B3$  and  $B5$ , drain, environment conditions, and  $K.Resp$  coverage remain separate premises. Corollary K.2c first establishes the lowered  $A5L\_Sys\_K$  result and  $K.Low.A5$  transports it to A5-L on  $Sys\_ref$ . Appendix R does not weaken, combine, or replace any of those premises.

#### R.6 Mechanization decomposition

A faithful mechanization should preserve the same module boundaries: (1)  $Core$  types and operational semantics; (2) Appendix I local replay/preparation; (3) Appendix J finite-ideal composition and  $KObs$  reconstruction; (4) Appendix K snapshot factoring, admissible continuation, serial reduction, and certificate coverage; and (5) this appendix as generated documentation or a compact theorem-export layer. In particular, the mechanization should not define a second  $Sys\_ref$ , a second admissibility relation, or a second  $KObs$  correspondence inside the companion layer.

#### R.7 Audit checklist

A structural audit passes when: every shared definition is located in the  $Core$  before first use; Appendix I depends only on the  $Core$  and its explicit premises; Appendix J imports I and the  $Core$ ; Appendix K imports the  $Core$ , I/J results, and its named liveness premises; no earlier section cites Appendix R for a required definition or lemma; and Appendix R's compact statements are traceable to named normative results without adding stronger claims.

## Appendix S – Lean Proof Strategy Document

A separate Lean proof strategy document is under development. It should import, rather than duplicate, the normative interfaces fixed here: `Core.ElabProj`; Appendix G R1–R5/E1–E5, including the R2C fixed-point semantics/matching used by R2/E2; `Core.RegSeq` and `ReturnedAtCycle`; `Core.IC/Core.SyncFence`; `Core.WFG`;

`Core.16/Core.Drain/Core.CycleState/Core.CycleResult/Core.CompleteCycle/Core.SilentCycle/Core.RTLCycleCarrier/Core.CycleClosure/Core.CycleChoiceExt/Core.FairPick/Core.FairCycleDovetail/Core.EnvStep/Core.EnvTerminal/DeclaredTerminal/Core.TerminalIdle/Core.LAdm`; `I.DeferredNBHole/NoDependentUse`; `J.ConstructionAction` and the field-sensitive location vocabulary;

`J.ReqSnapshot/J.NBDecision/J.BufSameCycle`; `Dep_J` semantic-versus-D5

stratification/`J.HorizonOrderSafe/FR/acyclicity/canonical order`; `J.RegPhaseNF/J.RegPhaseReach`;

`PCert/CCert/LocalGWR`; `frame/local-lift/commutation/enablement/Horizon-batched UnitExt`; `J.GWR-Comp/J.GWR`; `J.CycleReach_post/J.QSyncTraceState/J.QSyncTraceInd/J.QSyncCertInd-Trace/J.TraceCertCov-QSyncCycle/J.1-Safe-QSyncCycle/J.CycleIdealCover/J.1-Safe-QSync`;

`J.TraceCertCov/J.RefProjCover/J.IdealPrefixClosed/J.1-Safe`; `J.HCanon/OfferReach/OfferConf/KObsConf`;

`J.KObs-Proc/J.KObs-Chan/J.KObs-Offers/J.KObs-WFG`; `K.KObs-Dead/K.KObs-Ext`;

`K.DrainReach/K.RLAdmReach`; `K.OpKey/K.CVPred/K.CV-WF/K.2b.P0/K.2b.P1`; `K.Low.C/K.Low.A5`;

`K.DomResetScope/DomEpoch`; `K.JointFreeze-Ord/K.JointFreeze/K.EnvMF`;

`K.CompareAdvanceLegal/K.CompareAdvanceLegal-Ord`; `K.InterruptionGate` and its exhaustive `K.BF`

`K.ResetEpoch/K.TerminalDisposition` instantiations with `Continue/DirectFail` branch typing and reset-first nesting; `Core.SettleIsKCut/Core.KCutClosure/Core.KCutObserve`; `K.CertifiedCutpoint/K.CertCov` and

`K.1A/K.1A-Total`; and Appendix R's deletion-safe, nonnormative derived-companion status. The exported interface must preserve the distinction between supplied local facts, theorem-derived order and phase normality, the residual drain bridge, the `KObs` observation boundary, and universal package coverage.

The separate Lean document should be updated in the same named revision before mechanization relies on the old stop-at-hole or unit-at-a-time interfaces.

Mechanization status rule. Encode `Core.IC` and `Core.SyncFence` as legality/invariant content of the `Sys_ref` operational semantics, not as unconstrained global axioms. Proofs about `RTL_B` shall consume separately named conformance evidence (directly or through the corresponding R/E and Appendix M.7a discharge records) before using the analogous lifecycle or fence facts on the implementation side.

## Appendix T – Scheduling-Specific Obligations Beyond Standard Simulation Theory

### T.1 Purpose and Nature of the Formal Result

**Status.** Appendix T is informative and non-normative. It highlights selected scheduling-specific library gaps but does not add, remove, redefine, or exhaustively enumerate formal obligations. Normative authority remains with the Normative Formal Core and the applicable theorem interfaces; Appendix M.7a supplies the definitive consolidated side-condition discharge record, and Appendix S records the Lean mechanization decomposition and selected import anchors.

Standard formal-methods libraries provide reusable foundations for labeled transition systems, weak or epsilon-abstracted transitions, forward simulation, finite and infinite executions, trace inclusion, well-founded induction, fixed-point reasoning, fairness, and related proof techniques. Those foundations can substantially reduce the generic infrastructure needed for a Lean mechanization of this document. They do not, however, establish the HLS scheduling contract or its safety and liveness guarantees by themselves.

The top-level safety result is not ordinary trace inclusion or trace equivalence. It is the document's witness-relative ordered trace-compatibility relation over the selected observable alphabet. Matching is defined over complete observable units and bucketed executions, preserves the required scheduling-specific ordering constraints, permits independent observable units to commute or to appear in different cycle groupings in Sys\_ref and RTL\_B, and treats the Post  $\rightarrow$  Ref and restricted Ref  $\rightarrow$  Post uses differently. A generic theorem such as "forward simulation implies trace inclusion" therefore does not directly yield the result proved here.

The liveness result is likewise not ordinary per-process progress. It is a conditional global preservation statement about complete admissible executions, explicit channel boundaries, persistent blocking requests, drainable work, wait-for components, fairness, environment behavior, and a conservative relationship between overlapped and serial source execution. Generic liveness and graph libraries can express parts of this reasoning, but the scheduling-specific bridge must still be supplied.

Accordingly, this appendix selectively highlights important semantic and proof obligations that remain specific to the scheduling contract after generic formal-methods infrastructure is imported. It is illustrative rather than exhaustive: omission of an obligation here does not make that obligation generic or remove it from a faithful mechanization. It intentionally does not restate complete theorem-premise inventories, certificate field layouts, or step-by-step proof chains. Those authoritative details remain in the Normative Formal Core and Appendices I, J, K, and K-P; Appendix M.7 and M.7b describe discharge and certificate packaging; Appendix R provides the compact derived architecture map; and Appendix S records the Lean proof strategy, mechanization decomposition, and selected import anchors; it is not the definitive side-condition ledger.

## **T.2 Safety-Specific Obligations**

### **T.2.1 Initial-State and Reset-Epoch Correspondence**

A generic simulation theorem can use an initial-state relation once one is supplied, but it does not establish the particular source/RTL correspondence required here. The post-reset RTL\_B state must correspond to the prescribed Sys\_ref state at the selected abstraction boundary, including the process, channel, request, synchronization, and source-visible state that the theorem instance treats as relevant. Dynamic reset makes this more than a one-time startup obligation. Each affected reset epoch must re-establish the applicable source/RTL relationship, including the reset well-formedness, channel and request accounting, and any direct-input anchoring contract. The RESET observable unit, RW1-RW5, and the Appendix-J reset-splice construction supply the scheduling-specific semantics; generic libraries supply only the surrounding induction and transition-system machinery.

### **T.2.2 Dynamic Source Enablement and Operation Attribution**

When RTL\_B exhibits an observable unit, the proof must establish that the corresponding specific dynamic source operation can legally occur from the currently constructed Sys\_ref state. Equality of labels is not enough: the proof may need the process control position, branch and loop-instance identity, source-order predecessors, synchronization anchors, channel state, witness-selected outcomes, and dynamic-operation attribution.

This obligation is specific to HLS because the implementation may contain added FSM states, overlapped loop iterations, eager issue, or internal staging that do not appear as source control steps. Appendix I supplies process-local preparation/replay machinery and Appendix J combines it with channel, atomic-interaction, and global extension evidence. A generic simulation library can host that construction, but it cannot infer the HLS-specific enablement relation or the correct dynamic occurrence.

One especially important example is the R4/E3 safe message-issue discipline. The obligation is prefix-closed: at each applicable comparison cutpoint, if a source-later message operation has issued, every required source-earlier operation in the dynamic source order must already have issued. This is stronger than comparing timestamps of events that are already present, and Issue is proof-semantic state rather than merely a  $\Sigma$ -visible committed-trace event. A generic trace-inclusion or forward-simulation library

can represent the resulting relation once it is defined, but it does not supply the dynamic source order, operation attribution, or downward-closed issued-set invariant required by this scheduling contract.

### **T.2.3 Same-Cycle Independence, Atomic Units, and Buffered Sequentialization**

A hardware cycle may contain multiple observable events for which no meaningful order exists, while other collections of events must be treated as one indivisible observable unit. The formalization must therefore preserve scheduling-specific independence and required ordering rather than impose an arbitrary total order. Rendezvous transfers, synchronization handshakes, and the selected combinational fixed-point observation are examples of atomic units whose internal pieces may not be split across matching steps.

Positive-capacity channels add another scheduling-specific case. Under E4, same-cycle Push and Pop decisions are made from the common pre-cycle occupancy rather than by fall-through. In the interior-occupancy case the corresponding source effects can be sequentialized without changing the channel result; at empty or full occupancy the same-cycle complementary operation cannot create availability that was absent at the boundary. Generic concurrency or trace libraries can represent independence once specified, but they do not establish these E4-specific commutation and atomicity facts.

### **T.2.4 Witness Compatibility, Common Snapshots, and Deterministic Replay**

The source reference model may admit internal choices that are not independently visible in the observable trace, including non-blocking outcomes, arbitration choices, epsilon-state choices, and replay paths. The proof witness must therefore be tied to the behavior actually represented by RTL\_B rather than chosen merely because it makes a source execution possible. WC is the scheduling-specific obligation that supplies this connection.

Witness-significant non-blocking decisions also require a common request/channel snapshot. A process-local replay path cannot decide such a call in isolation if its outcome depends on requests or availability produced elsewhere in the same construction horizon. Appendix J therefore reconstructs the relevant shared snapshot and orders request writes against the decision that reads them. The generic library contribution is ordinary state and dependency reasoning; the snapshot contents and legal non-blocking outcome are consequences of the scheduling semantics.

Once the fixed stimulus, WC-compliant witness, dynamic-operation attribution, and complete process-local observable history are fixed, the logical replay checkpoint must be determined well enough to justify later source behavior. The document deliberately distinguishes that logical replay state from raw operational preparation state and from currently asserted uncommitted offers. A generic determinism theorem does not choose this quotient or prove that it is the correct replay boundary for this scheduling model.

### **T.2.5 Explicit Boundary Semantics, Capacity, and B-Faithfulness**

The safety proof depends on the behavior of selected explicit abstract communication boundaries, not merely on abstract message values. E4 supplies the legal Push/Pop behavior for rendezvous and positive-capacity channels, including availability and occupancy updates. When staged, bundled, joined, coupled, or sync-gated implementation structure cannot be faithfully represented by one ordinary E4 channel, Appendix Q supplies explicit elaborated boundary structure instead.

Back-annotated capacity is therefore not just a numeric upper bound. B-faithfulness requires the selected RTL boundary and its represented internal storage or credit to exhibit the same boundary-visible acceptance, retention, backpressure, and transfer behavior as the declared abstract channel model. Generic FIFO or simulation libraries do not identify the correct physical boundary, prove that hidden storage is completely represented, or establish the required source/RTL availability correspondence.

### **T.2.6 Proved Local-to-Global Witness Construction**

Process-local replay and preparation facts do not by conjunction produce one globally reachable source execution. The global proof must reconcile shared channel state, non-blocking snapshots, atomic

interactions, reset/environment effects, field-sensitive footprints, and cross-process dependencies while allowing genuinely independent work to commute.

Appendix J supplies the scheduling-specific construction that derives a validated proof order and a phase-correct reachable source witness from those local facts. Generic graph algorithms, topological ordering, well-founded induction, and commutation lemmas are reusable once the dependency relation has been defined and proved sound. They do not establish dependency completeness, prove that omitted edges correspond to commuting actions, or show that the resulting order respects the document's registered-cycle semantics.

This is also why the formal development distinguishes the constructive J.GWR route from a direct route. A direct package may supply an independently validated order and extension proof, but it must satisfy the same scheduling-specific interface; an unexplained time order or an arbitrary legal source trace is not sufficient.

#### **T.2.7 Qualified Synchronous Settling, Canonical Cutpoints, and KObs Correspondence**

Generic epsilon-closure or fixed-point machinery does not by itself establish the cycle structure required by the standard qualified-synchronous scope. The proof must identify the registered K-origin, hold the required registered and exogenous state fixed during same-cycle settling, establish finite source and RTL settling to the selected endpoints, and on the RTL side recover the correct cycle origin for each canonical K-facing cutpoint. Core.QSync and the associated source/post closure results supply those scheduling-specific hypotheses and theorems.

Trace compatibility alone is also insufficient for Appendix K. A process that has not yet reached a blocking call can have the same committed observable history as one that currently has an attributed uncommitted boundary request. The KObs interface therefore carries the additional residual-offer information needed to reconstruct process frontier, enablement, wait-for structure, and the selected deadlock predicate at the comparison cutpoint.

The corresponding implementation obligation is completeness at the selected request-abstraction boundary: every relevant asserted RTL request must be represented and uniquely attributed, not merely the requests for which a convenient certificate record was produced. The details of KObs, OfferReach/OfferConf, K.CertifiedCutpoint, and the checker/certificate split are defined in Appendices J and K and summarized in Appendix M.7; Appendix T needs only the point that this observation boundary is scheduling-specific and is not supplied by ordinary trace inclusion.

#### **T.2.8 Coverage Beyond One Prefix: Cycle Induction and Ideal Closure**

A theorem that constructs a matching source witness for one supplied RTL prefix does not by itself prove that suitable packages exist for every reachable implementation history. The standard QSync route therefore needs an implementation-wide base-and-step correspondence argument across reachable real cycles, with exhaustive successor coverage rather than evidence from one observed simulation path.

For finite-history safety properties, cycle-aligned package coverage must then be related to finite observable prefixes that may end inside a cycle. The Appendix-J ideal-cover result supplies that scheduling-specific history relation for properties satisfying the stated ideal-prefix-closure condition. Generic induction and prefix reasoning are reusable, but they cannot supply the implementation carrier, QSync-silent settling facts, atomic-unit boundaries, or the document-specific notion of an admissible observable ideal.

The full definitions and proof obligations are in Appendix J, with the safety and K-facing certificate triggers summarized in Appendix M.7. Appendix T therefore does not duplicate the QSyncTraceInd/QSyncCertInd field structure or their individual sub-obligations.

#### **T.2.9 Finite Local Traces Inside Infinite Global Executions**

A process may stop producing local observable units while the complete system continues indefinitely. A correct infinite witness cannot be created by simply appending local idle cycles as though the whole

design had terminated. The proof must instead select or construct an admissible global continuation and then project that execution onto the locally inactive process.

Only when the entire system satisfies the document's true terminal fixed-point conditions may indefinite global idle stutter be justified directly. This distinction between local trace finiteness, temporary silent cycles, and genuine global terminal idleness is part of the scheduling model; generic infinite-trace libraries provide the execution machinery but not the hardware-specific terminal criterion.

### **T.3 Liveness-Specific Global Obligations**

The safety obligations above are largely local or compositional. The liveness and deadlock-preservation obligations are global: they depend on the interaction of processes, channels, persistent requests, fairness assumptions, environment behavior, drainable work, reset/termination behavior, and the wait-for structure. They cannot generally be discharged by proving each process locally live.

#### **T.3.1 Admissible Global Executions and Fair Progress**

Generic temporal-logic and fairness libraries can state that continuously enabled actions eventually occur, but the document must first define what counts as a legal and admissible Policy-L execution. Core.LLegal, Core.Phase, and Core.LAdm bind the scheduling policy, fixed instance choices, cycle phases, request nonwithdrawal, settling and drain discipline, and the distinction between finite admissible prefixes used for reachability and complete executions used for fairness and limit arguments. B2, B3, and the applicable environment assumptions then express the required scheduler/channel progress over complete admissible executions. B5 supplies the stronger global fixed-point condition where genuine terminal quiescence is required. Standard libraries can support fairness proofs for already-supplied executions once these predicates are supplied, but they do not define the HLS-specific enabledness, persistence, channel-progress, or terminal semantics.

Some proof sites require more than checking fairness of an execution that is already given. In particular, Lemma K.2b.E constructs a complete fair continuation from a finite prefix. Core.FairPick packages the prefix-local issue and boundary selections and starvation-freedom evidence used for that construction, while Core.FairCycleDovetail proves that the selected fairness actions can be embedded in typed complete cycles before dependent choice extends them indefinitely. Generic fairness and dependent-choice libraries can host this construction, but they do not supply the HLS-specific selection functions, phase legality, cycle-completion compatibility, or fairness exposure needed to build the continuation.

#### **T.3.2 Drain Normalization and Persistent Blocking Structure**

Epsilon-quiescence is not the same as drain closure. A state can have no enabled internal epsilon step while already-issued downstream transfers can still commit and remove transient in-flight work. A deadlock argument that inspects the wait-for graph too early can therefore mistake a temporary pipeline/channel condition for a persistent blocking structure.

The scheduling-specific drain discipline permits already-issued work to complete fairly while preventing new source-later work from replenishing the region beyond the blocked frontier. The proof must show that the relevant blocking requests and closed wait-for component survive this normalization and that the resulting cutpoint still carries the correspondence facts needed by Appendix K. Generic normalization and termination lemmas can be reused once the finite drain measure and no-replenishment relation are established; they do not establish those hardware-specific facts.

#### **T.3.3 Conservative Serial-Source Deadlock Reduction**

The central non-generic liveness step is the relationship between overlapped/liberal source execution and the selected conservative serial source execution. The document does not assume that ordinary trace inclusion or local progress makes these schedules interchangeable. It proves, on a stated applicability fragment, that a relevant liberal internal deadlock would force a permanent process-facing obstruction in the selected deterministic Policy-S semantics and therefore a source-side bounded-response failure.

That reduction requires scheduling-specific facts about dynamic operation identity and provenance across the two policies, source-order comparability, persistent blocking-request lifecycle, point-to-point boundary ownership, legal comparison/reach steps, preservation of frozen blocking structure, and explicit handling of reset or declared-terminal interruptions. Lean can reuse finite maps, graph theory, well-founded induction, and rank arguments, but the relation being ranked and the proof that it faithfully represents these HLS operations are part of this document, not a standard library result.

At the conceptual level the argument is: expose a persistent liberal blocking component at a valid drain-closed cutpoint; relate the frozen liberal blocking component to the selected serial execution through the K.2b.P/K.2b.R dominance comparison and K.2b.Q frontier extraction, without releasing the obstruction; and use complete finite-bound response monitoring to contradict a clean Policy-S run. The exact K.2b sublemmas, interruption-gate structure, provenance interfaces, environment/frontier applicability limits, and response-certificate fields are specified in Appendix K/K-P and Appendix M.7. Repeating them here would not add a distinct library-gap argument.

The bounded-response certificate is source-side proof machinery for this reduction. It does not assert that the RTL completes within the same bound. This distinction is scheduling-specific and remains important even when generic temporal or response-property libraries are used to encode the monitor.

#### **T.3.4 Universal Reachable-Cutpoint Coverage**

A per-cutpoint preservation theorem and a proof that one serial source execution is clean still do not establish a universal RTL deadlock-freedom claim. The proof must also establish that every reachable K-facing RTL cutpoint belongs to the intended canonical domain and carries the complete source/RTL correspondence package consumed by Appendix K.

In the standard QSync normal-compositional scope, the document derives the canonical cutpoint domain from the qualified cycle structure and derives package totality from an implementation-wide base-and-step correspondence invariant. Alternate independently checked coverage routes remain possible. Generic invariant induction can support this proof, but it cannot supply the QSync origin/cutpoint relation, KObs correspondence, or the exact package whose existence is being propagated.

Certificate records and checker modules are evidence interfaces for these obligations, not substitutes for the mathematical obligations themselves. Their detailed field ownership, trigger conditions, and packaging are therefore maintained in Appendix M.7/M.7b and the normative J/K definitions rather than repeated in Appendix T.

#### **T.4 Summary**

Reusable Lean and formal-methods libraries can provide much of the generic proof infrastructure: transition systems, finite and infinite executions, epsilon closure, simulation, trace lifting, partial-order and graph utilities, well-founded induction, fixed-point reasoning, fairness, and temporal properties. The document should reuse those foundations wherever they match the required abstraction.

Among the scheduling-specific obligations highlighted here are the semantics and bridges that connect generic formal foundations to HLS behavior: dynamic source-operation attribution and enablement; prefix-closed dynamic message-issue ordering; same-cycle independence and atomic observable units; witness-compatible replay and common non-blocking snapshots; explicit channel-boundary and back-annotated-capacity semantics; local-to-global source-witness construction; qualified cycle settling and canonical K-facing cutpoints; KObs request correspondence; cycle-wide package coverage and ideal closure; proof-built fair continuation from typed cycle choices; drain-aware persistent-blocking reasoning; and the conservative reduction from liberal overlapped deadlock to a bounded-response failure in the selected serial source execution.

These examples explain why importing a standard simulation or temporal-logic library is necessary but not sufficient. The normative definitions and theorem interfaces remain in the Core and Appendices I-K/K-P, the evidence and qualification interfaces remain in Appendix M.7/M.7b, and the mechanization

structure remains in Appendices R and S. Appendix T is therefore a selective map of important scheduling-specific proof burdens, not an exhaustive mechanization checklist or a second copy of the detailed proof architecture.

## Appendix U – Revision Record

### **19 August 2026 — Appendix T selective-scope clarification and residual guide/reference cleanup.**

Appendix T is explicitly classified as informative, non-normative, selective, and non-exhaustive: it highlights important scheduling-specific obligations that standard Lean/formal-methods libraries do not supply, while the Normative Formal Core and applicable Appendices G–K remain authoritative for definitions and theorem interfaces, Appendix M.7a remains the consolidated discharge record, and Appendix S remains the mechanization-facing strategy. T.2.2 highlights the R4/E3 prefix-closed dynamic issue-order obligation, T.3.1 distinguishes proof-built fairness evidence through `Core.FairPick/Core.FairCycleDovetail` from fairness of an already-supplied execution, and T.4 remains deliberately selective. The Guide safety reading path is aligned with the J.2b standard QSync all-finite-prefix safety route as well as its richer K-facing package-totality route. The duplicate K.2.1 Standing Assumptions label is removed without renumbering the KObs subsection sequence, the M.7 H1/H2/H3 historical shorthand is replaced by current-interface terminology, and T.3.3 states the serial comparison without implying a reversed K.2b.P/K.2b.R mapping direction. The former T.2.9a organization referenced by the 18 August revision record was consolidated during the Appendix T refocusing into the current T.2.9/T.3 structure. A follow-on scope repair adds a first-class M.7a R/E scheduling-contract conformance entry covering R1–R5/E1–E5 and R2C where applicable, aligns the M.7b instance/QSync bundle and Appendix S import anchors to that definitive ledger, and clarifies in the Guide/T.1 that Appendix S supplies mechanization decomposition/import anchors rather than the exhaustive side-condition inventory. No theorem premise, safety/deadlock scope, or certificate-coverage requirement is changed.

### **18 August 2026 — operational-invariant/conformance split and abstract Commit cleanup.**

`Core.IC` and `Core.SyncFence` are reclassified from “Axiom” headings to normative operational definitions/invariants of `Sys_ref`, with an explicit status note preventing a mechanization from treating them as unconstrained trusted assumptions. Appendix M.7a adds a first-class `Core.IC` request-lifecycle/Issue–Commit conformance entry alongside the existing `Core.SyncFence` entry, and the M.7b qualification bundle, K.CertPkg accounting, Appendix R, and Appendix S are aligned; RTL use therefore remains evidence-backed while source-side use is constitutive transition legality. Definition `Core.8` now defines a boundary commit by the selected proof model’s legal E4 channel-commit transition and treats simultaneous mapped valid/ready observation plus payload sampling as the `RTL_B` realization/conformance fact. The two introductory informal commit descriptions are narrowed correspondingly to the RTL-facing ready/valid reading. The existing process-facing `Commit(q)/CommitCycle(q)` distinction and later L4/`Core.RegSeq` result-availability rule are unchanged. No theorem premise, safety/deadlock scope, or certificate-coverage requirement is weakened.

18 August 2026 — QSync finite-prefix ideal-cover closure and residual carrier/closure repairs.

Appendix J.2b adds Lemma `J.CycleIdealCover` and Corollary `J.1-Safe-QSync`. The existing `J.QSyncTraceInd` / `J.TraceCertCov-QSyncCycle` route remains the package-producing result on the cycle-aligned safety domain; the ideal-cover lemma extends the safety conclusion to all reachable finite outer observable prefixes without asserting a separate `J.TraceCertCov/J.GWR` package for each non-cycle-aligned ideal. A residual carrier assumption is made explicit by new Lemma `Core.RTLCycleCarrier`: a reachable RTL prefix that has already entered a legal real cycle and contains a complete selected unit inherits the enclosing `Core.CycleResult` and may be carried through that same cycle to its successor K-origin, without adding generic cross-cycle progress, fairness, or totality. `J.CycleIdealCover` now cites that fact and its case split

explicitly includes a nonempty current-cycle ideal that may equal the full observable bucket before the registered update. New Definition J.IdealPrefixClosed defines the safety closure condition as downward closure over finite required-order ideals, not merely suffix deletion from a linearized trace, and the safety corollaries and practical scope text distinguish naturally ideal-closed invariants from cross-interface scoreboards that may require direct J.TraceCertCov coverage. The one-off “cycle-aligned carriers” wording is replaced by “prefixes ending at reachable K-origins,” Appendix S imports Core.RTLCycleCarrier and J.IdealPrefixClosed, and the then-current T.2.9a bold-heading formatting is restored; the 19 August Appendix T refocusing later consolidates that material into the current T.2.9/T.3 structure. No K-facing certificate, deadlock premise, or per-prefix correspondence obligation is weakened.

**17 August 2026 — Commit typing/process-facing completion, Appendix C alignment, and K.1a operational-domain cleanup.**

Definition Core.8 now makes Commit(q) the source operation’s process-facing commit occurrence and CommitCycle(q) its cycle index, while preserving separate boundary-local cycle indices for proof-internal E4 commit occurrences; it also states explicitly that commit/release is distinct from later result availability governed by Core.Phase L4/Core.RegSeq. Appendix C now gives only an informal operational reading of Commit and defers the formal Commit(q)/CommitCycle(q) meaning to Core.8; its blocking-request linkage uses CommitCycle(q) explicitly. Lemma K.1a is retargeted from an operational RTL microstate statement to the source-side/lowered KCut domain on which Definition K.Dead is defined; its buffered and rendezvous conclusions now cite Core.12/Core.13 directly and its proof is correspondingly shortened. No theorem premise, coverage obligation, or user-facing theorem scope is changed.

**17 August 2026 — Dead\_K consolidation, Core A9/A10 authority cleanup, and introductory deadlock-scope clarification.**

Definition K.Dead now has one authoritative statement: the duplicated bullets and explanatory paragraph are removed, while the environment-only exclusion is retained in the authoritative definition. Core and pre-K references now cite the Core scalar endpoint and reset-empty conventions directly; Appendix K A9/A10 are instance assumptions referring back to those Core definitions rather than restating them as competing definitions. Formal Guarantees at a Glance adds one short plain-language sentence noting the conservative applicability boundary for communication overlap and clock-timed/reactive testbench behavior. No theorem statement, theorem premise, or proof obligation is otherwise changed.

**17 August 2026 — standard QSync certificate-profile packaging.**

Appendix M adds M.7b, a normative packaging profile for the standard normal-compositional QSync flow. It groups the detailed M.7a discharge ledger into four logical interfaces: instance/QSync qualification, J correspondence/refinement, K applicability/provenance, and the primary Policy-S response bundle. The section states explicitly that these are packaging interfaces only: M.7a remains definitive, no theorem premise or qualification obligation is weakened or made optional, theorem-derived facts may be referenced rather than reasserted, and every artifact remains bound through K.CertHdr/K.CertPkg. The profile distinguishes the safety-only J.QSyncTraceInd use from the richer J.QSyncCertInd K-facing use, records the J.QSyncCertInd-Trace reuse rule, explains the standard universal K.CertCov-QSync composition, and permits alternate K.A5Cert routes in place of the Policy-S response bundle when A5-L is discharged another way. No theorem statement, proof obligation, or theorem scope is changed.

**16 August 2026 — QSync cycle-aligned safety coverage projection.**

Appendix J.2b now exposes the safety content already carried by the QSync base-and-step construction without importing the K-only correspondence/drain burden into safety. New Definition J.CycleReach\_post names exact reachable RTL prefixes ending at K-origins; J.QSyncTraceState retains the exact-prefix J.LocalGWR/J.GWR-Comp evidence and the J.HCanon component needed by the safety

transfer; and J.QSyncTraceInd gives base, one-cycle preservation, and exhaustive-successor obligations that explicitly do not consume J.OfferReach/J.OfferConf/J.KObsConf/K.DrainReach/K.RLAdmReach. Lemma J.QSyncCertInd-Trace proves the richer K-facing induction projects to this safety induction, with the C11 proof recording that the dropped facts are downstream outputs rather than inputs and with J.RegPhaseReach re-derivable from J.GWR-Comp. Theorem J.TraceCertCov-QSyncCycle derives J.TraceCertCov on CycleReach\_post, and Corollary J.1-Safe-QSyncCycle instantiates the existing safety theorem on that domain. M.3, M.7a, Appendix S, Appendix T, the appendix map, and the front coverage summary are aligned. The revision deliberately does not claim arbitrary intra-cycle ideal coverage: QSync settling is  $\Sigma$ -silent and R2C fixed-point units are atomic, but the separate question whether every later-in-cycle complete-unit ideal is covered by a reachable succeeding cycle-aligned prefix is left for a possible cycle-completion/ideal-cover lemma.

**16 August 2026 — R2C/RESET carrier, request attribution, memory-order audit, and strict same-endpoint issue cleanup.**

A focused formal-consistency pass makes the following narrow repairs. Core.2 now describes combinational observations as emitted R2C fixed-point units matched by component identity and local emitted-occurrence ordinal rather than by global cycle bucket. The Core.17 observable-unit carrier is clarified to include explicitly selected proof-internal RESET units without adding RESET to  $\Sigma$ . Appendix C and Core.IC replace the ambiguous back-to-back Issue=t+1 wording with a continued-request attribution rule that permits inserted latency, and the strict same-endpoint issue rule is limited to distinct blocking instances on one scalar A9-governed explicit boundary/direction; R4/E3 cite that stricter special case while retaining their general non-strict inequalities. Core.MemSyncOrder and M.7a bind the main-text array/synchronization commit-time rule to a separate local schedule/mapping-layer qualification check without adding memory events to  $\Sigma$  or extending the J.1 replay state; it is not an Appendix I or Theorem J.1 replay premise, while J.Mem remains the additional cross-process shared-memory value-flow condition for J.1. Core Commit/Issue/BoundaryReq/IC terminology and the mirrored Appendix C wording now use “explicit abstract boundary” for the operational endpoint object, without redefining proof-internal  $\Sigma$  or outer  $\Sigma_{\text{DUT}}$  trace visibility. K.DomResetScope D3 now covers identities consumed by K.2b.PO as well as K.2b.R/K.2b.P, matching the stronger uniform U1 scope. No additional Appendix F design rule is introduced.

**16 August 2026 — deadlock observation-boundary disclosure and uniform interruption-gate discharge.**

Two qualification/workflow clarifications are added without changing the existing safety theorem or H1 coverage architecture. First, the front summary and Appendix M.6 now state that Appendix K deadlock qualification uses every WFG-relevant explicit abstract message-passing boundary, while RTL micro-staging already absorbed into B(c) remains internal; the existing E4/B-faithfulness and KObs residual-offer obligations are unchanged. Second, Appendix K adds a normative uniform instance-level sufficient discharge for the candidate-relative reset/terminal interruption gates: one conservative scope satisfying U1-U3 proves all required K.ResetEpoch.Continue and nested K.TerminalDisposition.Continue branches at once. CF-S, K.PSF, Corollary K.2c, M.6, and M.7a accept that single record as an alternative to per-candidate gate enumeration; the original per-candidate schema remains the fallback and the theorem's universal quantification is not weakened.

**16 August 2026 — certified-singleton proof restoration and recursive predecessor qualification test.**

A small regression in Lemma K.CompareAdvanceLegal-Ord is repaired: the proof now covers both branches of its own singleton hypothesis, using Core.5a/Core.OrdFrontSingleton in the ordinary case and the explicit total-order/source-comparability certificate in the certified-singleton extension. The K.CVPred recursive Core.16 disposition is also tightened so pointwise reuse of K.CompareAdvanceLegal-Ord for a lower-ranked blocking OpFact is permitted only when every Policy-S-required source-earlier blocking transfer for that predecessor is already present in the matched liberal transfer prefix with the

required keyed identity/ordinal before materialization; the predecessor-materialization extension may not create any such prerequisite. K-O0/K-O4 and the M.7a K.CV/K.CompareAdvanceLegal audit entries are aligned. No other H1/H2/H3, QSync, FairPick, JointFreeze, or ordinary-instance burden rule is changed.

**16 August 2026 — recursive predecessor Core.16 disposition and ordinary-certificate simplification.**

Two residual Core.16 coverage details are tightened without changing the H1/H2/H3 architecture. Lemma K.JointFreeze-Ord now proves the full JF1/JF2 ordinary/singleton no-release result, including phase-legal work owned outside D via A6 point-to-point ownership plus closed-WFG component closure. The K.CVPred comparison-availability recursion now requires an explicit Core.16 reach-legality disposition for every recursively materialized non-transfer predecessor: qualifying ordinary/singleton blocking OpFact predecessors reuse K.CompareAdvanceLegal-Ord pointwise, pure observation/settling reaches are certified Core.16-inert, and other launch-carrying reaches supply equivalent owner/frontier evidence; “phase-legal” alone is insufficient. K-O0/K-O3/K-O4, CF-S, M.7a, and the mechanization-facing summaries are aligned. K.PSF and M.6 now state the instance-level burden reduction explicitly: an ordinary non-elaborated singleton-frontier instance needs neither a separate K.JointFreeze certificate nor a separate main K.CompareAdvanceLegal certificate, because both frontier facts are theorem-derived. No QSync/H3, FairPick/H2, or other Appendix-J interface is changed.

16 August 2026 — Core.16 launch/issue symmetry and ordinary-freeze lemma.

The serial-reduction interfaces are tightened in two small but load-bearing ways. New Lemma K.JointFreeze-Ord proves the ordinary/non-elaborated or otherwise certified singleton-frontier no-release case from Core.OrdFrontSingleton, E3/source-prefix closure, Core.16, component closure, and Core.Drain/K.Drain, leaving certificate Definition K.JointFreeze only for admitted StandardEnv multi-frontier components. The former K.CompareIssueLegal interface is superseded by Definition K.CompareAdvanceLegal and Lemma K.CompareAdvanceLegal-Ord, which establish the Core.16 comparison fact before K.2b.P0 and explicitly discharge both distinct Core.16(2) obligations: P0’s finite Core.ReachPrep/source-evaluation launch and K.2b.R’s later request Issue/assertion. Coverage ranges over every K.2b.P0/K.2b.R-consumed owner/key in K.DomResetScope, inside or outside D; explicit Sys\_B^elab multi-frontiers require checked elaboration/order evidence, and the JointFreeze/CompareAdvanceLegal consistency rejection rule is retained. K.2b.P0/R/E/Q, K.2b/K.PSF/CF-S, K-O2/K-O4, M.6/M.7a, Appendix S, and Appendix T are aligned. No change is made to the QSync/H3 certificate-existence architecture or the H1 provenance repair.

**16 August 2026 — K.CompareIssueLegal scope separation and outside-D repair (intermediate; superseded).**

The serial-reduction comparison interface is tightened to remove the remaining  $P \notin D$  Core.16 shortcut. Definition K.JointFreeze is returned to its no-release role (JF1/JF2 only). New Definition K.CompareIssueLegal separately covers Core.16 issue compatibility for every K.2b.R-consumed owner/key in K.DomResetScope, whether inside or outside D; Lemma K.CompareIssueLegal-Ord derives the ordinary/singleton case from the least-missing/first-overtake invariant, E3/source-order prefix closure, and matched earlier transfers, while explicit Sys\_B^elab multi-frontiers require checked elaboration/order evidence. The interface includes an explicit consistency rule rejecting any prefix/key simultaneously classified as JF1-suppressed and comparison-required. K.2b.R no longer claims that  $P \notin D$  escapes Core.16. K.2b/K.PSF/CF-S, K-O2/K-O4, M.6/M.7a, Appendix S, and Appendix T are aligned. The preceding JF3 revision entry is retained as historical record but is superseded by this cleaner separation; no QSync/H3 or H1 provenance interface is changed. This intermediate interface is superseded by the later K.CompareAdvanceLegal/K.JointFreeze-Ord revision above.

16 August 2026 — K.JointFreeze propagation and comparison-legality cleanup.

Definition K.JointFreeze is tightened in three ways: JF1 now states the full-frontier condition under which Core.16 remains active; notation is normalized to  $K.JointFreeze\_I(D, \sigma\_L, \{f\_P\})$ ; and new JF3 makes

the StandardEnv multi-frontier K.2b.R issue-legality fact an explicit paper-level certificate clause rather than content deferred to K-O2. K.2b.R now consumes JF3 directly. K.2b.Q Step 1 consumes the joint freeze already established by K.2b.E/K.EnvMF and no longer re-derives the obsolete unqualified source-later claim; K-O2 is explicitly only the mechanization obligation for JF1–JF3. Appendix M.6, the M.7a roll-up and K.EnvMF entry, the K-P mechanization obligations, and the mechanization-facing summary are aligned. No change is made to the QSync/H3 certificate-existence architecture or to the H1 provenance repair.

**16 August 2026 — residual H1/H2/H3 formal and audit tightening.**

The StandardEnv multi-frontier branch now consumes explicit Definition K.JointFreeze, whose JF1/JF2 obligations close the remaining K.2b.E freeze seam for incomparable  $\text{Sys\_B}^{\text{elab}}$  frontiers; ordinary/single-frontier cases continue to use Core.OrdFrontSingleton/Core.16/Core.Drain. Definition Core.QSync gains QS2a RTL-KCut origin recoverability, and Core.QSync-KCutCanonical now derives canonical-domain coverage from that structural cycle decomposition rather than assuming an origin for an arbitrary quiescent KCut. K.CVPred clarifies that cross-process predecessor materialization must preserve every frozen member of D. Core.FairPick is re-described as a stronger constructive evidence form for the existing behavioral fairness obligations, with proof-selectable issue latitude separated from instance-fixed scheduler/arbitration/environment choices. M.6/M.7/M.7a, the formal-appendix guide, Appendix N, and Appendices S/T are aligned; J.2b now states explicitly why its cycle-aligned LocalGWR induction does not by itself discharge arbitrary-prefix J.TraceCertCov.

**16 August 2026 — QSync canonical-cycle certificate-existence repair (H3).**

The former universal package-totality gap is closed at the paper-theorem level for the standard QSync normal-compositional scope. Definition Core.QSync now includes QS2a RTL-KCut origin recoverability, and Lemma Core.QSync-KCutCanonical uses that structural clause to identify every reachable standard-scope RTL KCut with the unique Core.QSync-Post endpoint of its recovered cycle K-origin; quiescence alone is no longer treated as supplying the origin. New Definition J.QSyncCertState packages the substantive per-cutpoint J/K correspondence data without assuming K.CertCov; Definition J.QSyncCertInd gives an anti-circular base-and-universal-one-cycle-step refinement obligation; and Theorem J.CertExist-QSync proves total J-side package existence by finite cycle induction. New Corollary K.CertCov-QSync binds those tuples through K.CertHdr and derives K.CertCov. Appendix K.1A-Total/K.1, A7, K.4e, M.6/M.7a, and Appendix T are aligned. A single observed run still cannot prove package totality, and a concrete flow must mechanize or independently check QSync plus the QSyncCertInd invariant (or use an alternate K.CertCov route). H1 and H2 repairs are unchanged.

**16 August 2026 — K.2b.E constructive fair-continuation repair (H2).**

The fair-extension step used by the Policy-S reduction is now constructive rather than schematic. Definition Core.FairPick packages the existing Core.IssueFair and B2/B3 discharges as prefix-local issue/boundary selectors with local finiteness, fixed-scheduler legality, EnvStep-compatible per-cycle selections, starvation-freedom, SyncFence-prerequisite, and B3-exposure evidence; it preserves the same behavioral Core.IssueFair/B2/B3 property but requires a stronger constructive evidence form than an abstract fairness assumption. Lemma Core.FairCycleDovetail proves phase legality with two applications of Core.CycleChoiceExt: first obtain a typed completion of the selected L1 segment through an actual finite L2 SettlePoint, then choose the fair L3 segment at that settled state and invoke C2 again for the final CycleResult. Lemma K.2b.E now states the Core.FairPick constructive discharge explicitly and proves an explicit frozen-prefix invariant, applies that construction cycle by cycle, and invokes dependent choice only over already-typed invariant-preserving successors; FP4/FP5 then establish Core.IssueFair and B2/B3, while B5 remains a separate closure premise. Lemma K.2b now names B2/B3/B5 and the constructive fairness discharge explicitly. K.PSF, CF-S, K-O7, Appendix M.7a, Appendix N, and the mechanization-facing Appendices S/T are aligned; the later residual repair also promotes the StandardEnv multi-frontier freeze condition to explicit K.JointFreeze rather than leaving it only to K-O2.

That H2 repair itself does not change the Appendix J finite-prefix theorem, H1 K.CV provenance repair, or user-facing scheduling rule; the separate H3 revision recorded immediately above adds the QSync package-totally theorem route.

**16 August 2026 — K.2b.P cross-policy provenance-availability repair (H1).**

Lemma K.2b.P now carries a joint least-missing-transfer/provenance invariant rather than inferring all K.CV predecessor occurrences from earlier endpoint transfers. Definition K.CVPred gains a normative serial-route comparison-availability closure: after the liberal comparison has matched the source-earlier blocking operations and serial-transfer facts required before a Policy-S endpoint may issue, every  $\text{Pred\_CV}(\text{OpFact}(\kappa))$  fact needed by K.2b.P0/K.2b.R must be available with the same keyed value after a finite phase-legal extension that preserves the frozen component/DomEpoch and introduces no unmatched dominance-domain endpoint transfer. Transfer predecessors are discharged by the earlier-transfer induction; lower-ranked non-transfer predecessors are discharged recursively by CVRank and their certified reach rules, with K.2b.P1 used only after occurrence. K.2b.P0, K.2b.R, CF-S, K-O0/K-O3, Appendix M.7a, and the mechanization-facing summaries are aligned. No Appendix J finite-prefix result, K.2b.E fairness argument, or user-facing scheduling rule is changed.

**15 August 2026 — minimal mechanization-driven semantic clarification pass.**

A narrow clarification pass incorporates only semantic distinctions exposed by pre-mechanization, without changing theorem architecture or introducing Lean-specific representation choices. Core.18 now defines cross-model KObs correspondence through the common typed record and leaves J.KObsConf as the downstream theorem establishing it. The historical-name lookup marks Core.15a retired in favor of Core.WFG. Appendix G states explicitly that function-like Issue/Commit notation abbreviates the Appendix-C relational existence/uniqueness facts. R2C now identifies the cycle-entry valuation and the intended unique-settled-result meaning of determinism while allowing internal  $\epsilon$ /stutter re-evaluation; it also states that Sys\_ref and RTL\_B need not share an internal combinational model or absolute cycle buckets. The cycle-locked R2C clause now makes per-side participation/completeness, implementation/certificate justification, and the Core.17/E2 single-clause relationship explicit. Core.16 clarifies that “launch” is intentionally broader than message Issue. Existing reactive-environment replay wording and the conditional certification status of K.Ser are unchanged.

**14 August 2026 — deliberate proof-audit repair: cross-policy keys, lowering transport, dependency stratification, and cycle-extension interface.**

A focused pre-mechanization audit produced eight repairs without changing the user-facing scheduling contract. (1) Definition K.OpKey now supplies a theorem-instance common dynamic-operation key universe, partial per-execution realization maps, and a key-level cross-policy source order; execution-local Core.5 universes remain unchanged. (2) New Lemma K.2b.P0 separates cross-policy occurrence/reachability from K.2b.P1’s conditional-on-occurrence value/identity determinacy, and K.2b.R now consumes P0 explicitly. (3) Definition K.A5L model-indexes the scheduler-robust premise; the Policy-S route proves the lowered Sys\_K form and new Lemma K.Low.A5 transports it to the Sys\_ref A5-L consumed by K.1A, with waiver pullback made explicit. (4) Definition J.DepJ now distinguishes semantic D1–D4 prerequisites from D5 Horizon proof stratification; J.DepSound is limited to semantic edges and new Lemma J.HorizonOrderSafe validates D5-only commutation. (5) Core.CycleClosure C2 now includes finite partial-cycle extension, exported by new Lemma Core.CycleChoiceExt and consumed by K.2b.E’s fair continuation construction. (6) K.2b.R’s Policy-S premise is corrected from commitment before q is “reached” to commitment before q’s request may issue. (7) J.RefProjCover/J.1-Safe consistently use finite canonical ideals/down-closed prefixes. (8) the Appendix K finite-drain note now cites Core.16(6)/K.Drain as the actual finiteness premise rather than appearing to derive finiteness from capacity alone. M.7a, K-P mechanization obligations, and Appendices S/T are aligned to these interfaces.

**14 August 2026 — Core.17  $\Sigma$ Match, persistent-deadlock scope, and QSync adequacy repairs.**

Definition Core.17 now includes the canonical  $\Sigma$ -event/value occurrence matching as an explicit  $\Sigma$ Match conjunct of the normative observable-compatibility relation, and Appendix G is identified as an expansion rather than a redefinition of that convention. Appendix K's persistent-deadlock implication note is narrowed to the closed internal wait-cycle class actually captured by Dead\_K and explicitly excludes starvation/perpetual under-supply cases such as K.CE-5/H2 from the broader "never hangs permanently" reading. Definition Core.QSync QS3 now includes a transition-relation-to-settling-network realization/adequacy condition, and the QSync-Ref proof sketch and Appendix M.7a evidence checklist are aligned to that condition. No M.7a premise-entry expansion is made because the current revision already contains B1, B2/B3, B4/FPD, B5/B6-Terminal, A6, A8/A9/A10, K.BF, and Core.IssueFair audit entries.

**14 August 2026 — K.CV audit-trigger and provenance-closure cleanup.**

Appendix M.7a now separates the unconditional serial-route K.CVPred/K.CV-WF trigger from the additional design-level checks for cross-process observations beyond blocking Pop transfer values, so a pure blocking Push/Pop Policy-S route may not mark the P1 provenance obligation inapplicable. Definition K.CVPred now states predecessor occurrence closure explicitly and K-O0 checks it. Lemma K.2b.R now attributes dynamic identity, payload, and typed boundary/direction to K.2b.P1/K.CV while deriving reset-epoch agreement from K.DomResetScope and fixed DomEpoch. No Appendix J finite-prefix or serial-dominance architecture is otherwise changed.

**14 August 2026 — K.2b.P1 well-founded value/identity provenance repair (H2 closed).**

Lemma K.2b.P1 no longer invokes an undefined "causal order of commits." New Definition K.CVPred/K.CV-WF gives the K.CV serial fragment a common semantic fact-key/provenance relation with deterministic per-fact evaluators and an explicit WellFounded certificate or checked rank. P1 now proves the strengthened all-facts determinacy claim by well-founded induction on  $\prec_{CV}$ . Same-cycle rendezvous enablement and the broader Appendix-G causal-dependency graph  $\rightarrow$  are explicitly not predecessor edges merely by causal or temporal proximity, so legal cycles in  $\rightarrow$  do not undermine the proof. K.BF, K-O0 and the K-P mechanization obligations, CF-S, K.PSF, M.7a K.CV, Appendix R, Appendix S, and Appendix T are aligned. No Appendix J construction order is imported into K.2b.P1, and finite-prefix Post  $\rightarrow$  Ref is unchanged.

**14 August 2026 — M.6 reset-scope disclosure and DomResetScope certificate typing cleanup.**

M.6 now states the actual K.ResetEpoch applicability rule: Continue excludes every RESET whose RW scope intersects the candidate's K.DomResetScope, RW5 permits disjoint resets, and an intersecting RESET that does not satisfy the independently mapped ResetEpoch.DirectFail conditions lies outside the serial route and requires another A5-L discharge. Definition K.DomResetScope is retyped as a certificate-supplied finite process/boundary scope with explicit D1–D4 coverage, endpoint-closure, and anti-circularity conditions; larger conservative scopes are permitted and minimality is not required. K-O4, CF-S, M.7a, and Appendix T are aligned. No finite-prefix Appendix J result is changed.

**14 August 2026 — ReachPrep coverage-direction and rendezvous-example clarification.**

Core.ReachPrep now states the Post  $\rightarrow$  Ref coverage direction explicitly: every covered RTL\_B prefix must have selected Sys\_ref ReachPrep semantics permissive enough for the dependency-independent preparation/reach segments required by its accepted J.LocalGWR or J.GWR construction, while more-permissive source preparation remains conservative. The Appendix K blocking-rendezvous A5-L example is named as a straight-line ReachPrep-dependent example, states that its Push operands are independent of the preceding Pop results, and makes the eager Policy-L alternative conditional on the selected instance admitting that ReachPrep case. Appendix M.7a carries the same coverage and example-dependency obligations.

**14 August 2026 — K.BF gate exhaustiveness and dominance-reset-scope cleanup.**

K.InterruptionGate now states explicitly that, for the K.BF serial-reduction fragment, Core.IC and Lemma K.BF-Persist make Reset and Terminal the complete non-commit interruption-gate instantiations; any

third gate is a theorem-model extension requiring re-proof. New Definition K.DomResetScope binds the processes and explicit boundaries whose dynamic transfer identities can participate in K.2b.R/P, with DomEpoch denoting their reset-epoch vector. K.ResetEpoch.Continue now excludes RESETs intersecting that dominance scope while RW5-framed disjoint resets remain permitted. K.2b.R and K.2b.P are correspondingly scoped to the fixed DomEpoch, eliminating the former use of K.ResetEpoch.Continue to forbid resets of  $P \notin D$  that the old gate definition did not cover. K-O3/K-O4/K-O7, K.CertHdr, CF-S, K.PSF, M.7a, Appendices R/S/T, and Corollary K.2c are aligned. Core.CycleState is also added to the Appendix R and Appendix S Core-interface inventories. No finite-prefix Appendix J result or step-0–2 semantic interface is changed.

**14 August 2026 — shared nested interruption-gate schema (step 3 complete).**

Definition K.InterruptionGate now factors the common Policy-S comparison-interruption machinery into a prefix-local Continue branch and an independently mapped DirectFail branch with a single anti-circularity rule. K.ResetEpoch and K.TerminalDisposition are retained as gate-specific instantiations rather than parallel ad hoc definitions: reset is evaluated first; only K.ResetEpoch.Continue permits evaluation of the terminal gate; and only the two nested Continue branches enter K.2b.E/R/P/Q. ResetEpoch.DirectFail remains the matched-RESET/RW2 cancelled-pending route, while TerminalDisposition.DirectFail remains the DeclaredTerminal maximal-pending route. Lemma K.2b/K.2c, K-O4/K-O6/K-O7, K.CertHdr, CF-S, K.PSF, Theorem K.1, Appendix M.7a, Appendix R, Appendix S, and Appendix T are aligned. Steps 0–2 are unchanged, and no flat interruption enum or new third cancellation gate is introduced.

**14 August 2026 — full-state CycleResult and cycle-closure consolidation (step 2 complete).**

Definition Core.CycleResult is introduced as the primary typed real-cycle object. Its nextState is the full theorem-instance state, including registered/internal machine state, time-aware environment state, global cycle index, and cycle-valued auxiliary/monitor state; committedUnits records selected ordinary complete observable units /  $\Sigma$  commits and resetUnits records each dynamic scoped RESET occurrence. Core.CompleteCycle is retained as the nextState projection of CycleResult for compatibility with existing theorem statements, and Core.SilentCycle is the no-selected- $\Sigma$ -commit case; a reset-only cycle may therefore be silent at the selected outer alphabet while still carrying the proof-internal RESET occurrence. DeclaredTerminal remains a predicate on states rather than a CycleResult outcome. RESET is explicitly an emitted in-cycle unit, not a sum-type branch: RW4/RW5 and Lemma J.RS are aligned so partial resets preserve unaffected-scope continuation and the J induction remains indexed by the RESET unit itself. The former standalone environment-totality and source-cycle-closure premises are folded into the normative Core.CycleClosure record together with full-state step typing, source-cycle existence/finiteness, environment-step definedness, auxiliary/monitor advancement, and reset-emission fidelity. K.2b.E/K.2b, K-O7, M.7a, K.CertHdr/CF-S/K.CompleteS, Appendix R, Appendix S, and Appendix T are aligned. Step 3 interruption-gate unification is not included.

**14 August 2026 — finite-prefix Reach interface and generic Exec/J.FE removal (step 1 complete).**

Appendices I/J now use Reach\_post/Reach\_ref as the sole generic finite-prefix execution domain. The restricted Ref  $\rightarrow$  Post clause and Definition J.RTLConsistentRefPrefix are retyped over annotated reachable prefixes; because the supplied  $\tau_{\text{post}}^*$  is already a reachable RTL witness, the reverse proof still takes  $\tau_{\text{post}} := \tau_{\text{post}}^*$  without any fair-extension argument. The generic Exec\_post/Exec\_ref definitions and Lemma J.FE are removed. Core.PostCycleTotal is deleted after the dependency audit found no remaining semantic consumer; Core.CycleClosure is narrowed to the source complete-cycle/environment closure used by Core.LAdm/K.2b.E. Appendix I.5 is explicitly local/finite and makes no global infinite-execution existence claim. Appendix G, M.7a, Appendix R, Appendix T, B6 explanatory text, Core.IssueFair/B2-B3 scope, and the J.1 proof prose are aligned. Complete/fair execution obligations remain only where later liveness results explicitly require them, including

Exec\_L^adm/K.2b.E and K.CompleteS. No CycleResult or interruption-gate unification refactor is included in this revision.

**14 August 2026 — blocking-request lifecycle closure and frozen-frontier interruption-completeness repair (step 0 complete).**

Core.IC now states an exhaustive ordinary blocking-request lifecycle for the G–K theorem model: after issue, an outstanding blocking dynamic instance remains asserted and stably attributed until its process-facing commit or an affecting RW2 RESET; no other nonterminal transition may deassert, retire, reattribute, cancel, or replace it, and DeclaredTerminal may instead end the execution with the request still pending. Core.SilentCycle, the Appendix C/G/K/N no-deassert-before-commit statements, and Core.IssueFair are aligned to that reset-epoch-scoped rule. Definition K.BF and Lemma K.BF-Persist now consume the Core lifecycle rather than introducing a separate cancellation exception; K.BF-Persist therefore closes the frozen-frontier interruption enumeration used by Lemma K.2b.E/K.2b.R. Definition K.RespFail retains cancelled-pending only for an affecting RW2 RESET; terminating abort/end-of-test behavior is represented by DeclaredTerminal and uses maximal-pending. K.TerminalDisposition clause (2), K.2b.Q Step 3, the K.2b proof, K-O4/K-O6/K-O7, CF-S, K.PSF, Appendix M.7a, Appendix R, and Appendix T are aligned. The K.2b.R heading-format regression is repaired. Protocols with independent nonterminal cancellation/withdrawal of outstanding blocking requests require a separately defined operational extension and are outside the present G–K theorem stack. No J.FE/CycleResult/gate-unification refactor is included in this revision.

**14 August 2026 — K.ResetEpoch prefix-local applicability repair.**

K.ResetEpoch clause (1) is revised from a universal property over already-frozen complete extensions to a non-vacuous prefix/state-local reset-contract condition: at every finite admissible comparison prefix while all selected frozen requests remain pending in the same epoch, no affecting RESET is permitted as the next transition, so RW2 cannot clear those requests before the gate applies. K.TerminalDisposition clause (1) is put in the parallel prefix/state-local form by requiring every such frozen-prefix final state to be non-DeclaredTerminal. Lemma K.2b.R, the proof of K.2b.E, K.2b.Q Step 1, K-O4/K-O7, and the Appendix M.7a audit entries are aligned. K.2b.E continues to construct the infinite fair continuation from machine-closure, drain, fairness, and issue-fairness premises; neither applicability gate assumes that continuation exists. All previously recorded H1, terminal-branch, watchdog-scope, and finite-prefix Post → Ref conclusions are unchanged.

**14 August 2026 — K.2b declared-terminal applicability, termination-contract scope, and audit-consistency repair.**

Definition K.TerminalDisposition now separates the reset-free serial route into a terminal-free dominance branch and an independently certified Policy-S terminal/cancellation failure branch, parallel to K.ResetEpoch. Lemma K.2b.E/K.2b.R/K.2b.P/K.2b.Q and their Appendix K-P proofs use only the infinite-fair continuation on the terminal-free branch; a declared terminal no longer substitutes for old finite-maximal reasoning. The direct terminal branch must independently bind the liberal frontier to a non-waived Policy-S response monitor and prove the actual Policy-S terminal/cancellation disposition while that monitor remains pending, avoiding circular dependence on K.2b.P/Q. The clause-(1) branches of both K.ResetEpoch and K.TerminalDisposition are stated as contract-level applicability properties rather than referring to a continuation constructed by K.2b.E; the later prefix-local repair below makes their mechanizable finite-prefix form precise. DeclaredTerminal scope is clarified: a modeled cycle-count/watchdog/end-of-test disposition inside the selected machine/test semantics counts, whereas an external harness watchdog that merely stops host simulation does not truncate Core.CompleteCycle or defeat the terminal-free gate. CF-S and M.7a identify clause (1) as the normal prospective clean-run applicability evidence and clause (2) as an independently certified direct-failure branch that may be conditional or instantiated but is not required as an observed witness on every clean run. The remaining Policy-S 'maximal finite' paraphrases are aligned to the normative declared-maximal-terminal wording.

Corollary K.2c, K.PSF, Theorem K.1, CF-S, K-O3–K-O7, Appendix M.6/M.7a, Appendix R, and Appendix T remain aligned. The legacy Core.PolicyL returned-result sentence is synchronized with Core.ReachPrep, the stale J.FE environment-receptiveness name is removed, Core.CycleClosure receives an explicit Core definition, and M.7a includes B1, B4/FPD, A6, A8/A9/A10, K.BF, and K.TerminalDisposition audit entries. Finite-prefix Post  $\rightarrow$  Ref remains unchanged.

#### **14 August 2026 — fair-extension, complete-cycle, and source-issue formalization repair.**

The machine-closure path has been revised to separate same-cycle settling, ordinary silent clock cycles, and terminal B6 stutter. Definition Core.ReachPrep now makes the intended Policy-L eager-reach semantics explicit, including dependency-independent ordinary straight-line successors used by K.CE/K.CE-2; Definition Core.IssueFair gives the previously unnamed source-issue weak-fairness predicate; and new Core.CompleteCycle/Core.SilentCycle, EnvCycleTotal/EnvTerminal, SourceCycleClosure, and PostCycleTotal definitions type cross-cycle progression. Lemma J.FE is rewritten as a phase-structured source construction plus a distinct RTL totality/progress discharge, so a presently idle state is no longer inferred to be a B5 fixed point and B6 is reserved for Core.TerminalIdle, which requires both machine-side and environment-side permanent no-future-enabling evidence. Core.LAdm, K.2b.E, K-O1/K-O3/K-O6/K-O7, K.Resp maximality/aging, Appendix N, and Appendix M.7a are aligned. The generic J.FE route is explicitly separated from Appendix K's stronger Exec\_L<sup>adm</sup> continuation route. These changes do not alter the finite-prefix Post  $\rightarrow$  Ref safety construction.

#### **13 August 2026 — Appendix Q compositional source-QSync repair.**

Appendix Q.1(i) now distinguishes reusable elaboration-template settling facts from the whole-source Core.QSync discharge. Template-side finite/deterministic/acyclic settling of staged/bundle/join/guard/coupler/epoch-gate logic no longer implies whole-source QS2–QS5 by itself. For Sys\_B<sup>elab</sup>(E), the concrete theorem instance must also supply the design-owned Core.RegSeq/direct-input/R2C/non-template-boundary settling facts and a checked template/design settling-composition condition covering every cross-partition dependency; QS4 is applied to the complete combined same-cycle settling graph. Appendix M.7a now names this composition obligation and lists a combined dependency-rank/topological-order certificate as a sufficient mechanically checkable discharge. The Core.QSync-Ref proof note and Sys\_B<sup>elab</sup> audit entry are aligned accordingly. This is a qualification/discharge-interface correction only: Definition Core.QSync, QS1–QS5, Theorems Core.QSync-Ref/Core.QSync-Post, J.1 scope, and H3 are unchanged.

#### **13 August 2026 — J.TraceCertCov source-settling propagation.**

Definition J.TraceCertCov and its Appendix M.7a discharge entry now state the route-sensitive source-settling evidence explicitly rather than leaving it only inside “remaining Post  $\rightarrow$  Ref premises”: the normal J.LocalGWR/J.GWR-Comp route carries QualifiedSynchronousModel\_Sys\_ref(I)/Core.QSync-Ref, while a direct J.GWR package may instead carry checked finite L2-settled segments for its represented Horizons. Corollary J.1-Safe and the abbreviated front-matter safety-signoff prose now point to that same route-sensitive requirement. This is a propagation/qualification clarification only; it adds no unconditional post-side Core.QSync premise to plain finite-prefix J.1 and does not change H3.

#### **13 August 2026 — qualified synchronous hardware interpretation.**

An informative hardware-interpretation paragraph now precedes Definition Core.QSync. It states the intended single-clock cycle-level model directly: registered state and cycle inputs are fixed, sequential process outputs are derived from registered/fixed operands without same-cycle interface-result through-paths, the complete same-cycle combinational/handshake network settles deterministically with pure combinational feedback outside the standard scope, and boundary actions precede the next registered update. It also clarifies that SettlePoint, BoundaryEvalPoint, and Core.KCut are proof views of the ordinary settled cycle snapshot rather than additional hardware states, while preserving explicit latency-sensitive/WC and Policy-L nondeterminism.

#### **13 August 2026 — Stage A terminology, L2 frame repair, and qualified synchronous-settling Stage B.**

Stage A adds an explicit determinism taxonomy and separates  $\delta$ -settling, real silent  $\tau$ -cycles, and B6 idle ticks without redefining the general  $\varepsilon$  relation. Definition Core.Settle now states the normative L2 frame: cycle index, registered process/FSM/pipeline and explicit-channel storage/occupancy/credit state, and selected cycle-t exogenous/environment values are frozen during  $\rightarrow \varepsilon, L2$ ; no  $\Sigma$  commit or next-cycle registered update occurs. SettlePoint existence is no longer implicit in the definition. Stage B adds Definition Core.QSync and Theorems Core.QSync-Ref/Core.QSync-Post. The standard K-facing/cutpoint scope now requires finite acyclic deterministic same-cycle settling over explicit state boundaries; QSync-Ref proves source/ Sys\_B^elab SettlePoint/KCut existence and uniqueness, and QSync-Post derives Core.KCutClosure\_post plus a canonical K-facing NormPost endpoint. Appendix Q adds the matching elaboration obligation; Appendix M.7a adds a first-class Core.QSync audit entry and treats Core.KCutClosure\_post as theorem-derived in the standard scope; K.CertHdr records the structural evidence, theorem provenance, and any trusted-base delta when a settling claim is accepted without independent artifact checking. The normal compositional finite-prefix J.1 route now explicitly consumes source-side QSync-Ref settling existence through J.RegPhaseNF/J.UnitExt, while a directly validated J.GWR package may carry equivalent per-Horizon settling evidence; plain J.1 does not thereby acquire post-side QSync unless a post KCut/NormPost interface is used. This revision intentionally narrows the standard Core.KCutClosure\_post discharge scope: the former independent inductive-invariant and checked-normalization alternatives are withdrawn as standard-scope routes and require an explicitly qualified non-QSync theorem extension. QS1 now states the single-clock semantic condition directly, with M.7a ClockScope serving as its evidence entry. Appendix I/J replay normalization remains separate. No change is made to Appendix H.3 in this revision.

### **13 August 2026 — Lean/scheduling alignment clarifications.**

The post-side interpretation of Core.SyncFence/E5 now states explicitly that same-cycle predecessor/synchronization accounting is a trace/certificate attribution convention and does not impose an observable physical sub-cycle ordering within one RTL clock edge. Appendix M.7a adds named ClockScope and Core.SyncFence discharge entries, and K.CertPkg records their references where applicable. Theorem J.1's restricted Ref  $\rightarrow$  Post clause now exhibits its existential witness explicitly as  $\tau\_post := \tau\_post^*$ . Lemma K.2b and K.2b.E now state that drain-closedness is obtained from the Dead\_K / DeadKRec^W hypothesis itself rather than imposed as an additional narrowing of A5-L's source-KCut domain. Definition L.1 remains intentionally stated over generic  $\varepsilon$ -quiescent source witness-choice cutpoints rather than K-facing Core.KCut points.

### **12 August 2026 — synchronization-completion fence formalization (H4).**

The previously prose-only rule that a synchronization call cannot complete while an earlier blocking Push/Pop in its source interval remains uncommitted is promoted into the formal contract. New Core.SyncFence gates source synchronization commits, including same-cycle message/synchronization completion and Sys\_B^elab process-facing commit semantics; Core.Phase L3 and the operational transition relation consume the fence. R3 and E5 now contain explicit blocking-commit-before-sync clauses rather than relying on a parenthetical appeal to blocking-call control flow. Appendix C, Core.14, Core.18e, the drain-closed definition, the compact E1–E5 table, the system-level E5 proof, and Lemma J.KObs-Offers are aligned. This makes the interval scoping of Pending\_P/Front\_P/drain candidates lossless and closes the residual-offer versus PendingRec reconstruction seam without changing the intended deadlock predicate.

### **12 August 2026 — L2 settle / global $\varepsilon$ -quiescence, KCut closure, and K-cutpoint-domain correction.**

Core.4 defines model-specific global  $\varepsilon$ -quiescence; Core.Settle defines the complete L2 phase-restricted  $\varepsilon$  relation and BoundaryEvalPoint; Remark Core.1a classifies phase advancement as meta-level rather than  $\varepsilon/\Sigma$  activity; and Lemma Core.SettleIsKCut proves that every phase-bearing source L2 SettlePoint is a source KCut. Core.Phase L2/Core.16 use SettlePoint for activation, while WFG, KObs, Dead\_K, Corollary J.1, and K.1a use the Core.KCut comparison domain. Core.SettleKBridge remains the named activation-

to-later-drain-closed-KCut obligation and is consumed by K.DrainReach/Core.LAdm. RTL\_B retains its own Quiescent/NormPost/KCut\_post semantics without importing L0–L4, while Core.KCutClosure\_post plus Core.KCutObserve supply the non-vacuity/observability obligation ensuring that every RTL K-facing cycle-evaluation origin reaches a finite KCut and that persistent relevant blockers cannot remain forever outside that domain. K.CertCov remains certificate totality over KCutReach\_post and is paired with KCutClosure\_post for universal conclusions. Stale Core-table, Appendix I.2, K $\epsilon$ , and K.Low.C cutpoint terminology is aligned; Appendix T, M.7a, the front guarantee summary, and the certificate header are updated accordingly. B5 remains the stronger global fixed-point condition, not a synonym for Core.4 quiescence.

**1 August 2026 — logical replay-checkpoint clarification and B4/B5 unit correction.**

Definition I.ReplayObs now denotes the logical replay checkpoint represented at an operational state, not the raw post-preparation cursor/store. Replay-inert L1 preparation is separated from witness-fixed epsilon transitions that may update replay fields; normalized I.ReplayObs-Congruence and I.DR now use paired congruence of exact selected normalization paths rather than per-path neutrality or global confluence. I.CutpointOfferObs, I.DR', and Appendix T are aligned. B4/FPD use the per-process epsilon-step bound D\_P; B5 uses a separate cycle bound C\_sys, optionally derived through per-process C\_P charging lemmas. Lemma I.1 and A2 likewise keep the units distinct. Appendix Q records the equivalence of channel-first A\_E(c,  $\sigma$ ) and state-first A\_E( $\sigma$ , c) notation.

**31 July 2026 — replay/offer separation, R2C cycle-lock completeness, and B3 conclusion clarification.**

Appendix I now separates the replay-derived process slice I.ReplayObs from the issue-schedule-dependent current blocking-offer slice I.CutpointOfferObs. Lemmas I.ReplayObs-Congruence, I.DR, and I.DR' no longer infer Pending\_P, Front\_P, or BoundaryReq from committed observable-unit history; Appendix J continues to construct and audit those facts through O, J.OfferReach, J.OfferConf, and J.KObs-Offers. Core.18d, Appendix R, and Appendix T are aligned to the split. R2C now defines ComponentParticipatesInCycle and requires one emitted fixed-point unit for every participating cycle of a cycle-locked component, while forbidding units outside the participation domain. B3 now states explicitly that its buffered P\_push/P\_pop clauses conclude the complementary discharge; E4 plus B2 govern eventual commit of any original request that remains pending and continuously enabled afterward.

**31 July 2026 — ordinary-source frontier-uniqueness lemma.**

Lemma Core.OrdFrontSingleton now derives the at-most-one-member property of Front\_P for ordinary non-elaborated source intervals from the dynamic-program-order convention of Clarification Core.5a and the one-call/one-frontier-item ordinary-source mapping. Appendix B, Definition Core.15, and Clarification Core.WFGb now cite the lemma explicitly.

**26 July 2026 — reset-branch mapping and multi-frontier legality tightening.**

K.ResetEpoch clause (2) now supplies an independent, certificate-bound mapping from a selected liberal component frontier to a process-facing Policy-S serial frontier and binds the exact pending monitor and RW2 clearing event, rather than relying on the reset-free K.2b.Q mapping. K.2b.R now states Core.16 issue legality against the full frozen frontier set and tracks the StandardEnv multi-frontier case through K-O2/K-O4. The Formal Guarantees at a Glance section, the dynamic-reset main-body guidance, Appendix M.6/M.7a, CF-S, and Appendix T now disclose and name the dynamic-reset applicability condition consistently.

**26 July 2026 — serial-reduction seam and front-matter scope clarification.**

K.2b.R now proves explicitly that Core.16 does not block the reached serial-prefix request, removes the redundant independent-work disjunct, and scopes request persistence to one reset epoch. K.2b.P now states its elaborated-boundary coverage, explains why the same-cycle tie-break is harmless under begin-of-cycle non-fall-through E4 semantics, and includes an explicit composition note with K.2b.E. Definition K.ResetEpoch separates the reset-free dominance branch from an already-triggered matched-reset

cancelled-pending branch; pre-reset cutpoints that satisfy neither branch are explicitly outside the Policy-S reduction. K.4b, K-P, CF-S, K.PSF, Theorem K.1, Appendix M.7a, and Appendix T are aligned. The four-item Formal Guarantees at a Glance section now restores a compact scope exclusion, instance-qualification warning, and direct pointer to Appendix M.7 without reintroducing proof-interface labels or the detailed table.

**26 July 2026 — K.2b proof completion and user-facing responsibility/status clarification.**

Appendix K-P now contains explicit proofs of Lemma K.2b.R serial-prefix request reachability and Lemma K.2b.P deterministic serial progress dominance, including the least-missing-transfer argument and separate buffered-Pop, buffered-Push, and rendezvous cases. K-O3 now tracks K.2b.P mechanization and K-O4 tracks K.2b.R; K.4b and Appendix T are aligned. The Formal Guarantees at a Glance section has been reduced to four user-level statements and proof-interface labels have been removed from that section. The detailed responsibility/current-status table has been moved to Appendix M.7. Appendix M.6 continues to distinguish the design user's complete instrumented Policy-S execution from the full theorem discharge and to identify valid universal-coverage routes without claiming that a generic construction already exists.

**25 July 2026 — WFG, elaboration-map, and drain-interface completion.**

Definition Core.WFG now restores the edge-versus-blocked-process and explicit multi-frontier out-degree clarifications from the former Appendix K definition, while continuing to cover mixed buffered/rendezvous graphs. Definition Core.ElabProj promotes the four-role  $\Pi_{\text{elab}}$  interface—trace projection, occupancy projection, conservation/accounting, and proof-internal WFG visibility—into the Core; Appendix Q now normatively instantiates that interface. Definition Core.Drain removes the remaining Core-to-K definitional reference, and K.Drain is stated as the application evidence obligation that discharges it. Core.6, Core.7, Core.15, Clarification Core.5a, and Core.PolicyS now have navigable formal headings; Remark K.0-pre restores the derived-bridge framing. Definition Core.PolicyL now gives the liberal issue policy a citable heading parallel to Core.PolicyS. The K.Drain and Appendix Q normative run-in labels are restored to bold, and the A9–A11 assumption paragraph has normalized run boundaries so that only the assumption labels are emphasized.

**25 July 2026 — formal-core split attribution and side-condition organization cleanup.**

Remark Core.1a and Appendix T now attribute Policy-L step legality, cycle-phase semantics, and admissibility to Core.LLegal, Core.Phase, and Core.LAdm respectively. J.Mem and RESET/RW1–RW5 now appear under a dedicated Core subsection, and the RESET label is restored as a bold run-in. The normative wait-for-graph definition is renumbered Core.15a to preserve ascending definition order at its logical location after Core.15, and Definition Core.17 now refers to the Core rather than to an appendix. The active historical-name lookup maps old R.20 to Core.15a; older preserved revision text is otherwise unchanged.

**25 July 2026 — formal-core dependency audit follow-up.**

The follow-up audit removes the remaining  $J \rightarrow K$  proof citation by limiting Corollary J.1 to equal-KObs and J-owned reconstruction consequences; Appendix K now performs Dead\_K and waiver/diagnostic factoring. Appendix K imports Definition Core.15a for WFG instead of restating a second normative-looking graph definition. The Core KObs layer now exports only Core-owned reconstruction functions; Definition K.EnvTerminalAdequate and Definition K.KObsDerived, together with K.StimX and K.WFG+, now reside in Appendix K before the factoring lemmas. The KObs architecture explanation has moved to downstream Appendix R, the side-condition discharge record to Appendix M.7a, and Lemma J.RS to Appendix J. Core section headings are now descriptive rather than reusing definition numbers, the WFG definition fills the former Core.15a gap, the reading paths follow document/dependency order, and the Core scope text distinguishes Appendix G's raw Sys presentation from the Sys\_ref comparison used by Appendices I–K.

Historical-name lookup for preserved earlier revision entries: unlabelled Policy L → Core.PolicyL; R.RegSeq → Core.RegSeq; R.LAdm → Core.LLegal/Core.Phase/Core.LAdm; R.16 → Core.16/Core.Drain; R.18e → Core.18e; R.20 and the interim Core.15a label → Core.WFG; former Definition R.1a → Core.ElabProj; R.21b–R.21e → J.KObs-Proc/J.KObs-Chan/J.KObs-Offers/J.KObs-WFG; and R.21f–R.21g → K.KObs-Dead/K.KObs-Ext. Core.15a is a retired/interim label: occurrences in preserved revision-history text describe the name in force at that time and shall not be used as current normative cross-references; the current primary name is Core.WFG.

#### **25 July 2026 — normative formal-core dependency reorder.**

The former “Common Formal Definitions for Appendices I–K” has been promoted and moved immediately before Appendix G as the “Normative Formal Core for Appendices G–K.” The operational Sys\_ref/Policy L/Policy S material, Core.RegSeq, Core.LAdm, the exact Core.16 drain discipline, the shared frontier/request definitions including Core.7a BlockingMsgNextFront, the normative Core.15a WFG definition, and the Core.18c KObs datatype/reconstruction functions now occur before first use. Appendix J now contains the source-side KObs reconstruction lemmas J.KObs-Proc/J.KObs-Chan/J.KObs-Offers/J.KObs-WFG; Appendix K contains K.KObs-Dead and K.KObs-Ext. Appendix R has been retained as a downstream compact/mechanization companion with an explicit deletion-safety rule and no normative upstream dependencies. Active cross-references have been redirected from the former late R-numbered definitions and lemmas to their new Core/J/K locations; historical revision entries are preserved unchanged. The dependency audit now has no Core/J forward reference to Appendix K for WFG, and Appendix R remains deletion-safe.

#### **24 July 2026 — Policy-S user workflow and status clarification.**

The Abstract, Introduction, Formal Guarantees at a Glance, Guide to the Formal Appendices, and Appendix M.6 now present the deterministic Policy-S route in user-workflow order: one selected, complete instrumented pre-HLS execution is the primary design-user activity, while the flow separately establishes SDet, applicability, completeness, total finite-bound response coverage, and certificate validity. The text now distinguishes the operational practicality of simulation monitors from the outstanding mechanization and qualification of the K-P reduction and certificate interfaces, and it states separately that K.CertCov is required for a universal reachable-RTL-cutpoint conclusion. The underlying theorem premises and conclusions are unchanged.

#### **24 July 2026 — formal cross-reference, deadlock naming, snooping, and K.2b.Q cleanup.**

The stale clause-based Corollary J.1 signal-grouping reference has been redirected to the J.RegPhaseNF/J.UnitExt unit treatment. Appendix K now names the ordinary deadlock predicate Definition K.Dead and uses Dead\_K/Dead\_K^W consistently instead of overloading “K.1” as a predicate or component name; the phantom Definition K.Dead citation is thereby made valid. The J.Mem extension note no longer reuses invariant clause number I4. Appendix G now states snooping alignment conditionally on existence of a WC/L.Dir witness and points failed realization to Appendix L triage. K.2b.Q Step 1 now limits its source-prefix commitment claim to the process-facing blocking Sys\_K/K.BF serial prefix and explicitly excludes failed non-blocking polls. Appendix C now contains the normative ReturnedAtCycle definition directly rather than a dangling main-body cross-reference, and Definition R.18e cites both K.Dead and K.DeadW for the ordinary and waiver-parameterized reconstructions.

#### **24 July 2026 — main-body formal-detail cleanup.**

The pre-appendix scheduling discussion has been simplified without changing the formal contract. The Basic Conceptual Model now explains overlapping issue without proof-internal multi-frontier or elaboration terminology; the R1/E1 and R3/E5 rationale has been removed from the pipelined-loop section; and the synchronization-boundary subsection now gives only the operational drain/next-interval rule with an Appendix B reference. ReturnedAtCycle and the detailed registered sequential-interface semantics have been removed from the main body and replaced by one plain-language registered-cycle rule, with the normative ReturnedAtCycle definition relocated to Appendix C and the

dependent-use rule retained in R.RegSeq. The two formal-architecture bullets at the end of the main Summary have been removed; their unique Policy-S coverage content and a concise statement of proof-layer separation are consolidated in Appendix M.

**24 July 2026 — introductory restructuring and formal-results navigation revision.**

The former theorem-by-theorem opening discussion has been replaced by a short front abstract, a practical Introduction, and a separate Formal Guarantees at a Glance section. Broad claims concerning RTL verification cost, customer experience, implementation coverage, and source-level signoff have been qualified to avoid implying unconditional tool or RTL certification. A new unlettered Guide to the Formal Appendices has been inserted immediately before Appendix G without changing any existing appendix identifier. Appendix M has been rewritten as Formal Results Overview, preserving the precise Post  $\rightarrow$  Ref safety direction, restricted Ref  $\rightarrow$  Post use, local-to-global J.GWR-Comp architecture, KObs/K.RLAdmReach cutpoint bridge, certified-cutpoint versus total-coverage distinction, Policy-S/A5-S serial-reduction route, applicability limits, and implementation-evidence obligations.

**18 July 2026 — M1 safety-transfer direction and common-case pre-HLS signoff clarification.**

The document now states explicitly that ordinary source-side safety transfer uses Theorem J.1 Post  $\rightarrow$  Ref: RTL\_B introduces no new canonical  $\Sigma_{\text{DUT}}$  safety behavior absent from Sys\_ref. Definition J.RefProjCover names the optional observable-prefix coverage fact under which one complete pre-HLS execution represents all covered source-reference observations, and Corollary J.1-Safe packages both the universal-source and one-run transfer cases. The Appendix G “No Surprises” discussion and Appendix L Note 1 cross-reference this corollary. The restricted Ref  $\rightarrow$  Post clause remains an annotation-dependent selected-behavior realization clause and is not presented as the common safety-signoff mechanism. The former “source-side pass only through the restricted reverse direction” wording in the Practical Implication section is replaced by this safety-versus-realization distinction.

**18 July 2026 — H4 eager-issue completeness boundary and H5 environment/frontier scope correction.**

H4 is made explicit: because Policy-S-shaped prefixes are Policy-L-admissible, the two-channel crossing example contains an admissible two-Pop deadlock prefix and therefore falsifies A5-L for the unchanged design. The misleading suggestion that another A5-L proof route could rescue that instance is removed; alternate proof routes remain available only when A5-L is true. H5 is closed by the named applicability condition K.EnvMF. Certified multi-frontier components are supported by K.2b.E/K.2b/K.2c only under StandardEnv/B1-avail; timed/reactive or otherwise nondefault environments are restricted to a certified single frontier consisting solely of the selected internal blocker. K.EnvX-Auto, A5-S/CF-S, K.PSF, Theorem K.1, K-O2/K-O7, the Appendix K-P proof, Appendix T, the side-condition record, and the summaries are aligned. No general future-environment no-rescue certificate is introduced.

**18 July 2026 — K.1A statement and interface-seam correction.**

Theorem K.1A’s statement now explicitly restores A5-L, conditional KObsClosed waiver closure, the point-to-point evaluated-boundary premise, and the certified-cutpoint conclusion that its proof already consumed. K.CertCov now quantifies over an explicit reachable prefix/cutpoint pair. ReturnedAtCycle has one normative main-text definition and an Appendix C pointer. Definition J.RegPhaseReach and Corollary J.RegPhaseReach-Comp are separate citable anchors. R.RegSeq requires checked identification before rejection or theorem-scope exclusion of zero-time dependent NB cascades. The historical 17 July sequential-result-availability entry is restored verbatim; supersession remains recorded only in the later 18 July H3 entry.

**18 July 2026 — consolidated H1/H2/H3 registered-phase revision.**

H1 remains split into Definition K.CertifiedCutpoint/Theorem K.1A for one cutpoint and K.CertCov/Corollary K.1A-Total for universal coverage. H3 is reclassified as a formalization-fidelity defect: the intended registered sequential-interface semantics has no same-Horizon decision-to-dependent-request edge, but the former stop-at-NB hole replay forced decision-before-independent-continuation. Definition R.RegSeq and ReturnedAtCycle now cover successful and failed NB results,

request and Push payload outputs, stable direct inputs, Cat#/source conformance, and later-Horizon dependent use. Lemma I.3 promotes NBHole to an enriched carried ghost obligation with result destinations and NoDependentUse. J.ConstructionAction adopts a normative field-sensitive location vocabulary; A3 is narrowed to the shared decision/result; Dep\_J/FR are revised; J.RegPhaseNF proves the R.LAdm L1/L2/L3 normal form modulo J.HCanon using J.BufSameCycle; and J.UnitExt/J.GWR-Comp use outer Horizon induction with an inner L3 pass. No generic micro-position or new action taxonomy is introduced. For H2, Corollary J.RegPhaseReach-Comp now derives J.RegPhaseReach on the compositional route and supplies L0–L4; K.DrainReach remains the explicit nonwithdrawal/K $\epsilon$ /R.16/K.Drain bridge, and K.RLAdmReach is their conjunction. CF-0, K.CertifiedCutpoint, K.1A, the side-condition record, Appendix S/T, summaries, and the no-unresolved-hole cutpoint invariant are aligned. The separate Lean strategy must propagate the same interface change before mechanization.

**18 July 2026 — H1 certificate coverage and H2 admissibility-interface revision.**

The Appendix K preservation theorem was split into a per-cutpoint result and a coverage corollary. The consolidated H1/H2/H3 revision above retains that H1 architecture but refines H2: the L0–L4 portion is now J.RegPhaseReach, derived from the revised Appendix I/J construction, while K.DrainReach carries the remaining nonwithdrawal/K $\epsilon$ /R.16 evidence. Their conjunction is the K.RLAdmReach consumed by K.1A.

**17 July 2026 — H2 bounded-response serial-route revision.**

The primary K.2b/K.2c/CF-S route now uses validated process-facing bounded-response monitoring rather than the incomplete missing-supply owner trichotomy and Dead\_KS terminal taxonomy. Definitions K.RespMon/K.RespCov/K.RespFail/K.RespClean require default-total internal-site proof coverage, process-facing trigger/response attribution, maximal-finite and cancellation handling, complete-run evidence, and waiver preservation. Lemma K.2b.E and mechanization obligation K-O7 explicitly construct the fair complete frozen Policy-L continuation from a bare A5-L counterexample prefix; K.2b.Q then proves that the liberal Dead\_K cutpoint forces a finite Policy-S response failure, and K.2c converts K.RespClean into A5-L. The perpetual under-supplying-owner example is added explicitly. Dead\_KS, EnvStop, DoneStop, K.EnvOrd, and K.Done remain optional diagnostics only. CF-S, K.PSF, Theorem K.1, Appendix R's KObs boundary, Appendix T, the side-condition record, and both the abstract and main-body Summary are aligned; no Policy-S response bound is transferred to RTL.

**17 July 2026 — Policy-L admissibility and settle-window clarification.**

Definition R.LAdm now supplies one normative execution domain for Appendix K. It distinguishes Policy-L step legality, finite-prefix admissibility for A5-L reachability, and complete admissible executions for fairness, stabilization, and limit arguments; binds the fixed stimulus/environment/witness/arbitration choices; and defines the L1 pre-settle issue window, the L2  $\epsilon$ -settle snapshot, the exact activation point of Definition R.16/K.Drain, and next-cycle availability of committed results. A5-L, K.1A, K.2b, K.Ser, Examples K.CE-2/K.CE-3/K.CE-4, Appendix R, and Appendix T now cite this single definition. The change resolves the ambiguity without changing which independent eager issues Policy L permits or making K.Drain part of the basic Policy-L issue rule.

**17 July 2026 — sequential-result-availability clarification.**

The Issue/Commit semantics now state explicitly that a result committed by a sequential process in cycle  $t$  may affect a dynamically succeeding dependent operation only in a strictly later cycle. The clarification is added to the main Issue definition, Appendix C linkage rules, the Sys\_ref/Policy-L operational interpretation, Lemma I.3 and Corollary I.4, J.DepSound, J.FoundationRank, and the D1 case of Lemma J.DepRank. Same-cycle rendezvous, buffered-channel updates, eager issue, and pipeline overlap remain permitted for requests whose reachability, control, and operands were established independently before the cycle boundary.

**15 July 2026 — J.GWR formal-ordering and SDet clarification revision.**

The J.DepAcyclic proof now uses one graph-independent semantic foundation rank and the new Lemma J.DepRank; the prior union-of-local-orders argument is removed, and Height/J.CanonicalRank are defined only after acyclicity. J.RequiredPrefixCandidate removes the circular reference from J.PCert to the not-yet-derived canonical order. J.ConstructionOrderProvenance/J.ValidatedConstructionOrder provide one interface for theorem-derived and directly validated J.GWR packages. SDet now states functional history determinism as prefix-comparability/no-incompatible-history for run artifacts, while K.CompleteS separately establishes maximality, fairness, and completeness. Remaining legacy time-chain terminology is replaced by the derived or directly validated construction-order ideal chain.

**15 July 2026 — J.GWR snapshot, same-cycle, and dependency-completeness refinement.**

The local-to-global construction now derives its proof order from a generated typed dependency graph rather than accepting J.UnitRank/phase metadata. Definition J.ConstructionAction introduces explicit ReadSet/WriteSet footprints; J.DepSound and J.DepComplete prove prerequisite soundness and non-commutation coverage; J.FoundationRank/J.DepRank prove every edge advances one graph-independent semantic rank; J.DepAcyclic follows; and J.CanonicalRank is reconstructed only afterward. Lemma J.BufSameCycle proves legal sequentialization of simultaneous positive-capacity Push/Pop commits without permitting empty/full fall-through. Definition J.ReqSnapshot and Lemma J.NBDecision move witness-significant non-blocking outcomes out of process-local replay paths and evaluate them from one common attributed boundary snapshot. Lemma I.3, J.PCert/J.CCert/J.LocalGWR, J.PrepareCommute/J.UnitEnable/J.UnitExt, the expanded J.GWR-Comp proof, the side-condition record, and Appendices S/T are updated accordingly.

**15 July 2026 — J.GWR compositional-construction revision.**

Definition J.GWR remains the stable semantic interface but is no longer a mandatory unexplained Post → Ref premise. J.LocalGWR supplies the local certificates, and Lemma I.3 supplies finite process-owned preparation scripts with shared-decision holes. J.Frame/J.LocalLift/J.PrepareCommute/J.UnitEnable/J.UnitExt provide generic gluing; J.GWR-Comp constructs a reachable Sys\_ref witness and J.GWR package. J.O remains the separate pairwise composition lemma.